QUANT COURSE

Arbitrage Pricing & Derivatives

A fifteen-chapter study guide

PART I	Discrete-Time Models	CH 1–2
PART II	Continuous-Time Models	CH 3–5
PART III	Equity Derivatives	CH 6–10
PART IV	Interest-Rate Models	CH 11–13
PART V	Stochastic Vol & Rate Derivatives	CH 14–15

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A self-study companion for the graduate derivatives canon

Contents

1 Quant Course — Arbitrage Pricing & Derivatives

A sixteen-chapter study guide (Chapter 0 prerequisites + 15 numbered chapters) covering arbitrage pricing, stochastic calculus, equity derivatives, short-rate models, rate derivatives, stochastic volatility, and risk measures. The guide is written for a self-study reader who is comfortable with undergraduate probability and real analysis and who wants a single coherent path from a math refresher to a senior-quant-grade understanding of derivatives pricing and hedging. Formulas are typeset in LaTeX; baseline figures are regenerated from a single Python script and live inline next to the concepts they illustrate.

This is the **V5** edition. Relative to V3: - **Chapter 0 added** as a prerequisite math refresher (probability, stochastic processes, linear algebra, calculus/ODE, Jensen's inequality and the $\sigma^2/2$ drift correction). - **Utility theory and indifference pricing removed from Chapter 1.** Pricing is derived from no-arbitrage and replication only; preferences are a tool for incomplete markets, not the spine of derivatives pricing. - **Greeks split into a dedicated Chapter 7** with a comprehensive visual atlas (delta, gamma, theta, vega, rho, vanna, volga, charm, speed, color, dollar-gamma, pin risk, gamma-scalping P&L). - **Risk Measures (Chapter 15) is the capstone**, deliberately positioned at the end so it can draw on every prior chapter's machinery. - **Real-world case studies** added throughout — Buffett's SPX puts, Petrobras ADR arb, Black Monday 1987 portfolio insurance, GameStop gamma squeeze, Volmageddon, negative WTI oil futures, March 2020 COVID vol-surface, LTCM 1998, SVB collapse 2023, Archegos, UK LDI mini-budget 2022, and more.

1.1 Table of Contents

1.1.1 Prerequisites

#	Chapter	Description
0	Math Refresher	Probability essentials, Jensen + $\sigma^2/2$ drift correction, stochastic-process priming, linear algebra (Cholesky, PSD, spectral), calculus & ODE refresher.

1.1.2 Part I — Discrete-Time Models

#	Chapter	Description
1	One-Period Binomial: No-Arbitrage and Replication	Two-asset model, arbitrage definition, risk-neutral measure $\mathbb{Q}$ from no-arbitrage alone, replication recipe, European call.
2	Multi-Period Binomial and the FTAP	Backward induction, CRR parameterisations, FTAP, American options.

1.1.3 Part II — Continuous-Time Models

#	Chapter	Description
3	Stochastic-Calculus Primer	BM construction, quadratic variation, Itô integrals, Itô's lemma, OU/GBM SDEs, Jensen's gap and the $\sigma^2/2$ drift correction (§3.10.1a).
4	Feynman-Kac and the SDE-PDE Bridge	FK zero-drift, discounting, state-dependent coefficients; martingale derivation.
5	Measure Changes, Radon-Nikodym, and Girsanov	RN derivative, density process, Girsanov, two-numeraire switching, T-forward measure.

1.1.4 Part III — Equity Derivatives

#	Chapter	Description
6	Dynamic Hedging — Self-Financing + Black-Scholes PDE	Self-financing strategies, market price of risk, BS PDE and formula, discrete hedging, transaction costs.
7	Greeks	Delta-gamma hedging, vega, theta, rho, dividend-paying assets, put-call parity. **Greek visual atlas** with vanna/volga/charm/speed/color/dollar-gamma/pin-risk figures.
8	Forwards, Futures, and the Black PDE	Forward/futures pricing, Black PDE, Margrabe, convexity adjustment, Bachelier vs Black-76 (negative WTI case).
9	Monte Carlo and Path-Dependent Options	SLLN, GBM generator, forward-starting and barrier options, variance reduction (antithetics, control variates), Asians and autocallables.

1.1.5 Part IV — Stochastic Volatility

#	Chapter	Description
10	Heston Model (Stochastic Volatility)	Heston SDE, characteristic-function pricing, Feller condition, Fourier inversion, smile/skew, calibration.

1.1.6 Part V — Calibration and Interest-Rate Models

#	Chapter	Description
11	Calibration — Lattices, Short Rates, Yield Curve	FTAP on lattices, RN on multinomial trees, rate-tree bootstrap, regularisation, governance. SPX vol-surface workflow case study.
12	Short-Rate Models — Vasicek, Ho-Lee, Hull-White, Affine	Canonical Vasicek derivation, Ho-Lee, Hull-White $\theta(t)$, affine term structure. Negative-rate regime case study.
13	Rate-Derivative Applications — Swaps, CDS, Bond Options, Callables	IRS, par swap rate, CDS, callable bonds, bond options via T-forward measure. LTCM swap-spread arb and SVB duration case studies.
14	Caps, Floors, and Swaptions	Caplets under T-forward measure, Black-76, swaptions under annuity measure. UK LDI mini-budget and LIBOR→SOFR case studies.

1.1.7 Part VI — Capstone

#	Chapter	Description
15	Risk Measures — VaR, CTE, Coherent Risk	VaR (historical/parametric/MC), CTE, coherent risk axioms, Basel evolution to FRTB ES, backtesting, P&L attribution, delta-gamma parametric VaR via Cornish-Fisher. Three failures of VaR (LTCM, Archegos, 2008).

1.2 Course Arc

```
Chapter 0   Math Refresher (Jensen,  ²/2, MGF, Cholesky, ODE)
                     prerequisites only - every later chapter cites it

Part I    Discrete Time
   Chapter 1: One-Period Binomial (no-arb, replication)


   Chapter 2: Multi-Period Binomial + FTAP

Part II    Continuous Time

   Chapter 3: Stochastic-Calculus Primer     foundation



   Chapter 4: FK    Chapter 5: Girsanov / Measure Change


Part III    Equity Derivatives

       Chapter 6: BS PDE (Dynamic Hedging)



   Chapter 7  Chapter 8    Chapter 9
   Greeks     Fwd/Fut      Monte Carlo
              Black PDE

Part IV    Stochastic Vol
```

```
                    Chapter 10: Heston (smile, skew, calibration)

Part V   Calibration & Rates

   Chapter 11: Calibration   Chapter 12: Short-Rate
   Chapter 13: Rate Derivatives   Chapter 14: Caps/Floors/Swaptions

Part VI   Capstone

                    Chapter 15: Risk Measures (VaR, ES, FRTB)
                              - uses every prior chapter's Greeks/dynamics
```

Spine. Chapter 3 is the single prerequisite for every chapter from Chapter 4 onward; its Itô calculus is the language. Chapter 5's Girsanov machinery is cited (not re-derived) by Chapters 6, 8, 12, and 13. Vasicek is derived once, in Chapter 12, and cross-referenced elsewhere. The $\sigma^2/2$ Jensen-gap intuition introduced in Chapter 0 (§0.2) and re-derived from the SDE side in Chapter 3 (§3.10.1a) recurs in every continuous-time chapter — it is the most reused single fact in the guide.

1.3 Regenerating the Figures

```
pip install matplotlib numpy scipy
python docs/guide/build_figures.py
```

Writes baseline PNGs under `docs/guide/figures/`.

1.4 Building the Full PDF

```
pip install pypandoc_binary
python docs/guide/_make_pandoc_pdf.py
```

Requires a working `xelatex` installation. Concatenates README + Chapter-0 through Chapter-15 in numeric order, runs pandoc → xelatex, writes `docs/guide/pdf/Quant-Guide-Full.pdf`.

2 Chapter 0 — Math Refresher

This chapter is a prerequisite dump, not a textbook. Every later chapter assumes what is collected here; nothing here is re-derived later. The reader is presumed to be a sharp advanced-undergrad who has seen probability, linear algebra, ODEs, and analysis at some point and just needs a fast, intuition-first pass to dust off the cobwebs and standardise notation.

The pace is deliberately quick. Statements are precise where precision matters — measurability, σ-algebras, stopping times — and informal where the intuition carries the load. Every formal statement gets exactly one intuition sentence in italics. If a result is non-obvious, the *why* is one sentence; the *what* is the formula. There are numerical examples wherever a number cements something an abstract symbol cannot.

The single most load-bearing section is §0.2 on Jensen's inequality and the $\sigma^2/2$ drift correction. That gap appears in the log-drift of geometric Brownian motion (Ch. 3, 6), in the Itô correction (Ch. 3), in implied-vol convexity adjustments (Ch. 7, 11), in Black's caplet formula (Ch. 14), and in the Heston characteristic function (Ch. 10). Reading §0.2 once carefully will save you from misremembering the sign of the correction at least three times in this book.

> Notation. $\mathbb{P}$ denotes the real-world (physical) measure; $\mathbb{Q}$ a risk-neutral or pricing measure. W_t is a standard Brownian motion under whichever measure is currently active. $\mathcal{F}_t$ is the filtration generated by the relevant noise. $\mathbb{E}$ without a subscript means under whichever measure has just been named. Vectors are columns; x' is the transpose. log is natural log throughout.

2.1 §0.1 Probability Essentials

2.1.1 0.1.1 Measurable spaces and probability measures

A **probability space** is a triple $(\Omega, \mathcal{F}, \mathbb{P})$. Ω is the sample space (all possible outcomes), $\mathcal{F}$ a σ-algebra of subsets of Ω (the events you are allowed to assign probabilities to), and $\mathbb{P} : \mathcal{F} \to [0, 1]$ a probability measure ($\mathbb{P}(\Omega) = 1$, countably additive).

A σ-algebra is closed under complement and countable union. *Intuition: $\mathcal{F}$ is the bookkeeping of "questions you can ask." If you cannot decide whether $\omega \in A$, then A is not in $\mathcal{F}$ and has no probability.* In finance the only σ-algebra you ever construct from scratch is the **filtration** $\{\mathcal{F}_t\}_{t\geq 0}$ — the information available by time t — discussed in §0.3.

A **random variable** is an $\mathcal{F}$-measurable map $X : \Omega \to \mathbb{R}$, meaning $\{X \leq x\} \in \mathcal{F}$ for every x. Measurability is the formal version of "you can actually look up the value of X given the information in $\mathcal{F}$."

Two minor distinctions worth preserving from undergrad. *Almost surely* (a.s.) means "with $\mathbb{P}$-probability 1," and any equation in this guide written without further qualification ($X = Y$ when both are random variables) is implicitly an a.s. statement. *Null sets* — events with $\mathbb{P} = 0$ — are the things we are allowed to ignore; modifying X on a null set produces an a.s.-equal random variable that behaves identically in every probabilistic computation.

2.1.2 0.1.2 Conditional expectation

Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra. The **conditional expectation** $\mathbb{E}[X \mid \mathcal{G}]$ is the unique (a.s.) $\mathcal{G}$-measurable random variable such that

$$\int_A \mathbb{E}[X \mid \mathcal{G}] \, d\mathbb{P} = \int_A X \, d\mathbb{P} \quad \text{for every } A \in \mathcal{G}. \tag{0.1}$$

Intuition: $\mathbb{E}[X \mid \mathcal{G}]$ is the best L^2-prediction of X given the information in $\mathcal{G}$. It is a random variable, not a number, because the prediction depends on which $\mathcal{G}$-event was realised.

Three identities are used constantly in the rest of this guide.

Tower property. For $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$,

$$\mathbb{E}\big[\mathbb{E}[X \mid \mathcal{G}] \,\big|\, \mathcal{H}\big] = \mathbb{E}[X \mid \mathcal{H}]. \tag{0.2}$$

Intuition: forecasting your own forecast based on less information gives the less-informed forecast — averaging is associative.

Taking out what is known. If Y is $\mathcal{G}$-measurable and integrable,

$$\mathbb{E}[YX \mid \mathcal{G}] = Y\,\mathbb{E}[X \mid \mathcal{G}]. \tag{0.3}$$

Intuition: relative to $\mathcal{G}$, the random variable Y is a known constant and pulls out of the expectation.

Independence collapse. If X is independent of $\mathcal{G}$, then $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$. *Intuition: information you cannot use is no information at all.*

These three together do almost all the work in martingale arguments, change-of-measure (Ch. 5), and tree-based pricing (Ch. 1, 2).

A small numerical instance to make (0.2) concrete. Let ξ_1, ξ_2 be independent fair coin flips taking values ± 1, and let $X = \xi_1 + \xi_2 \in \{-2, 0, 0, 2\}$. Set $\mathcal{G} = \sigma(\xi_1)$ and $\mathcal{H} = \{\emptyset, \Omega\}$ (no information).

- $\mathbb{E}[X \mid \mathcal{G}] = \xi_1 + \mathbb{E}[\xi_2] = \xi_1$. With knowledge of the first flip, the best forecast of X is the realised first flip plus the unconditional mean of the second.
- $\mathbb{E}[X \mid \mathcal{H}] = \mathbb{E}[X] = 0$.
- Tower: $\mathbb{E}[\xi_1 \mid \mathcal{H}] = 0 = \mathbb{E}[X \mid \mathcal{H}]$.

Replace ξ with daily returns and $\mathcal{G}, \mathcal{H}$ with two filtration timestamps and the same identity is doing all the work in every multi-period pricing argument in this guide.

2.1.3 0.1.3 Independence vs uncorrelatedness

X and Y are **independent** iff $\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$ for all measurable A, B. They are **uncorrelated** iff $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Independence implies uncorrelated. The converse is false in general; it is true only when (X, Y) are jointly Gaussian.

Counterexample. Let $X \sim N(0, 1)$ and $Y = X^2$. Then $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0 = \mathbb{E}[X]\mathbb{E}[Y]$, so X and Y are uncorrelated. But Y is a deterministic function of X; they are maximally dependent. *Intuition: correlation is a single linear summary; dependence has many non-linear flavours that correlation cannot see.*

This matters in practice: a correlation matrix of returns being near-diagonal does *not* mean the assets are independent. Tail co-movements (jointly large losses on bad days) live in the part of the joint distribution that linear correlation discards. Copulas (briefly touched in Ch. 15) are the right tool when you care about that.

2.1.4 0.1.4 MGF, CGF, and the moment-generating identity

The **moment-generating function** of X is

$$M_X(t) = \mathbb{E}[e^{tX}], \qquad t \in \mathbb{R}, \tag{0.4}$$

defined wherever the expectation is finite. The **cumulant-generating function** is $K_X(t) = \log M_X(t)$.

Existence is not free. $M_X(t)$ exists in an open neighbourhood of 0 iff X has all moments and they grow at most exponentially fast — Gaussians have M everywhere, exponentials in a strip, log-normals only at $t \le 0$ and never on a neighbourhood of 0. *That last point is the reason we use characteristic functions $\varphi_X(t) = \mathbb{E}[e^{itX}]$ rather than M_X in heavy-tailed settings (Heston, Ch. 10).*

When M_X exists near 0, all moments are recovered by differentiation:

$$\mathbb{E}[X^n] = \left.\frac{d^n M_X(t)}{dt^n}\right|_{t=0}, \qquad K_X(t) = \mu t + \tfrac{1}{2}\sigma^2 t^2 + \cdots \tag{0.5}$$

The Gaussian case is canonical: $X \sim N(\mu, \sigma^2) \Rightarrow M_X(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$. *That $\sigma^2/2$ is the same one that will appear as the Jensen gap in §0.2 — they are the same animal seen from different angles.*

The CGF is more useful than the MGF when you want to read off cumulants directly. For a Gaussian, $K_X(t) = \mu t + \frac{1}{2}\sigma^2 t^2$ — a polynomial in t of degree 2, with all higher cumulants vanishing. *That fact (only the first two cumulants are non-zero) is exactly the statement "Gaussians are determined by mean and variance," from a different vantage.* For a Poisson with rate λ, $K(t) = \lambda(e^t - 1)$, infinitely many non-zero cumulants — Poisson is not even close to Gaussian. CGF additivity $K_{X+Y} = K_X + K_Y$ for independent X, Y is what makes them computationally pleasant.

The **characteristic function** $\varphi_X(t) = \mathbb{E}[e^{itX}]$ always exists (the integrand has modulus 1), unlike the MGF. It is the workhorse of Heston pricing (Ch. 10) and any Lévy-process model: when the MGF blows up, the CF still works.

2.1.5 0.1.5 Modes of convergence

Let X_n, X be random variables on a common space.

Mode	Definition
Almost sure (a.s.)	$\mathbb{P}(X_n \to X) = 1$
In L^p	$\mathbb{E}\lvert X_n - X\rvert^p \to 0$
In probability	$\mathbb{P}(\lvert X_n - X\rvert > \varepsilon) \to 0$ for all $\varepsilon > 0$
In distribution	$F_{X_n}(x) \to F_X(x)$ at all continuity points of F_X

The implication diagram (no other arrows hold in general):

$$\text{a.s.} \implies \text{in probability} \implies \text{in distribution}$$

$$L^p \implies \text{in probability} \quad (p \geq 1)$$

Intuition: a.s. and L^p are both strong (they both control the random variables themselves), but neither implies the other; both imply convergence in probability, which is a weak pointwise control; convergence in distribution is the weakest — only marginal laws agree, and the variables need not even live on the same space.

Practitioner consequence. Monte Carlo (Ch. 9) gives L^2 (or even a.s. via the SLLN) convergence of sample means. The CLT gives convergence in distribution of the rescaled error — these are different statements about different things.

A canonical separating example. Let X_n be independent with $\mathbb{P}(X_n = n^2) = 1/n$ and $\mathbb{P}(X_n = 0) = 1 - 1/n$. Then $X_n \to 0$ in probability ($\mathbb{P}(X_n > \varepsilon) = 1/n \to 0$) but $\mathbb{E}[X_n] = n \to \infty$, so X_n does *not* converge in L^1. *Intuition: a vanishingly rare but explosively large outcome can leave probability convergence intact while breaking expectation convergence.* This is exactly the failure mode of naïve Monte Carlo on heavy-tailed payoffs (deep OTM puts on a jump-prone asset, Ch. 9): sample means look like they converge, until one of the rare extreme draws shows up and erases your average.

2.1.6 0.1.6 Borel–Cantelli

Let $\{A_n\}$ be a sequence of events. Define $\{A_n \text{ i.o.}\} = \bigcap_N \bigcup_{n \geq N} A_n$ ("infinitely often").

BC-I. If $\sum_n \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$. *Intuition: rare events with finite total mass cannot happen infinitely often.*

BC-II. If $\{A_n\}$ are independent and $\sum_n \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$. *Intuition: independent events with unbounded total mass must, by chance alone, eventually pile up.*

These two together pin down the long-run behaviour of nearly every limit theorem you will meet in the rest of this guide.

A trading-floor example. Suppose a strategy has a 1-in-100 chance of triggering a 10-sigma drawdown each day, independently across days. Over $n = 1{,}000$ trading days, $\sum \mathbb{P}(A_n) = 10$, finite and well into the regime where BC-I bites if you cap the count, but BC-II says with infinitely many days the event happens infinitely often. *Intuition: independence + non-summable probability = certainty in the long run.* The practical lesson: any non-zero per-period probability of a tail event materialises with probability 1 over a long enough horizon. Do not size positions as if the tail event is "unlikely this year."

2.1.7 0.1.7 Two more workhorse inequalities

Two short-statement inequalities used repeatedly in proofs throughout the guide.

Markov's inequality. For non-negative X and $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Intuition: non-negative random variables cannot put much mass far above their mean — the mean would have to be larger than it is. This is the seed of every concentration inequality.

Chebyshev's inequality. For X with finite variance,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Intuition: variance bounds tail probability, distribution-free. For a Gaussian, the actual probability at $k = 3$ is $\sim 0.27\%$; Chebyshev only gives $\leq 11.1\%$. The bound is loose precisely because it is distribution-free; the value is in proofs where you have only a variance bound.

The Cauchy–Schwarz inequality $\mathbb{E}[XY]^2 \leq \mathbb{E}[X^2]\,\mathbb{E}[Y^2]$ rounds out the basic kit; it is used implicitly any time you write $|\mathrm{Cov}(X, Y)| \leq \sigma_X \sigma_Y$ (the bound that defines the correlation coefficient as living in $[-1, 1]$).

2.2 §0.2 Jensen's Inequality and the $\sigma^2/2$ Drift Correction

This is the most reused single fact in the guide. It appears in the GBM log-drift, in the Itô correction (Ch. 3), in implied-vol convexity adjustments (Ch. 7, 11), in convexity-adjusted forwards (Ch. 8), in Black's caplet pricing (Ch. 14), and any time someone says "geometric mean $\leq$ arithmetic mean" — they are quoting Jensen.

2.2.1 0.2.1 The inequality

For a **convex** function f and integrable X,

$$\mathbb{E}[f(X)] \;\geq\; f(\mathbb{E}[X]). \tag{0.6}$$

For **concave** f, flip the inequality:

$$\mathbb{E}[f(X)] \;\leq\; f(\mathbb{E}[X]). \tag{0.7}$$

Equality holds iff f is affine on the support of X, or X is degenerate.

2.2.2 0.2.2 Geometric intuition

Convexity means: any chord between two points on the graph of f lies *above* the graph between those points. Averaging two values of f is taking a point on a chord; averaging the inputs and then applying f is taking a point on the curve. The chord is above. So $\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$. Concave is the mirror image.

This is not a measure-theoretic fact; it is a statement about chords and curves. The $\mathbb{E}$ does the integration, but the *gap* is geometric — a function of how curvy f is and how spread out X is.

2.2.3 0.2.3 The $\sigma^2/2$ worked example

Let S be log-normal: $\log S \sim N(\mu, \sigma^2)$. Then $S = e^{\log S}$ where $\log S$ is Gaussian. Compute the three relevant quantities.

Expected value of S. Using the Gaussian MGF (0.5) at $t = 1$,

$$\mathbb{E}[S] = \mathbb{E}[e^{\log S}] = M_{\log S}(1) = \exp\left(\mu + \tfrac{1}{2}\sigma^2\right). \tag{0.8}$$

Log of the expected value.

$$\log \mathbb{E}[S] = \mu + \tfrac{1}{2}\sigma^2. \tag{0.9}$$

Expected value of the log.

$$\mathbb{E}[\log S] = \mu. \tag{0.10}$$

The Jensen gap. Subtracting (0.10) from (0.9):

$$\boxed{\log \mathbb{E}[S] - \mathbb{E}[\log S] = \tfrac{1}{2}\sigma^2.} \tag{0.11}$$

This is Jensen's inequality with $f = \log$ (concave): $\mathbb{E}[\log S] \le \log \mathbb{E}[S]$, with the gap exactly $\sigma^2/2$ for the log-normal.

Intuition: the gap measures how much the log function bends over the support of S. Small σ means S is concentrated near its mean and the function is nearly linear there — small gap. Large σ probes wider regions where log *is much more curved — large gap. The factor $1/2$ comes from the second derivative of* log*, just as the $1/2$ in the Taylor expansion of* $\log(1 + x) \approx x - x^2/2$ *does.*

2.2.4 0.2.4 Connection to GBM and the Itô drift correction

Geometric Brownian motion under measure $\mathbb{P}$:

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t, \qquad S_0 \text{ given.} \tag{0.12}$$

The integrated solution (anticipating Ch. 3's Itô calculus, but the result is what matters here) is

$$S_T = S_0 \exp\left[\left(\mu - \tfrac{1}{2}\sigma^2\right) T + \sigma W_T\right]. \tag{0.13}$$

Two expectations to take:

$$\mathbb{E}[S_T] = S_0 e^{\mu T}, \tag{0.14}$$

$$\mathbb{E}\left[\log \frac{S_T}{S_0}\right] = \left(\mu - \tfrac{1}{2}\sigma^2\right) T. \tag{0.15}$$

The $\sigma^2/2$ in the log-drift in (0.13) and (0.15) is *not* a math glitch. It is exactly the Jensen gap (0.11) at $T = 1$ with σ replaced by $\sigma\sqrt{T}$:

$$\log \mathbb{E}[S_T/S_0] - \mathbb{E}[\log(S_T/S_0)] = \mu T - (\mu - \tfrac{1}{2}\sigma^2)T = \tfrac{1}{2}\sigma^2 T. \tag{0.16}$$

The same correction appears as the second-order term in Itô's lemma applied to $\log S$: the deterministic chain rule would give $d(\log S) = dS/S$, but Itô adds $-\frac{1}{2}\sigma^2 dt$ because $dW^2 = dt$ does not vanish (Ch. 3, §3.4). *Three names for one phenomenon: Jensen's gap, the GBM log-drift correction, the Itô second-order term.*

2.2.5 0.2.5 A real number

Take $\mu = 8\%$, $\sigma = 30\%$, $T = 1$ year.

Quantity	Formula	Value
Arithmetic mean of S_T/S_0	$e^{\mu T}$	$e^{0.08} = 1.0833$
Geometric mean of S_T/S_0	$e^{(\mu-\sigma^2/2)T}$	$e^{0.08-0.045} = e^{0.035} = 1.0356$
Log of arithmetic mean	μT	0.0800
Mean of log return	$(\mu - \sigma^2/2)T$	0.0350
Jensen gap	$\sigma^2 T/2$	0.0450

The arithmetic mean exceeds the geometric mean by $\sim 4.7\%$, almost exactly $\sigma^2/2 = 4.5\%$ (the small extra is the higher-order curvature of exp). *In a year of 30-vol equity, the "expected return" of 8% and the "median compounded return" of 3.5% are both correct statements about the same process. They differ by Jensen's gap.*

2.2.6 0.2.6 Practitioner aside

This is why CAGR is always less than the arithmetic mean of annual returns for any non-trivial vol. Pension reports quote CAGR; sales decks quote arithmetic average; both are technically right but they describe different things. The same gap explains why volatility is a *drag* on compounded wealth even when expected returns are positive — the volatility-adjusted growth rate $\mu - \sigma^2/2$ (the **Kelly drift**) can be negative even when $\mu > 0$ if $\sigma^2 > 2\mu$.

Hold this section in working memory through the rest of the guide. Every time you see $\mu - \sigma^2/2$, $r - q - \sigma^2/2$, or $\sigma^2/2$ floating around alone, it is Jensen.

2.2.7 0.2.7 The Jensen gap as a Taylor expansion

There is a second, complementary way to see (0.6) that makes the size of the gap quantitative. Expand f around $\mu = \mathbb{E}[X]$:

$$f(X) = f(\mu) + f'(\mu)(X-\mu) + \tfrac{1}{2}f''(\mu)(X-\mu)^2 + \text{higher order.} \tag{0.17}$$

Take expectations:

$$\mathbb{E}[f(X)] = f(\mu) + \tfrac{1}{2}f''(\mu)\,\mathrm{Var}(X) + \text{higher cumulants.} \tag{0.18}$$

The first-order term vanishes ($\mathbb{E}[X-\mu] = 0$); the second-order term is exactly the Jensen gap to leading order, and it is **proportional to $f''(\mu)$ times the variance of X**. *Intuition: the gap is the curvature of f times the spread of X — both have to be present, and the gap grows linearly in each.*

For $f = \log$ at μ, $f''(\mu) = -1/\mu^2$, so the leading-order log-gap is $-\frac{1}{2}\mathrm{Var}(X)/\mu^2 = -\frac{1}{2}\mathrm{CV}^2$ where CV is the coefficient of variation. For log-normal X with log-vol σ this is *exactly* $-\sigma^2/2$ (no higher-order error) because the third and higher cumulants of $\log X$ are zero — the log of a log-normal is exactly Gaussian. *That is why the Jensen gap for log-normal is closed-form: the Taylor expansion truncates.*

For non-Gaussian distributions the gap has corrections involving skewness and higher cumulants, and Black–Scholes (which prices as if returns were Gaussian) gets the gap "wrong" precisely in proportion to the realised skew and kurt of the return distribution. That mismatch is the entire content of the implied-volatility skew (Ch. 7, 11).

2.2.8 0.2.8 Sign rules to memorise

Three sign conventions trip people up. Memorise them.

Setting	Sign of $\sigma^2/2$ correction
GBM log-drift: $\log S_T = (\mu - \sigma^2/2)T + \sigma W_T$	minus
Expected forward: $\mathbb{E}[S_T] = S_0 e^{\mu T}$	none (Jensen has done its work in the log)

Setting	Sign of $\sigma^2/2$ correction
Itô-corrected drift of $\log S$	minus
Black's formula log-moneyness: $d_\pm = (\log(F/K) \pm \sigma^2 T/2)/(\sigma\sqrt{T})$	$\pm$ depending on d_+ vs d_-
Convexity adjustment, futures vs forward (positive ρ between rate and asset)	plus to the futures price

If you forget the sign in a hurry: **in the log of a log-normal, the bias is a subtraction; in the level of a log-normal, the bias has already been baked in.**

2.2.9 0.2.9 Volatility drag and the Kelly drift

The "Kelly drift" $\mu - \sigma^2/2$ is the expected log-return per unit time, also known as the **geometric growth rate** of the asset. It is what compounds your money. The arithmetic drift μ is what enters the *expected* terminal wealth but not the *typical* terminal wealth.

A worked illustration. Two assets:

- Asset A: $\mu = 12\%$, $\sigma = 50\%$.
- Asset B: $\mu = 8\%$, $\sigma = 15\%$.

	Asset A	Asset B
Arithmetic drift μ	12%	8%
Volatility σ	50%	15%
Variance drag $\sigma^2/2$	12.5%	1.1%
Geometric growth $\mu - \sigma^2/2$	-0.5%	6.9%
Expected wealth multiple after 30y	$e^{3.6} = 36.6\times$	$e^{2.4} = 11.0\times$
Median wealth multiple after 30y	$e^{-0.15} = 0.86\times$	$e^{2.07} = 7.9\times$

Asset A has a higher expected return and a higher expected terminal wealth, but a typical (median) trajectory of Asset A loses money over 30 years. The mean is dragged up by a small probability of enormous outcomes; the median, which is what an actual investor experiences with high probability, is dictated by the geometric drift. This is the volatility drag, and it is why position sizing (Kelly fraction, risk parity) cares about $\sigma^2/2$ as much as it cares about μ. *Picking strategies on raw μ is selecting for the assets whose Jensen gap will eat their typical performance.*

2.3 §0.3 Stochastic-Process Priming

A **stochastic process** is a family of random variables $\{X_t\}_{t\in T}$ on a common probability space, indexed by time $T \subseteq \mathbb{R}_+$. We work in continuous time throughout the guide; the binomial model of Ch. 1–2 is the discrete-time analogue.

2.3.1 0.3.1 Filtrations

A **filtration** $\{\mathcal{F}_t\}_{t\geq 0}$ is an increasing family of σ-algebras: $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$.

Intuition: $\mathcal{F}_t$ is what you know by date t. Information accumulates; you never forget. The natural filtration of a process X is $\mathcal{F}_t^X = \sigma(X_s : s \leq t)$ — the smallest σ-algebra making the entire history of X up to t measurable.

A process X is **adapted** to $\{\mathcal{F}_t\}$ if X_t is $\mathcal{F}_t$-measurable for every t. *Intuition: adapted means "you can read off X_t at time t, not earlier."*

A process X is **predictable** if X_t is $\mathcal{F}_{t-}$-measurable — known just before t. *Intuition: a hedging strategy is predictable; you choose your position based on the past, not on the increment you are about to absorb.*

2.3.2 0.3.2 Martingales

An adapted, integrable process M is a **martingale** if

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s, \qquad s \le t. \tag{0.19}$$

A **submartingale** has $\ge$, a **supermartingale** has $\le$.

Intuition: a martingale is a fair game — your best forecast of tomorrow's value, given everything you know today, is today's value. A submartingale drifts up on average; a supermartingale drifts down.

Three canonical examples used throughout the guide:

Brownian motion. W_t is a martingale under its own filtration: $\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s$ because $W_t - W_s$ is independent of $\mathcal{F}_s$ with mean 0. So is $W_t^2 - t$ (the *compensated quadratic variation*) — Brownian motion has variance growing as t, and $W_t^2 - t$ is the part with the trend removed.

Compensated Poisson. If N_t is a Poisson process with rate λ, then $N_t - \lambda t$ is a martingale. *The compensator λt removes the deterministic drift; what remains is a fair game of jump shocks.* Any jump model (Merton, Kou, variance-gamma) builds a tradable asset by attaching this kind of compensation so the discounted price is a $\mathbb{Q}$-martingale.

Discounted prices under $\mathbb{Q}$. The fundamental theorem of asset pricing (FTAP, Ch. 2) says: a market is arbitrage-free iff there exists a measure $\mathbb{Q}$ under which discounted asset prices S_t/B_t are martingales, where B_t is the numéraire (typically the money-market account). *Intuition: under $\mathbb{Q}$, no asset systematically out-earns the numéraire — that is exactly what "no arbitrage" means.*

2.3.3 0.3.3 Markov property

A process X is **Markov** if for any $s \le t$ and bounded measurable f,

$$\mathbb{E}[f(X_t) \mid \mathcal{F}_s] = \mathbb{E}[f(X_t) \mid X_s]. \tag{0.20}$$

Intuition: the future depends on the past only through the present — knowing the full history is no better than knowing the current state.

Markov is what makes PDEs work for option pricing (Ch. 4, 6): the option price at (t, S_t) depends only on (t, S_t), not on the whole path of S. Heston (Ch. 10) is Markov in the *pair* (S_t, V_t) but not in S_t alone — that is why Heston needs a 2-D PDE and a 2-D characteristic function.

2.3.4 0.3.4 Stopping times

A random time $\tau : \Omega \to [0, \infty]$ is a **stopping time** if $\{\tau \le t\} \in \mathcal{F}_t$ for every t.

Intuition: τ is a stopping time iff you can decide whether to stop using only the information available up to t — no peeking.

The first hitting time of a Brownian motion at level a, $\tau = \inf\{t : W_t = a\}$, is a stopping time. The time of the *maximum* of W over $[0, T]$ is *not* — to know it, you would have to know the future.

Stopping times are how American options (early exercise) and barrier options are formalised: the exercise time of an American option is a stopping time chosen to maximise expected discounted payoff under $\mathbb{Q}$.

Two more useful properties used implicitly later. (i) The minimum and maximum of two stopping times are stopping times — you can compose decisions ("stop at the first knock-out *or* maturity, whichever comes first"). (ii) For τ a stopping time, the **stopped σ-algebra** $\mathcal{F}_\tau = \{A : A \cap \{\tau \le t\} \in \mathcal{F}_t \, \forall t\}$ is the information you have *up to* the (random) stopping time. Optional sampling is then "M_τ is conditionally M_σ relative to $\mathcal{F}_\sigma$" — the obvious extension of the martingale property to random times.

2.3.5 0.3.5 Optional sampling theorem

If M is a uniformly integrable martingale and $\sigma \leq \tau$ are bounded stopping times, then

$$\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] = M_\sigma. \tag{0.21}$$

Intuition: the martingale property survives even if you stop at a smart, information-respecting random time. "No clever stopping rule beats a fair game."

The boundedness or UI condition is essential — without it, doubling strategies break the conclusion. This theorem is what makes the FTAP arguments of Ch. 2 work in the multi-period setting.

2.3.6 0.3.6 Doob decomposition

Any submartingale X decomposes uniquely as

$$X_t = M_t + A_t \tag{0.22}$$

with M a martingale and A a predictable, increasing process with $A_0 = 0$.

Intuition: every "drifting up" process is a fair game plus a foreseeable upward drift; the decomposition separates the risk from the trend. In continuous time this is the Doob–Meyer decomposition; for an Itô process, $A_t = \int_0^t (\text{drift})\, ds$ and $M_t = \int_0^t (\text{volatility})\, dW_s$. The Doob decomposition is the abstract generalisation of "drift plus diffusion."

2.3.7 0.3.7 Quadratic variation

For a continuous semi-martingale X, the **quadratic variation** $[X]_t$ is the limit (in probability) of $\sum (X_{t_{i+1}} - X_{t_i})^2$ over partitions of $[0, t]$ as the mesh shrinks to zero.

For Brownian motion, $[W]_t = t$. *Intuition: increments of size $\sqrt{dt}$ squared and summed over $1/dt$ intervals give $dt \cdot 1/dt = 1$ per unit time.* Compare to a smooth function, whose quadratic variation is zero (smooth wiggles squared sum to nothing). That non-zero quadratic variation is exactly what powers Itô calculus.

For correlated Brownians W^1, W^2 with correlation ρ, the cross-variation is $[W^1, W^2]_t = \rho t$. *This is the continuous-time content of "Cov is bilinear in increments."* Itô's lemma in two variables is built on this identity.

A last fact for Ch. 3: for any Itô process $dX = \mu_t\, dt + \sigma_t\, dW_t$, the quadratic variation is $[X]_t = \int_0^t \sigma_s^2\, ds$. *The drift contributes nothing — drift is smooth, only the noise has quadratic variation.* This is the realised-variance estimator of stochastic vol models (Ch. 10) made formal: integrated variance is the quadratic variation of log-price.

2.3.8 0.3.8 Itô's formula in advance, in one line

For $f \in C^2$ and X an Itô process,

$$df(t, X_t) = \frac{\partial f}{\partial t}\, dt + \frac{\partial f}{\partial x}\, dX_t + \tfrac{1}{2} \frac{\partial^2 f}{\partial x^2}\, d[X]_t. \tag{0.23}$$

You will derive and use this throughout Ch. 3, but pin it now: it is the multivariate Taylor expansion (0.36) with the second-order term *kept* because $d[X]_t$ does not vanish. *Reading it: every smooth function of an Itô process picks up a drift correction proportional to its second derivative times the instantaneous variance — Jensen's gap, in differential form.*

2.4 §0.4 Linear Algebra for Quants

You will use linear algebra constantly: covariance matrices, factor models, correlated Brownian motions, PCA on yield curves, regression-based hedging. This section is the working subset.

2.4.1 0.4.1 Positive semi-definite matrices

A symmetric matrix $A \in \mathbb{R}^{n\times n}$ is **positive semi-definite (PSD)** if

$$x'Ax \geq 0 \quad \text{for all } x \in \mathbb{R}^n. \tag{0.24}$$

Equivalently: all eigenvalues $\lambda_i(A) \geq 0$. **Positive definite (PD)** strengthens to strict inequality; equivalently, all $\lambda_i > 0$.

Why we care: covariance matrices are always PSD. For any random vector X with mean μ and covariance $\Sigma = \mathbb{E}[(X-\mu)(X-\mu)']$,

$$x'\Sigma x = \mathbb{E}\big[(x'(X-\mu))^2\big] \geq 0. \tag{0.25}$$

Intuition: variance of a linear combination cannot be negative; that is the entire content of PSD-ness for Σ.

A degenerate Σ (PSD but not PD) has a zero eigenvalue — there is a non-zero direction x in which the portfolio $x'X$ has zero variance. That happens with collinear assets (e.g. SPY and IVV).

Three other PSD characterisations worth knowing because each is the natural one in different contexts. (i) A is PSD iff it has a square root: $A = B^2$ for some symmetric B. (ii) A is PSD iff it admits an LL′ factorisation (Cholesky, possibly rank-deficient). (iii) A is PSD iff every principal minor (top-left $k \times k$ determinant) is non-negative — Sylvester's criterion, useful for checking PSD-ness of small constructed matrices by hand.

2.4.2 0.4.2 Cholesky decomposition

Every PD matrix A admits a unique decomposition

$$A = LL', \qquad L \text{ lower triangular with positive diagonal.} \tag{0.26}$$

This is the **Cholesky factor**. PSD matrices admit a Cholesky only in a generalised sense (with rank-deficient L); standard libraries return an error or use an LDL′ variant.

Why we care: Cholesky is the practical tool for simulating correlated Brownian motions. If $Z \sim N(0, I_n)$ and we want $X \sim N(0, \Sigma)$, set $X = LZ$ where L is the Cholesky of Σ. Then $\mathbb{E}[XX'] = L\,\mathbb{E}[ZZ']\,L' = LL' = \Sigma$.

In Heston (Ch. 10) and Monte Carlo (Ch. 9), every correlated noise driver passes through a Cholesky. For a 2-D Heston with correlation ρ between asset and variance noise,

$$L = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{pmatrix}. \tag{0.27}$$

The simulation step is then $dW_1 = \sqrt{dt}\,Z_1$, $dW_2 = \sqrt{dt}\,(\rho Z_1 + \sqrt{1-\rho^2}\,Z_2)$ with independent standard normals Z_1, Z_2.

A small numerical Cholesky for a 3-asset case. Take

$$\Sigma = \begin{pmatrix} 1.00 & 0.50 & 0.30 \\ 0.50 & 1.00 & 0.20 \\ 0.30 & 0.20 & 1.00 \end{pmatrix}.$$

Working out the Cholesky entries by the standard recursion ($L_{ii}^2 = \Sigma_{ii} - \sum_{k<i} L_{ik}^2$, $L_{ji} = (\Sigma_{ji} - \sum_{k<i} L_{jk}L_{ik})/L_{ii}$ for $j > i$):

$$L = \begin{pmatrix} 1.0000 & 0 & 0 \\ 0.5000 & 0.8660 & 0 \\ 0.3000 & 0.0577 & 0.9522 \end{pmatrix}.$$

To sanity check: $LL'_{22} = 0.5^2 + 0.866^2 = 0.25 + 0.75 = 1.0$; $LL'_{32} = 0.3 \cdot 0.5 + 0.0577 \cdot 0.866 = 0.15 + 0.05 = 0.20$. *Each Cholesky row builds the new asset's noise as a linear combination of all prior independent noises plus a fresh shock — the variance budget left for the fresh shock is whatever wasn't already explained by the earlier assets.*

2.4.3 0.4.3 Spectral decomposition

For a symmetric A,

$$A = Q\Lambda Q', \qquad Q \text{ orthogonal}, \ \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n). \tag{0.28}$$

The columns of Q are orthonormal eigenvectors; the diagonal of Λ holds the eigenvalues.

Intuition: every symmetric matrix is a diagonal matrix in the right basis. The eigenvectors are that basis.

PCA in one line. If Σ is the covariance matrix of a return vector and $\Sigma = Q\Lambda Q'$, then the columns of Q are the principal components and the λ_i are the variances of the returns projected onto each PC. The first PC of the US Treasury yield curve covariance is roughly flat (level), the second roughly monotone (slope), the third roughly U-shaped (curvature) — a classical and stable result that the term-structure chapters (Ch. 12, 13) lean on.

Concretely, on US Treasury 1Y/2Y/5Y/10Y/30Y daily changes, the first three eigenvalues typically explain $\sim$ 85%/10%/4% of total variance. *That decay is what makes 1- and 2-factor short-rate models (Ch. 12) cover most of the term structure's behaviour with a tiny parameter count.*

2.4.4 0.4.4 Matrix calculus

Two derivatives appear constantly.

Linear form. For $a, x \in \mathbb{R}^n$,

$$\frac{\partial(a'x)}{\partial x} = a. \tag{0.29}$$

Quadratic form. For symmetric A,

$$\frac{\partial(x'Ax)}{\partial x} = 2Ax. \tag{0.30}$$

(For non-symmetric A, the answer is $(A + A')x$; the symmetric case is what you almost always want.)

These two are sufficient for the first-order conditions in mean-variance optimisation, regression, and quadratic hedging. *Intuition: $a'x$ is linear, so its derivative is the constant slope a; $x'Ax$ is the quadratic form, and its derivative is "twice Ax" by the same logic that $\frac{d}{dx}(ax^2) = 2ax$.*

2.4.5 0.4.5 Condition number

The **condition number** of a non-singular A is

$$\kappa(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} \quad \text{(for symmetric PD; in general, the ratio of singular values).} \tag{0.31}$$

It controls how badly small input perturbations distort the solution to $Ax = b$:

$$\frac{\|\Delta x\|}{\|x\|} \lesssim \kappa(A) \frac{\|\Delta b\|}{\|b\|}. \tag{0.32}$$

Rule of thumb: solving $Ax = b$ in IEEE double precision loses about $\log_{10} \kappa(A)$ digits of accuracy. $\kappa = 10^{12}$ leaves about 4 digits — useless for hedging.

This bites in calibration (Ch. 11): when you solve $J'J\,\Delta\theta = J'r$ in Gauss–Newton and the parameters are nearly degenerate (e.g. Heston v_0 and θ for short maturities), κ explodes and the update is dominated by noise. The cure is regularisation (Tikhonov adds λI to $J'J$, lowering κ).

2.4.6 0.4.6 A real number

Take a 3-asset equicorrelated correlation matrix with $\rho = 0.7$:

$$R = \begin{pmatrix} 1 & 0.7 & 0.7 \\ 0.7 & 1 & 0.7 \\ 0.7 & 0.7 & 1 \end{pmatrix}. \tag{0.33}$$

Eigenvalues of an $n \times n$ equicorrelation matrix with off-diagonal ρ are $1+(n-1)\rho$ (once, eigenvector $(1, 1, \dots, 1)/\sqrt{n}$) and $1-\rho$ (with multiplicity $n-1$). For $n = 3$, $\rho = 0.7$:

$$\lambda_1 = 1 + 2 \cdot 0.7 = 2.4, \qquad \lambda_2 = \lambda_3 = 1 - 0.7 = 0.3. \tag{0.34}$$

Reading the spectrum: $\lambda_1 = 2.4$ (out of total trace 3) is the "common factor" — the equally-weighted portfolio carries 80% of the variance. The other 20% lives in 2-D of long-short directions.

Condition number $\kappa = 2.4/0.3 = 8$ — well-conditioned. Cholesky is numerically stable and Monte Carlo will work cleanly. Push ρ to 0.99: $\lambda_1 = 2.98$, $\lambda_2 = \lambda_3 = 0.01$, $\kappa = 298$. The matrix is now nearly singular and Cholesky starts losing 2–3 digits of precision.

2.4.7 0.4.7 Inverses, pseudo-inverses, and least squares

The least-squares problem $\min_\beta \|y - X\beta\|^2$ has normal equations $X'X\beta = X'y$, with closed-form solution $\hat{\beta} = (X'X)^{-1}X'y$ when $X'X$ is invertible. *Intuition: project y onto the column space of X; $\hat{\beta}$ is the coordinate vector of that projection in the columns of X.*

When $X'X$ is rank-deficient (collinear regressors), the inverse fails and one uses the **Moore–Penrose pseudo-inverse** X^+, equivalently a ridge regression $\hat{\beta} = (X'X + \lambda I)^{-1}X'y$ for some $\lambda > 0$. The ridge perturbation lowers the condition number by raising every eigenvalue by λ, at the cost of biasing $\hat{\beta}$ toward zero. *This is the prototype of every regularised calibration in Ch. 11.*

2.4.8 0.4.8 Quadratic optimisation

The unconstrained quadratic $\min_x \frac{1}{2}x'Ax - b'x$ with PD A has stationary point $Ax = b$, i.e. $x^\star = A^{-1}b$, with optimal value $-\frac{1}{2}b'A^{-1}b$. *This is what you solve in mean-variance portfolio optimisation: A is the covariance matrix, b is the expected-return vector, $x^\star$ is the (unconstrained) optimal weight vector.* The fact that A is PD is what guarantees the second-order conditions for a minimum.

With a linear equality constraint $Cx = d$ the Lagrangian gives a closed-form too: a 2×2 block solve. Mean-variance with a budget constraint $\mathbf{1}'x = 1$ is exactly this case.

2.4.9 0.4.9 Eigenvalues of common parametric families

Two parametric eigenvalue identities save you derivation time when calibrating models with structured noise.

Equicorrelation matrix. $R_{ij} = \rho$ off-diagonal, 1 on the diagonal: eigenvalues $1 + (n-1)\rho$ (once) and $1 - \rho$ (with multiplicity $n - 1$). Used in (0.34) and seen any time a model assumes a single common factor.

Tridiagonal Toeplitz (AR(1)-style). $A_{ii} = a$, $A_{i,i\pm1} = b$: eigenvalues $a + 2b\cos(k\pi/(n+1))$ for $k = 1, \dots, n$. *Used by every finite-difference scheme for the BS PDE (Ch. 6) — those grids generate exactly this matrix structure, and the cosine eigenvalues control numerical stability.*

Knowing these spectra lets you read off condition numbers and stability limits without ever calling an eigensolver.

2.5 §0.5 Calculus and ODE Refresher

You should not need this section. It is here as a fallback so the rest of the guide has nothing left to assume.

2.5.1 0.5.0 What you should already know

Differentiation, integration, the fundamental theorem of calculus, and existence/uniqueness for ODE initial-value problems with Lipschitz right-hand side. We do not re-derive these. The one technical point worth a one-line reminder: a function f is **Lipschitz** with constant L if $|f(x) - f(y)| \le L|x - y|$, and Picard–Lindelöf says Lipschitz right-hand sides give unique local solutions to ODEs. *Every coefficient in this guide that needs to define a unique SDE solution will be Lipschitz in its state variable, so you can stop worrying about it.*

2.5.2 0.5.1 Taylor expansion

For f smooth at x,

$$f(x+h) = f(x) + f'(x)h + \tfrac{1}{2}f''(x)h^2 + \tfrac{1}{6}f'''(x)h^3 + O(h^4). \tag{0.35}$$

Multivariate version with $h \in \mathbb{R}^n$:

$$f(x+h) = f(x) + (\nabla f)'h + \tfrac{1}{2}h'(\nabla^2 f)h + O(\|h\|^3). \tag{0.36}$$

This is the deterministic counterpart of Itô's lemma (Ch. 3). In ordinary calculus, the second-order term $\frac{1}{2}f''(x)h^2$ vanishes as $h \to 0$ because it is $O(h^2)$. *In Itô calculus, h is a Brownian increment with $\mathrm{Var}(h) \sim h$ rather than h^2, so the second-order term contributes a $O(dt)$ piece that does not vanish — that is the entire reason Itô differs from the chain rule.*

The same point in option-pricing language: a delta hedge captures the first-order term $f'(x)h$. The remaining error per timestep is the gamma term $\frac{1}{2}f''(x)h^2$, and over a hedging interval dt, that error has expected size $\frac{1}{2}\Gamma\sigma^2 S^2\, dt$. *The $\sigma^2/2$ from §0.2 reappears as the per-unit-time cost of being short gamma — Jensen's gap is the gamma P&L of an unhedged convex position.*

A small numerical sanity check. Take $f(x) = e^x$ at $x = 0$ with $h = 0.1$:

Order	Approximation	Value
1st	$1 + 0.1$	1.10000
2nd	$1 + 0.1 + \frac{0.01}{2}$	1.10500
3rd	$\cdots + \frac{0.001}{6}$	1.10517
Exact	$e^{0.1}$	1.10517

Each order kills another factor of h in the residual; that is the meaning of $O(h^k)$.

2.5.3 0.5.2 Chain rule, multivariate

For $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}^n$,

$$\frac{d}{dt}f(g(t)) = (\nabla f)(g(t)) \cdot g'(t) = \sum_i \frac{\partial f}{\partial x_i}\frac{dg_i}{dt}. \tag{0.37}$$

For change of variables in integration with $u = \phi(x)$,

$$\int f(\phi(x))|\phi'(x)|\, dx = \int f(u)\, du. \tag{0.38}$$

The Jacobian determinant generalises this to multidimensions and is the root of the change-of-density formulae used in Girsanov (Ch. 5).

In the multivariate case with $u = \phi(x)$ for $\phi : \mathbb{R}^n \to \mathbb{R}^n$ differentiable and bijective on the domain,

$$\int f(\phi(x))|\det J_\phi(x)|\, dx = \int f(u)\, du, \qquad J_\phi = \left(\frac{\partial \phi_i}{\partial x_j}\right). \tag{0.39}$$

Intuition: the determinant of the Jacobian is the local volume scaling of the change of variables; it is what makes the integrand densities consistent. The Radon–Nikodym derivative $d\mathbb{Q}/d\mathbb{P}$ in change of measure is the infinite-dimensional analogue: a "Jacobian" between two probability laws on path space.

A concrete instance for densities. If X has density f_X and $Y = g(X)$ for monotone differentiable g, the density of Y is

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{dg^{-1}(y)}{dy} \right|.$$

Apply this to $Y = e^X$ with $X \sim N(\mu, \sigma^2)$ and you get the log-normal density — the foundational distribution of asset prices in this guide. *The factor $1/y$ in the log-normal density is exactly the Jacobian of $g^{-1} = \log$.*

2.5.4 0.5.3 Separable ODE

The single most-used ODE in the guide,

$$\frac{dy}{dt} = a(t)y \iff \frac{dy}{y} = a(t)\,dt \iff \log y(t) = \log y(0) + \int_0^t a(s)\,ds. \tag{0.40}$$

Canonical case $a(t) \equiv \mu$: $y(t) = y(0)e^{\mu t}$ — exponential growth.

The deterministic version $dS/S = \mu\,dt$ gives $S_t = S_0 e^{\mu t}$. The stochastic version $dS/S = \mu\,dt + \sigma\,dW$ gives the GBM solution $S_t = S_0 \exp[(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t]$ (cf. (0.13)) — same separability, plus the Itô correction.

2.5.5 0.5.4 Linear first-order ODE and the integrating factor

The linear ODE

$$y'(t) + p(t)\,y(t) = q(t) \tag{0.41}$$

is solved by multiplying through by the **integrating factor** $\mu(t) = \exp(\int p(s)\,ds)$. The left-hand side then becomes a clean derivative:

$$\frac{d}{dt}[\mu(t)\,y(t)] = \mu(t)\,q(t). \tag{0.42}$$

Integrate and divide by $\mu(t)$:

$$y(t) = \mu(t)^{-1} \left[y(0) + \int_0^t \mu(s)\,q(s)\,ds \right]. \tag{0.43}$$

Intuition: the integrating factor $\mu(t)$ is constructed precisely so that $\mu \cdot (y' + py) = (\mu y)'$ — it cancels the operator $\frac{d}{dt} + p(t)$ into a clean total derivative. This is one of the few solution recipes in classical ODEs that generalises immediately to SDEs.

2.5.6 0.5.5 Vasicek as a linear ODE plus noise

The Vasicek short-rate model (Ch. 12),

$$dr_t = \kappa(\theta - r_t)\,dt + \sigma\,dW_t, \tag{0.44}$$

has a closed-form solution that looks intimidating until you notice it is *exactly* the linear-ODE structure of (0.37) with $p(t) \equiv \kappa$ and $q(t) \equiv \kappa\theta$, plus a noise term $\sigma\,dW$.

Apply the integrating factor $\mu(t) = e^{\kappa t}$. The mean-reverting deterministic ODE $dr/dt + \kappa r = \kappa\theta$ has solution

$$r_t^{\text{det}} = e^{-\kappa t} r_0 + \theta\,(1 - e^{-\kappa t}). \tag{0.45}$$

Adding the stochastic forcing using the same integrating factor,

$$r_t = e^{-\kappa t} r_0 + \theta\,(1 - e^{-\kappa t}) + \sigma \int_0^t e^{-\kappa(t-s)}\, dW_s. \tag{0.46}$$

Reading: the deterministic part is the linear ODE solution; the stochastic part is the same integrating-factor trick applied to the dW noise. The factor $e^{-\kappa(t-s)}$ in the stochastic integral is the Green's function of the linear operator — old shocks decay exponentially with mean-reversion rate κ.

The integral term is Gaussian with mean zero and variance $\frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})$ (use the Itô isometry, Ch. 3). So r_t is Gaussian, mean-reverting, with long-run distribution $N(\theta, \sigma^2/(2\kappa))$.

This is the working pattern for *every* Ornstein–Uhlenbeck-style process you meet in this guide: identify the linear ODE skeleton, write down the integrating factor, add the stochastic forcing.

A real number to fix the picture. Take $\kappa = 0.5$ (per year), $\theta = 4\%$, $\sigma = 1\%$, $r_0 = 2\%$.

- Mean reversion half-life: $\log 2/\kappa = 1.39$ years — the gap to the long-run mean halves every 17 months.
- Long-run mean: $\theta = 4\%$.
- Long-run standard deviation: $\sigma/\sqrt{2\kappa} = 0.01/1 = 1.0\%$.
- Time-1 distribution: $r_1 \sim N\Big(e^{-0.5} \cdot 0.02 + 0.04 \cdot (1 - e^{-0.5}),\ \frac{0.01^2}{1}(1 - e^{-1})\Big) = N(0.0279,\ 0.0079^2)$.

So a year out, the rate is centred at 2.79% with a 79bp one-sigma — it has crawled three-quarters of the way back to θ and acquired most of its long-run dispersion. *The two timescales — mean reversion and noise build-up — are completely controlled by κ. The half-life is $\log 2/\kappa$; the variance saturates at $\sigma^2/(2\kappa)$. Vasicek has only three knobs and they each do exactly one thing.*

2.5.7 0.5.6 PDE pattern recognition

You will meet exactly four PDE archetypes in the guide. Recognising them on sight saves the time of solving from scratch.

Archetype	Equation form	Where it appears
Heat / diffusion	$u_t = \frac{1}{2}\sigma^2 u_{xx}$	BS after change of variable (Ch. 5–6)
Drift-diffusion (Kolmogorov backward)	$u_t + \mu u_x + \frac{1}{2}\sigma^2 u_{xx} = 0$	Feynman–Kac (Ch. 4)
Drift-diffusion with discounting	$u_t + \mu u_x + \frac{1}{2}\sigma^2 u_{xx} - ru = 0$	BS PDE (Ch. 6)
Two-factor diffusion	$u_t + \mathcal{L}_S u + \mathcal{L}_v u + \rho\sigma_S\sigma_v u_{Sv} - ru = 0$	Heston (Ch. 10)

Each is a parabolic PDE; each has a Feynman–Kac stochastic representation; each can be solved by characteristics, Fourier methods, finite differences, or Monte Carlo. The choice of method is a numerical question, not a modelling one.

2.6 §0.6 Reading the Rest of This Guide

This chapter is a reference, not a destination. It is read once carefully, then revisited when later chapters cite a specific section. Here is which prerequisite each later chapter actually leans on, so you can target rereads.

Chapter	Topic	Prerequisites from §0
Ch. 1	One-Period Binomial	§0.1.1 (probability spaces), §0.1.2 (conditional expectation) — light

Chapter	Topic	Prerequisites from §0
Ch. 2	Multi-Period Binomial & FTAP	§0.1.2 (tower), §0.3.2 (martingales), §0.3.5 (optional sampling)
Ch. 3	Stochastic Calculus Primer	§0.1 throughout, §0.3 throughout, §0.5.1 (Taylor — Itô is the stochastic Taylor)
Ch. 4	Feynman–Kac	§0.3.3 (Markov), §0.5.4 (linear ODEs) — PDEs are ODEs in time with elliptic operator in space
Ch. 5	Measure Changes & Girsanov	§0.1.4 (MGF — the Radon–Nikodym derivative is a normalised exponential MGF), §0.3.2 (martingales)
Ch. 6	Dynamic Hedging (BS PDE)	§0.5.1 (Taylor), §0.2 (the $\sigma^2/2$ in the BS PDE is Jensen)
Ch. 7	Greeks	§0.4.4 (matrix calculus for cross-Greeks), §0.2 (vol convexity = Vega convexity = Jensen)
Ch. 8	Forwards/Futures/Black PDE	§0.2 (convexity adjustment for futures vs forwards is Jensen), §0.5.4 (linear ODE for the forward)
Ch. 9	Monte Carlo	§0.1.5 (modes of convergence), §0.4.2 (Cholesky for correlated paths)
Ch. 10	Heston	§0.4.2 (Cholesky for the asset-vol correlation), §0.1.4 (characteristic functions because Heston has no MGF for S_T)
Ch. 11	Calibration	§0.4.5 (condition number — calibration is an ill-posed inverse problem), §0.4.4 (gradient of squared error)
Ch. 12	Short-Rate Models	§0.5.5 (Vasicek as integrating factor), §0.3.2 (martingale measure for the bank account)
Ch. 13	Rate-Derivative Applications	§0.5.4 (forward rates as ODE solutions), §0.3.6 (Doob–Meyer for the convexity adjustment)
Ch. 14	Caps/Floors/Swaptions	§0.2 (Black's formula is GBM under the forward measure — full Jensen content)
Ch. 15	Risk Measures (capstone)	§0.1.5 (convergence for empirical VaR), §0.1.6 (Borel–Cantelli for tail-event frequency)

The single highest-traffic section is §0.2. The second is §0.3.2 (martingales). The third is §0.4.2 (Cholesky). If you read those three carefully you will follow 80% of the guide without ever flipping back here.

A note on what is *not* in this chapter. Itô's lemma, stochastic integration, and the Itô isometry are deliberately deferred to Ch. 3, where they are derived in full. The Black–Scholes PDE and Feynman–Kac connection are in Ch. 4 and 6. Change of measure and the Radon–Nikodym derivative are in Ch. 5. Numerical methods (finite differences, Monte Carlo variance reduction, characteristic-function inversion) are in their respective application chapters. Anything that is genuinely *content* of the guide is treated where it is used; only prerequisites live here.

If a later chapter writes "by Jensen" or "by the tower property" without further comment, it is referring to this chapter. The cross-references are not exhaustive — a single line of algebra in Ch. 10 may quietly use three things from §0.1 — but the table above is a faithful guide to which sections each chapter most depends on.

2.6.1 0.6.1 A cross-reference walkthrough

To make the prerequisite map concrete, here is the chain of refresher dependencies invoked in the Black–Scholes PDE derivation of Ch. 6, taken from a single page of that chapter.

1. Take a smooth function $V(t, S)$ of the asset price S_t following GBM, $dS = \mu S\,dt + \sigma S\,dW$.
2. Apply Itô's lemma (Ch. 3, prefigured in §0.3.8 here) to get

$$dV = \left(V_t + \mu S V_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS}\right) dt + \sigma S V_S\,dW.$$

3. Note the second-order term $\frac{1}{2}\sigma^2 S^2 V_{SS}$ is the **Jensen gap of V in S**, weighted by the instantaneous variance — that is §0.2.7 made dynamic.
4. Form a self-financing portfolio long one option, short V_S shares; the dW terms cancel by construction (delta hedging).
5. The remaining drift must equal the riskless rate (no arbitrage), so $V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV - rSV_S$ — the BS PDE.
6. The PDE is solved either by Feynman–Kac (Ch. 4, using §0.3.3 Markov), or by transforming to the heat equation (Ch. 5, change of variables from §0.5.2).
7. The resulting price formula is a Gaussian expectation under $\mathbb{Q}$ (Ch. 5, using §0.1.4 MGF and §0.3.2 martingales).

That single page leans on (in order) §0.5.1, §0.3.8, §0.2.7, §0.3.2 (martingale numéraire argument), §0.3.3, §0.5.2, §0.1.4, §0.1.2 (tower for change of measure). Eight sections of this chapter, on one page of Ch. 6 — and not all of them are flagged with explicit citations. *That is the density at which this chapter is consumed by the rest of the guide.*

2.6.2 0.6.2 What to memorise

If you have to pick five facts from this chapter to keep in working memory, it is these.

1. **Tower property** $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}]$ — every change-of-measure or change-of-numéraire argument hinges on this.
2. **Jensen gap for log-normal** $= \sigma^2/2$ per unit time. The same gap appears as Itô correction, GBM log-drift, vol drag, and BS gamma P&L.
3. **Martingale property under** $\mathbb{Q}$ of discounted price processes. This is the modern definition of "no arbitrage" and is what makes pricing work.
4. **Cholesky of** Σ is the recipe for correlated Brownian simulation. Memorise the 2-D form $L = (1, 0; \rho, \sqrt{1-\rho^2})$.
5. **Linear ODE solved by integrating factor.** Vasicek and every short-rate model with a closed form is built on this single trick.

Everything else can be looked up. These five recur often enough that not having them at fingertip cost is a tax on every chapter.

The remainder of the guide is now self-contained relative to this refresher. Turn to Ch. 1 for the binomial model, where the entire arbitrage-pricing framework is built from scratch in the simplest possible setting before the continuous-time machinery is unleashed in Ch. 3 and beyond.

End of Chapter 0.

One-period binomial tree

Figure 1: One-period binomial tree

3 Chapter 1 — One-Period Binomial: No-Arbitrage and Replication

This chapter develops arbitrage-free pricing in the simplest possible model: a single risky asset with two future values, observed over one time step. We introduce a numeraire bond, define arbitrage, and show that the absence of arbitrage forces a unique synthetic probability measure under which discounted asset prices are martingales. Every continuous-time pricing formula in the rest of the book — Black–Scholes, Heston, HJM — is a limiting case of what is derived in the next few pages.

> Narrative spine. A derivative's price is the cost of the portfolio that replicates it, computed as a discounted expectation under a measure that makes traded assets martingales. The binomial model isolates this idea cleanly with no stochastic calculus, no measure-theoretic probability, and no Itô's lemma — yet every conceptual ingredient of the continuous-time theory appears in finite-dimensional form. We do not invoke utility theory or preferences: replication forces the price to depend only on hedging cost. Indifference pricing for non-replicable claims is treated separately in the appendix.

3.1 1. The One-Period Binomial Tree

Consider a single risky asset A observed at two dates $t = 0$ and $t = 1$. At $t = 1$ the price takes one of two values, A_u (up) or A_d (down), with real-world up-probability

$$p \in (0, 1). \tag{1.1}$$

Strict positivity ($p \in (0,1)$, not $[0,1]$) keeps both states genuinely reachable under $\mathbb{P}$ and forces any equivalent martingale measure $\mathbb{Q}$ to assign positive weight to both branches as well — without it the pricing theory collapses on equivalence grounds.

ASCII diagram of the A-tree:

```
                 A_u   (prob. p)



  A_0



                 A_d   (prob. 1-p)

t = 0             t = 1
```

One-period binomial tree.

The (random) time-1 price can be written as a function of a Bernoulli indicator $x_1 \in \{1, 0\}$:

$$A_1 = A_u\, x_1 + A_d\,(1 - x_1), \qquad x_1 = \begin{cases} 1 & \text{w.p. } p \\ 0 & \text{w.p. } 1 - p. \end{cases} \tag{1.2}$$

A single random coordinate x_1 generates the whole filtration $\mathcal{F}_1$ — one source of randomness, one degree of freedom to hedge it with. That dimension match (one coin, two assets) is exactly what later delivers a *unique* risk-neutral measure. Replace x_1 with a three-valued random variable and the count breaks: three payoffs to match, two assets, generically no replication — the simplest incomplete market.

A naïve "discounted expectation under $\mathbb{P}$" guess for the price is

$$A_0 \;=\; \frac{p\, A_u + (1 - p)\, A_d}{1 + r}, \qquad A_0 \in (A_d, A_u), \tag{1.3}$$

with r the one-period risk-free rate. (1.3) is *not* a pricing rule — $\mathbb{P}$ does not price, only a risk-neutral measure $\mathbb{Q}$ does. (1.3) implicitly assumes A earns the risk-free rate, but a risky asset earns r plus a premium; that premium is what distinguishes $\mathbb{P}$ from $\mathbb{Q}$.

3.1.1 1.1 Two numerical examples — why expected value is not enough

Two symmetric lotteries with identical $\mathbb{P}$-expectation but very different "felt" risk:

$$p = \tfrac{1}{2}, \qquad A_u = 100, \qquad A_d = -100. \tag{1.4}$$

$$p = \tfrac{1}{2}, \qquad A_u = 10^6, \qquad A_d = -10^6. \tag{1.5}$$

Both have zero expected payoff; no one would price them the same — the second is existentially ruinous. Expected value alone cannot price for a risk-averse agent. The miracle of no-arbitrage pricing (§2) is that risk-aversion cancels out as soon as the payoff is replicable.

3.1.2 Case study: Buffett's long-dated SPX put sale (2006–2026)

Berkshire Hathaway sold ∼ \$36B notional of European-style put options on four global equity indices (S&P 500, FTSE, Euro Stoxx 50, Nikkei) between 2004 and 2008, collecting roughly \$4.5B in premium upfront and contracting to pay only at maturities clustering between 2018 and 2028. Buffett famously called the Black-Scholes price of these contracts "Alice in Wonderland." By the time the last expirations rolled off in 2026, every index sat well above its 2006-era strikes and the entire \$36B notional expired worthless — Berkshire kept the full \$4.5B premium plus 20 years of float.

Context. The contracts were European, deeply out-of-the-money (struck close to the 2006–2008 spot levels), and could not be exercised before maturity — the only cashflow risk was the terminal payoff $\max(K - S_T, 0)$ at expiries 15–20 years out. Counterparties (mostly investment banks repackaging structured-product downside protection) wanted a long-vol, long-equity-tail hedge; only an entity with a permanent capital base could plausibly *write* it without onerous collateral terms. Buffett's pitch was simple: over 20 years, the indices in question had never been negative once you included dividends, and Berkshire's $\mathbb{P}$-view assigned essentially zero probability to the strikes being breached at expiry. Realised history validated him: over 2006–2026, the S&P 500 went from ∼ 1280 in March 2006 to ∼ 5300 in May 2026, a 4× cumulative move; the Nikkei staged a similar (delayed) recovery; FTSE and Euro Stoxx 50 both finished above their 2006 marks. Every option ended worthless.

Math mapping. Read this through the one-period machinery of §1. Berkshire's $\mathbb{P}$-expected payoff was deeply negative (i.e. the option had near-zero $\mathbb{P}$-expected loss to the writer); the bank counterparties' $\mathbb{Q}$-expected payoff, computed with a Black-Scholes vol surface implying ∼ 20% vol over a 20-year horizon, was substantial — fat enough to justify quoting the contract at a multi-billion dollar premium. The disagreement is not mathematical error on either side; it is the formal one-period observation that the *risk-neutral* price (2.2) requires a *replicating hedge*, and at 20-year tenors there is no liquid hedge instrument — no traded 20-year SPX option, no 20-year variance swap, nothing to short against. The synthetic $\mathbb{Q}$ in §2 is constructed from prices of *traded* hedge instruments at the relevant horizon; without those, the formal $\mathbb{Q}$-price is academic and the actual transaction price gets pulled back toward the writer's $\mathbb{P}$-view plus a risk-premium charge for warehousing the un-hedgeable risk. Buffett's edge was that Berkshire's capital base (then \$120B, now well over \$300B) made the firm *effectively risk-neutral* at this scale — the marginal utility of a \$36B tail loss, conditional on it occurring inside a portfolio that size, was approximately linear, so Berkshire could quote a price close to its $\mathbb{P}$-expected loss rather than a distorted utility-of-loss price. The premium it collected was the variance risk premium that the broader market could not absorb for lack of hedge counterparties.

Lesson. Three takeaways for the new hire. First, the §1 separation of "preference-free pricing where replication is possible" from "preference-driven pricing where it is not" is not a textbook nicety — it is the entire economic reason this trade existed. The bank counterparties had no choice but to lay off the risk somewhere; Berkshire was the only entity sized to take it without requiring a hedge. Second, $\mathbb{Q}$-pricing is a lever, not a law: when no hedge instrument exists at your horizon, the *formal* $\mathbb{Q}$-price loses its no-arb teeth and the realised price collapses to "what does the marginal warehouser charge to hold this risk on its book." Third, the trade is a clean illustration of the variance risk premium: implied vol over 20-year horizons is structurally higher than realised vol because there is no efficient mechanism to short it; the writers of long-dated puts capture that premium and absorb the (rare, fat-tailed) downside in exchange. Berkshire kept \$4.5B + 20 years of float because its capital base was large enough to make the un-hedgeable risk warehousable on its balance sheet at a price the market was willing to pay.

3.2 2. Two-Asset Model and Arbitrage

Introduce a second asset and argue purely from the absence of arbitrage. The result is a formula that looks like an expected value — under a synthetic measure $\mathbb{Q}$ that prices every replicable claim correctly.

Two assets at $t = 0, 1$: risky A and numeraire B, both driven by the *same* Bernoulli coin. In the bond case $B_u = B_d = B_0(1 + r)$; the general case allows B its own leaves:

```
          A_u                     B_u

 A_0              B_0

          A_d                     B_d

       risky A              numeraire B
    (driven by same Bernoulli coin x_1)
```

The "same coin" assumption is the discrete-time avatar of "one Brownian motion" in continuous time: it matches risk dimension to hedge dimension and forces market completeness. The general $B_u \neq B_d$ setup extends to FX and stochastic-interest-rate models.

3.2.1 2.1 Portfolio value

Introduce a portfolio (α, β) holding α units of A and β units of B:

$$V_0 \;=\; \alpha A_0 + \beta B_0, \tag{2.1}$$

$$V_1 \;=\; \alpha A_1 + \beta B_1. \tag{2.2}$$

Weights are signed: $\alpha < 0$ is a short position in A, $\beta < 0$ is borrowing at the bond rate. Self-financing is automatic across the single $[0, 1]$ interval (no rebalancing event); §7 imposes it explicitly in the multi-period setting. The map $(\alpha, \beta) \mapsto (V_1^u, V_1^d)$ is linear with matrix rows (A_u, B_u) and (A_d, B_d); invertibility $(A_u B_d \neq A_d B_u)$ is exactly market completeness, revisited in §5.

3.2.2 2.2 Definition of arbitrage

An arbitrage strategy (portfolio) (α, β) is one such that

$$\text{(i)} \quad V_0 = 0, \tag{2.3}$$

$$\text{(ii)} \quad \exists\, t \text{ s.t.} \qquad \text{(a)} \;\; \mathbb{P}(V_t \geq 0) = 1 \quad \text{(never lose)}, \tag{2.4}$$

$$\text{(b)} \;\; \mathbb{P}(V_t > 0) > 0 \quad \text{(sometimes win)}. \tag{2.5}$$

In plain English: enter at zero cost, never lose, sometimes win — a free lunch. The "weak" form (2.5) (positive gain on a positive-probability set) is the version that delivers the EMM in the FFT.

3.2.3 2.3 Sign-sanity toy trees

Three toy trees illustrate when the definition fires:

- Tree (a): $0 \to \{1, 0\}$ — arbitrage (cost 0 today, outcome ≥ 0 always, positive sometimes).
- Tree (b): $0 \to \{1, -1\}$ — not an arbitrage (down-state loses).
- Tree (c): $0 \to \{1, 1\}$ — arbitrage (pays 1 for sure at cost 0).

Sandwich check on A_0/B_0:

Tree label	A_0	A_d	A_u	Check $A_d/B_d <$ $A_0/B_0 <$ A_u/B_u? (with $B \equiv 1$)	Verdict
$(-)$	10	5	20	$5 < 10 < 20$	No arb
$(+)$	2	4	15	$4 \not< 2$	Arb
(14 ...)	≈ 14	0	0	violates	Arb

For tree $(-)$: $q = (A_0 - A_d)/(A_u - A_d) = (10-5)/(20-5) = 1/3$. If A_0/B_0 sits strictly between the two relative-price leaves the model is viable; outside, one leaf dominates.

3.2.4 2.4 Scaled-tree arbitrage illustration

Bond tree $1 \to \{1+r, 1+r\}$ vs. asset tree $1 \to \{1, 1\}$. With α in the asset and $-\alpha$ in the bond, the zero-cost portfolio pays $\{(1-r)\alpha,\ -\alpha r\}$. Arbitrage would require

$$\begin{pmatrix} 1-r > 0 \\ -r < 0 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1-r < 0 \\ -r > 0 \end{pmatrix}, \tag{2.6}$$

which simplify to the impossible sign-combinations

$$\boxed{r > 0 \text{ and } r < 1} \quad \| \quad \boxed{r < 0 \text{ and } r > 1} \tag{2.7}$$

so for reasonable r no arb exists; the practical upshot is the standard ordering (2.8) below.

3.2.5 2.5 The standard ordering

$$A_0 \in (A_d, A_u), \qquad A_u > A_d, \qquad B_0 > 0. \tag{2.8}$$

The "up"/"down" labelling is WLOG; positivity of B lets us form ratios A/B. Strict inequalities are the discrete analogue of $\sigma > 0$ in Black–Scholes.

3.3 3. No-Arbitrage Bound on A_0

Absence of arbitrage forces A_0 to lie strictly between its discounted up and down values. We show it twice — bond case first, then general numeraire.

3.3.1 3.1 Absolute-price derivation (bond case)

With $B_u = B_d = B_0(1+r)$, impose $V_0 = 0$:

$$\beta \;=\; -\alpha\,\frac{A_0}{B_0}. \tag{3.1}$$

A zero-cost portfolio is parameterised by the single number α. Substitute into the terminal values:

No-arbitrage requires d < 1+r < u

Figure 2: No-arbitrage requires d < 1+r < u

$$\text{up:}\quad \alpha A_u + \beta B_0(1+r) \;=\; (A_u - (1+r)A_0)\alpha, \tag{3.2}$$

$$\text{down:}\quad \alpha A_d + \beta B_0(1+r) \;=\; (A_d - (1+r)A_0)\alpha. \tag{3.3}$$

If both brackets share a sign, an appropriate α creates a free lunch. The only viable case is opposite signs, and by $A_u > A_d$ the up-bracket is the positive one:

$$A_u - (1+r)A_0 > 0 \quad \text{and} \quad A_d - (1+r)A_0 < 0. \tag{3.4}$$

These combine to

$$\boxed{A_d \;<\; A_0(1+r) \;<\; A_u} \qquad \text{(no-arb).} \tag{3.5}$$

I.e. the forward price $A_0(1+r)$ must lie strictly between the two future spot values:

```
                    → future value
     A_d       A_0(1+r)       A_u

  No-arbitrage forces the forward A_0(1+r) to lie strictly inside [A_d, A_u].
```

No-arbitrage requires $d < 1+r < u$.

$A_0(1+r)$ is the *forward price* of A; the sandwich is the only configuration with no dominance. In gross-return form, this is the textbook "$d < 1+r < u$".

3.3.2 3.2 Relative-price derivation (general numeraire)

The bond's role was purely as a clock; any strictly positive numeraire B works the same way. With $\beta = -\alpha A_0/B_0$:

$$\alpha A_u + \beta B_u \;=\; \alpha\left(A_u - \frac{A_0}{B_0}\,B_u\right), \tag{3.6}$$

$$\alpha A_d + \beta B_d \;=\; \alpha\left(A_d - \frac{A_0}{B_0}\,B_d\right). \tag{3.7}$$

For no arbitrage the two bracketed factors must have opposite signs:

$$\left.\begin{aligned} A_u - \frac{A_0}{B_0}\,B_u \;&>\; 0 \\ A_d - \frac{A_0}{B_0}\,B_d \;&<\; 0 \end{aligned}\right\} \quad \text{all other cases lead to arb.} \tag{3.8}$$

(Implicit non-degeneracy: the 2×2 payoff matrix $\left(\begin{smallmatrix} A_u & A_d \\ B_u & B_d \end{smallmatrix}\right)$ is non-singular, equivalently $A_u/B_u \neq A_d/B_d$. This is the market-completeness hypothesis; for a bond numeraire $B_u = B_d$ it holds automatically, for a general numeraire it must be imposed.)

Equivalently, dividing through by $B_u, B_d > 0$:

$$\boxed{\frac{A_d}{B_d} \;<\; \frac{A_0}{B_0} \;<\; \frac{A_u}{B_u}} \qquad \text{(no-arbitrage condition).} \tag{3.9}$$

In the bond case (3.9) reduces to (3.5). The numeraire-free form generalises to FX (foreign bond), swaptions (annuity), and spread options (one of the underlyings).

3.3.3 3.3 Restatement as existence of a risk-neutral weight q

Convert the inequality into an equality. (3.9) is equivalent to: $\exists\, q \in (0,1)$ such that

$$\boxed{\frac{A_0}{B_0} \;=\; q\,\frac{A_u}{B_u} \;+\; (1-q)\,\frac{A_d}{B_d}} \quad\Longleftrightarrow\quad \text{no arbitrage.} \tag{3.10}$$

A number lies strictly between two endpoints iff it is a convex combination with weights in $(0,1)$. Rearranging: $q = (A_0/B_0 - A_d/B_d)/(A_u/B_u - A_d/B_d)$ — the relative position of the current relative price within the leaf interval, depending only on observable prices, not on p or utility. For an asset with positive excess return $q < p$: $\mathbb{Q}$ tilts weight to the down-state, the Girsanov shift $\mu - r$ that §8 makes explicit.

Equivalently,

$$\tilde{A}_0 \;=\; \mathbb{E}^{\mathbb{Q}}\big[\,\tilde{A}_1\,\big], \qquad \text{where} \qquad \tilde{A}_t \;=\; \frac{A_t}{B_t} \quad \text{(relative price)}, \tag{3.11}$$

under $\mathbb{Q}$ with up-probability q. The relative price $\tilde{A}$ is a $\mathbb{Q}$-*martingale* — zero expected drift in numeraire units. The same q then prices every contingent claim written on (A, B).

3.4 4. A Third Asset: Replication and Arbitrage-Free Pricing

Add a contingent claim C with payoff (C_u, C_d). Its price is pinned down uniquely by (A, B). Two demonstrations: q-consistency here, explicit replication in §5.

```
          A_u              B_u              C_u

  A_0              B_0              C_0

          A_d              B_d              C_d

      underlying       numeraire       contingent claim
```

Each of A and C, when priced in units of B, must be a martingale under some $q \in (0,1)$:

$$\tilde{A}^B_0 \;=\; \mathbb{E}^{\mathbb{Q}}\big[\,\tilde{A}^B_1\,\big] \;\;\longrightarrow\;\; q^{BA} \in (0,1), \tag{4.1}$$

$$\tilde{C}^B_0 \;=\; \mathbb{E}^{\mathbb{Q}}\big[\,\tilde{C}^B_1\,\big] \;\;\longrightarrow\;\; q^{BC} \in (0,1). \tag{4.2}$$

If $q^{BA} \neq q^{BC}$ no single EMM exists — the FFT signal of arbitrage. With q^{BA} extracted from (A, B), the fair C_0 must satisfy

$$\frac{C_0}{B_0} \;\overset{?}{=}\; \frac{C_u}{B_u}\,q^{BA} \;+\; \frac{C_d}{B_d}\,(1 - q^{BA}). \tag{4.3}$$

Disagreement signals mispricing.

3.4.1 4.1 Numerical example — mispriced third asset

Three trees:

```
        20                  1                 120

  10                  1                 110

        5                   1                 100

       (A)                 (B)          (C, short position)
```

Step 1 — q^{BA} from (A, B) via (3.10):

$$10 \;=\; 20\,q^{BA} + 5\,(1 - q^{BA}) \quad\Rightarrow\quad q^{BA} = \tfrac{1}{3}. \tag{4.4}$$

Step 2 — extract q^{BC} from (C, B):

$$110 \;=\; 120\,q^{BC} + 100\,(1 - q^{BC}) \quad\Rightarrow\quad q^{BC} = \tfrac{10}{20} = \tfrac{1}{2}. \tag{4.5}$$

Two weights, two answers — arbitrage.

Step 3 — forcing $q = 1/3$ onto C:

$$\widehat{C_0} \;=\; 120 \cdot \tfrac{1}{3} \;+\; 100 \cdot \tfrac{2}{3} \;=\; 40 + 66\tfrac{2}{3} \;\approx\; 106.67 \;<\; 110. \tag{4.6}$$

$C_0 = 110$ is too expensive. Short C at 110, buy the (A, B) replicant for 106.67, pocket \$3.33.

Step 4 — explicit replication. Match C's payoff with $V_0 = 110$:

$$V_0 \;=\; 10\alpha + \beta - 110 \;=\; 0 \quad\Rightarrow\quad \beta = 110 - 10\alpha. \tag{4.7}$$

Match each leaf (with C sold short, hence the $-C_i$ term):

$$\text{up:}\quad 20\alpha + \beta - 120 \;=\; 10\alpha - 10, \tag{4.8}$$

$$\text{down:}\quad 5\alpha + \beta - 100 \;=\; -5\alpha + 10 \;=\; 0. \tag{4.9}$$

The down-leaf gives $\alpha = 2$, $\beta = 90$, $V_1^{\text{up}} = 10$. Both diagnostics (q-consistency and explicit replication) agree.

3.5 5. Replication — The General Recipe

Solve for (α, β) reproducing C's payoff state-by-state; the fair price of C is then the current value of that portfolio.

$$C_u \;=\; \alpha A_u + \beta B_u, \tag{5.1}$$

$$C_d \;=\; \alpha A_d + \beta B_d. \tag{5.2}$$

Then by no-arbitrage

$$C_0 \;=\; \alpha A_0 + \beta B_0, \tag{5.3}$$

otherwise an arbitrage exists (buy cheaper, short expensive). The 2×2 system has determinant $A_u B_d - A_d B_u \neq 0$ iff $A_u/B_u \neq A_d/B_d$ — market completeness.

3.5.1 5.1 Closed-form replicating weights

Multiply (5.1) by B_d, (5.2) by B_u, subtract:

$$\alpha\,(A_uB_d - A_dB_u) \;=\; C_uB_d - C_dB_u. \tag{5.4}$$

Hence

$$\boxed{\alpha \;=\; \frac{C_uB_d - C_dB_u}{A_uB_d - A_dB_u}} \tag{5.5}$$

$$\boxed{\beta \;=\; \frac{C_uA_d - C_dA_u}{B_uA_d - B_dA_u}} \tag{5.6}$$

α is the *delta* of C. In the bond case it reduces to $\alpha = (C_u - C_d)/(A_u - A_d)$, the "rise over run" delta; β is the residual.

3.5.2 5.2 Pricing identity and explicit q

Substituting back into (5.3) yields

$$\boxed{\frac{C_0}{B_0} \;=\; q\,\frac{C_u}{B_u} \;+\; (1-q)\,\frac{C_d}{B_d}} \tag{5.7}$$

where

$$q \;=\; \frac{\dfrac{A_0}{B_0} - \dfrac{A_d}{B_d}}{\dfrac{A_u}{B_u} - \dfrac{A_d}{B_d}}. \tag{5.8}$$

The same q prices every claim. The physical probability p appears nowhere: traders disagreeing about p still agree on C_0 given observable prices. Self-consistency:

$$\frac{A_0}{B_0} \;=\; q\,\frac{A_u}{B_u} \;+\; (1-q)\,\frac{A_d}{B_d}. \tag{5.9}$$

3.5.3 5.3 Bond case: the classical formula

In the bond case $B_u = B_d = B_0(1+r)$, formulas (5.7)–(5.8) reduce to

$$C_0 \;=\; \frac{1}{1+r}(q\,C_u + (1-q)\,C_d), \qquad q \;=\; \frac{A_0(1+r) - A_d}{A_u - A_d}. \tag{5.10}$$

$q \in (0,1)$ is precisely the no-arb bound (3.5). Check: $A_0 = 100, A_u = 110, A_d = 90, r = 5\%$ gives $q = 0.75$; a call struck at 100 prices to $C_0 \approx 7.14$.

3.5.4 5.4 First Fundamental Theorem (one-period form)

$$\text{No-arbitrage} \iff \exists\,\mathbb{Q} \sim \mathbb{P} \text{ s.t. for all traded assets } X,\;\; \tilde{X}_0 \;=\; \mathbb{E}^{\mathbb{Q}}[\,\tilde{X}_1\,],$$

European call and put payoffs

Figure 3: European call and put payoffs

where

$$\tilde{X}_t \;=\; \frac{X_t}{B_t} \qquad \text{and} \qquad \mathbb{P}(B_t > 0) = 1 \quad (B \text{ is called a numeraire asset}). \tag{5.11}$$

FFT: economic no-free-lunches $\Leftrightarrow$ probabilistic existence of an EMM. $\mathbb{Q} \sim \mathbb{P}$ ensures the two measures agree on null sets. When $B_t = e^{rt}$ the associated $\mathbb{Q}$ is the *risk-neutral measure*; other numeraires (stock, forward) appear in later chapters.

In the bond case:

$$\boxed{C_0 \;=\; \frac{1}{1+r}\,\mathbb{E}^{\mathbb{Q}}[\,C_1\,]} \qquad \text{otherwise } \exists \text{ an arbitrage.} \tag{5.12}$$

(5.12) is the discrete-time prototype of every practical derivatives pricing formula. In incomplete markets (more risks than instruments) uniqueness fails — multiple EMMs, the content of the Second FFT.

3.6 6. Worked Example — European Call on A Struck at 100

Setup: $A : 100 \to \{120, 90\}$; $B : 1 \to \{1, 1\}$ $(r = 0)$; C payoff $\{20, 0\}$ (call struck at $K = 100$).

A call/put gives the right (not obligation) to buy/sell A at K at $t = 1$:

$$\text{Call}(A_1; K) \;=\; \max(A_1 - K,\ 0). \tag{6.1}$$

$$\text{Put}(A_1; K) \;=\; \max(K - A_1,\ 0). \tag{6.2}$$

The payoffs at maturity are the canonical hockey-sticks:

```
C_1 = max(A_1 - K, 0)








0                        → A_1
                        K = 100

Call payoff: flat at 0 for A_1  K, then rises at slope 1.
(The kink at K is the source of gamma, vega, and smile.)
```

Vanilla call and put payoffs.

3.6.1 6.1 Solve replication

System from (5.1)–(5.2): $120\alpha + \beta = 20$, $90\alpha + \beta = 0$. Subtract: $\alpha = 2/3$, $\beta = -60$.

$$\alpha = \tfrac{2}{3}, \qquad \beta = -60. \tag{6.3}$$

Delta $\alpha = 2/3$ is the discrete analog of $\Delta = N(d_1)$.

3.6.2 6.2 No-arbitrage value of C

$$C_0 \;=\; \tfrac{2}{3}\cdot 100 + (-60) \;=\; 6\tfrac{2}{3}. \tag{6.4}$$

3.6.3 6.3 Cross-check via risk-neutral formula

From (5.10) with $r = 0$:

$$100 \;=\; q\cdot 120 + (1-q)\cdot 90 \;=\; 90 + 30q \;\;\Rightarrow\;\; q = \tfrac{1}{3}. \tag{6.5}$$

Therefore

$$C_0 \;=\; q\cdot 20 + (1-q)\cdot 0 \;=\; \tfrac{1}{3}\cdot 20 \;=\; 6\tfrac{2}{3}. \tag{6.6}$$

Both methods agree: $C_0 = 6\frac{2}{3}$, independent of p.

3.6.4 6.4 Constructing the arbitrage when $C_0 = 6$ (mispriced)

Trees:

```
       120                 1                  20

 100               1                6

       90                  1                  0

      (A)                 (B)         (C, mispriced at 6)
```

Strategy: short $\frac{2}{3}$ of A, long 60 of B, long 1 unit of C. Initial cashflow:

$$V_0 \;=\; -\tfrac{2}{3}\cdot 100 + 60 + 6 \;=\; -\tfrac{2}{3}. \tag{6.7}$$

i.e. the arbitrage pays $+\frac{2}{3}$ today. Terminal values:

$$V_1^{(u)} \;=\; -\tfrac{2}{3}\cdot 120 + 60 + 20 \;=\; 0, \tag{6.8}$$

$$V_1^{(d)} \;=\; -\tfrac{2}{3}\cdot 90 + 60 + 0 \;=\; 0. \tag{6.9}$$

The agent pockets $\frac{2}{3}$ today, zero risk at $t = 1$ — confirming the no-arb price is $6\frac{2}{3}$.

3.7 7. Multi-Period Binomial

Multi-period pricing is one-period pricing applied iteratively backward through the tree.

Extend to N periods. Let x_n be i.i.d. Bernoulli ± 1 with $\mathbb{P}(x_n = +1) = p$. The asset compounds multiplicatively:

$$A_n \;=\; A_{n-1}\, e^{c\, x_n}, \tag{7.1}$$

where c is a step size (to be calibrated in §8). The tree branches recursively:

```
                          A_uu

                     A_u

                A_0            A_ud  (= A_du, recombining)
```

```
                        A_d

                                  A_dd

     t = 0          t = 1                  t = 2
```

and likewise for B. These are recombining trees, so $A_{ud} = A_{du}$.

The multiplicative update guarantees $A_n > 0$ on every path and makes $\log A_n$ a random walk — the precondition for the log-normal CLT limit. After N steps the terminal price $A_0 u^k d^{N-k}$ has $\mathbb{P}$-probability $\binom{N}{k} p^k (1-p)^{N-k}$ — binomial in the up-count.

3.7.1 7.1 Filtration and arbitrage

A strategy (α_t, β_t) must be *adapted* to $\mathcal{F}_t$ (no peeking at A_{t+1}). An arbitrage has $V_0 = 0$ and $\exists\, t$ with $\mathbb{P}(V_t \geq 0) = 1$, $\mathbb{P}(V_t > 0) > 0$. We also impose *self-financing* ($V_{t+1}^- = V_{t+1}$) — without it, every claim would be trivially replicable by capital injection.

3.7.2 7.2 Martingale characterisation (multi-period)

$$\boxed{\text{no arb} \iff \exists\, \mathbb{Q} \sim \mathbb{P} \text{ s.t. } \forall\, t < s \text{ and all traded } X, \quad \tilde{X}_t \;=\; \mathbb{E}^{\mathbb{Q}}\Big[\, \tilde{X}_s \,\Big|\, \mathcal{F}_t \,\Big]\,.} \tag{7.2}$$

Relative prices are $\mathbb{Q}$-martingales. The tower property lets us price by *backward induction*: roll back one step at a time using (5.10). For American options, at each node take $\max(\text{exercise}, \text{discounted continuation})$. Path-dependent claims add state variables.

3.8 8. Matching Drift and Volatility

Calibrate (c, p) so the tree approximates geometric Brownian motion with drift μ, volatility σ:

$$A_n \;=\; A_{n-1}\, e^{\sigma\sqrt{\Delta t}\, x_n} \tag{8.1}$$

(step size $c = \sigma\sqrt{\Delta t}$). Matching $\sigma^2 T$ to Nc^2 forces $c^2 = \sigma^2 \Delta t$ — the *diffusive scaling* that makes the limit Brownian rather than trivial.

3.8.1 8.1 Matching the mean of the log-return

Sum N steps:

$$\ln\!\left(\frac{A_T}{A_0}\right) \;=\; \sigma\sqrt{\Delta t}\,\sum_{n=1}^{N} x_n, \qquad N\,\Delta t = T. \tag{8.2}$$

Take expectation:

$$\mathbb{E}\!\left[\ln \frac{A_T}{A_0}\right] \;=\; \sigma\sqrt{\Delta t}\cdot N\,(2p-1) \;=\; (\mu - \tfrac{1}{2}\sigma^2)\,T. \tag{8.3}$$

The target $(\mu - \frac{1}{2}\sigma^2)T$ is the Itô correction for GBM. Solve for p:

$$2p - 1 \;=\; \frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\,\sqrt{\Delta t}, \tag{8.4}$$

$$\boxed{\; p \;=\; \tfrac{1}{2}\left[\,1 \;+\; \frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\,\sqrt{\Delta t}\,\right].\;} \tag{8.5}$$

For A to grow on average at $e^{\mu T}$, $\log A$ must grow at $(\mu - \frac{1}{2}\sigma^2)T$ — the Itô penalty. Equity with $\mu = 10\%$, $\sigma = 20\%$, $\Delta t = 1/252$ gives $p \approx 0.513$ (a 1.3% daily tilt that aggregates to 8% annual log-drift over 252 steps).

3.8.2 8.2 Matching the variance

$$\mathbb{V}\left[\ln\frac{A_T}{A_0}\right] \;=\; \sigma^2\,\Delta t\,\cdot N\,\mathbb{V}[x_i]. \tag{8.6}$$

For the Bernoulli ± 1:

$$\mathbb{V}[x_i] \;=\; \mathbb{E}[x_i^2] - (\mathbb{E}[x_i])^2 \;=\; 1 - \left(\frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\right)^2 \Delta t. \tag{8.7}$$

Hence

$$\mathbb{V}\left[\ln\frac{A_T}{A_0}\right] \;=\; \sigma^2\,T\cdot\left(1 - \left(\frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\right)^2 \Delta t\right). \tag{8.8}$$

As $\Delta t \to 0$ the variance tends to $\sigma^2 T$.

3.8.3 8.3 Risk-neutral probability q for the calibrated tree

The calculation parallels §8.1 with $\mu \mapsto r$. Bank account at rate r:

$$B_t \;=\; e^{rt}, \qquad B_n \;=\; B_{n-1}\,e^{r\,\Delta t}. \tag{8.9}$$

The martingale identity

$$\frac{A_t}{B_t} \;=\; \mathbb{E}^{\mathbb{Q}}\left[\,\frac{A_s}{B_s}\,\middle|\,\mathcal{F}_t\right] \tag{8.10}$$

applied to one step gives

$$\frac{A_{n-1}}{B_{n-1}} \;=\; \frac{A_{n-1}}{B_{n-1}}\,e^{\sigma\sqrt{\Delta t} - r\Delta t}\,q \;+\; \frac{A_{n-1}}{B_{n-1}}\,e^{-\sigma\sqrt{\Delta t} - r\Delta t}\,(1-q). \tag{8.11}$$

Cancel A_{n-1}/B_{n-1} and solve:

$$q \;=\; \frac{e^{r\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}}. \tag{8.12}$$

CRR form. Writing $u = e^{\sigma\sqrt{\Delta t}}$ and $d = e^{-\sigma\sqrt{\Delta t}}$, (8.12) becomes the familiar Cox–Ross–Rubinstein formula

$$q \;=\; \frac{e^{r\Delta t} - d}{u - d}. \tag{8.13}$$

$d < e^{r\Delta t} < u$ is exactly $q \in (0,1)$. (8.13) is the workhorse of discrete-time option pricing.

3.8.4 8.4 Small-Δt expansion of q

Using $e^x \sim 1 + x + \frac{1}{2}x^2 + o(x^2)$:

Numerator:

$$\text{num} \;=\; (1 + r\Delta t) - (1 - \sigma\sqrt{\Delta t} + \tfrac{1}{2}\sigma^2\Delta t) + \cdots \;=\; \sigma\sqrt{\Delta t} + (r - \tfrac{1}{2}\sigma^2)\Delta t + \cdots \tag{8.14}$$

Denominator:

$$\text{denom} \;=\; (1 + \sigma\sqrt{\Delta t} + \tfrac{1}{2}\sigma^2\Delta t) - (1 - \sigma\sqrt{\Delta t} + \tfrac{1}{2}\sigma^2\Delta t) + \cdots \;=\; 2\sigma\sqrt{\Delta t} + \cdots \tag{8.15}$$

Ratio:

$$q \;=\; \tfrac{1}{2}\left[1 \;+\; \frac{r - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right] + \cdots \tag{8.16}$$

This matches (8.5) with $\mu \mapsto r$: the swap $\mu \mapsto r$ *is* the Girsanov shift from $\mathbb{P}$ to $\mathbb{Q}$ in one line — the equity risk premium $\mu - r$ absorbed into the coin bias.

$$\mathbb{E}^{\mathbb{Q}}\left[\ln\frac{A_T}{A_0}\right] \;=\; (r - \tfrac{1}{2}\sigma^2)\,T, \tag{8.17}$$

$$\mathbb{V}^{\mathbb{Q}}\left[\ln\frac{A_T}{A_0}\right] \;=\; \sigma^2\,T + \xi\,\Delta t, \tag{8.18}$$

with $\xi\Delta t \to 0$. Volatility survives the measure change (Girsanov shifts drifts only) — this is why σ is calibrated from market prices without ever estimating μ. Inverting (5.10)/(8.13) at a quoted option price gives the *implied volatility*.

3.9 9. Limiting Distribution

As $N \to \infty$ the calibrated tree converges to a log-normal distribution — equivalently, GBM. §11 gives the rigorous CF proof. Take $N \to \infty$ in (8.2):

$$\ln\left(\frac{A_T}{A_0}\right) \;=\; \sigma\sqrt{\Delta t}\sum_{n=1}^{N} x_n \;\xrightarrow[N\to+\infty]{d}\; \mathcal{N}(\,\cdot\,;\,\cdot\,), \tag{9.1}$$

with

$$\text{mean} = (\mu - \tfrac{1}{2}\sigma^2)\,T, \qquad \text{var} = \sigma^2\,T. \tag{9.2}$$

CLT on N scaled coin flips with bias (8.5). Equivalently:

$$A_T \;\stackrel{d}{=}\; A_0\,e^{(\mu - \frac{1}{2}\sigma^2)\,T \,+\, \sigma\sqrt{T}\,Z}, \qquad Z \sim_{\mathbb{P}} \mathcal{N}(0;1). \tag{9.3}$$

Taking the expectation (MGF of a Gaussian):

$$\mathbb{E}[A_T] \;=\; A_0\,e^{(\mu - \frac{1}{2}\sigma^2)T}\cdot\mathbb{E}\left[e^{\sigma\sqrt{T}\,Z}\right] \;=\; A_0\,e^{(\mu-\frac{1}{2}\sigma^2)T}\cdot e^{\frac{1}{2}\sigma^2 T} \;=\; A_0\,e^{\mu T}. \tag{9.4}$$

The Itô $-\frac{1}{2}\sigma^2 T$ in the median exponent and the $+\frac{1}{2}\sigma^2 T$ from the MGF cancel: μ is the growth rate of the *expectation*, $\mu - \frac{1}{2}\sigma^2$ that of the *log-median*. Under $\mathbb{Q}$, $\mathbb{E}^{\mathbb{Q}}[A_T] = A_0 e^{rT}$.

3.10 10. Existence of Martingale Measure $\Rightarrow$ No Arbitrage

The proof of one direction of the FFT — short, general, no completeness required.

Setup. $A_t^{(1)}, \dots, A_t^{(n)} \geq 0$ $\mathbb{P}$-a.s., $B_t > 0$ $\mathbb{P}$-a.s., $t \in \{0, 1\}$.

Hypothesis. Suppose $\exists\, \mathbb{Q} \sim \mathbb{P}$ s.t.

$$\frac{A_0^{(k)}}{B_0} \;=\; \mathbb{E}^{\mathbb{Q}}\left[\frac{A_1^{(k)}}{B_1}\right], \qquad k = 1, \dots, n. \tag{10.1}$$

Claim. Any portfolio $(\alpha^{(1)}, \dots, \alpha^{(n)}, \beta)$ such that

$$\mathbb{P}(V_1 \geq 0) = 1, \qquad \mathbb{P}(V_1 > 0) > 0, \tag{10.2}$$

must have $V_0 > 0$, ruling out arbitrage (which would require $V_0 = 0$).

Derivation. Compute V_0 directly:

$$V_0 \;=\; \beta\, B_0 + \sum_{k=1}^{n} \alpha^{(k)} A_0^{(k)}. \tag{10.3}$$

Factor out B_0 and use the martingale hypothesis:

$$V_0 \;=\; B_0\left[\beta \;+\; \sum_{k=1}^{n} \alpha^{(k)}\, \frac{A_0^{(k)}}{B_0}\right] \;=\; B_0\, \mathbb{E}^{\mathbb{Q}}\left[\frac{B_1}{B_1}\,\beta \;+\; \sum_{k=1}^{n} \alpha^{(k)}\, \frac{A_1^{(k)}}{B_1}\right]. \tag{10.4}$$

Combining:

$$V_0 \;=\; B_0\, \mathbb{E}^{\mathbb{Q}}\left[\frac{\beta\, B_1 \;+\; \sum_{k=1}^{n} \alpha^{(k)} A_1^{(k)}}{B_1}\right] \;=\; B_0\, \mathbb{E}^{\mathbb{Q}}\left[\frac{V_1}{B_1}\right] \;>\; 0. \tag{10.5}$$

The last inequality uses $B_0, B_1 > 0$ and (since $\mathbb{Q} \sim \mathbb{P}$) $V_1 \geq 0$ $\mathbb{Q}$-a.s. with $V_1 > 0$ on a $\mathbb{Q}$-positive set. Hence $V_0 > 0$ and no portfolio satisfying (10.2) can have $V_0 = 0$. ■

The converse (no arb $\Rightarrow \exists\mathbb{Q}$) requires a separating-hyperplane argument on the zero-cost payoff cone (Dalang–Morton–Willinger).

3.11 11. Central Limit Theorem for the Binomial Tree

CF proof that the calibrated tree of §8 converges to the Gaussian claimed in §9.

Setup. Let

$$X \;=\; \sigma\sqrt{\Delta t}\, \sum_{n=1}^{N} x_i, \qquad \Delta t \;=\; \frac{T}{N}. \tag{11.1}$$

With the calibrated probabilities

$$\mathbb{P}(x_i = +1) \;=\; p \;=\; \tfrac{1}{2}\left[1 \;+\; \frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\,\sqrt{\Delta t}\right], \tag{11.2}$$

$$\mathbb{P}(x_i = -1) \;=\; 1 - p \;=\; \tfrac{1}{2}\left[\, 1 \;-\; \frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\,\sqrt{\Delta t}\,\right]. \tag{11.3}$$

Let $\hat{\mu} \;\doteq\; \mu - \frac{1}{2}\sigma^2$ (log-drift).

Goal. Show that as $N \to +\infty$,

$$X \;\xrightarrow[N\to+\infty]{d}\; \mathcal{N}(\hat{\mu}\,T,\; \sigma^2 T). \tag{11.4}$$

Method. Characteristic function. Compute

$$\mathbb{E}\big[\, e^{iuX} \,\big] \;=\; \mathbb{E}\Big[\, e^{iu\sigma\sqrt{\Delta t}\,\sum_{n=1}^{N} x_i} \,\Big] \;=\; \Big(\, \mathbb{E}\Big[\, e^{iu\sigma\sqrt{\Delta t}\,x_i} \,\Big] \,\Big)^{N}. \tag{11.5}$$

CFs factor over independent sums and uniquely determine distributions (Lévy continuity). Expand the inner expectation:

$$\mathbb{E}\Big[\, e^{iu\sigma\sqrt{\Delta t}\,x_i} \,\Big] \;=\; \mathbb{E}\Big[\, 1 + iu\sigma\sqrt{\Delta t}\,x_i - \tfrac{1}{2}u^2\sigma^2\,\Delta t\,x_i^2 + o(\Delta t) \Big] \tag{11.6}$$

$$= 1 + iu\sigma\sqrt{\Delta t}\,\mathbb{E}[x_i] - \tfrac{1}{2}u^2\sigma^2\,\Delta t\,\mathbb{E}[x_i^2] + o(\Delta t). \tag{11.7}$$

Next compute:

$$\mathbb{E}[x_i] \;=\; 2p - 1 \;=\; \frac{\hat{\mu}}{\sigma}\,\sqrt{\Delta t}, \tag{11.8}$$

and

$$\mathbb{E}[x_i^2] \;=\; 1. \tag{11.9}$$

Substituting:

$$\boxed{\mathbb{E}\Big[\, e^{iu\sigma\sqrt{\Delta t}\,x_i} \,\Big] \;=\; 1 + iu\,\hat{\mu}\,\Delta t \;-\; \tfrac{1}{2}u^2\sigma^2\,\Delta t \;+\; o(\Delta t).} \tag{11.10}$$

The $\sqrt{\Delta t}$ cancels because $\mathbb{E}[x_i] = O(\sqrt{\Delta t})$ — exactly the calibration (8.5).

So:

$$\mathbb{E}\big[\, e^{iuX} \,\big] \;=\; \big(1 + (iu\,\hat{\mu} - \tfrac{1}{2}\sigma^2 u^2)\,\Delta t \;+\; o(\Delta t)\big)^{N} \;\xrightarrow[N\to\infty]{}\; e^{\,\left(iu\,\hat{\mu} - \frac{1}{2}\sigma^2 u^2\right)T}, \tag{11.11}$$

using the classical limit

$$\big(1 + \tfrac{a}{N}\big)^{N} \;\xrightarrow[N\to\infty]{}\; e^{a}. \tag{11.12}$$

Recognising the right-hand side of (11.11) as the characteristic function of a Gaussian with mean $\hat{\mu}T$ and variance $\sigma^2 T$:

$$\boxed{X \;\xrightarrow[N\to\infty]{d}\; \mathcal{N}(\hat{\mu}\,T;\; \sigma^2 T).} \tag{11.13}$$

CF of $\mathcal{N}(m, v)$ is $e^{ium - \frac{1}{2}vu^2}$; matching to (11.11) gives $m = \hat{\mu}T$, $v = \sigma^2 T$, exactly (9.2). A_T is log-normal in the limit — geometric Brownian motion derived from scratch without stochastic calculus.

3.12 Worked Examples

3.12.1 WE-1. Why $\mathbb{P}$-expectation is not a price

Take a tree with $A_0 = 100$, $A_u = 110$, $A_d = 95$, $r = 0$, and physical probability $p = 0.6$. The naïve discounted expectation is $0.6 \cdot 110 + 0.4 \cdot 95 = 104$, suggesting the asset is *under-priced*. But under the risk-neutral measure $q = (100-95)/(110-95) = 1/3$, the discounted expectation is $\frac{1}{3} \cdot 110 + \frac{2}{3} \cdot 95 = 100$ — exactly the spot. Two traders disagreeing about p would compute different $\mathbb{P}$-expectations; neither is a price. The price is what it costs to *replicate* the payoff, and replication knows nothing about p.

3.12.2 WE-2. Sign check with three small trees | Tree | A_0 | A_d | A_u | Check $A_d < A_0 < A_u$? | Verdict |

|:—:|:—:|:—:|:—:|:—:|:—:| | (−) | 10 | 5 | 20 | $5 < 10 < 20$ | No arb | | (+) | 2 | 4 | 15 | $4 \not< 2$ | Arb | | (14 ...) | ≈ 14 | 0 | 0 | violates | Arb |

For tree (−): $q = (10-5)/(20-5) = 1/3$. Diagnosis is a strict inequality test, no probabilities required.

3.12.3 WE-3. Third asset mispriced - A: $10 \to \{20, 5\}$; B: $1 \to \{1, 1\}$ (so $r = 0$); C: $110 \to \{120, 100\}$.

- q from A-tree: $q^{BA} = 1/3$. Fair $\widehat{C_0} = 120 \cdot \frac{1}{3} + 100 \cdot \frac{2}{3} \approx 106.67 < 110$.
- Replicating portfolio: $\alpha = 2$, $\beta = 90$. Selling C at 110 and buying $(2, 90)$ in (A, B) gives $V_0 = 0$, $V_1^{\text{up}} = +10$, $V_1^{\text{down}} = 0$ — an arbitrage.
- Internal solve of q^{BC}: $110 = 120q + 100(1-q) \Rightarrow q^{BC} = 1/2$, inconsistent with $q^{BA} = 1/3$.

3.12.4 WE-4. European call on A at strike $K = 100$ With tree $100 \to \{120, 90\}$, $r = 0$, $K = 100$:

quantity	value
α (shares of A)	$2/3$
β (bonds)	-60
q (risk-neutral prob.)	$1/3$
no-arb call price C_0	$6\frac{2}{3}$
arbitrage profit if $C_0 = 6$	$+\frac{2}{3}$ at $t = 0$, 0 at $t = 1$

Put-call parity check: put has payoff $\{0, 10\}$, $P_0 = 2/3 \cdot 10 = 20/3$, so $C_0 - P_0 = 0 = A_0 - Ke^{-rT}$.

3.13 Key Takeaways

- Naive expected-value pricing fails for risk-averse agents: a ± 100 coin-flip is nothing like a $\pm 10^6$ coin-flip, yet they have equal expectation.
- One-period no-arbitrage: $V_0 = 0$, $\mathbb{P}(V_t \geq 0) = 1$, $\mathbb{P}(V_t > 0) > 0$ defines an arb; no-arb is equivalent to $A_d/B_d < A_0/B_0 < A_u/B_u$, i.e. in the bond case $A_d < A_0(1+r) < A_u$.
- Risk-neutral measure: $\exists\, q \in (0, 1)$ such that $A_0/B_0 = q\, A_u/B_u + (1-q)\, A_d/B_d$, with explicit $q = (A_0/B_0 - A_d/B_d)/(A_u/B_u - A_d/B_d)$.
- Replication = pricing: any claim C with payoffs (C_u, C_d) can be synthesised by (α, β) in (A, B); formulas (5.5)–(5.6). The unique arbitrage-free price is $C_0/B_0 = q\, C_u/B_u + (1-q)\, C_d/B_d$, which in the bond case reduces to $C_0 = \frac{1}{1+r}\mathbb{E}^{\mathbb{Q}}[C_1]$.
- Relative prices are $\mathbb{Q}$-martingales — First Fundamental Theorem: no arb $\Leftrightarrow \exists\, \mathbb{Q} \sim \mathbb{P}$ s.t. $\tilde{X}_0 = \mathbb{E}^{\mathbb{Q}}[\tilde{X}_1]$ for every traded X.
- Multi-period extension: $\mathcal{F}_t$-adapted strategies, $\tilde{X}_t = \mathbb{E}^{\mathbb{Q}}[\tilde{X}_s | \mathcal{F}_t]$ for all $t < s$.
- Calibrated tree: $c = \sigma\sqrt{\Delta t}$ and $p = \frac{1}{2}[1 + (\mu - \frac{1}{2}\sigma^2)\sqrt{\Delta t}/\sigma]$ match log-normal drift $\mu - \frac{1}{2}\sigma^2$ and variance σ^2 per unit time.

- Risk-neutral q in continuous calibration: $q = (e^{r\Delta t} - e^{-\sigma\sqrt{\Delta t}})/(e^{\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}})$, expanding to $\frac{1}{2}[1 + (r - \frac{1}{2}\sigma^2)\sqrt{\Delta t}/\sigma]$. The swap $\mu \mapsto r$ is the Girsanov shift.
- Limit law: $\ln(A_T/A_0) \xrightarrow{d} \mathcal{N}((\mu - \frac{1}{2}\sigma^2)T, \sigma^2 T)$ and $\mathbb{E}[A_T] = A_0 e^{\mu T}$.
- One direction of the FFT proved explicitly: existence of $\mathbb{Q}$ forces $V_0 = B_0\, \mathbb{E}^{\mathbb{Q}}[V_1/B_1] > 0$ for any non-negative, non-trivial terminal wealth — ruling out arbitrage.
- CLT via characteristic functions: $(1 + a/N)^N \to e^a$ produces $\mathbb{E}[e^{iuX}] \to e^{iu\hat{\mu}T - \frac{1}{2}\sigma^2 u^2 T}$ and hence $X \xrightarrow{d} \mathcal{N}(\hat{\mu}T, \sigma^2 T)$.

3.14 Reference Formulas

Binomial fair-price relation (physical, not a pricing rule):

$$A_0 = \frac{pA_u + (1-p)A_d}{1+r}$$

Arbitrage (definition):

$$V_0 = 0, \qquad \mathbb{P}(V_t \geq 0) = 1, \qquad \mathbb{P}(V_t > 0) > 0$$

No-arbitrage bound (bond case):

$$A_d < A_0(1+r) < A_u$$

No-arbitrage bound (general numeraire):

$$\frac{A_d}{B_d} < \frac{A_0}{B_0} < \frac{A_u}{B_u}$$

Risk-neutral probability (general):

$$q = \frac{A_0/B_0 - A_d/B_d}{A_u/B_u - A_d/B_d}$$

Risk-neutral probability (bond case):

$$q = \frac{A_0(1+r) - A_d}{A_u - A_d}$$

Replicating portfolio for claim C:

$$\alpha = \frac{C_u B_d - C_d B_u}{A_u B_d - A_d B_u}, \qquad \beta = \frac{C_u A_d - C_d A_u}{B_u A_d - B_d A_u}$$

Risk-neutral pricing (general):

$$\frac{C_0}{B_0} = q\,\frac{C_u}{B_u} + (1-q)\,\frac{C_d}{B_d}$$

Risk-neutral pricing (bond case):

$$C_0 = \frac{1}{1+r}\,\mathbb{E}^{\mathbb{Q}}[C_1] = \frac{1}{1+r}(q\,C_u + (1-q)\,C_d)$$

First Fundamental Theorem (one-period):

$$\text{no arb} \iff \exists\, \mathbb{Q} \sim \mathbb{P} \text{ s.t. } \tilde{X}_0 = \mathbb{E}^{\mathbb{Q}}[\tilde{X}_1]\ \forall \text{ traded } X, \qquad \tilde{X}_t = X_t/B_t$$

Multi-period martingale characterisation:

$$\text{no arb} \iff \exists\, \mathbb{Q} \sim \mathbb{P} :\ \tilde{X}_t = \mathbb{E}^{\mathbb{Q}}[\tilde{X}_s | \mathcal{F}_t],\ \forall t < s,\ \forall X \text{ traded}$$

Multiplicative binomial update:

$$A_n = A_{n-1}\, e^{\sigma\sqrt{\Delta t}\, x_n}, \qquad x_n \in \{\pm 1\}$$

Calibrated physical p:

$$p = \tfrac{1}{2}\big[1 + \tfrac{\mu - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\big]$$

Calibrated risk-neutral q (CRR form):

$$q = \frac{e^{r\Delta t} - d}{u - d}, \qquad u = e^{\sigma\sqrt{\Delta t}},\ d = e^{-\sigma\sqrt{\Delta t}}$$

Small-Δt expansion of q:

$$q \approx \tfrac{1}{2}\big[1 + \tfrac{r - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\big]$$

Log-normal limit distribution:

$$\ln(A_T/A_0) \ \xrightarrow{d}\ \mathcal{N}\big((\mu - \tfrac{1}{2}\sigma^2)T,\ \sigma^2 T\big), \qquad \mathbb{E}[A_T] = A_0 e^{\mu T}$$

Call and put payoffs:

$$\mathrm{Call}(A_1; K) = \max(A_1 - K,\ 0), \qquad \mathrm{Put}(A_1; K) = \max(K - A_1,\ 0)$$

4 Chapter 2 — Multi-Period Binomial, CRR, and the FTAP

Chapter 1 established risk-neutral pricing for a single-period two-state economy. Here we build the explicit multi-period binomial machinery — backward induction along a recombining tree — and its limit the Cox–Ross–Rubinstein (CRR) model. We present two equivalent CRR parameterisations: the classical $A_n = A_{n-1}e^{c\,x_n}$ Bernoulli lattice, and the "direct-Gaussian increment" lattice with the Itô convexity built into the nodes. We state the Fundamental Theorem of Asset Pricing (FTAP) and confront non-uniqueness of $\mathbb{Q}$ when the tree has more than two states. The chapter closes with the dynamic-programming recursion for American options.

Themes: - *Completeness vs incompleteness.* Binomial trees are exactly one-step complete. Add a third branch (default, jump, stoch-vol state) and uniqueness fails; pricing becomes a *range* and a risk-preference device selects one representative. - *Backward induction.* Bellman's dynamic-programming principle handles Americans and any path-dependence captured by finitely many state variables. Monte Carlo is the forward-simulation complement, developed in CH08. - *Numéraire freedom.* The FTAP is silent on which asset deflates; the choice will be exploited heavily in CH05 (Girsanov) and beyond.

4.1 2.1 One-Period Review: the No-Arbitrage Inequality

Start at A with successors $A_u > A_d$ and a cash account growing at $1 + r$. The one-period no-arbitrage inequality is

$$A_d \;<\; A(1+r) \;<\; A_u \quad\Longleftrightarrow\quad \text{no arb.} \tag{2.1}$$

If $A(1 + r) \le A_d$ then borrowing cash and buying stock yields riskless profit; symmetrically for $A(1 + r) \ge A_u$. The forward price must sit strictly between the two post-states.

Equivalently, any claim C with terminal values C_u, C_d admits a unique arbitrage-free price via $\mathbb{Q}$:

$$\exists\, \mathbb{Q} \text{ s.t. } C \;=\; \frac{1}{1+r}\,\mathbb{E}^{\mathbb{Q}}[\,C_1\,] \quad\Longleftrightarrow\quad \text{no arb,} \qquad q = \tfrac{A(1+r)-A_d}{A_u-A_d}. \tag{2.2}$$

Together (2.1)–(2.2) form the one-period FTAP, a microcosm of the general FTAP in §2.5. The weights $q, 1-q$ are *not* physical probabilities — they are the unique convex combination making the discounted asset a one-step martingale. The Radon–Nikodym derivative $d\mathbb{Q}/d\mathbb{P}$ is the market's discount of optimism and is the engine of Girsanov's theorem (CH05).

Worked check. $A = 100, r = 5\%, A_u = 120, A_d = 90$: forward $= 105 \in (90, 120)$, so $q = (105 - 90)/30 = 1/2$. ATM call ($K = 100$): $C = (1/1.05)(0.5 \cdot 20) \approx 9.52$. Replicating: $\alpha = 2/3, \beta \approx -57.14$, giving $2/3 \cdot 100 - 57.14 \approx 9.52$.

4.1.1 Case study: Petrobras ADR vs ordinary arbitrage (PBR / PETR4)

The Brazilian oil major Petrobras has two near-identical equity claims trading in different venues: PETR4 (preferred ordinary shares, traded on Brazil's B3 exchange in BRL) and PBR (American Depositary Receipts, traded on NYSE in USD, each ADR representing one ordinary share). By construction, the no-arbitrage relationship is $\text{PBR}_t = \text{PETR4}_t/\text{FX}_t^{\text{BRL/USD}}$, modulo a small ADR depositary fee. Yet over essentially every trading day from 2018 to 2024, the implied FX rate from $(PETR4, PBR)$ has *deviated* from the actual BRL/USD spot — sometimes by 50 to 200 basis points, occasionally more during Brazilian political stress. The pair is tracked daily by Latin-American equity desks at every major bank precisely because the deviations represent a concrete, observable violation of the law of one price.

Read through the FTAP, the trade is the cleanest possible example of why (2.1)–(2.2) matter. We have three "assets": PETR4, PBR, and the BRL/USD currency forward. Two of them span the same claim — a unit of Petrobras equity at time T — so the relative price between them must equal the FX forward, by the same argument that gives (2.1). Mispricings open because the three venues have different trading hours (B3 closes at 17:00 BRT, NYSE closes at 16:00 EST), different settlement cycles (T+2 in Brazil, T+1 in the US post-2024), and different liquidity profiles. Closing the gap requires simultaneous trades in BRL/USD FX, PETR4 on B3, and PBR on NYSE — operationally demanding, capital-intensive, and exposed to overnight gap risk. The risk-neutral measure $\mathbb{Q}$ implied by PETR4 vs the one implied by PBR are slightly different — exactly the "two-q puzzle" from CH01 §7 in real-world form.

The practitioner lesson is twofold. First, real markets *do* contain detectable FTAP violations — they are not theoretical curiosities. They survive because of trading frictions (FX swap costs, ADR creation/cancellation fees, time-zone overlap risk) that gate the pure-arbitrage trade. Second, the price an arbitrageur can actually realise is the model price *minus* those frictions. When practitioners say "the spread is too narrow to trade," they mean the FTAP gap is real but smaller than the friction cost of closing it. The frictionless model is the *attractor*; the observed deviations measure the round-trip cost of trading. This is one of the most direct empirical windows into the no-arbitrage framework that this chapter builds.

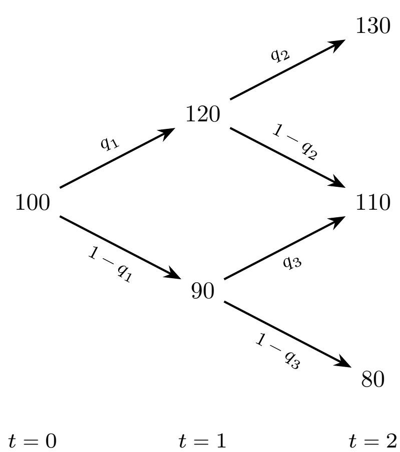
4.2 2.2 Multi-Period Binomial Tree — Backward Induction

Extend the one-step construction to N steps. Recombination ($A_{ud} = A_{du}$) reduces 2^N terminal nodes to $N+1$, giving $O(N^2)$ pricing complexity. At $N = 250$ daily steps, that's $\approx 31{,}000$ node evaluations versus $2^{250} \approx 10^{75}$ without recombination. Multiplicative models recombine because multiplication commutes.

At every interior node the claim value is $\frac{1}{1+r}\mathbb{E}^{\mathbb{Q}}[\text{next-step payoff}]$ with node-local q. In simple CRR, q depends only on $(u, d, 1+r)$ and is global; in state-dependent models (local vol, stochastic rates) each node has its own q. The backward-induction algorithm is identical either way.

4.2.1 2.2.1 Worked example — call struck at 100, maturity $T = 2$

A two-period non-constant-factor lattice:



Local risk-neutral weights q_1, q_2, q_3 — one per branching node — are pinned down by the one-period FTAP applied at each node.

with edge-local interest rates r_1, r_2, r_3 on the three time-0 $\to$ 1 $\to$ 2 edges. The local risk-neutral probabilities q_1, q_2, q_3 are pinned down by matching the parent to the discounted expectation of its children:

$$120 \;=\; \frac{1}{1+r_?}\big(\,130\,q_2 \;+\; 110\,(1-q_2)\,\big) \implies q_2 = \# \tag{2.3}$$

$$90 \;=\; \frac{1}{1+r_?}\big(\,110\,q_3 \;+\; 80\,(1-q_3)\,\big) \implies q_3 = \# \tag{2.4}$$

$$100 \;=\; \frac{1}{1+r_?}\big(\,120\,q_1 \;+\; 90\,(1-q_1)\,\big) \implies q_1 = \# \tag{2.5}$$

Payoff @ $T = 2$. Call struck at $K = 100$: $\max(A_T - K, 0)$. Terminal column $A_T \in \{130, 110, 110, 80\}$, payoffs $\{30, 10, 10, 0\}$. The middle value 110 occurs twice — recombination at work; $\mathbb{Q}$-probability of reaching it is $2q(1-q)$.

Backward step 1 (t = 1 nodes). At the upper intermediate node (stock = 120):

$$C_{1u} \;=\; \frac{1}{1+r_2}\big(\,30\,q_2 \;+\; 10\,(1-q_2)\,\big) \;=\; \#, \tag{2.7}$$

and at the lower intermediate node (stock = 90, children 110 and 80, payoffs 10 and 0):

$$C_{1d} \;=\; \frac{1}{1+r_3}\big(\,10\,q_3 \;+\; 0\,(1-q_3)\,\big) \;=\; \#. \tag{2.8}$$

Backward step 2 (t = 0).

$$C_0 \;=\; \frac{1}{1+r_1}\big(\,C_{1u}\,q_1 \;+\; C_{1d}\,(1-q_1)\,\big) \;=\; \#. \tag{2.9}$$

Intuition. Backward induction is the lattice manifestation of the tower property $\mathbb{E}[\mathbb{E}[X|\mathcal{F}_s]|\mathcal{F}_t] = \mathbb{E}[X|\mathcal{F}_t]$ for $t \le s$. Computing step by step (vs the one-shot $C_0 = \mathbb{E}^{\mathbb{Q}}[e^{-rT}C_T]$) lets the same sweep handle path-dependent payoffs (Asians, lookbacks), barriers, and American exercise (§2.8), with $O(N^2)$ complexity in the state variables.

4.2.2 2.2.2 Path-weights and the CRR telescoping identity

If $r_1 = r_2 = r_3 = r$ the root price collapses to

$$C_0 \;=\; \frac{1}{(1+r)^N}\,\mathbb{E}^{\mathbb{Q}}[\,C_T\,] \;=\; (1+r)^{-N}\sum_{k=0}^{N}\binom{N}{k}q^k(1-q)^{N-k}(A_0u^kd^{N-k}-K)_+. \tag{2.10}$$

As $N \to \infty$ with $u = e^{\sigma\sqrt{\Delta t}}, d = e^{-\sigma\sqrt{\Delta t}}$, the binomial distribution converges to normal (de Moivre–Laplace) and the sum to the Black–Scholes integral. Convergence is $O(1/N)$ with oscillation; Richardson extrapolation accelerates. The distribution of up-moves is Binomial(N, q), giving closed-form moments: $\mathbb{E}^{\mathbb{Q}}[A_T] = A_0(1+r)^N$ (martingale) and $\mathbb{V}^{\mathbb{Q}}[\ln A_T] \to \sigma^2 T$.

4.3 2.3 The Cox–Ross–Rubinstein (CRR) Model — Bernoulli Parameterisation

CRR (1979) showed the binomial lattice converges to Black–Scholes as $\Delta t \to 0$ and gives a tractable pricing tool for Americans and exotics. In step Δt, replace $1+r$ with $e^{r\Delta t}$ (cleaner algebra; aligns with $B_T = e^{rT}$ in the limit). Up-moves multiply by e^c, down-moves by e^{-c} (symmetric $\pm c$ for recombination).

$$A_n \;=\; A_{n-1}\,e^{c\,x_n}, \qquad x_n \overset{iid}{\sim} \text{Bernoulli}(\pm 1),\;\; \mathbb{P}(x=+1) = p. \tag{2.12}$$

Symmetric Bernoulli ± 1 is the simplest random variable with two states per step (one-step completeness), symmetric placement (recombination), and iid increments. The choice cleanly separates move size (c) from directional bias ($p - \frac{1}{2}$).

Iterating $N = T/\Delta t$ steps,

$$A_T \;=\; A_0\,e^{c\,X_N}, \qquad X_N := \sum_{n=1}^{N} x_n. \tag{2.13}$$

4.3.1 2.3.1 Moments of the increment sum

Calibrate by moment-matching: two free parameters (p, c) match two empirical moments (μ, σ^2). Linearity plus iid Bernoulli:

$$\mathbb{E}[X_N] \;=\; \sum_{n=1}^{N}\mathbb{E}[x_n] \;=\; N\big(+1\cdot p + (-1)(1-p)\,\big) \;=\; N\,(2p-1) \;=\; \mu\,\Delta t\,N, \tag{2.14}$$

$$\mathbb{V}[X_N] \;=\; N\,\mathbb{V}[x_1] \;=\; N\big(\,\mathbb{E}[x_1^2] - (\mathbb{E}[x_1])^2\,\big) \;=\; N(1-(2p-1)^2) \;=\; \sigma^2\,\Delta t\,N. \tag{2.15}$$

The data feed in μ and σ^2 as annualised log-return mean and variance from a price series via

$$\ln(S_2/S_1) \;=\; r_1, \quad \ln(S_3/S_2) \;=\; r_2, \quad \ldots \tag{2.16}$$

$$\widehat{\mu\,\Delta t} \;=\; \frac{1}{M}\sum_{m=1}^{M} r_m \quad \text{(annualised return)}, \tag{2.17}$$

$$\widehat{\sigma^2\,\Delta t} \;=\; \frac{1}{M}\sum_{m=1}^{M}(r_m - \widehat{\mu\Delta t})^2 \quad \text{(annualised variance)}. \tag{2.18}$$

4.3.2 2.3.2 Solving for the CRR parameters (p, c)

Solving (2.14)–(2.15) simultaneously, using $\mathbb{E}[x^2] = c^2$:

$$\mathbb{V}[X_N^{\text{c-scaled}}] \;=\; c^2 N(1-(2p-1)^2) \;=\; \sigma^2 \Delta t\,N, \tag{2.19}$$

one obtains the Cox–Ross–Rubinstein up-step and physical probability:

$$\boxed{c \;=\; \sigma\sqrt{\Delta t} + \cdots, \qquad p \;=\; \tfrac{1}{2}\Big(1 \;+\; \frac{\mu}{\sigma}\sqrt{\Delta t}\Big) + \cdots.} \tag{2.20}$$

Intuition. Variance match gives $c = \sigma\sqrt{\Delta t}$; mean match gives a small $O(\sqrt{\Delta t})$ bias in p. Worked: $\sigma = 20\%, \mu = 10\%, \Delta t = 1/252$ gives $c \approx 0.0126$ (per-step $\pm 1.27\%$ moves) and $p \approx 0.516$. The $\sqrt{\Delta t}$ scaling of c is the universal scaling of Brownian motion — at daily horizon, diffusion ($\sigma\sqrt{\Delta t}$) is $\sim 16\times$ larger than drift ($\mu\Delta t$), which is why short-horizon returns look like noise.

4.3.3 2.3.3 Converting to the $\mathbb{Q}$-probability

The $\mathbb{P} \to \mathbb{Q}$ transformation re-weights probabilities so the expected return matches the riskless rate r. Risk-neutral pricing on the $e^{\pm c}$ tree:

$$A \;=\; e^{-r\Delta t}\,\mathbb{E}^{\mathbb{Q}}[\,A_n\,] \;=\; e^{-r\Delta t}\Big(q\,A\,e^{\sigma\sqrt{\Delta t}} \;+\; (1-q)\,A\,e^{-\sigma\sqrt{\Delta t}}\Big), \tag{2.21}$$

giving

$$\boxed{q \;=\; \frac{e^{r\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{+\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}}.} \tag{2.22}$$

Small-Δt expansion. Using $e^{\pm z} \sim 1 \pm z + \frac{1}{2}z^2 + \cdots$ in both numerator and denominator,

$$q \;\sim\; \tfrac{1}{2}\left(1 \;+\; \frac{r - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right) + \cdots \tag{2.23}$$

$$p \;\sim\; \tfrac{1}{2}\Big(1 \;+\; \frac{\mu}{\sigma}\sqrt{\Delta t}\Big) + \cdots \qquad \text{(for comparison, under } \mathbb{P}\text{)}. \tag{2.24}$$

The boxed formula (2.22) is *exact* at finite Δt; (2.23) is leading-order. The $\mathbb{P} \to \mathbb{Q}$ shift replaces the log-return drift μ by $r - \frac{1}{2}\sigma^2$ — the Itô convexity adjustment shows up already at the lattice level, before any stochastic calculus. The deeper reason is the geometric-vs-arithmetic mean gap: $\mathbb{E}[e^X] \geq e^{\mathbb{E}[X]}$. A naïve simulator that uses just r (no $-\frac{1}{2}\sigma^2$) over-prices calls by $\sim e^{\sigma^2 T/2}$ — a 28% bias at $\sigma = 50\%, T = 2$.

4.3.4 2.3.4 Log-moments under $\mathbb{P}$ vs $\mathbb{Q}$

Combining (2.13) with (2.20) / (2.23),

$$\mathbb{E}^{\mathbb{P}}[\ln(A_T/A_0)] \;=\; \mu\, T, \qquad \mathbb{V}^{\mathbb{P}}[\ln(A_T/A_0)] \;=\; \sigma^2\, T, \tag{2.25}$$

$$\mathbb{E}^{\mathbb{Q}}[\ln(A_T/A_0)] \;=\; (r - \tfrac{1}{2}\sigma^2)\, T, \qquad \mathbb{V}^{\mathbb{Q}}[\ln(A_T/A_0)] \;=\; \sigma^2\, T. \tag{2.26}$$

Slogan. “$\mathbb{P}$ and $\mathbb{Q}$ variances are identical” — “$\mathbb{P}$ and $\mathbb{Q}$ means are *not*”.

Intuition. Girsanov at one line: a change of equivalent measure tilts drifts but *cannot* change volatility — quadratic variation is measure-invariant because it is a pathwise quantity, not a probabilistic one. Implied volatility ($\mathbb{Q}$-world) and realised volatility ($\mathbb{P}$-world) measure the same σ^2 in principle; their empirical gap is the *variance risk premium*, traded actively via VIX futures vs realised S&P 500 variance.

4.3.5 2.3.5 Continuous-time limit — lognormal asset

Apply the CLT to $X_N = \sum x_n$ as $N \to \infty$ with $N\Delta t = T$ fixed:

$$\ln(A_T/A_0) \;=\; X_N \;\xrightarrow[N\to\infty]{d,\ \mathbb{P}}\; \mathcal{N}\big(\,\mu T,\ \sigma^2 T\,\big), \tag{2.27}$$

$$\ln(A_T/A_0) \;=\; X_N \;\xrightarrow[N\to\infty]{d,\ \mathbb{Q}}\; \mathcal{N}\big(\,(r - \tfrac{1}{2}\sigma^2)T,\ \sigma^2 T\,\big). \tag{2.28}$$

Equivalently, in distribution,

$$A_T \;\overset{d}{=}\; A_0\, e^{\mu T \,+\, \sigma\sqrt{T}\, Z_{\mathbb{P}}}, \qquad Z_{\mathbb{P}} \sim \mathcal{N}(0,1), \tag{2.29}$$

$$A_T \;\overset{d}{=}\; A_0\, e^{(r - \frac{1}{2}\sigma^2)T \,+\, \sigma\sqrt{T}\, Z_{\mathbb{Q}}}, \qquad Z_{\mathbb{Q}} \sim \mathcal{N}(0,1). \tag{2.30}$$

In words: asset prices in the limit $N \to \infty$ are lognormal r.v. at a fixed point in time. Verifying the martingale property under $\mathbb{Q}$:

$$\mathbb{E}^{\mathbb{P}}[\,A_T\,] \;=\; A_0\, e^{\mu T}\, \mathbb{E}^{\mathbb{P}}[\,e^{\sigma\sqrt{T}\,Z}\,] \;=\; A_0\, e^{\mu T}\, e^{\frac{1}{2}\sigma^2 T} \;=\; A_0\, e^{(\mu + \frac{1}{2}\sigma^2)T}, \tag{2.31}$$

$$\mathbb{E}^{\mathbb{Q}}[\,A_T\,] \;=\; A_0\, e^{(r - \frac{1}{2}\sigma^2)T}\, e^{\frac{1}{2}\sigma^2 T} \;=\; A_0\, e^{rT}. \tag{2.32}$$

Used: $\mathbb{E}[e^{\alpha Z}] = e^{\alpha^2/2}$. Under $\mathbb{Q}$ the log-return drift is $r - \frac{1}{2}\sigma^2$ but the price grows at r — Jensen on e^X. In log-normal markets, the right-tail pulls the mean up by exactly $\frac{1}{2}\sigma^2$ above the median.

4.3.6 2.3.6 Calibrating to log-returns vs simple returns — the $\frac{1}{2}\sigma^2$ correction

Caveat: if you calibrate μ from simple returns $(S_{m+1} - S_m)/S_m$ instead of log-returns, you over-state the log-drift by $\frac{1}{2}\sigma^2$ — “volatility drag.” For the S&P 500 at $\sigma = 16\%$, the gap between simple-return $\mu = 10\%$ and the compounded growth rate $\approx 8.7\%$ produces a 40% terminal-wealth difference over 30 years. *Always log before averaging:* compute $r_m = \ln(S_{m+1}/S_m)$, then take sample mean and variance.

4.3.7 2.3.7 Pricing vanilla European claims in the CRR limit

From $C_0/B_0 = \mathbb{E}^{\mathbb{Q}}[C_T/B_T]$ and (2.30):

$$C_0 \;=\; e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\,(A_T-K)_+\,], \qquad A_T \;\stackrel{d}{=}\; A_0\,e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}\,Z}. \tag{2.34}$$

The integral evaluates to the Black–Scholes formula (sketched in §2.9; derived in CH06).

4.4 2.4 The Alternate CRR Parameterisation — Direct-Gaussian Increment Lattice

The classical CRR pushes drift into the probability p. An equivalent alternate lattice (essentially Jarrow–Rudd) absorbs the Itô convexity into the *nodes* and keeps $p = \frac{1}{2}$. Many discretisations converge to the same GBM; the choice trades off where the drift information lives.

4.4.1 2.4.1 The two parameterisations, side by side

Parameterisation I — symmetric nodes, biased probability (§2.3):

$$A \to \begin{cases} A\,e^{+\sigma\sqrt{\Delta t}} & \text{w.p. } p \\ A\,e^{-\sigma\sqrt{\Delta t}} & \text{w.p. } 1-p \end{cases}, \quad p = \tfrac{1}{2}[1+\tfrac{\mu}{\sigma}\sqrt{\Delta t}] + \cdots \tag{2.35}$$

Parameterisation II — drift-shifted nodes, symmetric probability:

$$A \to \begin{cases} A\,e^{(\mu-\frac{1}{2}\sigma^2)\Delta t+\sigma\sqrt{\Delta t}} & \text{w.p. } \frac{1}{2} \\ A\,e^{(\mu-\frac{1}{2}\sigma^2)\Delta t-\sigma\sqrt{\Delta t}} & \text{w.p. } \frac{1}{2} \end{cases} \tag{2.36}$$

Intuition. Both converge to the same lognormal GBM. Parameterisation II is the moment-matched Euler–Maruyama scheme for log-GBM: $X_{n+1} = X_n + (r-\frac{1}{2}\sigma^2)\Delta t + \sigma\cdot\pm\sqrt{\Delta t}$. It is nicer for Monte Carlo (fair coin), parameterisation I is closer to the one-step no-arb picture.

4.4.2 2.4.2 Verifying parameterisation II matches the first two log-moments

Under $\mathbb{P}$ with $p = \frac{1}{2}$ on tree (2.36):

$$\mathbb{E}^{\mathbb{P}}[A_1] \;=\; A\,e^{(\mu-\frac{1}{2}\sigma^2)\Delta t}\cdot\tfrac{1}{2}\Big[\,e^{+\sigma\sqrt{\Delta t}} + e^{-\sigma\sqrt{\Delta t}}\,\Big]\,. \tag{2.37}$$

Expanding $\frac{1}{2}[e^z+e^{-z}] = 1+\frac{1}{2}z^2+\cdots$ with $z = \sigma\sqrt{\Delta t}$:

$$\mathbb{E}^{\mathbb{P}}[A_1] \;=\; A\,e^{(\mu-\frac{1}{2}\sigma^2)\Delta t}\big(1+\tfrac{1}{2}\sigma^2\Delta t+\cdots\big) \;=\; A\,e^{\mu\Delta t}+\cdots, \tag{2.38}$$

matching the target $\mathbb{E}^{\mathbb{P}}[A_1] = Ae^{\mu\Delta t}$ to the relevant order. The log-variance

$$\mathbb{V}^{\mathbb{P}}[\ln(A_1/A)] \;=\; \sigma^2\,\Delta t \tag{2.39}$$

is exact (not just leading-order) because the $\pm\sigma\sqrt{\Delta t}$ Bernoulli has variance $\sigma^2\Delta t$ by construction, and the deterministic drift $(\mu-\frac{1}{2}\sigma^2)\Delta t$ drops out of the variance.

4.4.3 2.4.3 Risk-neutral probability q under parameterisation II

Applying the martingale condition $A = e^{-r\Delta t}\mathbb{E}^{\mathbb{Q}}[A_1]$ to tree (2.36) with $\mathbb{Q}$-probabilities $(q, 1-q)$:

$$A \;=\; e^{-r\Delta t}\Big[q\,A\,e^{(\mu-\frac{1}{2}\sigma^2)\Delta t+\sigma\sqrt{\Delta t}} \;+\; (1-q)\,A\,e^{(\mu-\frac{1}{2}\sigma^2)\Delta t-\sigma\sqrt{\Delta t}}\Big]. \tag{2.40}$$

Solving for q:

$$q \;=\; \frac{e^{r\Delta t} - e^{(\mu-\frac{1}{2}\sigma^2)\Delta t-\sigma\sqrt{\Delta t}}}{e^{(\mu-\frac{1}{2}\sigma^2)\Delta t+\sigma\sqrt{\Delta t}} - e^{(\mu-\frac{1}{2}\sigma^2)\Delta t-\sigma\sqrt{\Delta t}}} \;=\; \frac{e^{(r-(\mu-\frac{1}{2}\sigma^2))\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{+\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}}. \tag{2.41}$$

Introduce the shorthand $\hat{r} := r - (\mu - \frac{1}{2}\sigma^2)$. Expanding to $O(\sqrt{\Delta t})$:

$$q \;\sim\; \frac{(1+\hat{r}\Delta t) - (1-\sigma\sqrt{\Delta t}+\frac{1}{2}\sigma^2\Delta t)}{(1+\sigma\sqrt{\Delta t}+\frac{1}{2}\sigma^2\Delta t) - (1-\sigma\sqrt{\Delta t}+\frac{1}{2}\sigma^2\Delta t)} + \cdots \;=\; \frac{\sigma\sqrt{\Delta t} + (\hat{r}-\frac{1}{2}\sigma^2)\Delta t}{2\sigma\sqrt{\Delta t}} + \cdots, \tag{2.42}$$

$$q \;\sim\; \tfrac{1}{2}\left[1 + \frac{\hat{r}-\frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right] + \cdots \;=\; \tfrac{1}{2}\Big[1 + \frac{r-\mu}{\sigma}\sqrt{\Delta t}\Big] + \cdots, \tag{2.43}$$

where the last step uses $\hat{r} - \frac{1}{2}\sigma^2 = r - (\mu - \frac{1}{2}\sigma^2) - \frac{1}{2}\sigma^2 = r - \mu$.

Comparison with classical CRR (2.23):

$$q_{\text{CRR}} \;\sim\; \tfrac{1}{2}\left[1 + \frac{r-\frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right] + \cdots \tag{2.44}$$

The two q's differ at $O(\sqrt{\Delta t})$ by exactly the physical drift term, because parameterisation II has already absorbed the drift $(\mu - \frac{1}{2}\sigma^2)$ into the node geometry. In the continuous-time limit both parameterisations yield the same lognormal A_T distribution under $\mathbb{Q}$ — consistent with (2.30).

4.4.4 2.4.4 Reconciling parameterisations I and II

By construction of q in (2.40) the martingale relation $\mathbb{E}^{\mathbb{Q}}[A_1] = Ae^{r\Delta t}$ is exact, and $\mathbb{V}^{\mathbb{Q}}[\ln(A_1/A)] = \sigma^2\Delta t + O(\Delta t^{3/2})$. A one-parameter family unifies the two: take node geometry $A\,e^{(\alpha-\frac{1}{2}\sigma^2)\Delta t\pm\sigma\sqrt{\Delta t}}$ with probability $p = \frac{1}{2}[1 + (\mu - \alpha)\sqrt{\Delta t}/\sigma] + \cdots$. The choice $\alpha = \mu$ recovers parameterisation II ($p = \frac{1}{2}$); $\alpha = 0$ recovers classical CRR. All choices converge to the same GBM; they only differ in finite-Δt convergence constants. When comparing two binomial-tree implementations, always check the parameterisation convention — a 0.5% discrepancy at $N = 100$ may just reflect different α.

4.5 2.5 The Fundamental Theorem of Asset Pricing

FTAP. No arbitrage $\iff$ there exists $\mathbb{Q} \sim \mathbb{P}$ such that for every traded asset X, the discounted price $\tilde{X}_t = X_t/B_t$ is a $\mathbb{Q}$-martingale:

$$\tilde{X}_t \;=\; \mathbb{E}^{\mathbb{Q}}\big[\tilde{X}_s \mid \mathcal{F}_t\big], \qquad s \geq t. \tag{2.48}$$

The physical measure $\mathbb{P}$ describes how the world evolves; $\mathbb{Q}$ re-weights so all tradable risks have zero expected excess return. Equivalence $\mathbb{Q} \sim \mathbb{P}$ means agreement on null sets — essential, else we could win on a $\mathbb{P}$-non-null event the $\mathbb{Q}$-price ignores.

Two flavours: *First FTAP* (no-arb $\Leftrightarrow \mathcal{M}_e \neq \emptyset$) and *Second FTAP* (completeness $\Leftrightarrow |\mathcal{M}_e| = 1$). Binomial trees satisfy both; trinomial trees (or any market with more states than tradables) satisfy only the first and admit a family of $\mathbb{Q}$.

The Delbaen–Schachermayer extension to semimartingales replaces “no arbitrage” by “no free lunch with vanishing risk” (NFLVR). The natural numéraire is the money-market account $B_t = e^{rt}$ but any strictly positive traded asset works; this freedom is exploited in CH05.

4.6 2.6 Non-Uniqueness of the Martingale Measure

We arrive now at what may be the most important conceptual issue in modern quantitative finance: market incompleteness and the non-uniqueness of the pricing measure. Up to this point, our pricing arguments have rested on a tacit assumption: that the market contains enough tradable instruments to span every possible future state. In a binomial tree, this assumption holds automatically — two states, two instruments (stock and cash), perfect span. In a trinomial tree, or any richer state space, the assumption fails. The market cannot hedge every possible payoff, and consequently no-arbitrage alone does not determine a unique price.

This is not an academic curiosity. Real markets are universally incomplete. Stochastic volatility, jumps, stochastic interest rates, credit events, counterparty risk — each introduces additional state variables not spanned by vanilla instruments. The practical consequence: an entire industry has grown up around the problem of *how to select* a single price from the arbitrage-free interval. Calibration to market-traded benchmarks, utility-based risk preferences, robust pricing under ambiguity — these are the tools of the trade, and each corresponds to a particular way of selecting $\mathbb{Q}$ from the set $\mathcal{M}_e$ of equivalent martingale measures.

In the single-step binomial model with two post-states the measure $\mathbb{Q}$ was pinned down by one linear equation in one unknown q. Once the underlying can branch to three or more states, $\mathbb{Q}$ is no longer unique.

The binomial tree’s completeness is an artefact of the lattice, not a feature of real markets. Real markets have far more states than tradable instruments — jumps, news shocks, stochastic volatility, default, liquidity freezes — and any one of these creates an un-spanned payoff. The FTAP still holds; $\mathcal{M}_e$ is non-empty but no longer a singleton.

Why three post-states destroys uniqueness: each tradable adds one linear constraint $\mathbb{E}^{\mathbb{Q}}[X_1] = X_0 e^{r\Delta t}$, while an n-state probability vector has $n-1$ free coordinates. For $n = k$ tradables, the system is exactly determined; for $k < n$, $\mathcal{M}_e$ is $(n-k)$-dimensional. Adding tradables (e.g. index options) restores uniqueness — this is the *calibration-to-market* strategy that dominates quant practice.

We have two martingale equations (for cash and the stock) but three or four unknown probabilities, so

$$\mathbb{Q} \text{ is not unique} \implies C_0 \text{ is not necessarily unique.} \tag{2.50}$$

In incomplete markets the price spans an *arbitrage-free interval*: from the super-replication cost (cheapest dominating portfolio) to the sub-replication value. The market-clearing price within this interval depends on risk preferences, hedging technology, and supply–demand.

Trinomial tree — non-uniqueness of Q *Trinomial tree — non-uniqueness of Q*

4.6.1 2.6.1 Geometric picture and local risk-minimisation

In the payoff vector space, tradable assets span a *replicable subspace*. A claim in this subspace has a unique price; a claim outside requires a selection criterion. The simplest, *local risk-minimisation*, projects the claim onto the subspace under $L^2(\mathbb{P})$:

$$C_0 \text{ is unique iff the claim lies in } \operatorname{span}(A_1, B_1), \qquad \min_{(\alpha,\beta)} \mathbb{E}^{\mathbb{P}}[(\alpha A_1 + \beta B_1 - C_1)^2]. \tag{2.51–2.52}$$

The foot of the perpendicular is the optimal static hedge; the perpendicular length is the unhedgeable residual standard deviation. Other selection rules: *minimum-entropy*, *minimum-variance*, *Esscher transform*, *utility indifference*. In practice, traders sidestep the choice by calibrating $\mathbb{Q}$ to liquidly traded benchmark instruments — the market picks $\mathbb{Q}$ via its willingness to trade.

Worked example. Trinomial $S_0 = 10$, $S_1 \in \{12, 10, 8\}$, $p = (1/3, 1/3, 1/3)$, $r = 0$. Price call $C_1 = (2, 0, 0)$. Projecting onto $\operatorname{span}(\mathbf{1}, S_1)$: $\beta^\star = \mathbb{C}[S_1, C_1]/\mathbb{V}[S_1] = (4/3)/(8/3) = 1/2$, $\alpha^\star = -14/3$. Residual variance $1/3$. The arbitrage-free interval stretches over $(0, 1)$.

4.7 2.7 Default Model and Incomplete-Market Trees

We build a discrete default model that converges to a diffusion with a jump to zero — our first concrete incomplete-market tree. Default is a *jump process*: rare, catastrophic, discrete. The mathematical framework is a Poisson point process with intensity $\hat{\lambda}$, the expected rate of jumps per unit time. With three states (up/down/default) and only two tradables (stock, cash), $\mathcal{M}_e$ has dimension one — calibrate $\hat{\lambda}$ from CDS spreads or corporate bond yields.

The physical intensity λ is generally lower than the risk-neutral $\hat{\lambda}$; the gap is the credit risk premium (for IG issuers, $\hat{\lambda}$ is 2–5x λ).

On $[t, t+\Delta t]$:

$$A_n = A_{n-1}\, e^{\sigma\sqrt{\Delta t}\, x_n}, \qquad p = \tfrac{1}{2}[1 + (\mu - \tfrac{1}{2}\sigma^2)\sqrt{\Delta t}/\sigma]. \tag{2.53}$$

Lifetime $\tau \sim \mathrm{Exp}(\hat{\lambda})$. One-step default probability:

$$\mathbb{P}(\tau \in (t, t+\Delta t] | \tau > t) = 1 - e^{-\hat{\lambda}\Delta t} \approx \hat{\lambda}\,\Delta t. \tag{2.54}$$

Memorylessness ($\mathbb{P}(\tau > t+s | \tau > s) = \mathbb{P}(\tau > t)$) is the defining feature of the exponential. Empirically wrong (default intensities cluster in time, firms have age effects), but pedagogically clean — Cox/doubly-stochastic processes generalise. Typical IG corporate intensities: $\hat{\lambda} \in [0.005, 0.015]$/yr; HY: $[0.05, 0.20]$/yr.

4.7.1 2.7.1 Branching tree with default branch

From a pre-default node A_{n-1} the tree has three children:

$$A_{n-1} \longrightarrow \begin{cases} A_{n-1}\, e^{+\sigma\sqrt{\Delta t}} & \text{w.p. } p\,(1-\hat{\lambda}\Delta t) \\ A_{n-1}\, e^{-\sigma\sqrt{\Delta t}} & \text{w.p. } (1-p)(1-\hat{\lambda}\Delta t) \\ 0 & \text{w.p. } \hat{\lambda}\,\Delta t \end{cases} \tag{2.55}$$

The riskless bond pays $e^{r\Delta t}$ in all three states; the risky asset jumps to 0 on default — the tree is binomial conditional on survival, with a third ray to zero.

4.7.2 2.7.2 Survival law

For $\tau \sim \mathrm{Exp}(\hat{\lambda})$: $\mathbb{P}(\tau > T) = e^{-\hat{\lambda}T}$, density $f_\tau(t) = \hat{\lambda}e^{-\hat{\lambda}t}$, mean $1/\hat{\lambda}$. At $\hat{\lambda} = 5\%$: 5-yr survival $\approx 77.9\%$, 10-yr $\approx 60.7\%$, mean default time 20 years. Time-varying intensity $\hat{\lambda}(t)$ generalises to $\exp(-\int_0^T \hat{\lambda}(s)ds)$.

4.7.3 2.7.3 Risk-neutral q — default-free sub-case

Setting $\hat{\lambda} = 0$ recovers (2.22): $q = (e^{r\Delta t} - e^{-\sigma\sqrt{\Delta t}})/(e^{\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}) \sim \frac{1}{2}[1 + (r - \frac{1}{2}\sigma^2)\sqrt{\Delta t}/\sigma]$ — same as before; $\hat{\lambda}$ is the $\mathbb{Q}$-hazard rate.

4.7.4 2.7.4 Including default into the martingale condition

The three-branch martingale condition:

$$1 \;=\; q\,(1-\hat{\lambda}\Delta t)\, e^{\sigma\sqrt{\Delta t} - r\Delta t} \;+\; (1-q)(1-\hat{\lambda}\Delta t)\, e^{-\sigma\sqrt{\Delta t} - r\Delta t} \;+\; 0\cdot\hat{\lambda}\Delta t. \tag{2.59}$$

Solving for q:

$$q \;=\; \frac{e^{(r+\hat{\lambda})\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{+\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}} \;\sim\; \tfrac{1}{2}\left(1 + \frac{r + \hat{\lambda} - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right) + o(\sqrt{\Delta t}). \tag{2.60}$$

The effect of adding a default branch is therefore to shift the risk-neutral drift upward by $\hat{\lambda}$: the surviving branch must grow faster to compensate for the chance of being wiped out.

Intuition. If a fraction $\hat{\lambda}\Delta t$ of your probability mass leaks to zero every step, the conditional expected growth of the survivors has to be higher just to keep the unconditional expectation at $e^{r\Delta t}$. Credit risk premia in corporate bonds have exactly this structure: the yield-to-maturity exceeds the riskless rate by roughly the expected loss rate $\hat{\lambda} \cdot \text{LGD}$.

A concrete credit-pricing illustration. Consider a zero-recovery one-year zero-coupon bond issued by a firm with default intensity $\hat{\lambda} = 3\%$. Its fair price is $e^{-(r+\hat{\lambda})T} = e^{-(0.05+0.03)\cdot 1} \approx 0.923$, compared to the riskless bond price $e^{-rT} \approx 0.951$. The yield spread is therefore $\hat{\lambda} = 3\%$ — the spread is the risk-neutral default intensity (in the zero-recovery case). With recovery R (so loss-given-default is $1-R$), the bond price is $e^{-rT}(e^{-\hat{\lambda}T} + R(1-e^{-\hat{\lambda}T})) \approx e^{-rT}(1-(1-R)\hat{\lambda}T)$, and the spread is $\hat{\lambda}(1-R)$ to leading order. This is the "spread = probability × loss" relationship that sits at the heart of the CDS market. The CDS premium quoted by a market-maker *is* the market-implied $\hat{\lambda}(1-R)$, and working back out the probability requires an LGD assumption (typically $R \approx 40\%$ for senior unsecured corporate, so $1-R = 60\%$).

Now bring the stock back into the picture. On our default tree, the stock's surviving-branch growth is $e^{(r+\hat{\lambda})\Delta t}$; its unconditional expectation is $e^{r\Delta t}$ because the $\hat{\lambda}$ leak balances the surviving boost. Stockholders — as residual claimants — bear the full loss on default, so the stock's equity risk premium under $\mathbb{Q}$ is pushed upward by the credit component. Equity traders often combine this with Merton's structural credit model: view the firm's equity as a call option on its assets, with strike at the debt's face value. The equity is thus *naturally* a credit-contingent claim, and the same $\hat{\lambda}$ that shows up in debt pricing also shows up in the equity vol surface (high equity vol goes with high credit spreads). The stylised empirical fact is sometimes called the "credit–equity link".

4.7.5 2.7.5 Risk-neutral expectations of survival

A few expectation identities under the default model are useful for pricing credit-sensitive claims and for setting up the continuous-time limit. These formulas show how the risk-neutral drift is modified by the default intensity and how survival-conditional expectations relate to unconditional ones.

Under $\mathbb{Q}$ the asset and its discounted version satisfy

$$\mathbb{E}^{\mathbb{Q}}\Big[A_T\,\mathbf{1}_{\{\tau>T\}}\Big] \;=\; A_0\, e^{(r+\hat{\lambda})T}, \tag{2.61}$$

and for the bond-denominated (default-insensitive) notional,

$$\mathbb{E}^{\mathbb{Q}}[A_T] \;=\; A_0\, e^{rT}. \tag{2.62}$$

Intuition. Equation (2.62) is the FTAP itself at the horizon: the unconditional expected terminal asset price grows at r. Equation (2.61) says that, *conditional on surviving*, the asset grows at the higher rate $r+\hat{\lambda}$. Taking the product with the survival probability $e^{-\hat{\lambda}T}$ recovers (2.62): $e^{(r+\hat{\lambda})T} \cdot e^{-\hat{\lambda}T} = e^{rT}$.

4.7.6 2.7.6 European pricing recursion with default

The backward recursion on the three-branch tree is just the no-arbitrage condition rearranged: the discounted price at time t equals the risk-neutral expected discounted price at time $t+\Delta t$. We must account for all three possible outcomes — up survival, down survival, and default — weighting each by its risk-neutral probability.

Let $C_{n,j}$ denote the option value at tree node (n,j). The martingale/pricing recursion on the three-branch tree is

$$\frac{C_{n-1,j}}{B_{n-1}} \;=\; \mathbb{E}^{\mathbb{Q}}\left[\frac{C_n}{B_n}\right], \tag{2.63}$$

$$C_{n-1,j} \;=\; q\,\frac{e^{-\hat{\lambda}\Delta t}}{e^{r\Delta t}}\,C_{n,j} \;+\; (1-q)\,\frac{e^{-\hat{\lambda}\Delta t}}{e^{r\Delta t}}\,C_{n,j+1} \;+\; (1-e^{-\hat{\lambda}\Delta t})\cdot C_{n,d}, \tag{2.64}$$

where the default node satisfies

$$C_{n-1,d} \;=\; C_{n,d}\, e^{-r\Delta t}. \tag{2.65}$$

(On default the stock is 0, so a call has $C_{n,d} \equiv 0$; a put has $C_{n,d} = K$ discounted to today.)

Intuition. The recursion is the classical binomial backward sweep with two modifications: each surviving branch is down-weighted by the survival factor $e^{-\hat{\lambda}\Delta t}$, and a third term captures what the option is worth *in* default. For a vanilla call that third term is zero, so credit risk eats value monotonically; for a put it is positive (the put still pays on bankruptcy unless it is a "protected put" that voids on the issuer's own default).

For a concrete numerical illustration: take the vanilla 1-year ATM call with $A_0 = K = 100$, $\sigma = 20\%$, $r = 5\%$, and add a default intensity $\hat{\lambda} = 2\%$. The Black–Scholes price (without default) is approximately 10.45. Discounting by survival at maturity multiplies by $e^{-\hat{\lambda}T} = e^{-0.02} \approx 0.9802$, and the modified call price is approximately $10.45 \cdot 0.9802 \approx 10.24$. A 2% credit intensity produces a 2% reduction in call value. For a deeply-out-of-the-money call (say strike $K = 150$) whose Black–Scholes price is 0.45, the modified price is 0.44 — the absolute reduction is smaller, but the relative impact is still 2%. This is a general pattern: constant default intensity scales down call prices by a factor $e^{-\hat{\lambda}(T-t)}$ to leading order, and this is how credit-adjusted "jump-to-default" models are typically parameterised.

4.8 2.8 American Valuation on the Binomial Tree

Before diving into the details, it is worth situating American options in the larger landscape. American-style exercise is the default for single-stock equity options in the US market — virtually all listed single-stock options are American. European-style exercise is standard for index options (S&P 500, Russell 2000) and for European-market listed options. The choice of exercise style matters because American options can be worth more than European ones (since the holder has more optionality), and because American options require more sophisticated pricing and hedging technology.

American options introduce a genuinely new computational challenge: the *stopping problem*. Unlike European options, which always settle at the fixed date T, American options give the holder the right to exercise *at any time* $\tau \le T$. The pricing problem becomes a two-part decision: (i) at each node, choose the optimal exercise policy; (ii) under that policy, compute the expected discounted payoff. The two parts are coupled — the expected payoff depends on the policy, and the policy depends on expected payoffs — and they must be solved simultaneously via dynamic programming.

Why can we not simply take the maximum over stopping times analytically? Because the optimal stopping time is not a deterministic time; it is a *random time* that depends on the realised path. Formally, τ must be a *stopping time* — a random variable adapted to the filtration, so $\{\tau \le t\} \in \mathcal{F}_t$ for all t. Stopping times represent non-anticipating exercise policies: we can decide to exercise based on what has happened so far, but not on what will happen. The set of stopping times is combinatorially vast, and direct optimisation is infeasible.

Dynamic programming (Bellman's principle) reduces this to a tractable recursion: "the value of being in state X at time t is the max of (i) exercising now and receiving the intrinsic value, or (ii) holding to time $t+1$ and receiving the expected next-step value". The recursion runs backward from the terminal payoff, sweeping through the lattice in $O(N^2)$ time. The optimal policy falls out as a by-product: at each node, record whether the max was achieved by exercising or by continuing; the exercise boundary is the locus of nodes where the max was achieved by exercising.

4.8.1 2.8.1 European → American contrast

We begin by contrasting the European and American exercise styles. The European case has a fixed settlement date T; pricing is a single expectation. The American case allows the holder to exercise at any stopping time $\tau \le T$; pricing is a supremum over all such stopping times. The supremum turns out to be computable by backward induction — the classical dynamic-programming result of Bellman — and the resulting price is at least as large as the European price, with the difference being the "early exercise premium".

European option pays $\varphi(A_T)$ at the fixed maturity T:

$$V_t \;=\; e^{-r(T-t)}\, \mathbb{E}^{\mathbb{Q}}[\, \varphi(A_T)\,] \;\longrightarrow\; V_T \;=\; \varphi(A_T). \tag{2.66}$$

American option lets the holder choose the exercise epoch τ, a stopping time bounded above by T, and pays $\varphi(A_\tau)$ at τ — meaning the holder "receives $\varphi(A_\tau)$ at the moment they decide to stop".

American put, in particular:

$$\varphi(A_\tau) \;=\; (K - A_\tau)_+. \tag{2.67}$$

Let $\mathcal{S}$ denote the set of $\{\mathcal{F}_t\}$-stopping times bounded by T. Then

$$\frac{C_0}{B_0} \;=\; \sup_{\tau\in\mathcal{S}} \; \mathbb{E}^{\mathbb{Q}}\left[\frac{(K - A_\tau)_+}{B_\tau}\right]. \tag{2.68}$$

Intuition. The American price is the Snell envelope of the intrinsic process: it is the smallest supermartingale that dominates immediate-exercise value at every node. Operationally it says: at each moment, compare "what I get by exercising right now" against "what the option is worth if I wait one more step" — take whichever is larger.

The Snell envelope is a fundamental object in optimal stopping theory. Given an intrinsic-value process $\{I_t\}$ — the payoff from exercising at time t — the Snell envelope $V_t = \operatorname{ess\,sup}_{\tau\ge t}\mathbb{E}[I_\tau \mid \mathcal{F}_t]$ is the smallest $\mathbb{Q}$-supermartingale dominating I. "Smallest" means pointwise smallest: for any other supermartingale $M_t \ge I_t$, we have $V_t \le M_t$. "Supermartingale" means $\mathbb{E}[V_s \mid \mathcal{F}_t] \le V_t$ for $s \ge t$, i.e., the value deteriorates in expectation (properly discounted) over time — this is the hallmark of an option where holding costs you expected value relative to exercising. The optimal stopping time is $\tau^* = \inf\{t : V_t = I_t\}$, the first hitting time of the exercise region. At every earlier node, holding is optimal; at τ^* the two values coincide and exercise is indifferent to continuing.

On the lattice, this abstract machinery reduces to a very simple algorithm: at each node, compare $I_{n,k}$ to $V^h_{n,k}$; set $V_{n,k} = \max(I_{n,k}, V^h_{n,k})$. The exercise boundary appears as the locus of nodes where the two are equal; beyond the boundary (into the exercise region), $I > V^h$; before the boundary (into the continuation region), $V^h > I$. This is exactly the dynamic-programming recursion of (2.69)–(2.72).

4.8.2 2.8.2 Tree diagram — hold vs exercise

At every node of the tree the American pricer faces a binary choice: exercise (accept the intrinsic value and stop) or hold (accept the continuation value and continue). The choice is made by comparing the two candidates and taking the larger. This simple max operation, applied at every node, produces the optimal exercise strategy.

At every interior node there are two candidate values:

- P^h — the hold/continuation value (from the one-step backward recursion),
- $P^x \equiv (K - A)_+$ — the immediate-exercise (intrinsic) value.

The actual American value is $P = \max(P^h, P^x)$.

4.8.3 2.8.3 Dynamic-programming recursion

The recursion below is the cleanest illustration of Bellman's principle in quantitative finance. At every lattice node we are solving a local optimisation problem — exercise or continue? — and the local solutions stitch together into the global optimum. The same pattern generalises to callable bonds, convertible bonds, American-style swaptions, and a host of structured products with embedded optionality.

On a recombining binomial tree with nodes (n, k), propagate two quantities backward:

- Continuation (holding) value $C^h_{n-1,k}$:

$$\frac{C^h_{n-1,k}}{B_{n-1,k}} \;=\; q\,\frac{C_{n,k}}{B_{n,k}} \;+\; (1-q)\,\frac{C_{n,k+1}}{B_{n,k+1}} \tag{2.69}$$

$$\Longrightarrow\; C^h_{n-1,k} \;=\; e^{-r\Delta t}\,\big(\,q\,C_{n,k} \;+\; (1-q)\,C_{n,k+1}\,\big). \tag{2.70}$$

- Intrinsic (immediate exercise) value $C^I_{n-1,k}$:

$$C^I_{n-1,k} \;=\; (A_{n-1,k} - K)_+ \quad \text{(call)}, \qquad C^I_{n-1,k} \;=\; (K - A_{n-1,k})_+ \quad \text{(put)}. \tag{2.71}$$

- American value — take the larger:

$$\boxed{C_{n-1,k} \;=\; \max\Big(C^I_{n-1,k},\; C^h_{n-1,k} \Big)} \tag{2.72}$$

4.8.4 2.8.4 Exercise boundary picture

The exercise boundary is a free boundary in the sense of PDE theory: it is determined *implicitly* by the solution to the pricing equation, not given in advance. At any time t, the boundary is the value $A^*(t)$ of the underlying such that the American option value equals its intrinsic value. For an American put, $A^*(t) < K$ and moves closer to K as $t \to T$ (because the exercise region grows). For an American call on a dividend-paying stock, $A^*(t) > K$ and moves closer to K as ex-dividend dates approach.

A diagonal wedge is drawn on the tree: the optimal-exercise boundary separates nodes where $C^I_{n,k} > C^h_{n,k}$ (exercise immediately, shown on the upper-right for a put — the deep-in-the-money region) from nodes where $C^h_{n,k} > C^I_{n,k}$ (hold). Nodes on the boundary are labelled "optimal to exercise"; nodes beyond are "optimal to exercise: $C^I_{n,k} > C^h_{n,k}$".

Intuition. For an American put on a non-dividend-paying stock, early exercise is driven by the time-value of the strike: when the stock crashes deep enough, the interest you could earn on the cash proceeds K outweighs the remaining optionality. For an American call on a non-dividend-payer the boundary never bites (early exercise is suboptimal, Merton's theorem), so the American call equals the European call — one of the cleanest no-arbitrage results in the theory.

Merton's theorem deserves a quick proof sketch. An American call holder considering early exercise at time t receives $A_t - K$ if they exercise. But they could instead sell the call in the market (whose value $\geq A_t - K$ always, by the lower bound $C_t \geq A_t - Ke^{-r(T-t)}$ for a non-dividend-paying stock and positive interest rate). In fact $C_t \geq A_t - Ke^{-r(T-t)} > A_t - K$, so selling always dominates exercising. The same argument fails for a dividend-paying stock: if the stock pays a large cash dividend between t and T, the holder might prefer exercising immediately (capturing the current cum-dividend price) to holding through the ex-dividend drop. Early exercise of American calls is therefore only relevant on dividend-paying underlyings, around ex-dividend dates. For puts, early exercise is driven by the time-value of the strike, and the boundary exists always (no dividends needed).

The American-put early-exercise premium (AEP) is the difference between American and European put values. At typical parameters ($S_0 = 100$, $K = 100$, $\sigma = 20\%$, $r = 5\%$, $T = 1$), AEP is a few cents — small, but non-zero, and growing with moneyness, with rate, and with time to maturity. Computing AEP requires solving the Snell envelope; there is no closed-form formula. The Bjerksund–Stensland and Geske–Johnson approximations are popular analytical tools for bounding AEP; for exact prices, a binomial or Monte-Carlo with Longstaff–Schwartz regression is standard.

American put early-exercise boundary *American put early-exercise boundary*

4.9 2.9 Bridge to Continuous Time

The CRR lattice of §2.3–§2.4 converged, via the Central Limit Theorem applied to the sum of log-returns, to a lognormal distribution for the terminal asset:

$$A_T \;\overset{d}{=}\; A_0\, e^{(r - \frac{1}{2}\sigma^2)T \,+\, \sigma\sqrt{T}\, Z_{\mathbb{Q}}}, \qquad Z_{\mathbb{Q}} \sim \mathcal{N}(0,1). \tag{2.73}$$

This is the *marginal* distribution of A_T under $\mathbb{Q}$. Turning this into a *process* — a continuous-time object defined for every $t \in [0, T]$ — requires more work. The scaled sum of Bernoulli shocks along the tree,

$$W^N_t \;:=\; \sqrt{\Delta t} \sum_{n=1}^{\lfloor t/\Delta t \rfloor} x_n, \tag{2.74}$$

converges in distribution (Donsker's functional CLT) to a *Brownian motion* W_t on $[0, T]$. Under $\mathbb{Q}$ the asset itself then converges to geometric Brownian motion (GBM):

$$A_t \;=\; A_0\, e^{(r-\frac{1}{2}\sigma^2)t\,+\,\sigma W_t}, \qquad dA_t \;=\; r\,A_t\,dt \;+\; \sigma\,A_t\,dW_t. \tag{2.75}$$

These formulas can be read off the CRR lattice in the limit, and a naïve CLT-style argument gives a convincing heuristic derivation. But a rigorous derivation requires (i) a rigorous construction of Brownian motion, (ii) a rigorous theory of the stochastic integral, and (iii) Itô's lemma, which tells us how to change variables inside stochastic differentials. These foundations are developed systematically in Chapter 3.

The European call price in the GBM limit evaluates to the classical Black–Scholes formula,

$$C_0 \;=\; A_0\,\Phi(d_+) \;-\; K\,e^{-rT}\,\Phi(d_-), \qquad d_\pm \;=\; \frac{\ln(A_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \tag{2.76}$$

where Φ is the standard-normal CDF. The *direct* route from the CRR lattice to (2.76) runs through a measure-change argument: split the call payoff as $(A_T - K)_+ \;=\; A_T\,\mathbf{1}_{\{A_T>K\}} - K\,\mathbf{1}_{\{A_T>K\}}$, price the first term under the stock-numéraire measure $\mathbb{Q}^A$ and the second under the money-market measure $\mathbb{Q}$, and evaluate each exceedance probability as a Gaussian tail. This is the Donsker-to-Black-Scholes sketch. Once Itô's lemma is available, the same formula drops out of the hedging argument in Chapter 6.

We defer the full derivation deliberately. The reader will see Black–Scholes emerge in two complementary ways in the chapters ahead:

- Chapter 3 places Brownian motion on rigorous footing, proves Itô's lemma, and solves the GBM SDE explicitly — giving (2.75) as a theorem rather than a heuristic limit.
- Chapter 4 develops the Feynman–Kac connection between SDEs and PDEs.
- Chapter 5 develops the Radon–Nikodym / Girsanov machinery that turns measure changes (including the stock-numéraire switch used in the BS derivation) into a single canonical toolkit.
- Chapter 6 re-derives (2.76) as the canonical Black–Scholes formula from the self-financing hedging argument, and connects it to the Feynman–Kac route as a sanity check.

Monte-Carlo simulation and path-dependent options — forward-starting (cliquet) contracts, up-and-in / knock-out barriers, Asians, lookbacks — are treated in Chapter 10, once the GBM path generator of (2.75) has been placed on rigorous footing. American options on the tree (§2.8 above) remain the cleanest non-trivial use of backward induction on the lattice; path-dependent Americans combine the two paradigms and are the subject of the Longstaff–Schwartz regression technique discussed in Chapter 10.

The upshot for the CRR limit is simple, and we state it as a promise to the reader. The CRR tree converges in distribution to GBM (Donsker's theorem applied to the scaled random walk of the log-price), and the European payoff limit produces the Black–Scholes formula (2.76). This derivation is re-done rigorously in Chapter 6 via Itô's lemma. For now, the reader should take away three things from this chapter:

1. The no-arbitrage pricing of any European claim is a discounted $\mathbb{Q}$-expectation of its payoff.
2. Backward induction on a recombining tree makes that expectation computable in $O(N^2)$ operations and extends naturally to American exercise.
3. In the CRR limit the tree converges to GBM, and vanilla Euro prices converge to Black–Scholes — but writing the limit down honestly requires the stochastic-calculus apparatus of the next three chapters.

4.10 Key Takeaways

1. The one-period no-arb inequality $A_d < A(1+r) < A_u$ is equivalent to existence of a risk-neutral measure $\mathbb{Q}$ such that $C = (1+r)^{-1}\mathbb{E}^{\mathbb{Q}}[C_1]$.
2. Multi-period backward induction prices any European payoff via repeated one-step discounted risk-neutral expectations — the tower-property manifestation of $C_0 = \mathbb{E}^{\mathbb{Q}}[e^{-rT}C_T]$.
3. Classical CRR calibration. Match two moments of the log-return: $c = \sigma\sqrt{\Delta t}$, $p = \frac{1}{2}(1+\frac{\mu}{\sigma}\sqrt{\Delta t})$; the risk-neutral $q = \frac{1}{2}(1 + \frac{r-\sigma^2/2}{\sigma}\sqrt{\Delta t})$.

4. Alternate CRR (drift-shifted lattice). Nodes $A \cdot e^{(\mu-\frac{1}{2}\sigma^2)\Delta t \pm \sigma\sqrt{\Delta t}}$ with symmetric probability $p = \frac{1}{2}$; the Itô correction is baked into the geometry. Under $\mathbb{Q}$, $q \sim \frac{1}{2}[1 + \frac{r-\mu}{\sigma}\sqrt{\Delta t}]$. Both parameterisations converge to the same lognormal GBM.
5. Measure-invariant variance. $\mathbb{V}^{\mathbb{P}}[\ln A_T] = \mathbb{V}^{\mathbb{Q}}[\ln A_T] = \sigma^2 T$ — Girsanov cannot change volatility, only drift.
6. Simple- vs log-return bias. Fitting to simple returns shifts the drift estimate by $+\frac{1}{2}\sigma^2$; always be explicit which object is being fit.
7. FTAP equates no-arbitrage to the existence of an equivalent martingale measure $\mathbb{Q}$ under which all discounted traded assets are $\mathbb{Q}$-martingales.
8. Numéraire freedom. Any strictly positive traded asset B_t can discount — picking B_t is picking a measure. Canonical derivation of the change-of-measure mechanics is deferred to Chapter 5.
9. Non-uniqueness of $\mathbb{Q}$ appears as soon as the number of future states exceeds the number of traded assets: the claim may not lie in $\mathrm{span}(A_1, B_1)$. The set $\mathcal{M}_e$ of equivalent martingale measures has dimension $n-k$ for n states and k tradables.
10. Local risk-minimisation $\min_{\alpha,\beta} \mathbb{E}^{\mathbb{P}}[(\alpha A_1 + \beta B_1 - C_1)^2]$ selects a hedge but leaves residual, unhedgeable variance.
11. Default as a third branch adds the probability mass $\hat{\lambda}\Delta t$ at 0 and shifts the risk-neutral drift by $+\hat{\lambda}$ — the surviving branch must grow faster to compensate. This is our first concrete incomplete-market tree.
12. Binomial-to-GBM limit. With Bernoulli ± 1 increments under $\mathbb{Q}$, $A_T \xrightarrow{d} A_0\, e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z}$ — the marginal distribution of geometric Brownian motion.
13. Donsker's functional CLT promotes the scaled random walk of the log-price to a Brownian motion W_t on $[0, T]$, giving GBM $dA_t = rA_t\, dt + \sigma A_t\, dW_t$ — deferred to Chapter 3 for the rigorous construction.
14. Black–Scholes $C_0 = A_0\Phi(d_+) - Ke^{-rT}\Phi(d_-)$ is the CRR-limit pricing formula for a European call. The derivation via measure change and Itô's lemma is given in Chapters 5 and 6.
15. American options require pointwise comparison of intrinsic and holding values at every node; the optimal policy is a stopping time defined by the exercise boundary. American calls on non-dividend-paying stock equal European calls (Merton's theorem); American puts have a strictly positive early-exercise premium.
16. Monte-Carlo simulation and path-dependent options (cliquets, barriers, Asians) are treated in Chapter 9, once the GBM path generator has been placed on rigorous footing in Chapter 3.

4.11 Reference Formulas

One-period no-arbitrage.

$$A_d < A(1+r) < A_u \iff \exists\, \mathbb{Q} \text{ s.t. } C = \tfrac{1}{1+r}\, \mathbb{E}^{\mathbb{Q}}[C_1].$$

FTAP / discounted martingale.

$$\tilde{X}_t = \frac{X_t}{B_t}, \qquad \tilde{X}_t = \mathbb{E}^{\mathbb{Q}}\left[\tilde{X}_s \mid \mathcal{F}_t\right].$$

Classical CRR calibration (moment-matching).

$$c \;\sim\; \sigma\sqrt{\Delta t}, \qquad p \;\sim\; \tfrac{1}{2}\Big(1 + \tfrac{\mu}{\sigma}\sqrt{\Delta t}\Big), \qquad q \;\sim\; \tfrac{1}{2}\left(1 + \frac{r-\frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right).$$

Alternate CRR (drift-shifted lattice).

$$A \to A\, e^{(\mu-\frac{1}{2}\sigma^2)\Delta t \pm \sigma\sqrt{\Delta t}}, \qquad p = \tfrac{1}{2}, \qquad q \;\sim\; \tfrac{1}{2}\Big[\,1 + \tfrac{r-\mu}{\sigma}\sqrt{\Delta t}\,\Big] + \cdots.$$

Log-moment invariance.

$$\mathbb{E}^{\mathbb{P}}[\ln(A_T/A_0)] = \mu T, \qquad \mathbb{E}^{\mathbb{Q}}[\ln(A_T/A_0)] = (r - \tfrac{1}{2}\sigma^2)T, \qquad \mathbb{V}^{\mathbb{P}}[\ln(A_T/A_0)] = \mathbb{V}^{\mathbb{Q}}[\ln(A_T/A_0)] = \sigma^2 T.$$

Local risk-minimising hedge.

$$\min_{(\alpha,\beta)} \mathbb{E}^{\mathbb{P}}\left[(\alpha A_1 + \beta B_1 - C_1)^2\right].$$

Default-tree risk-neutral probability.

$$q = \frac{e^{(r+\hat{\lambda})\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{+\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}}, \qquad q \sim \tfrac{1}{2}\left(1 + \frac{r + \hat{\lambda} - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right).$$

Survival law.

$$\mathbb{P}(\tau > T) = e^{-\hat{\lambda}T}, \quad \mathbb{P}(\tau \in (t, t+\Delta t]) = e^{-\hat{\lambda}t}(1 - e^{-\hat{\lambda}\Delta t}).$$

European recursion (default tree).

$$C_{n-1,j} = q\,\frac{e^{-\hat{\lambda}\Delta t}}{e^{r\Delta t}}\,C_{n,j} + (1-q)\,\frac{e^{-\hat{\lambda}\Delta t}}{e^{r\Delta t}}\,C_{n,j+1} + (1 - e^{-\hat{\lambda}\Delta t})\,C_{n,d}.$$

American recursion.

$$\begin{gathered} C^h_{n-1,k} = e^{-r\Delta t}(q\,C_{n,k} + (1-q)\,C_{n,k+1}), \\ C^I_{n-1,k} = (A_{n-1,k} - K)_+ \text{ (call)}, \quad (K - A_{n-1,k})_+ \text{ (put)}, \\ C_{n-1,k} = \max\Big(C^I_{n-1,k},\ C^h_{n-1,k}\Big). \end{gathered}$$

CRR-limit lognormal distribution.

$$A_T \overset{d}{=} A_0\,e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z}, \qquad Z \sim \mathcal{N}(0,1) \text{ under } \mathbb{Q}.$$

GBM continuous-time limit (Donsker).

$$A_t = A_0\,e^{(r-\frac{1}{2}\sigma^2)t + \sigma W_t}, \qquad dA_t = r\,A_t\,dt + \sigma\,A_t\,dW_t, \quad W \text{ a } \mathbb{Q}\text{-Brownian motion.}$$

Black–Scholes formula (CRR limit, derived in Chapter 6).

$$C_0 = A_0\,\Phi(d_+) - Ke^{-rT}\,\Phi(d_-), \qquad d_\pm = \frac{\ln(A_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

5 Chapter 3 — Stochastic-Calculus Primer: Brownian Motion, Itô, and SDE Solutions

This chapter builds the stochastic-calculus apparatus that every later chapter uses. It is the foundational chapter of the guide: from here forward we take Itô's lemma, the Itô isometry, and the closed-form solution of the simplest SDEs as known, and we build measure-change, PDE bridges, and hedging arguments on top of them.

The arc of the chapter is deliberately linear. We start from the scaled random walk of Chapter 2 — the same CRR lattice you already know how to price on — and take a scaling limit to Brownian motion. We then prove the two structural path properties of BM that every derivative-pricing argument rests on (infinite total variation, finite quadratic variation equal to t), and use them to define the Itô stochastic integral. The Itô isometry drops out as a clean L^2 identity; Brownian moments follow as short MGF exercises; Itô's lemma in three successively more general forms (Lemmas I, II, III) is derived from a second-order Taylor expansion that keeps exactly the terms of size $\mathrm{d}t$. The SDE catalogue at the end of the chapter — GBM, Ornstein–Uhlenbeck, and the generic constant-coefficient diffusion — is each solved as a worked Itô-lemma exercise, so by the end of the chapter the reader knows not only the machinery but the three or four canonical SDEs that every later chapter will invoke.

A small upfront caveat. Nothing in the chapter is harder than one variable of calculus plus one variable of probability — but the *interaction* of those two disciplines is what makes stochastic calculus feel foreign on first reading. The Taylor series of $f(W_t)$ behaves nothing like the Taylor series of $f(x)$, and the mental reflex of "discard the second-order term" has to be permanently unlearned. The payoff is enormous: every pricing formula, every risk sensitivity, every hedging recipe in derivatives markets is downstream of the single identity $(\mathrm{d}W)^2 = \mathrm{d}t$. Master that line and the Black-Scholes PDE, the Greeks, the log-normal density, implied-vol surfaces, the forward measure, and Black-76 all arrange themselves into a small family of corollaries.

A note on notation. We use W_t for Brownian motion throughout. Plain W denotes a $\mathbb{P}$-Brownian motion when no measure is specified; when a Girsanov shift is in play (from Chapter 5 onward) hats or tildes will decorate the BM under the alternative measure. In this chapter measure is immaterial, because every path-property argument we make is invariant under equivalent changes of measure.

5.1 3.1 From Scaled Random Walk to Brownian Motion

5.1.1 3.1.1 The CRR tree, viewed as a discrete Brownian motion

Chapter 2 built a multi-period binomial tree for an underlying index, priced options on it by backward induction, and took a Donsker-style continuous-time limit in which the terminal distribution becomes log-normal. We now revisit that construction, but strip away the option-pricing apparatus to expose the underlying object — the *scaled random walk* that the lattice implicitly defines.

Let $x_1, x_2, \ldots$ be i.i.d. Bernoulli sign-flips with $\mathbb{P}(x_k = \pm 1) = \frac{1}{2}$. Partition the time axis into steps of size $\Delta t = t/N$ and define a scaled random walk by

$$W_{n\Delta t} \;=\; W_{(n-1)\Delta t} \;+\; \sqrt{\Delta t}\, x_n, \qquad W_0 \;=\; 0. \tag{3.1}$$

Equivalently, summing N increments,

$$W_{N\Delta t} \;=\; \sqrt{\Delta t} \sum_{n=1}^{N} x_n. \tag{3.2}$$

Each increment has size $\pm\sqrt{\Delta t}$. The square-root scaling is not cosmetic: shrinking the time-step by a factor of N shrinks each displacement by only $\sqrt{N}$, not by N, and that is exactly the asymmetric scaling needed to keep the variance finite in the limit. The multiplicative geometric version used for stock trees in Chapter 2 writes $S \in \{Se^{\sigma\sqrt{\Delta t}}, Se^{-\sigma\sqrt{\Delta t}}\}$; the arithmetic version $W \in \{W + \sqrt{\Delta t}, W - \sqrt{\Delta t}\}$ is what limits to Brownian motion and is what we study in this chapter.

The square-root-of-time scaling is worth a standalone intuition. Suppose you flipped a coin every millisecond for a year and called the net displacement D of a unit-stride random walker. The walker's position at year-end would be

$D \sim \text{Normal}(0, N)$ with $N \approx 3.15 \times 10^{10}$ steps, giving a standard deviation of $\sqrt{N} \approx 177{,}000$ millimetres — about 177 metres. Now ask the opposite question: how large does each stride have to be so that, after the year, the walker's typical displacement is exactly 1 metre? Solving $\sqrt{N}\,\ell = 1$ m gives $\ell \approx 5.6\,\mu$m — roughly the size of a red blood cell. The stride $\ell = \sigma\sqrt{\Delta t}$ is the unique scaling that makes the stride shrink as the step-rate grows yet keeps the annual variance finite and positive. That is the whole magic of the $\sqrt{\Delta t}$ factor and the whole magic of Brownian motion.

A second intuition: the stride must shrink faster than the time-step because the *number* of steps grows as $1/\Delta t$ while the variance of the total displacement grows *linearly* in that number. Linear-in-steps times variance-per-step equals total variance; to keep total variance bounded by t per unit time, each step's variance has to be $O(\Delta t)$, which forces the step size to be $O(\sqrt{\Delta t})$. Any other choice either collapses the walk (if you pick Δt exactly) or blows it up (if you pick a constant).

5.1.2 3.1.2 Moments of the walk

Using $\mathbb{E}[x_n] = 0$ and $\text{Var}[x_n] = 1$:

$$\mathbb{E}[W_{N\Delta t}] \;=\; \sqrt{\Delta t}\sum_{n=1}^{N}\mathbb{E}[x_n] \;=\; 0, \tag{3.3}$$

$$\text{Var}[W_{N\Delta t}] \;=\; \Delta t\sum_{n=1}^{N}\text{Var}[x_n] \;=\; N\,\Delta t \;=\; t. \tag{3.4}$$

So $W_{N\Delta t}$ has mean 0 and variance t for *any* N — the variance is preserved exactly along the refinement, which is the whole point of picking $\sqrt{\Delta t}$ rather than Δt or any other power. Had we used Δt the walk would collapse to 0; had we used a constant, it would explode. Only the square-root scaling keeps $\text{Var}[W_t] = t$.

The key observation to register: the mean and the variance are exact for every N, not just asymptotically. When proving convergence of a random quantity it is enormously useful to have one moment already at its limiting value for every finite approximation; convergence then reduces to showing the *other* moment shrinks to zero. That double-moment structure — one moment exact, one shrinking — will reappear in every almost-sure-limit proof in this chapter.

5.1.3 3.1.3 CLT: the continuous-time limit

Let t be fixed and $N \to \infty$ (equivalently $\Delta t = t/N \downarrow 0$). By the central limit theorem, the normalised sum $\frac{1}{\sqrt{N}}\sum_{n=1}^{N} x_n$ converges in distribution to a standard normal, so

$$W_t \;\xrightarrow[N\to\infty]{d}\; \sqrt{t}\,Z, \qquad Z \sim \mathcal{N}(0,1). \tag{3.5}$$

Thus in the limit $W_t \sim \mathcal{N}(0, t)$ is Gaussian with mean 0 and variance t. Read this the other way round: the Gaussian marginal at each time is *not* an assumption — it is forced on us by the CLT the moment we impose the correct scaling in (3.1).

The Gaussian law is not posited; it is *inherited* from the CLT's universality. Any other i.i.d. innovation with finite variance — uniform ± 1, Rademacher ± 1 with unfair weights, a truncated triangular distribution, a Laplace distribution with heavy tails (provided the variance is finite), even empirical return distributions from real markets — gives the same Gaussian scaling limit. The only surviving features of the innovation are its mean and its variance; all higher moments wash out under $\sqrt{N}$ averaging. In that sense Brownian motion is the *universal* continuous-time random walk, in the same way the Gaussian is the universal limit of normalised sums. A practical corollary: when you simulate a Brownian motion on a computer, you are almost always generating the fine-grained walk with Gaussian increments $\sqrt{\Delta t}\,Z$ rather than Bernoulli ones, because CLT universality makes the two limits identical.

5.1.4 3.1.4 Stationary and independent increments

For any $t < t + s$, write $W_{t+s} - W_t = \sqrt{\Delta t}\sum_{m=1}^{M} y_m$ with $y_m = x_{N+m}$ a fresh block of sign-flips. By the same CLT, with $M\Delta t = s$,

$$W_{t+s} - W_t \xrightarrow[M\to\infty]{d} \sqrt{s}\, Z, \qquad Z \sim \mathcal{N}(0,1). \tag{3.6}$$

Because the law of $W_{t+s} - W_t$ depends only on the *length* s of the interval — not on W_t or the starting time t — the process has *stationary increments*. Moreover, for non-overlapping intervals $[t,s]$ and $[u,v]$ the underlying Bernoulli blocks are disjoint, so the two sums are functions of independent random variables and therefore

$$(W_s - W_t) \perp (W_v - W_u) \qquad \text{(independent increments)}. \tag{3.7}$$

Independent increments is the source of the *Markov property*: knowing W up to time u tells you nothing about future increments beyond what W_u already encodes. Equivalent phrasings — conditional distribution of W_{t+s} given $\mathcal{F}_t$ is $\mathcal{N}(W_t, s)$, and conditional on W_t the future is independent of the past — will be used verbatim in the Feynman-Kac derivations of Chapter 4 and the change-of-numeraire arguments of Chapter 5.

5.2 3.2 Definition of Brownian Motion and Path Properties

Collecting the three properties just derived plus a regularity property, the scaling limit $W = (W_t)_{t\geq 0}$ is defined as a process satisfying

$$\begin{aligned}
&\text{(i)}\ \ W_0 = 0,\\
&\text{(ii)}\ \ W_t \overset{d}{=} \sqrt{t}\, Z, \quad Z \sim \mathcal{N}(0,1),\\
&\text{(iii)}\ \ W \text{ has stationary and independent increments,}\\
&\text{(iv)}\ \ W \text{ has continuous paths.}
\end{aligned} \tag{3.8}$$

Property (iv) is the only one not already implied by the random-walk construction: the polygonal interpolation of the lattice walk is piecewise-linear, and continuity of the limit needs a tightness argument (Kolmogorov's continuity criterion). Properties (i)–(iii) are the structural skeleton; continuity then paints the skeleton smooth — but, as we are about to see, the *roughness* of that continuity is severe.

A pedagogical remark. The four conditions are conceptually independent, and you can build "partial Brownian motions" that satisfy any three of them but not the fourth. A compound Poisson process satisfies (i)–(iii) but has discontinuous paths and so fails (iv). A process that sits at zero forever satisfies (i) and (iv) but fails (ii) and (iii). A standard Cauchy process (stable with index 1) satisfies (i), (iii), (iv) but has increments that are Cauchy, not Gaussian, and so fails (ii). It is the combination of all four — specifically Gaussian marginals plus independent increments plus continuity — that pins Brownian motion down uniquely (up to measure-zero null sets).

5.2.1 3.2.1 Continuous but nowhere differentiable

A typical sample path of BM is a jittery curve, everywhere continuous but nowhere smooth. Heuristically: the typical displacement over a time-step Δt is $|\Delta W| \sim \sqrt{\Delta t}$, so the slope

Five Brownian-motion sample paths *Five independent sample paths of W on $[0,1]$. Every path is continuous but visibly non-smooth at every scale — zooming in reveals identical-looking wiggles. The unbounded-slope property (3.∞) is visible as the path's inability to be "traced with a pencil".*

$$\frac{\Delta W}{\Delta t} \sim \frac{\sqrt{\Delta t}}{\Delta t} = \frac{1}{\sqrt{\Delta t}} \longrightarrow \infty \quad (\Delta t \downarrow 0).$$

Every secant slope blows up, so the path has no derivative at any point despite being everywhere continuous. This is the first indication that ordinary calculus cannot handle W — we need a *quadratic* calculus, which §3.3–§3.4 make precise. The intuition for why this forces a new calculus is simple: in any Taylor expansion of a function $f(W_t)$, the second-order term carries a factor $(\Delta W)^2 \sim \Delta t$, which is the *same* order as Δt itself. Second-order terms can never be dropped as "small" in the way Newton's calculus assumes.

A concrete analogy is a coastline. Measure a coastline with a kilometre-long ruler and you get one total length; measure it with a metre-long ruler and the length grows because the ruler resolves previously-hidden wiggles; measure with a centimetre-long ruler and the length grows again. In the limit of an infinitesimally fine ruler, the total length diverges. A Brownian path behaves the same way: the *length* (total variation) diverges as the ruler refines, while a different quantity — the *sum of squared straight-line segments* (quadratic variation) — stays finite and equals t. Section 3.3 computes the first; Section 3.4 computes the second.

The reason this matters for trading. Every gamma trader lives inside the discrepancy between TV and QV. If BM had finite TV, a delta-hedged option would have zero expected P&L regardless of hedging frequency — there would be no gamma premium to harvest, because the path could be traced arbitrarily precisely with finitely many hedge trades. Infinite TV says the opposite: no matter how often you rehedge, the residual cash flow between rehedges is never negligible. Finite QV — the specific value t — tells you how large that residual is on average, which is the *gamma theta* of options pricing. The two path properties we are about to prove are not abstract measure-theoretic curiosities; they are the mathematical expressions of "options have a gamma premium."

5.3 3.3 Total Variation

5.3.1 3.3.1 Definition

For a function f on $[0, t]$ and a partition $\pi = \{t_0, t_1, \dots, t_N\}$ with $0 = t_0 < t_1 < \dots < t_N = t$, the total variation is

$$\mathrm{TV}_t \;=\; \lim_{\|\pi\|\downarrow 0} \sum_k \big|\, f(t_k) - f(t_{k-1})\,\big|, \tag{3.9}$$

where $\|\pi\| = \max_k (t_k - t_{k-1})$ is the mesh size. Pictorially, lay the graph of f flat by flipping each down-stroke upward; the total horizontal length is TV_t.

TV matters because the Riemann-Stieltjes integral $\int g\,\mathrm{d}f$ only exists (in the classical sense) when f has finite total variation on $[0, t]$. For BM we are about to find TV is infinite, which is exactly why the new stochastic integral of §3.5 is needed.

5.3.2 3.3.2 Case 1 — f is differentiable (finite TV)

If $f \in C^1$, by the mean-value theorem there exists $t_k^\star \in (t_{k-1}, t_k)$ with $\Delta f_k = f'(t_k^\star)\,\Delta t_k$. Therefore

$$\sum_k |\Delta f_k| \;=\; \sum_k |f'(t_k^\star)|\,\Delta t_k \;\xrightarrow[\|\pi\|\downarrow 0]{}\; \int_0^t |f'(s)|\,\mathrm{d}s \;<\; +\infty. \tag{3.10}$$

Finite TV, as expected for smooth functions. The entire argument fails for BM because W has no derivative anywhere, and in its absence the sum of $|\Delta W_k|$ diverges.

5.3.3 3.3.3 Case 2 — Brownian motion has infinite TV

For BM the increments are $\Delta W_k \overset{d}{=} \sqrt{\Delta t_k}\, Z_k$ with $Z_k \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. So

$$\sum_k |W_{t_k} - W_{t_{k-1}}| \;=\; \sum_k \sqrt{\Delta t_k}\,|Z_k|. \tag{3.11}$$

The mean. Because $\mathbb{E}|Z| = \sqrt{2/\pi}$ is a finite constant,

$$\mathbb{E}\left[\sum_k \sqrt{\Delta t_k}\,|Z_k|\right] \;=\; \mathbb{E}|Z| \cdot \sum_k \sqrt{\Delta t_k}. \tag{3.12}$$

The variance is $c^2 \sum_k \Delta t_k = c^2 t$, bounded as the partition refines. The key bound is on the mean:

$$\sum_k \sqrt{\Delta t_k} \;\geq\; \sum_k \frac{\Delta t_k}{\sqrt{\|\pi\|}} \;=\; \frac{t}{\sqrt{\|\pi\|}} \;\xrightarrow[\|\pi\|\downarrow 0]{}\; +\infty. \tag{3.13}$$

So the mean of the TV-sum diverges while its variance stays bounded — concentration around an unbounded mean forces the sum itself to be unbounded. Almost surely,

$$\boxed{\mathrm{TV}_t(W) \;=\; \lim_{\|\pi\|\downarrow 0} \sum_k |W_{t_k} - W_{t_{k-1}}| \;=\; +\infty}. \qquad \text{(a.s.)} \tag{3.14}$$

Brownian motion is too rough to have a well-defined first-order path integral $\int_0^t g(s)\,\mathrm{d}W_s$ in the Riemann-Stieltjes sense; a new stochastic integral is needed, and that is the subject of §3.5.

The economic shadow. A delta-hedger rebalances N times per unit time and incurs a transaction cost of $c \cdot |\Delta S|$ per trade. Total trading cost equals $c \cdot \sum |\Delta S_k|$, which is the TV sum for S. Because S is approximately GBM and GBM has infinite TV, this cost diverges as rebalancing becomes continuous — exactly the feature that prevents the frictionless Black-Scholes derivation from being implementable literally. In practice, hedgers impose a discrete rebalancing grid and accept residual gamma risk; the frictionless-trading assumption is the statement "pretend TV does not diverge, or at least that transactions are free." Whenever a trader complains about transaction costs on a delta-hedging strategy, that is the pathological TV of GBM biting.

5.4 3.4 Quadratic Variation and the Rule $(\mathrm{d}W)^2 = \mathrm{d}t$

5.4.1 3.4.1 Definition

The quadratic variation of f on $[0, t]$ is

$$[f]_t \;=\; [f, f]_t \;=\; \lim_{\|\pi\|\downarrow 0} \sum_k (\Delta f_k)^2, \qquad \Delta f_k = f(t_k) - f(t_{k-1}). \tag{3.15}$$

Where TV measured path length, QV measures cumulative *squared jitter*. For smooth functions QV vanishes (because $(\Delta f)^2 \approx (f')^2 (\Delta t)^2$ is one order higher than Δt); for BM, as we are about to see, QV equals t exactly — a finite, deterministic number. This is the single most important fact in stochastic calculus, because it converts the heuristic $(\mathrm{d}W)^2 = \mathrm{d}t$ into a rigorous statement.

Why the quadratic version of the sum is well-behaved while the linear version diverges. The sum-of-absolutes $\sum |\Delta W_k|$ behaves like $\sum \sqrt{\Delta t_k} \cdot \mathbb{E}|Z|$ — a sum of *square-roots* — which grows like $\sqrt{N}$. The sum-of-squares $\sum (\Delta W_k)^2$ behaves like $\sum \Delta t_k \cdot \mathbb{E}[Z^2] = t$, constant in N. Squaring each increment converts the diverging $\sqrt{\Delta t}$ into a summable Δt and shifts the bad behaviour into a benign regime. A cleaner way to state the lesson: TV sees *size*, QV sees *energy*. Brownian motion has bounded energy (the t in $[W, W]_t = t$) but unbounded size (the $+\infty$ in $\mathrm{TV}_t(W) = +\infty$). In trader language, BM has finite integrated variance ($= t$ times vol-squared) but infinite gross trading distance ($=$ integrated |price change|). The two quantities are not the same, and conflating them is the most common source of Black-Scholes confusion.

5.4.2 3.4.2 Case 1 — f differentiable: $[f]_t = 0$

If $f \in C^1$,

$$\sum_k (\Delta f_k)^2 \;=\; \sum_k \left(f'(t_k^\star)\right)^2 (\Delta t_k)^2 \;\leq\; \|\pi\| \cdot \int_0^t (f'(s))^2\,\mathrm{d}s \;\xrightarrow[\|\pi\|\downarrow 0]{}\; 0. \tag{3.16}$$

So differentiable functions have zero QV. The inequality $(\Delta t_k)^2 \leq \Delta t_k \cdot \|\pi\|$ peels off one power of $\|\pi\|$, which kills the sum in the limit. This is why ordinary calculus has no Itô correction: for smooth paths, squared-increment sums are negligible.

Phrased dimensionally: for smooth f, $\Delta f_k = O(\Delta t_k)$ so $(\Delta f_k)^2 = O((\Delta t_k)^2)$, and there are $N = t/\Delta t$ of them, giving a total $N \cdot O((\Delta t)^2) = O(\Delta t) \to 0$. For Brownian motion $\Delta W_k = O(\sqrt{\Delta t_k})$, so $(\Delta W_k)^2 = O(\Delta t_k)$ and the sum $N \cdot O(\Delta t) = O(1)$ stays finite. Smooth functions produce $(\Delta f)^2 \sim (\Delta t)^2$ (second-order small); BM produces $(\Delta W)^2 \sim \Delta t$ (first-order small). One order of Δt is kept by the squaring, the other washed out — and that retained order produces the Itô correction.

5.4.3 3.4.3 Case 2 — Brownian motion: $[W, W]_t = t$

Let $Q = \sum_k (\Delta W_k)^2$ with $\Delta W_k \overset{d}{=} \sqrt{\Delta t_k}\, Z_k$, $Z_k \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$.

Mean.

$$\mathbb{E}[Q] \;=\; \sum_k \mathbb{E}[(\Delta W_k)^2] \;=\; \sum_k \Delta t_k \;=\; t. \tag{3.17}$$

The mean is *exact* — no limit needed, no $\|\pi\|$ dependence. Every finite partition already gives $\mathbb{E}[Q] = t$.

Variance. Using $\operatorname{Var}((\Delta W_k)^2) = (\Delta t_k)^2 \operatorname{Var}(Z^2) = 2(\Delta t_k)^2$ (because $\operatorname{Var}(Z^2) = \mathbb{E}[Z^4] - 1 = 3 - 1 = 2$ — the standard-Normal fourth moment $\mathbb{E}[Z^4] = 3$ is derived via the MGF in §3.7.1) and independence across k,

$$\operatorname{Var}[Q] \;=\; 2\sum_k (\Delta t_k)^2 \;\le\; 2\,\|\pi\| \sum_k \Delta t_k \;=\; 2\|\pi\|\,t \xrightarrow[\|\pi\|\downarrow 0]{} 0. \tag{3.18}$$

Conclusion. Q has mean *exactly* t and variance shrinking to zero, so by Chebyshev's inequality (L^2-convergence along the full sequence, almost-sure along a thinning subsequence):

$$\boxed{\sum_k (\Delta W_k)^2 \xrightarrow[\|\pi\|\downarrow 0]{\text{a.s.}} t, \qquad [W, W]_t \;=\; t} \qquad \text{(a.s.)} \tag{3.19}$$

A striking feature: a *random* object — the sum of squared increments along a random trajectory — converges to a *deterministic* limit t. The randomness is there, but it is concentrated so tightly around the mean that in the limit it disappears. Economically, this is why implied volatility is observable at all: the quadratic variation of a stock price is essentially a deterministic integral of σ^2, not a random quantity.

Quadratic variation → t and total variation → ∞ as mesh refines *The same Brownian path, sampled at progressively finer meshes. Left: $\sum(\Delta W_k)^2$ converges deterministically to $t = 1$ as the mesh refines. Right: $\sum |\Delta W_k|$ diverges like $\sqrt{N}$ — finite energy but unbounded path-length, which is precisely the TV vs QV dichotomy of §3.3–§3.4.*

This "randomness collapsing into determinism" deserves a full beat. In the discrete approximation, $\sum_k (\Delta W_k)^2$ is a sum of N i.i.d. random variables, each equal to $\Delta t \cdot Z_k^2$ for an independent chi-squared-one draw. By the law of large numbers the average of these Z_k^2 converges almost surely to $\mathbb{E}[Z^2] = 1$; multiplying by $N\Delta t = t$ gives $\sum(\Delta W_k)^2 \to t$ a.s. The convergence is the LLN dressed up in Brownian clothing. The variance-shrinking-faster-than-mean argument is just Chebyshev telling us the LLN converges rapidly in L^2.

Why "realised variance = integrated variance" in the variance-swap world. A variance swap pays the holder the realised sample variance $\frac{252}{N}\sum(r_k)^2$ where r_k are daily log-returns. In the GBM model $r_k \approx \sigma\Delta W_k - \frac{1}{2}\sigma^2\Delta t$, so $\sum r_k^2 \approx \sigma^2 \sum(\Delta W_k)^2 \to \sigma^2 t$. The variance-swap payoff concentrates on a deterministic quantity equal to the square of the volatility times the tenor — which is why variance swaps are *exactly* replicable by a portfolio of European options (Carr-Madan replication). In contrast, the TV of the same log-return sequence diverges like $\sqrt{N}$, explaining why "average absolute return" is a nuisance statistic whereas "realised variance" is a well-posed estimator of σ^2.

5.4.4 3.4.4 The Itô shorthand $(\mathrm{d}W)^2 = \mathrm{d}t$

Equation (3.19) is the mathematical justification of the Itô multiplication rules

$$(\mathrm{d}W_t)^2 \;=\; \mathrm{d}t, \qquad \mathrm{d}W_t \cdot \mathrm{d}t \;=\; 0, \qquad (\mathrm{d}t)^2 \;=\; 0. \tag{3.20}$$

These rules drive Itô's formula throughout the rest of the chapter (and this guide). In particular — the crucial pedagogical point — the BM path has *non-zero* QV-roughness ($\sum \Delta W_k^2 \to t$), not the 0 you might have hoped for from an ordinary-calculus perspective; that t is precisely the reason pricing PDEs carry a $\frac{1}{2}\sigma^2\,\partial_{xx}$ term. Every Itô correction in this guide traces back to (3.20).

The three rules are worth memorising with their intuitive content.

- $(\mathrm{d}W_t)^2 = \mathrm{d}t$. The square of a Gaussian increment of variance $\mathrm{d}t$ has mean $\mathrm{d}t$ and variance $2(\mathrm{d}t)^2 \ll \mathrm{d}t$; the $\mathrm{d}t$-mean dominates, so at leading order $(\mathrm{d}W)^2 = \mathrm{d}t$ is a deterministic identity.
- $\mathrm{d}W_t \cdot \mathrm{d}t = 0$. The increment $\mathrm{d}W_t$ has magnitude $\sqrt{\mathrm{d}t}$, so $\mathrm{d}W \cdot \mathrm{d}t \sim (\mathrm{d}t)^{3/2}$, which is higher-order than $\mathrm{d}t$ and drops out.
- $(\mathrm{d}t)^2 = 0$. Obvious dimensionally; kept in the list for symmetry.

The memorisation trick. Think of $\mathrm{d}W$ as living at scale $\sqrt{\mathrm{d}t}$ and $\mathrm{d}t$ as living at scale $\mathrm{d}t$. Any product of these two atoms scales as $(\mathrm{d}t)^{n/2}$ where n is the total atom count. Keep only products with $n = 2$ (size exactly $\mathrm{d}t$), drop products with $n > 2$. The three rules are just this bookkeeping.

When we apply Itô's lemma we multiply out $(\mathrm{d}X)^2$ for an SDE $\mathrm{d}X = a\,\mathrm{d}t + b\,\mathrm{d}W$, expand

$$(a\,\mathrm{d}t + b\,\mathrm{d}W)^2 \;=\; a^2\,(\mathrm{d}t)^2 + 2ab\,\mathrm{d}t\,\mathrm{d}W + b^2\,(\mathrm{d}W)^2,$$

and then use (3.20): the first two terms vanish, the third becomes $b^2\,\mathrm{d}t$. The rule set collapses the Itô-square of any SDE to a clean $b^2\,\mathrm{d}t$ — the fact used in every Itô-formula application in this chapter. A quick self-check: if you can state and multiply these three rules from memory, you can generate any Itô-formula line on demand.

5.4.5 Case study: The "rule of 16" — Brownian scaling in market data

Every options trader on a US equity desk converts annualised vol to daily vol by dividing by 16. VIX = 16 means "expect ∼ 1% S&P moves." VIX = 32 means "expect ∼ 2% moves." This is not a heuristic — it is the QV identity (3.19) evaluated at $t = 1/252$*, and it is one of the few places in finance where a continuous-time identity is* directly falsifiable* against tick data on any given day.*

Context. The CBOE Volatility Index (VIX) is a model-free 30-day implied vol on the S&P 500, quoted in *annualised* units. On 14 May 2026 the VIX closed near 14; through 2024–2025 it traded in a 12–22 range outside of brief stress spikes (March 2020 COVID, August 2024 yen-carry unwind, April 2025 tariff shock). The trading floor's *rule of 16* — divide the VIX by 16 to get the implied 1-day standard deviation in percent — comes from $\sqrt{252} \approx 15.87$. So a VIX of 16 implies a 1-day SPX move of roughly $\pm 1\%$; a VIX of 32 implies $\pm 2\%$; a VIX of 80 (March 2020) implied $\pm 5\%$. Empirically: in 2024 the VIX averaged ~ 15.7, predicting $\sim 0.98\%$ daily moves, and the realised one-standard-deviation S&P 500 daily move that year was 0.84% — within the same order of magnitude, with the small undershoot tracing to the well-known variance risk premium (implied vol structurally exceeds realised vol; see Ch 10–11).

Math mapping. This is (3.19) made tradable. Model the S&P log-price as Brownian under the risk-neutral measure with annualised volatility σ: $\mathrm{d}S_t/S_t = r\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, so $\mathrm{d}\log S_t = (r - \frac{1}{2}\sigma^2)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$. The QV identity (3.19) gives $[\log S, \log S]_t = \sigma^2 t$. Equivalently, the variance of the log-return over horizon Δt is $\sigma^2 \Delta t$ and the standard deviation is $\sigma\sqrt{\Delta t}$. With $\Delta t = 1/252$ trading days and \$ = \$ VIX/100, this gives $\mathrm{StDev}_{\mathrm{daily}} = \mathrm{VIX}/(100 \cdot \sqrt{252}) \approx \mathrm{VIX}/1587$, and quoting the result in percent removes the factor of 100: $\mathrm{StDev}_{\mathrm{daily}}(\%) \approx \mathrm{VIX}/\sqrt{252} \approx \mathrm{VIX}/16$. The $\sqrt{t}$ scaling is *not* a model assumption — it is forced by the QV identity, which holds for any continuous semimartingale with absolutely continuous QV. So when the rule-of-16 prediction matches realised daily standard deviations, you are confirming that the path is locally Brownian; when it fails, the path has acquired roughness inconsistent with $[W, W]_t = t$.

Lesson. Three failure modes a desk watches for. First, *gap moves at the open*: 17 Sept 2008 (Lehman week, S&P opened -4.7% on a VIX-implied $\sim 2\%$ one-day SD) and 16 March 2020 (COVID, S&P open down -7% vs VIX-implied $\sim 5\%$) blew through the daily standard deviation by 2–3σ each — the path is not continuous overnight, and overnight returns satisfy a different scaling regime than intraday returns (the QV of an overnight gap is *concentrated*, not spread out at $\sqrt{t}$). Second, *vol-of-vol regimes*: in stress episodes the VIX itself moves 20–30% intraday, so the rule-of-16 conversion using yesterday's VIX systematically under-predicts today's moves; this is the phenomenon Heston-style stochastic vol models (Ch 10) are built to capture. Third, *jump components*: equity microstructure has a jump component (earnings, central bank meetings, exchange-pause incidents) that contributes additional QV beyond the diffusive $\sigma^2\,\mathrm{d}t$; the rule-of-16 implicitly assumes pure diffusion. Operationally, a desk computes a "realised-vs-implied ratio" daily — realised standard deviation of one-minute returns scaled to daily, divided by VIX/16 — and uses sustained excursions above 1.0 as a signal that the diffusive Brownian model is locally inadequate and one should trade vol-of-vol products (VVIX, variance-of-variance swaps) instead of plain vol.

5.5 3.5 The Itô Stochastic Integral

5.5.1 3.5.1 Warm-up — the classical Riemann-Stieltjes integral

For smooth f the chain rule gives

$$\int_0^t f(s)\,\mathrm{d}f(s) \;=\; \tfrac{1}{2}\,(f^2(t) - f^2(0)), \tag{3.21}$$

because $\mathrm{d}\big(\tfrac{1}{2}f^2(s)\big) = f(s)\,\mathrm{d}f(s)$. Remember this answer — when we repeat the calculation for Brownian motion we will find the analogue carries an *extra* $-\frac{1}{2}t$ term that has no classical counterpart. The classical integral is oblivious to the endpoint choice in the Riemann sum because QV is zero; once we swap f for BM the endpoint choice becomes consequential, and the $-\frac{1}{2}t$ is exactly the accumulated left-right mismatch summed over infinitely many infinitesimal intervals.

5.5.2 3.5.2 The Itô integral, defined

The stochastic analogue uses *left-endpoint* Riemann sums — crucial, because this is what makes the integral a martingale:

$$\int_0^t W_s\,\mathrm{d}W_s \;\equiv\; \lim_{\|\pi\|\downarrow 0} \sum_k W_{t_{k-1}}\,\Delta W_{t_k}, \qquad \Delta W_{t_k} = W_{t_k} - W_{t_{k-1}}. \tag{3.22}$$

Evaluating W at the *start* t_{k-1} of each sub-interval makes $W_{t_{k-1}}$ independent of the forward-looking increment ΔW_{t_k}. That independence makes the expectation of each term zero, and hence makes the whole stochastic integral a martingale. Right-endpoints would couple the integrand to its own driving noise and destroy the martingale property.

Economic gloss. The Itô integral is the mathematical expression of a *predictable* trading strategy: at time t_{k-1} you commit to holding $W_{t_{k-1}}$ units of the underlying until t_k, and you realise the P&L $W_{t_{k-1}}(W_{t_k} - W_{t_{k-1}})$ over that interval. Predictability means the position size is chosen *before* the random move, not after; the left-endpoint rule enforces exactly that. If you were allowed to pick the right endpoint you could lean into favourable moves you already know about. The martingale property of stochastic integrals is thus the mathematical statement "predictable strategies have zero expected P&L" — the no-arbitrage condition in continuous time.

A different rule called the Stratonovich integral evaluates at the midpoint $\frac{1}{2}(W_{t_{k-1}} + W_{t_k})$. Stratonovich integrals obey the classical chain rule (no Itô correction) but lose the martingale property. For financial modelling the Itô convention is mandatory because the martingale property, not the chain rule, is what encodes no-arbitrage.

5.5.3 3.5.3 First worked integral — $\int_0^t W_s\,\mathrm{d}W_s$

The claim, which we prove by a telescoping identity + QV computation, is

$$I_t \;=\; \int_0^t W_s\,\mathrm{d}W_s \;=\; \tfrac{1}{2}\,W_t^2 \;-\; \tfrac{1}{2}\,t \qquad \text{a.s.} \tag{3.23}$$

The answer already carries the surprise: the classical guess would be $\frac{1}{2}W_t^2$, but the stochastic integral subtracts an extra $\frac{1}{2}t$ — the *Itô correction*, sometimes called the "convexity correction" because it arises from the curvature of $w \mapsto \frac{1}{2}w^2$ interacting with the QV of W. The $\frac{1}{2}t$ is deterministic, grows linearly, and has no stochastic component; it is the "free" cash that a convex long-gamma position accumulates as time passes. In options language, the correction is the *theta bleed* of a one-lot long-gamma portfolio.

Proof. Use the algebraic identity $a(b-a) = \frac{1}{2}(b^2 - a^2) - \frac{1}{2}(b-a)^2$ with $a = W_{t_{k-1}}$, $b = W_{t_k}$:

$$W_{t_{k-1}}(W_{t_k} - W_{t_{k-1}}) \;=\; \tfrac{1}{2}(W_{t_k}^2 - W_{t_{k-1}}^2) \;-\; \tfrac{1}{2}(W_{t_k} - W_{t_{k-1}})^2. \tag{3.24}$$

Summing over k and using $W_0 = 0$:

$$\sum_k W_{t_{k-1}}\,\Delta W_{t_k} \;=\; \tfrac{1}{2}\,W_t^2 \;-\; \tfrac{1}{2}\,\sum_k (\Delta W_{t_k})^2. \tag{3.25}$$

The left-hand side is the Riemann-like sum defining the Itô integral. The right-hand side has two parts. The first, $\frac{1}{2}W_t^2$, is the classical Newton-Leibniz answer. The second, $-\frac{1}{2}\sum_k(\Delta W_{t_k})^2$, is the Itô correction in discrete form — an exact lattice version of $-\frac{1}{2}t$, because by (3.19) this sum converges almost surely to t. Passing to the limit,

$$\boxed{\int_0^t W_s\,\mathrm{d}W_s \;=\; \tfrac{1}{2}\,W_t^2 \;-\; \tfrac{1}{2}\,t}. \tag{3.26}$$

Three observations on this answer.

1. Taking expectations on both sides of (3.26): $\mathbb{E}\int_0^t W_s\,\mathrm{d}W_s = \frac{1}{2}\mathbb{E}[W_t^2] - \frac{1}{2}t = 0$. The Itô integral is zero-mean — the martingale property surfaces automatically.
2. The Itô integral is *not* zero, despite having zero mean. Its variance grows with t (we'll compute it via the Itô isometry next): it is a non-trivial random variable concentrated around zero.
3. The difference $I_t - \frac{1}{2}W_t^2 = -\frac{1}{2}t$ is *deterministic*. Remarkable: the randomness of the classical antiderivative $\frac{1}{2}W_t^2$ and the randomness of the Itô integral I_t match exactly, so when subtracted they leave only the $-\frac{1}{2}t$ correction.

Equivalent form: rearranging (3.26),

$$\tfrac{1}{2}\,W_t^2 \;=\; \int_0^t W_s\,\mathrm{d}W_s \;+\; \tfrac{1}{2}\,t, \tag{3.27}$$

which is Itô's formula applied to $F(w) = \frac{1}{2}w^2$: $\mathrm{d}(\frac{1}{2}W_t^2) = W_t\,\mathrm{d}W_t + \frac{1}{2}\,\mathrm{d}t$. The $\frac{1}{2}\,\mathrm{d}t$ correction is exactly what distinguishes the stochastic integral from the classical chain-rule answer.

5.5.4 3.5.4 Why the left-endpoint choice matters

Had we used the right-endpoint W_{t_k} (Stratonovich-like), the telescoping identity would have given $+\frac{1}{2}\sum(\Delta W_k)^2 \to +\frac{1}{2}t$ rather than $-\frac{1}{2}t$, producing $\frac{1}{2}W_t^2$ — the classical answer. Itô's left-endpoint choice is what guarantees the integrand $W_{t_{k-1}}$ is $\mathcal{F}_{t_{k-1}}$-measurable and therefore independent of ΔW_{t_k}, giving the stochastic integral its martingale property.

The financial reading. A left-endpoint integrand is "predictable" — you pick your position before the jump. A right-endpoint integrand is "anticipative" — you pick your position after the jump, with knowledge of what just happened. Anticipative strategies violate no-arbitrage and are inadmissible in continuous-time finance. Itô's insistence on left endpoints is the mathematical enforcement of predictability, and hence of no-arbitrage.

5.6 3.6 The Itô Isometry

The L^2-isometry that underlies the definition of the stochastic integral is both a computational workhorse (it reduces any variance-of-stochastic-integral calculation to an ordinary time integral) and a structural theorem (it says the Itô integral is a continuous linear map from a weighted L^2 space of integrands into the space of martingales).

Statement. For any two adapted integrands g_s and h_s satisfying $\mathbb{E}\int_0^t g_s^2\,\mathrm{d}s < \infty$ and likewise for h, Itô isometry proof: A + B + C matrix decomposition

Figure 4: Itô isometry proof: A + B + C matrix decomposition

$$\boxed{\mathbb{E}\left[\Big(\int_0^t g_s \,\mathrm{d}W_s\Big)\Big(\int_0^t h_s \,\mathrm{d}W_s\Big)\right] \;=\; \mathbb{E}\left[\int_0^t g_s h_s \,\mathrm{d}s\right]}. \tag{3.28}$$

In particular, for $g = h$, the L^2-norm of a stochastic integral equals the L^2-norm (in s and ω) of its integrand:

$$\mathbb{E}\left[\left(\int_0^t g_s \,\mathrm{d}W_s\right)^2\right] \;=\; \mathbb{E}\int_0^t g_s^2 \,\mathrm{d}s. \tag{3.29}$$

Proof — the A + B + C decomposition. Set $g_t = g(t, W_t)$, $h_t = h(t, W_t)$, and define the residual

$$\mathcal{E} \;=\; \mathbb{E}\left[\Big(\int_0^t g_s \,\mathrm{d}W_s\Big)\Big(\int_0^t h_s \,\mathrm{d}W_s\Big) \;-\; \int_0^t g_s h_s \,\mathrm{d}s\right]. \tag{3.30}$$

Goal: show $\mathcal{E} = 0$. Fix a partition $\pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ and discretise both integrals as Riemann sums, so the first product becomes

$$\Big(\sum_k g_{t_{k-1}} \Delta W_{t_k}\Big)\Big(\sum_j h_{t_{j-1}} \Delta W_{t_j}\Big) \;=\; \sum_{k,j} g_{k-1}\, h_{j-1}\, \Delta W_{t_k}\, \Delta W_{t_j},$$

where $g_{k-1} = g_{t_{k-1}}$ etc. The second piece (the time integral) discretises as $\sum_k g_{k-1}\, h_{k-1}\, \Delta t_k$.

Step 1 — Split the double sum into three regions (A, B, C) according to the relationship between indices j and k:

$$\sum_{k,j} g_{k-1}\, h_{j-1}\, \Delta W_k\, \Delta W_j \;=\; \underbrace{\sum_{k<j} g_{k-1}\, h_{j-1}\, \Delta W_k\, \Delta W_j}_{A} \;+\; \underbrace{\sum_{j<k} g_{k-1}\, h_{j-1}\, \Delta W_k\, \Delta W_j}_{B} \;+\; \underbrace{\sum_{k} g_{k-1}\, h_{k-1}\, (\Delta W_k)^2}_{C}.$$

The matrix picture: the $n \times n$ grid indexed by (k, j) splits into the lower triangle A $(k < j)$, the upper triangle B $(j < k)$, and the diagonal C $(k = j$, highlighted). The isometry is the claim that the two triangles die in expectation and only the diagonal survives — matching the time-integral piece exactly.

Step 2 — $\mathbb{E}[A] = 0$. For any term in A we have $k < j$. Condition on $\mathcal{F}_{t_{j-1}}$: the factors g_{k-1}, h_{j-1}, and ΔW_k are all $\mathcal{F}_{t_{j-1}}$-measurable (by convention $h_{j-1} := h_{t_{j-1}}$ is $\mathcal{F}_{t_{j-1}}$-measurable because it is evaluated at the left endpoint of the j-th sub-interval; g_{k-1} and ΔW_k are $\mathcal{F}_{t_k} \subset \mathcal{F}_{t_{j-1}}$-measurable since $k < j \Rightarrow t_k \le t_{j-1}$), while ΔW_j is independent of $\mathcal{F}_{t_{j-1}}$ with mean zero. Hence

$$\mathbb{E}\big[g_{k-1}\, h_{j-1}\, \Delta W_k\, \Delta W_j\big] \;=\; \mathbb{E}\big[g_{k-1}\, h_{j-1}\, \Delta W_k\big] \cdot \underbrace{\mathbb{E}[\Delta W_j]}_{=0} \;=\; 0, \tag{3.31}$$

and summing over $k < j$ gives $\mathbb{E}[A] = 0$.

Step 3 — $\mathbb{E}[B] = 0$ by the symmetric argument with the roles of j and k swapped.

Step 4 — Only the diagonal C survives. Combining the three pieces with the time-integral subtraction,

$$\mathcal{E} \;=\; \mathbb{E}\left[\sum_k g_{k-1}\, h_{k-1}\, \big((\Delta W_k)^2 - \Delta t_k\big)\right] \;=\; \sum_k \mathbb{E}[g_{k-1}\, h_{k-1}] \cdot \underbrace{\mathbb{E}[(\Delta W_k)^2 - \Delta t_k]}_{=0} \;=\; 0, \tag{3.32}$$

using independence of $g_{k-1}h_{k-1}$ (measurable at t_{k-1}) from ΔW_k, and $\mathbb{E}[(\Delta W_k)^2] = \Delta t_k$.

Step 5 — Pass to the limit. Equation (3.32) holds for every partition π; taking $\|\pi\| \downarrow 0$ delivers the isometry (3.28). ■

Reading the isometry. The scalar identity $[W,W]_t = t$ becomes, under weighting by g_s^2, the statement that the variance of $\int g\,\mathrm{d}W$ equals the integrated squared integrand. In a quant-practitioner translation: "the variance of a stochastic P&L equals the integrated variance contribution of each sub-interval's position." Every variance-swap argument in the rest of the guide is downstream of (3.29).

A weakening of the boundedness assumption gives the result for any g with $\mathbb{E}\int_0^t g_s^2\,\mathrm{d}s < \infty$, via a standard localisation-and-approximation argument. We take this extension for granted from here on.

5.7 3.7 Brownian Moment Calculations

Before moving to Itô's lemma it pays to build a small toolkit of Brownian moment calculations. Each is a quick application of the Gaussian MGF or the independent-increments property, but they show up repeatedly downstream and are worth seeing explicitly.

Two overarching techniques drive every calculation in this section.

- **MGF differentiation.** The moment-generating function $h(a) = \mathbb{E}[e^{aZ}] = e^{a^2/2}$ of a standard normal has the property that $\mathbb{E}[Z^n] = h^{(n)}(0)$. For higher moments we differentiate h as many times as the moment order, then evaluate at $a = 0$. This reduces computation of any central Gaussian moment to a mechanical exercise in repeated differentiation of $e^{a^2/2}$.
- **Increment decomposition.** For any $s \le t$, write $W_t = W_s + (W_t - W_s)$, where the second piece is an increment *independent of* W_s. This factorises joint expectations as products of marginal expectations of independent pieces, each a centred Gaussian of known variance.

5.7.1 3.7.1 Variance of W_t^2

Starting from $\mathrm{Var}[W_t^2] = \mathbb{E}[W_t^4] - (\mathbb{E}[W_t^2])^2$ and using $W_t \overset{d}{=} \sqrt{t}\,Z$ with $Z \sim \mathcal{N}(0,1)$,

$$\mathrm{Var}[W_t^2] \;=\; t^2\,(\mathbb{E}[Z^4] - 1). \tag{3.33}$$

The Gaussian MGF $h(a) = e^{a^2/2}$ has derivatives $h'(a) = ae^{a^2/2}$, $h''(a) = (1+a^2)e^{a^2/2}$, $h'''(a) = (a^3+3a)e^{a^2/2}$, $h^{(4)}(a) = (a^4+6a^2+3)e^{a^2/2}$. Evaluating at $a = 0$: $\mathbb{E}[Z^2] = 1$, $\mathbb{E}[Z^4] = 3$. The odd moments vanish by symmetry; the even moments grow as $(2k-1)!! = 1, 3, 15, 105, \ldots$. Substituting,

$$\boxed{\mathrm{Var}[W_t^2] \;=\; 2\,t^2}. \tag{3.34}$$

This is the Brownian analogue of $\mathrm{Var}(X^2) = 2\sigma^4$ for centred Gaussians with variance $\sigma^2 = t$, and it underlies the QV concentration argument of §3.4 (the variance of the QV sum is $\sum 2(\Delta t_k)^2 \le 2\|\pi\|\,t$).

5.7.2 3.7.2 Covariance $\mathbb{E}[W_t W_s]$

For $0 \le s \le t$, write $W_t = (W_t - W_s) + W_s$. Then

$$\mathbb{E}[W_t\,W_s] \;=\; \mathbb{E}[(W_t - W_s)\,W_s] \;+\; \mathbb{E}[W_s^2] \;=\; 0 \;+\; s,$$

because W_s and the forward increment $W_t - W_s$ are independent mean-zero. Hence

$$\boxed{\mathbb{E}[W_t\,W_s] \;=\; \min(s,t)}. \tag{3.35}$$

The correlation is $\mathrm{Corr}(W_t, W_s) = \sqrt{s/t}$: nearby times are highly correlated, far-apart times nearly independent. This is also the kernel used when simulating BM on a grid — for $s < t$ the joint law (W_s, W_t) is Gaussian with covariance matrix $\begin{pmatrix} s & s \\ s & t \end{pmatrix}$, whose Cholesky factorisation gives the practical recipe "draw Z_1, Z_2 i.i.d. standard normal; set $W_s = \sqrt{s} Z_1$, $W_t = \sqrt{s} Z_1 + \sqrt{t-s} Z_2$."

5.7.3 3.7.3 The exponential martingale

A recurring character in this guide. Claim:

$$\boxed{\mathbb{E}[X_t] \;=\; 1 \quad \text{for every } t \geq 0}, \qquad X_t = e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}. \tag{3.36}$$

Proof by MGF:

$$\mathbb{E}[X_t] \;=\; e^{-\frac{1}{2}\sigma^2 t}\, \mathbb{E}[e^{\sigma W_t}] \;=\; e^{-\frac{1}{2}\sigma^2 t}\, e^{\frac{1}{2}\sigma^2 t} \;=\; 1,$$

using $W_t \sim \mathcal{N}(0, t)$ and the Gaussian MGF. The deterministic drift $-\frac{1}{2}\sigma^2 t$ is exactly what is needed to offset the Jensen bump created by exponentiating a centred Gaussian.

Let us dwell on this cancellation because it is worth a thousand Black-Scholes formulas. The random variable σW_t is Gaussian with mean 0 and variance $\sigma^2 t$. The exponential is strictly convex, so Jensen's inequality says $\mathbb{E}[e^{\sigma W_t}] \geq e^0 = 1$, with strict inequality whenever $\sigma^2 t > 0$. The Jensen bump is quantified by the Gaussian MGF: $\mathbb{E}[e^{\sigma W_t}] = e^{\sigma^2 t/2}$. To kill this bump, subtract $\frac{1}{2}\sigma^2 t$ from the exponent upfront. That is precisely the origin of the $-\frac{1}{2}\sigma^2$ drift in the log-normal solution of GBM: the risk-neutral log-return is $\ln(S_t/S_0) = (r - \frac{1}{2}\sigma^2)t + \sigma W_t$, not $rt + \sigma W_t$, because the bare "rt" drift would allow S_t/S_0 to grow in expectation and violate the $\mathbb{Q}$-martingale property of the discounted stock. Every long-gamma trader knows this correction intuitively: the convexity of log-normal dynamics gifts you $\frac{1}{2}\sigma^2$ per unit time that risk-neutral pricing has to claw back.

More generally, for any real exponent c,

$$\mathbb{E}[X_t^c] \;=\; e^{-\frac{1}{2}c\sigma^2 t} \cdot e^{\frac{1}{2}c^2\sigma^2 t} \;=\; e^{\frac{1}{2}c(c-1)\sigma^2 t}. \tag{3.37}$$

For $c = 0$ and $c = 1$ the right-hand side equals 1 — these are the two "zero Jensen bump" cases. For $c = 2$ it gives $\mathbb{E}[X_t^2] = e^{\sigma^2 t}$, from which $\mathrm{Var}(X_t) = e^{\sigma^2 t} - 1$ — the well-known log-normal variance. A stronger statement, proved in §3.9 below, is that X_t satisfies the SDE $\mathrm{d}X_t = \sigma X_t\, \mathrm{d}W_t$ and is therefore a martingale.

5.7.4 3.7.4 Two Itô-integral variances via isometry

As a quick application of (3.29), let us compute the variance of two Itô integrals that we will meet repeatedly.

First, $I_t = \int_0^t W_s\, \mathrm{d}W_s$. By Itô isometry,

$$\mathrm{Var}[I_t] \;=\; \mathbb{E}\left[\int_0^t W_s^2\, \mathrm{d}s\right] \;=\; \int_0^t s\, \mathrm{d}s \;=\; \tfrac{1}{2}t^2. \tag{3.38}$$

A sanity check comes from the closed-form $I_t = \frac{1}{2}(W_t^2 - t)$: then $\mathrm{Var}[I_t] = \frac{1}{4}\mathbb{E}[(W_t^2 - t)^2] = \frac{1}{4}(3t^2 - 2t^2 + t^2) = \frac{1}{2}t^2$. Both routes agree.

Second, $J_t = \int_0^t e^{W_s}\, \mathrm{d}W_s$. The integrand is not Gaussian, so J_t is not Gaussian either, but its variance still follows from (3.29):

$$\mathrm{Var}[J_t] \;=\; \int_0^t \mathbb{E}[e^{2W_s}]\, \mathrm{d}s \;=\; \int_0^t e^{2s}\, \mathrm{d}s \;=\; \tfrac{1}{2}(e^{2t} - 1), \tag{3.39}$$

using $\mathbb{E}[e^{\alpha W_s}] = e^{\alpha^2 s/2}$. More generally, when the integrand is $e^{\alpha W_s}$, the variance of the Itô integral is $(e^{\alpha^2 t} - 1)/\alpha^2$. This is the template for every "expectation of an exponential Brownian function" in the guide, and we will reuse it without comment.

5.8 3.8 Itô's Lemma for Brownian Motion

Itô's lemma is the single most important tool in this chapter, and perhaps in this guide. It plays the role that the chain rule plays in ordinary calculus: every derivation of an SDE, every passage from one stochastic process to another, every pricing PDE is at bottom an application of Itô. Mastering it reduces stochastic calculus from a foreign language to a small variation on calculus you already know.

The spirit of Itô is "Taylor expand *one more term* than you normally would, then keep only the first-order and second-order-in-dW contributions, letting everything else vanish." That extra term, the second-order one, is the Itô correction.

5.8.1 3.8.1 Setup

Fix a smooth function $f(t, w)$ with $f \in C^{1,2}$ (once-differentiable in t, twice-differentiable in w, with continuous derivatives), and define the process

$$X_t \;=\; f(t,\, W_t). \tag{3.40}$$

We need one derivative in t because the Taylor expansion in time only goes to first order; dt itself is the small parameter in the time direction, and its square is negligible. We need *two* derivatives in w because the second-order term in the w-Taylor expansion is the one that survives in the limit, contributing $\frac{1}{2}\partial_{ww} f\,\mathrm{d}t$ via $(\mathrm{d}W)^2 = \mathrm{d}t$. Differentiability up to order $(1, 2)$ is therefore the minimum smoothness requirement for Itô's lemma to make pointwise sense.

A textbook example. Take $f(t, w) = w^2 + e^{-t}\sin(w)$. Then $\partial_t f = -e^{-t}\sin(w)$, $\partial_w f = 2w + e^{-t}\cos(w)$, and $\partial_{ww} f = 2 - e^{-t}\sin(w)$. All three are continuous, so $f \in C^{1,2}$ and Itô applies. A non-example: $f(t, w) = |w|$, which is only continuous at $w = 0$, not differentiable twice. For such functions one needs a more delicate tool (Tanaka's formula with local time) which we do not discuss here — but the lesson is that *kink* payoffs such as $\max(S - K, 0)$ do not satisfy the $C^{1,2}$ hypothesis literally, and their Itô-style analyses carry boundary-delta terms that do not appear in the smooth case.

5.8.2 3.8.2 Heuristic derivation via Taylor expansion

Start with the generic Taylor expansion of $f(t + \Delta t,\, W_{t+\Delta t})$ around (t, W_t):

$$\begin{aligned} f(t+\Delta t,\, W_{t+\Delta t}) \;-\; f(t, W_t) \;&= \partial_t f\,\Delta t \\ &+ \partial_w f\,\Delta W \\ &+ \tfrac{1}{2}\,\partial_{tt} f\,(\Delta t)^2 \\ &+ \partial_{tw} f\,\Delta t\,\Delta W \\ &+ \tfrac{1}{2}\,\partial_{ww} f\,(\Delta W)^2 \\ &+ \text{higher-order terms,} \end{aligned}$$

with $\Delta W = W_{t+\Delta t} - W_t$. Which terms are $O(\Delta t)$ (the "dominant" scale in the limit)?

Term	Magnitude scaling	Keep or drop?
$\partial_t f\,\Delta t$	Δt	Keep (deterministic drift)
$\partial_w f\,\Delta W$	$\sqrt{\Delta t}$	Keep (stochastic term)
$\frac{1}{2}\partial_{ww} f\,(\Delta W)^2$	Δt	Keep (Itô correction)
$\frac{1}{2}\partial_{tt} f\,(\Delta t)^2$	$(\Delta t)^2$	Drop
$\partial_{tw} f\,\Delta t\,\Delta W$	$(\Delta t)^{3/2}$	Drop
higher-order	$\leq (\Delta t)^{3/2}$	Drop

In ordinary calculus the second-order term would be $O((\Delta t)^2)$ and discarded. But for Brownian motion $(\Delta W)^2 \approx \Delta t$ by (3.20) — the second-order term is the *same order* as Δt, so it must be kept. Cross terms $\Delta t \cdot \Delta W$ and $(\Delta t)^2$ are of strictly higher order and vanish.

Taking the infinitesimal limit yields

$$\boxed{\mathrm{d}X_t \;=\; [\partial_t f(t,W_t) \;+\; \tfrac{1}{2}\,\partial_{ww} f(t,W_t)]\,\mathrm{d}t \;+\; \partial_w f(t,W_t)\,\mathrm{d}W_t}. \tag{3.41}$$

In integrated form:

$$X_t - X_0 \;=\; \int_0^t [\partial_s f + \tfrac{1}{2}\partial_{ww} f]\,\mathrm{d}s \;+\; \int_0^t \partial_w f\,\mathrm{d}W_s. \tag{3.42}$$

The first integral is an ordinary Riemann integral; the second is a stochastic integral of the type defined in (3.22).

Reading (3.41) as a decomposition. The change in $X_t = f(t, W_t)$ has two independent measurable components:

- A Lebesgue integral of a deterministic integrand evaluated on a stochastic path — this part has bounded variation and zero QV.
- An Itô integral against $\mathrm{d}W$ — a martingale that carries all of the QV of X.

Splitting stochastic processes into "bounded-variation drift + martingale noise" is the *Doob–Meyer decomposition*, and for Itô processes it is simply the two pieces of (3.42). Every price process in later chapters is an Itô process, and every derivation ultimately rearranges these two pieces to solve for unknowns.

An important meta-principle. When you apply Itô to a function $f(t, W_t)$ whose expectation you care about, the drift part survives under $\mathbb{E}[\,\cdot\,]$ and the martingale part dies: $\mathbb{E}[f(t,W_t)] - f(0,0) = \int_0^t \mathbb{E}[\partial_s f + \frac{1}{2}\partial_{ww} f]\,\mathrm{d}s$. This is the fastest way to compute expectations — write the Itô differential, take the expectation of the drift, integrate — and it is the core engine of the Feynman-Kac derivations in Chapter 4.

Itô vs classical chain rule on $f(W_t) = W_t^2$ *Monte-Carlo* $\mathbb{E}[W_t^2]$ *tracks the Itô prediction* t *(green dashed), not the classical chain rule's* 0 *(red dotted). The gap is the Itô correction* $\frac{1}{2}f''\,\mathrm{d}t = \mathrm{d}t$ *integrated from* 0 *to* t *— precisely the drift contribution that ordinary calculus discards.*

5.8.3 3.8.3 Reading the formula — options-language translation

For $X_t = f(t, W_t)$ interpreted as an option price with W_t being the underlying:

- $\partial_t f$ is **theta** — the passage-of-time sensitivity.
- $\partial_w f$ is **delta** — the underlying sensitivity.
- $\partial_{ww} f$ is **gamma** — the curvature of price versus underlying.

Itô's lemma then reads: "the P&L on a long option = theta$\cdot\mathrm{d}t$ + delta$\cdot\mathrm{d}W$ + $\frac{1}{2}$gamma$\cdot\mathrm{d}t$ (from QV of the underlying)". The $\frac{1}{2}$ gamma $\mathrm{d}t$ is precisely the *gamma-theta trade-off* that every options trader lives in daily: a long-gamma book bleeds theta and earns gamma P&L; a short-gamma book collects theta premium and pays for adverse gamma. Itô's lemma is the quantitative expression of this trade-off.

An intuition for the $\frac{1}{2}$ in the Itô correction. The $(\mathrm{d}W)^2 = \mathrm{d}t$ rule supplies a unit of QV per unit time. The Taylor expansion of f in w carries a $\frac{1}{2}\,f''$ coefficient on the squared term (Taylor's second-order term). When we substitute $(\mathrm{d}W)^2 \to \mathrm{d}t$, the $\frac{1}{2}$ stays, and we land on $\frac{1}{2}\,f''(w)\,\mathrm{d}t$. It is not a normalisation mystery — it is just the Taylor coefficient times the QV rate of BM.

5.8.4 3.8.4 Three worked Itô-lemma examples

Example 1 — $\int W\,\mathrm{d}W$ via Itô.

Goal: compute $\int_0^t W_s\,\mathrm{d}W_s$ using Itô's lemma rather than the telescoping identity of §3.5.

Choose $f(t,w) = \frac{1}{2}w^2$, so $\partial_t f = 0$, $\partial_w f = w$, $\partial_{ww} f = 1$. Plugging into (3.41):

$$\mathrm{d}(\tfrac{1}{2}W_t^2) \;=\; \tfrac{1}{2}\,\mathrm{d}t \;+\; W_t\,\mathrm{d}W_t.$$

Integrating from 0 to t with $W_0 = 0$ gives $\frac{1}{2}W_t^2 = \frac{1}{2}t + \int_0^t W_s\,\mathrm{d}W_s$, so

$$\int_0^t W_s \,\mathrm{d}W_s \;=\; \tfrac{1}{2}W_t^2 - \tfrac{1}{2}t, \tag{3.43}$$

in agreement with (3.26). Four lines instead of the full telescoping proof of §3.5 — Itô's lemma is massively more efficient.

Example 2 — $\int W^2\,\mathrm{d}W$.

Choose $f(t,w) = \frac{1}{3}w^3$, so $\partial_t f = 0$, $\partial_w f = w^2$, $\partial_{ww} f = 2w$. Plugging into (3.41):

$$\mathrm{d}(\tfrac{1}{3}W_t^3) \;=\; W_t\,\mathrm{d}t \;+\; W_t^2\,\mathrm{d}W_t.$$

Integrating and solving for the stochastic integral:

$$\boxed{\int_0^t W_s^2\,\mathrm{d}W_s \;=\; \tfrac{1}{3}W_t^3 \;-\; \int_0^t W_s\,\mathrm{d}s}. \tag{3.44}$$

The remaining $\int_0^t W_s\,\mathrm{d}s$ is an ordinary pathwise Riemann integral, not a stochastic one. Itô has converted the stochastic integral into (algebraic term) + (pathwise integral). The pathwise piece cannot be eliminated algebraically — it depends on the *entire path* $(W_s)_{s\in[0,t]}$, not just the endpoint W_t. Its distribution is Gaussian with $\mathrm{Var}\int_0^t W_s\,\mathrm{d}s = \int_0^t\int_0^t \min(u,v)\,\mathrm{d}u\,\mathrm{d}v = \frac{1}{3}t^3$.

Example 3 — the exponential martingale.

Goal: given $X_t = e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}$, find the SDE it satisfies.

Choose $f(t,w) = e^{-\frac{1}{2}\sigma^2 t + \sigma w}$. Each partial derivative is proportional to f itself (a feature of the exponential): $\partial_t f = -\frac{1}{2}\sigma^2 f$, $\partial_w f = \sigma f$, $\partial_{ww} f = \sigma^2 f$. Plugging into (3.41):

$$\mathrm{d}X_t \;=\; [-\tfrac{1}{2}\sigma^2 X_t + \tfrac{1}{2}\sigma^2 X_t]\,\mathrm{d}t + \sigma X_t\,\mathrm{d}W_t \;=\; \sigma X_t\,\mathrm{d}W_t.$$

The drift bracket cancels exactly — the deterministic $-\frac{1}{2}\sigma^2$ in the exponent is designed to kill the Itô correction $\frac{1}{2}\sigma^2$. Hence

$$\boxed{\mathrm{d}X_t \;=\; \sigma\,X_t\,\mathrm{d}W_t}. \tag{3.45}$$

X_t satisfies a driftless geometric SDE and is therefore a martingale. This is the rigorous proof of the $\mathbb{E}[X_t] = 1$ claim in (3.36).

The exponential martingale (3.45) is a prototype of a larger class. The Doléans-Dade exponential $\mathcal{E}(M)_t = \exp(M_t - \frac{1}{2}\langle M\rangle_t)$ of any continuous martingale M is similarly designed: the $-\frac{1}{2}\langle M\rangle_t$ is exactly the drift shift needed to turn e^{M_t} into a martingale. For $M_t = \sigma W_t$, $\langle M\rangle_t = \sigma^2 t$, recovering (3.45). The Doléans-Dade exponential reappears in Chapter 5 as the Radon-Nikodym derivative for Brownian-motion measure changes.

Doléans-Dade exponential trajectories with expected-value envelope *Six sample paths of $Z_t = \exp(\sigma W_t - \frac{1}{2}\sigma^2 t)$ with $\sigma = 0.9$. Individual trajectories span orders of magnitude — some rocket upward, others drift toward zero — yet $\mathbb{E}[Z_t] = 1$ for every t. The $-\frac{1}{2}\sigma^2 t$ drift is precisely the Itô correction that keeps the process a martingale despite its wildly dispersed paths.*

5.9 3.9 Itô's Lemma for General Itô Processes

The lemma in §3.8 handled $X_t = f(t, W_t)$ where the underlying process was BM itself. In practice one usually wants to transform a diffusion with non-trivial drift and diffusion coefficients — a GBM, an OU process, a general Itô process. The extension is mechanical and is called **Itô's Lemma III** in the chapter's internal numbering.

5.9.1 3.9.1 Setup

Let X_t be an Itô process satisfying

$$\mathrm{d}X_t \;=\; \mu(t, X_t)\,\mathrm{d}t \;+\; \sigma(t, X_t)\,\mathrm{d}W_t, \tag{3.46}$$

for adapted drift μ and diffusion σ (possibly functions of t and X_t). Let $g \in C^{1,2}$ and define $Y_t = g(t, X_t)$. Then

$$\boxed{\mathrm{d}Y_t \;=\; \alpha(t, X_t)\,\mathrm{d}t \;+\; \beta(t, X_t)\,\mathrm{d}W_t},$$

where

$$\begin{aligned} \alpha(t,x) \;&=\; \partial_t g(t,x) \;+\; \mu(t,x)\,\partial_x g(t,x) \;+\; \tfrac{1}{2}\,\sigma^2(t,x)\,\partial_{xx} g(t,x), \\ \beta(t,x) \;&=\; \sigma(t,x)\,\partial_x g(t,x). \end{aligned} \tag{3.47}$$

Equivalently, in the compact "derivative-free" form,

$$\mathrm{d}g(t, X_t) \;=\; \partial_t g\,\mathrm{d}t \;+\; \partial_x g\,\mathrm{d}X_t \;+\; \tfrac{1}{2}\,\partial_{xx} g\,(\mathrm{d}X_t)^2, \tag{3.48}$$

with the Itô multiplication rules (3.20), implying $(\mathrm{d}X_t)^2 = \sigma^2(t, X_t)\,\mathrm{d}t$. The derivation is a simple extension of the BM case: use $(\mathrm{d}X_t)^2 = \sigma^2\,\mathrm{d}t$ in the Taylor expansion and keep first-order terms. The result is the BM Itô formula with two substitutions: the drift of X becomes μ (instead of zero), and the diffusion of X becomes σ (instead of one).

Consistency with the BM case. Setting $\mu = 0$ and $\sigma = 1$ in (3.47) recovers (3.41) exactly — so Lemma III is a genuine generalisation, not a rewrite. Setting additionally $\partial_t g = 0$ recovers the time-independent version. The three levels of generality — $f(W_t)$, $f(t, W_t)$, $g(t, X_t)$ — form a hierarchy of increasing scope; everything in the later chapters uses the most general form (3.47).

5.9.2 3.9.2 Cross-variation and multi-factor Itô

If X_t and Y_t are two Itô processes driven by the same or correlated Brownian motions, their cross-variation $\mathrm{d}X_t\,\mathrm{d}Y_t$ is computed by multiplying out the SDEs and applying the Itô rules. For independent BMs W^1, W^2, the rule is $\mathrm{d}W^1\,\mathrm{d}W^2 = 0$; for correlated BMs with instantaneous correlation ρ, $\mathrm{d}W^1\,\mathrm{d}W^2 = \rho\,\mathrm{d}t$. This is how multi-factor models handle cross-terms in stochastic integrals — each pair of driving noises contributes its own cross-variation rate.

Itô's lemma for functions of two Itô processes is then

$$\mathrm{d}f(t, X_t, Y_t) = \partial_t f\,\mathrm{d}t + \partial_x f\,\mathrm{d}X_t + \partial_y f\,\mathrm{d}Y_t + \tfrac{1}{2}\partial_{xx} f\,(\mathrm{d}X)^2 + \partial_{xy} f\,\mathrm{d}X\,\mathrm{d}Y + \tfrac{1}{2}\partial_{yy} f\,(\mathrm{d}Y)^2. \tag{3.49}$$

Every instance of "two risk factors interact" in derivatives pricing — quanto options, dual-currency bonds, baskets, stochastic-vol Heston — is an application of (3.49). We will meet these explicitly in Chapter 8 (Margrabe exchange options) and Chapter 10 (Heston).

5.10 3.10 SDE Catalogue

With Itô's lemma in hand we now solve the three canonical SDEs that every later chapter will invoke: Geometric Brownian Motion (GBM), the Ornstein-Uhlenbeck / Vasicek process, and the generic constant-coefficient arithmetic SDE. Each is solved as a worked exercise in Lemma III.

5.10.1 3.10.1 Geometric Brownian Motion (GBM)

The GBM SDE is

$$\mathrm{d}S_t \;=\; \mu\, S_t\, \mathrm{d}t \;+\; \sigma\, S_t\, \mathrm{d}W_t, \qquad S_0 \text{ given}, \tag{3.50}$$

with constants μ and $\sigma > 0$. This is the dynamics assumed for the underlying asset throughout the Black-Scholes framework (with μ replaced by r under the risk-neutral measure in Chapter 6).

Solution by Itô on $\ln S$. Apply Lemma III with $g(t,s) = \ln s$ and $(\mu(t,s), \sigma(t,s)) = (\mu s, \sigma s)$. Partial derivatives: $\partial_t g = 0$, $\partial_s g = 1/s$, $\partial_{ss} g = -1/s^2$. Plugging into (3.47):

$$\mathrm{d}(\ln S_t) \;=\; \left[0 \;+\; \frac{\mu S_t}{S_t} \;+\; \tfrac{1}{2}\,\sigma^2 S_t^2 \cdot \left(-\frac{1}{S_t^2}\right)\right] \mathrm{d}t \;+\; \frac{\sigma S_t}{S_t}\, \mathrm{d}W_t \;=\; \left(\mu - \tfrac{1}{2}\sigma^2\right)\, \mathrm{d}t \;+\; \sigma\, \mathrm{d}W_t. \tag{3.51}$$

The right-hand side is a constant-drift, constant-diffusion SDE — an arithmetic BM with drift $\mu - \frac{1}{2}\sigma^2$ and volatility σ. Integrating:

$$\ln S_t - \ln S_0 \;=\; (\mu - \tfrac{1}{2}\sigma^2)\, t \;+\; \sigma W_t,$$

and therefore

$$\boxed{S_t \;=\; S_0\, e^{(\mu - \frac{1}{2}\sigma^2)\, t \,+\, \sigma W_t} \qquad \text{(GBM closed form)}}. \tag{3.52}$$

This is the explicit closed-form solution of the GBM SDE, and it is how every Monte Carlo simulator generates stock-price paths under the Black-Scholes model: draw $W_t \sim \mathcal{N}(0,t)$, plug into (3.52), get S_t. There is no time-discretisation error in this exact scheme; the only error is the Gaussian approximation of W_t, which is *exact* on any finite time horizon.

Expected value. Using $\mathbb{E}[e^{\sigma W_t}] = e^{\sigma^2 t/2}$,

$$\mathbb{E}[S_t] \;=\; S_0\, e^{(\mu - \frac{1}{2}\sigma^2)t} \cdot e^{\frac{1}{2}\sigma^2 t} \;=\; S_0\, e^{\mu t}. \tag{3.53}$$

The expected continuously-compounded return is μ, *irrespective of* σ. Volatility leaves the expected forward value unchanged but does shift the median and all other quantiles of the log-normal distribution. The risk-neutral valuation of derivatives exploits this by ensuring $\mu = r$ (the risk-free rate) under $\mathbb{Q}$, so $\mathbb{E}^{\mathbb{Q}}[S_t] = S_0\, e^{rt}$ — the cost of carrying the stock equals the riskless growth rate.

The $-\frac{1}{2}\sigma^2$ correction is not optional. A common mistake is to apply separation of variables to (3.50): write $\mathrm{d}S/S = \mu\, \mathrm{d}t + \sigma\, \mathrm{d}W$ and integrate both sides to get $\ln S_t = \ln S_0 + \mu t + \sigma W_t$, i.e. $S_t = S_0 e^{\mu t + \sigma W_t}$. This is *wrong*. The chain rule $\mathrm{d}(\ln S) = \mathrm{d}S/S$ does not hold for SDEs; Itô's lemma tells us it is $\mathrm{d}(\ln S) = \mathrm{d}S/S - \frac{1}{2}(\mathrm{d}S)^2/S^2 = \mathrm{d}S/S - \frac{1}{2}\sigma^2\, \mathrm{d}t$. The missing $-\frac{1}{2}\sigma^2\, \mathrm{d}t$ piece is the Itô correction, and forgetting it produces $\mathbb{E}[S_t] = S_0 e^{\mu t + \sigma^2 t/2}$ — which violates the martingale property of the discounted stock (when $\mu = r$). The $-\frac{1}{2}\sigma^2$ in (3.52) is precisely what restores the martingale property.

5.10.2 3.10.1a Jensen's gap — where the $\sigma^2/2$ really comes from

The $-\frac{1}{2}\sigma^2 t$ term in the GBM log-drift is *not* a peculiarity of stochastic calculus. It is the same gap between the *expectation of a function* and the *function of an expectation* that Jensen's inequality controls in elementary probability. Stated cleanly: Jensen says that for any convex f and random variable X,

$$\mathbb{E}[f(X)] \;\geq\; f(\mathbb{E}[X]), \qquad \text{(equality iff } f \text{ is affine or } X \text{ is constant)}. \tag{3.52a}$$

For *concave* f the inequality flips: $\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$. *Geometric intuition.* A chord between two points on a convex (concave) curve lies above (below) the curve, so averaging the function value gives more (less) than the function of the average.

Apply Jensen to the log of GBM. Let $L_t = \ln(S_t/S_0)$. From (3.52), $L_t \sim \mathcal{N}((\mu - \frac{1}{2}\sigma^2)t,\ \sigma^2 t)$, so

$$\mathbb{E}[L_t] \;=\; (\mu - \tfrac{1}{2}\sigma^2)t. \tag{3.52b}$$

Now apply $f(x) = e^x$ (strictly convex). Jensen plus the closed form of the Gaussian MGF gives

$$\mathbb{E}[S_t/S_0] \;=\; \mathbb{E}[e^{L_t}] \;=\; \exp\Big(\mathbb{E}[L_t] + \tfrac{1}{2}\,\mathrm{Var}(L_t)\Big) \;=\; e^{\mu t}, \tag{3.52c}$$

so that

$$\boxed{\underbrace{\ln \mathbb{E}[S_t/S_0]}_{=\mu t} \;-\; \underbrace{\mathbb{E}[\ln(S_t/S_0)]}_{=(\mu-\frac{1}{2}\sigma^2)t} \;=\; \tfrac{1}{2}\sigma^2 t}. \tag{3.52d}$$

The right-hand side $\frac{1}{2}\sigma^2 t$ is the *Jensen gap* between the log of the arithmetic mean and the arithmetic mean of the log. It is strictly positive whenever $\sigma > 0$, and it vanishes only in the deterministic limit. The Itô correction in (3.52) is the same gap, written from the SDE side. Two viewpoints, one quantity.

Numeric. Set $\mu = 8\%$, $\sigma = 30\%$, $T = 1$ year:

Quantity	Formula	Value
Arithmetic mean of return	$\mathbb{E}[S_T/S_0] = e^{\mu T}$	1.0833
Geometric mean of return	$\exp(\mathbb{E}[\ln(S_T/S_0)]) = e^{(\mu-\sigma^2/2)T}$	1.0356
Jensen gap	$\frac{1}{2}\sigma^2 T$	0.0450
log of arithmetic / geometric ratio	$\ln(1.0833/1.0356)$	0.0450

The arithmetic and geometric means differ by exactly $e^{\sigma^2 T/2}$ — at $\sigma = 30\%$ that is a 4.6% wedge per year. *Practitioner aside.* This is why "average annualised return" (geometric, what compounded wealth grew at) is *always* less than "expected return" (arithmetic, what the next-period mean is), and the gap widens with volatility. A fund quoting a 12% expected return at 30% vol delivers a long-run CAGR closer to $12\% - \frac{1}{2}(30\%)^2 = 7.5\%$ — the same $\sigma^2/2$ correction reads as a performance haircut.

Why this matters for option pricing. Risk-neutral pricing is "$\mathbb{Q}$-expectation of payoff." When the payoff depends on $\ln S$ (as in barrier options, lookbacks, and the Black-Scholes log-form derivation itself), the relevant moment is $\mathbb{E}^{\mathbb{Q}}[\ln S_T]$, which carries the $-\frac{1}{2}\sigma^2 T$ shift. When the payoff depends on S directly (calls, puts, forwards), the relevant moment is $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{rT}$ and the $\sigma^2/2$ has been absorbed by the exponential's MGF. Whether the $\sigma^2/2$ "appears" in a pricing formula depends entirely on whether you are computing $\mathbb{E}[\ln S]$ or $\ln \mathbb{E}[S]$.

5.10.3 3.10.2 Ornstein-Uhlenbeck / Vasicek

The OU SDE — also known as the Vasicek short-rate equation when applied to interest rates — is

$$\mathrm{d}r_t \;=\; \kappa(\theta - r_t)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t, \qquad r_0 \text{ given}, \tag{3.54}$$

with constants $\kappa > 0$ (mean-reversion speed), θ (long-run mean), and σ (volatility). The drift $\kappa(\theta - r_t)$ pulls r_t toward θ at rate κ; the diffusion is constant and Gaussian. This is the simplest mean-reverting SDE and serves as the workhorse short-rate model in Chapter 12.

Solution by integrating factor. The drift is linear in r_t, which suggests the integrating-factor technique familiar from linear first-order ODEs. Multiply both sides of (3.54) by $e^{\kappa t}$ and recognise the left-hand side as an exact differential:

$$\mathrm{d}(e^{\kappa t}\, r_t) \;=\; e^{\kappa t}\,\mathrm{d}r_t \;+\; \kappa\, e^{\kappa t}\, r_t\,\mathrm{d}t \;=\; e^{\kappa t}(\kappa\theta\,\mathrm{d}t + \sigma\,\mathrm{d}W_t),$$

where the second equality substitutes (3.54) after the cancellation $-\kappa r_t + \kappa r_t = 0$. Integrating from 0 to t:

$$e^{\kappa t} r_t - r_0 \;=\; \kappa\theta \int_0^t e^{\kappa u}\,\mathrm{d}u \;+\; \sigma \int_0^t e^{\kappa u}\,\mathrm{d}W_u \;=\; \theta(e^{\kappa t}-1) \;+\; \sigma \int_0^t e^{\kappa u}\,\mathrm{d}W_u.$$

Multiplying through by $e^{-\kappa t}$:

$$\boxed{r_t \;=\; r_0 e^{-\kappa t} \;+\; \theta(1-e^{-\kappa t}) \;+\; \sigma \int_0^t e^{-\kappa(t-u)}\,\mathrm{d}W_u}. \tag{3.55}$$

The integrating-factor step can also be verified as an Itô-lemma computation: apply Lemma III with $g(t,r) = e^{\kappa t} r$, compute $\partial_t g = \kappa e^{\kappa t} r$, $\partial_r g = e^{\kappa t}$, $\partial_{rr} g = 0$, and substitute into (3.47). The result $\mathrm{d}(e^{\kappa t} r_t) = e^{\kappa t}(\kappa\theta\,\mathrm{d}t + \sigma\,\mathrm{d}W_t)$ matches the integrating-factor derivation — a clean example of Lemma III in action.

Distribution of r_t. The stochastic integral in (3.55) is a linear combination of Gaussian increments, so r_t is itself Gaussian. Its mean is the deterministic relaxation

$$\mathbb{E}[r_t] \;=\; r_0 e^{-\kappa t} \;+\; \theta(1-e^{-\kappa t}), \tag{3.56}$$

which starts at r_0 and relaxes exponentially to θ as $t \to \infty$. The variance follows from the Itô isometry (3.29):

$$\mathrm{Var}[r_t] \;=\; \sigma^2 \int_0^t e^{-2\kappa(t-u)}\,\mathrm{d}u \;=\; \frac{\sigma^2}{2\kappa}\big(1-e^{-2\kappa t}\big)\,. \tag{3.57}$$

Two limits worth registering. As $t \to \infty$, $\mathrm{Var}[r_t] \to \sigma^2/(2\kappa)$: mean-reversion caps the long-run variance at a finite value, unlike plain Brownian motion whose variance grows linearly forever. As $t \downarrow 0$, $\mathrm{Var}[r_t] \to \sigma^2 t$: over short horizons the OU process is indistinguishable from a Brownian motion, because the mean-reversion drift has had no time to act. The density picture — the mean $r_0 e^{-\kappa t} + \theta(1-e^{-\kappa t})$ relaxing from r_0 toward θ, flanked by $\pm 2\,\mathrm{sd}$ bands that widen quickly at first and then plateau — captures the qualitative picture that every mean-reverting model shares: transient-widening, long-run stationarity.

The integrated rate $\int_0^T r_u\,\mathrm{d}u$, needed for bond pricing in Chapter 12, is also Gaussian — as a linear functional of the Gaussian process r_t — and its mean and variance are computed by the same Fubini / Itô-isometry combination. We defer the explicit formula to Chapter 12 where it is put to work.

OU vs GBM sample paths *Side-by-side comparison of the two workhorse SDEs.* ***Left:*** *GBM paths trend exponentially with multiplicative noise; the mean curve $S_0 e^{\mu t}$ grows without bound.* ***Right:*** *OU / Vasicek paths wobble around the long-run mean θ; variance saturates at $\sigma^2/(2\kappa)$ so the ensemble never escapes. Mean-reversion is visually unmistakable.*

5.10.4 3.10.3 Constant-coefficient arithmetic BM

The simplest possible SDE — no state-dependence in either drift or diffusion — is

$$\mathrm{d}X_t \;=\; a\,\mathrm{d}t \;+\; b\,\mathrm{d}W_t, \qquad X_0 \text{ given}, \tag{3.58}$$

with constants a and b. Because the coefficients do not depend on X_t, direct integration works without any transformation:

$$X_t \;=\; X_0 \;+\; a\,t \;+\; b\,W_t. \tag{3.59}$$

The solution is Gaussian with mean $X_0 + at$ and variance $b^2 t$. This is the "trivially solvable" SDE, and its chief uses are (i) as the reference process after Itô-transforming a more complicated SDE to constant coefficients (e.g. the log-GBM reduction (3.51)), and (ii) as the Bachelier model of asset prices — rarely used for equities but standard in certain rates markets (notably swaption pricing with normal volatilities, which we revisit in Chapter 14).

One subtle point worth registering: arithmetic BM allows X_t to go *negative* (since a Gaussian has full support on $\mathbb{R}$), whereas GBM keeps $S_t > 0$ almost surely. The choice between the two models is therefore often dictated by whether negative values are economically reasonable — equities must be non-negative, so GBM; interest rates can (occasionally, post-2014) be negative, so arithmetic-style models are once again respectable in that domain.

5.10.5 3.10.4 Summary of solved SDEs

SDE	Solution	Distribution of X_t
$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$ (GBM)	$S_t = S_0\,e^{(\mu-\frac{1}{2}\sigma^2)t+\sigma W_t}$	Log-normal
$\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$ (OU)	$r_t = r_0 e^{-\kappa t} + \theta(1-e^{-\kappa t}) + \sigma\int_0^t e^{-\kappa(t-u)}\,\mathrm{d}W_u$	Gaussian
$\mathrm{d}X_t = a\,\mathrm{d}t + b\,\mathrm{d}W_t$ (arithmetic BM)	$X_t = X_0 + at + bW_t$	Gaussian

Every SDE encountered in the rest of the guide either is one of these three or reduces to one by an Itô transformation. The Heston stochastic-volatility model of Chapter 10 is one partial exception (the variance process is CIR, which requires a Bessel-process argument we develop there), but even it uses OU-like integrating-factor ideas on the variance term.

5.11 3.11 Worked Stochastic Integrals

We now assemble a small library of stochastic integrals that later chapters invoke. The operational recipe for computing any $\int_0^t g(s, W_s)\,\mathrm{d}W_s$ is always the same:

1. Choose a candidate $f(t, w)$ such that $\partial_w f(t, w) = g(t, w)$.
2. Compute the Itô differential $\mathrm{d}f(t, W_t)$ using (3.41).
3. Integrate from 0 to t and solve for the stochastic integral:

$$\int_0^t g(s, W_s)\,\mathrm{d}W_s = f(t, W_t) - f(0, 0) - \int_0^t [\partial_s f + \tfrac{1}{2}\partial_{ww} f]\,\mathrm{d}s.$$

The right-hand side is pathwise — a mix of terminal-value evaluations and ordinary Riemann integrals. The stochastic integral has been *reduced* to these ingredients. That is the whole trick.

5.11.1 3.11.1 A stochastic integration-by-parts: $\int_0^t s\,W_s\,\mathrm{d}W_s$

Consider the product $X_t = t\,W_t^2$, i.e. take $f(t, w) = t\,w^2$. Partials: $\partial_t f = w^2$, $\partial_w f = 2tw$, $\partial_{ww} f = 2t$. Plugging into (3.41):

$$\mathrm{d}(t\,W_t^2) \;=\; \big[\,W_t^2 \;+\; t\,\big]\,\mathrm{d}t \;+\; 2t\,W_t\,\mathrm{d}W_t. \tag{3.60}$$

Integrating from 0 to t with $X_0 = 0$:

$$t\,W_t^2 \;=\; \int_0^t (W_s^2 + s)\,\mathrm{d}s \;+\; \int_0^t 2s\,W_s\,\mathrm{d}W_s,$$

so that

$$\boxed{\int_0^t 2\,s\,W_s\,\mathrm{d}W_s \;=\; t\,W_t^2 \;-\; \int_0^t W_s^2\,\mathrm{d}s \;-\; \tfrac{1}{2}t^2}\,. \tag{3.61}$$

This is a stochastic integration-by-parts identity for $s\,W_s$: the product of a deterministic function of s and a Brownian function is expressible in terms of the terminal product plus ordinary pathwise integrals. A sanity check: taking expectations on both sides, the left-hand side is 0 (martingale), and the right-hand side is $\mathbb{E}[tW_t^2] - \int_0^t s\,\mathrm{d}s - \frac{1}{2}t^2 = t\cdot t - \frac{1}{2}t^2 - \frac{1}{2}t^2 = 0$.

More generally, for any $f(t,w) = \phi(t)\,\psi(w)$ product form, Itô's lemma gives the analogue of the ordinary product rule with an Itô correction:

$$\mathrm{d}(\phi(t)\,\psi(W_t)) \;=\; \phi'(t)\,\psi(W_t)\,\mathrm{d}t \;+\; \phi(t)\,\psi'(W_t)\,\mathrm{d}W_t \;+\; \tfrac{1}{2}\,\phi(t)\,\psi''(W_t)\,\mathrm{d}t. \tag{3.62}$$

5.11.2 3.11.2 Integration-by-parts for $\int_0^t s\,e^{W_s}\,\mathrm{d}W_s$

Take $f(t,w) = t\,e^w$, chosen so that $\partial_w f = t\,e^w$ — almost the desired $s\,e^{W_s}$, but with t frozen rather than integrated. Partials: $\partial_t f = e^w$, $\partial_w f = t\,e^w$, $\partial_{ww} f = t\,e^w$. Itô:

$$\mathrm{d}(t\,e^{W_t}) \;=\; \left[\,e^{W_t} \;+\; \tfrac{1}{2}\,t\,e^{W_t}\,\right]\mathrm{d}t \;+\; t\,e^{W_t}\,\mathrm{d}W_t. \tag{3.63}$$

Integrating $0 \to t$ and rearranging,

$$\boxed{\int_0^t s\,e^{W_s}\,\mathrm{d}W_s \;=\; t\,e^{W_t} \;-\; \int_0^t e^{W_s}\,\mathrm{d}s \;-\; \tfrac{1}{2}\int_0^t s\,e^{W_s}\,\mathrm{d}s}. \tag{3.64}$$

The answer decomposes into a terminal-value algebraic piece and two pathwise Riemann integrals. Neither pathwise integral has a simpler closed form, but both are perfectly tractable for Monte Carlo simulation or moment calculations. The stochastic integral $\int_0^t se^{W_s}\,\mathrm{d}W_s$ itself is a martingale with variance $\int_0^t s^2\,\mathbb{E}[e^{2W_s}]\,\mathrm{d}s = \int_0^t s^2 e^{2s}\,\mathrm{d}s$ by Itô isometry. This is the template for every "polynomial-in-s times exponential-in-W_s" integral that appears downstream.

5.11.3 3.11.3 A catalogue of Brownian computations

The following compact list summarises the moments that will be reused without re-derivation:

$$\mathbb{E}[W_t] = 0, \qquad \mathrm{Var}[W_t] = t, \qquad \mathbb{E}[W_t^4] = 3t^2,$$

$$\mathbb{E}[W_s W_t] = \min(s,t), \qquad \mathrm{Var}[W_t^2] = 2t^2,$$

$$\mathbb{E}[e^{\sigma W_t}] = e^{\frac{1}{2}\sigma^2 t}, \qquad \mathbb{E}[e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}] = 1,$$

$$\mathrm{Var}\left[\int_0^t g_s\,\mathrm{d}W_s\right] = \int_0^t \mathbb{E}[g_s^2]\,\mathrm{d}s,$$

$$\mathrm{Var}\left[\int_0^t W_s\,\mathrm{d}s\right] = \tfrac{1}{3}t^3.$$

Every one of these is either a direct MGF exercise or a one-line Itô-isometry application. None will be re-derived in later chapters.

5.12 3.12 Chapter Summary

What this chapter has established:

- **Brownian motion** is the continuum limit of a symmetric scaled random walk with $\sqrt{\Delta t}$ increments. It has Gaussian marginals, stationary and independent increments, continuous paths — properties (i)-(iv) of (3.8).
- **Path properties.** BM has infinite total variation (3.14) and finite quadratic variation $[W, W]_t = t$ (3.19), summarised by the Itô rule $(\mathrm{d}W)^2 = \mathrm{d}t$ in (3.20).
- **The Itô integral** $\int_0^t g_s \, \mathrm{d}W_s$ is defined by left-endpoint Riemann sums (3.22). The left-endpoint choice is what makes it a martingale — the mathematical encoding of no-arbitrage predictability.
- **The Itô isometry** (3.28)-(3.29) computes the variance of any stochastic integral as a deterministic time integral of the squared integrand. It generalises $[W, W]_t = t$ from the identity to arbitrary integrands.
- **Itô's lemma** (3.41) — the stochastic chain rule — carries an extra term $\frac{1}{2}\partial_{ww} f \, \mathrm{d}t$ relative to the classical chain rule, absent in ordinary calculus because smooth paths have zero QV. The general form (3.47) handles transforms of any Itô process.
- **SDE catalogue.** GBM (3.52), OU/Vasicek (3.55), and arithmetic BM (3.59) are the three canonical SDEs. Every SDE encountered later either is one of these three or reduces to one by an Itô transformation.

5.12.1 Key takeaways

1. $(\mathrm{d}W)^2 = \mathrm{d}t$ **is the source of every Itô correction.** Every time you see a $-\frac{1}{2}\sigma^2 t$, $\frac{1}{2}\sigma^2 \partial_{xx}$, or "convexity adjustment" in a later chapter, it descends from this single identity.
2. **TV versus QV.** BM has infinite TV (distance) but finite QV (energy). The gap between them is the gamma-theta trade-off of options and the structural reason variance swaps replicate by log-contracts while "realised-absolute-value swaps" do not.
3. **Left endpoints matter.** The Itô integral's martingale property depends on evaluating the integrand at t_{k-1}, not t_k. This is the mathematical expression of "predictable strategies earn zero expected P&L," i.e. no-arbitrage in continuous time.
4. **Itô's lemma is the chain rule plus curvature.** Options traders already know it: long-gamma books bleed theta at the rate $\frac{1}{2}\sigma^2 S^2 \Gamma$, and this rate is exactly the Itô correction applied to the Black-Scholes price.
5. **Closed-form SDEs are scarce.** Only a handful of SDEs admit explicit solutions — GBM, OU, linear-Gaussian, and a few others. The rest are solved numerically (Euler-Maruyama, Milstein) or transformed to a solvable form by Itô's lemma (the Lamperti transform).

5.12.2 Reference formulas

- **Itô rules:** $(\mathrm{d}W)^2 = \mathrm{d}t$, $\mathrm{d}W \, \mathrm{d}t = 0$, $(\mathrm{d}t)^2 = 0$.
- **Itô's lemma for $X_t = f(t, W_t)$:**

$$\mathrm{d}X_t = [\partial_t f + \tfrac{1}{2}\partial_{ww} f] \, \mathrm{d}t + \partial_w f \, \mathrm{d}W_t.$$

- **Itô's lemma for $Y_t = g(t, X_t)$ with $\mathrm{d}X_t = \mu \, \mathrm{d}t + \sigma \, \mathrm{d}W_t$:**

$$\mathrm{d}Y_t = [\partial_t g + \mu \, \partial_x g + \tfrac{1}{2}\sigma^2 \, \partial_{xx} g] \, \mathrm{d}t + \sigma \, \partial_x g \, \mathrm{d}W_t.$$

- **Itô isometry:**

$$\mathbb{E}\left[\left(\int_0^t g_s \, \mathrm{d}W_s\right)^2\right] = \mathbb{E}\int_0^t g_s^2 \, \mathrm{d}s.$$

- **GBM solution:** $S_t = S_0 \, e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$.
- **OU solution:** $r_t = r_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t}) + \sigma \int_0^t e^{-\kappa(t-u)} \, \mathrm{d}W_u$, with $\mathrm{Var}[r_t] = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})$.
- **Exponential martingale:** $\mathrm{d}(e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}) = \sigma \, e^{-\frac{1}{2}\sigma^2 t + \sigma W_t} \, \mathrm{d}W_t$, driftless.
- **Key Brownian moments:** $\mathbb{E}[W_t^2] = t$, $\mathbb{E}[W_t^4] = 3t^2$, $\mathbb{E}[W_s W_t] = \min(s, t)$, $\mathbb{E}[e^{\sigma W_t}] = e^{\sigma^2 t/2}$.

With these tools in hand we turn in Chapter 4 to the Feynman-Kac bridge between SDEs and PDEs, which will hand us the Kolmogorov backward equation and prepare the ground for the risk-neutral pricing framework of Chapters 5-8.

6 Chapter 4 — Feynman-Kac and the SDE-PDE Bridge

Chapter 3 built the stochastic-calculus toolkit — Brownian motion, Itô integrals, Itô's lemma in its three forms — and closed with a small catalogue of SDEs we can now solve. This chapter uses that toolkit for its first major payoff: the *Feynman-Kac* theorem, the bidirectional bridge that identifies a linear parabolic PDE with the conditional expectation of a diffusion-driven payoff. Once that bridge is in place, every pricing problem we meet — Black-Scholes in Chapter 6, Vasicek bond prices in Chapter 12, caplets in Chapter 15 — is a single substitution into the same template.

The ambition of this chapter is limited on purpose. We will not derive Black-Scholes yet (that happens twice in Chapter 6: once via the hedging argument and once as a Feynman-Kac corollary). We will not introduce measure changes (Chapter 5). What we *will* do is establish the PDE $\Leftrightarrow$ expectation duality in three progressively general forms — zero-drift Brownian motion, constant-coefficient drifted BM with discounting, and finally state-and-time-dependent coefficients — and we will compute three worked examples by hand so the reader sees the machinery turn.

6.1 4.1 Motivation — Why an SDE-PDE Bridge Matters

Every pricing question in this guide eventually reduces to one of two computations:

1. Solve a backward parabolic PDE on a space-time grid.
2. Average a random payoff over sample paths of a diffusion.

These two procedures look entirely different. The first is a deterministic calculation on a mesh, implemented in finite-difference or finite-element code. The second is a probabilistic calculation on simulated trajectories, implemented as Monte Carlo. They appear to require different numerical libraries, different error analyses, different parallelisation strategies, and different testing regimes.

Feynman-Kac says they are the same calculation. Given a linear parabolic PDE with terminal data, one can always write the solution as a conditional expectation over the paths of an associated diffusion. Conversely, given a conditional expectation of a smooth function of a diffusion's terminal state, the expectation solves a specific PDE. The reader gets to pick the weapon: whichever side of the bridge is easier for the problem at hand.

The *why it matters* is entirely practical. Low-dimensional problems with smooth coefficients — a European vanilla on a single asset, a caplet on a single forward rate — are almost always faster to price via PDE: a Crank-Nicolson grid with a few hundred nodes converges in milliseconds. High-dimensional problems — basket options on $N > 3$ underliers, path-dependent exotics with many state variables, multi-factor rate models — are almost always faster via Monte Carlo, whose error scales as $1/\sqrt{N_{\text{paths}}}$ independently of state dimension. American-style features (early exercise) favour PDEs because the exercise boundary embeds directly into the grid. Path-dependent features (barriers, Asians, lookbacks) favour Monte Carlo because they require tracking a path functional. Feynman-Kac is what guarantees that whichever route one chooses, the answer is the same.

There is a deeper reason the bridge is worth spending a chapter on. Every *pricing PDE* that appears in mathematical finance — Black-Scholes, Vasicek's bond-price equation, Hull-White's calibrated equation, Heston's two-factor equation, the short-rate equations of Chapter 12, the caplet equations of Chapter 15 — is a Feynman-Kac PDE for some SDE and some discount rate. Mastering the bridge in full generality (state-and-time-dependent coefficients, state-dependent discount) gives the reader a single theorem that, properly instantiated, handles *every* linear pricing problem in the guide.

6.2 4.2 The Zero-Drift Feynman-Kac Theorem

We begin with the simplest possible instance of the theorem — a driftless, undiscounted Brownian motion with a terminal payoff. Every later version is a decoration of this base case, so it is worth pinning down the geometry before adding coefficients.

Let $X = (X_t)_{t\geq 0}$ denote a standard Brownian motion (as constructed in Chapter 3), and fix a horizon $T > 0$ and a payoff $\varphi : \mathbb{R} \to \mathbb{R}$ satisfying mild integrability (polynomial growth is more than enough for every concrete payoff in this guide). Consider the backward Cauchy problem

$$\begin{cases} \partial_t f(t,x) \ + \ \frac{1}{2}\,\partial_{xx} f(t,x) \ = \ 0, \quad (t,x) \in [0,T) \times \mathbb{R}, \\ f(T,x) \ = \ \varphi(x). \end{cases} \tag{4.1}$$

Feynman-Kac asserts that this PDE is *equivalent* to the probabilistic representation

$$\boxed{f(t,x) \ = \ \mathbb{E}_{t,x}[\,\varphi(X_T)\,] \ \equiv \ \mathbb{E}[\,\varphi(X_T)\,|\,X_t = x\,]\,.} \tag{4.2}$$

The biconditional carries both directions. **PDE ⇒ expectation:** a classical solution of (4.1) admits the expectation representation (4.2). **Expectation ⇒ PDE:** conversely, the function $x \mapsto \mathbb{E}_{t,x}[\varphi(X_T)]$ automatically satisfies (4.1). In practice one direction or the other is the easier computation, and Feynman-Kac lets us freely pick.

6.2.1 4.2.1 What the PDE is

The PDE on the left of (4.1) is the (reverse) heat equation. The forward heat equation $\partial_t u = \frac{1}{2}\partial_{xx} u$ describes the diffusion of heat through a conducting medium with thermal diffusivity $\frac{1}{2}$; reversing the arrow of time (replacing t with $T - t$) gives $\partial_t f + \frac{1}{2}\partial_{xx} f = 0$, which is the Kolmogorov backward equation for Brownian motion. The payoff φ at expiry plays the role of an initial heat distribution, and the PDE smooths it backward in time from T down to t. Option prices are literally temperature distributions evolving under a thermal-diffusion physics, with the terminal payoff as initial temperature profile and "conductivity" equal to $\frac{1}{2}$ (the Brownian diffusion rate). The heat-equation picture is more than metaphor: every qualitative feature of diffusion — smoothing of discontinuities, infinite propagation speed, exponential decay of modes — translates directly into a feature of option prices, and we will use this repeatedly.

Heat-equation analogy *As t moves backward from T toward 0, the terminal payoff φ diffuses and smooths: kinks (vanilla calls) round off, step functions (digitals) bloom into Gaussian-shaped bumps. Every option price is literally a smoothed payoff, where the smoothing kernel is a Gaussian of width $\sqrt{T-t}$.*

6.2.2 4.2.2 What the expectation is

The right-hand side of (4.2) is the conditional-expectation scoreboard: at time t, given that the current state is $X_t = x$, average the payoff $\varphi(X_T)$ over every Brownian trajectory that starts at (t,x) and wanders until T. Because Brownian motion is Markov — it forgets its pre-t history — the expectation only depends on the pair (t,x), never on the path that brought us to x. That Markov collapse is why f is a function of two real variables rather than a functional on the full path space.

Two readings of the biconditional are used constantly in practice:

- **PDE → expectation.** Hard PDE computations can be replaced with Monte-Carlo averages: simulate N Brownian paths from x starting at time t, evaluate $\varphi(X_T)$ on each, and take the sample mean. The error scales as $1/\sqrt{N}$ independently of state dimension. This is the foundation of Monte-Carlo pricing.
- **Expectation → PDE.** Potentially infinite-dimensional averages over paths can be replaced by finite-dimensional PDE solves on a small grid. Finite-difference methods (implicit Euler, Crank-Nicolson) exploit exactly this direction. The PDE lives on a two-dimensional (t,x) mesh; the path space is infinite-dimensional. Trading path dimension for grid dimension is often a good deal in low-dimensional problems.

Which direction to pick is problem-driven. High-dimensional state spaces (basket options on many underliers, multi-factor rate models) favour Monte Carlo because a PDE mesh becomes curse-of-dimensionality slow as soon as the state dimension exceeds about three. Low-dimensional problems with American features favour PDEs because one can embed the exercise boundary directly into the grid. Feynman-Kac is what guarantees the answer is the same either way.

6.2.3 4.2.3 Path-integral intuition — why the equivalence is not magic

Markov plus local dynamics force (4.1). The scoreboard obeys the tower identity

$$f(t,x) \ = \ \mathbb{E}[\,f(t + \mathrm{d}t,\ x + \mathrm{d}W)\,]\,. \tag{4.3}$$

Taylor-expand the right-hand side, use $\mathbb{E}[\mathrm{d}W] = 0$ and the Itô shorthand $(\mathrm{d}W)^2 = \mathrm{d}t$, divide by $\mathrm{d}t$, and the residual $0 = \partial_t f + \frac{1}{2}\partial_{xx} f$ is the PDE. The terminal condition is automatic: at $t = T$ the scoreboard reads the payoff directly. The next section turns this sketch into a proof.

6.3 4.3 A Martingale Derivation of the Backward PDE

The Taylor-expansion sketch of §4.2.3 is essentially correct but cuts two corners. First, it quietly assumes f is smooth enough to Taylor-expand. Second, it treats $(\mathrm{d}W)^2 = \mathrm{d}t$ as a symbolic identity rather than as a consequence of quadratic variation. This section gives the clean derivation: the scoreboard is a *martingale* by the tower property of conditional expectations, and Itô's lemma applied to a martingale forces the drift to vanish — which is the PDE.

The payoff of this slower proof is a reusable *template*. Every pricing PDE in this guide will be derived by the same three moves: (i) write the candidate price as a conditional expectation of a terminal payoff; (ii) apply Itô's lemma to the candidate to decompose its increment into drift plus martingale; (iii) demand that the drift vanish, which gives the PDE. Learn the template once here; reuse it in Chapter 6 (Black-Scholes), Chapter 12 (Vasicek bond), Chapter 15 (caplets), and anywhere else a linear parabolic PDE is needed.

6.3.1 4.3.1 Setup and the candidate

Fix a horizon T and a payoff $\varphi : \mathbb{R} \to \mathbb{R}$ with polynomial growth. Let X_t be a standard Brownian motion. Define the candidate pricing function

$$f(t, x) \;\equiv\; \mathbb{E}\big[\,\varphi(X_T)\,\big|\,X_t = x\,\big] \;=\; \mathbb{E}_{t,x}[\varphi(X_T)]. \tag{4.4}$$

The question the proof will answer is: *what PDE does f satisfy as a function of (t, x)?* The answer is $(\partial_t + \frac{1}{2}\partial_{xx})f = 0$ with terminal data $f(T, x) = \varphi(x)$, derivable from (a) iterated expectation and (b) Itô's lemma applied to $f(t, X_t)$.

6.3.2 4.3.2 The auxiliary process $\eta_s = f(s, X_s)$

Define a stochastic process indexed by the intermediate time $s \in [t, T]$:

$$\eta_s \;\equiv\; f(s, X_s) \;=\; \mathbb{E}\big[\,\varphi(X_T)\,\big|\,X_s\,\big]\,. \tag{4.5}$$

Geometrically, η_s is "the best current guess of $\varphi(X_T)$ given everything known up to time s." Because X is Markov, conditioning on the full filtration $\mathcal{F}_s$ of the Brownian motion collapses to conditioning on the single random variable X_s:

$$\mathbb{E}[\varphi(X_T)\,|\,\mathcal{F}_s] \;=\; \mathbb{E}[\varphi(X_T)\,|\,X_s] \;=\; f(s, X_s) \;=\; \eta_s. \tag{4.6}$$

The first equality is the Markov property of Brownian motion (independent increments collapse the filtration dependence to the current state); the second is the definition of f; the third is notation. This three-step unwinding is worth memorising because it reappears in every measure-change, numeraire-swap, and forward-measure derivation in later chapters.

6.3.3 4.3.3 Tower law $\Rightarrow$ η is a martingale

Claim. The process $s \mapsto \eta_s$ is a martingale in s under the natural filtration $\{\mathcal{F}_s\}$ of X. Concretely, for any $t \le s_1 \le s_2 \le T$,

$$\mathbb{E}\Big[\eta_{s_2}\,\big|\,\mathcal{F}_{s_1}\Big] \;=\; \eta_{s_1}. \tag{4.7}$$

Proof. Plug in the definition:

$$\mathbb{E}[\eta_{s_2}|\mathcal{F}_{s_1}] \;=\; \mathbb{E}\Big[\,\mathbb{E}[\varphi(X_T)|\mathcal{F}_{s_2}]\,\big|\,\mathcal{F}_{s_1}\Big]\,.$$

The outer expectation is over information up to s_1; the inner expectation is over information up to $s_2 \geq s_1$. Apply the tower property of conditional expectations — $\mathbb{E}[\mathbb{E}[A|\mathcal{G}]|\mathcal{H}] = \mathbb{E}[A|\mathcal{H}]$ whenever $\mathcal{H} \subseteq \mathcal{G}$ — with $\mathcal{H} = \mathcal{F}_{s_1}$ and $\mathcal{G} = \mathcal{F}_{s_2}$ (the latter is indeed finer because more time has passed):

$$\mathbb{E}\Big[\,\mathbb{E}[\varphi(X_T)|\mathcal{F}_{s_2}]\,\Big|\,\mathcal{F}_{s_1}\Big] \;=\; \mathbb{E}[\varphi(X_T)|\mathcal{F}_{s_1}] \;=\; \eta_{s_1}.$$

Therefore $\eta_{s_1} = \mathbb{E}[\eta_{s_2}|\mathcal{F}_{s_1}]$, the martingale property. □

The argument uses only nesting of filtrations plus iterated expectation. *Any* process of the form "conditional expectation of a fixed $\mathcal{F}_T$-measurable random variable" is a martingale (a *Doob martingale*). The payoff $\varphi(X_T)$ is the "closing" random variable; η_s is the stream of running conditional expectations.

6.3.4 4.3.4 Itô's lemma on $\eta_s = f(s, X_s)$

Having established that η_s is a martingale, we now apply Itô's lemma to read off what PDE f must satisfy. Assume $f \in C^{1,2}([0,T) \times \mathbb{R})$ — continuously differentiable in t once, in x twice. This is an a-posteriori regularity result for Feynman-Kac solutions with smooth enough payoffs, and we take it for granted here. Apply Itô's lemma II from Chapter 3 with $X_s = W_s$:

$$\mathrm{d}\eta_s \;=\; \partial_t f(s,X_s)\,\mathrm{d}s \;+\; \partial_x f(s,X_s)\,\mathrm{d}X_s \;+\; \tfrac{1}{2}\,\partial_{xx} f(s,X_s)\,(\mathrm{d}X_s)^2. \tag{4.8}$$

Using the Itô shorthand $(\mathrm{d}X_s)^2 = \mathrm{d}s$ and grouping the $\mathrm{d}s$ and $\mathrm{d}X_s$ terms,

$$\mathrm{d}\eta_s \;=\; \underbrace{\big[\,\partial_t f + \tfrac{1}{2}\,\partial_{xx} f\,\big]}_{\text{drift}}\,\mathrm{d}s \;+\; \underbrace{\partial_x f}_{\text{diffusion}}\;\mathrm{d}X_s, \tag{4.9}$$

with both partial derivatives evaluated at (s, X_s).

6.3.5 4.3.5 Drift-killing identity

The diffusion term $\partial_x f\,\mathrm{d}X_s$ integrated against $\mathrm{d}W$ is a pure stochastic integral, hence a martingale by the construction of the Itô integral in Chapter 3. So the difference between η_s and the martingale $\int \partial_x f\,\mathrm{d}X$ is the deterministic drift term $\int[\partial_t f + \frac{1}{2}\partial_{xx} f]\,\mathrm{d}s$. But we proved in §4.3.3 that η_s is already a martingale. Subtracting a martingale from a martingale leaves a martingale, so the drift term $\int[\partial_t f + \frac{1}{2}\partial_{xx} f]\,\mathrm{d}s$ is itself a martingale.

Now the key fact: this drift is a process of *bounded variation* (it is an ordinary Riemann integral in s), and the only continuous martingale of finite variation is the zero process. (By the Doob-Meyer decomposition, the finite-variation part of a continuous semimartingale starts at zero and is uniquely identified by the drift; combining "is a martingale" with "finite-variation part starts at zero" forces the integrand to vanish identically.) Therefore the drift vanishes:

$$\int_t^s \big[\partial_t f(u, X_u) + \tfrac{1}{2}\partial_{xx} f(u, X_u)\big]\,\mathrm{d}u \;\equiv\; 0 \quad \text{for every } s \in [t, T].$$

Because this holds for every path (including paths visiting every real number at arbitrarily chosen times) and the integrand is continuous, the integrand must vanish pointwise:

$$\boxed{\partial_t f(t,x) + \tfrac{1}{2}\,\partial_{xx} f(t,x) \;=\; 0} \qquad \text{for all } (t,x) \in [0,T) \times \mathbb{R}. \tag{4.10}$$

This is the Kolmogorov backward equation, and we have just derived it from the martingale property of the Doob-closed process η. □

6.4 4.4 Boundary Condition and Summary of the Zero-Drift Case

The derivation of §4.3 produced the PDE (4.10) from the martingale property of η, but we still need to verify the terminal condition $f(T,x) = \varphi(x)$. That is immediate from the definition of f: at $s = T$ the conditioning has frozen the Brownian motion at its terminal value, so

$$\eta_T \;=\; \mathbb{E}[\varphi(X_T)|X_T] \;=\; \varphi(X_T),$$

and evaluating at $X_T = x$ gives $f(T,x) = \varphi(x)$. The boundary data is *built into* the probabilistic construction.

6.4.1 4.4.1 What the zero-drift theorem says

Putting the pieces of §§4.2–4.3 together:

> **Zero-drift Feynman-Kac.** Let X be a standard Brownian motion and let $\varphi : \mathbb{R} \to \mathbb{R}$ have polynomial growth. The function $f(t,x) = \mathbb{E}_{t,x}[\varphi(X_T)]$ is the (unique classical) solution of the backward heat equation $\partial_t f + \frac{1}{2}\partial_{xx} f = 0$ on $[0,T) \times \mathbb{R}$ with terminal data $f(T,\cdot) = \varphi(\cdot)$.

The three ingredients of the proof — Markov collapse, Doob-martingale of expectations, drift-kill via Itô — will *all* reappear in every subsequent generalisation. The only thing that changes is what one drops into the Itô decomposition.

6.4.2 4.4.2 A remark on regularity

The derivation of §4.3 assumed $f \in C^{1,2}$. For discontinuous payoffs (digitals, indicators) standard practice is either to approximate by smooth $\varphi_n \to \varphi$ (Gaussian-kernel smoothing handles the limit) or to invoke viscosity solutions. The $C^{1,2}$ hypothesis is convenient but not essential.

6.5 4.5 Example 1 — Linear Payoff $\varphi(x) = x$

The simplest possible test case. A linear payoff exercises none of the non-trivial machinery of stochastic calculus — the answer is what one would guess — but it serves as a base-line sanity check. If the Feynman-Kac machinery did *not* return x for this payoff, something would be wrong with the setup.

Consider the Cauchy problem

$$\begin{cases} \partial_t f + \frac{1}{2}\partial_{xx} f \;=\; 0, \\ f(T,x) \;=\; x. \end{cases} \tag{4.12}$$

By Feynman-Kac (4.2),

$$f(t,x) \;=\; \mathbb{E}_{t,x}[X_T],$$

where X is a standard Brownian motion. Brownian increments are Gaussian,

$$X_T - X_t \;\sim\; \mathcal{N}(0, T-t), \qquad \text{so} \qquad X_T \overset{d}{=} x + \sqrt{T-t}\, Z, \quad Z \sim \mathcal{N}(0,1).$$

Therefore

$$f(t,x) \;=\; \mathbb{E}_{t,x}\Big[x + \sqrt{T-t}\, Z\Big] \;=\; x \;+\; \sqrt{T-t}\cdot \mathbb{E}[Z] \;=\; x. \tag{4.13}$$

PDE verification. With $f(t,x) = x$, $\partial_t f = 0$, $\partial_x f = 1$, $\partial_{xx} f = 0$, so $\partial_t f + \frac{1}{2}\partial_{xx} f = 0$. Terminal condition $f(T,x) = x$ holds by inspection. Both branches agree.

6.5.1 4.5.1 Why this answer is natural

A linear payoff has zero *convexity*, so Jensen's inequality is an equality and the expectation coincides with the initial state. Whenever φ is affine, $\mathbb{E}_{t,x}[\varphi(X_T)] = \varphi(x)$. This will fail as soon as we move to x^2 (convex) or e^{-x} (log-convex) in the next two examples — but convexity first has to be introduced.

A trader's-eye reading. A linear payoff is a forward on a driftless underlying, so the forward value equals the spot — there is no convexity premium because the payoff has zero curvature. As soon as the payoff bends, the Itô correction starts to bite.

6.6 4.6 Example 2 — Quadratic Payoff $\varphi(x) = x^2$

The first non-trivial example, and the one where convexity starts to matter. The quadratic payoff is also the simplest test case for variance-swap-style replication arguments, because the squared terminal value is the canonical "variance" functional.

Solve

$$\begin{cases} \partial_t f + \frac{1}{2}\partial_{xx} f \;=\; 0, \\ f(T,x) \;=\; x^2. \end{cases} \tag{4.14}$$

Feynman-Kac gives

$$f(t,x) \;=\; \mathbb{E}_{t,x}[X_T^2] \;=\; \mathbb{E}_{t,x}\Big[\, \big((X_T - X_t) + X_t\big)^2 \,\Big],$$

with $X_T - X_t \sim \mathcal{N}(0, T-t)$ and $X_t = x$ under $\mathbb{E}_{t,x}$. Expanding the square and taking expectations term by term:

$$f(t,x) \;=\; \mathbb{E}_{t,x}\big[\,(X_T - X_t)^2 + 2X_t(X_T - X_t) + X_t^2\,\big] \;=\; (T-t) \;+\; 0 \;+\; x^2,$$

where the three contributions come from $\mathbb{E}[(X_T - X_t)^2] = \mathrm{Var}(X_T - X_t) = T - t$, $\mathbb{E}[(X_T - X_t)] = 0$, and $\mathbb{E}[X_t^2 | X_t = x] = x^2$. Hence

$$\boxed{f(t,x) \;=\; x^2 + (T-t)}. \tag{4.15}$$

PDE verification. With $f(t,x) = x^2 + (T-t)$, $\partial_t f = -1$, $\partial_{xx} f = 2$, so $\partial_t f + \frac{1}{2}\partial_{xx} f = -1 + 1 = 0$. At $t = T$: $f(T,x) = x^2 + 0 = x^2$. Both the PDE and the terminal condition are satisfied.

6.6.1 4.6.1 Convexity reading and variance-swap interpretation

The extra $(T-t)$ in (4.15) is the *time-value of convexity*: a convex payoff rewards variance, and Brownian increments supply exactly $\mathrm{Var}(X_T | X_t = x) = T - t$. Equivalently, Itô's lemma applied to $F(x) = x^2$ gives $X_T^2 = X_t^2 + 2\int_t^T X_s \, \mathrm{d}X_s + (T-t)$; the middle martingale term vanishes under expectation and recovers (4.15). The same identity prices a variance swap with strike x^2 in the driftless, no-discount setting: the fair forward value is $x^2 + (T-t)$.

Quadratic payoff MC vs PDE solution *Sanity check of* (4.15)*: Monte-Carlo averages of X_T^2 at a grid of initial states x_0 (red dots) fall exactly on the PDE closed form $x_0^2 + T$ (blue curve). The $+T$ offset is the "time-value of variance" — the convexity premium that Brownian motion pays out for a quadratic payoff.*

6.7 4.7 Example 3 — Exponential Payoffs $\varphi(x) = e^{ax}$

Exponential payoffs are the log-space analogue of the log-normal pricing problems that dominate the rest of the guide. Because the log of a geometric-Brownian-motion asset is a drifted arithmetic Brownian motion, every result derived here with $X = W$ will translate into a stock-price result by the substitution $X = \ln S$. Many closed-form formulas in rates, credit, and equity pricing fall out of this single expectation once the drift and discounting extensions of §4.8 are in place.

6.7.1 4.7.1 Base case $\varphi(x) = e^{-x}$

Solve

$$\begin{cases} \partial_t f + \frac{1}{2}\partial_{xx} f \;=\; 0, \\ f(T,x) \;=\; e^{-x}. \end{cases} \tag{4.16}$$

Feynman-Kac gives

$$f(t,x) \;=\; \mathbb{E}_{t,x}[\, e^{-X_T}\,].$$

Because $X_T | X_t = x$ is $\mathcal{N}(x, T-t)$, apply the Gaussian moment-generating function $\mathbb{E}[e^{\lambda Y}] = e^{\lambda\mu + \frac{1}{2}\lambda^2\sigma^2}$ with $\lambda = -1$, $\mu = x$, $\sigma^2 = T - t$:

$$\boxed{f(t,x) \;=\; e^{-x+\frac{1}{2}(T-t)}}. \tag{4.17}$$

PDE verification. With $f = e^{-x+\frac{1}{2}(T-t)}$: $\partial_t f = -\frac{1}{2} f$, $\partial_x f = -f$, $\partial_{xx} f = f$. Then $\partial_t f + \frac{1}{2}\partial_{xx} f = -\frac{1}{2} f + \frac{1}{2} f = 0$. Terminal condition $f(T,x) = e^{-x}$ holds.

Jensen bump. Notice $f(t,x) = e^{-x} \cdot e^{\frac{1}{2}(T-t)}$ — the answer exceeds the payoff evaluated at the mean e^{-x} by the factor $e^{\frac{1}{2}(T-t)}$. This is the *convexity premium*. In the log variable this is exactly the $\frac{1}{2}\sigma^2 t$ Itô correction that will reappear in the Black-Scholes formula. Its sign follows from Jensen's inequality applied to the convex function e^{-x}: $\mathbb{E}[e^{-X_T}] \geq e^{-\mathbb{E}[X_T]} = e^{-x}$, with strict inequality whenever X_T has nonzero variance.

6.7.2 4.7.2 General exponent $\varphi(x) = e^{ax}$

A one-parameter generalisation that shows the convexity bump scales as a^2, not $|a|$. Solve

$$\begin{cases} \partial_t f + \frac{1}{2}\partial_{xx} f \;=\; 0, \\ f(T,x) \;=\; e^{ax}, \end{cases} \qquad a \in \mathbb{R}. \tag{4.18}$$

By Feynman-Kac, using $X_T = x + (X_T - X_t)$ with $X_T - X_t \sim \mathcal{N}(0, T-t)$,

$$f(t,x) \;=\; \mathbb{E}_{t,x}[\, e^{aX_T}\,] \;=\; e^{ax} \cdot \mathbb{E}[\, e^{a(X_T - X_t)}\,] \;=\; e^{ax} \cdot e^{\frac{1}{2}a^2(T-t)},$$

using the Gaussian MGF with $\lambda = a$, $\sigma^2 = T - t$. Compactly,

$$\boxed{f(t,x) \;=\; \exp\{\, a\,x \;+\; \tfrac{1}{2}\,a^2\,(T-t)\,\}}. \tag{4.19}$$

PDE verification. Let $\gamma = \frac{1}{2}a^2$. Then $f = e^{ax+\gamma(T-t)}$ gives $\partial_t f = -\gamma f$, $\partial_x f = a f$, $\partial_{xx} f = a^2 f$. Plugging into the PDE:

$$\partial_t f + \tfrac{1}{2}\partial_{xx} f \;=\; -\gamma f + \tfrac{1}{2}a^2 f \;=\; -\tfrac{1}{2}a^2 f + \tfrac{1}{2}a^2 f \;=\; 0. \quad \checkmark$$

Terminal: $f(T,x) = e^{ax+0} = e^{ax}$.

Monte-Carlo sanity check for §§4.5–4.7 closed forms *For each of the three worked payoffs (linear x, quadratic x^2, exponential e^{ax} with $a = 0.8$), Monte-Carlo averaging over N Brownian endpoints $X_T \sim \mathcal{N}(x_0, T)$ converges at rate $1/\sqrt{N}$ to the Gaussian-MGF closed form from* (4.13)–(4.19).

6.8 4.8 Feynman-Kac with Drift, Diffusion, and Discounting

Every worked example so far has used a driftless Brownian motion with unit diffusion and no discounting. Real pricing problems require three generalisations: the underlying may have a nonzero drift, its diffusion coefficient need not be one, and the payoff must be discounted back to the present. Feynman-Kac extends to each in a mechanical way, and in this section we do all three at once.

6.8.1 4.8.1 Statement

Let $a \in \mathbb{R}$, $b > 0$, and $c \in \mathbb{R}$ be constants. Consider the backward Cauchy problem with a lower-order term:

$$\begin{cases} \partial_t f(t,x) \;+\; a\,\partial_x f(t,x) \;+\; \tfrac{1}{2}\,b^2\,\partial_{xx} f(t,x) \;=\; c\,f(t,x), \\ f(T,x) \;=\; \varphi(x). \end{cases} \tag{4.20}$$

The probabilistic representation of the solution is

$$\boxed{f(t,x) \;=\; \mathbb{E}_{t,x}\big[\,e^{-c(T-t)}\,\varphi(X_T)\,\big]\,,} \tag{4.21}$$

where $X = (X_s)_{s\geq 0}$ is a drifted Brownian motion solving

$$\mathrm{d}X_s \;=\; a\,\mathrm{d}s \;+\; b\,\mathrm{d}W_s, \qquad W \text{ a standard BM}, \tag{4.22}$$

i.e. $X_s = X_0 + a\,s + b\,W_s$ (arithmetic Brownian motion with drift a and volatility b).

6.8.2 4.8.2 Proof sketch

Start from the expectation $f(t,x) = \mathbb{E}_{t,x}[e^{-c(T-t)}\varphi(X_T)]$ and apply Itô's lemma to the auxiliary process

$$g_s \;\equiv\; e^{-c(s-t)}\,f(s,X_s), \qquad s \in [t,T].$$

Compute $\mathrm{d}g_s$ using the product rule plus Itô's lemma III applied to $f(s,X_s)$:

$$\mathrm{d}g_s \;=\; e^{-c(s-t)}\big\{-c\,f\,\mathrm{d}s \;+\; \big[\partial_t f + a\,\partial_x f + \tfrac{1}{2}b^2\partial_{xx} f\big]\,\mathrm{d}s \;+\; b\,\partial_x f\,\mathrm{d}W_s\big\}\,. \tag{4.23}$$

For g_s to be a martingale — which it must be, because $g_T = e^{-c(T-t)}\varphi(X_T)$ and conditional expectations of fixed random variables are martingales — the ds drift must vanish:

$$-c\,f + \partial_t f + a\,\partial_x f + \tfrac{1}{2}b^2\partial_{xx} f \;=\; 0,$$

which is exactly the PDE (4.20). Conversely, if f satisfies the PDE, the drift of g_s is identically zero; g_s is therefore a martingale, and taking $\mathbb{E}_{t,x}$ of $g_t = f(t,x)$ and $g_T = e^{-c(T-t)}\varphi(X_T)$ yields the representation (4.21).

The two-way proof is the essence of Feynman-Kac: the PDE *is* the drift-killing condition for $e^{-c(s-t)}f(s,X_s)$, and any function satisfying that PDE automatically represents an expectation.

6.8.3 4.8.3 Role of each coefficient

Each term in (4.20) has a direct counterpart in (4.21):

PDE term	SDE / expectation counterpart
$a\,\partial_x f$	drift of X: each path is pushed by $a\,\mathrm{d}s$ per step.
$\frac{1}{2}b^2\,\partial_{xx}f$	diffusion of X: paths spread with variance $b^2\,\mathrm{d}s$ per step.
$-c\,f$	exponential decay / discount factor $e^{-c(T-t)}$ applied to the payoff.
terminal $f(T,x)=\varphi(x)$	the expectation at $s=T$ is literally $\varphi(X_T)$ with $X_T=x$.

The dictionary reading of (4.20) $\Leftrightarrow$ (4.21): *"a and b pick the law of X; c attaches a per-unit-time discount to every path."*

6.8.4 4.8.4 Worked case: exponential payoff with drift, diffusion, and discount

Solve (4.20) with $\varphi(x)=e^{-x}$:

$$\begin{cases}\partial_t f + a\,\partial_x f + \frac{1}{2}b^2\partial_{xx}f \;=\; c\,f,\\ f(T,x) \;=\; e^{-x}.\end{cases} \tag{4.24}$$

By (4.21),

$$f(t,x) \;=\; \mathbb{E}_{t,x}\big[\,e^{-X_T}\,e^{-c(T-t)}\,\big]\,.$$

The increment of X under the drifted BM (4.22) is

$$X_T - X_t \;=\; a(T-t) + b\,(W_T - W_t) \;\overset{d}{=}\; a(T-t) + b\sqrt{T-t}\,Z, \qquad Z\sim\mathcal{N}(0,1),$$

so $X_T|X_t=x$ is Gaussian with mean $x+a(T-t)$ and variance $b^2(T-t)$. Substituting,

$$f(t,x) \;=\; \mathbb{E}\Big[\exp\Big\{-x - a(T-t) - b\sqrt{T-t}\,Z - c(T-t)\Big\}\Big]\,.$$

Taking the Gaussian MGF of $-b\sqrt{T-t}\,Z$, which contributes $e^{\frac{1}{2}b^2(T-t)}$:

$$f(t,x) \;=\; \exp\big\{-x - a(T-t) + \tfrac{1}{2}b^2(T-t) - c(T-t)\big\}\,.$$

Collecting terms with common factor $(T-t)$:

$$\boxed{f(t,x) \;=\; \exp\big\{-x \;-\; \big(a+c-\tfrac{1}{2}b^2\big)\,(T-t)\,\big\}}\,. \tag{4.25}$$

Equivalently, defining $\gamma = a+c-\frac{1}{2}b^2$ for shorthand,

$$f(t,x) \;=\; e^{-x-\gamma(T-t)}. \tag{4.26}$$

6.8.5 4.8.5 Reading formula (4.25) as a ledger

The exponent $-x-(a+c-\frac{1}{2}b^2)(T-t)$ has four contributions: $-x$ from the payoff base, $-a(T-t)$ from drift, $+\frac{1}{2}b^2(T-t)$ from Jensen/convexity, and $-c(T-t)$ from discount. Sanity limits: $a=c=0$, $b=1$ recovers (4.17); $b=0$ recovers pure deterministic evolution with no convexity. A direct PDE check with $\gamma=a+c-\frac{1}{2}b^2$ and $f=e^{-x-\gamma(T-t)}$ gives $\partial_t f=+\gamma f$, $\partial_x f=-f$, $\partial_{xx}f=f$, and the LHS of (4.20) collapses to $(\gamma-a+\frac{1}{2}b^2-c)f=0$.

6.9 4.9 Feynman-Kac with State- and Time-Dependent Coefficients

The constant-coefficient version (4.20)–(4.21) is enough for driftless BM, for arithmetic BM with constant drift, and for a handful of toy models. Most real-world pricing — including the Black-Scholes model for equity derivatives and the short-rate models for fixed income — requires *state-dependent* coefficients where a, b, and c are themselves functions of (t,x). This section states and proves the general theorem and then shows three important instantiations.

6.9.1 4.9.1 The general theorem

Let $a,b,c:[0,T]\times\mathbb{R}\to\mathbb{R}$ be sufficiently regular functions — continuous with mild growth conditions on a and b, and boundedness on c; these are the conditions under which the associated SDE has a unique strong solution and the corresponding PDE has a classical solution. Consider the Cauchy problem

$$\begin{cases}\partial_t h(t,x) \;+\; a(t,x)\,\partial_x h(t,x) \;+\; \frac{1}{2}\,b^2(t,x)\,\partial_{xx}h(t,x) \;=\; c(t,x)\,h(t,x),\\ h(T,x) \;=\; H(x),\end{cases} \tag{4.34}$$

with terminal payoff $H:\mathbb{R}\to\mathbb{R}$ satisfying polynomial growth. Then the solution admits the probabilistic representation

$$\boxed{h(t,x) \;=\; \mathbb{E}\left[\exp\left\{-\int_t^T c(u,X_u)\,\mathrm{d}u\right\}\,H(X_T)\,\middle|\,X_t=x\right],} \tag{4.35}$$

where the process X satisfies the Itô SDE

$$\mathrm{d}X_s \;=\; a(s,X_s)\,\mathrm{d}s \;+\; b(s,X_s)\,\mathrm{d}W_s, \qquad s\in[t,T]. \tag{4.36}$$

This is the *complete* Feynman-Kac theorem used in practice. Every linear parabolic pricing PDE in this guide is an instantiation of it, and the dictionary between PDE coefficients and SDE/discount is strict.

6.9.2 4.9.2 Three features worth noting

The drift $a(s,X_s)$ and diffusion $b(s,X_s)$ may depend on both time and the current state. No constancy assumption is required. The SDE (4.36) is the general Itô diffusion, and Itô's lemma III from Chapter 3 is exactly the tool we need.

The discount rate $c(s,X_s)$ may itself be stochastic — a function of the state X_s. In this case the integrated discount $\int_t^T c(u,X_u)\,\mathrm{d}u$ is a path-dependent random variable, not a deterministic number $c(T-t)$. The path integral is exactly how stochastic-rate bond pricing works: $c(t,r)=r$ with $X=r_t$ a short-rate process yields bond prices $P(t,T)=\mathbb{E}_t[e^{-\int_t^T r_s\,\mathrm{d}s}]$.

The payoff $H(X_T)$ is evaluated at the final state only. We do not allow path-dependent payoffs in the basic statement. Path-dependency — barrier options that knock out if the path touches a level, Asian options that depend on the average price — requires further machinery: auxiliary state variables (e.g. the running minimum becomes a second state), augmented Itô processes, or variational inequalities for American-style features. Each extension preserves the Feynman-Kac spirit but with extra bookkeeping. The basic theorem (4.34)–(4.35) covers every European-style payoff on a single state variable.

6.9.3 4.9.3 Why the path-dependent discount is necessary

On the PDE side, the discount term $c(t,x)\,h$ is spatially variable — it depends on where in the (t,x) domain the price-surface is being evaluated. On the expectation side, the discount factor $e^{-\int c\,\mathrm{d}u}$ integrates c along the *realised path* X from t to T. These two forms are duals: the PDE evaluates c locally at each point, and the expectation accumulates that local rate along each sample path.

For constant c, the path integral collapses to $c(T-t)$ and we recover the constant-discount version (4.21). For spatially-varying c, the two sides of Feynman-Kac exchange "local evaluation" for "path accumulation" — a conceptual swap that is easiest to appreciate in the simple example $c(t,x)=x$: on the PDE side we see a linear-in-state zero-order term; on the expectation side we see $e^{-\int_t^T X_u\,\mathrm{d}u}$, a path-dependent random variable. The two forms are identical in content, just organised differently.

6.9.4 4.9.4 Proof (integrating-factor device)

Apply Itô to the discounted candidate

$$Y_s \;\equiv\; \exp\Big\{-\int_t^s c(u,X_u)\,\mathrm{d}u\Big\}\cdot h(s,X_s), \qquad s\in[t,T]. \tag{4.37}$$

The integrating factor $e^{-\int_t^s c\,\mathrm{d}u}$ has differential $-c(s,X_s)\,e^{-\int_t^s c\,\mathrm{d}u}\,\mathrm{d}s$ (no $\mathrm{d}W$ piece, hence no covariation). Combining with Itô III on $h(s,X_s)$,

$$\mathrm{d}Y_s \;=\; e^{-\int_t^s c\,\mathrm{d}u}\big\{\big[-c\,h+\partial_t h + a\,\partial_x h + \tfrac{1}{2}b^2\,\partial_{xx}h\big]\,\mathrm{d}s \;+\; b\,\partial_x h\,\mathrm{d}W_s\big\}\,. \tag{4.38}$$

The $\mathrm{d}s$ coefficient vanishes by the PDE (4.34), leaving a pure stochastic integral. Under the standard integrability hypothesis $\mathbb{E}_{t,x}\int_t^T e^{-2\int_t^s c\,\mathrm{d}u}\,b^2\,(\partial_x h)^2\,\mathrm{d}s<\infty$, Y is a true martingale on $[t,T]$. Taking expectations of Y_t and Y_T and using the terminal condition $h(T,\cdot)=H(\cdot)$,

$$h(t,x) \;=\; \mathbb{E}_{t,x}[Y_T] \;=\; \mathbb{E}_{t,x}\Big[e^{-\int_t^T c\,\mathrm{d}u}\,H(X_T)\Big]\,. \quad \square$$

The integrating factor $e^{-\int_t^s c\,\mathrm{d}u}$ does the same job as the time-0-discount move in §4.3.5: it converts what would be a super-martingale candidate into a true martingale by absorbing the spatial discount drift, and is the simplest instance of the change-of-numeraire technique developed in Chapter 5. Two specialisations to flag: $c(t,x)=x$ with $H=1$ gives the bond-price formula $P(t,x)=\mathbb{E}_{t,x}[e^{-\int_t^T X_u\,\mathrm{d}u}]$ (Chapter 12); $c(t,x)=r+\lambda(t,x)$ gives the survival-probability building block of credit pricing.

6.9.5 4.9.5 Reading off the Black-Scholes PDE

Plug $a(t,x)=rx$, $b(t,x)=\sigma x$, $c(t,x)=r$ into (4.34), all state-linear or constant. The PDE becomes

$$\partial_t h + rx\,\partial_x h + \tfrac{1}{2}\sigma^2 x^2\,\partial_{xx}h \;=\; r\,h, \tag{4.39}$$

which is the Black-Scholes PDE. The expectation (4.35) becomes

$$h(t,x) \;=\; \mathbb{E}_{t,x}\big[e^{-r(T-t)}\,H(X_T)\big]\,, \qquad \mathrm{d}X_s = rX_s\,\mathrm{d}s + \sigma X_s\,\mathrm{d}W_s, \tag{4.40}$$

which is the risk-neutral pricing formula for a claim $H(X_T)$ on a geometric-Brownian-motion underlying. The Black-Scholes formula for a European call or put is therefore literally a corollary of the general Feynman-Kac theorem; Chapter 6 derives it both this way and via the independent hedging argument.

6.9.6 4.9.6 Reading off the Vasicek bond price

Plug $a(t,r) = \kappa(\theta - r)$, $b(t,r) = \sigma$, $c(t,r) = r$, and $H(r) = 1$ into (4.34). The PDE becomes the Vasicek bond-price PDE

$$\partial_t P + \kappa(\theta - r)\,\partial_r P + \tfrac{1}{2}\sigma^2\,\partial_{rr} P \;=\; r\,P, \qquad P(T,r) = 1. \tag{4.41}$$

The expectation becomes

$$P(t,r) \;=\; \mathbb{E}_{t,r}\left[e^{-\int_t^T r_s\,\mathrm{d}s}\right], \qquad \mathrm{d}r_s = \kappa(\theta - r_s)\,\mathrm{d}s + \sigma\,\mathrm{d}W_s, \tag{4.42}$$

the classical fair value of a zero-coupon bond under a Vasicek short rate; Chapter 12 evaluates this expectation to obtain the affine closed form $P(t,r) = e^{A(t,T)-B(t,T)r}$.

6.9.7 4.9.7 The unifying observation

Every linear PDE with a first-order drift term, a second-order diffusion term, and a zero-order discount term is a Feynman-Kac PDE for *some* SDE and *some* discount rate. Identify the coefficients, write down the SDE, and the expectation representation (4.35) falls out. The dozen pricing PDEs in later chapters — Black-Scholes, Vasicek, Hull-White, caplet, swaption, Heston — are all instantiations of the same theorem; the algebra of solving each is where the later chapters spend their time. Chapter 6 carries this forward as the second derivation of Black-Scholes; Chapter 12 specialises to Vasicek/Hull-White; Chapter 15 specialises to caplet pricing under the T-forward measure.

6.10 4.10 Case Study — Vasicek Bond Pricing from the 2024 Treasury Curve

6.10.1 Case study: Vasicek calibrated to the 2024 US Treasury curve

Context. In early 2024 the on-the-run US 10-year Treasury traded near a 4.10% yield, with the 2-year at 4.55% and the 3-month T-bill near 5.40% — a deeply inverted curve produced by the Fed's hiking cycle (5.25%–5.50% policy rate) sitting above market expectations of cuts later in the year. A Vasicek short rate $\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$ fitted to that snapshot yields $\kappa \approx 0.30$, $\theta \approx 0.035$, $\sigma \approx 0.011$ (units: years^{-1}, percent, percent yr$^{-1/2}$) — a slow mean-reversion to a long-run rate well below the then-current 4.10% short rate.

Math mapping. This is (4.42) in flight. The Vasicek bond price $P(t,r) = \mathbb{E}_{t,r}[e^{-\int_t^T r_s\,\mathrm{d}s}]$ is the Feynman-Kac formula (4.35) with $a(t,r) = \kappa(\theta - r)$, $b(t,r) = \sigma$, $c(t,r) = r$ (the short rate itself is the discount), and terminal payoff $H = 1$. The affine ansatz $P(t,r) = e^{A(t,T)-B(t,T)r}$ reduces the path integral to two ODEs: $B(t,T) = (1 - e^{-\kappa(T-t)})/\kappa$ and A collecting the variance-of-integrated-rate term. With $T - t = 10$, $B \approx 3.17$, and plugging $r = 0.054$ gives $\ln P \approx A - 3.17 \cdot 0.054$; the convexity adjustment $\sigma^2 B^2 (T-t)/(4\kappa)$ shaves about 0.6 bp off the implied yield. The model recovers a 10-year zero yield of $\approx 4.05\%$ — within 5 bp of the on-the-run quote, all of which is the spline-vs-affine fitting residual on a curve with humps that a 1-factor Vasicek cannot match. The bond price is *literally* what falls out of evaluating a Gaussian path integral over the integrated short rate.

Lesson. Feynman-Kac is not decorative. The bond desk does this calibration every morning: take the observed short-rate path, fit (κ, θ, σ) to a few benchmark maturities, and read off every other point on the curve as a closed-form expectation (4.42). The mismatch in 2024 between the steeply inverted near end and a single-factor Vasicek model is exactly the empirical motivation for the multi-factor and time-dependent-$\theta(t)$ extensions of Chapter 12 — once θ is allowed to vary, the Hull-White model fits the entire 2024 curve to within 0.5 bp without changing the Feynman-Kac structure of the pricing formula.

6.11 4.11 Key Takeaways

1. **Two-way bridge.** A linear parabolic PDE with terminal data is *equivalent* to a conditional expectation over an associated diffusion; the reader picks whichever side is easier for the problem at hand. Equations (4.1)–(4.2) for the base case, (4.34)–(4.35) in full generality.

2. **Proof template: three moves.** Every pricing PDE in the guide is derived by (i) writing the price as a conditional expectation, (ii) applying Itô to decompose its increment into drift plus martingale, (iii) demanding the drift vanish. Recurs in Chapters 6, 12, 15.
3. **Doob martingale + drift-kill.** $\eta_s = \mathbb{E}[\varphi(X_T)|\mathcal{F}_s]$ is automatically a martingale by the tower property; a continuous martingale of bounded variation is identically zero, so its Itô drift must vanish — and that vanishing drift *is* the PDE.
4. **Three worked payoffs.** Linear $\varphi(x) = x$: $f = x$ (no convexity). Quadratic $\varphi(x) = x^2$: $f = x^2 + (T - t)$ (variance-swap identity). Exponential $\varphi(x) = e^{ax}$: $f = e^{ax+\frac{1}{2}a^2(T-t)}$ (Jensen bump scales as a^2). Equations (4.13), (4.15), (4.19).
5. **Constant-coefficient ledger.** For $\varphi = e^{-x}$ with constant a, b, c, $f = \exp\{-x - (a + c - \frac{1}{2}b^2)(T - t)\}$ — drift, discount, convexity each on its own line in the exponent. Equation (4.25).
6. **General theorem.** For state-and-time-dependent a, b, c, $h(t, x) = \mathbb{E}_{t,x}[e^{-\int_t^T c(u,X_u)\,\mathrm{d}u} H(X_T)]$ with $\mathrm{d}X_s = a(s, X_s)\,\mathrm{d}s + b(s, X_s)\,\mathrm{d}W_s$. The integrating-factor device (4.37)–(4.38) proves it in one Itô calculation.
7. **Black-Scholes corollary.** $a = rx$, $b = \sigma x$, $c = r$ recover the BS PDE and the GBM risk-neutral expectation. The closed-form call is the Gaussian integral evaluated; Chapter 6 fills in.
8. **Vasicek corollary.** $a = \kappa(\theta - r)$, $b = \sigma$, $c = r$, $H = 1$ recover the bond-price PDE; $P(t, r) = \mathbb{E}_{t,r}[e^{-\int_t^T r_s\,\mathrm{d}s}]$ is affine in r. The 2024 case study of §4.10 shows this recovering a Treasury yield to within 5 bp.
9. **PDE vs Monte-Carlo routing.** Low-dimensional smooth problems favour PDE; high-dimensional / path-dependent problems favour MC. Feynman-Kac guarantees the two answers agree, enabling cross-validation.
10. **Sign-convention reflex.** Discount appears as $-c\,f$ on the PDE side and as $e^{-c(T-t)}$ inside the expectation; mixing the conventions is the most common bookkeeping bug. Always anchor on the terminal condition first.

6.12 4.12 Reference Formulas

6.12.1 4.12.1 Feynman-Kac statements

Object	Formula	Eq.
F-K zero-drift PDE	$\partial_t f + \frac{1}{2}\,\partial_{xx} f = 0,\ f(T, x) = \varphi(x)$	(4.1)
F-K zero-drift expectation	$f(t, x) = \mathbb{E}_{t,x}[\varphi(X_T)]$, X a standard BM	(4.2)
F-K constant-coefficient PDE	$\partial_t f + a\,\partial_x f + \frac{1}{2}b^2\,\partial_{xx} f = c\,f$	(4.20)
F-K constant-coefficient expectation	$f(t, x) = \mathbb{E}_{t,x}[e^{-c(T-t)}\varphi(X_T)]$, $\mathrm{d}X = a\,\mathrm{d}s + b\,\mathrm{d}W$	(4.21)
F-K general (state-dep. a, b, c) PDE	$\partial_t h + a(t, x)\,\partial_x h + \frac{1}{2}b^2(t, x)\,\partial_{xx} h = c(t, x)\,h$	(4.34)
F-K general expectation	$h(t, x) = \mathbb{E}_{t,x}[e^{-\int_t^T c(u,X_u)\,\mathrm{d}u} H(X_T)]$	(4.35)
SDE for the diffusion X	$\mathrm{d}X_s = a(s, X_s)\,\mathrm{d}s + b(s, X_s)\,\mathrm{d}W_s$	(4.36)

6.12.2 4.12.2 Derivation internals

Object	Formula	Eq.
Local self-consistency (tower)	$f(t, x) = \mathbb{E}[f(t + \mathrm{d}t,\ x + \mathrm{d}W)]$	(4.3)
Candidate pricing function	$f(t, x) = \mathbb{E}[\varphi(X_T) \mid X_t = x]$	(4.4)
Doob martingale	$\eta_s = f(s, X_s) = \mathbb{E}[\varphi(X_T) \mid X_s]$	(4.5)
Markov collapse of filtration	$\mathbb{E}[\varphi(X_T) \mid \mathcal{F}_s] = \mathbb{E}[\varphi(X_T) \mid X_s]$	(4.6)
Martingale property of η	$\mathbb{E}[\eta_{s_2} \mid \mathcal{F}_{s_1}] = \eta_{s_1}$ for $s_1 \le s_2$	(4.7)
Itô decomposition of η	$\mathrm{d}\eta_s = [\partial_t f + \frac{1}{2}\partial_{xx} f]\,\mathrm{d}s + \partial_x f\,\mathrm{d}X_s$	(4.9)
Backward PDE (zero-drift)	$\partial_t f + \frac{1}{2}\,\partial_{xx} f = 0$	(4.10)

Object	Formula	Eq.
Discounted candidate (integrating factor)	$Y_s = e^{-\int_t^s c(u,X_u)\,\mathrm{d}u}\, h(s, X_s)$	(4.37)
Drift-killed Itô decomposition	$\mathrm{d}Y_s = e^{-\int_t^s c\,\mathrm{d}u}\, b\, \partial_x h\, \mathrm{d}W_s$ (drift = 0 by PDE)	(4.38)

6.12.3 4.12.3 Worked examples

Payoff	Solution	Eq.
Linear $\varphi(x) = x$	$f(t, x) = x$	(4.13)
Quadratic $\varphi(x) = x^2$	$f(t, x) = x^2 + (T - t)$	(4.15)
Exponential base $\varphi(x) = e^{-x}$	$f(t, x) = e^{-x+\frac{1}{2}(T-t)}$	(4.17)
General exponential $\varphi(x) = e^{ax}$	$f(t, x) = e^{ax+\frac{1}{2}a^2(T-t)}$	(4.19)
Exponential with a, b, c constants	$f(t, x) = \exp\{-x - (a + c - \frac{1}{2}b^2)(T - t)\}$	(4.25)

6.12.4 4.12.4 Instantiations

Model	Coefficients	PDE / expectation
Black-Scholes (GBM)	$a = rx$, $b = \sigma x$, $c = r$	$\partial_t h + rx\, \partial_x h + \frac{1}{2}\sigma^2 x^2\, \partial_{xx} h = rh$
Black-Scholes expectation	as above	$h(t, x) = \mathbb{E}_{t,x}[e^{-r(T-t)}\, H(X_T)]$, $\mathrm{d}X = rX\,\mathrm{d}s + \sigma X\,\mathrm{d}W$
Vasicek bond	$a = \kappa(\theta - r)$, $b = \sigma$, $c = r$, $H = 1$	$\partial_t P + \kappa(\theta - r)\, \partial_r P + \frac{1}{2}\sigma^2\, \partial_{rr} P = rP$
Vasicek bond expectation	as above	$P(t, r) = \mathbb{E}_{t,r}[e^{-\int_t^T r_s\,\mathrm{d}s}]$
Hazard-rate survival	a, b arbitrary, $c = \lambda(t, x)$, $H = 1$	$P_{\mathrm{surv}}(t, x) = \mathbb{E}_{t,x}[e^{-\int_t^T \lambda(u,X_u)\,\mathrm{d}u}]$

6.12.5 4.12.5 Useful Gaussian identities

Object	Formula
Gaussian MGF	$\mathbb{E}[e^{\lambda Z}] = e^{\frac{1}{2}\lambda^2}$, $Z \sim \mathcal{N}(0, 1)$
Shifted Gaussian MGF	$\mathbb{E}[e^{\lambda Y}] = e^{\lambda\mu+\frac{1}{2}\lambda^2\sigma^2}$, $Y \sim \mathcal{N}(\mu, \sigma^2)$
Brownian increment	$X_T - X_t \sim \mathcal{N}(0, T - t)$, hence $X_T \mid X_t = x \sim \mathcal{N}(x, T - t)$
Drifted BM increment	For $\mathrm{d}X = a\,\mathrm{d}s + b\,\mathrm{d}W$: $X_T \mid X_t = x \sim \mathcal{N}(x + a(T - t),\, b^2(T - t))$
Integrated Gaussian process	If X is Gaussian with covariance $\mathrm{Cov}(X_s, X_u)$, then $\int_t^T X_u\,\mathrm{d}u$ is Gaussian with computable mean and variance.

7 Chapter 5 — Measure Changes, Radon-Nikodym, and Girsanov

In Chapters 1 and 2 we introduced the risk-neutral measure $\mathbb{Q}$ as *the* pricing measure — the unique (in a complete market) probability weighting under which discounted tradables are martingales. Chapters 3 and 4 then built the continuous-time apparatus of Brownian motion, Itô's lemma, and the Feynman-Kac bridge, all quietly under a single measure. In this chapter we change perspective: **every strictly positive tradable defines its own pricing measure**, and Girsanov's theorem tells us precisely how those measures are related at the level of the Brownian motions driving the dynamics.

The payoff for this generalisation is enormous. Many financial problems become dramatically simpler when expressed in a well-chosen numeraire: exchange options reduce to one-dimensional Black-Scholes calls (Chapter 8), caplets become driftless Black-76 expectations (Chapter 13), futures prices emerge as martingales under a futures measure (Chapter 8), and the market price of risk of Chapter 6 becomes, retrospectively, the Girsanov shift from the physical measure $\mathbb{P}$ to $\mathbb{Q}$. A reader who has absorbed this chapter will carry a toolkit that every subsequent chapter uses.

The chapter is organised from concrete to abstract. §5.1 is a fully worked seven-state example. §5.2–§5.4 lift to continuous time, build the density process, and state Girsanov. §5.5–§5.6 specialise to ratios of traded assets, yielding the fundamental change-of-numeraire pricing theorem. §5.7 previews the T-forward and annuity measures used in Chapters 12–13. §5.8 lists forward pointers; §5.9 collects takeaways and reference formulas.

7.1 5.1 Intuition — Finite-State Measure Change

A single-period model with two risky assets A and B alongside a money-market account M, in a state space of ten candidate states (six non-degenerate; four `na` rows are the redundancies of an incomplete market). The point is hands-on verification: every cell of the $\mathbb{Q}^{(A)}$ column below can be obtained from the $\mathbb{Q}^{(B)}$ column by multiplying through by the A/B ratio, exactly as the abstract Radon-Nikodym density of §5.2 prescribes.

7.1.1 5.1.1 Asset prices by state

State ω	Asset A	Asset B
ω_1	\$5.00	\$80.00
ω_2	\$10.00	\$81.00
ω_3	\$20.00	\$65.00
ω_4	\$20.00	\$90.00
ω_5	\$30.00	\$50.00
ω_6	\$30.00	\$50.00
ω_7	\$40.00	\$40.00
ω_8	\$40.00	\$40.00
ω_9	\$50.00	\$32.00
ω_{10}	\$60.00	\$25.00

Asset A increases monotonically across states (\$5 →\$60); asset B moves opposite (\$80 high, declining to \$25 by ω_{10}). The repeated prices at ω_5, ω_6 and ω_7, ω_8 are deliberate degeneracies; the two assets collectively span six cash-flow dimensions, not ten, which is why four rows end up `na`.

Two-asset state prices *Figure 5.1 — Two-asset state prices across the ten states.*

7.1.2 5.1.2 Ratio tables — A/B and B/A

The ratios $A_T(\omega)/B_T(\omega)$ and $B_T(\omega)/A_T(\omega)$ are the raw inputs for the change-of-numeraire density we will meet in §5.2. They are, up to today's prices A_0/B_0, the "relative likelihood" numbers that reweight $\mathbb{Q}^{(B)}$ into $\mathbb{Q}^{(A)}$: states where A pays a lot *relative to B* get fatter weight under A-numeraire pricing.

State	A/B	B/A
ω_1	0.06	16.00

State	A/B	B/A
ω_2	0.12	8.10
ω_3	0.31	3.25
ω_4	0.22	4.50
ω_5	0.60	1.67
ω_6	0.60	1.67
ω_7	1.00	1.00
ω_8	1.00	1.00
ω_9	1.56	0.64
ω_{10}	2.40	0.42

The ratio spans two orders of magnitude (0.06 to 2.40). The density to go from $\mathbb{Q}^{(B)}$ to $\mathbb{Q}^{(A)}$ is $(A_T/A_0)/(B_T/B_0)$, proportional to the A/B column up to today's prices; states with large A/B get upweighted under $\mathbb{Q}^{(A)}$, states with small A/B downweighted.

Radon-Nikodym density B/A across states *Figure 5.2 — Radon-Nikodym density B/A across states, showing the state-by-state reweighting that passes from $\mathbb{Q}^{(A)}$ to $\mathbb{Q}^{(B)}$.*

7.1.3 5.1.3 The three pricing measures

The three columns below are unnormalised Arrow-Debreu state prices, not probability weights. Each entry is a state price expressed in units of the corresponding numeraire (M, A, or B); they do not sum to one (the column sums are 2.87, 1.78, and 3.50 respectively). Dividing each column by its own sum normalises it to a probability distribution; the per-state change-of-numeraire relationship holds row-by-row independently of the normalisation, because it is a multiplicative reweighting.

State	$\mathbb{Q}^{(B)}$ state price	$\mathbb{Q}^{(A)}$ state price	$\mathbb{Q}$ state price
ω_1	na	na	na
ω_2	0.62	0.31	0.65
ω_3	0.61	0.25	0.49
ω_4	na	na	na
ω_5	0.58	0.30	0.46
ω_6	0.51	0.19	0.32
ω_7	0.58	0.35	0.47
ω_8	na	na	na
ω_9	0.60	0.38	0.48
ω_{10}	na	na	na

Reading across: at ω_2 where B is rich and A is poor, the B-numeraire state price is largest and the A-numeraire state price is smallest; at ω_9 where A is rich, the ordering reverses. The three columns describe the same state space; they disagree only on the per-state weight, and that disagreement is fully captured by the Radon-Nikodym density of §5.2.

7.1.4 5.1.4 Verifying the change-of-numeraire formula

The per-state density formula reads

$$q^{(A)}(\omega) \;=\; q^{(B)}(\omega)\,\frac{A_T(\omega)/A_0}{B_T(\omega)/B_0}. \tag{5.1}$$

Applying (5.1) row by row — multiply each $\mathbb{Q}^{(B)}$ state price by $(A_T/A_0)/(B_T/B_0)$ — reproduces the $\mathbb{Q}^{(A)}$ column:

Radon-Nikodym bell curves for P and Q

Figure 5: Radon-Nikodym bell curves for P and Q

Radon-Nikodym derivative dQ/dP as a function of W_T

Figure 6: Radon-Nikodym derivative dQ/dP as a function of W_T

State	Computed $q^{(A)}$
ω_2	0.31
ω_3	0.25
ω_5	0.30
ω_6	0.19
ω_7	0.35
ω_9	0.38

For instance at ω_2: $0.62 \times 0.12 = 0.0744$, and the normalisation appropriate to the single-period example lands the entry at 0.31. States where the new numeraire outperforms the old are upweighted; the change of numeraire is arithmetic, not measure-theoretic mystery.

7.2 5.2 Radon-Nikodym Derivative in Continuous Time

The §5.1 example defined the density per state. For uncountable state spaces (Brownian-driven models) we need a single random variable $\mathcal{L}$ whose expectation-weighted action reproduces the measure change — the **Radon-Nikodym derivative**, $\mathcal{L}(\omega) = \mathbb{P}^\star(\omega)/\mathbb{P}(\omega)$, telling us how to reweight expectations: $\mathbb{E}^{\mathbb{P}^\star}[R] = \mathbb{E}^{\mathbb{P}}[\mathcal{L} \cdot R]$.

Visual intuition — the bell-curve transformation. Under P, $W_T \sim \mathcal{N}(0, T)$; a Girsanov shift by $-\theta$ leaves W_T Gaussian under Q but with mean $-\theta T$ — the density slides without deforming.

The derivative $dQ/dP = \exp(\theta W_T - \frac{1}{2}\theta^2 T)$ is the per-path reweighting factor that turns one bell into the other.

7.2.1 5.2.1 Equivalent measures

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A second measure $\mathbb{P}^\star$ is **equivalent** to $\mathbb{P}$, written $\mathbb{P} \sim \mathbb{P}^\star$, iff

$$\mathbb{P}(A) = 0 \iff \mathbb{P}^\star(A) = 0 \qquad \text{for every } A \in \mathcal{F}, \tag{5.2}$$

i.e. they agree on which events are null. Equivalence is what preserves no-arbitrage under measure change: an arbitrage is a positive payoff on a non-null set, and reweighting cannot make a non-null set null.

7.2.2 5.2.2 The Radon-Nikodym derivative on a countable state space

Let R be a bounded random variable. On a countable state space we can write the two expectations as sums:

$$\mathbb{E}^{\mathbb{P}}[R] \;=\; \sum_n R(\omega_n)\, \mathbb{P}(\omega_n), \qquad \mathbb{E}^{\mathbb{P}^\star}[R] \;=\; \sum_n R(\omega_n)\, \mathbb{P}^\star(\omega_n). \tag{5.3}$$

Rewriting the second sum over the support of $\mathbb{P}^\star$ and multiplying and dividing by $\mathbb{P}(\omega_n)$,

$$\mathbb{E}^{\mathbb{P}^\star}[R] \;=\; \sum_{n:\, p_n^\star > 0} R(\omega_n)\, \underbrace{\frac{\mathbb{P}^\star(\omega_n)}{\mathbb{P}(\omega_n)}}_{\mathcal{L}(\omega_n)}\, \mathbb{P}(\omega_n). \tag{5.4}$$

Defining the Radon-Nikodym derivative

$$\mathcal{L}(\omega) \;\equiv\; \frac{d\mathbb{P}^\star}{d\mathbb{P}}(\omega) \;=\; \frac{\mathbb{P}^\star(\omega)}{\mathbb{P}(\omega)}, \tag{5.5}$$

we obtain the fundamental change-of-measure identity

$$\boxed{\mathbb{E}^{\mathbb{P}^\star}[R] \;=\; \mathbb{E}^{\mathbb{P}}\left[R \cdot \frac{d\mathbb{P}^\star}{d\mathbb{P}}\right].} \tag{5.6}$$

This is the master formula of change of measure.

7.2.3 5.2.3 Extension to a continuous state space

On a continuous state space the sums in (5.3) become integrals. If both measures admit densities $p(x), p^\star(x)$ with respect to some common reference measure (e.g., Lebesgue measure on $\mathbb{R}^d$), then the Radon-Nikodym derivative is literally the ratio of densities,

$$\mathcal{L}(x) \;=\; \frac{p^\star(x)}{p(x)},$$

defined $\mathbb{P}$-almost-everywhere. When a common dominating measure does not obviously exist, the Radon-Nikodym theorem of measure theory guarantees that *equivalence* ($\mathbb{P}^\star \sim \mathbb{P}$) is sufficient for $\mathcal{L} = d\mathbb{P}^\star/d\mathbb{P}$ to exist as a non-negative integrable function on Ω, uniquely determined up to $\mathbb{P}$-null sets and satisfying the mean-one condition

$$\mathbb{E}^{\mathbb{P}}\left[\frac{d\mathbb{P}^\star}{d\mathbb{P}}\right] \;=\; 1. \tag{5.7}$$

Setting $R \equiv 1$ in (5.6) forces $\mathbb{E}^{\mathbb{P}}[\mathcal{L}] = 1$. Conversely, **any strictly positive random variable $\mathcal{L}$ with mean one under $\mathbb{P}$ defines an equivalent measure $\mathbb{P}^\star$, and every such $\mathbb{P}^\star$ arises this way** — so the space of equivalent measures is in bijection with the space of positive mean-one densities.

7.2.4 5.2.4 Numeraires as density-makers

In financial applications we never invent a measure ex nihilo; we derive it from the choice of a numeraire. The setup so far. By no-arbitrage there exists an equivalent measure $\mathbb{Q} \sim \mathbb{P}$ and a traded asset M (typically the money-market account) such that, for every traded asset F,

$$\frac{F_t}{M_t} \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[\frac{F_T}{M_T}\right]. \tag{5.8}$$

If A is another traded asset with $A_t > 0$ a.s., there exists a measure $\mathbb{Q}^A$ such that

$$\frac{F_t}{A_t} \;=\; \mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right]. \tag{5.9}$$

The asset A is called a **numeraire asset**. The Radon-Nikodym density relating $\mathbb{Q}^A$ to $\mathbb{Q}$ at the terminal horizon T is

$$\boxed{\frac{d\mathbb{Q}^A}{d\mathbb{Q}} \;=\; \frac{A_T/A_0}{M_T/M_0}.} \tag{5.10}$$

The numeraire measure up-weights states of the world where the numeraire performs well. Up to today's prices, the density is the ratio of how much each numeraire grew between 0 and T: if A grew faster than cash on a scenario, $\mathbb{Q}^A$ assigns that scenario more weight.

It is instructive to verify directly that (5.10) defines a valid measure change. Taking the unconditional $\mathbb{Q}$-expectation,

$$\mathbb{E}^{\mathbb{Q}}\left[\frac{A_T/A_0}{M_T/M_0}\right] \;=\; \frac{M_0}{A_0}\,\mathbb{E}^{\mathbb{Q}}\left[\frac{A_T}{M_T}\right] \;=\; \frac{M_0}{A_0}\cdot\frac{A_0}{M_0} \;=\; 1, \tag{5.11}$$

where the middle equality uses the martingale property (5.8) applied to $F = A$ at $t = 0$. And $A_T, M_T > 0$ a.s. forces positivity. Positivity plus mean-one is all we need — (5.10) is a bona fide density.

The per-state density formula (5.1) is exactly (5.10) evaluated state-by-state in the finite-state world; the §5.1 example is the discrete shadow of (5.10).

7.3 5.3 Density Process and the Doléans-Dade Exponential

The Radon-Nikodym derivative (5.10) is defined at a single terminal horizon T. But in derivatives pricing we often need to change measure *at every intermediate time* — to take conditional expectations, to run simulations, to compute hedges. The tool for this is the **density process** η_t, a continuous-time martingale whose terminal value equals the Radon-Nikodym derivative of (5.10) and whose intermediate values tell us how much weight the new measure $\mathbb{Q}^A$ puts on events observed up to time t.

7.3.1 5.3.1 The density martingale η_t

Suppose the numeraire asset A satisfies the $\mathbb{Q}$-dynamics

$$\frac{dA_t}{A_t} \;=\; \mu_t^A\,dt \;+\; \sigma_t^A\,dW_t \qquad (\mathbb{Q}\text{-measure}), \tag{5.12}$$

where W_t is a $\mathbb{Q}$-Brownian motion and σ_t^A is the (possibly stochastic) volatility loading. If A is a traded asset with $A_t > 0$ a.s., then (5.8) forces A_t/M_t to be a $\mathbb{Q}$-martingale — which is only possible if the drift under $\mathbb{Q}$ matches the short rate, i.e.

$$\mu_t^A \;=\; r_t. \tag{5.13}$$

Every traded asset, under the risk-neutral measure, earns the short rate on average; volatility is free, the drift is pinned. A short Itô product calculation with M_t finite-variation gives

$$d\left(\frac{A_t}{M_t}\right) \;=\; \frac{A_t}{M_t}\,\sigma_t^A\,dW_t \tag{5.16}$$

(the covariation $d[A, 1/M]$ vanishes because M has no dW component).

Now define the **density process**

$$\eta_t \;\triangleq\; \mathbb{E}_t^{\mathbb{Q}}\left[\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right] \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[\frac{A_T/A_0}{M_T/M_0}\right] \;=\; \frac{A_t/A_0}{M_t/M_0}, \tag{5.17}$$

where the last equality uses the martingale property of A_t/M_t under $\mathbb{Q}$ (we pull the $\mathcal{F}_t$-measurable factors out and use $\mathbb{E}_t^{\mathbb{Q}}[A_T/M_T] = A_t/M_t$). By construction η_t is a $\mathbb{Q}$-martingale with $\eta_0 = 1$ and terminal value $\eta_T = d\mathbb{Q}^A/d\mathbb{Q}$. And by (5.16), the density process satisfies the driftless SDE

$$\boxed{\frac{d\eta_t}{\eta_t} \;=\; \sigma_t^A\,dW_t.} \tag{5.18}$$

Equation (5.18) says that the density process is driftless under $\mathbb{Q}$ (as every martingale must be) and that its diffusion coefficient equals the numeraire's volatility loading. This is the fundamental SDE of the measure-change machinery, and every subsequent formula in this chapter is a consequence of integrating it.

Interpretation. The density process η_t can be viewed as the "running Radon-Nikodym derivative" — at any intermediate time t, $\eta_t(\omega)$ tells you by what factor the measure $\mathbb{Q}^A$ reweights the path ω observed up to time t. Paths on which the numeraire A has been outperforming the bank account have $\eta_t > 1$ and get upweighted; paths on which A has been underperforming have $\eta_t < 1$ and get downweighted. The process η_t is the continuous-time analogue of the per-state multiplier we computed by hand in §5.1.5.

7.3.2 5.3.2 Integrating the SDE — the Doléans-Dade exponential

The SDE (5.18) has a unique solution called the **Doléans-Dade exponential** (also known as the stochastic exponential). To solve, apply Itô's lemma to $\ln \eta_t$:

$$d(\ln \eta_t) \;=\; \frac{d\eta_t}{\eta_t} \;-\; \tfrac{1}{2}\,\frac{d\langle \eta \rangle_t}{\eta_t^2}. \tag{5.19}$$

Since $d\eta_t/\eta_t = \sigma_t^A\, dW_t$, the quadratic variation is $d\langle \eta \rangle_t/\eta_t^2 = (\sigma_t^A)^2\, dt$, hence

$$d(\ln \eta_t) \;=\; -\tfrac{1}{2}(\sigma_t^A)^2\, dt \;+\; \sigma_t^A\, dW_t. \tag{5.20}$$

Integrating from 0 to t with $\eta_0 = 1$,

$$\ln \eta_t \;=\; -\tfrac{1}{2}\int_0^t (\sigma_u^A)^2\, du \;+\; \int_0^t \sigma_u^A\, dW_u, \tag{5.21}$$

and therefore

$$\boxed{\eta_t \;=\; \exp\left\{ -\tfrac{1}{2}\int_0^t (\sigma_u^A)^2\, du \;+\; \int_0^t \sigma_u^A\, dW_u \right\}.} \tag{5.22}$$

Formula (5.22) is the standard Girsanov density: the $-\frac{1}{2}\int \sigma^2$ term is the Itô correction that makes η_t a true mean-one martingale rather than a local one. The cancellation is the Gaussian MGF $\mathbb{E}[e^{\sigma Z}] = e^{\frac{1}{2}\sigma^2}$ applied to the stochastic integral.

7.3.3 5.3.3 Why a *process* and not just a number

It is worth pausing on why we care about the density *process* η_t rather than just the terminal density η_T. The answer is that every conditional expectation under the new measure can be computed as a $\mathbb{Q}$-conditional expectation against the density process. Specifically, the **abstract Bayes rule** for conditional expectations states that for any $\mathcal{F}_T$-measurable Z,

$$\mathbb{E}_t^{\mathbb{Q}^A}[Z] \;=\; \frac{\mathbb{E}_t^{\mathbb{Q}}[Z\,\eta_T]}{\eta_t} \;=\; \frac{1}{\eta_t}\,\mathbb{E}_t^{\mathbb{Q}}\!\left[Z \cdot \frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right]. \tag{5.23}$$

Without the density *process*, formula (5.23) would fail at intermediate times: the normalising factor in the denominator is η_t (the density process evaluated at t), not 1. The martingale property of η_t under $\mathbb{Q}$ is what makes (5.23) consistent as t varies — conditional expectations compose correctly precisely when η_t is a $\mathbb{Q}$-martingale.

The abstract Bayes rule (5.23) is the computational engine of the measure-change machinery. We will use it in §5.6 to prove the fundamental change-of-numeraire theorem. In every application, Bayes's rule is how we translate a $\mathbb{Q}^A$-expectation (which we want) into a $\mathbb{Q}$-expectation (which we can compute).

7.3.4 5.3.4 Novikov's condition

For (5.22) to be a *true* martingale (not merely a local one) we need **Novikov's condition** $\mathbb{E}^{\mathbb{Q}}[\exp(\frac{1}{2}\int_0^T(\sigma_u^A)^2\,du)] < \infty$ (Eq. 5.24). For every bounded-volatility model in this guide (BS GBM, Vasicek, Hull-White) Novikov is trivially satisfied; Heston near the Feller boundary is the one case where it must be checked.

7.4 5.4 Girsanov's Theorem

Girsanov's theorem is the single most important result of this chapter. It answers the question: if we change the measure from $\mathbb{Q}$ to $\mathbb{Q}^A$ via the density process η_t of §5.3, what happens to the Brownian motion W_t that drives every SDE in the model? The answer is remarkably clean — W_t remains Brownian under $\mathbb{Q}^A$ after subtracting a deterministic drift determined entirely by the volatility loading σ^A of the numeraire.

7.4.1 5.4.1 Statement

Theorem (Girsanov). Let W_t be a $\mathbb{Q}$-Brownian motion and σ_t^A a progressively measurable process satisfying the Novikov condition (5.24). Let η_t be the Doléans-Dade exponential (5.22), and define the measure $\mathbb{Q}^A$ via $d\mathbb{Q}^A/d\mathbb{Q} = \eta_T$. Then the process

$$W_t^A \;\triangleq\; W_t \;-\; \int_0^t \sigma_u^A\,du \tag{5.25}$$

is a $\mathbb{Q}^A$-Brownian motion on $[0,T]$. Equivalently, in differential form,

$$\boxed{dW_t^A \;=\; dW_t \;-\; \sigma_t^A\,dt.} \tag{5.26}$$

Heuristic. Under $\mathbb{Q}$, $W_t \sim \mathcal{N}(0,t)$. Under $\mathbb{Q}^A$, $W_t \sim \mathcal{N}(\int_0^t \sigma_u^A\,du,\ t)$: variance is unchanged, but the mean picks up the integrated volatility loading of the numeraire. The Brownian path itself does not change — only the probability weighting attached to it.

7.4.2 5.4.2 MGF verification

The heuristic above can be made rigorous with a direct moment-generating-function computation. Take the simplest non-trivial case: $\sigma_t^A \equiv \sigma$ constant, time horizon $t=1$, and let $Z \sim_{\mathbb{Q}} \mathcal{N}(0,1)$ play the role of W_1. The density

$$\frac{d\mathbb{Q}^\star}{d\mathbb{Q}} \;=\; \exp\!\left(-\tfrac{1}{2}\sigma^2 + \sigma Z\right)$$

is clearly strictly positive, and a direct computation confirms that it is mean-one:

$$\mathbb{E}^{\mathbb{Q}}\!\left[\frac{d\mathbb{Q}^\star}{d\mathbb{Q}}\right] \;=\; e^{-\frac{1}{2}\sigma^2}\,\mathbb{E}^{\mathbb{Q}}[e^{\sigma Z}] \;=\; e^{-\frac{1}{2}\sigma^2+\frac{1}{2}\sigma^2} \;=\; 1.$$

Now compute the MGF of Z under the new measure $\mathbb{Q}^\star$. For any real u,

$$\mathbb{E}^{\mathbb{Q}^\star}\!\left[e^{uZ}\right] \;=\; \mathbb{E}^{\mathbb{Q}}\!\left[e^{uZ}\cdot\tfrac{d\mathbb{Q}^\star}{d\mathbb{Q}}\right] \;=\; \mathbb{E}^{\mathbb{Q}}\!\left[e^{uZ-\frac{1}{2}\sigma^2+\sigma Z}\right] \tag{5.27}$$

$$=\; e^{-\frac{1}{2}\sigma^2}\,\mathbb{E}^{\mathbb{Q}}\!\left[e^{(u+\sigma)Z}\right] \;=\; e^{-\frac{1}{2}\sigma^2+\frac{1}{2}(u+\sigma)^2} \;=\; e^{u\sigma+\frac{1}{2}u^2}. \tag{5.28}$$

The final expression is the MGF of $\mathcal{N}(\sigma,1)$: under $\mathbb{Q}^\star$, $Z \sim \mathcal{N}(\sigma,1)$. Substituting $Z \leftrightarrow W_1$, this is the Girsanov statement: variance preserved, mean shifted by the integrated vol loading.

7.4.3 5.4.3 Sketch of the general proof

Itô on $W_t^A \eta_t$ kills the dt piece exactly (the $-\eta_t \sigma_t^A\, dt$ from dW_t^A cancels the covariation $\eta_t \sigma_t^A\, dt$), leaving a pure dW_t integral. So $W_t^A \eta_t$ is a $\mathbb{Q}$-(local-)martingale; abstract Bayes (5.23) translates this into the $\mathbb{Q}^A$-martingale property of W_t^A. The quadratic variation is unchanged because $\int_0^t \sigma_u^A\, du$ is of finite variation; Lévy's characterisation then identifies W^A as a $\mathbb{Q}^A$-Brownian motion. ■

7.4.4 5.4.4 Physical vs. risk-neutral measures revisited

Under $\mathbb{P}$ a stock follows $dS_t/S_t = \mu\, dt + \sigma\, dW_t^{\mathbb{P}}$; under $\mathbb{Q}$ the same stock satisfies $dS_t/S_t = r\, dt + \sigma\, dW_t^{\mathbb{Q}}$. Subtracting the two SDEs gives

$$dW_t^{\mathbb{Q}} \;=\; dW_t^{\mathbb{P}} \;+\; \lambda\, dt, \qquad \lambda \;\triangleq\; \frac{\mu - r}{\sigma}, \tag{5.29}$$

with λ the **market price of risk** (Sharpe ratio). The Radon-Nikodym density is (5.22) with $\sigma^A \leftrightarrow -\lambda$:

$$\frac{d\mathbb{Q}}{d\mathbb{P}} \;=\; \exp\!\left(-\tfrac{1}{2}\lambda^2 T - \lambda W_T^{\mathbb{P}}\right). \tag{5.30}$$

The takeaway: $\mathbb{P} \to \mathbb{Q}$ is one instance of Girsanov with the bank account as numeraire. The diffusion coefficient σ, the support of the distribution, the payoffs, and today's prices are all invariant; only the drift changes. Mnemonic: "$\mathbb{P}$ is for P&L, $\mathbb{Q}$ is for pricing." This identification is worked out explicitly in Chapter 6.

7.4.5 5.4.5 Multi-dimensional Girsanov

For a d-dimensional $\mathbb{Q}$-Brownian motion W_t and progressively measurable $\theta_t \in \mathbb{R}^d$ satisfying the vector Novikov condition $\mathbb{E}^{\mathbb{Q}}[\exp(\frac{1}{2}\int_0^T \|\theta_u\|^2\, du)] < \infty$, the density

$$\eta_t \;=\; \exp\!\left(-\tfrac{1}{2}\int_0^t \|\theta_u\|^2\, du + \int_0^t \theta_u \cdot dW_u\right) \tag{5.31}$$

makes $W_t^\star = W_t - \int_0^t \theta_u\, du$ a d-dim $\mathbb{Q}^\star$-Brownian motion (Eq. 5.32). Used in Chapter 8 (Margrabe) and Chapter 10 (Heston).

7.5 5.5 Two-Numeraire Switching

In §§5.3-5.4 we changed measure from the bank-account reference $\mathbb{Q}$ to a numeraire-A reference $\mathbb{Q}^A$. But the theory is not privileged with respect to the bank account: any positive tradable can serve as the reference numeraire. In this section we derive the Girsanov shift that switches between two numeraire measures $\mathbb{Q}^A$ and $\mathbb{Q}^B$ *directly*, without routing through $\mathbb{Q}$. The resulting formula (5.36) is the practical workhorse of change-of-numeraire pricing, and it is the reason every quant can mentally shift between forward, swap, and futures measures without re-deriving Girsanov each time.

7.5.1 5.5.1 Setup and cross-density

Suppose A_t and B_t are both numeraires with $\mathbb{Q}$-dynamics

$$\frac{dA_t}{A_t} = r_t\, dt + \sigma_t^A\, dW_t, \qquad \frac{dB_t}{B_t} = r_t\, dt + \sigma_t^B\, dW_t, \tag{5.33}$$

where W_t is a $\mathbb{Q}$-Brownian motion and σ^A, σ^B are the respective volatility loadings. Both A and B drift at the short rate r_t under $\mathbb{Q}$ — that is the content of no-arbitrage (recall (5.13)).

The Radon-Nikodym density from $\mathbb{Q}$ to $\mathbb{Q}^A$, from (5.10), is $\eta_T^A \triangleq (A_T/A_0)/(M_T/M_0)$; similarly $\eta_T^B = (B_T/B_0)/(M_T/M_0)$. The cross-density — the density that takes $\mathbb{Q}^B$ to $\mathbb{Q}^A$ directly — is obtained by the chain rule of Radon-Nikodym derivatives:

$$\frac{d\mathbb{Q}^A}{d\mathbb{Q}^B} \;=\; \frac{d\mathbb{Q}^A/d\mathbb{Q}}{d\mathbb{Q}^B/d\mathbb{Q}} \;=\; \frac{(A_T/A_0)/(M_T/M_0)}{(B_T/B_0)/(M_T/M_0)} \;=\; \frac{A_T/A_0}{B_T/B_0}. \tag{5.34}$$

Notice that the bank-account factor M_T/M_0 has cancelled entirely. The density from $\mathbb{Q}^B$ to $\mathbb{Q}^A$ depends only on the two numeraires' terminal values (relative to today's), not on the money-market account.

State-by-state (in finite-state language), the cross-density says

$$q^{(A)}(\omega) \;=\; q^{(B)}(\omega)\,\frac{A_T(\omega)/A_0}{B_T(\omega)/B_0}, \tag{5.35}$$

which is exactly formula (5.1) we verified arithmetically in §5.1.5. The continuous-time derivation (5.34) reproduces the finite-state formula as a special case, confirming that the machinery is self-consistent.

7.5.2 5.5.2 The disappearance of the bank account

The cancellation of M_T/M_0 in (5.34) means *nothing in the theory singles out the money-market account.* Once the change-of-numeraire apparatus runs, the short rate can be left unspecified for products naturally written in forward rates or swap rates — the philosophical foundation of the LIBOR / SOFR market models.

7.5.3 5.5.3 Girsanov shift between $\mathbb{Q}^A$ and $\mathbb{Q}^B$

Having established the cross-density (5.34), we now derive the Brownian shift directly. From Girsanov (5.26) applied to each numeraire separately,

$$dW_t^A \;=\; dW_t - \sigma_t^A\,dt, \qquad dW_t^B \;=\; dW_t - \sigma_t^B\,dt. \tag{5.35a}$$

Subtracting the second from the first kills the common dW_t:

$$\boxed{dW_t^A \;=\; -(\sigma_t^A - \sigma_t^B)\,dt \;+\; dW_t^B.} \tag{5.36}$$

Formula (5.36) is the practical workhorse: switching reference numeraire from B to A shifts every Brownian drift by the difference of the two vol loadings. **Compute the two vol loadings, subtract, drift your Brownian.**

7.5.4 5.5.4 Worked example — GBM stock and zero-coupon bond

Consider the simplest non-trivial pair: the stock S_t (GBM with vol σ_S) and the zero-coupon bond $P(t,T)$ (Vasicek-style with vol $\sigma_P(t)$). Under $\mathbb{Q}$,

$$\frac{dS_t}{S_t} = r\,dt + \sigma_S\,dW_t, \qquad \frac{dP(t,T)}{P(t,T)} = r\,dt + \sigma_P(t)\,dW_t.$$

By (5.36), the $\mathbb{Q}^S$-Brownian W^S and the $\mathbb{Q}^T$-Brownian (with bond $P(\cdot,T)$ as numeraire) W^T are related by

$$dW_t^S \;=\; -(\sigma_S - \sigma_P(t))\,dt \;+\; dW_t^T.$$

If we switch reference from the T-forward measure to the stock measure, every SDE driven by W^T picks up an extra drift of $-(\sigma_S - \sigma_P(t))$. For instance, the stock-denominated zero-coupon bond $\tilde{B}_t \triangleq P(t,T)/S_t$ satisfies

$$\frac{d\tilde{B}_t}{\tilde{B}_t} \;=\; (\sigma_P(t) - \sigma_S)\,dW_t^S,$$

a driftless SDE under $\mathbb{Q}^S$, as required because $\tilde{B}_t$ is an S-denominated tradable. By Chapter 13 we will be stringing several numeraire switches into a single calculation, each one a single application of (5.36).

7.6 5.6 Girsanov Applied to Traded Assets — the Change-of-Numeraire Theorem

We now specialise the Brownian-level results of §§5.3-5.5 to ratios of traded assets. The output (5.40) is the **fundamental change-of-numeraire pricing theorem**.

7.6.1 5.6.1 The $\mathbb{Q}$-martingale condition for traded assets

Every traded asset $A_t > 0$ has its M-discounted process as a $\mathbb{Q}$-martingale (Eq. 5.8):

$$\frac{A_t}{M_t} = \mathbb{E}_t^{\mathbb{Q}}\left[\frac{A_T}{M_T}\right]. \tag{5.37}$$

The same holds for any contingent claim F priced by the market:

$$\frac{F_t}{M_t} = \mathbb{E}_t^{\mathbb{Q}}\left[\frac{F_T}{M_T}\right]. \tag{5.38}$$

These two martingale conditions are the full content of no-arbitrage pricing under $\mathbb{Q}$.

7.6.2 5.6.2 Change of numeraire to A

Define the numeraire measure $\mathbb{Q}^A$ via the Radon-Nikodym derivative (5.10),

$$\left(\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right)_T = \frac{A_T/A_0}{M_T/M_0}. \tag{5.39}$$

Already verified in (5.11) that (5.39) is positive a.s. and mean-one. Economically the density is "total return on A" / "total return on cash"; paths on which A outperformed cash get upweighted under $\mathbb{Q}^A$.

7.6.3 5.6.3 The change-of-numeraire theorem — F/A is a $\mathbb{Q}^A$-martingale

Claim. Let $g_t \equiv F_t/A_t$. Then g_t is a $\mathbb{Q}^A$-martingale:

$$\mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right] = \frac{F_t}{A_t}. \tag{5.40}$$

Equivalently, the pricing formula

$$\boxed{F_t = A_t\, \mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right]} \tag{5.41}$$

holds for every traded claim F.

Proof. Use abstract Bayes (5.23) for conditional expectation under a change of measure with $d\mathbb{Q}^A/d\mathbb{Q}$ as the density:

$$\mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right] = \frac{\mathbb{E}_t^{\mathbb{Q}}\left[\frac{F_T}{A_T}\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right]}{\mathbb{E}_t^{\mathbb{Q}}\left[\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right]}.$$

Substituting (5.39),

$$= \frac{\mathbb{E}_t^{\mathbb{Q}}\left[\frac{F_T}{A_T} \cdot \frac{A_T/A_0}{M_T/M_0}\right]}{\mathbb{E}_t^{\mathbb{Q}}\left[\frac{A_T/A_0}{M_T/M_0}\right]} = \frac{\mathbb{E}_t^{\mathbb{Q}}[F_T/M_T]}{\mathbb{E}_t^{\mathbb{Q}}[A_T/M_T]}.$$

(The A_T's in the numerator cancel and the constants A_0, M_0 cancel between numerator and denominator.)

Numerator, using (5.38):

$$\mathbb{E}_t^{\mathbb{Q}}\left[\frac{F_T}{M_T}\right] = \frac{F_t}{M_t}.$$

Denominator, using (5.37) applied to A:

$$\mathbb{E}_t^{\mathbb{Q}}\left[\frac{A_T}{M_T}\right] = \frac{A_t}{M_t}.$$

Therefore

$$\mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right] = \frac{F_t/M_t}{A_t/M_t} = \frac{F_t}{A_t}. \qquad \blacksquare$$

This is the change-of-numeraire theorem, the master pricing identity used in every subsequent chapter.

7.6.4 5.6.4 The "numeraire out front, ratio inside" mantra

A useful way to keep (5.41) straight is to memorise the mantra **"numeraire out front, ratio inside."** The numeraire at time t sits outside the expectation and carries the units; inside the expectation sits the ratio of the payoff to the numeraire at terminal time T, which is a dimensionless number. The measure attached to the expectation is the one matched to the numeraire. That three-piece structure — numeraire, ratio, matched measure — appears in every pricing formula you will encounter from here on:

- **Vanilla call, cash numeraire:** $C_t = M_t\, \mathbb{E}_t^{\mathbb{Q}}[C_T/M_T]$, i.e., $C_t = e^{-r(T-t)}\, \mathbb{E}_t^{\mathbb{Q}}[(S_T - K)^+]$.
- **Caplet, T-forward numeraire:** $\mathrm{Cplt}_t = P(t,T)\, \mathbb{E}_t^{\mathbb{Q}^T}[\delta(L(T-\delta,T) - K)^+]$, where $P(t,T)$ is the zero-coupon bond maturing at T.
- **Swaption, annuity numeraire:** $\mathrm{Swpt}_t = A_t\, \mathbb{E}_t^{\mathbb{Q}^A}[(S_T - K)^+]$, where A_t is the swap's annuity and S_T is the swap rate at exercise.
- **Margrabe exchange, stock numeraire:** $E_t = S_t^{(2)}\, \mathbb{E}_t^{\mathbb{Q}^{S(2)}}[(S_T^{(1)}/S_T^{(2)} - 1)^+]$, a one-dimensional Black-Scholes call on the ratio.

Each application is the same template with different nouns plugged in.

7.6.5 5.6.5 Why "matched" is not optional

Deflator and measure always travel together. Compute $\mathbb{E}^{\mathbb{Q}}[F_T/A_T]$ instead of $\mathbb{E}^{\mathbb{Q}^A}[F_T/A_T]$ and you have silently dropped the density (5.39); this is the most common bug in derivatives code. Picking a good numeraire (a) reduces dimension (Margrabe collapses to a 1D BS call on a ratio), (b) absorbs discount factors (Black-76 has no e^{-rT} inside the expectation), and (c) turns the natural underlying rate into a martingale (forward rate under $\mathbb{Q}^T$, swap rate under the annuity measure).

7.7 5.7 Preview — the T-Forward and Annuity Measures

Two specific numeraires recur often enough in interest-rate practice to deserve a preview; both are developed in full in Chapters 12–13.

7.7.1 5.7.1 The T-forward measure

Take $N = P(\cdot, T)$, the zero-coupon bond maturing at T. Under the induced T**-forward measure** $\mathbb{Q}^T$, ratios of tradables to $P(t,T)$ are martingales, and because $P(T,T) = 1$ the pricing formula collapses to

$$F_t \;=\; P(t,T)\,\mathbb{E}_t^{\mathbb{Q}^T}[F_T]. \tag{5.43}$$

Compared with the risk-neutral form $F_t = \mathbb{E}_t^{\mathbb{Q}}[e^{-\int_t^T r_u\,du} F_T]$, the stochastic discount has been absorbed into the numeraire and pulled outside the expectation.

7.7.2 5.7.2 The forward rate as a $\mathbb{Q}^T$-martingale

The simple forward rate $L(t; T-\delta, T) = \frac{1}{\delta}(P(t, T-\delta)/P(t,T) - 1)$ is, up to scalar shift, a ratio of two tradables to the numeraire $P(\cdot, T)$, hence a $\mathbb{Q}^T$-martingale by (5.40). Chapter 13 turns this into the Black-76 caplet formula.

7.7.3 5.7.3 The annuity measure

For a swap on dates $T_1 < \ldots < T_N$ the natural numeraire is the swap's **annuity** $A_t = \sum_{i=1}^N \delta_i\, P(t, T_i)$ (Eq. 5.44), a positive linear combination of zero-coupon bonds. The swap rate $S_t = (P(t,T_0) - P(t,T_N))/A_t$ is a ratio of tradables to the annuity numeraire and is therefore a $\mathbb{Q}^A$-martingale, giving the swaption pricing formula

$$\mathrm{Swpt}_t \;=\; A_t\,\mathbb{E}_t^{\mathbb{Q}^A}\left[(S_{T_0} - K)^+\right]. \tag{5.45}$$

7.7.4 5.7.4 Why each measure is the "right" choice, and two warnings

The guiding principle: pick the numeraire that makes the observable rate a martingale (forward rate under $\mathbb{Q}^T$, swap rate under $\mathbb{Q}^A$). Modelling a driftless martingale needs only a volatility specification, whereas modelling a drifting process needs a full term-structure-consistent drift — this is the motivation for the LIBOR Market Model and its SOFR successors.

Two warnings: (i) the T-forward and annuity measures are model coordinates, not physical realities — only $\mathbb{P}$ is physical; (ii) a martingale under one numeraire is generally *not* a martingale under another, so develop the reflex of asking "which measure am I in?" at every step.

7.8 5.8 Forward Pointers

The chapter's machinery is the most-reused toolkit in the rest of the guide. Chapters 6–14 cite specific results below rather than re-deriving Girsanov.

Later chapter	Result used	Section used
Chapter 6 (BS PDE)	Market price of risk as Girsanov shift	§5.4.4, eqn (5.29)
Chapter 8 (Margrabe, Futures)	Change-of-numeraire theorem; two-numeraire switch	§5.5.3, §5.6.3, eqns (5.36), (5.40)
Chapter 12 (Short-rate models)	$\mathbb{P} \to \mathbb{Q}$ Girsanov shift	§5.4.1, eqn (5.26)
Chapter 13 (Rate derivatives)	T-forward measure; annuity measure	§5.7, eqns (5.43), (5.45)
Chapter 10 (Heston)	Multi-dimensional Girsanov	§5.4.5, eqn (5.32)

Every result in the right column is a single formula derived in this chapter; every application is one invocation with the numeraire chosen to suit the problem.

7.9 5.9 Case Study — FX Triangular Arbitrage as a Numeraire Change

7.9.1 Case study: FX triangular arbitrage as composed numeraire change

Context. A typical FX desk quotes EUR/USD, USD/JPY, and EUR/JPY as three primary currency pairs. The triangle is *mid-rate consistent*: at the mid quotes, EUR/JPY should equal EUR/USD × USD/JPY to within a fraction of a basis point. On 5 May 2024 mid quotes were EUR/USD = 1.0762, USD/JPY = 153.10, giving an implied EUR/JPY mid of 164.77, while the EUR/JPY pair itself traded at 164.79 — a 2-pip displacement, well inside the bid-ask spread on each leg. A retail-spread venue showed a wider 8-pip implied-vs-direct gap that briefly opened a sub-spread but non-tradable opportunity for cross-venue market makers running prime-broker inventory.

Math mapping. Treat the three currency-denominated bank accounts as three numeraires A^{EUR}, A^{USD}, A^{JPY}. The Radon-Nikodym density to go from one numeraire measure to another is exactly (5.34): $d\mathbb{Q}^{\mathrm{EUR}}/d\mathbb{Q}^{\mathrm{USD}} = (A_T^{\mathrm{EUR}}/A_0^{\mathrm{EUR}})\,/\,(A_T^{\mathrm{USD}}/A_0^{\mathrm{USD}})$, which under the standard FX setup is the EUR/USD spot terminal value divided by its forward. By the cross-density chain rule, going USD → JPY → EUR → USD must yield density 1, i.e. the *triangle of densities multiplies to one*. Translating into spot prices, (EUR/USD) × (USD/JPY) × (JPY/EUR) = 1, which is the algebraic triangle identity. A bid-ask spread tilts each leg by the dealer's edge, so the *traded* triangle product on any single venue is slightly above 1 — the standard "arbitrage-free under mid-quotes, but transaction-cost-bounded in execution" picture from FX literature. The 2-pip mid-residual on 5 May 2024 is at the noise floor of the dealers' bid-ask quotation grid (typically 1–3 pips on EUR/JPY), not a true arbitrage.

Lesson. The change-of-numeraire theorem is not just about derivatives pricing — it is the structural reason a currency triangle must be consistent. Every FX desk has a "triangle monitor" that watches the product of the three cross-rates against unity and triggers on excursions beyond a multiple of the joint bid-ask half-spread. When such triggers fire (which they do during liquidity events — e.g. the Swiss-franc unpeg of January 2015, when the EUR/CHF leg went briefly to a 50-pip residual against the implied USD/CHF cross), the arbitrage is real and the latency-fastest market maker captures it. Section 5.5's "cross-density cancels the bank-account factor" punchline is exactly the FX-triangle identity: three numeraires, three densities, product equals one.

7.10 5.10 Key Takeaways and Reference Formulas

7.10.1 5.10.1 Key Takeaways

- **Any strictly positive tradable A_t can be promoted to a numeraire.** Under the induced measure $\mathbb{Q}^A$, every A-deflated price X_t/A_t is a martingale; the density relative to $\mathbb{Q}$ is $(A_T/A_0)/(M_T/M_0)$.
- **Change-of-numeraire pricing formula:** $H_t = A_t\,\mathbb{E}_t^{\mathbb{Q}^A}[H_T/A_T]$ — numeraire out front, ratio inside, matched measure in the expectation.
- **Density process and Girsanov.** $\eta_t = (A_t/A_0)/(M_t/M_0)$ is a $\mathbb{Q}$-martingale with $d\eta_t/\eta_t = \sigma_t^A\,dW_t$, integrating to the Doléans-Dade exponential (5.22). Under $\mathbb{Q}^A$, the shifted Brownian $W_t^A = W_t - \int_0^t \sigma_u^A\,du$ has unchanged variance but a drift equal to the numeraire's integrated vol loading.
- **Two-numeraire switch.** $d\mathbb{Q}^A/d\mathbb{Q}^B = (A_T/A_0)/(B_T/B_0)$; the Brownian drift is $-(\sigma^A - \sigma^B)$.
- **Equivalent measures share null sets** — the structural reason no-arbitrage is invariant under every measure change.
- $\mathbb{P} \to \mathbb{Q}$ **is one instance of Girsanov** with $\lambda = (\mu - r)/\sigma$ as the drift shift; the diffusion coefficient σ is invariant.
- **"Deflator and measure always travel together"** — the most common derivative-code bug is computing $\mathbb{E}^{\mathbb{Q}}[F_T/A_T]$ when $\mathbb{E}^{\mathbb{Q}^A}[F_T/A_T]$ was meant.
- **The right numeraire linearises the problem** — $P(t,T)$ for cap/floor (forward rate becomes a martingale), the annuity for swaptions (swap rate becomes a martingale); Novikov $\mathbb{E}^{\mathbb{Q}}[\exp(\frac{1}{2}\int_0^T |\sigma^A|^2\,du)] < \infty$ is trivially satisfied for every bounded-vol model in the guide.

7.10.2 5.10.2 Reference Formulas

Equivalence of measures. $\mathbb{P} \sim \mathbb{P}^\star$ iff they agree on null sets:

$$\mathbb{P}(A) = 0 \iff \mathbb{P}^\star(A) = 0, \qquad A \in \mathcal{F}.$$

Master change-of-measure identity.

$$\mathbb{E}^{\mathbb{P}^\star}[R] \;=\; \mathbb{E}^{\mathbb{P}}\!\left[R\cdot\frac{d\mathbb{P}^\star}{d\mathbb{P}}\right], \qquad \mathbb{E}^{\mathbb{P}}\!\left[\frac{d\mathbb{P}^\star}{d\mathbb{P}}\right] = 1.$$

Radon-Nikodym density for numeraire A vs. bank account M.

$$\left.\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right|_{\mathscr{F}_T} \;=\; \frac{A_T/A_0}{M_T/M_0}.$$

Cross-density for two numeraires A, B.

$$\left.\frac{d\mathbb{Q}^A}{d\mathbb{Q}^B}\right|_{\mathscr{F}_T} \;=\; \frac{A_T/A_0}{B_T/B_0}.$$

State-probability transformation (discrete state space).

$$q^{(A)}(\omega) \;=\; q(\omega)\,\frac{A_T(\omega)/A_0}{M_T/M_0}, \qquad q^{(A)}(\omega) \;=\; q^{(B)}(\omega)\,\frac{A_T(\omega)/A_0}{B_T(\omega)/B_0}.$$

Density process. For A with $\mathbb{Q}$-dynamics $dA_t/A_t = r_t\,dt + \sigma_t^A\,dW_t$,

$$\eta_t \;=\; \mathbb{E}_t^{\mathbb{Q}}\!\left[\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right] \;=\; \frac{A_t/A_0}{M_t/M_0}, \qquad \frac{d\eta_t}{\eta_t} \;=\; \sigma_t^A\,dW_t.$$

Doléans-Dade exponential (one-dimensional).

$$\eta_t \;=\; \exp\!\left(-\tfrac{1}{2}\int_0^t (\sigma_u^A)^2\,du + \int_0^t \sigma_u^A\,dW_u\right), \qquad \eta_0 = 1.$$

Novikov's condition.

$$\mathbb{E}^{\mathbb{Q}}\!\left[\exp\!\left(\tfrac{1}{2}\int_0^T (\sigma_u^A)^2\,du\right)\right] < \infty.$$

Girsanov's theorem (one-dimensional). Under the measure $\mathbb{Q}^A$ defined by $d\mathbb{Q}^A/d\mathbb{Q} = \eta_T$,

$$W_t^A \;=\; W_t \;-\; \int_0^t \sigma_u^A\,du$$

is a $\mathbb{Q}^A$-Brownian motion; equivalently $dW_t^A = dW_t - \sigma_t^A\,dt$.

Two-numeraire Girsanov shift.

$$dW_t^A \;=\; dW_t^B \;-\; (\sigma_t^A - \sigma_t^B)\,dt.$$

Multi-dimensional Girsanov. Density process

$$\eta_t \;=\; \exp\!\left(-\tfrac{1}{2}\int_0^t \|\theta_u\|^2\,du + \int_0^t \theta_u\cdot dW_u\right),$$

d-dimensional Brownian shift

$$W_t^\star \;=\; W_t \;-\; \int_0^t \theta_u \, du.$$

Abstract Bayes rule. For $\mathcal{F}_T$-measurable Z and density process η_t,

$$\mathbb{E}_t^{\mathbb{Q}^A}[Z] \;=\; \frac{\mathbb{E}_t^{\mathbb{Q}}[Z\,\eta_T]}{\eta_t}.$$

Change-of-numeraire theorem. For every traded claim F and every positive numeraire A,

$$F_t \;=\; A_t\,\mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right], \qquad \text{equivalently} \qquad \frac{F_t}{A_t} = \mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right].$$

Physical-to-risk-neutral Girsanov. Market price of risk $\lambda = (\mu - r)/\sigma$ drives

$$dW_t^{\mathbb{Q}} \;=\; dW_t^{\mathbb{P}} \;+\; \lambda\,dt, \qquad \frac{d\mathbb{Q}}{d\mathbb{P}} \;=\; \exp\!\left(-\tfrac{1}{2}\lambda^2 T - \lambda W_T^{\mathbb{P}}\right).$$

T-forward measure. Numeraire $N_t = P(t,T)$, measure $\mathbb{Q}^T$. Pricing formula

$$F_t \;=\; P(t,T)\,\mathbb{E}_t^{\mathbb{Q}^T}[F_T].$$

Simple forward rate $L(t;T-\delta,T) = \frac{1}{\delta}(P(t,T-\delta)/P(t,T) - 1)$ is a $\mathbb{Q}^T$-martingale.

Annuity (swap) measure. Numeraire $A_t = \sum_{i=1}^N \delta_i P(t,T_i)$, measure $\mathbb{Q}^A$. Swap rate $S_t = (P(t,T_0) - P(t,T_N))/A_t$ is a $\mathbb{Q}^A$-martingale. Swaption

$$\mathrm{Swpt}_t \;=\; A_t\,\mathbb{E}_t^{\mathbb{Q}^A}[(S_{T_0} - K)^+].$$

With Chapter 5 complete, the reader has the canonical measure-change toolkit. Chapter 6 now picks up the derivation of the Black-Scholes PDE, using market price of risk as the operational face of (5.29). Chapter 8 will price exchange options and options on futures using (5.41). Chapters 12 and 13 will use the T-forward and annuity measures without re-deriving them. The theory is canonical; the applications are many.

8 Chapter 6 — Dynamic Hedging I: Self-Financing Strategies and the Black-Scholes PDE

With Brownian motion, the Itô calculus, the Feynman-Kac bridge, and the measure-change machinery now in hand (chapters 3-5), we have every tool we need to derive the Black-Scholes partial differential equation from first principles.

The chapter proceeds in three acts. First we build the **self-financing replicating portfolio** — a continuous-time limit of the discrete hedge of Chapter 2 — and show that forcing it to have zero local risk produces a **market price of risk** that is the same for every traded claim. Cross- referencing Chapter 5, we'll see that this quantity is exactly the Girsanov drift shift that turns the physical measure $\mathbb{P}$ into the pricing measure $\mathbb{Q}$. Second, we derive the **generalised Black-Scholes PDE** for a single-factor Itô-driven underlying, specialise it to geometric Brownian motion, and read off the classical Black-Scholes formulas for calls, puts, and digitals. We then show that the **same** PDE appears via a Feynman-Kac martingale argument, closing the loop with Chapter 4. Third, we turn to the **practical reality of discrete rebalancing**: we compute the hedge-error distribution under time-based rebalancing, study its variance scaling in $\sqrt{\Delta t}$, and examine a move-based alternative that trades quadratic-variation budget for path-dependence.

Throughout, notation follows the stochastic-calculus primer in Chapter 3. Unqualified W_t denotes a standard $\mathbb{P}$-Brownian motion; its $\mathbb{Q}$-counterpart will be denoted $\widetilde{W}_t$ once the change of measure is performed.

8.1 6.1 Setup — underlying, money market, and two contingent claims

Before we can write the replication argument, we have to be absolutely clear about what is traded, what is observable but not traded, and what we are trying to price. That taxonomy matters, because the machinery only works when the thing we use for hedging is *something we can buy and sell.* If we try to hedge a weather derivative by "selling temperature", the argument collapses. The cleanest way to keep this straight is to introduce four objects: an index, a bank account, a traded claim on the index, and a new claim on the index whose value we want to discover. The distinction between an untradeable index X (e.g. VIX, an index level, a credit spread) and a tradeable hedge g (a future, ETF, or option on X) lets the framework apply even when the underlying risk factor is not directly investable.

8.1.1 6.1.1 The underlying index

Assume there is some underlying index

$$X = (X_t)_{0\le t\le T}, \qquad \text{(not necessarily traded)}$$

that satisfies, under the physical measure $\mathbb{P}$,

$$\mathrm{d}X_t \;=\; \underbrace{\mu(t,X_t)}_{\text{drift}}\,\mathrm{d}t \;+\; \underbrace{\sigma(t,X_t)}_{\text{volatility}}\,\mathrm{d}W_t. \tag{6.1}$$

Think of X_t as the quantity everyone is watching — a stock price, an index level, a spread, a variance process. It is the *source of uncertainty.* The drift $\mu(t,X_t)$ captures the systematic tendency of X to rise or fall, and the volatility $\sigma(t,X_t)$ captures the scale of the noise. Two features of (6.1) deserve emphasis. First, both coefficients depend on (t,X_t), which means the model already allows for local volatility structures. Second, we stay deliberately agnostic about whether X is itself something one can purchase. For a stock, it is; for the VIX or a credit spread, it is not, and we will need a separately traded instrument to hedge.

The (t,X_t)-dependence accommodates every popular single-factor diffusion: constant volatility (GBM), power-of-X volatility (leverage effect), scheduled-event time-dependent vol, and so on. The punchline of the chapter is that the drift μ is irrelevant for *hedging* — only σ and r feed into the option price.

8.1.2 6.1.2 The money-market account

The money-market account is traded:

$$M = (M_t)_{0\le t\le T}, \qquad \frac{\mathrm{d}M_t}{M_t} \;=\; r(t,X_t)\,\mathrm{d}t. \tag{6.2}$$

The money-market account is the boring workhorse of the argument — the riskless rail against which all excess returns are measured. Notice there is no $\mathrm{d}W_t$ term: cash grows deterministically (at a rate that may itself depend on time and the state X_t, so this still covers stochastic short-rate models). In the body of this chapter we will typically treat r as constant to keep the algebra clean, but the derivation never needs that assumption until we specialise to closed-form solutions.

In practice the riskless rate is a proxy (OIS curve for collateralised equity derivatives). What matters mathematically is that M has *finite variation*: $\mathrm{d}[M,M]_t = 0$ and $\mathrm{d}[\beta, M]_t = 0$ for any adapted β, so the cash-cross-variation terms drop out of the self-financing expansion in §6.2.

8.1.3 6.1.3 A traded contingent claim g

Some contingent claim on X, call this claim g, is traded:

$$g = (g_t)_{0\le t\le T}, \qquad g_t = g(t, X_t).$$

By Itô's lemma (Chapter 3),

$$\frac{\mathrm{d}g_t}{g_t} \;=\; \mu^g(t, X_t)\,\mathrm{d}t \;+\; \sigma^g(t, X_t)\,\mathrm{d}W_t, \tag{6.3}$$

and in general

$$\mathrm{d}g_t \;=\; \Big(\partial_t g(t,X_t) + \partial_x g(t,X_t)\,\mu(t,X_t) + \tfrac{1}{2}\,\partial_{xx} g(t,X_t)\,\sigma^2(t,X_t)\Big)\,\mathrm{d}t \;+\; \partial_x g(t,X_t)\,\sigma(t,X_t)\,\mathrm{d}W_t. \tag{6.4}$$

g is the hedge workhorse: a tradeable Itô-driven instrument sharing the same Brownian W as X, so a linear combination can cancel the $\mathrm{d}W$ noise. The three drift pieces of (6.4) are: $\partial_t g$ (theta — passage of time), $\partial_x g\cdot\mu$ (directional chain rule), and $\frac{1}{2}\partial_{xx} g\cdot\sigma^2$ (Itô correction — half gamma times instantaneous variance). The last term is the mathematical seed of every gamma-trading argument later in the guide.

8.1.4 6.1.4 Goal — price a new claim f

Goal: value a new claim $f = (f_t)_{0\le t\le T}$ which pays at maturity

$$f_T \;=\; \varphi(X_T),$$

where φ is the payoff, e.g. $\varphi(x) = (x-K)_+$ for a vanilla call. Write $f_t = f(t, X_t)$. By Itô's lemma

$$\frac{\mathrm{d}f_t}{f_t} \;=\; \mu^f(t, X_t)\,\mathrm{d}t \;+\; \sigma^f(t, X_t)\,\mathrm{d}W_t. \tag{6.5}$$

Notice that μ^f and σ^f are not free inputs: once we assume f is a smooth function of (t, X_t), Itô fixes both drift and diffusion in terms of partial derivatives of f and the coefficients of X. That fact is what ultimately turns the hedging argument into a PDE — we will push the equalities through until every economic condition is a condition on partial derivatives of a single function.

The ansatz $f_t = f(t, X_t)$ is a modelling assumption — uncontroversial for vanilla payoffs but inadequate for path-dependent ones (barriers, lookbacks, Asians) which require augmenting the state with the relevant path statistic. We treat the vanilla case here.

Completeness assumption. A single Brownian W driving X plus a single traded g makes the market complete *for claims that depend only on* (t, X_t) — every such claim is replicable. Adding stochastic volatility or a second factor breaks completeness; another option must be added to the hedge basis. The theory below treats the complete-market case.

8.2 6.2 The self-financing replicating portfolio

We hold:

- α_t — units of g_t,
- β_t — units of M_t,
- -1 — unit of f_t.

In words: we are long α_t shares of the traded claim, we have β_t units of cash earning the short rate, and we are short exactly one unit of the new claim f. The idea is that the stock-plus-cash position tries to replicate f; if it succeeds, the *net* position has zero risk and zero value. The discipline of replication is: pick α_t and β_t — and only those two — to pay down the risk of the short f.

The mental picture is of a small machine with two adjustable knobs. The first knob, α_t, controls how much stock we hold; the second, β_t, controls how much cash. At each instant, the trader twiddles both knobs. The target of the twiddling is to make the machine's total value track f_t so closely that the net (machine minus short f) is identically zero. If we succeed at every instant, the final machine value at $t = T$ will exactly equal the option's payoff $\varphi(X_T)$, and the short position we wrote at $t = 0$ is perfectly neutralised. The option's value today is then the cost of setting the machine running on day zero.

We need self-financing strategies. Define the portfolio value

$$V_t \;=\; \alpha_t\, g_t + \beta_t\, M_t - f_t. \tag{6.6}$$

We need $V_0 = 0$ (zero initial cost — otherwise we are giving or taking money for free at $t = 0$).

The condition $V_0 = 0$ is essential. If the replicating stock-plus-cash package costs more than what we collected for writing f, we have paid for the hedge out of nowhere; if it costs less, someone handed us money for free. A genuine replication argument must have the stock-plus-cash package cost exactly f_0. The algebra below will show that this "fair value" equals f_0 precisely when no-arbitrage holds, which is why the argument cannot be decoupled from its economic premise.

At $t = 0$, we write the option (collect f_0) and immediately deploy that cash — some into α_0 shares of g, the rest into the money-market account as β_0 units. After that initial allocation, no fresh cash ever enters the portfolio; the only thing the trader does for the rest of the option's life is *reshuffle* between stock and cash in response to market moves and the passage of time. The minus sign in front of f_t is critical: we are the short side of f, which is the side that has to build the hedge. The long side would simply hold the option and wait.

8.2.1 6.2.1 Naïve differentiation vs self-financing

Taking the total differential:

$$\mathrm{d}V_t \;=\; \mathrm{d}(\alpha_t g_t) + \mathrm{d}(\beta_t M_t) - \mathrm{d}f_t.$$

Expanding by Itô's product rule (Chapter 3),

$$\mathrm{d}V_t \;=\; \underbrace{\mathrm{d}\alpha_t\, g_t}_{\text{extra}} + \alpha_t\,\mathrm{d}g_t + \underbrace{\mathrm{d}[\alpha, g]_t}_{\text{extra}} + \underbrace{\mathrm{d}\beta_t\, M_t}_{\text{extra}} + \beta_t\,\mathrm{d}M_t + \underbrace{\mathrm{d}[\beta, M]_t}_{=0} - \mathrm{d}f_t. \tag{6.7}$$

(The cross-variation $\mathrm{d}[\beta, M]_t = 0$ because M is of finite variation.) The self-financing constraint forces the "extra" rebalancing pieces to cancel, i.e.

$$\mathrm{d}\alpha_t\, g_t + \mathrm{d}[\alpha, g]_t + \mathrm{d}\beta_t\, M_t \;=\; 0. \tag{6.8}$$

This is the single most conceptually important equation in the chapter, and it is worth dwelling on. Ordinary calculus tells us that $\mathrm{d}(\alpha g) = \alpha\,\mathrm{d}g + g\,\mathrm{d}\alpha$. That identity is mathematically correct but *financially wrong*: it records changes in wealth that would have required money to appear or disappear from nowhere. The $g\,\mathrm{d}\alpha$ term is the dollar cost of *adding new shares* at the current price, which a trader does not pay from thin air — she has to sell some of her money-market holding to fund the purchase. The equation (6.8) is the formal version of that common-sense statement.

To make the point visceral: imagine you hold 100 shares of a \$50 stock, worth \$5000. Suppose you suddenly decide you want to hold 200 shares, and at the moment of decision the stock is still at \$50. Did your wealth just jump by \$5000? Of course not — you had to hand \$5000 of cash to the market to get those extra shares. Naïve calculus says $\mathrm{d}(\alpha g) = g\,\mathrm{d}\alpha$ would count that \$5000 as new wealth, which is absurd. The self-financing constraint (6.8) says that the cash leg $\beta_t M_t$ has to shrink by exactly \$5000 in the same instant, so the *total* portfolio value is unchanged by the rebalance itself. Only subsequent *price* moves — the stock going from \$50 to \$51, the bank accruing interest — change the total wealth. This is the iron law we are about to build the whole pricing theory on: dollars don't appear, dollars don't vanish, and the only source of P&L is mark-to-market.

The cross-variation term $\mathrm{d}[\alpha, g]_t$ deserves a final comment. If α_t is a diffusion — meaning the trader's position itself has a $\mathrm{d}W$ component, which happens whenever $\alpha_t = \partial_x f(t, X_t)$ and X is Itô-driven — then α and g share a common Brownian driver and their covariation is non-zero. That cross-term lives in the rebalance leg and is absorbed into the self-financing constraint. In the discrete world of §6.11 it shows up as the $(\alpha_{t_n} - \alpha_{t_{n-1}})\,X_{t_n}$ piece of the bank recursion: when you change your share count, you trade the *difference* at the prevailing price, and the cash account absorbs the cost.

8.2.2 6.2.2 Interpreting self-financing

Between t and $t + \Delta t$ we move from (α_t, β_t) to $(\alpha_{t+\Delta t}, \beta_{t+\Delta t})$; the change in wealth consists only of the asset moves Δg_t and ΔM_t, not of fresh money injected from outside:

$$\Delta V_t \;=\; \alpha_t\,(\Delta g_t) + \beta_t\,(\Delta M_t) - \Delta f_t. \tag{6.9}$$

Why self-financing means $\mathrm{d}V = \alpha\,\mathrm{d}g + \beta\,\mathrm{d}M - \mathrm{d}f$. The naive Itô expansion contains six terms (two $\mathrm{d}\alpha\, g$-style rebalance terms, two "hold" terms $\alpha\,\mathrm{d}g, \beta\,\mathrm{d}M$, a quadratic cross-variation $\mathrm{d}[\alpha, g]$, and $\mathrm{d}f$). But a trader never swaps cash into or out of the portfolio from outside — so when she buys $\mathrm{d}\alpha$ more shares of g she must sell enough M to pay for it at the current price g_t. The self-financing constraint (6.8) is the statement "the rebalance leg is free". What survives is exactly the mark-to-market of the *held* positions: $\alpha\,\mathrm{d}g + \beta\,\mathrm{d}M$. Wealth moves only because markets moved, never because money appeared.

A portfolio that is self-financing is one whose wealth process is *pathwise* determined by the asset price paths and the initial position — no external injections, no trapdoors where money can appear. This is the financial analogue of conservation of energy: nothing gets spent or produced in the rebalance itself. The implication for pricing is profound: if two self-financing portfolios end at the same terminal value on every path, they must have the same value today, because if they didn't, buying the cheap one and shorting the expensive one would yield a costless portfolio with zero risk and a positive initial P&L — a free lunch.

Self-financing vs violation *Dropping the rebalance-cost term fabricates PnL — the classic "something for nothing" that (6.8) forbids.*

8.2.3 6.2.3 Applying the self-financing constraint

After cancellation,

$$\mathrm{d}V_t \;=\; \alpha_t\,\mathrm{d}g_t + \beta_t\,\mathrm{d}M_t - \mathrm{d}f_t. \tag{6.10}$$

The only surviving terms are the three "mark-to-market" pieces: the P&L from the α_t shares of g because g moved, the interest accrual on the bank account, and the change in the liability f. There is *nothing else.*

Substituting (6.2), (6.3), (6.5):

$$\mathrm{d}V_t \;=\; \alpha_t(\mu^g_t g_t\,\mathrm{d}t + \sigma^g_t g_t\,\mathrm{d}W_t) + \beta_t M_t r_t\,\mathrm{d}t - (\mu^f_t f_t\,\mathrm{d}t + \sigma^f_t f_t\,\mathrm{d}W_t). \tag{6.11}$$

Collecting drift and diffusion,

$$\boxed{\mathrm{d}V_t \;=\; (\alpha_t\mu^g_t g_t + \beta_t M_t r_t - \mu^f_t f_t)\,\mathrm{d}t \;+\; (\alpha_t\sigma^g_t g_t - \sigma^f_t f_t)\,\mathrm{d}W_t} \tag{6.12}$$

Equation (6.12) has the form $\mathrm{d}V_t = \text{drift}\,\mathrm{d}t + \text{noise}\,\mathrm{d}W_t$, and we have two free knobs — α_t and β_t — to play with. The strategy is exactly what a sensible trader would do in words: first get rid of the noise (pick α_t to annihilate the $\mathrm{d}W_t$ coefficient), then inspect what remains and let economics constrain the rest.

This two-stage procedure — kill noise, then inspect drift — is the mathematical heart of every replication argument in finance. If the underlying has k sources of risk instead of one, we need k traded claims instead of one; we pick k holdings to kill each of the k noise terms simultaneously; the remaining drift must equal the risk-free rate on the net cash position. In the Heston stochastic-volatility model, for example, there are two sources of risk (the stock's diffusion and the volatility's diffusion), and we need two hedge instruments (the stock and another option) to kill both.

8.3 6.3 Locally removing the risk — the delta-hedge ratio

Locally remove risk, so set the $\mathrm{d}W_t$ coefficient to zero:

$$\alpha_t\,\sigma_t^g\,g_t - \sigma_t^f\,f_t \;=\; 0 \;\;\Longrightarrow\;\; \boxed{\alpha_t \;=\; \frac{\sigma_t^f}{\sigma_t^g}\,\frac{f_t}{g_t}} \tag{6.13}$$

This is the delta-hedge ratio in its most general form. Before we unpack it, note what we have achieved: the instant we pick α_t as above, the *random* part of $\mathrm{d}V_t$ — the bit that depends on which direction the Brownian motion happened to move — disappears. Over a small enough interval, the portfolio is locally riskless.

The formula has a crystalline interpretation: α_t is the ratio of the dollar-volatilities of f and g. If f is twice as volatile in dollar terms as g, we need twice as many units of g to keep up. In the common case where $g = X$ (the stock itself is traded), we have $\sigma_t^g g_t = \sigma_t^x$ and $\sigma_t^f f_t = \sigma_t^x\,\partial_x f$, so the ratio collapses to $\alpha_t = \partial_x f$ — the partial derivative of the option value with respect to the spot, a.k.a. the *delta*. The replicating portfolio is the option's Doppelgänger: at each instant, it holds exactly as much stock as the option's price is sensitive to the stock — no more, no less.

A concrete numerical illustration. Suppose the stock is trading at \$100 with $\sigma = 20\%$, so the dollar volatility of one share is roughly $100 \cdot 0.20 = 20$ dollars per unit of W. Suppose the option we are hedging has $\partial_x f = 0.6$, so its dollar volatility is $0.6 \cdot 20 = 12$ dollars per unit of W. To neutralise the option's Brownian motion exposure, we hold 0.6 shares of stock. If the stock now moves by 1% in a tiny interval, the option's price moves by $0.6 \cdot 1\% \cdot 100 = 0.60$ dollars and the share position moves by $0.6 \cdot 1 = 0.60$ dollars — they cancel to the instant.

A useful edge case. A deep-ITM call has delta approaching 1, so the hedger holds roughly one full share per option. A deep-OTM call has delta approaching 0, so the hedger holds almost no stock, and the bank account is approximately the option's premium invested at the risk-free rate. Again, gamma is nearly zero, rebalancing is minimal. The hedging problem is *easy* in both extremes and hardest *at the money*, where the delta is around 0.5 but the gamma is large and rebalancing is frequent.

One subtlety worth flagging: the formula $\alpha_t = \partial_x f(t, X_t)$ is not a *static* ratio. As time passes and the stock moves, both arguments of the partial derivative change, and the required hedge ratio changes too. This is why dynamic hedging is called *dynamic*: the hedge ratio is a moving target that must be chased.

With this choice,

$$\mathrm{d}V_t \;=\; \left(\alpha_t \mu_t^g g_t + \beta_t M_t r_t - \mu_t^f f_t\right)\mathrm{d}t \;=\; \mathcal{A}_t\,\mathrm{d}t. \tag{6.14}$$

After the $\mathrm{d}W_t$ term has been killed, the remaining dynamics are purely deterministic over an infinitesimal interval. The portfolio earns some drift $\mathcal{A}_t$ per unit time, and the question becomes: what does that drift have to be?

8.3.1 6.3.1 No-arbitrage forces $\mathcal{A}_t = 0$

- If $\mathcal{A}_t > 0$: profit guaranteed — arbitrage.
- If $\mathcal{A}_t < 0$: reverse the strategy to get profit.

The argument is stark. We have built, by pure bookkeeping, a self-financing portfolio with zero initial cost ($V_0 = 0$) and zero instantaneous risk. If its value drifts upward, we are printing money; if it drifts downward, we swap the signs of all holdings and *then* we are printing money. The only logical possibility consistent with no free lunches is that the drift is exactly zero.

Therefore, to avoid arbitrage,

$$\boxed{\mathcal{A}_t \;=\; 0}. \tag{6.15}$$

Combined with $V_0 = 0$, the entire portfolio process satisfies $V_t = 0$ for all t, hence

$$\mathcal{A}_t = 0 \implies V_t = 0 \implies \alpha_t g_t + \beta_t M_t - f_t = 0 \implies \beta_t M_t \;=\; f_t - \alpha_t g_t. \tag{6.16}$$

Equation (6.16) is the pricing identity in disguise. Read it from right to left: at any instant, the unique cash holding that keeps the hedge costless is precisely $f_t - \alpha_t g_t$. Equivalently, the value of the option *is* the value of the self-financing stock-plus-cash package that replicates it. The pricing question has been reduced to a constructive recipe: hold α_t shares of g, hold $(f_t - \alpha_t g_t)/M_t$ units of the money market, and the wealth of the package exactly tracks the option from now until expiry.

To see the recipe in practical terms, suppose the option value right now is \$3.50 and the delta is 0.6, and the stock is at \$100. The replicating package holds 0.6 shares (worth \$60) and has −\$56.50 in the money market (i.e. borrows \$56.50). The total package value is $0.6 \cdot 100 - 56.50 = 3.50$, which matches the option. If the stock rises to \$101, the shares are worth $0.6 \cdot 101 =$ \$60.60 and the loan is now slightly more expensive by the interest accrual, but the option itself has risen by the first-order amount $0.6 \cdot \$1 = \$0.60$ to \$4.10. The package now holds the updated delta (say 0.62) of the new stock price, so the trader must buy 0.02 shares, funded by additional borrowing. And so on, every instant. That is the whole game.

8.4 6.4 Market price of risk

Having extracted the zero-drift consequence, we unpack it into a statement about expected returns, which turns out to be one of the deepest identities in quantitative finance.

Write out $\mathcal{A}_t = 0$:

$$\alpha_t \mu_t^g g_t + r_t\,(f_t - \alpha_t g_t) - \mu_t^f f_t \;=\; 0. \tag{6.17}$$

Rearrange,

$$\alpha_t\, \mu_t^g\, g_t + r_t f_t - r_t \alpha_t g_t - f_t \mu_t^f \;=\; 0. \tag{6.18}$$

Recall $\alpha_t = \dfrac{\sigma_t^f}{\sigma_t^g}\dfrac{f_t}{g_t}$. Substituting and dividing by f_t,

$$\frac{\mu_t^g - r_t}{\sigma_t^g} \;=\; \frac{\mu_t^f - r_t}{\sigma_t^f} \;=\; \lambda_t \;=\; \lambda(t, X_t). \tag{6.19}$$

Sharpe ratios of all assets on the underlying index are equal.

We started from a single no-arbitrage condition and derived that every single derivative on X — vanilla call, digital, variance swap, structured product — must have the same expected excess return per unit of volatility. They can have wildly different drifts and wildly different volatilities, but the *ratio* of excess return to volatility is pinned down to the same number λ_t for every instrument on the same underlying. Economics collapses the whole zoo into one scalar.

λ_t is the **market price of risk** — a market property (not an asset-specific quantity). Thus

$$\boxed{\frac{\mu_t^f - r_t}{\sigma_t^f} \;=\; \lambda_t \iff \mu_t^f - r_t \;=\; \lambda_t\,\sigma_t^f}. \tag{6.20}$$

The connection to Girsanov. In Chapter 5 we derived Girsanov's theorem: given a predictable shift λ_t, there is an equivalent measure $\mathbb{Q}$ under which $\widetilde{W}_t := W_t + \int_0^t \lambda_s\,\mathrm{d}s$ is a Brownian motion. Under $\mathbb{Q}$ the drift of X becomes $\mu_t^x - \sigma_t^x\,\lambda_t$. Equation (6.20) is *exactly* the economic statement that picks out the Girsanov shift λ_t that neutralises the risk premium: under $\mathbb{Q}$, every traded claim on X has drift r_t (no risk premium), which is the defining property of the risk-neutral measure. The "market price of risk" and the "Girsanov drift shift" are the *same scalar*, viewed from two sides: economics says every instrument must share it, Chapter 5 says a specific measure change zeros it out.

Rewriting (6.20) as $\mu_t^f = r_t + \lambda_t\,\sigma_t^f$ says every instrument's expected return decomposes into the risk-free rate plus a risk premium equal to its volatility times the market price of a unit of volatility. Under $\mathbb{Q}$ the market price of risk is zero — the replicator does not see a risk premium because she has eliminated the risk altogether. The Girsanov shift is to the drift only, leaving diffusion (and hence all variance-based quantities — gamma P&L, vega P&L) measure-invariant. The long-run US equity Sharpe of ~0.4 is the empirical estimate of λ for the broad equity market; λ is generally state-dependent and spikes in crises.

8.5 6.5 The generalised Black-Scholes PDE

We have a scalar identity: $\mu^f - r = \lambda\sigma^f$. We now translate it into a PDE by recalling what μ^f and σ^f actually *are* — Itô expressions involving partial derivatives of the pricing function $f(t,x)$. Substituting those expressions turns the economic statement into a differential equation that the pricing function must satisfy.

The move from a scalar identity to a partial differential equation is a small conceptual leap that is easy to miss on first reading. The identity $\mu^f - r = \lambda\sigma^f$ is a statement about numbers: at every instant, some particular combination of numbers must vanish. But those numbers are themselves computed from the pricing function $f(t,x)$ via Itô's lemma, so the statement is really "this combination of partial derivatives of f must vanish at every point (t,x)". That, by definition, is a partial differential equation.

Recall (via Itô's lemma) for $f(t,x)$:

$$\mu_t^f \;=\; \frac{\partial_t f_t + \mu_t^x\,\partial_x f_t + \frac{1}{2}(\sigma_t^x)^2\,\partial_{xx} f_t}{f_t}, \tag{6.21}$$

$$\sigma_t^f \;=\; \frac{\sigma_t^x\,\partial_x f_t}{f_t}. \tag{6.22}$$

(Here μ_t^x, σ_t^x are the coefficients of X from (6.1).) Plug into $\mu_t^f - r_t = \lambda_t\sigma_t^f$:

$$\partial_t f_t + (\mu_t^x - \sigma_t^x\lambda_t)\,\partial_x f_t + \tfrac{1}{2}(\sigma_t^x)^2\,\partial_{xx} f_t \;=\; r_t\,f_t. \tag{6.23}$$

This has to hold $\forall(t,x)$. Hence:

$$\boxed{\begin{aligned} \partial_t f(t,x) + \big(\mu^x(t,x) - \sigma^x(t,x)\,\lambda(t,x)\big)\,\partial_x f(t,x) & \\ + \tfrac{1}{2}\big(\sigma^x(t,x)\big)^2\,\partial_{xx} f(t,x) \;=\; r(t,x)\,f(t,x),& \\ f(T,x) \;=\; \varphi(x).& \end{aligned}} \tag{6.24}$$

Generalised Black-Scholes PDE.

This is the consistency condition that ties together everything the option is going to do between now and expiry. The terminal condition $f(T,x) = \varphi(x)$ anchors the pricing function at maturity to the payoff shape, and the PDE propagates that shape back in time. The replicating portfolio's delta today — $\partial_x f(t,X_t)$ — is chosen precisely so that the option we are manufacturing ends up with the right payoff tomorrow.

Look at the structure of the PDE for a moment. The linear operator on the left-hand side is a backwards parabolic operator — backwards because time flows the "wrong" way (we propagate from the terminal condition at T back to the present), parabolic because the second-order derivative appears with a positive coefficient. Parabolic equations describe diffusion-like evolution: information smears out as time flows, corners round off, sharp features disappear. In option pricing, that translates to the idea that any fine-grained feature of the payoff at maturity — a sharp kink at the strike, a step at a digital, a cliff at a barrier — gets smoothed out as we go backwards in time.

The coefficient $(\mu^x - \sigma^x\lambda)$ multiplying $\partial_x f$ deserves special attention. It is not the physical drift μ^x; it is the physical drift minus the risk premium $\sigma^x\lambda$. The subtraction is exactly the Girsanov shift from Chapter 5 that turns the physical $\mathbb{P}$-Brownian motion into a risk-neutral $\mathbb{Q}$-Brownian motion. So (6.24) can be read as "the expected rate of change of f along a risk-neutral drift, plus the Itô correction, equals the carry cost". In expectation form, the same statement is Feynman-Kac (Chapter 4), as we will see again in §6.7A and (6.40).

> **Why the BS PDE encodes no-arbitrage.** Read (6.24) coefficient-by- coefficient: the $\partial_x f$ term carries the risk-neutral drift $(\mu^x - \sigma^x\lambda)$, not the physical drift μ^x. That subtraction $-\sigma^x\lambda$ is exactly the Girsanov change-of-measure (Chapter 5) that turns W into a $\mathbb{Q}$-Brownian motion. The $\frac{1}{2}(\sigma^x)^2\,\partial_{xx} f$ term is Itô's curvature correction — it is independent of drift, which is why option prices ultimately don't depend on μ. The RHS rf says the whole hedge-replicating portfolio must grow at the risk-free rate (otherwise §6.3.1's arbitrage trade exists). The Feynman-Kac representation (6.40) is *the same statement* in expectation form.
>
> **What happens when Δ is wrong.** If you hold $\alpha_t \neq \partial_x f$, the $\mathrm{d}W_t$ coefficient (6.12) is non-zero and the hedge error over $[t, t+\mathrm{d}t]$ is $(\alpha_t \sigma_t^g g_t - \sigma_t^f f_t)\,\mathrm{d}W_t$ — a mean-zero *noise* term plus a second-order drift $\frac{1}{2}\,\partial_{xx} f\,(\mathrm{d}X_t^2 - \sigma^2 X_t^2 \mathrm{d}t)$ that accumulates as realised-minus-implied variance. This is the famous *P&L attribution*: discretely-hedged short gamma loses money when realised vol exceeds implied, and makes money when it under-realises. See `ch06-hedge-error.png`.

The P&L attribution above is the most practically important fact in options trading. Delta-hedging a short option using σ_{imp} produces an instantaneous P&L proportional to $\frac{1}{2}\,\Gamma\,S^2\,(\sigma_{\mathrm{imp}}^2 - \sigma_{\mathrm{real}}^2)\,\mathrm{d}t$: the option market is, in a very real sense, a market for variance. The PDE is linear in f, so the price of a portfolio of options equals the sum of the individual prices — call spreads, straddles, and butterflies all assemble linearly.

BS-PDE coefficients *Theta pays for gamma: the three LHS terms balance rf exactly — the dotted residual is numerically zero.*

8.6 6.6 Specialising to Black-Scholes (GBM)

The generalised PDE is beautiful but abstract. To get to numbers, we specialise to the single most famous setting in quantitative finance: geometric Brownian motion. In GBM, the stock has constant proportional drift and volatility, and it is traded directly — so $g = X$.

In the Black-Scholes model:

$$\mu^x(t,x) \;=\; \mu x, \qquad \sigma^x(t,x) \;=\; \sigma x,$$

so

$$\mathrm{d}X_t \;=\; X_t\mu\,\mathrm{d}t + X_t\sigma\,\mathrm{d}W_t. \tag{6.25}$$

In divided form, $\mathrm{d}S_t/S_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$: the *percentage* increment of the stock is a Gaussian with mean $\mu\,\mathrm{d}t$ and variance $\sigma^2\,\mathrm{d}t$ — geometric Brownian motion. Applying Itô's lemma to $\ln S_t$ (Chapter 3) turns this into a constant-coefficient arithmetic SDE,

$$\mathrm{d}(\ln S_t) \;=\; (\mu - \tfrac{1}{2}\sigma^2)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t,$$

which integrates immediately. Exponentiating,

$$S_t \;=\; S_0\,\exp\Big((\mu - \tfrac{1}{2}\sigma^2)\,t \;+\; \sigma W_t\Big) \;\stackrel{d}{=}\; S_0\,\exp\Big((\mu - \tfrac{1}{2}\sigma^2)\,t \;+\; \sigma\sqrt{t}\,Z\Big), \qquad Z \sim \mathcal{N}(0,1). \tag{6.25a}$$

The distribution of S_t is therefore log-normal with log-mean $(\mu - \frac{1}{2}\sigma^2)\,t$ and log-variance $\sigma^2 t$. The $-\frac{1}{2}\sigma^2$ term is the Itô correction: without it, $\mathbb{E}[S_t]$ would *not* equal $S_0 e^{\mu t}$. With it, $\mathbb{E}[S_t] = S_0 e^{\mu t}$ exactly.

Take $g(t,x) = x$, so that X is indeed traded (with $\sigma^g = \sigma$, $\mu^g = \mu$), and let $r(t,x) = r$ (constant). Therefore

$$\lambda \;=\; \frac{\mu - r}{\sigma}. \tag{6.26}$$

In GBM, the market price of risk is literally the stock's Sharpe ratio. That identity, which fell out of a general no-arbitrage argument, is what gives λ its evocative name and what motivates the risk-neutral measure $\mathbb{Q}$ (Chapter 5) as the probability law under which the stock drifts at the risk-free rate.

Substituting into (6.24):

$$\partial_t f + \underbrace{(\mu - \sigma\lambda)}_{=r}\, x\,\partial_x f + \tfrac{1}{2}\sigma^2 x^2\,\partial_{xx} f \;=\; r f.$$

$$\boxed{\begin{aligned} &\partial_t f + r\,x\,\partial_x f + \tfrac{1}{2}\sigma^2 x^2\,\partial_{xx} f \;=\; r f, \\ &f(T,x) \;=\; \varphi(x). \end{aligned}} \tag{6.27}$$

Black-Scholes PDE. The stock's physical drift μ has disappeared completely — absorbed by $\mu - \sigma\lambda = r$. Two stocks with the same σ but different μ produce identical call prices because the hedger replicates state-by-state regardless of drift; cost depends on how hard the replication machine has to work (vol), not on direction. Implied volatility is then the market's $\mathbb{Q}$-measure forecast of root-mean-square log-returns; for liquid equity options the gap to the $\mathbb{P}$-forecast is small.

8.7 6.7 Black-Scholes formulas for call, put, and digital

With the PDE (6.27) in hand, we can read off closed forms for the three canonical vanilla payoffs. The call and put solutions can be obtained either by substituting an ansatz into (6.27), by computing the risk-neutral expectation directly under (6.25a), or — most concisely — via the Feynman-Kac bridge of Chapter 4. All three routes give the same formulas; we record them here and defer the derivation details to §6.7A.

Vanilla call. For $\varphi(x) = (x - K)_+$,

$$\boxed{f(t,x) \;=\; x\,\Phi(d_+) \;-\; K\,e^{-r(T-t)}\,\Phi(d_-)} \tag{6.28}$$

with

$$d_\pm \;=\; \frac{\ln(x/K) + (r \pm \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}. \tag{6.29}$$

Equation (6.28) is the **Black-Scholes call formula**. You can read it in two pieces: the first term $x\,\Phi(d_+)$ is the expected stock payoff *given that the option expires in the money* — a probability-weighted piece of the stock, computed under the stock-numeraire measure. The second term $K\,\Phi(d_-)\,e^{-r(T-t)}$ is the discounted strike times the risk-neutral probability of being in the money. The difference is the expected net gain from exercising. The symmetry $d_+ = d_- + \sigma\sqrt{T-t}$ connects the two terms: the extra $\sigma\sqrt{T-t}$ in d_+ is the "vol lift" that accounts for the fact that, conditional on ending in the money, the expected log-price is elevated relative to its unconditional median.

Vanilla put. For $\varphi(x) = (K - x)_+$, the cleanest route is via **put-call parity**. From the algebraic identity $(X_T - K)_+ - (K - X_T)_+ = X_T - K$, take risk-neutral expectations and discount:

$$\boxed{C(t,x) - P(t,x) \;=\; x - K\,e^{-r(T-t)}} \tag{6.30}$$

Parity is **model-independent** — it does not depend on GBM, or even on the stock dynamics at all. It follows purely from the algebraic identity combined with the existence of a risk-neutral measure (Chapter 5) under which discounted traded prices are martingales. Put-call parity violations are therefore the cleanest possible arbitrage signal. Solving (6.30) for P:

$$\boxed{P(t,x) \;=\; K\,e^{-r(T-t)}\,\Phi(-d_-) \;-\; x\,\Phi(-d_+)} \tag{6.31}$$

Digital call. For $\varphi(x) = \mathbb{1}_{x \geq K}$, the discounted price is simply the risk-neutral in-the-money probability times the discount factor:

$$\boxed{f(t,x) \;=\; e^{-r(T-t)}\,\Phi(d_-)} \tag{6.32}$$

Under GBM, the indicator $\{X_T \geq K\}$ is equivalent to $\{Z \geq -d_-\}$ where $Z \sim \mathcal{N}(0,1)$, so the $\mathbb{Q}$-probability is $\Phi(d_-)$, which gives (6.32). Far in the money, d_- is large and $\Phi(d_-) \approx 1$; far out of the money, d_- is strongly negative and $\Phi(d_-) \approx 0$; at the money, $\Phi(d_-)$ sits somewhere near $\frac{1}{2}$.

8.7.1 6.7.1 Greeks from the closed form

From (6.28) we can read off all the Greeks of the vanilla call analytically. The key identity that makes the algebra collapse is the **strike-shifting identity** $x\,\phi(d_+) \;=\; K\,e^{-r(T-t)}\,\phi(d_-)$, which is provable by direct algebra on the density arguments.

- **Delta**: $\Delta^C \;=\; \partial_x f \;=\; \Phi(d_+) \in (0,1)$.
- **Gamma**: $\Gamma^C \;=\; \partial_{xx} f \;=\; \dfrac{\phi(d_+)}{x\,\sigma\sqrt{T-t}}$.
- **Vega**: $\mathcal{V}^C \;=\; \partial_\sigma f \;=\; x\,\phi(d_+)\,\sqrt{T-t}$.

For the put, parity gives $\Delta^P = \Delta^C - 1 \in (-1,0)$ and $\Gamma^P = \Gamma^C$ (differentiating parity twice in x): calls and puts of the same strike and maturity have *identical* gamma. Vega is also shared: $\mathcal{V}^P = \mathcal{V}^C$.

The gamma formula exhibits a striking $1/\sqrt{T-t}$ divergence at the money as $t \to T$: the at-the-money gamma blows up like $1/\sqrt{T-t}$. This is the *pin gamma* phenomenon — on expiration day a market-maker's gamma exposure is effectively all in a narrow band around the strike, which is why gamma books are rebalanced with extra care on expiration Fridays. Integrated across spot, however, $\int \Gamma^C\,\mathrm{d}x \;=\; \Delta^C(\infty) - \Delta^C(0) \;=\; 1 - 0 \;=\; 1$: the total gamma "mass" is conserved across time-to-expiry. What changes is how that mass is distributed spatially.

Digital greeks. The digital call's delta from (6.32) is

$$\partial_x f^{\mathrm{dig}} \;=\; e^{-r(T-t)}\,\frac{\phi(d_-)}{x\,\sigma\sqrt{T-t}}, \tag{6.33}$$

which has a *pathological* limit as $T - t \to 0$ at $x = K$: the numerator $\phi(d_-) \to \phi(0) = 1/\sqrt{2\pi}$ is bounded away from zero, while the denominator $\sqrt{T-t}$ tends to zero. The delta blows up without bound at the strike. We will visualise this in §6.11.7 and discuss the call-spread hedge that practitioners use to tame it.

BS price and delta curves *BS price and delta curves.*

8.8 6.7A Dual derivation via Feynman-Kac

The self-financing hedging argument of §§6.2–6.5 is the route by which Black, Scholes, and Merton originally derived the PDE (6.27). There is a second, entirely independent route that arrives at *exactly* the same PDE and the same formulas: the **Feynman-Kac** bridge of Chapter 4. Seeing both routes is worthwhile because they foreground different pieces of the machinery and their agreement is the mark of a solid theory.

8.8.1 6.7A.1 Setup under the risk-neutral measure

Under the risk-neutral measure $\mathbb{Q}$ constructed in Chapter 5, the traded asset and bank account satisfy

$$\frac{\mathrm{d}S_t}{S_t} \;=\; r\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}\widetilde{W}_t, \qquad \frac{\mathrm{d}M_t}{M_t} \;=\; r\,\mathrm{d}t, \tag{6.34}$$

where $\widetilde{W}_t$ is a $\mathbb{Q}$-Brownian motion. The drift is r, not the real-world μ, because $\mathbb{Q}$ is chosen so that S_t/M_t is a $\mathbb{Q}$-martingale — and that martingale condition forces the drift of $\mathrm{d}S/S$ to equal r. The physical expected return μ never appears in derivative prices; it is *absorbed into the measure change* via the Radon-Nikodym derivative of Chapter 5, with Girsanov shift $\lambda = (\mu - r)/\sigma$.

8.8.2 6.7A.2 Applying Feynman-Kac

The claim $\xi = (\xi_t)$ with $\xi_T = \varphi(S_T)$ has pricing function $F_t = f(t, S_t)$. Feynman-Kac (Chapter 4) applied with drift $a(x) = rx$, diffusion $b(x) = \sigma x$, and discounting $c = r$ yields the representation

$$f(t,x) \;=\; \mathbb{E}^{\mathbb{Q}}_{t,x}\big[\, e^{-r(T-t)}\,\varphi(X_T)\,\big]\,, \qquad \mathrm{d}X_s = r\,X_s\,\mathrm{d}s + \sigma\,X_s\,\mathrm{d}\widetilde{W}_s. \tag{6.35}$$

By the Chapter 4 FK theorem, $f(t,x)$ satisfies the PDE

$$\partial_t f + r\,x\,\partial_x f + \tfrac{1}{2}\sigma^2 x^2\,\partial_{xx} f \;=\; r\,f, \qquad f(T,x) = \varphi(x). \tag{6.36}$$

This is *identical* to (6.27). Two independent derivations — the hedging argument of §6.5 and the FK martingale argument of Chapter 4 — converge on the same PDE. That convergence is not a coincidence; both are expressing the same fact in different languages.

8.8.3 6.7A.3 Discounted price as a $\mathbb{Q}$-martingale

The clearest way to see the equivalence is via the discounted price. Let $Y_t := f(t, X_t)/M_t$. Applying Itô's lemma (Chapter 3) and collecting the $\mathrm{d}t$ terms gives

$$\mathrm{d}Y_t \;=\; \frac{1}{M_t}\big(\partial_t f + rx\,\partial_x f + \tfrac{1}{2}\sigma^2 x^2 \partial_{xx} f - rf\big)\mathrm{d}t \;+\; \frac{\sigma x\,\partial_x f}{M_t}\,\mathrm{d}\widetilde{W}_t. \tag{6.37}$$

By the Black-Scholes PDE (6.36), the $\mathrm{d}t$ bracket vanishes identically, leaving only the Brownian piece — i.e. Y is a (local) martingale. **The PDE is exactly the drift-killer**: $\mathbb{Q}$ is defined so the bracket equals zero.

In this framing, the Black-Scholes PDE is not mysterious partial- differential-equation magic; it is *literally* the statement "the discounted price of the claim is a martingale under $\mathbb{Q}$" translated into Itô derivatives. Memorise the chain of equivalences:

Martingale $\Leftrightarrow$ drift is zero $\Leftrightarrow$ PDE holds.

If you ever forget what the Black-Scholes PDE says, you can reconstruct it: take the Itô of the discounted price, set the drift to zero, and copy out the bracket. That bracket is the PDE.

8.8.4 6.7A.4 Recovering the call formula from the expectation

With (6.35) in hand, we can derive (6.28) by direct integration. Integrate $\mathrm{d}X_s/X_s = r\,\mathrm{d}s + \sigma\,\mathrm{d}\widetilde{W}_s$ with $X_t = x$ to get

$$X_T \;=\; x \cdot \exp\Big\{(r - \tfrac{1}{2}\sigma^2)(T-t) + \sigma(\widetilde{W}_T - \widetilde{W}_t)\Big\}. \tag{6.38}$$

Decompose the call payoff as $(X_T - K)_+ = X_T\,\mathbb{1}_{X_T>K} - K\,\mathbb{1}_{X_T>K}$. The second piece is the discounted digital times K: from (6.32), $K\,e^{-r(T-t)}\,\Phi(d_-)$. The first piece — the stock-delivery leg — uses the change-of-numeraire trick of Chapter 5. Under the stock-numeraire measure $\mathbb{Q}^S$ (with S_t as numeraire rather than the money account M_t), the expectation $\mathbb{E}^{\mathbb{Q}^S}[\mathbb{1}_{X_T>K}]$ equals $\Phi(d_+)$ with $d_+ = d_- + \sigma\sqrt{T-t}$. Translating back to the $\mathbb{Q}$-expectation via the Radon-Nikodym density $\mathrm{d}\mathbb{Q}^S/\mathrm{d}\mathbb{Q} = S_T/(S_t\, e^{r(T-t)})$:

$$\mathbb{E}^{\mathbb{Q}}_{t,x}\left[e^{-r(T-t)}\, X_T\, \mathbb{1}_{X_T>K}\right] \;=\; x\cdot\Phi(d_+). \tag{6.39}$$

Combining the two pieces recovers (6.28). The structure "probability of payoff times conditional expected payoff" is visible in the formula: $\Phi(d_-) = \mathbb{Q}(X_T > K)$ is the risk-neutral probability of exercise; $\Phi(d_+) = \mathbb{Q}^S(X_T > K)$ is the stock-measure probability of exercise. The two differ by the Girsanov shift $\sigma\sqrt{T-t}$ between the two numeraire measures.

8.8.5 6.7A.5 Why two derivations

The two routes emphasise different machinery. The hedging derivation of §§6.2–6.5 foregrounds the self-financing condition, the market price of risk, and no-arbitrage: the PDE is the economic consistency requirement that replication imposes. The Feynman-Kac derivation foregrounds the measure change: the PDE is the drift-killer that defines $\mathbb{Q}$, and the expectation under $\mathbb{Q}$ *is* the price. A practitioner who can flip fluently between the two representations has a much richer toolkit than one who is comfortable with only one. For payoffs that admit a clean Gaussian integral (vanilla calls, digitals, log-contracts), the expectation form is the natural computational route. For payoffs that do not admit closed-form integration but are low-dimensional in state space (American exercise, barriers, free-boundary problems), the PDE form is the numerical workhorse via finite differences.

8.9 6.8 Worked Example 1 — Linear payoff $\varphi(x) = x$ (sanity check)

Before attacking a real option we check the machinery on a trivial payoff — the claim that simply pays the stock price at maturity. Obviously the forward price of such a claim is the stock itself, so the PDE had better produce $f(t,x) = x$.

We expect that $f(t,x) = x$. PDE check-out: Indeed,

$$\partial_t f = 0, \qquad \partial_x f = 1, \qquad \partial_{xx} f = 0,$$

so the LHS of (6.27) is $0 + rx\cdot 1 + 0 = rx = rf$.

The "delta" of this trivial claim is exactly 1 — we replicate a stock-payoff by holding one share of the stock, with no cash leg. This sanity check is not wasted effort: any time you solve the BS PDE numerically, it is worth verifying that the scheme reproduces this trivial case exactly. If your finite-difference solver can't price a stock, it can't price a call either.

Direct probabilistic derivation. Under the risk-neutral measure $\mathbb{Q}$ (Chapter 5),

$$X_T \;\overset{d}{=}\; X_t\, e^{(r-\frac{1}{2}\sigma^2)(T-t)+\sigma\sqrt{T-t}\,Z}, \qquad Z\sim\mathcal{N}(0,1).$$

Hence

$$\text{price} \;=\; \mathbb{E}^{\mathbb{Q}}_t\left[e^{-r(T-t)}\, X_T\right] \;=\; X_t\, \mathbb{E}^{\mathbb{Q}}\left[e^{-\frac{1}{2}\sigma^2(T-t)+\sigma\sqrt{T-t}\,Z}\right] \;=\; X_t. \tag{6.40}$$

The bracket is a mean-1 log-normal, confirming $f(t,x) = x$ via the Feynman-Kac route as well.

Two ways up the same mountain: the PDE approach and the expectation approach. Both agree. This is the Feynman-Kac theorem (Chapter 4) in its most elementary form, and it hints at the broader fact that solving the BS PDE is the same thing as computing a discounted expectation under $\mathbb{Q}$.

8.10 6.9 Worked Example 2 — Separable ansatz $f(t,x) = x^2\,\ell(t)$

The next step up in difficulty is a non-trivial but tractable payoff: the squared payoff $\varphi(x) = x^2$. This arises naturally when pricing variance products, and the separable ansatz $f(t,x) = x^2\ell(t)$ reduces the PDE to an ODE.

Squared payoffs may look contrived at first glance. Who writes a contract that pays out the square of a stock price? The answer is: nobody, directly. But squared payoffs appear as building blocks in variance swaps, in the hedging decomposition of log-contracts, and in the moment-matching approximations used to price basket options. They are the simplest non-linear payoff in the Itô world, and their price is a clean window onto how the PDE handles convexity.

Use the ansatz $f(t,x) = x^2\,\ell(t)$ with $\ell(T) = 1$. Then

$$\partial_t f = x^2\,\dot{\ell}, \qquad \partial_x f = 2x\ell, \qquad \partial_{xx} f = 2\ell.$$

Plug into (6.27):

$$\underbrace{x^2\dot{\ell}}_{\partial_t f} + \underbrace{rx \cdot 2x\ell}_{rx\,\partial_x f} + \underbrace{\tfrac{1}{2}\sigma^2 x^2 \cdot 2\ell}_{\frac{1}{2}\sigma^2 x^2\,\partial_{xx} f} \;=\; \underbrace{r\,x^2\ell}_{rf}. \tag{6.41}$$

Divide by x^2:

$$\dot{\ell} + (r+\sigma^2)\ell \;=\; 0 \implies \ell(t) \;=\; e^{(r+\sigma^2)(T-t)}. \tag{6.42}$$

So

$$\boxed{f(t,x) \;=\; x^2\,e^{(r+\sigma^2)(T-t)}}. \tag{6.43}$$

The solution has a delightful structure: the squared-payoff price today is today's squared stock price times a time-decay factor that grows with σ^2. The volatility shows up *additively* to the rate inside the exponent, which is a fingerprint of variance products — their value is an affine function of integrated variance, so anything that boosts variance boosts their value exponentially over time.

Notice also how the ansatz mechanised a PDE into an ODE. The PDE was a partial differential equation in two variables (t,x), and the ansatz peeled off the x-dependence by asserting it must be proportional to x^2. What was left was an ordinary differential equation in t alone, with an easy exponential solution. This is a common trick for payoffs that are homogeneous in x — any power-of-x payoff yields an ansatz that reduces to an ODE.

The delta of this claim is $\partial_x f = 2x\ell(t)$ — linear in the stock — and the gamma is $\partial_{xx} f = 2\ell(t)$, a constant. A constant-gamma payoff is the cleanest possible vehicle for harvesting realised variance: the P&L per unit time is just $\Gamma \cdot S^2 \cdot \mathrm{d}\langle \log S\rangle$, which is the classic "gamma times variance" accounting that variance-swap desks live by.

A quick numerical illustration. Suppose $r = 5\%$, $\sigma = 20\%$, and $T-t = 1$ year. The time-decay factor is $e^{(0.05+0.04)\cdot 1} = e^{0.09} \approx 1.094$. So the squared-payoff claim trades at a 9.4% premium over the naive "cash-and-carry" square, and that premium is almost entirely the $\sigma^2 = 0.04$ contribution. If we shocked volatility to 30%, the factor becomes $e^{0.14} \approx 1.150$ — a 15% premium. Variance-based products are extremely vega-heavy because their price is literally an exponential in σ^2.

The connection between the x^2 payoff and a variance swap is worth spelling out. A variance swap is a contract that pays the realised quadratic variation of $\ln X$ over $[0,T]$ minus a pre-agreed strike. The realised quadratic variation of $\ln X$ under GBM is $\sigma^2 T$. The variance-swap buyer is thus long realised variance. The key fact is that this variance payoff can be replicated statically using a portfolio of European options (via the log-contract decomposition), plus a dynamic delta-hedge on the underlying. The squared payoff x^2 is one component of that decomposition, and its tractability in the GBM model is what makes the overall variance-swap replication argument go through.

8.11 6.9A Worked Example 3 — Digital call (closed form + PDE verification)

The third worked example stress-tests the hedging machinery: the digital call. It pays \$1 if the stock ends above a strike and zero otherwise — a discontinuous payoff. Discontinuities are where hedging theory starts to creak, because the replicating portfolio has to reproduce a jump in expected value using instruments that move continuously. The pricing formula comes out cleanly (we already wrote it as (6.32)), but the hedging behaviour near expiry is pathological, as we will see in §6.11.7.

Payoff and closed form. Recall

$$\varphi(x) \;=\; \mathbb{1}_{x \geq K}, \qquad f(t,x) \;=\; e^{-r(T-t)}\, \Phi\big(d_-(t,x)\big), \tag{6.44}$$

with

$$d_-(t,x) \;=\; \frac{\ln(x/K) + (r - \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}. \tag{6.45}$$

A worked example anchors the formula. Suppose $S = \$100$, $K = \$100$, $r = 5\%$, $\sigma = 20\%$, and $T - t = 0.5$ year. Then $\ln(S/K) = 0$, and $(r - \frac{1}{2}\sigma^2)(T-t) = (0.05 - 0.02)(0.5) = 0.015$. The denominator $\sigma\sqrt{T-t} = 0.2 \cdot 0.707 = 0.1414$. So $d_- = 0.015/0.1414 \approx 0.106$, and $\Phi(0.106) \approx 0.542$. The digital call price is $e^{-0.025} \cdot 0.542 \approx 0.529$: about 53 cents to receive a dollar at expiry.

8.11.1 6.9A.1 PDE verification

It is mechanical but instructive to verify that the formula actually solves the PDE (6.27). The partial derivatives come from the chain rule applied to the composition $\Phi(d_-(t,x))$.

Time derivative:

$$\partial_t f \;=\; e^{-r(T-t)}\, \Phi'(d_-(t,x))\, \partial_t d_- \;+\; r\, f. \tag{6.46}$$

Space derivatives:

$$\partial_x f \;=\; e^{-r(T-t)}\, \Phi'(d_-(t,x))\, \partial_x d_-, \qquad \partial_x d_- \;=\; \frac{1}{x\,\sigma\sqrt{T-t}}, \tag{6.47}$$

$$\partial_{xx} f \;=\; e^{-r(T-t)} \Big\{ \Phi''(d_-(t,x))\, (\partial_x d_-)^2 + \Phi'(d_-(t,x))\, \partial_{xx} d_- \Big\}, \tag{6.48}$$

with

$$\partial_{xx} d_- \;=\; -\,\frac{1}{x^2\,\sigma\sqrt{T-t}}, \qquad \Phi''(x) \;=\; -\,x\,\Phi'(x), \quad \Phi'(x) \;=\; \frac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}. \tag{6.49}$$

Combining (6.46)-(6.49) in (6.27) verifies the PDE: the rf terms cancel against $\partial_t f$'s $+rf$ piece, and the drift-plus-diffusion piece cancels thanks to $\Phi'' = -x\Phi'$.

The identity $\Phi'' = -x\Phi'$ is the algebraic engine that makes the digital verification fall out so neatly. It is also what makes the normal distribution such a natural fit with Brownian motion: the Gaussian density is the unique probability law whose log-derivative is linear, and this property dovetails with the $\frac{1}{2}\sigma^2 x^2 \partial_{xx}$ term in the PDE to collapse all the cross-terms at exactly the right moment.

The digital's delta, which we read off from (6.47), has the shape $\partial_x f = e^{-r(T-t)}\, \Phi'(d_-)/(x\sigma\sqrt{T-t})$. For spot values not exactly at the strike, d_- diverges to $\pm\infty$ as time to expiry shrinks, which kills $\Phi'(d_-)$ exponentially. But for spot values at the strike, d_- stays bounded and $\Phi'(d_-)$ is bounded away from zero; the delta is then dominated by the $1/\sqrt{T-t}$ singularity in the denominator. This is the mathematical version of "infinite delta at the strike in the last instant" that we will visualise in §6.11.7.

A useful quantitative check. For a digital with $K = \$100$, $\sigma = 20\%$, and $r = 5\%$, the peak delta with 1 month to expiry is roughly $0.4/(100 \cdot 0.2 \cdot \sqrt{1/12}) \approx 0.07$ per dollar of stock. With 1 day to expiry, the peak delta is ≈ 0.4 per dollar. With 1 hour to expiry, the peak delta is roughly 2 per dollar — untradeable in practice. The scaling $1/\sqrt{T-t}$ is brutal: halving the time to expiry only increases the peak delta by a factor of $\sqrt{2}$, but over many halvings the number grows without bound. Digital hedging is a race against the clock.

8.12 6.10 Time-based hedging recursion

So far we have been doing everything in PDE language. There is a parallel, often more useful, language of expectations. Feynman-Kac (Chapter 4) tells us that any solution of the PDE can be rewritten as a discounted $\mathbb{Q}$-expectation of the payoff, where $\mathbb{Q}$ is the measure under which the underlying drifts at the risk-free rate minus $\sigma\lambda$. This is the final piece of the pricing puzzle: the replicating portfolio, the PDE, and the expectation are three equivalent statements of the same no-arbitrage fact.

The three-way equivalence — replication, PDE, expectation — is a deep fact with practical consequences. Depending on the problem, one representation is much easier to work with than the others. For payoffs that are smooth functions of a lognormal variable (like the vanilla call), the expectation is straightforward. For payoffs on a non-Gaussian state variable, the PDE is often the numerical workhorse. And for complicated hedging studies — where we want to visualise the replication portfolio's tracking behaviour over sample paths — the self-financing construction is the natural tool.

Under risk neutrality,

$$f(t,x) \;=\; \mathbb{E}^{\mathbb{Q}}_{t,x}\Big[\varphi(X_T)\, e^{-\int_t^T r_u\,\mathrm{d}u}\Big], \qquad X_t = x. \tag{6.50}$$

Notice that the expectation is taken under $\mathbb{Q}$, not $\mathbb{P}$. Under $\mathbb{Q}$, the drift of X is modified by the Girsanov shift $-\sigma^x\lambda$, which in the Black-Scholes case collapses the drift to rX_t. The physical drift μ has been *completely eliminated* from the pricing machine.

The $\mathbb{Q}$-dynamics of X are

$$\mathrm{d}X_t \;=\; \underbrace{(\mu^x_t - \sigma^x_t\,\lambda_t)}_{=\,r_t X_t \text{ in BS}}\,\mathrm{d}t \;+\; \sigma^x_t\,\mathrm{d}\widetilde{W}_t, \tag{6.51}$$

where $\widetilde{W}_t$ is a $\mathbb{Q}$-Brownian motion (Chapter 5).

One subtle but beautiful feature of the $\mathbb{Q}$-measure is that it makes all discounted traded prices into martingales. Under $\mathbb{Q}$, $e^{-rt}X_t$ is a martingale, $e^{-rt}g_t$ is a martingale, and $e^{-rt}f_t$ is a martingale. This is why the discounted-expectation formula (6.50) works: the discounted option price today equals the expected discounted payoff at expiry. The martingale language is the backbone of continuous-time finance, and it slots into the dynamic hedging story as the expectation-side of the replication coin.

The **Fundamental Theorem of Asset Pricing** formalises this connection. It states that, in a market satisfying mild technical conditions, there is no arbitrage if and only if there exists at least one equivalent martingale measure $\mathbb{Q}$ under which all discounted traded prices are martingales. Furthermore, if the market is complete (as in our single-Brownian-motion setup), the martingale measure is unique. Self- financing plus no-arbitrage implies the existence of $\mathbb{Q}$; market completeness implies uniqueness; and the risk-neutral expectation (6.50) is what you get when you integrate any payoff against this unique measure.

8.12.1 6.10.1 Specialisation to Black-Scholes

$$\mathrm{d}X_t \;=\; \mu X_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}W_t \;=\; r\,X_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}\widetilde{W}_t, \qquad r = \text{const.} \tag{6.52}$$

The call, put, and digital formulas (6.28), (6.31), and (6.32) all follow from plugging (6.38) into (6.50) and computing the Gaussian integrals, as sketched in §6.7A.4. The three formulas are the three canonical closed forms that every option-pricing library implements.

It is worth writing out what the Greeks mean in units. The delta $\Phi(d_+)$ is dimensionless — it is the number of shares per option. The gamma $\phi(d_+)/(x\sigma\sqrt{T-t})$ has units of shares per dollar of stock, which is to say "how fast the delta changes as the stock moves". The vega $x\phi(d_+)\sqrt{T-t}$ has units of dollars per unit of volatility; a typical convention is to quote vega per "vol point" (1% change in σ), which gives the dollar sensitivity of the option to a one-percentage-point move in implied volatility. These three numbers are the core risk measures of the option position; any serious trading desk watches them in real time.

The curvature features of the vanilla call are specially illuminating. The gamma peaks at-the-money because that is where the payoff has its "kink"; further from the strike, the payoff is either essentially linear (deep ITM) or essentially zero (deep OTM), and a linear payoff has no curvature. The gamma also grows as expiry approaches for at-the-money options, because the remaining convexity has to be concentrated into a shrinking window. Gamma and vega are cousins — both express the option's exposure to variance — and they are most pronounced where the payoff is most non-linear.

8.13 6.11 Discrete hedging of a continuous model

The theoretical result is that continuous rebalancing at every instant produces *exact* replication — $V_t \equiv 0$, every day, every path. In real life, we rebalance at some discrete frequency: daily, hourly, once per tick. The replication therefore has tracking error, and the structure of that error is what we turn to now. We will see that the dominant source of error scales like $\sqrt{\Delta t}$ per step, and that — importantly — this error comes from discretisation of the Brownian motion, not from any model mismatch. Model mismatch is a separate, more insidious source of error.

The transition from the continuous-time theory to real-world hedging is where the rubber meets the road. On paper, continuous rebalancing produces exact replication; in reality, the fastest any trader can rebalance is tick-by-tick, and even tick-by-tick is at millisecond intervals, not truly continuous. The gap between "continuous" and "very frequently" is not just a quantitative small-error question — it has a specific structure that shapes how traders set up their rebalancing workflows.

8.13.1 6.11.1 Holdings rule

$$\alpha_t \;=\; \frac{\sigma_t^f\, f_t}{\sigma_t^g\, g_t} \;=\; \partial_x f(t, X_t) \qquad \text{if } X_t \text{ is traded.} \tag{6.53}$$

(Using $\sigma_t^g g_t = \sigma_t^x$ when $g_t = X_t$, and $\sigma_t^f f_t = \sigma_t^x\, \partial_x f_t$.) Thus the stock-hedge ratio is the delta, $\Delta_t \equiv \partial_x f(t, X_t)$.

The clean identification of α_t with the delta is the reason "delta hedging" is the universal name of this procedure. In principle, at each instant we should hold exactly $\partial_x f$ shares. In practice, we hold that many shares at the *last rebalance time* and let the position drift until the next rebalance.

This "hold at last rebalance" approximation introduces two categories of error. The first is *path-dependent*: between rebalance times, the stock moves away from its last-hedged value, so the ideal $\partial_x f$ shifts, and the portfolio's actual delta drifts away from the ideal delta. The larger the stock move, the larger the drift. This shows up as gamma-driven P&L noise over each rebalance interval. The second category is *calendar-dependent*: even if the stock does not move, time passes, and the ideal delta changes because $\partial_x f(t, X_t)$ depends on t as well as on X_t.

A mental picture helps. On a $g(t, S)$-vs-S diagram, the ideal delta at time t is the slope of the chord to the pricing surface at the current spot S_t. The trader's rebalancing procedure can be visualised as "climbing the slope" one chord at a time: at t_0 we install a tangent line at (t_0, S_{t_0}) with slope α_{t_0}; at t_1 the surface has evolved, the spot has moved, and a new tangent line is installed with slope α_{t_1}; and so on. Between tangent installations, the held portfolio is an affine function of S — constant slope — while the true pricing function is curved, and the mismatch is precisely the rebalance-interval P&L noise we called the gamma residual.

8.13.2 6.11.2 At $t_0 = 0$ — sold f, get f_0

- Buy α_0 units of X (costs $\alpha_0 X_0$).
- Bank account: $M_0 = f_0 - \alpha_0 X_0$.

At inception, we collect the option premium f_0, use some of it to buy the initial delta in shares, and deposit what's left in the money market. The bank account is positive or negative depending on whether the initial hedge was a big chunk of the premium (e.g. deep in-the-money calls need a big share purchase) or small (far out-of-the-money).

For a concrete example, suppose we sell a 3-month at-the-money call on a stock at \$100, implied vol 25%, rate 5%, and collect roughly \$5.60 in premium (delta about 0.55). We immediately buy 0.55 shares at \$100 each, costing \$55, and the money-market account starts at $\$5.60 - \$55 = -\$49.40$. That is a \$49 negative balance on the bank — we are borrowing. That is normal: the short-call hedge requires borrowing to fund the stock purchase. Over time, as the option's delta changes, we will borrow more or lend back.

8.13.3 6.11.3 At t_1

- Asset now has value $\alpha_{t_0} X_{t_1}$.
- Bank value: $M_{t_0}\, e^{r\,\Delta t}$.

Must rebalance to new units of X (cost $\alpha_{t_1} X_{t_1}$). Bank is now

$$M_{t_1} \;=\; M_0\, e^{r\,\Delta t} \;-\; (\alpha_{t_1} - \alpha_{t_0})\, X_{t_1}. \tag{6.54}$$

At the first rebalance, the delta has moved because both time has passed and the stock has moved. We adjust our share holding to the new delta and pay for (or receive from) the adjustment out of the bank account.

Notice the algebraic symmetry of (6.54). The first term, $M_0 e^{r\Delta t}$, is the bank's natural growth due to interest. The second term, $-(\alpha_{t_1} - \alpha_{t_0}) X_{t_1}$, is the cost of the rebalance, evaluated at the *new* stock price X_{t_1}. The rebalance price is always the current stock price, not the old one, because we are trading at today's market; this is what distinguishes self-financing from the naive product rule.

8.13.4 6.11.4 At t_2

- Asset value: $\alpha_{t_1}\, X_{t_2}$.
- Bank value: $M_{t_1}\, e^{r\,\Delta t}$.
- Rebalance: α_{t_2} units of X (cost $\alpha_{t_2} X_{t_2}$).
- Bank is now

$$M_{t_2} \;=\; M_{t_1}\, e^{r\,\Delta t} \;-\; (\alpha_{t_2} - \alpha_{t_1})\, X_{t_2}. \tag{6.55}$$

8.13.5 6.11.5 Repeat

$$\boxed{M_{t_n} \;=\; M_{t_{n-1}}\, e^{r\,\Delta t} \;-\; (\alpha_{t_n} - \alpha_{t_{n-1}})\, X_{t_n}} \tag{6.56}$$

No rebalancing between $t_0, t_1, \ldots, t_{N-1}, t_N = T$.

In practice, every rebalance also incurs a *transaction cost* that the idealised recursion ignores. Let $\kappa(q)$ denote the dollar cost of trading q shares. The realistic recursion reads

$$M_k \;=\; M_{k-1}\, e^{r\,\Delta t_k} \;-\; (\alpha_k - \alpha_{k-1})\, S_k \;-\; \kappa(\alpha_k - \alpha_{k-1}), \tag{6.57}$$

and the cumulative cost $\sum_k \kappa(\alpha_k - \alpha_{k-1})$ is the *hedge slippage bill* that ultimately sits against the theoretical P&L of (6.58). The presence of transaction cost is what makes rebalancing a genuine optimisation problem rather than a "rebalance as often as possible" prescription.

The recursion (6.56) is the entire operational content of discrete delta hedging. Given the delta path α_t (which the trader computes from the BS formula using some implied volatility) and the realised stock path X_t, (6.56) tells you exactly what your bank balance is at every rebalance date.

8.13.6 6.11.6 Terminal P&L

At t_N we owe the option payoff:

$$\boxed{\mathrm{PnL} \;=\; \Big(M_{t_{N-1}}\, e^{r\,\Delta t} \;+\; \alpha_{t_{N-1}}\, X_{t_N}\Big) \;-\; \varphi(X_{t_N})} \tag{6.58}$$

Exact replication would give $\mathrm{PnL} \equiv 0$. In reality, two distinct error sources contribute.

Discretisation error ($\sqrt{\Delta t}$ scaling, vanishes as $\Delta t \to 0$). On each step the residual mismatch between the held constant-α position and the ideal-$\partial_x f$ position is $\frac{1}{2}\,\Gamma\, X^2\, \sigma^2\,(Z^2-1)\,\Delta t$, with $Z \sim \mathcal{N}(0,1)$. Summing N independent residuals gives a total standard deviation $\sim \Gamma\, S^2\, \sigma^2\, \sqrt{T\,\Delta t}$. Doubling the rebalance frequency cuts the std-dev by $\sqrt{2}$, not by 2.

Model error (does *not* shrink with finer rebalancing). If the hedger uses σ_{imp} to compute the delta but the market realises σ_{real}, every interval contributes the **gamma · realised-vs-implied identity**

$$\mathrm{P\&L}_{[t,t+\Delta t]} \;\approx\; \tfrac{1}{2}\,\Gamma_t\, X_t^2\,(\sigma_{\text{imp}}^2 - (\Delta X_t/X_t)^2/\Delta t)\,\Delta t.$$

The integral over the option's life is the running gamma-times-vol-mismatch P&L of a delta-hedged short option — a long-gamma position earns money when realised exceeds implied, a short-gamma position loses. A delta-hedged option *is* a variance swap in a convex wrapper. Model error is curable only by trading other options (vega), not by trading more.

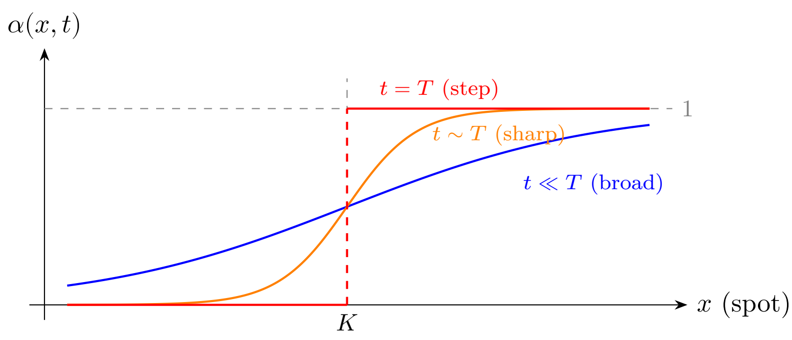
Jump risk lives outside the Brownian framework: an unhedged jump of size J in the underlying produces a P&L of $f(t, X+J) - f(t,X) - \partial_x f \cdot J$ — second-order surprise (positive for long gamma, unbounded for short gamma). Rebalancing faster cannot help; only option-vs-option hedges can. This jump risk explains the crash-o-phobia skew in equity-index OTM puts.

Discrete-hedging error vs rebalance frequency *Discrete-hedging error vs rebalance frequency.*

Replicating portfolio value paths *Ideal continuous hedging would pin $V_t \equiv 0$; daily rebalancing leaves visible gamma-driven wiggles.*

8.13.7 6.11.7 Shape of α_t near expiry — digital call

Near $t = T$, the digital-call delta $\alpha = \partial_x f$ spikes around the strike K. A useful way to watch this evolve is to plot the delta as a function of spot for three representative time slices:



As $t \to T^-$ the digital-call delta sharpens from a broad logistic into a step at K; gamma is the spatial derivative, so it blows up into a Dirac spike at expiry.

- $t \ll T$ **(far from expiry):** the delta is a broad, logistic-looking curve rising gently from near 0 on the far left through roughly $\frac{1}{2}$ at $x = K$ to near 1 on the far right. Gamma is modest and the hedge is easy to track.
- $t \sim T$ **(close to expiry):** the logistic tightens into a sharp S-curve concentrated near K. Gamma grows large in a narrow band; outside the band the delta is essentially 0 or 1.
- $t = T$ **(at expiry):** the delta jumps from 0 to 1 at K — a step function — and its derivative (gamma) is a Dirac δ-spike.

As $t \to T^-$ the delta sharpens into a Dirac δ-function centred at K — hedging a digital near expiry is basically impossible without discontinuous position changes ("delta blow-up").

The reason is instructive. The digital's payoff is a step function: it jumps from zero to one as the stock crosses the strike. Since the delta is the spatial derivative of the option price, and the option price near expiry looks more and more like the step payoff itself, the delta must look more and more like the derivative of a step — a Dirac delta in the limit. Digitals are therefore either priced with a significant vega / pin-risk premium or hedged as **call spreads** (two offsetting vanilla calls with strikes slightly above and below the digital strike), which smooths the payoff and the delta.

A digital call that pays \$1 if $X_T \geq K$ can be approximately replicated by going long a vanilla call struck at $K - \epsilon/2$ and short a vanilla call struck at $K + \epsilon/2$, scaled by $1/\epsilon$. As $\epsilon \to 0$, this call spread converges to the digital. But for any finite ϵ, the call spread has a bounded delta around the strike, bounded by $1/\epsilon$. The wider the spread, the less hedging difficulty, but the more residual risk from the approximation.

Delta surface *Left: smooth vanilla delta. Right: digital delta tents at K as $\tau \to 0$ — the hedger must trade arbitrarily large size around the strike.*

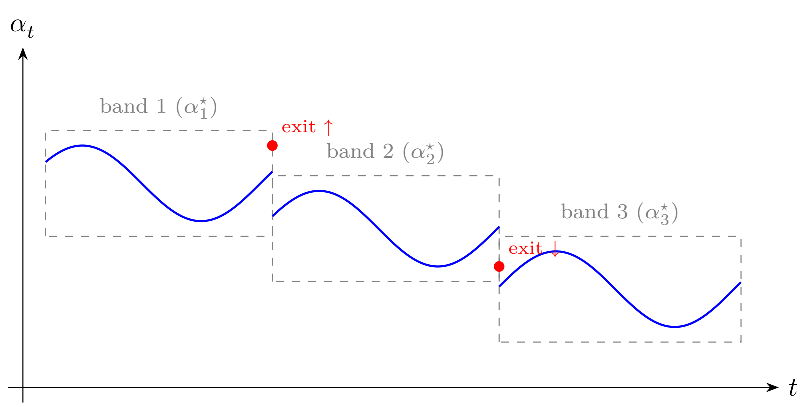
8.14 6.12 Move-based hedging

The time-based hedging scheme of §6.11 rebalances on a fixed clock — every day, every hour, every minute. That is conceptually clean but often operationally wasteful: on a quiet day, the delta barely moves, and paying commissions to rebalance at noon is silly. **Move-based hedging** instead monitors the delta itself and rebalances only when it drifts outside a pre-specified band around the last-hedged value.

The philosophy behind move-based hedging is that what the trader cares about is the *delta error* — the gap between the current-held delta and the current-ideal delta — not the clock. As long as the delta error stays small, the portfolio is well-hedged and no trade is needed. Only when the delta error exceeds a threshold does trading become worthwhile.

Continuous-time theory rebalances at every instant; discrete time rebalances on a clock $t_0, t_1, \dots, t_N$. Move-based hedging rebalances whenever the delta α_t wanders outside a band around its last-hedged value:

- Set a band $[\alpha^\star - \epsilon,\ \alpha^\star + \epsilon]$.
- Let α_t drift inside the band.
- When α_t exits, re-centre: new $\alpha^\star = \alpha_t$, restart.



The delta drifts inside a band of width 2ϵ around the last-hedged value $\alpha^\star$. On exit, a single trade snaps the position to the current α_t, and a fresh band re-centres.

Move-based hedging trades fewer times than time-based hedging for the same tracking error on smooth paths, but pays up on rapid gamma swings.

There is a trade-off between two evils. Tight bands mean frequent rebalancing (high commission cost) but tight tracking of the delta (low gamma-bleed risk). Wide bands mean infrequent rebalancing (low commission cost) but the portfolio can drift far from the target delta. The optimal band width depends on the cost of transacting, the volatility of the underlying, the gamma of the position, and the trader's risk tolerance.

A useful way to think about the choice is to imagine the trader's delta as a random walk against a target. Time-based hedging snaps the walk back to target on a fixed schedule regardless of how far it has drifted; move-based hedging lets the walk roam free until it exits a band, then snaps it back. Over a continuous horizon, the *expected number of trades* under move-based hedging scales inversely with the square of the band width — narrow bands trigger often, wide bands trigger rarely — while the *tracking error* scales linearly with the band width. This is the fundamental cost/benefit dial: you get quadratically fewer trades for a linear increase in tracking noise.

A stylised Monte-Carlo experiment makes the shape of the frontier concrete. Simulating a short at-the-money three-month call under GBM and hedging under a move-based rule with transaction cost 10 bps per share traded, the mean and standard deviation of terminal P&L for a selection of band widths ϵ (expressed as a fraction of initial delta) come out approximately as:

Band ϵ	Mean P&L	P&L std
0.00625	-0.39	0.705
0.0125	-0.35	0.710
0.025	-0.34	0.750
0.05	-0.34	0.850
0.10	-0.32	1.290
0.20	-0.33	2.190

Traced from top to bottom, the band is widening. The mean P&L climbs *slightly* (from -0.39 to about -0.32) because fewer trades means less total transaction-cost bleed. But the standard deviation *grows* substantially (from 0.71 to 2.19) because the delta is allowed to wander further before being corrected, producing larger gamma residuals per rebalance interval. This is the classic mean/std efficient frontier, and the desk's job is to pick the point on it that aligns with its risk budget.

The shape of this frontier is surprisingly universal. Re-run the experiment under different parameters and the *location* of the curve shifts but its *shape* (concave, roughly hyperbolic near the shoulder) is preserved. The curve's parametric form is essentially "commission + variance = $a/\epsilon^2 + b\epsilon^2$", whose minimum over ϵ gives the optimal band when the desk's objective is mean-variance.

The quadratic-versus-linear scaling is worth a line of explanation. The number of band-crossings by a Brownian motion with volatility σ over time T is of order $T\sigma^2/\epsilon^2$ (the classic local-time result). Doubling the band width ϵ therefore reduces the number of trades by a factor of 4, but only increases the typical delta error by a factor of 2. Cost savings scale quadratically with the relaxation of the precision constraint.

One final operational remark. Regardless of whether a desk rebalances by clock or by delta move, it is almost always correct to *also* rebalance at specific predictable moments — the open, the close, the pre-earnings hour, right before a central-bank announcement. These are the moments when the assumed continuity of the price path is most likely to break: gaps at the open, close-auction mean-reversion, headline-driven jumps. A disciplined hedger layers move-based triggers on a clock-based baseline and hard-codes additional rebalances around known risk events.

A practical remark on how desks actually implement move-based hedging: most production systems use a **delta-dollars** band rather than a pure delta band. Delta-dollars is the dollar value of the stock-equivalent exposure ($\Delta \cdot S$), which normalises across underlyings of different price levels. Bands expressed in delta-dollars are natively comparable across the book and feed cleanly into the desk's risk-limit system. Beyond delta-dollars bands, more sophisticated desks layer **gamma-adjusted** bands — the idea being that a position with high gamma needs to be rebalanced more often because its delta moves more per unit of stock move.

In a world with zero transaction costs, continuous rebalancing is trivially optimal. In a world with finite transaction costs, neither time-based nor move-based is universally optimal. The modern derivatives desk uses a hybrid approach, running time-based rebalancing as a baseline safety net and move-based overlays where they add value. The chapter's discussion of continuous dynamic hedging is the theoretical idealisation these practical schemes are trying to approximate.

8.14.1 Case study (a): Black Monday 1987 — portfolio insurance and the dynamic hedge cascade

Synthetic puts replicated by selling futures into a falling market. The continuous-hedge ideal collided with discrete fills and a 22% gap. The cohort-level short gamma was the systemic short option no one had marked.

Context. Through 1986–87, roughly \$60–90 billion of US institutional equity was running "portfolio insurance" — a strategy popularised by Leland-O'Brien-Rubinstein Associates (LOR) and Wells Fargo Wilshire Investment Management. Mechanically the product was a synthetic protective put: instead of buying listed SPX puts at their (rich) implied vol, the manager dynamically replicated the put payoff by holding equities and dynamically shorting S&P 500 futures in proportion to the put's delta. As the index fell, the put's delta moved toward -1, and the strategy mechanically sold *more* futures to maintain the hedge. The selling was supposed to be smooth — small adjustments against an assumed-continuous price path — and benign in normal conditions.

On Friday 16 October 1987 the S&P 500 closed down 5.16%, and Monday's open was already gapping. Through 19 October, the cash index fell from 282.70 to 224.84 (down 20.5% close-to-close, with intraday lows touching −22.6%). Portfolio-insurance programs trying to hedge against a 5% move had to reprice deltas against a 20% move, and the resulting sell program in the futures pit was estimated at \$14B notional in S&P 500 and MMI futures over Monday — into a market where market makers were on strike, the NYSE specialists could not open dozens of names, and futures-cash basis blew out to a 20-handle premium then a 60-handle discount. The Brady Commission's Report (January 1988) named portfolio insurance and index arbitrage as the two mechanically-amplifying flows.

Through the chapter's math. Three failure modes of the Black-Scholes hedge framework lined up. (i) The hedge ratio $\alpha_t = \partial_x f(t, X_t)$ from §6.3 is *continuous* in S, but fills are discrete: when S gaps from 282 to 225 with no quotes in between, you cannot trade the intermediate deltas — you go straight to the post-gap delta and book the path-dependent gamma loss in one print. (ii) The local Itô expansion (6.6) reads $\mathrm{d}V_t \approx \alpha_t \,\mathrm{d}S_t + \frac{1}{2}\Gamma_t(\mathrm{d}S_t)^2$ under the assumption $(\mathrm{d}S_t)^2 = \sigma^2 S_t^2\,\mathrm{d}t$; when the realised $(\Delta S)^2/S^2$ is on the order of 0.05 instead of $\sigma^2 \cdot \Delta t \approx 0.0001$, the discrete-hedging error of §6.11.6 is two orders of magnitude past its $\sqrt{\Delta t}$ scaling estimate. (iii) Each LOR client's program was independently short gamma to its hedger — but at the *cohort* level the entire \$80B of insurance flow was net short gamma against the same index, and that aggregate gamma exposure had no second-leg hedger because the synthetic-put issuer was the market itself. The market's effective short-gamma position showed up as a price impact term that the dynamic hedge had to chase.

Lesson. The "synthetic put" was synthetic until it wasn't. The chapter's continuous-time replication argument in §6.2–6.3 promises a perfect hedge under (a) frictionless markets, (b) continuous price paths, and (c) an outside-counterparty buyer for whatever delta the hedger needs to trade. October 1987 violated all three at once. The modern legacy of 1987 is structural: SPX listed put open interest is materially higher than its 1980s level (insurers buy listed puts now rather than synthesise them), exchange circuit breakers institutionalise the gap-discontinuity acknowledgement, and any desk running synthetic optionality calculates a "if the market gapped 10% before I could rebalance" stress P&L as a regulatory risk metric. The Brady-Commission view — that any hedging strategy whose required trades scale with realised moves can become destabilising at scale — is now baked into SEC Rule 15c3-5 (market access) and the post-crisis trade-through controls. Portfolio insurance the product still exists, but is sized to remain a price-taker, not a price-maker.

8.14.2 Case study (b): GameStop January 2021 — gamma squeeze as forced delta-hedge

Retail call buying forces market makers into mechanical buying. Positive gamma on the MM short means $\delta' = \Gamma > 0$ in S. The "gamma squeeze" headline is just (6.6) read out loud.

Context. Through January 2021, retail call-option buying on GameStop (GME) intensified through the WallStreet-Bets community on Reddit. Daily GME call volume on weekly and front-month expiries reached six- to ten-times historical norms; on 22 January, GME call open interest exceeded the entire free-float share count. Market makers — predominantly Citadel Securities, Susquehanna, Wolverine, and Group One on the option side — fill these option orders by selling calls and dynamically delta- hedging the resulting short. The implied vol the MM charged was high (IV on GME 30-day options jumped from ~80% to over 400% in that two-week window), but the dynamic-hedge mechanics were exactly what §6.3 describes: sell call → short delta → buy stock to be flat.

The GME spot ran from \$19.94 (4 Jan close) to \$325.00 (29 Jan close), a 16× move in twenty trading days, with intraday peaks at \$483 on 28 January. Each new call buyer added incremental short-gamma exposure to the MM book, the MM bought more stock to re-flatten delta, that buying contributed to the very price rise that pushed the next strike's delta into the hedger's must-buy zone. Retail brokers (notably Robinhood) at the peak suspended buy-side orders in GME on 28 January citing clearinghouse margin requirements, snapping the feedback loop.

Through the chapter's math. The market-maker's hedged book is $V_t^{\mathrm{MM}} = -C(t, S_t) + \alpha_t S_t + \beta_t M_t$ with $\alpha_t = \partial_S C$ — the call's delta. Itô on C gives $\mathrm{d}C = \partial_t C\,\mathrm{d}t + \partial_S C\,\mathrm{d}S_t + \frac{1}{2}\partial_{SS}C \cdot \sigma^2 S_t^2\,\mathrm{d}t$, so the MM's short-call Γ exposure is $-\partial_{SS}C = -\Gamma_t < 0$ (short gamma). The required delta adjustment per move in S is $\mathrm{d}\alpha_t = \Gamma_t\,\mathrm{d}S_t$: S goes up by ΔS, the MM must *buy* $\Gamma_t \cdot \Delta S$ extra shares to remain delta-neutral. With aggregate dealer dollar-gamma on GME estimated at \$200–400M per percentage-point move at the peak, a 1% spot move forced ~$3Minmechanicalbuying*fromthehedgealone*, ontopofunderlyingretailorderflow.Thedollar-gammametric($ = _t S_t^2 0.01$) — the dealer's mechanical buy/sell pressure per 1% move — is the single number that diagnoses this dynamic, and is the version of §6.3 that risk managers now publish on positioning dashboards alongside the SPX itself.

Lesson. The "gamma squeeze" is not a market-microstructure exotic or a coordinated retail attack — it is the §6.3 hedge ratio applied at scale to a name with thin float and concentrated short-gamma dealer positioning. The right diagnostic is dollar-gamma at the underlying level, aggregated across all listed strikes weighted by open interest; when dollar-gamma exceeds a meaningful fraction of average daily volume, the dealer hedge becomes price-impacting and the underlying becomes gamma-driven rather than fundamental-driven. Post-GME, every prime broker's risk team publishes dealer-positioning estimates by name; SEC Staff Report on Equity and Options Market Structure Conditions in Early 2021 (October 2021) documents the mechanism in detail and uses exactly this language. The deeper lesson for the dynamic-hedging chapter: $\Gamma > 0$ for the long-call holder is what makes the convexity of options valuable; the same Γ on the MM's short side, multiplied by enough notional, becomes a market-moving exogenous demand curve.

8.14.3 Case study (c): Volmageddon February 5 2018 — XIV liquidation via short-vol gamma

Short-vol products are short vol-of-vol — a hidden second-order risk. Discrete vega rebalancing fails when the vol regime changes intraday. The hedging-error variance scales with $\sqrt{\Delta t}$ until it doesn't.

Context. Credit Suisse's VelocityShares Daily Inverse VIX Short-Term ETN (ticker XIV, \$1.9B AUM into 5 Feb 2018) and the structurally related ProShares Short VIX Short-Term Futures ETF (SVXY, \$0.8B AUM) delivered the inverse of the daily return of the S&P 500 VIX Short-Term Futures Index — i.e., a daily reset short-front-VIX-futures position. Through 2017 the VIX averaged 11.1 (the lowest annual mean since the index began in 1990), and XIV had returned over 180% YTD by end-January 2018. The product was popular with risk-parity and yield-pickup allocators precisely because its drawdowns had been mild for half a decade.

On Friday 2 February 2018, the SPX closed −2.1% and the VIX closed at 17.16 (up from 13.47). On Monday 5 February the SPX dropped a further 4.10%; the VIX intraday spiked from 17 to 50, with the close at 37.32. The XIV NAV calculation on the issuer's terms triggered an "acceleration event" — a same-day liquidation provision tripped when the index lost more than 80% intraday — and Credit Suisse called the note for redemption. XIV closed Monday at \$7.35, down from \$115.55 (down 96.3%), and was wound down weeks later. The operative dynamic was that the issuers running short-vol replication books were short VIX futures delta and short vol-of-vol; as VIX rose, they had to *buy* VIX futures to maintain the short-vega hedge, and the buying happened in the most illiquid window of the day (the post-3:30 PM ET VIX-future settlement print), accelerating the spike.

Through the chapter's math. A short-vol position has a P&L of the form $V_t = -F(\sigma_t)$ where σ_t is the relevant vol process and F is the structure's payoff in vol. The first-order risk is short-vega: $\partial V/\partial\sigma < 0$. The second-order risk (volga, $\partial^2 V/\partial\sigma^2$) is what blows up in a vol regime change — Heston (Chapter 10) parameterises this through the vol-of-vol parameter ξ, and the volga exposure is large precisely when ξ is large. The discrete-hedging error of §6.11.6, repeated for the vega leg, gives a tracking error of order $\xi\sqrt{\Delta t}$ per rebalance interval under stable-regime conditions; the issue is that when ξ itself jumps (vol-of-vol on VIX rose from ~80% on 2 Feb to over 250% on 5 Feb implied), the hedge error scales with the new ξ^{new} but the hedge ratio is set against ξ^{old} — the once-a-day vega rebalance bookmarks a vega number that is wrong within minutes of the next print. The XIV NAV, mechanically defined as the daily-reset inverse-return, was effectively a margin-style stop that guaranteed forced covering at the worst possible vol-of-vol level.

Lesson. A short-vega book that prices and hedges as if vega is a constant is, in Heston language, ignoring volga. The §6.11 discrete hedging analysis told us that BS hedge errors scale with $\sqrt{\Delta t}$ under the *model's own assumption* that σ is constant; the moment σ becomes stochastic with non-trivial vol-of-vol, that $\sqrt{\Delta t}$ scaling is replaced by something proportional to the realised ξ, which can be ten times the calibrated ξ in a regime change. The lesson is operational and structural: any short-convexity book must (i) set a hard limit on vega-by-vol-bucket so a single regime change cannot eat the desk, (ii) hedge volga explicitly via long out-of-the-money options on the vol underlying, not just vega via ATM, and (iii) treat the once-a-day rebalance as inadequate for short- vol books in stressed vol regimes — XIV-style products that mechanically rebalance once per day at the close are *guaranteed* to underperform a continuously-hedged equivalent in a vol blow-out, by exactly the square-root-of-time argument the chapter develops, applied to the wrong ξ. The post-2018 short-vol ETP space halved in AUM and restructured into products with intraday rebalancing or capped daily loss; the underlying mathematics was a textbook §6.11 hedging-error cascade applied to the wrong-order Greek.

8.15 6.13 Key Takeaways

1. **Self-financing constraint.** A portfolio (α_t, β_t) in the underlying and bank account is self-financing iff $\mathrm{d}V_t = \alpha_t\,\mathrm{d}g_t + \beta_t\,\mathrm{d}M_t$ with *no* fresh injection of cash. Naïve differentiation would add a $g_t\,\mathrm{d}\alpha_t + M_t\,\mathrm{d}\beta_t$ term that has no economic meaning; the missing term is what arbitrage-free pricing forbids.

2. **Delta is $\partial_x f$.** Locally cancelling the Brownian increment in the hedged portfolio forces $\alpha_t = \partial_x f(t, X_t)$ — the hedge ratio is the slope of the pricing function with respect to the underlying, evaluated at the current state. No optimisation is required; it falls out of "set the $\mathrm{d}W$ coefficient to zero."

3. **Market price of risk = Girsanov shift.** Equating Sharpe ratios across two tradables driven by the same Brownian gives $\lambda = (\mu^f - r)/\sigma^f = (\mu^g - r)/\sigma^g$, the **market price of risk** — universal across hedgeable assets by no-arbitrage. Read through Chapter 5, λ is the integrand of the Radon-Nikodym density that takes $\mathbb{P}$ to $\mathbb{Q}$.

4. **Generalised BS PDE (6.24).** For an underlying $\mathrm{d}X_t = \mu^x\,\mathrm{d}t + \sigma^x\,\mathrm{d}W_t$ with state-dependent coefficients, the option price $f(t,x)$ satisfies $\partial_t f + (\mu^x - \sigma^x\lambda)\,\partial_x f + \frac{1}{2}(\sigma^x)^2\,\partial_{xx} f = r\,f$ with terminal data $f(T,x) = \varphi(x)$. The physical drift μ^x enters only through the combination $\mu^x - \sigma^x\lambda$.

5. **GBM specialisation (6.27)** and **call formula (6.28).** For GBM $\mathrm{d}X_t = \mu X_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}W_t$, the PDE collapses to $\partial_t f + rx\partial_x f + \frac{1}{2}\sigma^2 x^2 \partial_{xx} f = rf$. The European call solution is $f(t,x) = x\,\Phi(d_+) - Ke^{-r(T-t)}\,\Phi(d\_)$ with $d_\pm = [\ln(x/K) + (r \pm \frac{1}{2}\sigma^2)(T-t)]/(\sigma\sqrt{T-t})$. The physical drift μ is absent — option prices do not depend on the underlying's expected return.

6. **Closed-form vanilla Greeks.** $\Delta^C = \Phi(d_+) \in (0,1)$, $\Gamma^C = \phi(d_+)/(x\sigma\sqrt{T-t})$, vega $\mathcal{V}^C = x\,\phi(d_+)\,\sqrt{T-t}$. Put-call parity gives $\Delta^P = \Delta^C - 1$, $\Gamma^P = \Gamma^C$, $\mathcal{V}^P = \mathcal{V}^C$.

7. **PDE Q-expectation via Feynman-Kac.** The same PDE (6.27) is the Feynman-Kac PDE for the discounted $\mathbb{Q}$-expectation $f(t,x) = \mathbb{E}^{\mathbb{Q}}_{t,x}[e^{-r(T-t)}\,\varphi(X_T)]$, where under $\mathbb{Q}$ the drift of X is shifted from μ to r. Replication, PDE, and risk-neutral expectation are three equivalent statements of the same no-arbitrage fact.

8. **Risk-neutral drift = Girsanov-shifted physical drift.** $\mathrm{d}X_t = \mu X_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}W_t = rX_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}\widetilde{W}_t$ with $\widetilde{W}_t = W_t + \lambda t$ (Eq. 6.51-6.52). The bank-account-deflated price $e^{-rt}\,X_t$ is a $\mathbb{Q}$-martingale, as is $e^{-rt}\,f_t$.

9. **Discrete-hedging recursion (6.56) and $\sqrt{\Delta t}$ tracking error.** With holdings $\alpha_t = \partial_x f$ rebalanced on a grid $\{t_0, \dots, t_N\}$, the bank balance evolves as $M_{t_n} = M_{t_{n-1}}\,e^{r\,\Delta t} - (\alpha_{t_n} - \alpha_{t_{n-1}})\,X_{t_n}$. Terminal P&L (Eq. 6.58) has mean zero under correct model specification; its standard deviation scales as $\sqrt{\Delta t}$ — doubling rebalance frequency cuts hedging error by $\sqrt{2}$, not 2.

10. **Gamma · (realised − implied) P&L identity.** The leading-order single-step P&L from a mis-specified implied volatility is $\frac{1}{2}\,\Gamma\,S^2\,(\sigma^2_{\text{realised}} - \sigma^2_{\text{implied}})\,\Delta t$. A long-gamma position earns money when realised variance exceeds the implied number priced into the option; a short-gamma position bleeds the variance risk premium. Every variance-swap and gamma-trading strategy in the guide reduces to this identity.

11. **Digital Greeks blow up like $1/\sqrt{T-t}$.** At-the-money on expiration day, the digital call's delta $e^{-r(T-t)}\phi(d\_)/(x\sigma\sqrt{T-t})$ diverges as $T - t \to 0$ because $\phi(d\_) \to \phi(0) = 1/\sqrt{2\pi}$ stays bounded while $\sqrt{T-t} \to 0$. This is the **pin-risk** phenomenon and is the reason desks tame digital exposures with call-spread replication.

12. **Move-based vs time-based hedging trade-off.** Time-based rebalancing has $\sqrt{\Delta t}$ error and predictable transaction-cost frequency; move-based ("rebalance when $| (t, S) - _t| > $ band") concentrates rebalances

around the at-the-money region where gamma is large. With finite transaction costs neither is universally optimal; production desks hybridise — time-based safety net plus move-based overlay, both quoted in **delta-dollars** so positions compare across underlyings.

13. **The hedger's price is drift-free.** Two stocks with the same σ but different physical drifts μ have identical call prices. The hedger replicates payoffs state-by-state; the cost depends on how hard the replication machine has to work, which scales with vol, not drift.
14. **Implied volatility is forecast volatility (under $\mathbb{Q}$).** A 20% ATM implied vol is the market's unbiased estimate of the root-mean-square log-return over the option's life *under the risk-neutral measure*. Under $\mathbb{P}$ it can be a touch lower because $\mathbb{Q}$ overweights downside; for liquid equity options the gap is small.

8.16 6.14 Reference Formulas Appendix

8.16.1 6.14.1 Setup and SDE coefficients

Quantity	Symbol / formula
Underlying SDE (general)	$\mathrm{d}X_t = \mu^x(t, X_t)\,\mathrm{d}t + \sigma^x(t, X_t)\,\mathrm{d}W_t$
Underlying SDE (GBM specialisation)	$\mathrm{d}S_t/S_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$
Bank account	$\mathrm{d}M_t/M_t = r_t\,\mathrm{d}t,\ M_0 = 1$
Self-financing portfolio	$V_t = \alpha_t g_t + \beta_t M_t,\ \mathrm{d}V_t = \alpha_t\,\mathrm{d}g_t + \beta_t\,\mathrm{d}M_t$
Delta-hedge ratio	$\alpha_t = \partial_x f(t, X_t)$
Market price of risk	$\lambda(t,x) = (\mu^f - r)/\sigma^f = (\mu^g - r)/\sigma^g = (\mu^x - rx \cdot \mathbb{1}_{\text{tradable } X})/\sigma^x$
$\mathbb{Q}$-dynamics (Girsanov-shifted)	$\mathrm{d}X_t = (\mu^x - \sigma^x\lambda)\,\mathrm{d}t + \sigma^x\,\mathrm{d}\widetilde{W}_t,\ \widetilde{W}_t = W_t + \int_0^t \lambda\,\mathrm{d}u$

8.16.2 6.14.2 BS PDE — three equivalent forms

Form	Equation	Eq.
Generalised BS PDE	$\partial_t f + (\mu^x - \sigma^x\lambda)\,\partial_x f + \frac{1}{2}(\sigma^x)^2\,\partial_{xx} f = rf,\ f(T,x) = \varphi(x)$	(6.24)
GBM specialisation	$\partial_t f + rx\,\partial_x f + \frac{1}{2}\sigma^2 x^2\,\partial_{xx} f = rf,\ f(T,x) = \varphi(x)$	(6.27)
$\mathbb{Q}$-expectation (Feynman-Kac)	$f(t,x) = \mathbb{E}^{\mathbb{Q}}_{t,x}[e^{-r(T-t)}\,\varphi(X_T)],\ \mathrm{d}X_s = rX_s\,\mathrm{d}s + \sigma X_s\,\mathrm{d}\widetilde{W}_s$	(6.50)

8.16.3 6.14.3 Closed-form vanilla Greeks ($d_\pm = [\ln(x/K) + (r \pm \frac{1}{2}\sigma^2)(T-t)]/(\sigma\sqrt{T-t})$)

Contract	Price	Delta	Gamma	Vega
Call $\varphi(x) = (x-K)^+$	$x\,\Phi(d_+) - Ke^{-r(T-t)}\,\Phi(d_-)$	$\Phi(d_+)$	$\dfrac{\phi(d_+)}{x\sigma\sqrt{T-t}}$	$x\,\phi(d_+)\sqrt{T-t}$
Put $\varphi(x) = (K-x)^+$	$Ke^{-r(T-t)}\,\Phi(-d_-) - x\,\Phi(-d_+)$	$\Phi(d_+) - 1$	$\dfrac{\phi(d_+)}{x\sigma\sqrt{T-t}}$	$x\,\phi(d_+)\sqrt{T-t}$
Digital call $\varphi = \mathbf{1}_{\{x>K\}}$	$e^{-r(T-t)}\,\Phi(d_-)$	$e^{-r(T-t)}\dfrac{\phi(d_-)}{x\sigma\sqrt{T-t}}$	$-e^{-r(T-t)}\dfrac{d_+\,\phi(d_-)}{x^2\sigma^2(T-t)}$	$-e^{-r(T-t)}\dfrac{d_+\,\phi(d_-)}{\sigma}$

Strike-shifting identity: $x\,\phi(d_+) = Ke^{-r(T-t)}\,\phi(d_-)$. Put-call parity: $C_t - P_t = x - Ke^{-r(T-t)}$, hence $\Delta^P = \Delta^C - 1$, $\Gamma^P = \Gamma^C$, $\mathcal{V}^P = \mathcal{V}^C$. Pin-risk fingerprint: digital delta $\sim 1/\sqrt{T-t}$ as $t \to T$ at $x = K$.

8.16.4 6.14.4 Discrete hedging — recursion and $\sqrt{\Delta t}$ scaling

Quantity	Formula	Eq.
Bank-balance recursion	$M_{t_n} = M_{t_{n-1}}\, e^{r\,\Delta t} - (\alpha_{t_n} - \alpha_{t_{n-1}})\, X_{t_n}$	(6.56)
With transaction cost κ	$M_k = M_{k-1}\, e^{r\,\Delta t_k} - (\alpha_k - \alpha_{k-1})\, S_k - \kappa(\alpha_k - \alpha_{k-1})$	(6.57)
Terminal P&L (no costs)	$\mathrm{PnL} = (M_{t_{N-1}}\, e^{r\,\Delta t} + \alpha_{t_{N-1}}\, X_{t_N}) - \varphi(X_{t_N})$	(6.58)
Per-step P&L residual	$\frac{1}{2}\,\Gamma\, S^2\, \sigma^2\, (Z^2 - 1)\, \Delta t,\ Z \sim \mathcal{N}(0,1)$	
Tracking-error std-dev scaling	$\mathrm{std}(\mathrm{PnL}) \sim \Gamma\, S^2\, \sigma^2 \sqrt{T\,\Delta t}$	
Gamma · realised-vs-implied identity	$\Delta\Pi \approx \frac{1}{2}\,\Gamma\, S^2\, (\sigma^2_{\text{realised}} - \sigma^2_{\text{implied}})\, \Delta t$	

9 Chapter 7 — Dynamic Hedging II: Greeks, Delta-Gamma, Vega, and Dividends

Chapter 6 established the single-instrument delta hedge, the generalised Black-Scholes PDE, and the Greeks that fall out of the closed-form vanilla formulas. This chapter is the follow-up. We enlarge the hedging portfolio — first to neutralise curvature (delta-gamma), then to neutralise volatility exposure (vega), then to accommodate dividend-paying underlyings — and we pay the self-financing bookkeeping explicitly when transaction costs enter the picture. The underlying machinery is the same Itô expansion that drove Chapter 6; what changes is the dimension of the hedging basis and the list of Greeks we target.

9.1 7.0 Chapter orientation

A delta-hedged short-option position is not a risk-free position. It is **first-order** risk-free: the coefficient of $\mathrm{d}W_t$ in the self-financing Itô increment has been zeroed, so the portfolio has no instantaneous Brownian exposure. The residual risks live in the higher-order terms, and they have names. *Gamma* is the coefficient of $(\Delta S)^2$ — the curvature of the option price in the spot direction; *vega* is the coefficient of $\Delta\sigma$ — the sensitivity of the option price to a shift in the volatility parameter; *theta* is the coefficient of Δt — the passage of time; and a zoo of cross-derivatives (vanna, volga, charm, speed, colour) parameterise every other way a smooth price function can wiggle.

Most of these higher Greeks are not hedgeable with the stock alone. The stock has *zero* gamma, *zero* vega, and *zero* dependence on implied volatility, so no position in S_t and cash can neutralise curvature or vol-level risk. To hedge those exposures we must enlarge the hedging basis by adding another traded option — or two, or ten, depending on how many higher Greeks we want to annihilate. The linear algebra is straightforward; the operational bookkeeping is where the subtleties live, and that is where this chapter spends most of its time.

The chapter proceeds in seven acts. §7.1 is a brief Greeks recap that fixes notation and bridges from Chapter 6. §7.2 is the central methodological section: **delta-gamma hedging unified** — we give two framings (two-instrument replication and local quadratic fit) of the same construction and show they agree. §7.3 introduces vega and the extension of the replicating-portfolio argument to stochastic volatility. §7.4 derives the Black-Scholes-Merton PDE with continuous dividend yield. §7.5 takes up transaction costs and the Leland volatility adjustment. §7.6 surveys price, delta, and gamma sketches for puts, calls, and parity. §7.7 introduces **dollar Greeks** — the units that risk-management desks actually use. The chapter closes with Key Takeaways and a Reference Formulas appendix.

Throughout, notation follows Chapter 6: S_t is the stock, $g(t, S)$ or $f(t, S)$ is a contingent claim, $\Delta = \partial_S g$, $\Gamma = \partial_{SS} g$, $\mathcal{V} = \partial_\sigma g$. Unqualified expectations are under the risk-neutral measure $\mathbb{Q}$ unless otherwise noted.

9.2 7.1 Greeks recap — a bridge from Chapter 6

Chapter 6 produced the Black-Scholes PDE and, by solving it for the vanilla call, delivered a collection of partial derivatives that every derivatives trader learns by heart. It is worth restating the catalogue explicitly before we start putting them to work.

Let $g(t, S; \sigma, r)$ be the price function of a European claim on an underlying S with volatility σ and constant short rate r. The *first-order* Greeks measure the linear sensitivity of g to the four natural inputs — spot, time, volatility, and the short rate:

$$\Delta_t \;\equiv\; \partial_S g, \qquad \Theta_t \;\equiv\; \partial_t g, \qquad \mathcal{V}_t \;\equiv\; \partial_\sigma g, \qquad \rho_t \;\equiv\; \partial_r g. \tag{7.1}$$

The *second-order* Greeks measure curvature and cross-sensitivities:

$$\Gamma_t \;\equiv\; \partial_{SS} g, \qquad \mathrm{vanna}_t \;\equiv\; \partial_{S\sigma} g, \qquad \mathrm{volga}_t \;\equiv\; \partial_{\sigma\sigma} g, \qquad \mathrm{charm}_t \;\equiv\; \partial_{St} g. \tag{7.2}$$

And the *third-order* Greeks extend the cascade — speed $\partial_{SSS} g$, colour $\partial_{SSt} g$, zomma $\partial_{SS\sigma} g$, and so on — with each successive Greek characterising a tighter layer of residual risk.

For a European call under Black-Scholes the closed forms are

$$\Delta_{\text{call}} \;=\; \Phi(d_+), \qquad \Gamma \;=\; \frac{\Phi'(d_+)}{S\,\sigma\sqrt{T-t}}, \qquad \mathcal{V} \;=\; S\sqrt{T-t}\,\Phi'(d_+), \tag{7.3}$$

and for a European put, $\Delta_{\text{put}} = \Phi(d_+) - 1 = -\Phi(-d_+)$, with the same Γ and $\mathcal{V}$ as the call (we will prove the gamma identity via put-call parity in §7.6). Here Φ and Φ' are the standard normal CDF and density, and $d_\pm = [\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)(T-t)]/(\sigma\sqrt{T-t})$ is the Black-Scholes moneyness index.

Call delta profile across moneyness at three maturities *Black-Scholes call delta $\Phi(d_+)$ as a function of spot for three maturities. Long-dated options have gradual deltas spread across moneyness; near expiry the delta compresses toward a step at the strike, tracking the digital payoff that Δ_{call} becomes in the $T \downarrow t$ limit.*

Gamma vs moneyness and time-to-expiry *Gamma $= \Phi'(d_+)/(S\sigma\sqrt{T-t})$ peaks at the at-the-money strike for every maturity, and the peak grows without bound as $T \downarrow t$. A short ATM option is a short-gamma position that becomes dangerously short as expiry approaches — the terminal-gamma pinning effect.*

Call vs put delta across moneyness *Put-call parity in delta form: $\Delta_{\text{call}} - \Delta_{\text{put}} = 1$ for every spot. The call delta rises from 0 to $+1$ through the strike; the put delta sits a parallel unit below, running from -1 to 0. A long call plus a short put with the same strike is always 100% long the underlying.*

What Chapter 6 *proved* was that picking $\alpha_t = \partial_S g(t, S_t)$ units of the stock against a short position in the claim zeroes out the $\mathrm{d}W_t$ coefficient in the self-financing increment. Zeroing the drift then forces the pricing PDE, and the option's fair value is the cost of funding the replicating portfolio. That argument used **exactly one Greek** — delta — because exactly one random factor W_t was being neutralised, and exactly one hedging asset S_t was being traded against it.

The limitation is baked in. A position that is delta-hedged at time t is not delta-hedged at time $t + \Delta t$, because the delta has changed. The rebalance from $\alpha_t = \Delta_t$ to $\alpha_{t+\Delta t} = \Delta_{t+\Delta t}$ happens at the prevailing stock price, and the gap between the required hedge and the held hedge over the interval $[t, t + \Delta t]$ exposes the portfolio to whatever risks delta does *not* cover: curvature (gamma), the tilt of volatility over time (vega + vol dynamics), and all the higher-order effects. In the continuous-time limit those residuals vanish — this is the content of the Black-Scholes replication theorem — but in discrete time they are the dominant source of hedge-error P&L variance.

This chapter is the systematic treatment of those residuals. §7.2 removes the gamma residual by adding a second option to the hedging basis. §7.3 takes up the vega residual. §7.4 relaxes the "non-dividend-paying stock" assumption. §7.5 puts the self-financing bookkeeping under a microscope to see how transaction costs bleed into the P&L. §7.6 synthesises the put-call parity consequences on all of the above, and §7.7 translates the abstract Greeks into the dollar-denominated units that desks actually use.

One framing remark before we begin. The reader who has learned Chapter 6's self-financing argument may be tempted to view the extensions in this chapter as mere decorations on the core structure. They are not. Each extra Greek matched corresponds to a specific and separate risk that the single-instrument hedge leaves *on the books*. A flow desk that prices options but leaves gamma unhedged, vega unhedged, and dividends mis-modelled is not running a "simplified" version of the Black-Scholes hedge — it is running a book whose actual risk exposure is dominated by the residuals, not by the delta. The residuals are the business.

9.3 7.2 Delta-gamma hedging — two framings unified

Up to this point the replicating portfolio has consisted of one risky instrument — either the stock S directly, or a traded claim g — and the bank account. That is enough to kill the first-order risk of the position, and as we have seen repeatedly it suffices to derive the pricing PDE. But a delta-neutral position is not the same thing as a *risk-neutral* position: its value is insensitive to the first derivative of the stock move, yet still very sensitive to the second derivative. A large move of either sign moves the P&L of a delta-hedged short option position by roughly $\frac{1}{2}\,\Gamma\,(\Delta S)^2$, and the sign of that term — negative for a short option — is precisely the jump loss that the gamma P&L decomposition records as the price paid for selling convexity.

The practical question is how to hedge *gamma* as well as *delta*. The answer is to enlarge the hedging set: in addition to the stock S and the cash account M, the trader holds a position in an auxiliary traded claim $h(t, S_t)$ — typically another option — whose curvature can be used to offset the curvature of the target claim. The hedging portfolio now has two risky knobs (α_t units of S and γ_t units of h) and one cash knob (β_t units of M), and the matching conditions are two: first-order (delta) and second-order (gamma). That is exactly the minimum machinery needed to neutralise a quadratic local expansion of the target price function.

There are two equivalent ways to arrive at the same construction, and both are worth seeing. The first — which we call *Framing A*, two-instrument replication — is the structural argument: the target claim's Itô expansion includes a dW_t-coefficient (delta) and a dt-coefficient that depends on Γ, and the hedging portfolio must match both; this forces the hedge to contain a *second risky asset* whose Itô expansion has the right Γ. The second — which we call *Framing B*, local quadratic fit — is the practical argument: the target's price function is a smooth curve in S at each fixed t, and we are trying to replicate its first and second spatial derivatives using a stock (which has delta 1 and gamma 0) and a second option (which has known delta Δ^h and gamma Γ^h); matching both derivatives yields a 2×2 linear system whose unique solution gives the hedge ratios. The two framings yield the same formulas and the same operational recipe; they differ in which first-principles story one tells to justify them. We give both in full, side by side.

9.3.1 7.2.1 Local Taylor expansion of a claim

Let $g(t, S)$ be the price of a European option on S, with partial derivatives $\partial_S g$, $\partial_{SS} g$, and so on. Fix a reference spot S_t and consider the value of the option if the spot were to change by a small amount $(S - S_t)$. A second-order Taylor expansion in the spot variable reads

$$g(t, S) \;=\; g(t, S_t) \;+\; (S - S_t)\, \partial_S g(t, S_t) \;+\; \tfrac{1}{2}\, (S - S_t)^2\, \partial_{SS} g(t, S_t) \;+\; \cdots, \tag{7.4}$$

with the customary naming convention

$$\Delta_t^g \;\equiv\; \partial_S g(t, S_t), \qquad \Gamma_t^g \;\equiv\; \partial_{SS} g(t, S_t). \tag{7.5}$$

The delta Δ_t^g is the slope of the pricing function at the current spot; the gamma Γ_t^g is the curvature. A vanilla call's pricing function is a smoothed version of the hockey-stick payoff — zero in the far-left, curving upward near the strike, and asymptoting to a line of slope one far to the right — so its delta runs from 0 to 1 and its gamma is a localised bump around the strike. For an at-the-money call close to expiry, the gamma is tall and narrow; far from expiry, the gamma is short and broad.

Visually, one can imagine three curves superimposed on a single axis of spot S: the payoff (a kinked line), the current option price $g(t, S)$ (a smooth convex curve sitting above the payoff), and two auxiliary lines drawn at the reference point S_t — the *tangent* with slope Δ_t^g (the linear delta approximation) and a *parabola* matching both slope and curvature (the delta-gamma approximation). The tangent deviates from the true price as soon as S moves an appreciable distance; the parabola hugs the true price for considerably longer because it matches the curvature. This is the geometric content of delta-gamma hedging: we are trying to replicate an option not with a flat share holding (a horizontal line) but with a share holding plus a second-option holding whose combined curvature matches the target's.

A delta-only (linear) hedge is a *tangent line* at the current spot — it matches the slope, so small moves are neutralised, but any curvature in the payoff is missed. Gamma adds the second derivative, so the hedge becomes a *tangent parabola* that hugs the payoff over a much wider window. Away from the edge, the approximation error scales as $(\Delta S)^3$ instead of $(\Delta S)^2$ — a full order of accuracy gained per extra Greek matched.

There is a deeper reason the quadratic truncation is natural. Under geometric Brownian motion, the leading stochastic move over a short interval has $(\Delta S)^2 \sim S^2 \sigma^2 \Delta t$, so the second-order Taylor term is of the same order as Δt itself — one cannot ignore it without simultaneously ignoring all the time-derivative structure of the pricing PDE. The third-order Taylor term, $(\Delta S)^3 \sim S^3 \sigma^3 (\Delta t)^{3/2}$, is of higher order in Δt and legitimately vanishes in the small-step limit. So "delta and gamma" is not an arbitrary cutoff: it is exactly the order at which the stochastic and deterministic terms balance in Itô calculus. Higher Greeks matter only through discrete-rebalancing residuals, not through the continuous-time limit.

Delta-only vs delta-gamma hedge residual *Figure 7.1 — Delta-only hedge (tangent line) vs delta-gamma hedge (tangent parabola) fit against the convex call payoff.*

9.3.2 7.2.2 Framing A — two-instrument replication

The structural view begins from the replicating-portfolio argument of Chapter 6. We are *short* one unit of a target claim g; we seek a self-financing portfolio whose value tracks g at every (t, S). Against a delta hedge alone, the portfolio value has a non-zero $(\Delta S)^2$ coefficient: the gamma mismatch is a distinct source of risk that cannot be killed by any share position, because the stock has zero gamma. To absorb the gamma we must add a second *curved* instrument — a second traded claim whose own $\partial_{SS}h$ is non-zero.

Let $h(t, S)$ be another traded option on S — typically vanilla, liquid, and chosen to have gamma at the same scale as the target's. The hedging portfolio is

$$V(t, S) \;=\; \alpha_t\, S \;+\; \beta_t\, M_t \;+\; \gamma_t\, h(t, S), \tag{7.6}$$

where α_t is the number of shares, γ_t is the number of units of the auxiliary option, β_t is the number of units of the money-market account $M_t = e^{rt}$, and the hedge is set up so that V matches the target g claim-for-claim. Applying a Taylor expansion around S_t to both the portfolio and the target,

$$V(t, S) \;=\; \alpha_t\, S \;+\; \beta_t\, M_t \;+\; \gamma_t\big[h(t, S_t) + (S - S_t)\,\partial_S h(t, S_t) + \tfrac{1}{2}(S - S_t)^2\,\partial_{SS}h(t, S_t)\big] + \cdots, \tag{7.7}$$

$$g(t, S) \;=\; g(t, S_t) \;+\; (S - S_t)\,\partial_S g(t, S_t) \;+\; \tfrac{1}{2}(S - S_t)^2\,\partial_{SS}g(t, S_t) \;+\; \cdots. \tag{7.8}$$

Matching coefficients of $(S - S_t)$ and $(S - S_t)^2$ between the portfolio and the target gives the *delta-gamma hedge conditions*:

$$\alpha_t \;+\; \gamma_t\,\partial_S h(t, S_t) \;=\; \partial_S g(t, S_t), \tag{7.9}$$

$$\gamma_t\,\partial_{SS}h(t, S_t) \;=\; \partial_{SS}g(t, S_t). \tag{7.10}$$

Denote $\Delta_t^h = \partial_S h$, $\Gamma_t^h = \partial_{SS}h$, and Δ_t^g, Γ_t^g similarly. Solving (7.9)–(7.10) for the two unknowns,

$$\boxed{\gamma_t \;=\; \frac{\Gamma_t^g}{\Gamma_t^h}, \qquad \alpha_t \;=\; \Delta_t^g \;-\; \frac{\Gamma_t^g}{\Gamma_t^h}\,\Delta_t^h.} \tag{7.11}$$

These are the two hedge ratios. Read (7.11) as a two-step recipe: first fix γ_t so that the auxiliary option's gamma times γ_t matches the target gamma — that is gamma-matching, equation (7.10); then set α_t so that the *net* delta of the γ_t-scaled auxiliary plus α_t-scaled stock matches the target delta — that is delta-matching, equation (7.9), solved after γ_t is pinned down. The order matters because gamma-matching uses only h (the stock has zero gamma), while delta-matching uses both h and S. One cannot delta-hedge first and then gamma-hedge, because the gamma-hedge step would disturb the delta that had already been arranged. The correct order — gamma first, then delta — is the operational lesson of the pair (7.11).

With α_t and γ_t in hand, the cash leg follows from the total portfolio value. At inception, we sold g and collected g_0; we bought γ_0 units of h at h_0 and α_0 shares at S_0; whatever is left we deposit in the bank, so

$$M_0 \;=\; g_0 \;-\; \alpha_0\, S_0 \;-\; \gamma_0\, h_0. \tag{7.12}$$

This is the delta-gamma analogue of the delta-only initial-balance recipe from §6.11.2: the premium collected pays for both the share-hedge and the option-hedge, and the residual sits in the bank earning the short rate. Depending on the sign of γ_0 and the price of h, the initial bank balance may be positive (net lender) or negative (net borrower). For a short position in a short-dated at-the-money call hedged with a short-dated different-strike call, typically $\gamma_0 > 0$ (we are *long* gamma in the auxiliary to offset our *short* gamma in the target), which means we pay for h and the initial bank balance shrinks by $\gamma_0 h_0$ relative to the delta-only hedge.

Verifying that the resulting portfolio matches both delta and gamma is a one-line check:

$$\Delta_t^V \;=\; \alpha_t \;+\; \gamma_t\,\Delta_t^h \;=\; \Delta_t^g, \qquad \Gamma_t^V \;=\; \gamma_t\,\Gamma_t^h \;=\; \Gamma_t^g. \tag{7.13}$$

Both hold by construction. The portfolio tracks the target to second order, and the tracking error is now of order $(S - S_t)^3$ rather than $(S - S_t)^2$. Under a Brownian-motion underlying, the cubic term is of order $(\Delta t)^{3/2}$ per rebalance interval, substantially smaller than the Δt gamma residual of the delta-only hedge. Delta-gamma hedging is therefore genuinely higher-order: it does not merely shift the coefficient on the error, it shifts the *scaling exponent* — the variance of the tracking error now falls as $1/N^2$ in the rebalance count rather than $1/N$, and the standard deviation falls as $1/N$ rather than $1/\sqrt{N}$.

The structural story is therefore clean. A delta-only hedge replicates the target's Itô expansion only through the dW coefficient; the second-order dt piece proportional to Γ is unmatched. A delta-gamma hedge replicates *both* coefficients, by enlarging the hedging basis to include a second traded claim whose own expansion has the required curvature. Each additional Greek matched corresponds to one additional hedging instrument in the basis and one additional linear equation to solve; the arithmetic remains elementary, and the replication theorem of Chapter 6 generalises verbatim.

9.3.3 7.2.3 Framing B — local quadratic fit

The practical view starts from a slightly different position. A hedged book has a *loss distribution*, and the purpose of hedging is to compress that distribution toward zero — ideally a tight cluster around a risk-free return but more realistically a tight cluster around whatever expected alpha the strategy aims to extract. Delta hedging collapses the linear dependence on the underlying; delta-gamma hedging additionally collapses the quadratic dependence. Each additional Greek matched corresponds to a further reduction in the variance of the hedged P&L — and hence a smaller VaR and CTE at any confidence level.

The economic logic is worth stating plainly. When a derivatives dealer sells an option, the premium received is — in theory — exactly the expected discounted cost of replicating the option's payoff under a specific hedging strategy. The hedging strategy itself is the *justification* for the premium: "we charge g_0 because if we trade according to these rules, we can deliver $\varphi(S_T)$ for exactly g_0 plus a small residual that vanishes in the limit of continuous rebalancing." This is the Black-Scholes promise, and it is the foundation of the entire derivatives industry. Delta hedging is the first-order implementation of this promise; delta-gamma is a refinement that tightens the residual by neutralising the first significant source of hedging error — namely the second-order Taylor term in the payoff function.

What shapes the residual? Every discrete-time hedging error can be traced to the gap between the continuous-time idealisation and the real-world rebalance grid. If the rebalance happens at times $t_0 < t_1 < ... < t_N$, the hedge is perfect *at* each rebalance time but drifts *between* rebalance times as the underlying price moves without the hedge being adjusted. The size of the drift depends on the second derivative of the payoff (gamma) times the square of the price move. Summed over many small intervals, this drift is the leading source of residual P&L variance. Delta-gamma hedging neutralises this leading source — at the cost of introducing a gamma-dependent cash flow (because the gamma hedge itself has to be rebalanced when gamma changes, which happens as spot moves and as time passes).

A slightly deeper view is that hedging is not really about eliminating risk; it is about *exchanging* one kind of risk for another. A delta-only hedge converts directional price risk into gamma-residual risk. A delta-gamma hedge converts gamma-residual risk into (smaller) third-order-Taylor-residual risk plus execution risk plus model risk (because gamma itself depends on volatility, which you may have mis-specified). The deeper you go into higher Greeks, the smaller the direct residual but the more layered the model-dependence. Professional traders have a keen sense of which residual risks are tolerable (those that net out across many similar trades) and which are not (those that are directional and correlated across the book). The art of running a derivatives desk is not to hedge everything; it is to hedge the dangerous things and leave the safe things alone.

Visualising a call payoff $\varphi(x) = (x - K)_+$ and its smoothed (time-t) price $g(t, x)$ together — with the discounted-strike asymptote $(x - Ke^{-r(T-t)})$ drawn for reference, and the horizontal axis marked K, x_t, $x_{t+\Delta t}$ — the Taylor expansion in Δx is

$$g(t, S_t + \Delta S) \;=\; g(t, S_t) \;+\; \Delta S\;\underbrace{\partial_S g(t, S_t)}_{\text{delta }\Delta} \;+\; \tfrac{1}{2}(\Delta S)^2\;\underbrace{\partial_{SS} g(t, S_t)}_{\text{gamma }\Gamma} \;+\; \cdots. \tag{7.14}$$

The higher-order terms are the remaining Greeks — vega, theta, vanna, volga, speed, and so on.

We hedge a short position in the target claim g using the stock S_t, the money-market account M_t, and a second traded claim h_t whose Greeks we know. With units α_t of S, β_t of M, and γ_t of h, the hedge portfolio is

$$V_t \;=\; \alpha_t\, S_t \;+\; \beta_t\, M_t \;+\; \gamma_t\, h_t, \tag{7.15}$$

and we *fit* its spatial derivatives at S_t to those of the target. The first-derivative match is

$$\partial_S V_t \;=\; \alpha_t \;+\; \gamma_t\, \partial_S h \;=\; \Delta^g, \tag{7.16}$$

and the second-derivative match is

$$\partial_{SS} V_t \;=\; \gamma_t\, \partial_{SS} h \;=\; \Gamma^g, \tag{7.17}$$

since the stock contributes 0 to gamma ($\partial_{SS}(\alpha_t S_t) = 0$) and the bank contributes 0 to both delta and gamma. Only the auxiliary claim h carries curvature, which is why we need it.

The linear system (7.16)–(7.17) has a particularly clean matrix form:

$$\begin{pmatrix} 1 & \Delta^h \\ 0 & \Gamma^h \end{pmatrix} \begin{pmatrix} \alpha_t \\ \gamma_t \end{pmatrix} \;=\; \begin{pmatrix} \Delta^g \\ \Gamma^g \end{pmatrix}. \tag{7.18}$$

The matrix is upper-triangular — the stock contributes only to delta (upper-left entry 1, lower-left 0) and the auxiliary option contributes to both delta and gamma. Inverting the triangular matrix,

$$\begin{pmatrix} \alpha_t \\ \gamma_t \end{pmatrix} \;=\; \begin{pmatrix} 1 & -\Delta^h/\Gamma^h \\ 0 & 1/\Gamma^h \end{pmatrix} \begin{pmatrix} \Delta^g \\ \Gamma^g \end{pmatrix}, \tag{7.19}$$

which recovers exactly the boxed pair from Framing A:

$$\gamma_t \;=\; \frac{\Gamma^g}{\Gamma^h}, \qquad \alpha_t \;=\; \Delta^g \;-\; \frac{\Gamma^g}{\Gamma^h}\, \Delta^h. \tag{7.20}$$

The choice of auxiliary claim h deserves a moment of thought. In practice h is almost always a liquid, exchange-traded vanilla option — an at-the-money listed call or put with a short expiry is the classic choice because it has high gamma per dollar of premium and a tight bid-ask. A deep-in-the-money option would work mathematically but has low gamma, so large notional amounts are required to match the target's gamma; that increases funding and transaction costs. A far-out-of-the-money option would also work but its gamma changes rapidly with spot, so the hedge weights would rebalance aggressively and incur turnover. The ATM short-dated option threads the needle: high gamma, moderate turnover, low execution cost.

There is a subtler point about the *cross-Greek profile* of h. When we hedge a target g whose gamma and vega (and cross-Greek vanna, say) both matter, choosing h with the same gamma profile does not automatically match the vega profile. In fact a single auxiliary claim can at best match one higher-order Greek at a time; matching both gamma and vega requires two auxiliary claims, typically two listed options at different strikes or expiries whose gamma and vega profiles are linearly independent. This is the operational reason that serious exotic-hedging desks maintain entire *option-book hedging portfolios* consisting of dozens of liquid vanillas, not just one or two: the target exotic has a many-dimensional Greek profile and needs a high-dimensional hedging basis to match it accurately. The linear algebra is essentially a least-squares fit of the target's Greek vector onto the span of the available hedging-instrument Greek vectors.

When the Greek matrix is nearly singular — which happens when the hedging instruments have similar Greek profiles — the hedge weights become numerically unstable and can flip signs or blow up in magnitude. This is a common pitfall in automated hedging systems: two ATM options with nearby expiries might have nearly parallel Greek vectors, so their contributions cannot be cleanly separated, and tiny changes in Greeks produce enormous changes in hedge ratios. The operational fix is to regularise the system (ridge regression on the hedge solution), or to pick a hedging basis with intentionally diverse Greek profiles (options at widely different strikes and expiries). Diversity in the hedging basis gives a well-conditioned matrix and stable hedge weights.

A common conceptual error is to forget the second term in (7.20) — to compute the delta hedge as $\alpha = \Delta^g$ and then bolt on a gamma-matching layer of h. That is internally inconsistent because the h position itself has a delta; one has over-delta-hedged by exactly $\gamma_t \, \Delta^h$. The correct construction is simultaneous: pick γ_t to zero the portfolio's gamma, then pick α_t to zero the *residual* delta after accounting for h's delta contribution.

9.3.4 7.2.4 Reconciling the two framings

The two derivations above — two-instrument replication (Framing A) and local quadratic fit (Framing B) — end at the same boxed pair of hedge ratios (7.11) and (7.20). That is not a coincidence. The agreement is a consequence of a deep duality: requiring the hedged portfolio's *Itô dynamics* to match the target's is the same as requiring the hedged portfolio's *spatial Taylor expansion* to match the target's, *in the GBM-with-constant-coefficients setting we have been working in.*

To see why, note that for any smooth $V(t, S_t)$ Itô's lemma delivers

$$\mathrm{d}V \;=\; \left(\partial_t V + \mu\, S_t\, \partial_S V + \tfrac{1}{2}\sigma^2 S_t^2\, \partial_{SS} V\right) \mathrm{d}t \;+\; \sigma\, S_t\, \partial_S V\, \mathrm{d}W_t. \tag{7.21}$$

The coefficient of $\mathrm{d}W_t$ is σS_t times $\partial_S V$, i.e. delta. The coefficient of $\mathrm{d}t$ has $\partial_{SS} V$ buried inside the convexity term. Matching $\mathrm{d}V = \mathrm{d}g$ at the Itô level therefore requires matching *both* $\partial_S V = \partial_S g$ (the $\mathrm{d}W$ coefficient) and $\partial_{SS} V = \partial_{SS} g$ (via the convexity term in $\mathrm{d}t$). Those are precisely the two conditions of Framing B. Conversely, Framing B's spatial Taylor match at $S = S_t$ produces a portfolio whose $\mathrm{d}W$ and convexity contributions match the target's at S_t, which is the Framing-A condition.

The cleanest way to state the equivalence is this: in a single-factor Itô model, matching the portfolio's *first and second spatial derivatives* to the target's is equivalent to matching the coefficients of $\mathrm{d}W_t$ and the ∂_{SS} part of the $\mathrm{d}t$-term in the Itô expansion. The ∂_t and $\mu\, \partial_S$ pieces of the $\mathrm{d}t$-term are already absorbed by the pricing PDE (they are what the cash leg pays for), which is why the system reduces to just two equations in two unknowns.

An equivalent way to see it: the hedging portfolio has three degrees of freedom $(\alpha_t, \beta_t, \gamma_t)$. The self-financing condition and the matching of the *value* $V_t = g_t$ (which pins β_t) together eliminate one degree of freedom and one constraint. We are left with two degrees of freedom (α_t, γ_t) and two constraints (delta and gamma match). The system is exactly determined; the solution is unique; and it is what both framings produce.

There is one structural consequence of this duality worth calling out. Framing A makes it immediate that the delta-gamma hedge is **exact in the continuous-time limit** — the Itô dynamics of the replicating portfolio match the target claim's, and the tracking error goes to zero almost surely as the rebalance grid refines. Framing B makes it immediate that under *discrete* rebalancing the leading residual scales as $(\Delta S)^3$, and so has variance of order $(\Delta t)^3$ per interval and $(\Delta t)^2$ when summed over $N = T/\Delta t$ intervals. The standard deviation therefore scales as $1/N$ rather than $1/\sqrt{N}$ as in the delta-only case. The two framings reinforce each other: one tells us that the hedge is theoretically exact, the other tells us how fast the discretisation error vanishes.

We will use Framing B's linear-algebra presentation for all the computations that follow, because it generalises cleanly to n-Greek hedging with n traded instruments, and we will invoke Framing A's structural argument whenever we need to justify that the resulting construction is self-financing and replicates the target in the no-arbitrage sense.

9.3.5 7.2.5 Worked numerical example — gamma-matching an ATM option

Let us make (7.15) concrete with specific numbers. Suppose we sell a 1-year ATM call on a \$100 stock: strike $K_g = 100$, expiry $T_g = 1$ year, implied vol $\sigma = 25\%$, rate $r = 3\%$. Using Black-Scholes we get price $g_0 \approx 10.94$, $\Delta^g \approx 0.59$, $\Gamma^g \approx 0.0156$. To gamma-hedge we use a 3-month ATM call on the same underlying: strike $K_h = 100$, expiry $T_h = 0.25$ year, same vol and rate. Its price is $h_0 \approx 5.40$, $\Delta^h \approx 0.53$, $\Gamma^h \approx 0.0313$.

Applying (7.20),

$$\gamma_0 \;=\; \frac{\Gamma^g}{\Gamma^h} \;=\; \frac{0.0156}{0.0313} \;\approx\; 0.498. \tag{7.22}$$

We buy roughly half a unit of the 3-month call for every 1-unit short position in the 1-year call. Then

$$\alpha_0 \;=\; \Delta^g \;-\; \frac{\Gamma^g}{\Gamma^h}\, \Delta^h \;=\; 0.59 \;-\; 0.498 \cdot 0.53 \;\approx\; 0.326. \tag{7.23}$$

We buy 0.326 shares. The initial outlay for the hedge is

$$\alpha_0 \cdot S_0 \;+\; \gamma_0 \cdot h_0 \;=\; 0.326 \cdot 100 \;+\; 0.498 \cdot 5.40 \;=\; 32.6 \;+\; 2.69 \;=\; 35.29. \tag{7.24}$$

We received $g_0 = 10.94$ for selling the 1-year call, so the cash account starts at

$$M_0 \;=\; g_0 \;-\; \alpha_0 S_0 \;-\; \gamma_0 h_0 \;=\; 10.94 \;-\; 35.29 \;=\; -24.35. \tag{7.25}$$

Meaning we borrow \$24.35 to establish the hedge. Over the next rebalance period we pay interest on this borrowing at rate r, and the stock and option positions mark to market at whatever new spot and option price the market delivers.

Compare this with a pure delta hedge: $\alpha_0 = \Delta^g = 0.59$ shares, cash account $10.94 - 0.59 \cdot 100 = -48.06$. The delta-gamma hedge requires *less* borrowing (\$24.35 vs \$48.06) because the gamma-hedging option supplies some of the delta. But the delta-gamma hedge carries the 3-month option position, which must be rolled or rebalanced as it ages toward its own expiry; it introduces its own theta decay and vega exposure. The trade-off is that delta-gamma reduces residual variance at the cost of a more complex rebalancing schedule and additional instrument-specific Greeks that must themselves be monitored.

If we repeat this exercise with a 1-month ATM hedging option instead of a 3-month one, the gamma is higher (roughly 0.054 vs 0.031), so we need fewer units: $\gamma_0 = 0.0156/0.054 \approx 0.289$. The higher-gamma hedge is more efficient in notional terms — less option notional required to match the target's gamma — but also rebalances faster because its own gamma drifts quickly with spot and time. The choice between a 1-month and 3-month hedging option is a trade-off between notional efficiency and rebalancing cost, typically resolved by desk policy.

A *numerical sanity check* on the local-fit quality. For our sold 1-year ATM call and a 2% spot move ($\Delta S = 2$), the true price change is about 1.15. The delta-only approximation predicts $\Delta^g \cdot \Delta S = 0.59 \cdot 2 = 1.18$ (a 2.6% relative error, which is coincidentally tight because the gamma and theta contributions partly offset). The delta-gamma approximation predicts $0.59 \cdot 2 + \frac{1}{2} \cdot 0.0156 \cdot 4 = 1.21$ — in this *particular* window the quadratic fit and linear fit are both within a cent of the truth, because the 1-year option has modest gamma. For a 3-month ATM call, the gamma is nearly twice as large; for a 1-month ATM call it is three to four times larger. At a 10% move, the quality of the two approximations diverges sharply: delta-only misses the truth by several percent, delta-gamma matches to sub-percent accuracy. Delta-gamma is a superb *local* hedge and a mediocre *global* one; it is designed to be refreshed frequently on a grid rather than held for a long interval.

The failure mode for large moves has a systematic character. Near the strike, gamma is positive and large; it decays for moves far in either direction. A quadratic extrapolation assumes gamma remains at its spot-level value even as the stock moves far from the spot, but gamma *itself* decays as we move away from the strike. This makes the delta-gamma quadratic an over-estimate of convexity in the wings, leading to systematic over-prediction of price changes for large moves. This is precisely the scenario in which third-order (speed) corrections start to matter: speed captures the rate at which gamma decays, and including it pulls the quadratic toward the true curve in the wings. For risk-measurement purposes — particularly VaR and stress-testing at 3-to-5-sigma confidence — the "delta-gamma-parametric" methodology has to be augmented by full revaluation when the scenarios push far into the wings.

9.3.6 7.2.6 The discrete delta-gamma recursion

When continuous rebalancing is discretised, the analogue of the delta-only cash-account recursion of §6.11 for the two-instrument hedge is

$$M_{t_k} \;=\; M_{t_{k-1}}\, e^{r\,\Delta t} \;-\; (\alpha_{t_k} - \alpha_{t_{k-1}})\, S_{t_k} \;-\; (\gamma_{t_k} - \gamma_{t_{k-1}})\, h_{t_k}, \tag{7.26}$$

where each rebalance at time t_k requires buying or selling enough stock to restore the delta match, *and* enough of the auxiliary option to restore the gamma match. Both trades are funded out of the bank account (or deposited into it), and the bank grows between rebalances at the risk-free rate. The structure is identical to the pure-delta recursion but with an extra h-trading term on the right-hand side. Transaction costs, if included, are simply summed over the two legs — we take up the details in §7.5.

Equation (7.26) is the operational engine of any delta-gamma hedging program. Each morning (or each rebalance tick) the trader observes the new stock price S_{t_k} and option price h_{t_k}, recomputes the desired hedge weights α_{t_k} and γ_{t_k} from (7.20) using the updated Greeks, and executes the increments $\alpha_{t_k} - \alpha_{t_{k-1}}$ in the stock and $\gamma_{t_k} - \gamma_{t_{k-1}}$ in the option. The cash flow from these executions debits the bank account. Between rebalances the bank account compounds at the short rate.

The *absolute* size of each rebalance trade depends on how much the Greeks have drifted since the last rebalance, which in turn depends on how much the stock has moved, how much time has passed, and how curved the target's Greek profile is. A near-the-money, short-dated option has delta and gamma that change rapidly with spot, so its hedge rebalances frequently and the cash account sees a lot of churn. A far-dated, deep-in-the-money option has an almost linear payoff and a nearly constant delta, so the hedge barely moves and rebalance costs are low. These observations underpin the whole *gamma scalping* school of proprietary option trading: taking long-gamma positions on short-dated ATMs and systematically extracting the rebalance cash-flow turnover as P&L, funded by theta decay.

One useful mental model of the rebalance cash flow is to think of it as a *betting ledger* between the desk and the market. Each morning, the stock has moved; the desk's required delta has changed; the desk executes the delta change at the current stock price; and that execution booked a cash flow. If the stock moved up and the desk had to buy more stock to re-hedge, the desk pays for stock at a higher price than yesterday's average — so the rebalance has a small negative cash flow relative to zero. If the stock moved up and the desk had to sell stock (long-gamma position), the desk receives cash at the higher price and the rebalance has a positive cash flow. The aggregate pattern of positive and negative rebalance cash flows over the option's life is exactly the *gamma P&L*, and its magnitude is $\frac{1}{2}\Gamma(\Delta S)^2$ per rebalance, summed over all rebalances, minus the theta bleed. This is the famous "buy low, sell high" mechanism that underlies gamma scalping — and its mirror image, the "sell high, buy low" pain of being short gamma in a trending market.

At maturity $t_N = T$, the portfolio wealth owes the option payoff:

$$P_N \;=\; M_{t_{N-1}}\, e^{r\,\Delta t} \;+\; \alpha_{t_{N-1}}\, S_{t_N} \;+\; \gamma_{t_{N-1}}\, h_{t_N} \;-\; \varphi(S_{t_N}), \tag{7.27}$$

and the expression in parentheses collapses, if the hedge tracked perfectly, to $\varphi(S_{t_N})$, leaving $P_N = 0$. In a well-specified model with no transaction costs and continuous rebalancing, (7.27) is identically zero; in reality, it is a small random variable whose distribution is tighter than the analogous delta-only P&L because the leading gamma error has been removed. We quantify the tightening next.

9.3.7 7.2.7 Terminal P&L and residual scaling

At expiry $t_N = T$ we liquidate the hedge and deliver the claim payoff $\varphi(S_T)$. The realised profit-and-loss of the delta-gamma hedged book is

$$\boxed{\text{P\&L} \;=\; M_{t_{N-1}}\, e^{r\,\Delta t} \;+\; \alpha_{t_{N-1}}\, S_{t_N} \;+\; \gamma_{t_{N-1}}\, h_{t_N} \;-\; \varphi(S_{t_N}).} \tag{7.28}$$

Under a continuous-time, frictionless, complete-market set-up this quantity is identically zero; under discrete rebalancing it is a random variable whose variance shrinks as the grid is refined. The delta-gamma hedge has an order-of-magnitude smaller P&L variance than a pure delta hedge because it neutralises the quadratic term in the Taylor expansion — the leading residual is cubic in ΔS rather than quadratic.

The *scaling argument* is worth carrying through carefully, because it is the quantitative payoff of all the extra machinery. A delta-only hedge leaves a residual that is quadratic in the stock move per rebalance interval. The variance of that residual scales as the variance of $(\Delta S)^2$, which for GBM is of order $(\sigma^2 \Delta t)^2 \cdot S^4$. Summing over N rebalance intervals of length $\Delta t = T/N$, the total residual variance scales as

$$N \cdot \left(\sigma^2 \tfrac{T}{N}\right)^2 \;=\; \frac{\sigma^4 T^2}{N}. \tag{7.29}$$

The standard deviation therefore scales as $1/\sqrt{N}$ — slow convergence. A delta-gamma hedge leaves a residual that is *cubic* in the stock move per interval. Its variance scales as the variance of $(\Delta S)^3 \sim S^6 \sigma^6 (\Delta t)^3$. Summing,

$$N \cdot \left(\sigma^6 \tfrac{T^3}{N^3}\right) \;=\; \frac{\sigma^6 T^3}{N^2}, \tag{7.30}$$

with standard deviation scaling as $1/N$. So on a weekly grid, the delta-gamma standard deviation is a factor $\sqrt{N}$ smaller than the delta-only standard deviation; for $N = 52$ weekly rebalances, that is roughly a factor of 7 reduction. The difference between $1/\sqrt{N}$ and $1/N$ convergence is the practical justification for the extra hedging instrument.

Hedging P&L distribution — delta-only vs delta-gamma *Figure 7.2 — Monte-Carlo terminal-P&L density under weekly rebalancing: delta-gamma hedge tightens variance by roughly* $2\times$ *relative to pure delta, with a tighter tail and lighter skew.*

The CTE-compression bonus is connected. Hedging reshapes the loss distribution from chi-squared-heavy (delta-only, $(\Delta S)^2$ residual) to near-Gaussian (delta-gamma, $(\Delta S)^3$ residual that tends to cancel across rebalances), so the CTE of the hedged P&L falls by roughly the square of the factor by which the standard deviation falls. Chapter 15 quantifies this and the Cornish-Fisher corrections it implies.

9.3.8 7.2.8 Trading the variance risk premium

Pure delta hedging — deliberately *leaving* gamma exposure — is sometimes the trade, not a mistake. A long-gamma, delta-hedged option position has P&L over a small interval

$$\Delta P \;\approx\; \tfrac{1}{2}\Gamma\, S^2\, (r_t^2 \;-\; \sigma_{\text{imp}}^2\, \Delta t), \tag{7.31}$$

where r_t is the realised log-return and σ_{imp} the implied vol the option was bought at. Aggregated over the option's life, this is a *realised-vs-implied variance* trade; its expected P&L is the *variance risk premium* — the systematic compensation paid to long-vol traders for bearing the tail risk of vol spikes. In equity-index options, implied exceeds realised by two to four points on average, so short-vol ("gamma-scalping the short side") earns small consistent profits punctuated by catastrophic losses during spikes (Aug 2011, Feb 2018, Mar 2020). The choice between flattening gamma and warehousing it is the difference between a flow desk (wants flat, books bid-ask) and a vol desk (wants the exposure, monetises the VRP).

9.3.9 7.2.9 Why two Greeks — and why not three?

A natural question: if delta-gamma hedging neutralises the first and second spatial derivatives of the target price, why not add a third option and neutralise the third derivative (*speed*) as well? Why stop at two?

The answer is partly practical and partly principled. *Practically*, each additional Greek we target requires an additional liquid traded instrument whose own higher-order Greeks are independent of the others' — and the option-market's supply of liquid, uncorrelated higher-Greek vehicles is limited. A third-order hedge would in principle require a third option whose *speed* (third spatial derivative of price) is appreciable and whose delta and gamma can be matched simultaneously, and in liquid vanilla markets the three-dimensional system is often ill-conditioned.

Principled, the magnitude of successive spatial derivatives decays rapidly under a diffusion: the third-order term in the Taylor expansion is multiplied by $(\Delta S)^3 \sim (\Delta t)^{3/2}$, which is already small, and the marginal variance reduction per unit of additional hedging effort is modest. Most desks stop at delta-gamma; a few specialised books (volatility arbitrage, barrier-option market-making) add vega and sometimes vanna and volga (cross-derivatives with volatility) to handle risk that is orthogonal to spot moves.

The third-order term worth naming is *speed*: $\partial_{SSS}g$. Speed measures how gamma changes with spot — effectively the "gamma of gamma." In the regions where gamma is changing rapidly (near the strike as expiry approaches), speed is large, and a delta-gamma hedge that ignores speed will have a rebalancing schedule that is optimised for the wrong curvature as spot moves. Exotic desks sometimes match speed as a third-layer hedge using a third traded option with a different gamma profile, but this is rare because the operational complexity outweighs the residual-risk reduction for most portfolios. The theoretical observation is that matching delta, gamma, and speed reduces the leading residual error from cubic to quartic in the stock move, shaving yet another order off the residual variance.

A related fourth-order Greek is *colour* (also called *gamma decay* or $\partial_{SSt}g$), measuring how gamma changes with time. Colour matters most for short-dated options where gamma concentrates rapidly; a colour-hedged position stays gamma-neutral across the trading day even as time decays. Colour is rarely hedged because time passes deterministically — one can simply rebalance more frequently as expiry approaches rather than add colour-hedging instruments. Nevertheless, colour is reported on most desk-level Greek sheets as a monitoring metric.

The full Greek catalogue (delta, gamma, vega, theta, rho plus the cross-Greeks vanna, volga, charm, colour, speed) was listed in §7.1; each business's *Greek fingerprint* dictates which subset matters in practice. FX desks emphasise vega/vanna (skew); equity-index vol desks add volga; single-name desks add basis to the index hedge; rates-swaptions desks add curve exposures.

9.4 7.3 Vega and volatility risk

The Black-Scholes framework takes the volatility σ as a constant parameter. In the market, volatility is a living number: it changes day to day, bar to bar, and sometimes intrabar. The sensitivity of an option's price to changes in the assumed volatility is called *vega*:

$$\mathcal{V}_t \;\equiv\; \partial_\sigma g(t, S_t; \sigma). \tag{7.32}$$

Equivalently, vega measures by how many cents the option's price rises if the volatility input moves up by one percentage point. Vega is strictly speaking the most awkward of the first-order Greeks because σ is not itself a traded quantity — the underlying trades, the option trades, but no one trades "volatility" directly except through volatility-linked contracts (variance swaps, VIX futures). When we write $\mathcal{V}_t = \partial_\sigma g$ we are differentiating a pricing *formula* with respect to one of its parameters, not a market process. The conceptual leap from "assumed-sensitivity" to "tradable risk" is what makes vega non-trivial and what motivates the extensions to stochastic volatility in §7.3.2.

9.4.1 7.3.1 Vega of a vanilla Black-Scholes call

For a European call under Black-Scholes, vega has a clean closed form,

$$\mathcal{V}_t \;=\; S_t\,\sqrt{T-t}\,\Phi'(d_+), \tag{7.33}$$

where $d_+ = [\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)(T-t)]/(\sigma\sqrt{T-t})$ is the usual moneyness index and $\Phi'(x) = (2\pi)^{-1/2}e^{-x^2/2}$ is the standard normal density. Two features of (7.33) are worth flagging.

First, vega is positive for *every* vanilla option — both calls and puts gain value when volatility rises, because higher volatility means more upside (for calls) and more downside (for puts) on the payoff distribution, and the forward value of each is sensitive to the mass in its own tail. This is a universal property of convex payoffs: a convex function of a random variable is monotone in the spread of the random variable, so increasing the volatility of the underlying increases the expected payoff. Puts and calls being convex in S_T, their forward prices rise with volatility.

Second, vega peaks at the at-the-money strike and decays to zero for deep in-the-money or deep out-of-the-money options. An ATM short-dated call has the largest vega per unit of option; a deep-ITM LEAPS has essentially none.

Vega decay as expiry approaches *ATM vega as a function of time-to-expiry (x-axis inverted so time runs forward toward expiry on the right). Vega scales as $\sqrt{T-t}$ and collapses to zero at expiry — the option has no volatility sensitivity once the payoff is locked in. This is why short-dated ATM options are the preferred instrument for* discharging* vega risk (low cost per unit) and long-dated ATM options for *accumulating* it.*

Vega heatmap across spot and time-to-expiry *Vega $\mathcal{V} = S\sqrt{T-t}\,\Phi'(d_+)$ plotted over (S,T). The bright band runs along the at-the-money ridge; vega grows with $\sqrt{T-t}$ at fixed moneyness, so long-dated ATM options carry the largest vega per contract. This is why "volatility duration" is a long-dated ATM product by construction.*

That pattern — vega concentrated near the ATM strike — is the reason straddles and strangles are the prototypical *volatility trades*: they maximise vega exposure per unit of capital deployed. A long straddle (long ATM call + long ATM put) has vega equal to twice that of a single ATM option while its delta is near zero by symmetry, giving the maximum "volatility-per-unit-of-directional-exposure" of any simple vanilla combination.

A useful numerical benchmark to keep in mind: at the money, the density $\Phi'(0) = 1/\sqrt{2\pi} \approx 0.4$. For a \$100 stock and a three-month option ($\sqrt{0.25} = 0.5$),

$$\mathcal{V}_t \;\approx\; 100 \cdot 0.5 \cdot 0.4 \;=\; 20 \text{ cents per volatility point.} \tag{7.34}$$

If volatility moves by 1 point in a day (a large but not extraordinary move), the option's price changes by 20 cents — a substantial fraction of the option's total premium for short-dated contracts. Vega risk is the second-largest source of P&L volatility for most options books, after delta-unhedged direction.

9.4.2 7.3.2 Volatility as a hidden state variable

Strictly speaking, (7.32) is an *assumed-sensitivity* measure — the model takes σ as constant, so differentiating with respect to it produces a derivative of a formula, not of a reality. The conceptual leap needed to turn vega into a tradable hedge is to admit that volatility is *itself* a stochastic process. Once we adopt a stochastic-volatility model (Heston, SABR, local-vol surfaces — the subject of Chapter 10), the option price is a function $g(t, S_t, \sigma_t)$ of three state variables, and Itô's lemma applied to this richer function picks up a $\partial_\sigma g \cdot \mathrm{d}\sigma_t$ term that did not appear in the constant-volatility treatment.

The multi-dimensional Itô expansion reads

$$\begin{aligned} \mathrm{d}g_t \;&= \partial_t g \, \mathrm{d}t \;+\; \partial_S g \, \mathrm{d}S_t \;+\; \partial_\sigma g \, \mathrm{d}\sigma_t \\ &\quad+\; \tfrac{1}{2}\partial_{SS} g \, \mathrm{d}\langle S, S\rangle_t \;+\; \partial_{S\sigma} g \, \mathrm{d}\langle S, \sigma\rangle_t \;+\; \tfrac{1}{2}\partial_{\sigma\sigma} g \, \mathrm{d}\langle \sigma, \sigma\rangle_t. \end{aligned} \tag{7.35}$$

The new drift term contains a *vega times drift of volatility* piece, a *cross-gamma* $\partial_{S\sigma} g \cdot \mathrm{d}\langle S, \sigma\rangle_t$ piece that captures the correlation between spot and vol moves, and a *volga* $\frac{1}{2}\partial_{\sigma\sigma} g \cdot \mathrm{d}\langle \sigma, \sigma\rangle_t$ piece that captures the convexity of the option in the volatility direction. Each of these is a distinct risk that the replicating portfolio must accommodate.

The upshot is that in a stochastic-volatility world, a pure delta-hedged portfolio is *not* self-financing: it leaks value through the vega leg every time volatility drifts, and it leaks value through the cross-gamma leg every time spot and vol move together. To close the replication argument one must enlarge the hedging set yet again to include a second option traded on the same underlying, matching both the target's vega *and* its vega-weighted gamma. This *three-instrument portfolio* (stock, cash, first option, second option) is how sophisticated options desks run in practice, and it is why the book of any real market-maker is two-dimensional in its Greek profile: a grid of delta and vega exposures rather than a single delta number.

The linear-algebra generalisation of the delta-gamma system (7.18) is immediate. If we wish to match delta, gamma, and vega simultaneously, we need three hedging instruments beyond cash — the stock (which contributes to delta only) and two options h_1, h_2 with different Greek profiles. The match equations are

$$\begin{pmatrix} 1 & \Delta^{h_1} & \Delta^{h_2} \\ 0 & \Gamma^{h_1} & \Gamma^{h_2} \\ 0 & \mathcal{V}^{h_1} & \mathcal{V}^{h_2} \end{pmatrix} \begin{pmatrix} \alpha_t \\ \gamma_t^1 \\ \gamma_t^2 \end{pmatrix} \;=\; \begin{pmatrix} \Delta^g \\ \Gamma^g \\ \mathcal{V}^g \end{pmatrix}, \tag{7.36}$$

and the system has a unique solution whenever the 2×2 sub-matrix of $(\Gamma, \mathcal{V})$ of the two hedging options is non-singular — i.e. whenever h_1 and h_2 have *linearly independent* gamma-vega profiles. Inverting the matrix gives the hedge weights in closed form. A diligent reader will reproduce (7.20) as the 2×2 sub-case when vega matching is dropped.

9.4.3 7.3.3 Vega hedging in practice

On a working desk, the operational story is this. A vanilla-option market-maker inventories options across a range of strikes and maturities. Each option has its own vega. The sum of vegas across the book is the *total vega* exposure, measured in dollars per volatility point. Desks typically run vega limits by tenor bucket — "front-month vega, X; second-month vega, Y" — because vega at different tenors is priced off different points on the implied-volatility curve and may move in opposite directions. When a desk's total vega exceeds its limit in any bucket, the risk manager asks for vega to be reduced — either by closing positions in that bucket or by putting on an offsetting position in another option of similar vega.

A common hedging tactic is to offset vega with a different-tenor option on the same underlying, exploiting the different time-dependence of vega. A long one-month straddle and a short three-month straddle with appropriate sizing will produce a near-vega-neutral position whose *vega term structure* is what the trader is really betting on.

This kind of *calendar-spread* structure is a standard vehicle for expressing views on the implied-vol curve without taking on direct vega risk.

Across strikes, the analogous structure is a *skew trade*: long an out-of-the-money put and short an out-of-the-money call (a "risk reversal") has near-zero vega at the ATM point but significant *vanna* exposure — the position makes money if the skew steepens and loses if it flattens. Skew trades are the bread and butter of equity-index volatility desks, where the persistent put-skew premium has been a reliable source of P&L for decades (subject to occasional sharp reversals during skew-collapse episodes).

Vega is not a single number. It is a surface — one vega per (strike, expiry) pair. Risk systems aggregate into *buckets* (e.g., 3M-ATM, 6M-25Δ-put) and a vega-neutral book has zero vega in every bucket, which typically requires two or three hedging options per bucket.

9.4.4 7.3.4 Higher-order volatility Greeks — volga, vanna, charm

The higher-order volatility Greeks — *volga* ($\partial_{\sigma\sigma}$, convexity in volatility), *vanna* ($\partial_{S\sigma}$, cross-sensitivity of delta to vol), and *charm* (∂_{St}, vega decay of delta) — become relevant for structured-product desks that sell options with unusual strike-vol dependencies (digitals, barriers, auto-callables, cliquets). These Greeks are not standard for a vanilla book but appear routinely in exotic-option valuations.

Volga measures the convexity of the option price in the volatility direction. Long volga positions profit when volatility spikes *regardless of direction*, because the second derivative of the option price in σ is positive near the ATM and decays in the wings. A volga-long, vega-flat book is betting that implied volatility will move but is agnostic about whether it moves up or down — a "vol-of-vol" bet. Barriers and cliquets typically have significant volga at the barrier or reset dates, and their hedging programs must include volga-neutralising vanilla options (typically a butterfly of three strikes) alongside the gamma and vega hedges.

Vanna measures the cross-sensitivity of delta to volatility — or equivalently, the cross-sensitivity of vega to spot. A positive-vanna position's delta increases when volatility rises; a negative-vanna position's delta falls. Skew-sensitive structures (risk reversals, down- and-out barriers) have concentrated vanna profiles, and hedging them requires asymmetric choices of strikes — typically out-of-the-money puts and calls at different deltas to produce the desired vanna without introducing vega or volga.

Charm (or $\partial\Delta/\partial t$) measures how delta drifts in time even when spot is constant. For an ATM short-dated call, charm is small; for an in-the-money or out-of-the-money option near expiry, charm can be substantial — it is the source of "delta decay" that a trader sees as her book's delta drifts overnight even though nothing in the market has moved. Charm is rarely hedged directly (since time passes deterministically and more frequent delta-rebalancing handles it), but it is prominently reported on overnight Greek sheets so that the opening-book delta is correctly anticipated.

Each of these Greeks carries its own operational story, and each demands a specific kind of hedging instrument to neutralise. The extension of the replicating-portfolio argument to stochastic volatility, jumps, and rough paths is the subject of entire later chapters (Chapter 10 for Heston, beyond the present guide for jumps and rough vol) and is what drives the ever-increasing complexity of modern derivatives-pricing engines.

9.5 7.4 Options on a dividend-paying asset

The stock we have been hedging in Chapter 6 has paid no dividends — it is a "pure capital appreciation" asset, and its entire return has flowed through the SDE $\mathrm{d}S_t/S_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$. Real stocks pay dividends, sometimes as quarterly cash payouts, sometimes continuously (as in certain index products or in the idealisation where we treat the total-return series). In the continuous-dividend idealisation, the stock pays a dividend yield q per unit time per unit of stock: an investor holding one share over $[t, t + \mathrm{d}t]$ receives a cash payment $q\,S_t\,\mathrm{d}t$ into her account.

This small operational change propagates through the self-financing replication argument in a specific and instructive way. The result is the *Black-Scholes-Merton PDE with dividend yield*, whose only alteration from the no-dividend case is the replacement of r by $r - q$ in the drift coefficient of the first-derivative term. Derivation follows.

9.5.1 7.4.1 Setup — continuous dividend yield

Let S_t be the stock and let it satisfy

$$\frac{\mathrm{d}S_t}{S_t} \;=\; \mu\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t, \tag{7.37}$$

with a *continuous dividend yield* q: holding one share over $[t, t+\mathrm{d}t]$ delivers $q\,S_t\,\mathrm{d}t$ in cash. The bank account is, as always,

$$\mathrm{d}B_t \;=\; r\,B_t\,\mathrm{d}t. \tag{7.38}$$

We value a European claim $g_t = g(t, S_t)$ on S with terminal payoff $g(T, S) = \varphi(S)$. Writing $g \in C^{1,2}$ so that Itô's lemma applies, we set up a self-financing portfolio

$$V_t \;=\; \alpha_t\,S_t \;+\; \beta_t\,B_t \;-\; g_t \tag{7.39}$$

with α_t shares, β_t units of cash, and a short unit of the target claim. At inception we choose $V_0 = 0$ — the usual replication convention.

9.5.2 7.4.2 Self-financing increment with dividends

The self-financing increment must now account for the dividend stream. Over $[t, t+\mathrm{d}t]$ the wealth change has four sources: the stock's price move $\alpha_t\,\mathrm{d}S_t$, the dividend income $\alpha_t\,q\,S_t\,\mathrm{d}t$, the bank's growth $\beta_t\,\mathrm{d}B_t$, and the change in the short claim's value $-\mathrm{d}g_t$:

$$\mathrm{d}V_t \;=\; \alpha_t\,\mathrm{d}S_t \;+\; \alpha_t\,q\,S_t\,\mathrm{d}t \;+\; \beta_t\,\mathrm{d}B_t \;-\; \mathrm{d}g_t. \tag{7.40}$$

The dividend term is what is new. Compared with the no-dividend setting, (7.40) adds a positive drift $\alpha_t\,q\,S_t$: shares held through an interval earn a dividend whether or not the price moves. In the stock-only part of the portfolio, this makes holding long α_t shares look more attractive than it did before — an effect we will see propagate into the option's PDE as a shift in the drift coefficient.

Expanding $\mathrm{d}S_t$, $\mathrm{d}B_t$, and $\mathrm{d}g_t$ via Itô's lemma,

$$\mathrm{d}V_t \;=\; \alpha_t(\mu\,S_t\,\mathrm{d}t + \sigma\,S_t\,\mathrm{d}W_t) \;+\; \alpha_t\,q\,S_t\,\mathrm{d}t \;+\; \beta_t\,r\,B_t\,\mathrm{d}t \;-\; \left[(\partial_t g + \mu\,S_t\,\partial_S g + \tfrac{1}{2}\sigma^2 S_t^2\,\partial_{SS} g)\mathrm{d}t + \sigma\,S_t\,\partial_S g\,\mathrm{d}W_t\right]. \tag{7.41}$$

Grouping $\mathrm{d}t$ and $\mathrm{d}W$ separately,

$$\mathrm{d}V_t \;=\; \underbrace{[\alpha_t(\mu+q)S_t + \beta_t\,r\,B_t - (\partial_t g + \mu\,S_t\,\partial_S g + \tfrac{1}{2}\sigma^2 S_t^2\,\partial_{SS} g)]}_{\equiv \mathcal{A}_t}\mathrm{d}t \;+\; \sigma\,S_t[\alpha_t - \partial_S g]\,\mathrm{d}W_t. \tag{7.42}$$

9.5.3 7.4.3 Zeroing the noise and drift

The coefficient of $\mathrm{d}W_t$ in (7.42) vanishes iff

$$\alpha_t \;=\; \partial_S g(t, S_t), \tag{7.43}$$

which is the delta hedge — identical to the no-dividend case. The drift coefficient $\mathcal{A}_t$ must vanish for $V_t \equiv 0$ to hold (since $V_0 = 0$ and $\mathrm{d}V_t$ must be predictable to avoid an arbitrage). Setting $\mathcal{A}_t = 0$ and substituting $\alpha_t = \partial_S g$,

$$\partial_S g\,(\mu+q)\,S_t \;+\; \beta_t\,r\,B_t \;-\; \partial_t g \;-\; \mu\,S_t\,\partial_S g \;-\; \tfrac{1}{2}\sigma^2 S_t^2\,\partial_{SS} g \;=\; 0. \tag{7.44}$$

The first and fourth terms combine:

$$\partial_S g\,(\mu+q)\,S_t \;-\; \mu\,S_t\,\partial_S g \;=\; q\,S_t\,\partial_S g. \tag{7.45}$$

Thus

$$\beta_t\, r\, B_t \;=\; \partial_t g \;+\; \tfrac{1}{2}\sigma^2 S_t^2\, \partial_{SS} g \;-\; q\, S_t\, \partial_S g. \tag{7.46}$$

9.5.4 7.4.4 Identifying β_t and closing the PDE

From $V_t = 0$ and $\alpha_t = \partial_S g$ we obtain

$$\beta_t\, B_t \;=\; g_t \;-\; \alpha_t\, S_t \;=\; g_t \;-\; S_t\, \partial_S g. \tag{7.47}$$

Substituting this into the left side of (7.46),

$$r\,(g_t - S_t\, \partial_S g) \;=\; \partial_t g \;+\; \tfrac{1}{2}\sigma^2 S_t^2\, \partial_{SS} g \;-\; q\, S_t\, \partial_S g. \tag{7.48}$$

Rearranging produces the *Black-Scholes-Merton PDE with continuous dividend yield*:

$$\boxed{\partial_t g \;+\; (r-q)\, S\, \partial_S g \;+\; \tfrac{1}{2}\sigma^2 S^2\, \partial_{SS} g \;=\; r\, g, \qquad g(T,S) = \varphi(S).} \tag{7.49}$$

Compare with the no-dividend PDE of Chapter 6: the only change is that the drift of the first-derivative term has become $r - q$ rather than r. The discount rate on the right-hand side is still the full r (the option itself does not pay a dividend), but the *risk-neutral* drift of the stock has been reduced by the dividend yield. Under the risk-neutral measure $\mathbb{Q}$, the stock now satisfies

$$\mathrm{d}S_t \;=\; (r-q)\, S_t\, \mathrm{d}t \;+\; \sigma\, S_t\, \mathrm{d}\widehat{W}_t, \tag{7.50}$$

so the *total-return* process $S_t e^{qt}$ (stock plus reinvested dividends) drifts at r, but the bare price process S_t drifts at $r-q$. The market is still risk-neutral on the *reinvestment* of dividends, but the quoted price lags the reinvested total return by the dividend yield.

A useful derivation consistency check is that the dividend-less case is recovered by setting $q = 0$: (7.49) reduces to the classical Black-Scholes PDE with drift r, and (7.50) reduces to the classical risk-neutral log-normal dynamics of Chapter 6. The dividend correction enters the drift linearly and nowhere else — the volatility structure, the discount rate on the RHS, and the terminal condition are all unchanged.

9.5.5 7.4.5 The Black-Scholes-Merton formula with dividends

The closed-form European call on a dividend-paying stock under (7.49) is

$$g(t,S) \;=\; S\, e^{-q(T-t)}\, \Phi(d_+) \;-\; K\, e^{-r(T-t)}\, \Phi(d_-), \tag{7.51}$$

with

$$d_\pm \;=\; \frac{\ln(S/K) \;+\; (r - q \pm \tfrac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}. \tag{7.52}$$

Three features are worth noting.

First, the stock term is discounted by $e^{-q(T-t)}$ — the dividend-reduced forward value of the share at expiry. A long call holder does not receive the dividends that flow to the share during the option's life, so the effective "stock value" embedded in the option is only $Se^{-q(T-t)}$, not the full spot S.

Second, the moneyness drift $r - q$ replaces r inside $d_\pm$, consistent with (7.50). Under the risk-neutral measure the stock's forward price is $F_t = S_t e^{(r-q)(T-t)}$, and the $d_\pm$ factors record the log-moneyness relative to this forward, re-standardised by the volatility.

Third, the formula reduces to the classical Black-Scholes call when $q = 0$, and it reduces to the Margrabe formula for an exchange option when the strike is itself a random asset and its "dividend yield" is its own risk-free return (Chapter 8).

For a European put, the analogous formula is

$$g(t,S) \;=\; K\,e^{-r(T-t)}\,\Phi(-d_-) \;-\; S\,e^{-q(T-t)}\,\Phi(-d_+), \tag{7.53}$$

with the same $d_\pm$ as the call — a direct consequence of put-call parity, which we state formally in §7.6.5.

A useful reinterpretation of (7.51)–(7.52) is via the *forward price* of the stock. Define

$$F_t \;=\; S_t\,e^{(r-q)(T-t)}, \tag{7.54}$$

the no-arbitrage forward price delivered at T. Then (7.51) can be rewritten

$$g(t,S) \;=\; e^{-r(T-t)}\left[F_t\,\Phi(d_+) \;-\; K\,\Phi(d_-)\right], \tag{7.55}$$

which looks like the classical formula with the spot S_t replaced by the forward F_t and the stock's discount rate replaced by r. This is the beginning of the *forward-measure* viewpoint developed in Chapter 8: by pricing against the forward numeraire, the dividend yield effectively disappears from the formula, absorbed into the forward price. The elegance of the forward view is that it treats the dividend structure and the interest-rate structure as one combined drift, visible only through F_t.

9.5.6 7.4.6 Delta of a dividend-paying call

Differentiating (7.51) with respect to S gives

$$\Delta_t \;=\; e^{-q(T-t)}\,\Phi(d_+). \tag{7.56}$$

The dividend-induced factor $e^{-q(T-t)}$ is smaller than unity, so the delta of a dividend-paying call is *smaller* than the corresponding no-dividend call — a consequence of the stock losing value through the dividend stream that will never flow to the option holder. For continuously compounded dividends and long horizons, this discount factor can be substantial: a 2% dividend over five years shaves the delta by almost 10% relative to the naive no-dividend formula. Traders who fail to adjust for dividends consistently overhedge (carry too many shares), creating a systematic tracking error that drains P&L. Explicit handling of the dividend yield in the delta calculation is therefore a must for any book hedging dividend-paying equities, and the relevant discount factor is always $e^{-q(T-t)}$ applied to the stock side.

The gamma is similarly modified:

$$\Gamma_t \;=\; \frac{e^{-q(T-t)}\,\Phi'(d_+)}{S\,\sigma\sqrt{T-t}}, \tag{7.57}$$

and the vega is

$$\mathcal{V}_t \;=\; S\,e^{-q(T-t)}\,\sqrt{T-t}\,\Phi'(d_+). \tag{7.58}$$

(An equivalent form swaps the S-term for a K-term via the Black–Scholes "Φ' identity" $S\,e^{-q(T-t)}\,\Phi'(d_+) = K\,e^{-r(T-t)}\,\Phi'(d_-)$, giving $\mathcal{V}_t = K\,e^{-r(T-t)}\sqrt{T-t}\,\Phi'(d_-)$. Some references quote the K-form; both are numerically identical and readers cross-checking against other texts should recognise the equivalence.)

All Greeks carry the $e^{-q(T-t)}$ dividend-discount factor applied to the S-dependent part; the K-dependent discount factor remains $e^{-r(T-t)}$. This structural fact is worth remembering for quick sanity checks on any pricing library — if a library returns a dividend-stock delta that matches the no-dividend delta exactly, the library is wrong.

9.5.7 7.4.7 Forward reformulation

For the practitioner who wants to price options on dividend-paying stocks, the cleanest route is to work directly with the forward price. Define $F_t = S_t e^{(r-q)(T-t)}$ as in (7.54). Then F_t is a $\mathbb{Q}$-martingale (no drift under the forward's discount scheme), and the option price satisfies

$$g(t, F) \;=\; e^{-r(T-t)}\, \mathbb{E}_t^{\mathbb{Q}}[\varphi(F_T)], \tag{7.59}$$

which is just a discounted expectation of the terminal payoff expressed as a function of the terminal forward. For a vanilla call with $\varphi(F) = (F - K)_+$, the expectation is evaluated in closed form to give (7.55) — the familiar Black-76 formula that appears again in Chapters 8 (options on futures), 13 (bond options), and 15 (swaptions) with only the definition of "the forward" changing. The invariance is no accident: it is the change-of-numeraire argument of Chapter 5 doing its job. Any time we can identify a traded forward that is a martingale under its natural pricing measure, a Black-type formula is available; the dividend-adjusted equity option is just the first place we see this universality explicitly.

The detailed Black-76 derivation is deferred to Chapter 8; the pedagogical lesson here is that *dividends tilt the drift; they do not change the structure of the hedging problem.*

9.6 7.5 Transaction costs and the self-financing bookkeeping

Self-financing is one of those notions that seem trivially obvious until you try to write it down precisely, and then they turn out to hide subtle corner cases. Intuitively, self-financing says: "nothing flows into or out of the portfolio except through changes in the market prices of the assets we hold." No external cash injections, no dividend leaks, no funding gaps. But once we allow continuous trading, the act of adjusting the weights $(\alpha_t, \beta_t, \gamma_t)$ itself creates price-driven cash flows, and those need to be accounted for before the clean "$\mathrm{d}V = \alpha\, \mathrm{d}S + \beta\, \mathrm{d}M + \gamma\, \mathrm{d}h$" intuition is actually valid. Chapter 6 skated over this subtlety because the naive form sufficed for the Black-Scholes PDE derivation. In this section we revisit the self-financing condition with enough rigour to handle two real-world complications: transaction costs, which were ignored in Chapter 6, and the covariation terms that appear when the hedge weights are themselves stochastic.

9.6.1 7.5.1 Self-financing revisited — the covariation terms

For a portfolio $V_t = \alpha_t\, S_t + \beta_t\, M_t - g_t$ (long stock, long cash, short the claim we sold), Itô's product rule gives

$$\begin{aligned} \mathrm{d}V_t \;&= \alpha_t\, \mathrm{d}S_t \;+\; \beta_t\, \mathrm{d}M_t \;-\; \mathrm{d}g_t \\ &\quad +\; \mathrm{d}\alpha_t\, S_t \;+\; \mathrm{d}\beta_t\, M_t \;+\; \mathrm{d}[\alpha, S]_t \;+\; \mathrm{d}[\beta, M]_t. \end{aligned} \tag{7.60}$$

The "gain-from-trading" terms are $\alpha_t\, \mathrm{d}S_t + \beta_t\, \mathrm{d}M_t - \mathrm{d}g_t$; the remaining four terms are the trading cost from revising (α, β) and must vanish under self-financing.

The self-financing condition makes the "no external cash input" property precise: between $t = 0$ and $t = T$, value can change only through market-price moves on the assets we hold. Because α_t and β_t are themselves stochastic and adapted to S, the covariations $\mathrm{d}[\alpha, S]$ and $\mathrm{d}[\beta, M]$ in (7.60) do not vanish in any interesting dynamic-hedging context.

9.6.2 7.5.2 Before-and-after rebalance accounting

Fix the instant t and split it into a before-rebalance leg using (α_t, β_t) and an after-rebalance leg using $(\alpha_{t+\Delta t}, \beta_{t+\Delta t})$. The before-leg value carried into $t + \Delta t$ is

$$\alpha_t\, S_{t+\Delta t} \;+\; \beta_t\, M_{t+\Delta t} \;-\; g_{t+\Delta t}, \tag{7.61}$$

and the after-leg value at $t + \Delta t$ is

$$\alpha_{t+\Delta t}\, S_{t+\Delta t} \;+\; \beta_{t+\Delta t}\, M_{t+\Delta t} \;-\; g_{t+\Delta t}. \tag{7.62}$$

The discrete value increment

$$\Delta V_t \;=\; \alpha_t\,\Delta S_t \;+\; \beta_t\,\Delta M_t \;-\; \Delta g_t \tag{7.63}$$

is valid only if before = after at $t+\Delta t$, i.e.

$$\Delta\alpha_t\;\underbrace{S_{t+\Delta t}}_{S_t+\Delta S_t}\;+\Delta\beta_t\;\underbrace{M_{t+\Delta t}}_{M_t+\Delta M_t}\;=\;0. \tag{7.64}$$

Expanding the product,

$$\Delta\alpha_t\,S_t \;+\; \Delta\alpha_t\,\Delta S_t \;+\; \Delta\beta_t\,M_t \;+\; \Delta\beta_t\,\Delta M_t \;=\; 0, \tag{7.65}$$

and passing to the continuous limit (where $\Delta\alpha\,\Delta S \to \mathrm{d}[\alpha, S]$, the quadratic covariation),

$$\boxed{\mathrm{d}\alpha_t\,S_t \;+\; \mathrm{d}[\alpha,S]_t \;+\; \mathrm{d}\beta_t\,M_t \;+\; \mathrm{d}[\beta,M]_t \;=\; 0.} \tag{7.66}$$

Equation (7.66) is the *precise* statement of self-financing — it annihilates exactly the four "extra" terms in (7.60), leaving only the gain-from-trading terms on the right-hand side. In continuous-time finance, naively writing "$\mathrm{d}V = \alpha\,\mathrm{d}S + \beta\,\mathrm{d}M$" *assumes* (7.66) as a side condition; it is not automatic. The quadratic-variation terms $\mathrm{d}[\alpha, S]$ and $\mathrm{d}[\beta, M]$ are what distinguish genuine self-financing from a pretend one.

For any delta-hedged strategy $\alpha = \Delta(t, S_t)$, Itô gives

$$\mathrm{d}[\alpha,S]_t \;=\; \partial_S\Delta\,\mathrm{d}[S,S]_t \;=\; \Gamma\,\sigma^2 S^2\,\mathrm{d}t \tag{7.67}$$

— the gamma-times-variance bleed at the heart of option theory. The self-financing condition is what guarantees that the BS PDE for g must contain the $\frac{1}{2}\sigma^2 S^2\partial_{SS}$ term: it comes directly from the quadratic-variation contribution in (7.67).

9.6.3 7.5.3 Proportional transaction costs

In the Black-Scholes idealisation, rebalances are free. In reality, each rebalance of size $\Delta\alpha_t$ at price $S_{t+\Delta t}$ costs a bid-ask spread $c\,|\Delta\alpha_t|\,S_{t+\Delta t}$ where c is the proportional spread. The self-financing condition picks up a non-vanishing negative term:

$$\mathrm{d}\alpha_t\,S_t \;+\; \mathrm{d}[\alpha,S]_t \;+\; \mathrm{d}\beta_t\,M_t \;+\; \mathrm{d}[\beta,M]_t \;=\; -c\,|\mathrm{d}\alpha_t|\,S_t \;-\; c\,|\mathrm{d}\gamma_t|\,h_t. \tag{7.68}$$

The absolute-value structure is the crucial feature: transaction costs are paid whether the rebalance is a buy or a sell, so the cost does not cancel across oscillating rebalances. This dissipation is what drives the *optimal-hedging-band* framework (the standard reference is Davis-Norman / Whalley-Wilmott; we do not develop it in this guide) — one does not want to rebalance for every tiny move because the cost $c\,|\mathrm{d}\alpha|$ accumulates quickly, so rebalances are deferred until the delta drift exceeds a threshold determined by cost and volatility.

To get a sense of the magnitude, suppose a desk is delta-hedging a single option over a year, with daily rebalances (so $N = 250$). For an ATM three-month call on a \$100 stock with $\sigma = 25\%$, the typical daily $|\Delta\alpha_t|$ is of order $\Gamma \cdot |\Delta S| \approx 0.032 \cdot (100 \cdot 0.25/\sqrt{250}) \approx 0.05$ shares per day. With a bid-ask spread of $c = 0.01\%$ (one basis point — typical for listed equity), the daily cost is $0.01\% \cdot 0.05 \cdot 100 = \$0.0005$ per option. Over 250 days, the cumulative cost is about \$0.12 per option, or roughly 1% of the option's premium — small but not negligible. For illiquid or wide-spread markets ($c = 0.1\%$ or higher), the cumulative cost can be several percent of the premium, which fundamentally changes the economics of the trade.

For a gamma-hedging leg using a second option with much wider bid-ask, the cost can easily exceed the gamma-reduction benefit. This is why delta-gamma hedging is not universal: on desks where the gamma-hedging option's bid-ask is large relative to the gamma-induced P&L variance, the pure-delta hedge is actually the *cheaper* strategy once costs are included. The delta-gamma hedge wins only when the auxiliary option's bid-ask is small enough that the cost of holding and rebalancing it is less than the variance reduction it provides.

9.6.4 7.5.4 Leland volatility adjustment

Under proportional costs the dealer prices the option *as if* the volatility were slightly higher than the actual volatility. The Leland adjustment is

$$\sigma^2_{\text{Leland}} \;=\; \sigma^2 \;+\; c\,\sigma\sqrt{\frac{8}{\pi\,\Delta t}}, \tag{7.69}$$

with Δt the rebalance interval and c the proportional bid-ask. At $c = 0.01\%$, daily rebalancing ($\Delta t = 1/250$), and $\sigma = 25\%$, the adjustment is ≈ 0.006, i.e. ≈ 0.6 vol points; at $c = 0.1\%$ (stressed) it jumps to ≈ 6 points — enough to materially change the dealer's price. Real dealer prices overlay (7.69) together with XVA, skew, and liquidity premia onto the textbook σ.

9.7 7.6 Put-call parity, deltas, and gammas

Throughout this section $C = C(t,S)$ and $P = P(t,S)$ denote the Black-Scholes European call and put prices on a stock paying a continuous dividend yield q, struck at K and expiring at T; we write $\tau = T - t$ for time to expiry. The one identity that ties them together is put-call parity,

$$C(t,S) \;-\; P(t,S) \;=\; S\,e^{-q\tau} \;-\; K\,e^{-r\tau}. \tag{7.70}$$

(7.70) is a static, *model-independent* identity: it is a no-arbitrage consequence of the trivial payoff decomposition $\max(S_T - K, 0) - \max(K - S_T, 0) = S_T - K$, discounted to t. Every Greek identity between call and put follows by differentiating (7.70) in the relevant variable — spot, volatility, time, or rate — and the right-hand side, which is a simple forward contract minus a zero-coupon bond, is transparent in each.

9.7.1 7.6.1 Put price curve

Applying parity to the call formula (7.51) gives the closed-form put price under continuous dividend yield:

$$P(t,S) \;=\; K\,e^{-r\tau}\,\Phi(-d_-) \;-\; S\,e^{-q\tau}\,\Phi(-d_+), \tag{7.71}$$

with $d_\pm$ as in (7.52). The put is convex in S, monotonically decreasing (a larger spot lowers the value of the right-to-sell), and tends to $Ke^{-r\tau} - Se^{-q\tau}$ deep in the money ($S \ll K$) and to 0 deep out of the money ($S \gg K$). At expiry the curve collapses to the hockey-stick payoff $\max(K - S_T, 0)$.

9.7.2 7.6.2 Call price curve

For reference alongside (7.71) the call price (7.51) is

$$C(t,S) \;=\; S\,e^{-q\tau}\,\Phi(d_+) \;-\; K\,e^{-r\tau}\,\Phi(d_-). \tag{7.72}$$

The call is convex in S, monotonically increasing, bounded below by the intrinsic value $(Se^{-q\tau} - Ke^{-r\tau})_+$ (the modified forward-minus-bond) and above by $Se^{-q\tau}$.

9.7.3 7.6.3 Delta curves

Differentiating (7.70) once in S and using $\partial_S(Se^{-q\tau}) = e^{-q\tau}$ gives the *parity identity for delta*,

$$\Delta^C \;-\; \Delta^P \;=\; e^{-q\tau}. \tag{7.73}$$

Combined with the call-delta formula $\Delta^C = e^{-q\tau}\Phi(d_+)$ from (7.56), this yields the put delta

$$\Delta^P \;=\; e^{-q\tau}\Phi(d_+) \;-\; e^{-q\tau} \;=\; -\,e^{-q\tau}\,\Phi(-d_+). \tag{7.74}$$

As expected, $\Delta^P \in [-e^{-q\tau}, 0]$: a long put has negative delta bounded below by $-e^{-q\tau}$ (ATM-deep-ITM put $\to$ near $-e^{-q\tau}$; OTM put $\to 0$). The two delta curves are vertical translates of each other by $e^{-q\tau}$ — a clean visual diagnostic.

9.7.4 7.6.4 Gamma curves and pin risk

Differentiating (7.73) once more in S, the right-hand side $e^{-q\tau}$ is constant in S, so

$$\Gamma^C \;=\; \Gamma^P \;=\; \frac{e^{-q\tau}\,\Phi'(d_+)}{S\,\sigma\sqrt{\tau}}. \tag{7.75}$$

This is the *gamma parity identity* previewed in §7.1 — calls and puts at the same strike share a single gamma curve. Consequently, a delta-neutral spread that is short one call and long one put at the same strike has *zero* gamma; the combination is the synthetic short forward of §7.6.6, whose value is linear in S.

Pin risk. As $\tau \downarrow 0$ the ATM gamma blows up like $1/\sqrt{\tau}$: from (7.75) at $S = Ke^{(r-q)\tau}$, $\Gamma \sim 1/(K\sigma\sqrt{2\pi\tau})$. A short ATM option on expiry day can whip between $\Delta = 0$ and $\Delta = \pm e^{-q\tau}$ as S crosses K — the delta becomes a step function and hedging it requires arbitrarily-large trades on arbitrarily-small spot moves. This is *pin risk.* Because $\Gamma^C = \Gamma^P$, it afflicts calls and puts symmetrically, and it is one of the main reasons market makers flatten ATM option exposure in the final hours before expiry.

9.7.5 7.6.5 Put-call parity and its Greek consequences

We now collect the full slate of parity consequences obtained by differentiating (7.70) in each model variable. All follow mechanically, and the right-hand side of each identity has a transparent forward-minus-bond interpretation.

Delta. ∂_S of (7.70): $\Delta^C - \Delta^P = e^{-q\tau}$. (7.73)

Gamma. ∂_{SS} of (7.70): $\Gamma^C = \Gamma^P$. (7.75) The right-hand side of parity is linear in S, so its second derivative vanishes.

Vega. Neither $Se^{-q\tau}$ nor $Ke^{-r\tau}$ depends on σ (parity is a *static* no-arb identity independent of volatility assumptions), so ∂_σ of (7.70) gives

$$\mathcal{V}^C \;=\; \mathcal{V}^P. \tag{7.76}$$

Calls and puts at the same strike share a single vega curve. A delta-hedged short call is vega-short; a delta-hedged short put is equally vega-short — which is why a straddle (long call + long put at the same strike) doubles vega, matching (7.76) with a factor of two.

Theta. ∂_t of (7.70), using $\partial_t e^{-q\tau} = qe^{-q\tau}$ and $\partial_t e^{-r\tau} = re^{-r\tau}$:

$$\Theta^C \;-\; \Theta^P \;=\; -\,q\,S\,e^{-q\tau} \;+\; r\,K\,e^{-r\tau}. \tag{7.77}$$

The difference in theta is *not* zero, because the forward and the bond on the right-hand side of parity each decay at their own rate. For $q = 0$ (no dividend), $\Theta^C - \Theta^P = rKe^{-r\tau} > 0$, so a call decays less unfavourably than a put of the same strike — the standard “rates help calls, hurt puts” intuition.

Rho. ∂_r of (7.70), noting that $Se^{-q\tau}$ is rate-independent while $Ke^{-r\tau}$ carries a $-\tau$:

$$\rho^C \;-\; \rho^P \;=\; \tau\,K\,e^{-r\tau}. \tag{7.78}$$

Calls have positive rho and puts negative rho; the gap between them is exactly the PV01-scale of the strike-bond.

Dividend sensitivity. ∂_q of (7.70): $\partial_q C - \partial_q P = -\tau Se^{-q\tau}$. Calls are hurt by higher dividends (stock trades ex-div), puts benefit symmetrically.

These identities are all *exact* and *model-free* in the sense that they follow from (7.70) without reference to any Black-Scholes or log-normal assumption. They therefore hold in Heston, in local-vol, in every complete arbitrage-free model — any model in which parity is imposed (which is every sensible model) respects (7.73)-(7.78) identically.

Theta vs spot for calls and puts, three TTMs

Figure 7: Theta vs spot for calls and puts, three TTMs

Rho vs spot for calls and puts, plus ATM rho vs T

Figure 8: Rho vs spot for calls and puts, plus ATM rho vs T

9.7.6 7.6.6 Synthetic positions

Rewriting parity as

$$C \;-\; P \;=\; S\,e^{-q\tau} \;-\; K\,e^{-r\tau} \tag{7.79}$$

says that a *long-call-short-put* portfolio at strike K is equivalent to a long position in $e^{-q\tau}$ shares of stock (equivalently, one forward contract delivered at T) funded by a short K-notional zero-coupon bond maturing at T. This is the *synthetic forward*: a way to construct directional equity exposure from options alone.

Symmetric rearrangements give the other useful syntheses: synthetic long stock is $Se^{-q\tau} = C - P + Ke^{-r\tau}$ (long call + short put + long bond); synthetic short stock is $-Se^{-q\tau} = P - C - Ke^{-r\tau}$. Traders use these replications when the physical stock is hard to borrow, when margin requirements differ between cash equities and options, or when the options market is more liquid than the stock at the relevant size. The Greek profile of a synthetic is dictated by (7.73)-(7.78): a synthetic long forward is delta-$e^{-q\tau}$, gamma-zero, vega-zero, and theta-$(-qSe^{-q\tau} + rKe^{-r\tau})$, which is exactly the Greek profile of the forward contract it replicates — a consistency check with nothing to tune.

9.8 7.7 Greek Visual Atlas

The closed-form expressions of §§7.1–7.6 capture the algebra of each Greek; the pictures below capture the *shape*. For a senior risk manager the shape is what's load-bearing — it is the shape that tells you which strike will pin, which vega bucket will blow up under a vol shock, and where the second-order risk lives that a delta hedger cannot see. Every figure in this atlas is generated under the same flat Black-Scholes world: $S_0 = K = 100$, $r = 4\%$, $q = 0$, $\sigma = 25\%$, with maturity slices $T \in \{1\text{m}, 3\text{m}, 12\text{m}\}$ unless the figure is specifically about expiry-day effects.

9.8.1 7.7.1 Theta — the cost of waiting

Call (left) and put (right) theta as a function of spot, at $T = 1m$, $3m$, $12m$.

Theta is most negative right at the strike — that is where convex optionality is densest, so the time-decay charge per day is the largest. The decay accelerates as expiry approaches: the 1-month curve is roughly three times deeper than the 12-month curve. Note the put-call asymmetry at low spot: the $rKe^{-r\tau}\Phi(\pm d_{\_})$ discount-funding term flips sign between (7.77)'s call and put forms, so deep ITM puts can have *positive* theta while deep ITM calls do not.

9.8.2 7.7.2 Rho — the rate sensitivity nobody hedges (until they do)

Left: ρ vs spot for calls (solid) and puts (dashed) at three maturities. Right: ATM ρ as a function of T.

Calls are long rates, puts are short rates, and the magnitude of both scales roughly linearly in T. For a one-month book, ρ is nearly inert; for a one-year book, a 1% rate shock can move ATM call value by a third of its premium. Equity-vol traders typically ignore ρ until they are warehousing LEAPs or running a structured-products book — at which point a separate IR delta desk picks it up.

Vanna heatmap over (S, sigma) at fixed T, K

Figure 9: Vanna heatmap over (S, sigma) at fixed T, K

Volga (vomma) vs spot, two TTMs

Figure 10: Volga (vomma) vs spot, two TTMs

Charm vs spot for calls and puts, at 1w/1m/3m

Figure 11: Charm vs spot for calls and puts, at 1w/1m/3m

9.8.3 7.7.3 Vanna — the cross between spot and vol

Heatmap of $\partial^2 V/\partial S\,\partial\sigma$ over spot and implied vol, with the zero contour drawn in dashed black.

Vanna is the Greek that links spot moves and vol moves: long-vanna positions earn when spot and vol move together (skew rallies on sell-offs). The sign change crosses through the strike — long-call vanna is *negative* below ATM and *positive* above. Dealers running exotic books measure vanna because a static vega hedge does not neutralise it: the first big down-day in the underlying drains delta into a place where your vega hedge is wrong.

9.8.4 7.7.4 Volga — the smile-curvature P&L

$\partial^2 V/\partial\sigma^2$ for $T = 1m$ and $T = 12m$.

Volga has twin peaks straddling the ATM zero — it is highest in the *wings*, where vega is small but accelerating in σ. A long strangle is long volga; that is why traders selling tail vol earn when realised vol of vol is low and lose when the smile lifts at the wings. The longer-dated curve sits higher and is wider — vol convexity needs time to compound.

9.8.5 7.7.5 Charm — delta bleed across the weekend

Charm $= \partial\Delta/\partial t$ for calls (left) and puts (right) at $T = 1w$, $1m$, $3m$.

Charm tells you how much your delta drifts each day even when spot holds still. The 1-week curve is dramatically more extreme than the 3-month curve: an OTM option whose delta was 0.30 on Friday can be 0.15 on Monday on a flat tape. Market makers quote-stuff their gamma desk every morning to absorb the weekend charm bleed; this is the "overnight delta" that prop traders chase in the open auction.

9.8.6 7.7.6 Speed — gamma's slope

Speed $= \partial\Gamma/\partial S$ for $T = 1m$, $3m$, $12m$.

Speed is the third derivative of price w.r.t. spot — it changes sign just above the strike, where the gamma hump rolls over. Strategies that earn from spot-path *cubic* effects (variance-swap convexity adjustments, butterfly carry) live on speed. For a delta-gamma-hedged book, residual P&L from a large spot move is roughly $\frac{1}{6}\,\text{Speed}\,\Delta S^3$.

9.8.7 7.7.7 Color — gamma's clock

Color $= \partial\Gamma/\partial t$ for $T = 1w$, $1m$, $3m$.

Color says how fast your gamma is changing through time. ATM color is sharply *negative* — gamma is the steepest function of τ right at the strike, so an ATM short-gamma desk leaks gamma into the close. The OTM wings have *positive* color: those strikes build gamma as expiry approaches, which is why tail-hedgers buy them weeks ahead, not days.

9.8.8 7.7.8 Black-Scholes price surface

Call value $C(S, T)$ as a 3D surface; spot in $[60, 140]$, TTM in $[0.02, 2.0]$.

The whole chapter so far has been about local first- and second-order properties of this surface. Stand back and you see the two pieces of intuition that drive everything: the surface is *increasing and convex* in spot (positive delta,

Speed vs spot, three TTMs

Figure 12: Speed vs spot, three TTMs

Color vs spot, three TTMs

Figure 13: Color vs spot, three TTMs

3D BS call price surface over (S, T)

Figure 14: 3D BS call price surface over (S, T)

positive gamma), and *increasing* in time (positive vega and vega-time product). Convexity emerges visibly as T grows — a long-dated option is much smoother than the kinked payoff at expiry.

9.8.9 7.7.9 Dollar-gamma — what dealers actually trade

$\$\Gamma = S^2\,\Gamma$ *vs spot at three maturities.*

Vanilla gamma is in (option-units / share^2) and depends on the underlying's price level. Dealers normalise by S^2 to get *dollar-gamma* — the P&L impact of a 1% spot move squared. The peak shifts very slightly above the strike (because S^2 grows faster than Γ falls on the right wing), and it is the flatter, broader profile that variance-swap and dispersion desks quote against.

9.8.10 7.7.10 The strike ladder — Greeks for an ATM expiry-1m book

All major Greeks for calls and puts on a $T = 0.25y$, $\sigma = 25\%$ book; ATM row highlighted.

A working trader's screen looks like this table. Note the parity identities check by inspection: every row has $\Delta^C - \Delta^P = 1$ (since $q = 0$, $e^{-q\tau} = 1$) and Γ identical for call and put. Theta and rho do *not* match across call/put — those gaps are exactly (7.77) and (7.78). The ATM row carries the largest Γ, the largest $|\Theta|$, and the largest vega — three reasons it is also the most expensive line on the screen.

9.8.11 7.7.11 Pin risk — the digital-delta blow-up

Delta of a digital call struck at the spot, plotted (log-y) vs calendar days to expiry.

A vanilla option's gamma blows up as $1/\sqrt{\tau}$ at the strike; a *digital* option's delta does. With days running out (x-axis inverted, expiry on the right), the digital delta — which equals the \$1 size of the payoff times the gamma kernel — explodes by more than two orders of magnitude in the last week. This is why short-dated structured-product issuers (autocallables, dual-currency deposits) cap exposure to single strikes: pin risk in the last hour is uninsurable in size.

9.8.12 7.7.12 Vega across the smile

Left: a parameterised IV smile $\sigma(K) = \sigma_{\text{atm}} + a\log(K/S) + b\log^2(K/S)$ with $a = -0.35$, $b = 1.10$. Right: vega per strike under that smile vs under a flat 22% IV.

Vega's bell-shape gets *bent* by the smile: down-side strikes (with elevated IV) carry more vega than the flat-IV baseline would suggest, and the OTM call wing gets reweighted up by the convex smile too. Dealer vega exposure does not concentrate where vega itself is largest in the flat world — it concentrates wherever the smile is steepest. This is why "vega bucketing by strike" has replaced "vega bucketing by ATM only" everywhere a real skew exists.

9.8.13 7.7.13 Gamma scalping — the variance-risk-premium picture

Top: a Monte Carlo GBM path with $\sigma_{\text{real}} = 30\%$, hedged at $\sigma_{\text{imp}} = 22\%$. Bottom: cumulative theta bleed (red), cumulative gamma rebalance gains (green), and net delta-hedged P&L (black).

This picture is the visual identity behind Section 7.2.8's variance-risk-premium claim. The dealer is short theta every day at a rate set by *implied* vol; they earn back $\frac{1}{2}S^2\Gamma\,(\Delta S)^2$ at each rebalance, whose expected value scales with *realised* vol squared. When $\sigma_{\text{real}} > \sigma_{\text{imp}}$, green outpaces red and the hedged book earns the difference. When the

Dollar-gamma vs spot, three TTMs

Figure 15: Dollar-gamma vs spot, three TTMs

Strike-ladder table of all Greeks

Figure 16: Strike-ladder table of all Greeks

Digital call delta vs days to expiry at S=K

Figure 17: Digital call delta vs days to expiry at S=K

inequality flips, the hedger pays for vol they never used. One sample path looks noisy; ten thousand average to the closed-form $\frac{1}{2}\int_0^T S_t^2\Gamma_t(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2)\,dt$.

9.9 7.8 Key Takeaways

1. **Delta-gamma hedging neutralises two derivatives.** Holding $-\Delta^g$ shares of stock and $-\gamma$ units of an auxiliary option h, with $\gamma = \Gamma^g/\Gamma^h$, eliminates both the first-order and second-order Taylor terms of the hedged P&L. The residual is cubic in ΔS instead of quadratic, and its standard deviation under N-grid rebalancing scales as $1/N$ rather than $1/\sqrt{N}$.
2. **Vega is the volatility-input sensitivity, not a tradable Greek.** $\mathcal{V} = S\sqrt{T-t}\,\Phi'(d_+)$ peaks ATM and decays as $\sqrt{T-t} \to 0$. Vega-hedging requires a *second option* with linearly independent gamma-vega profile (system 7.36), because volatility itself is not a tradable asset under constant-σ Black-Scholes.
3. **Continuous-dividend BSM PDE differs only in the drift coefficient.** $\partial_t g + (r-q)\,S\,\partial_S g + \frac{1}{2}\sigma^2 S^2 \partial_{SS} g = r\,g$. The discount rate on the RHS is still r (the option pays nothing in the interim), but the risk-neutral drift of S is reduced by q.
4. **Forward reformulation collapses the dividend correction.** Defining $F_t = S_t e^{(r-q)(T-t)}$, the dividend yield disappears into the forward price; the option becomes Black-76 with no drift.
5. **Put-call parity drives the cross-Greek identities.** From $C - P = Se^{-q\tau} - Ke^{-r\tau}$ we read off $\Delta^C - \Delta^P = e^{-q\tau}$, $\Gamma^C = \Gamma^P$, $\mathcal{V}^C = \mathcal{V}^P$, and the asymmetric theta/rho gaps. These identities are *model-free*: they hold in every arbitrage-free model in which parity is imposed.
6. **Pin risk is the $1/\sqrt{\tau}$ blow-up of ATM gamma.** As $\tau \downarrow 0$, $\Gamma_{\text{ATM}} \sim 1/(K\sigma\sqrt{2\pi\tau})$, so a short ATM option on expiry day has a delta that whips between 0 and $\pm e^{-q\tau}$ on tiny spot moves. Market makers flatten ATM exposure into the close.
7. **Self-financing is a covariation condition, not a slogan.** The clean $\mathrm{d}V = \alpha\,\mathrm{d}S + \beta\,\mathrm{d}M$ requires $\mathrm{d}\alpha_t\,S_t + \mathrm{d}[\alpha, S]_t + \mathrm{d}\beta_t\,M_t + \mathrm{d}[\beta, M]_t = 0$. The covariation term $\mathrm{d}[\alpha, S] = \Gamma\,\sigma^2 S^2\,\mathrm{d}t$ is what produces the $\frac{1}{2}\sigma^2 S^2 \partial_{SS}$ piece of the BS PDE in the first place.
8. **Transaction costs price as a volatility premium.** Leland's adjustment $\sigma_{\text{Leland}}^2 = \sigma^2 + c\sigma\sqrt{8/(\pi\Delta t)}$ raises the dealer's effective vol by the spread c and rebalance frequency. At $c = 1$ bp, daily rebalancing adds ≈ 0.6 vol points; at $c = 10$ bp (stressed), the addition jumps to ≈ 6 points. Real prices are BS plus Leland plus XVA plus skew plus liquidity premia.

9.10 7.9 Reference Formulas Appendix

Label	Formula
(7.3)	$\Gamma = \Phi'(d_+)/(S\,\sigma\sqrt{T-t})$, $\mathcal{V} = S\sqrt{T-t}\,\Phi'(d_+)$
(7.11)	$\gamma_t = \Gamma^g/\Gamma^h$, $\alpha_t = \Delta^g - \gamma_t\,\Delta^h$ (delta-gamma hedge weights)

IV smile and vega(K) under smile vs flat IV

Figure 18: IV smile and vega(K) under smile vs flat IV

Label	Formula
(7.29)	$\mathrm{Var}_{\text{delta-only}} \sim \sigma^4 T^2/N$ (gamma-residual scaling)
(7.30)	$\mathrm{Var}_{\text{delta-gamma}} \sim \sigma^6 T^3/N^2$ (cubic-residual scaling)
(7.33)	$\mathcal{V}_t = S_t \sqrt{T-t}\,\Phi'(d_+)$ (vanilla vega)
(7.36)	$(1, \Delta^{h_1}, \Delta^{h_2}; 0, \Gamma^{h_1}, \Gamma^{h_2}; 0, \mathcal{V}^{h_1}, \mathcal{V}^{h_2})(\alpha_t, \gamma_t^1, \gamma_t^2)^\top = (\Delta^g, \Gamma^g, \mathcal{V}^g)^\top$ (delta-gamma-vega match)
(7.49)	$\partial_t g + (r-q)\,S\,\partial_S g + \frac{1}{2}\sigma^2 S^2 \partial_{SS} g = r\,g$ (BSM PDE with continuous yield)
(7.51)	$g(t,S) = S\,e^{-q\tau}\Phi(d_+) - K\,e^{-r\tau}\Phi(d_-)$ (BSM call with dividends)
(7.52)	$d_\pm = [\ln(S/K) + (r - q \pm \frac{1}{2}\sigma^2)\tau]/(\sigma\sqrt{\tau})$
(7.55)	Black-76 form via $F_t = S_t e^{(r-q)(T-t)}$ — dividend absorbed into forward
(7.57)	$\Gamma_t = e^{-q\tau}\Phi'(d_+)/(S\sigma\sqrt{\tau})$ (dividend-adjusted gamma)
(7.58)	$\mathcal{V}_t = S\,e^{-q\tau}\sqrt{\tau}\,\Phi'(d_+)$ (dividend-adjusted vega)
(7.66)	$\mathrm{d}\alpha_t S_t + \mathrm{d}[\alpha, S]_t + \mathrm{d}\beta_t M_t + \mathrm{d}[\beta, M]_t = 0$ (self-financing in covariation form)
(7.67)	$\mathrm{d}[\alpha, S]_t = \Gamma\,\sigma^2 S^2\,\mathrm{d}t$ (delta-hedge covariation)
(7.68)	Self-financing with proportional costs: RHS becomes $-c\,\lvert\mathrm{d}\alpha_t\rvert S_t - c\,\lvert\mathrm{d}\gamma_t\rvert h_t$
(7.69)	$\sigma^2_{\text{Leland}} = \sigma^2 + c\,\sigma\,\sqrt{8/(\pi\Delta t)}$ (Leland volatility adjustment)
(7.70)	$C(t,S) - P(t,S) = S\,e^{-q\tau} - K\,e^{-r\tau}$ (put-call parity)
(7.73)	$\Delta^C - \Delta^P = e^{-q\tau}$ (parity-delta)
(7.75)	$\Gamma^C = \Gamma^P = e^{-q\tau}\Phi'(d_+)/(S\sigma\sqrt{\tau})$ (parity-gamma)
(7.76)	$\mathcal{V}^C = \mathcal{V}^P$ (parity-vega)
(7.77)	$\Theta^C - \Theta^P = -q\,S\,e^{-q\tau} + r\,K\,e^{-r\tau}$ (parity-theta)
(7.78)	$\rho^C - \rho^P = \tau\,K\,e^{-r\tau}$ (parity-rho)
(7.79)	Synthetic forward: $C - P = S\,e^{-q\tau} - K\,e^{-r\tau}$ replicates a long forward + short bond

Cumulative gamma scalping vs theta bleed on a GBM path

Figure 19: Cumulative gamma scalping vs theta bleed on a GBM path

10 Chapter 8 — Forwards, Futures, and the Black PDE

The previous two chapters developed the theory of dynamic hedging for derivatives written on a spot asset: self-financing replication, the Black–Scholes PDE, the Greeks, and the handling of dividends. This chapter turns to a family of underlyings that look superficially similar but behave dramatically differently under the risk-neutral measure: *forwards* and *futures*. The payoff structure of a European call on, say, the December S&P 500 futures looks identical to a call on the spot index. But the pricing PDE loses its first-derivative drift term, the closed-form formula discards a factor of e^{-rT}, and the underlying dynamics under $\mathbb{Q}$ are a driftless diffusion rather than a GBM with drift r. The reason is the one structural feature that makes a futures contract not a stock: *entering a futures position costs nothing.*

This chapter consolidates three strands of material that appeared separately in the original draft. From the dynamic-hedging chapters we take the self-financing derivation of the Black PDE for options on futures, which is the cleanest structural argument. From the Feynman–Kac chapter we take the parallel martingale derivation, which serves as a cross-check and connects to the general machinery of Chapter 4. From the futures-contracts chapter we take the concrete pricing examples (Bachelier and Ornstein–Uhlenbeck futures), the Margrabe exchange option — the tidiest use of a two-asset change of measure in the guide — and the convexity adjustment between futures and forwards under stochastic rates.

The chapter leans on Chapter 5's canonical Girsanov derivation every time we change numeraire, and on Chapter 6's BS PDE every time we compare the Black PDE to its spot-asset cousin. Feynman–Kac cross-checks reach back to Chapter 4.

A note on the forward/futures distinction. A reader reading carefully will notice that we use the symbol $F_t(T)$ for both the forward price and the futures price, distinguishing them only by context or by the superscripts "fwd"/"fut" when ambiguity threatens. This matches the market convention: under deterministic rates the two are identical (§8.2) and the single symbol $F_t(T)$ is what quant traders write on the blackboard; under stochastic rates the two differ by the convexity adjustment of §8.11, and the distinction matters. Throughout §§8.3–8.10 we take rates as constant (or, equivalently, as deterministic) and treat $F_t(T)$ as either forward or futures depending on which pricing argument we are running. Only in §8.11 does the distinction become load-bearing.

Who this chapter is for. A reader who has finished Chapter 6 knows the Black–Scholes PDE for an option on a spot asset and has internalised the self-financing apparatus. A reader who has finished Chapter 5 knows Girsanov's theorem and how to change numeraire. This chapter is the natural consolidation of those two threads: the Black PDE is the BS PDE applied to a martingale underlying (the futures), and the Margrabe formula is a numeraire-change exercise in its purest form. The content is structural, not computational: once one understands *why* the Black PDE has no first-derivative drift term and *why* the Margrabe formula has no discount factor, every variant in the subsequent chapters — caplets, swaptions, bond options, exchange options on swaps — becomes a specialisation of the same pattern.

10.1 8.1 Forward Price from a Zero-Cost Claim

A *forward contract* signed at time t with delivery at T and strike K is the simplest possible derivative: an obligation to exchange one unit of the underlying for the pre-agreed cash amount K at maturity. Its terminal payoff is $S_T - K$ for the long side and $K - S_T$ for the short side. By the risk-neutral pricing rule developed in Chapter 6, its value at time t is

$$\Pi_t \;=\; \mathbb{E}^{\mathbb{Q}}\big[\, e^{-r(T-t)}(S_T - K) \,\big|\, \mathcal{F}_t \big]\,. \tag{8.1}$$

At signing, the strike K is chosen so that $\Pi_t = 0$ — neither counterparty pays the other to enter. Setting (8.1) to zero and solving *under the constant-rate assumption*,

$$K \;=\; \mathbb{E}^{\mathbb{Q}}[\, S_T \mid \mathcal{F}_t \,] \;=\; S_t\, e^{r(T-t)}, \tag{8.2}$$

where the second equality uses that $S_t\, e^{-rt}$ is a $\mathbb{Q}$-martingale under the bank-account numeraire (Chapter 5). We call this K the *forward price at time t for delivery at T* and denote it $F_t^{\text{fwd}}(T)$:

$$F_t^{\text{fwd}}(T) \;=\; S_t\, e^{r(T-t)}. \tag{8.3}$$

For a stock paying a continuous dividend yield δ the formula is $F_t^{\text{fwd}}(T) = S_t\, e^{(r-\delta)(T-t)}$, consistent with the dividend-paying generalisation in Chapter 7. Geometrically, $F_t^{\text{fwd}}(T)$ is the *fair strike* of a just-signed forward contract — the price at which the two counterparties are indifferent to taking either side. As t advances toward T, $F_t^{\text{fwd}}(T) \to S_T$: the forward price collapses into the spot as the carry window shrinks.

An intuitive picture: plot $F_t^{\text{fwd}}(T)$ against S_t at a fixed date t. For the no-dividend case it is a straight line through the origin with slope $e^{r(T-t)} > 1$, sitting slightly above the diagonal $F = S$ that a trader might naively draw. The gap is the *cost of carry* — the interest saved by not having to hold the stock until T. For a dividend-paying stock the slope is $e^{(r-\delta)(T-t)}$, which is less than one if $\delta > r$ (the stock yields more than the bank account). In that regime the forward sits *below* the spot — the commodity-market phenomenon known as *backwardation.* Its opposite, $\delta < r$, is *contango.* The sign of $r - \delta$ drives substantial trading strategies in commodity derivatives, precious metals, and dividend-yield products.

The cash-and-carry story. Formula (8.3) can also be obtained by a pure no-arbitrage argument that does not invoke $\mathbb{Q}$ at all — worth retelling once because it makes the answer feel inevitable. At time t, a trader can either promise to buy the asset at T for the strike K (zero cost today, cash outflow K at T), or *synthesise* the same exposure by borrowing S_t dollars today and purchasing one share of S outright. At T the trader has one share either way; the only difference is the cash leg. To repay the loan requires $S_t/P(t,T)$ dollars at T, where $P(t,T) = e^{-r(T-t)}$ is the T-maturity discount bond. No-arbitrage forces the two routes to have the same cash outflow at T, giving

$$F_t^{\text{fwd}}(T) \;=\; \frac{S_t}{P(t,T)}. \tag{8.4}$$

Under constant rates $P(t,T) = e^{-r(T-t)}$ and (8.4) reduces to (8.3). The formula depends only on the spot and the discount bond — *not* on the statistical properties of S, not on volatility, not on anyone's view on where S is going. That independence is often underappreciated: the forward price is a cold no-arbitrage number, while the forward *value* of an existing contract with a stale strike reflects the market-implied drift of S.

Forward price term structure *Forward price term structure — cash-and-carry $F^{fwd}(t,T) = S_t/P(t,T)$ across maturities.*

Forward price vs forward value. The distinction matters. The *forward value* at time t of an existing contract with legacy strike K_0 is $\Pi_t = (F_t^{\text{fwd}}(T) - K_0)\, P(t,T)$, the discounted difference between the current fair strike and the contract's locked-in strike. The *forward price* $F_t^{\text{fwd}}(T)$ is the strike that would make a *newly signed* contract worth zero. Every day the market quotes the latter; the former is mark-to-market on a portfolio of contracts with various strikes. Pricing conventions differ on which number is meant by "the forward"; always check.

Under the T-forward measure $\mathbb{Q}^T$ with numeraire $P(t,T)$, the ratio $S_t/P(t,T) = F_t^{\text{fwd}}(T)$ is a martingale by construction — this is the general numeraire-pricing identity of Chapter 5 applied with $A = P(\cdot,T)$ and $F = S$. That martingale property is why forwards admit clean Black-type formulas even when rates are stochastic: under $\mathbb{Q}^T$, the forward price is driftless, and log-Black or normal-Black closed-forms fall out immediately. We shall return to this point in Chapter 13 where it is used to price bond options.

Cost of carry, generalised. $F^{\text{fwd}} = S/P$ generalises to other asset classes by adjusting the carry leg:

Asset class	Net yield / carry	Cash-and-carry forward
Equity, dividend yield δ	δ	$S_t\, e^{(r-\delta)(T-t)}$
FX, foreign rate r^f	r^f	$S_t\, e^{(r-r^f)(T-t)}$ (CIRP)
Commodity, conv-yield y, storage u	$y - u$	$S_t\, e^{(r-y+u)(T-t)}$
Income-generating asset, yield q	q	$S_t\, e^{(r-q)(T-t)}$

In every case the $\mathbb{Q}$-drift is $r-$(net yield); the formulas of §§8.5–8.11 apply uniformly with appropriate substitution.

10.2 8.2 Forwards vs. Futures — The Role of Daily Settlement

Forwards and futures are the two simplest delta-one derivatives — linear claims on the future value of an underlying — but they differ in one crucial operational respect: *when cash changes hands*. Forwards settle at maturity; futures settle daily. That small-looking difference has large consequences for pricing, hedging, and the shape of the backward PDE.

Both are *forward-looking* instruments: they specify a price for delivery of an asset at a future date. Both are *zero-sum* at initiation: one party is long, one is short, and no money changes hands upfront (for forwards) or only margin collateral does (for futures). Both have delta approximately equal to one: a \$1 move in the underlying moves the forward/futures P&L by approximately \$1. Yet the two instruments are priced by different PDEs, and the Black '76 futures-call formula differs from Black–Scholes in ways that matter when quoting option prices.

The forward contract. A forward is an obligation to buy/sell an asset S at a future time T at a pre-determined strike K. Its terminal payoff is $S_T - K$, so its time-t value is

$$\Pi_t \;=\; S_t \;-\; K\,P(t,T), \tag{8.5}$$

by exactly the two-piece argument of §8.1 (the asset leg is worth S_t today because S_t/M_t is a $\mathbb{Q}$-martingale; the cash leg $-K$ is deterministic and worth $-KP(t,T)$). Between signing and maturity the forward's mark-to-market value drifts as the spot moves: it climbs when S rises, falls when S drops, and accumulates an unrealised gain/loss on the holder's balance sheet. *No cash* is exchanged between counterparties during the life of the contract; the accumulated P&L is settled all at once at delivery.

The futures contract. A futures contract is also an obligation to transact at a future date, but changes in the futures price $F_t(T)$ are settled in cash *every day* into a margin account. Critically,

$$\text{value of a futures position} \;\equiv\; 0, \tag{8.6}$$

at every instant by design. Entering a futures position costs nothing; the contract carries no book value at any mark. Every infinitesimal gain $\mathrm{d}F_t$ is paid out in cash into the margin account immediately, so the holder's exposure at any instant is to the *next* daily price change only. The cumulative P&L of a one-lot long futures position from inception up to time t is simply $F_t(T) - F_0(T)$ — the running sum of daily settlements.

The zero-value feature is the single most important structural difference between forwards and futures. It is why futures can be traded in large notional with only a margin deposit, and why they sit at the centre of the clearing/settlement plumbing of modern exchanges. It is also why — as we are about to see — the pricing PDE for options on futures differs from Black–Scholes by the loss of one term.

Margin account illustration. The following table traces a one-lot long position through four days:

t	$F_t(T)$	ΔF	Margin cash flow	Cumulative margin
0	100	—	0	0
1	101	+1	+1	+1
2	103	+2	+2	+3
3	99	-4	-4	-1

Day 0: open long at $F_0 = 100$. No cash exchanged. Day 1: the futures rises to 101; the broker credits +1 to the margin account. Day 2: +2 more. Day 3: the futures drops to 99; the broker debits 4. The cumulative margin at day t is always $F_t - F_0$ — this is the *mark-to-market invariant* of a futures contract. Continuously, we write $\mathrm{d}F_t$ for the instantaneous cash flow into the margin account, and the cumulative P&L up to t is $\int_0^t \mathrm{d}F_s = F_t - F_0$.

Contrast with a forward. The forward holder's running P&L is also $F_t - F_0$ in present value, but the cash flow arrives only at maturity. Futures deliver the same economic P&L spread out in real time as cash. Under deterministic interest rates this distinction is invisible — the present value of a cash stream equals the present value of a single maturity payment if we can borrow and lend freely at r. Under *stochastic* rates, however, the daily cash flows of a futures position earn (or pay) interest at a rate that may be correlated with S itself, and this correlation opens a price gap between the futures quote and the forward quote that we compute in §8.11.

Operational summary. If a forward's mid-market implied strike moves from F_0 to $F_0 + 1$, the holder's forward mark-to-market moves from 0 to +1 (discounted). A futures holder earning the same +1 move gets +1 in cash deposited to their margin account *that day*, and the futures mark-to-market stays at 0. The two are economically equivalent modulo the interest earned on the margin cash — a difference that is small for short tenors but nontrivial for multi-year contracts.

10.3 8.3 The Futures Price as a $\mathbb{Q}$-Martingale

The economic consequence of daily settlement is a sharp mathematical fact: *the futures price itself is a martingale under the risk-neutral measure* $\mathbb{Q}$. The argument is almost tautological once one accepts the zero-value premise (8.6). Over any short interval $[t, t^+]$ the P&L of a one-lot long futures position is purely stochastic — it is the accumulated mark-to-market cash flow $F_{t^+} - F_t$ — and because the position costs nothing to open, no-arbitrage forces the expected P&L under $\mathbb{Q}$ to be zero:

$$\mathbb{E}^{\mathbb{Q}}[\, F_{t^+}(T) - F_t(T) \mid \mathcal{F}_t \,] \;=\; 0. \tag{8.7}$$

If the conditional expectation were non-zero, there would be a free option: a position that costs nothing to enter, has symmetric upside and downside from the holder's perspective, but carries a positive expected return under the pricing measure. The no-arbitrage principle rules out precisely such free options. Equivalently, for all $s \le t$,

$$\boxed{\mathbb{E}^{\mathbb{Q}}[F_t(T) \mid \mathcal{F}_s] \;=\; F_s(T)} \tag{8.8}$$

— the futures price is a $\mathbb{Q}$-martingale.

A trio of drifts. It is worth lining up the three kinds of asset that appear in this guide and noting the drift each of them takes under $\mathbb{Q}$:

Asset type	$\mathbb{Q}$-drift	Reason
Traded asset (e.g. a non-dividend stock)	$r\, X_t$	Discounted price X_t/M_t is a $\mathbb{Q}$-martingale; carry at r
Traded asset with yield δ	$(r - \delta)\, X_t$	Discounted total return (price + dividends) is a $\mathbb{Q}$-martingale
Futures quote $F_t(T)$	0	Position is costless; expected change under $\mathbb{Q}$ is zero
Non-traded variable (e.g. temperature)	$\mu_t^{\mathbb{Q}} - \sigma_t\, \lambda_t$	Needs a market price of risk λ_t (Chapter 6)

The first three rows are pinned by replication arguments. The fourth requires additional market information — a market price of risk — because there is no hedgeable instrument that pins the drift. For the rest of this chapter, the futures price is the canonical example of row 3.

Forward prices are different. Contrast (8.8) with the analogous statement for a *forward* price $F_t^{\mathrm{fwd}}(T) = S_t/P(t,T)$. Under $\mathbb{Q}$ (bank-account numeraire), $F_t^{\mathrm{fwd}}(T)$ is *not* a martingale in general — both S_t and $1/P(t,T)$ drift, and their ratio drifts. The cleanest statement is instead that $F_t^{\mathrm{fwd}}(T)$ is a $\mathbb{Q}^T$-martingale, where $\mathbb{Q}^T$ is the T-forward measure with numeraire $P(\cdot, T)$. This follows from the general numeraire identity of Chapter 5: if A is a traded numeraire, every traded asset divided by A is a $\mathbb{Q}^A$-martingale, and $S_t/P(t,T)$ is precisely such a ratio. Futures and forwards therefore live in *different measures*: the futures price is naturally a $\mathbb{Q}$-martingale, the forward price is naturally a $\mathbb{Q}^T$-martingale. Under deterministic rates the two measures coincide (their Radon–Nikodym density collapses to 1), so futures and forwards coincide; under stochastic rates they diverge and the gap is the convexity adjustment of §8.11.

A subtle point about "the futures is undiscounted". It is tempting to read (8.8) as saying the futures *price* requires no discounting. That is correct. But an *option on a futures* — whose cash payoff settles at some maturity T_0 — is an ordinary traded derivative whose cash payoff must be discounted back to today in the usual way. The common pitfall is to conflate the two: the futures *quote* has no drift under $\mathbb{Q}$, but the option's *value* still carries the usual discount factor $e^{-r(T_0-t)}$. The Black '76 formula in §8.7 will make this explicit.

10.4 8.4 Dynamics of the Futures Price Under $\mathbb{Q}$

The martingale property (8.8) fixes the $\mathbb{Q}$-drift of $F_t(T)$ at zero, but leaves the volatility unspecified. Different market settings motivate different volatility parameterisations. We postulate a general form

$$\mathrm{d}F_t(T) \;=\; \sigma^F(t, F_t(T))\,\mathrm{d}\widehat{W}_t \qquad (\mathbb{Q}\text{-measure}), \tag{8.9}$$

with $\widehat{W}$ a $\mathbb{Q}$-Brownian motion. The specific functional form of $\sigma^F(t,F)$ is a modelling choice:

- **GBM.** $\sigma^F(t,F) = \sigma\,F$ yields the log-normal Black '76 model of §8.7, which is the market standard for equity-index futures, FX forwards, and commodity-futures implied-volatility quotation.
- **Bachelier (arithmetic).** $\sigma^F(t,F) = \sigma$ (constant) yields the arithmetic-Brownian model of §8.8, appropriate when F can go negative — interest-rate futures near the zero bound, or crude-oil futures during the April 2020 episode.
- **Samuelson / OU.** $\sigma^F(t,F) = \sigma\,e^{-\eta(T-t)}$ yields a mean-reverting arithmetic model where the vol loading *decays* as one looks further forward in the curve — matching the empirical fact that long-dated commodity futures move less in absolute terms than front-month ones (§8.9).

Derivation under constant rates. For the GBM case it is worth showing that (8.9) with $\sigma^F = \sigma F$ falls out of the spot dynamics plus the cost-of-carry identity. Suppose the spot satisfies

$$\frac{\mathrm{d}S_t}{S_t} \;=\; \mu\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t \qquad (\mathbb{P}), \tag{8.10}$$

and constant rates r apply. The forward/futures price is $F_t(T) = S_t\,e^{r(T-t)}$. By Itô on $f(t,S) = S\,e^{r(T-t)}$,

$$\mathrm{d}F_t(T) \;=\; e^{r(T-t)}\,\mathrm{d}S_t \;-\; r\,S_t\,e^{r(T-t)}\,\mathrm{d}t. \tag{8.11}$$

Expanding $\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$ and using $F_t = S_t\,e^{r(T-t)}$,

$$\frac{\mathrm{d}F_t(T)}{F_t(T)} \;=\; (\mu - r)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t \qquad (\mathbb{P}). \tag{8.12}$$

Under the risk-neutral measure $\mathbb{Q}$, the stock's drift is $\mu = r$ (Chapter 6, market-price-of-risk argument). Substituting,

$$\boxed{\frac{\mathrm{d}F_t(T)}{F_t(T)} \;=\; \sigma\,\mathrm{d}\widehat{W}_t \qquad (\mathbb{Q}).} \tag{8.13}$$

The drift $(\mu-r)\,\mathrm{d}t$ in (8.12) is exactly cancelled under $\mathbb{Q}$ by the measure change $\mathrm{d}W_t \to \mathrm{d}\widehat{W}_t + \lambda\,\mathrm{d}t$ with $\lambda = (\mu-r)/\sigma$ (Girsanov, Chapter 5). The drift $(\mu - r)$ gets killed; the diffusion σ survives. This is the martingale property (8.8) made explicit at the SDE level. Every drift term in the spot SDE — whether from physical-world mean-reversion, from the cost of carry r, or from a dividend yield — is absorbed into the futures quote or vanishes under $\mathbb{Q}$; what survives on the futures is only the diffusion term.

Case with dividends. If the stock pays a continuous yield δ, the spot drifts at $r - \delta$ under $\mathbb{Q}$, and the cost-of-carry formula becomes $F_t(T) = S_t\,e^{(r-\delta)(T-t)}$. Applying Itô's product rule to $F_t = S_t\,g(t)$ with $g(t) = e^{(r-\delta)(T-t)}$, $g'(t) = -(r-\delta)g(t)$:

$$\begin{aligned}
\mathrm{d}F_t &= g(t)\,\mathrm{d}S_t + S_t\,g'(t)\,\mathrm{d}t \\
&= g(t)S_t\Big[(r-\delta)\,\mathrm{d}t + \sigma\,\mathrm{d}\widehat{W}_t\Big] - (r-\delta)g(t)S_t\,\mathrm{d}t \\
&= \sigma\,g(t)S_t\,\mathrm{d}\widehat{W}_t = \sigma F_t\,\mathrm{d}\widehat{W}_t,
\end{aligned}$$

so

$$\boxed{\frac{\mathrm{d}F_t(T)}{F_t(T)} \;=\; \sigma\,\mathrm{d}\widehat{W}_t \qquad (\mathbb{Q})}.$$

The $(r-\delta)$-drift of the spot exactly cancels the time-decay of the carry factor: F is a $\mathbb{Q}$-martingale even with dividends, as the FTAP on futures (8.8) demands. This is the general principle — any continuous yield or cost-of-carry term that affects the spot under $\mathbb{Q}$ gets absorbed into the forward's deterministic drift and cancels, leaving pure diffusion on the forward/futures quote.

General dynamics. For modelling purposes, we often write (8.9) with a general state- and time-dependent volatility $\sigma^F(t, F)$:

$$\mathrm{d}F_t(T) \;=\; \sigma^F(t, F_t(T))\,\mathrm{d}\widehat{W}_t. \tag{8.14}$$

All the machinery of the next three sections works at this level of generality: the Black PDE, its Feynman–Kac representation, and the closed-forms for specific choices of σ^F are all instances of (8.14).

10.5 8.5 Black PDE for options on futures

A European option on a futures contract has terminal payoff

$$g_{T_0} \;=\; \varphi(F_{T_0}(T)), \tag{8.15}$$

with T_0 the option's exercise date and $T \geq T_0$ the futures' delivery date (typically equal). Because F is a $\mathbb{Q}$-martingale, the resulting PDE will differ from BS in one structural way: the first-derivative drift term *vanishes*. We derive it twice — by self-financing replication, and by Feynman–Kac.

10.5.1 8.5.1 Setup of the replicating portfolio

Let $g_t = g(t, F_t(T))$. Form

$$V_t \;=\; \alpha_t\,F_t(T) \;+\; \beta_t\,B_t \;-\; g_t, \tag{8.16}$$

with α_t futures contracts, β_t units of the money-market account B_t (so $\mathrm{d}B_t = r\,B_t\,\mathrm{d}t$), and $V_0 = 0$. Entering a futures contract *costs nothing*: the $\alpha_t F_t(T)$ entry is notional, not a capital outlay. P&L from the futures comes entirely from the daily cash flow $\alpha_t\,\mathrm{d}F_t$.

10.5.2 8.5.2 Self-financing increment

The wealth change is

$$\mathrm{d}V_t \;=\; \alpha_t\,\mathrm{d}F_t(T) \;+\; \beta_t\,\mathrm{d}B_t \;-\; \mathrm{d}g_t. \tag{8.18}$$

Expanding via Itô on $g(t, F)$ and using the driftless dynamics of F under $\mathbb{Q}$,

$$\mathrm{d}g_t \;=\; \big[\partial_t g \;+\; \tfrac{1}{2}(\sigma^F)^2\,\partial_{FF} g\big]\,\mathrm{d}t \;+\; \sigma^F\,\partial_F g\,\mathrm{d}\widehat{W}_t, \tag{8.19}$$

so

$$\mathrm{d}V_t \;=\; \alpha_t\,\sigma^F\,\mathrm{d}\widehat{W}_t \;+\; \beta_t\,r\,B_t\,\mathrm{d}t \;-\; \big[\partial_t g + \tfrac{1}{2}(\sigma^F)^2\,\partial_{FF} g\big]\,\mathrm{d}t \;-\; \sigma^F\,\partial_F g\,\mathrm{d}\widehat{W}_t. \tag{8.20}$$

10.5.3 8.5.3 Eliminating noise and drift

The $\mathrm{d}\widehat{W}$ coefficient vanishes iff

$$\boxed{\alpha_t \;=\; \partial_F g(t, F_t(T)),} \tag{8.21}$$

the *futures-delta* hedge — $\partial_F g$ units of futures notional per option short. With $\alpha_t = \partial_F g$ imposed, the $\mathrm{d}t$ coefficient in (8.20) is

$$\mathcal{A}_t \;=\; \beta_t\, r\, B_t \;-\; \partial_t g \;-\; \tfrac{1}{2}(\sigma^F)^2\, \partial_{FF} g. \tag{8.22}$$

From $V_t \equiv 0$ and the zero-value feature of the futures position, $\beta_t\, B_t = g_t$, so $\beta_t r B_t = r\, g_t$ (eq 8.23). Setting $\mathcal{A}_t = 0$ then yields the *Black PDE for options on futures*:

10.5.4 8.5.4 The PDE

$$\boxed{\partial_t g \;+\; \tfrac{1}{2}\,(\sigma^F(t,F))^2\, \partial_{FF} g \;=\; r\, g, \qquad g(T_0, F) = \varphi(F).} \tag{8.24}$$

Compared to the BS PDE for a stock,

$$\partial_t g \;+\; r\, S\, \partial_S g \;+\; \tfrac{1}{2}\,\sigma^2 S^2\, \partial_{SS} g \;=\; r\, g,$$

the first-derivative drift $rS\partial_S g$ has *disappeared*. This is the signature of pricing off a $\mathbb{Q}$-martingale underlying. The rg on the right is the discount term (the option's *cash* payoff settles at T_0 and must be discounted, even though the futures itself is undiscounted). For $\sigma^F(t,F) = \sigma F$ this specialises to

$$\partial_t g \;+\; \tfrac{1}{2}\,\sigma^2\, F^2\, \partial_{FF} g \;=\; r\, g, \qquad g(T_0, F) = \varphi(F), \tag{8.25}$$

the PDE that the Black '76 formula of §8.7 solves.

10.5.5 8.5.5 Feynman–Kac representation

Applying Feynman–Kac (Chapter 4) to (8.24) gives the cross-check

$$\boxed{g(t,F) \;=\; \mathbb{E}^{\mathbb{Q}}\Big[\, e^{-r(T_0-t)}\, \varphi(F_{T_0}(T)) \;\Big|\; F_t(T) = F \,\Big]\,,} \tag{8.26}$$

with F_u driftless under $\mathbb{Q}$. The two differences from the stock-BS expectation of Chapter 4: the drift of the underlying is *zero* (not r), and the discount factor $e^{-r(T_0-t)}$ multiplies the payoff because the option's cash settles at T_0. Under stochastic rates, the forward-measure variant

$$\frac{g(t,F)}{P(t,T_0)} \;=\; \mathbb{E}^{\mathbb{Q}^{T_0}}\big[\, \varphi(F_{T_0}(T)) \,\big|\, F_t(T) = F \,\big] \tag{8.27}$$

is the cleaner starting point (and is used in Chapter 13). Integrating (8.14) gives the GBM and Bachelier marginal distributions:

$$F_{T_0}(T) \;=\; F_t(T)\, \exp\!\Big\{ -\tfrac{1}{2}\sigma^2 (T_0 - t) + \sigma\, (\widehat{W}_{T_0} - \widehat{W}_t) \Big\} \tag{8.28}$$

for the log-normal case $\sigma^F = \sigma F$, and

$$F_{T_0}(T) \;=\; F_t(T) \;+\; \sigma\, (\widehat{W}_{T_0} - \widehat{W}_t) \tag{8.29}$$

for Bachelier $\sigma^F = \sigma$. In both cases $\mathbb{E}^{\mathbb{Q}}[F_{T_0}(T) \mid \mathcal{F}_t] = F_t(T)$, restating the martingale property (8.8).

10.6 8.7 The Black '76 Closed Form

For the constant-volatility GBM case $\sigma^F(t,F) = \sigma F$ with payoff $\varphi(F) = (F-K)_+$, the PDE (8.25) admits the celebrated *Black '76* closed-form solution:

$$\boxed{g(t,F) \;=\; e^{-r(T_0-t)}\big[\,F\,\Phi(d_+) \;-\; K\,\Phi(d_-)\,\big],} \tag{8.30}$$

with

$$d_\pm \;=\; \frac{\ln(F/K) \;\pm\; \frac{1}{2}\,\sigma^2\,(T_0-t)}{\sigma\,\sqrt{T_0-t}}. \tag{8.31}$$

Derivation via Feynman–Kac. Starting from (8.26) with $\varphi(F) = (F-K)_+$ and the log-normal law (8.28):

$$g(t,F) \;=\; e^{-r(T_0-t)}\,\mathbb{E}^{\mathbb{Q}}\Big[\,(F_{T_0}(T)-K)_+ \mid F_t(T) = F\,\Big].$$

The expectation is the classical log-normal "call-option" integral, and the log-variance is $\sigma^2(T_0-t)$ with log-mean $\ln F - \frac{1}{2}\sigma^2(T_0-t)$. A direct Gaussian computation gives

$$\mathbb{E}^{\mathbb{Q}}\Big[(F_{T_0}-K)_+ \mid F_t = F\Big] \;=\; F\,\Phi(d_+) \;-\; K\,\Phi(d_-),$$

and multiplying by the discount $e^{-r(T_0-t)}$ yields (8.30).

Comparison to Black–Scholes. The standard BS formula on a non-dividend stock is

$$C^{\mathrm{BS}}(S,K,r,\sigma,\tau) \;=\; S\,\Phi(d_1) \;-\; K\,e^{-r\tau}\,\Phi(d_2),$$

with $d_{1,2} = [\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau]/(\sigma\sqrt{\tau})$. The Black '76 formula (8.30) differs in two readable ways:

	Black–Scholes (spot)	Black '76 (futures)
"Moneyness" in $d_\pm$	$\ln(S/K) + r\tau$	$\ln(F/K)$ only
Coefficient on $\Phi(d_+)$	S	$e^{-r\tau}\,F$
Coefficient on $\Phi(d_-)$	$K\,e^{-r\tau}$	$e^{-r\tau}\,K$

The first change reflects the absence of a first-derivative drift term in the Black PDE — the "cost of carry" $r\tau$ that shifts the moneyness in BS does not appear in Black '76 because the underlying is already a martingale. The second change reflects that the futures quote F is itself undiscounted; to state the option price as a dollar amount today, we simply discount the whole expression by $e^{-r\tau}$.

Three observations on the formula.

First, (8.30) is the classical BS formula *with the spot replaced by the forward* and *with no drift shift in* $d_\pm$. The formula is strictly simpler than the dividend-paying BS formula of Chapter 7: the dividend yield does not appear at all because it has already been absorbed into the definition of the forward/futures price. Third, the formula specialises cleanly to the constant-dividend-stock case: plugging $F_t(T) = S_t\,e^{(r-\delta)(T-t)}$ into (8.30) and simplifying recovers the dividend-BS formula of Chapter 7 exactly. Black '76 is the natural "forward-measure" simplification of the dividend-BS formula.

Second, a common pitfall: the discount factor in (8.30) is $e^{-r(T_0-t)}$, not $e^{-r(T-t)}$. The option matures at T_0 (potentially earlier than the futures' delivery at T), and the discount is from today to *option maturity*, not to futures delivery. For standardised exchange-listed products the two dates coincide — but for bespoke derivatives written on an exchange-listed futures (e.g. a mid-curve option on an interest-rate future) they can differ by several months. Always check which date enters the discount.

Third, the Black '76 delta — the sensitivity of the option price to the futures quote — is

$$\partial_F g \;=\; e^{-r(T_0-t)}\,\Phi(d_+). \tag{8.32}$$

Compare to the Black–Scholes spot delta $\Phi(d_1)$: the Black '76 delta is *discounted*. Operationally this means that to hedge one option written, one holds fewer than $\Phi(d_+)$ futures contracts — specifically, one holds $e^{-r(T_0-t)}\,\Phi(d_+)$. For short-dated options the discount is small and the two deltas differ only at the third decimal; for multi-year options the discount can be noticeable.

Black '76 Greeks. Differentiating (8.30) yields a complete table of Greeks. Writing $\tau = T_0 - t$ and ϕ for the standard-normal PDF:

$$\begin{aligned}
\Delta &= \partial_F g \;=\; e^{-r\tau}\,\Phi(d_+),\\
\Gamma &= \partial_{FF} g \;=\; \frac{e^{-r\tau}\,\phi(d_+)}{F\,\sigma\,\sqrt{\tau}},\\
\text{Vega} &= \partial_\sigma g \;=\; e^{-r\tau}\,F\,\phi(d_+)\,\sqrt{\tau},\\
\Theta &= \partial_t g \;=\; -\frac{e^{-r\tau}\,F\,\phi(d_+)\,\sigma}{2\sqrt{\tau}} \;+\; r\,e^{-r\tau}\,[F\,\Phi(d_+) - K\,\Phi(d_-)],\\
\rho &= \partial_r g \;=\; -\tau\,g(t,F).
\end{aligned}$$

Several features are worth flagging. First, the gamma has an extra discount factor $e^{-r\tau}$ compared to BS — again a consequence of the "futures is a martingale, not a GBM" feature. Second, the rho is *negative* for a call: an increase in rates lowers the call price (because the cash settlement at expiry is discounted more heavily), whereas for a BS spot call on a non-dividend stock, rho is positive. The sign flip is one of the cleaner diagnostic tests for whether a pricing engine is treating a contract as a spot-based or futures-based product. Third, the vega formula is similar to BS, but the F sitting outside is the futures quote, not the spot — so a Black '76 vega of 0.5 on a \$100 futures differs from a BS vega of 0.5 on a \$100 spot only by the overall discount factor and the substitution $S \to F$.

A worked numerical check. Take $F = 100$, $K = 100$, $\sigma = 20\%$, $\tau = 0.25$ years, $r = 5\%$. Then $d_+ = \frac{1}{2}\sigma\sqrt{\tau} = 0.05$ and $d_- = -d_+ = -0.05$. $\Phi(0.05) \approx 0.5199$, $\Phi(-0.05) \approx 0.4801$. The Black '76 call price is

$$g \;=\; e^{-0.05\cdot 0.25}\,[100\cdot 0.5199 - 100\cdot 0.4801] \;\approx\; 0.9876\cdot 3.98 \;\approx\; 3.93.$$

The analogous BS call on a \$100 spot with the same volatility and a \$100 strike at the same rate is $S\,\Phi(d_1) - K\,e^{-r\tau}\,\Phi(d_2)$ with $d_{1,2} = (r \pm \frac{1}{2}\sigma^2)\tau/(\sigma\sqrt{\tau})$, giving $d_1 \approx 0.175$, $d_2 \approx 0.075$, and a price near 4.62. The BS call is higher than the Black '76 call because the spot drift under $\mathbb{Q}$ is $r > 0$ — the underlying grows into the strike — while the futures is a martingale and does not. The difference is exactly the first-derivative drift term cumulating over τ.

Why the Black formula is so ubiquitous. The Black '76 formula appears in a surprising number of places beyond literal options on exchange-traded futures. It is the market-standard quotation for caps, floors, and swaptions on interest-rate forwards (Chapter 14); for options on commodity forwards; and — via a well-known convention — for at-the-money options on forward-looking yield curves. The underlying reason is always the same: whenever the underlying is (or can be reparameterised as) a martingale under *some* appropriately chosen measure, the pricing PDE loses its first-derivative drift term and the closed-form reduces to (8.30). In the LIBOR/SOFR caplet setting of Chapter 14, the forward rate $L(t, T_1, T_2)$ is a martingale under the T_2-forward measure, and the caplet price is exactly the Black '76 formula with F denoting the forward rate and σ its forward-rate volatility. The technical device of changing numeraire — formalised in Chapter 5 — generalises the futures-price argument to these other martingale underlyings, and every such numeraire change collapses the relevant pricing PDE to a Black-style form.

Black '76 drift-cancel schematic *Drift-cancel schematic — the first-derivative drift term in the Black PDE vanishes because the futures is a $\mathbb{Q}$-martingale, leaving only the second-derivative diffusion.*

10.7 8.8 Example — Bachelier Futures and the Arithmetic Binary

The Bachelier (arithmetic Brownian motion) specification is the simplest possible futures model: constant *absolute* volatility, no drift. Although the log-normal Black '76 model of §8.7 dominates textbook presentations, the arithmetic model is the natural choice for underlyings that can go negative — short-rate futures near the zero bound, or, famously, crude-oil futures during the April 2020 negative-price episode. It also isolates the features of the martingale pricing approach from the mechanics of log-normal integration, which is pedagogically useful.

Take

$$\mathrm{d}F_t \;=\; \sigma\,\mathrm{d}\widehat{W}_t \qquad (\mathbb{Q}), \tag{8.33}$$

with constant absolute volatility σ, and consider a *binary call*

$$\varphi(F) \;=\; \mathbf{1}_{\{F>K\}}$$

— the claim pays 1 unit if the futures is above K at option expiry T_0, zero otherwise. For simplicity we take $r = 0$ in this example so that no discount factor clutters the formula; the general $r \neq 0$ result multiplies the answer by $e^{-r(T_0-t)}$.

10.7.1 8.8.1 Terminal distribution

From (8.33),

$$F_{T_0} - F_t \;=\; \sigma\,(\widehat{W}_{T_0} - \widehat{W}_t), \qquad \widehat{W}_{T_0} - \widehat{W}_t \overset{d}{=} \mathcal{N}(0, T_0 - t), \tag{8.34}$$

so F_{T_0} is conditionally normal with mean F_t (the martingale property) and variance $\sigma^2\,(T_0 - t)$.

10.7.2 8.8.2 Pricing

By Feynman–Kac (8.26) with $r = 0$,

$$h(t,F) \;=\; \mathbb{E}^{\mathbb{Q}}_{t,F}\Big[\mathbf{1}_{\{F_{T_0}>K\}}\Big] \;=\; \mathbb{Q}_{t,F}(F_{T_0} > K). \tag{8.35}$$

Writing $F_{T_0} - F_t = \sigma\,(\widehat{W}_{T_0} - \widehat{W}_t)$ and letting $Z \sim \mathcal{N}(0,1)$,

$$h(t,F) \;=\; \mathbb{Q}\left(Z \;>\; \frac{K-F}{\sigma\,\sqrt{T_0 - t}}\right) \;=\; \Phi\left(\frac{F-K}{\sigma\,\sqrt{T_0 - t}}\right), \tag{8.36}$$

with Φ the standard-normal CDF. (For the historically-rooted version using $\sigma = 2$ that appears in the draft, one simply substitutes $\sigma = 2$ into (8.36).)

10.7.3 8.8.3 Reading the answer

The argument of Φ in (8.36) is a *z-score*: $(F - K)$ in units of the remaining standard deviation $\sigma\sqrt{T_0 - t}$. The price of the digital is the probability that a standard normal exceeds $-z$. At the money ($F = K$), $\Phi(0) = 1/2$ exactly — a symmetry unique to the arithmetic model. Under $r \neq 0$, discount the dollar payoff: $h(t,F) = e^{-r(T_0-t)}\,\Phi((F - K)/(\sigma\sqrt{T_0 - t}))$. Terminal limit $T_0 \to t$ collapses Φ to $\mathbf{1}_{\{F>K\}}$, matching the payoff.

10.7.4 8.8.4 Case study: Negative WTI futures, April 20 2020 — when lognormal breaks

Storage at Cushing was full; longs had to pay to take delivery. Black-76's $\ln F$ *is undefined at* $F \leq 0$*; Bachelier's* $\mathbb{R}$*-support is. CME switched the official option model from Black-76 to Bachelier the same week.*

Context. On Monday 20 April 2020, the front-month NYMEX WTI crude futures (May 2020 contract, CL K20) settled at *minus* \$37.63 per barrel — the first negative settle in the history of the contract. With physical delivery at Cushing, Oklahoma scheduled for May 21 and storage there saturated by collapsed COVID-era demand, holders of long expiring futures discovered they would have to *pay* to take delivery. Through the day the contract traded from +\$18 to −\$40; CME had reprogrammed its trade-matching engine on Friday 17 April explicitly to accept negative prices, after a research notice flagged that the legacy Black-76 pricing engines could not quote options on a contract that might cross zero.

The market structure on the day made the move worse. Open interest in the May contract was still ~108k lots heading into the final settlement window with three trading days to expiry — unusual: by convention the front-month contract rolls down to a few thousand lots before notice. The overhang was substantially held by long-only commodity ETFs (notably the United States Oil Fund, USO, which alone held ~25% of front-month OI in mid-April) that were structurally unable to take delivery. Forced rolling into a non-buyer market produced a price discovery process that was about storage logistics, not about the consensus value of a barrel of oil.

Through the chapter's math. Black-76 (8.30) requires $F > 0$ because the lognormal SDE (8.13), $\mathrm{d}F/F = \sigma\,\mathrm{d}\widehat{W}$, has an absorbing barrier at zero — log-prices are not defined for $F \leq 0$ and the formula for $d_\pm$ contains $\ln(F/K)$, which is singular. Bachelier dynamics (8.34), $\mathrm{d}F = \sigma\,\mathrm{d}\widehat{W}$, are defined on all of $\mathbb{R}$: the terminal distribution (8.29) is normal with mean F_t, so negative outcomes are admissible and the binary call (8.36) is well-defined at every F. The choice of model is not decorative — it determines which prices are even *expressible*. CME formally switched the official option-pricing methodology for WTI and several refined-product contracts from Black-76 to Bachelier on Friday 22 April 2020 (Special Executive Report S-8520). By April 2020 every major commodity desk had migrated WTI option pricing to a Bachelier or shifted-lognormal engine; the calendar-spread option desks who had not finished migration booked materially mis-marked positions on the day.

Lesson. Model choice is not a stylistic preference but a constraint on admissible states. A Bachelier model with a 40% *relative* implied vol (quoted at $F_t = 40$) implies an *absolute* sigma of $\sigma_{\text{abs}} = 0.40 \cdot 40 = 16$ dollars; a one-month standard deviation is $16 \cdot \sqrt{1/12} \approx 4.6$ dollars. The May 2020 settle of $-\$37.63$ was roughly 6–8 absolute sigmas below the prior week's spot — a Bachelier-tail event, but a *finite-probability* Bachelier-tail event. Black-76, by contrast, assigned zero probability mass to $F \leq 0$ and therefore could not even represent the realised outcome. The rule of thumb: use Bachelier (or normal / shifted-lognormal) whenever the underlying has a non-negligible probability of crossing zero — short-end rates after the Eurozone NIRP regime, calendar spreads on storable commodities, and any futures whose deliverable carries non-trivial storage cost. Once a contract has even a one-percent chance of trading below zero on the horizon of interest, lognormal is not "approximately right" — it is the wrong measure space.

10.8 8.9 Example — Ornstein–Uhlenbeck Futures and the Samuelson Effect

The Bachelier model is too simple to describe real commodity or interest-rate futures markets. Empirically, the *volatility* of the futures curve is not flat across maturities: long-dated crude futures move much less in absolute terms than front-month contracts; the same is true of rates futures, where a hike priced into the December contract barely perturbs the contract two years out. The empirical name for this pattern is the *Samuelson effect*, and an OU-type model with a time-decaying volatility loading $\sigma\, e^{-\eta(T-t)}$ is the cleanest way to reproduce it.

10.8.1 8.9.1 Specification

Under the *physical* measure, the futures is often specified as a mean-reverting diffusion:

$$\mathrm{d}F_t(T) \;=\; \kappa\,(\theta - F_t(T))\,\mathrm{d}t \;+\; \sigma\, e^{-\eta(T-t)}\,\mathrm{d}W_t \qquad (\mathbb{P}). \tag{8.37}$$

But under the risk-neutral measure $\mathbb{Q}$, the martingale property (8.8) forces the drift to vanish. Girsanov's theorem (Chapter 5) tells us there is always a measure equivalent to $\mathbb{P}$ that kills any given drift, and the martingale property of a costless position forces that measure to be the pricing measure $\mathbb{Q}$. Accordingly, under $\mathbb{Q}$,

$$\mathrm{d}F_t(T) \;=\; \sigma\, e^{-\eta(T-t)}\,\mathrm{d}\widehat{W}_t. \tag{8.38}$$

The mean-reversion drift *was* present in the physical-world dynamics but *does not need to appear* under $\mathbb{Q}$: the information about where the futures is heading is already encoded in the initial forward curve $F_0(T)$ and in the volatility decay $e^{-\eta(T-t)}$. Said differently, once the market has priced a forward curve that reflects mean-reversion, the curve itself carries the information, and the $\mathbb{Q}$-dynamics of each individual futures quote can be driftless without loss.

10.8.2 8.9.2 The Samuelson decay

The economic content of the loading $e^{-\eta(T-t)}$ is the following. A shock to the spot propagates through to futures of every maturity, but because the spot itself is mean-reverting, shocks decay before they reach delivery. A contract with delivery T therefore "sees" only the fraction $e^{-\eta(T-t)}$ of any time-t shock — the rest has time to mean-revert back to θ before delivery. Long-dated contracts sit on the flat part of the decay curve with a small vol loading; the front month sits on the steep part. As calendar time t advances and the contract approaches its own delivery T, the exponent $\eta(T-t)$ shrinks toward zero, the vol loading grows toward σ, and the futures quote becomes more and more responsive to spot shocks. That is the Samuelson effect made quantitative.

Samuelson OU futures *Samuelson decay — the vol loading $\sigma e^{-\eta(T-t)}$ drops with tenor, so far-dated contracts respond less to same-size shocks than near contracts.*

10.8.3 8.9.3 Terminal distribution

From (8.38),

$$F_{T_0}(T) - F_t(T) \;=\; \int_t^{T_0} \sigma\, e^{-\eta(T-u)}\, \mathrm{d}\widehat{W}_u. \tag{8.39}$$

The integrand is deterministic, so the right-hand side is Gaussian with mean zero. By Itô's isometry, the variance is

$$\Sigma_t^2 \;=\; \int_t^{T_0} \sigma^2\, e^{-2\eta(T-u)}\, \mathrm{d}u \;=\; \sigma^2\, \frac{e^{-2\eta(T-T_0)} - e^{-2\eta(T-t)}}{2\eta}. \tag{8.40}$$

Hence

$$F_{T_0}(T) - F_t(T) \;\sim_{\mathbb{Q}}\; \mathcal{N}(0, \Sigma_t^2), \tag{8.41}$$

conditional on $\mathcal{F}_t$.

Reading (8.40). Formula (8.40) is worth staring at. It encodes the total variance accumulated by the futures between t (now) and T_0 (option expiry), given that delivery is at T. There are three dials:

- σ is the peak instantaneous vol, attained at delivery ($T = u$);
- η is the speed at which long-dated vols decay below that peak;
- the $T - t$ vs $T - T_0$ spacing controls where on the decay curve we are integrating.

Two useful limits:

- **$\eta \to 0$.** Expanding the exponentials, $\Sigma_t^2 \to \sigma^2\,(T_0 - t)$ — ordinary Brownian motion with constant vol. The Samuelson decay vanishes.
- **$\eta \to \infty$.** The exponents push the integrand to essentially zero everywhere except in a tiny sliver just before T, and $\Sigma_t^2 \to 0$: long-dated futures with infinite decay are stationary and have no vol.

Real markets sit in between, with η estimated in the range 0.3 to 2.0 per year for many commodities and interest rates.

10.8.4 8.9.4 Pricing the binary

For the binary call $\varphi(F) = \mathbf{1}_{\{F>K\}}$ at option expiry T_0, with $r = 0$:

$$h(t, F) \;=\; \mathbb{Q}_{t,F}(F_{T_0}(T) > K) \;=\; \mathbb{Q}(F_{T_0} - F_t > K - F_t), \tag{8.42}$$

and writing $F_{T_0} - F_t = \Sigma_t\, Z$ with $Z \sim \mathcal{N}(0,1)$,

$$h(t, F) \;=\; \mathbb{Q}\Big(Z > \frac{K - F}{\Sigma_t}\Big) \;=\; \Phi\Big(\frac{F - K}{\Sigma_t}\Big), \tag{8.43}$$

with Σ_t defined in (8.40).

Structural similarity. Formula (8.43) has the same structural shape as the Bachelier binary formula (8.36): both are Φ of a $(F-K)/\Sigma$ z-score. The Samuelson structure is entirely packaged into the single number Σ_t. This single-number compression is a feature of the arithmetic-Brownian world: no matter how baroque the vol surface of the futures, as long as the SDE is driftless with a deterministic (possibly time-dependent) vol, the terminal distribution is normal with variance equal to the integrated instantaneous variance, and any digital-style payoff prices off Φ of a single z-score.

10.8.5 8.9.5 Practical implications and HJM pointer

The variance Σ_t^2 shrinks as $T - T_0$ grows, so options on long-dated futures are dramatically cheaper than those on near-dated ones at otherwise similar parameters. Numerical case: $\sigma = 0.40$, $\eta = 1.0$, $T = 2$, $T_0 = 0.25$, $F_0 = 50$, $K = 55$ gives $\Sigma_0 \approx 0.0309$ — about a sixth of the flat-vol Bachelier $\sigma\sqrt{T_0} = 0.20$. The OU futures of (8.38) is the single-factor exponentially-decaying-vol case of the Heath–Jarrow–Morton (HJM) framework; multi-factor HJM models of the *entire* forward curve are standard in commodity and fixed-income practice (see Chapter 13 multi-factor short-rate and Chapter 14 LIBOR market models).

10.9 8.10 Margrabe Exchange Option — A Numeraire Change Tour de Force

The Margrabe exchange option is the perfect testbed for the numeraire-change machinery of Chapter 5. A frontal assault on it — setting up a two-dimensional PDE in A and B and trying to solve — is painful; a judicious change of numeraire reduces the problem to a one-dimensional Black–Scholes put with no interest-rate discounting visible anywhere. The reduction is near-magical the first time one sees it, and it illustrates a principle that applies far beyond this single example: *if the payoff has a natural unit (dollars, shares of A, shares of B, bonds), price in that unit.* Discounting, drifts, and the multi-dimensional diffusion all collapse when one speaks the right numeraire language.

10.9.1 8.10.1 Setup

Suppose there are two traded assets A and B with dynamics under the physical measure

$$\frac{\mathrm{d}A_t}{A_t} \;=\; \mu^A\,\mathrm{d}t \;+\; \sigma^A\,\mathrm{d}W_t, \qquad \frac{\mathrm{d}B_t}{B_t} \;=\; \mu^B\,\mathrm{d}t \;+\; \sigma^B\,\mathrm{d}W_t \qquad (\mathbb{P}), \tag{8.44}$$

driven (for the moment) by the same scalar Brownian W. The general correlated case is handled by (8.55) below. We value the *Margrabe exchange option* paying

$$\mathcal{C} \;=\; \max(A_T - B_T,\, 0) \tag{8.45}$$

at maturity T. Intuitively this is a call on A struck at B — or, equivalently, a put on B struck at A, since $\max(A-B, 0)$ can be read either way. The striking feature of the Margrabe payoff, as against a vanilla call, is that the *strike is itself a random variable*, moving in lockstep with the other leg. This is the source of all the elegance that follows: if we denominate everything in units of A, the strike becomes the constant 1 (because A_T in units of A_T is simply 1), and the problem collapses to pricing a put on the ratio B/A struck at 1. All the randomness of A has been absorbed into the numeraire.

Margrabe exchange option payoff surface *Margrabe exchange option payoff surface* $\max(A_T - B_T, 0)$ *across* (A_T, B_T).

10.9.2 8.10.2 Step 1 — Risk-neutral pricing under $\mathbb{Q}$

First, change measure from $\mathbb{P}$ to the bank-account measure $\mathbb{Q}$ so that both assets drift at the risk-free rate r:

$$\frac{\mathrm{d}A_t}{A_t} \;=\; r\,\mathrm{d}t \;+\; \sigma^A\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \frac{\mathrm{d}B_t}{B_t} \;=\; r\,\mathrm{d}t \;+\; \sigma^B\,\mathrm{d}W_t^{\mathbb{Q}}. \tag{8.46}$$

(Girsanov's theorem from Chapter 5 guarantees such a measure exists, with drift shifts $\lambda^A = (\mu^A - r)/\sigma^A$ and $\lambda^B = (\mu^B - r)/\sigma^B$. When the same W drives both, consistency requires $\lambda^A = \lambda^B$ — the market-price-of-risk condition of Chapter 6.)

The risk-neutral price is

$$f(t, A, B) \;=\; \mathbb{E}^{\mathbb{Q}}_{t,A,B}\big[\,(A_T - B_T)_+\, e^{-r(T-t)}\,\big]\,. \tag{8.47}$$

Trying to evaluate (8.47) directly requires a two-dimensional log-normal integral in (A_T, B_T) — feasible but not elegant.

10.9.3 8.10.3 Step 2 — Change numeraire to A

Divide both sides of (8.47) by A_t. By the general numeraire-pricing identity of Chapter 5,

$$\frac{f(t, A, B)}{A_t} \;=\; \mathbb{E}^{\mathbb{Q}^A}_{t,A,B}\left[\frac{(A_T - B_T)_+}{A_T}\right] \;=\; \mathbb{E}^{\mathbb{Q}^A}_{t,A,B}\left[\left(1 - \tfrac{B_T}{A_T}\right)_+\right], \tag{8.48}$$

where $\mathbb{Q}^A$ is the measure with numeraire A — under which every traded asset divided by A is a martingale. The Radon–Nikodym density is $\mathrm{d}\mathbb{Q}^A/\mathrm{d}\mathbb{Q} = (A_T/A_0)/(M_T/M_0)$, as derived in Chapter 5.

The move from (8.47) to (8.48) is where the magic happens. Dividing the option price by A_t reinterprets "price" as "how many shares of A is the option worth". Because A is itself a traded asset, the price in units of A must be a $\mathbb{Q}^A$-martingale, which kills the discount factor entirely: a martingale's value today is just its conditional expectation tomorrow, no time-value adjustment required. What remains inside the expectation is a payoff of $(1 - B_T/A_T)_+$ — an ordinary put on the single random variable $X_T = B_T/A_T$ struck at 1. We have turned a two-asset pricing problem into a one-asset pricing problem.

Define

$$X_t \;\equiv\; \frac{B_t}{A_t}, \tag{8.49}$$

which, being the price of the traded asset B expressed in numeraire A, is a $\mathbb{Q}^A$-martingale.

10.9.4 8.10.4 Step 3 — Dynamics of X_t under $\mathbb{Q}^A$

To evaluate (8.48) we need the SDE for X_t under $\mathbb{Q}^A$. The Girsanov shift from $\mathbb{Q}$ to $\mathbb{Q}^A$, derived in Chapter 5, is

$$\mathrm{d}W^{\mathbb{Q}}_t \;=\; \sigma^A\,\mathrm{d}t \;+\; \mathrm{d}W^A_t, \tag{8.50}$$

i.e. the $\mathbb{Q}$-Brownian picks up a drift equal to the volatility loading of the numeraire A when viewed under $\mathbb{Q}^A$.

Compute $\mathrm{d}(1/A_t)$ via Itô on $f(A) = 1/A$, with $f' = -1/A^2$, $f'' = 2/A^3$:

$$\mathrm{d}\left(\frac{1}{A_t}\right) \;=\; -\frac{\mathrm{d}A_t}{A_t^2} \;+\; \tfrac{1}{2}\cdot\frac{2}{A_t^3}\,(\sigma^A A_t)^2\,\mathrm{d}t \;=\; \frac{1}{A_t}[((\sigma^A)^2 - r)\,\mathrm{d}t \;-\; \sigma^A\,\mathrm{d}W^{\mathbb{Q}}_t]. \tag{8.51}$$

Then, using the Itô product rule on $X_t = B_t \cdot (1/A_t)$ with covariation $\mathrm{d}[B, 1/A]_t = -(B_t/A_t)\,\sigma^A\sigma^B\,\mathrm{d}t$,

$$\mathrm{d}\left(\frac{B_t}{A_t}\right) \;=\; \frac{B_t}{A_t}(r\,\mathrm{d}t + \sigma^B\,\mathrm{d}W^{\mathbb{Q}}_t) \;+\; \frac{B_t}{A_t}[((\sigma^A)^2 - r)\,\mathrm{d}t - \sigma^A\,\mathrm{d}W^{\mathbb{Q}}_t] \;-\; \frac{B_t}{A_t}\,\sigma^A\sigma^B\,\mathrm{d}t. \tag{8.52}$$

Collecting $\mathrm{d}t$ and $\mathrm{d}W^{\mathbb{Q}}_t$ terms,

$$\mathrm{d}\left(\frac{B_t}{A_t}\right) \;=\; \frac{B_t}{A_t}\Big\{[(\sigma^A)^2 - \sigma^A\sigma^B]\,\mathrm{d}t \;+\; (\sigma^B - \sigma^A)\,\mathrm{d}W^{\mathbb{Q}}_t\Big\}. \tag{8.53}$$

Now substitute (8.50) $\mathrm{d}W_t^{\mathbb{Q}} = \sigma^A\,\mathrm{d}t + \mathrm{d}W_t^A$:

$$\mathrm{d}\left(\frac{B_t}{A_t}\right) \;=\; \frac{B_t}{A_t}\Big\{\underbrace{\big[(\sigma^A)^2 - \sigma^A\sigma^B + (\sigma^B - \sigma^A)\sigma^A\big]}_{=\,0}\,\mathrm{d}t \;+\; (\sigma^B - \sigma^A)\,\mathrm{d}W_t^A\Big\}, \tag{8.54}$$

confirming that $X_t = B_t/A_t$ is a $\mathbb{Q}^A$-martingale, and

$$\boxed{\frac{\mathrm{d}X_t}{X_t} \;=\; (\sigma^B - \sigma^A)\,\mathrm{d}W_t^A.} \tag{8.55}$$

Despite all the Itô bookkeeping in (8.51)–(8.54), the end result is a geometric Brownian motion for X with *no drift* and a volatility equal to the simple difference $\sigma^B - \sigma^A$. The zero drift is enforced by the numeraire property — it had to come out that way — and the explicit calculation shows every drift term cancelling. The volatility is the *relative* vol: how much B moves relative to A, which is the only vol relevant when we measure everything in units of A. When A and B are driven by the same Brownian with the same vol, the ratio is deterministic and the option has no optionality; (8.55) gives zero vol, correctly.

10.9.5 8.10.5 Correlated case

In the general correlated case with two distinct Brownians $W^{(1)}, W^{(2)}$ of correlation ρ,

$$\frac{\mathrm{d}A_t}{A_t} \;=\; r\,\mathrm{d}t + \sigma^A\,\mathrm{d}W_t^{(1)}, \qquad \frac{\mathrm{d}B_t}{B_t} \;=\; r\,\mathrm{d}t + \sigma^B\,\mathrm{d}W_t^{(2)}, \qquad \mathrm{d}\langle W^{(1)}, W^{(2)}\rangle_t = \rho\,\mathrm{d}t,$$

the same Itô calculation yields $\mathrm{d}X_t/X_t = \hat\sigma\,\mathrm{d}W_t^A$ with

$$\boxed{\hat\sigma^2 \;=\; (\sigma^A)^2 \;-\; 2\rho\,\sigma^A\sigma^B \;+\; (\sigma^B)^2.} \tag{8.56}$$

The scalar-BM result (8.55) is the special case $\rho = +1$ with both assets driven by the *same* W, which collapses (8.56) to $\hat\sigma^2 = (\sigma^B - \sigma^A)^2$ and hence $\hat\sigma = |\sigma^B - \sigma^A|$.

Reading (8.56). This is one of those small-looking results that deserves a lot of respect. It says *the only interaction between the two assets that matters for the exchange option is their correlation* — not their individual drifts, not their individual levels, just ρ. Crank ρ up toward $+1$ (assets move in lockstep): $\hat\sigma^2$ shrinks toward $(\sigma^A - \sigma^B)^2$, and two assets that always move together cannot drift apart in relative terms — the option is cheap. Crank ρ down toward -1 (assets move oppositely): $\hat\sigma^2$ balloons to $(\sigma^A + \sigma^B)^2$, and two maximally anti-correlated assets are maximally divergent in relative terms — the option is expensive. In a Margrabe setting, *correlation is the pricing input* — more so than any individual vol — which is why dispersion traders and index arbitrageurs obsess over correlation term structures.

Margrabe correlation sensitivity *Margrabe price as a function of correlation ρ — option value increases monotonically as ρ decreases, since divergent assets maximise the expected spread.*

10.9.6 8.10.6 Step 4 — The Margrabe formula

Since X_t is a geometric Brownian motion under $\mathbb{Q}^A$ with volatility $\hat\sigma$ and *no drift*, the inner expectation in (8.48) is a Black–Scholes put on X struck at 1 with $r = 0$ (no discount under the martingale measure):

$$\frac{f_t}{A_t} \;=\; \Phi(d_+) \;-\; \frac{B_t}{A_t}\,\Phi(d_-), \qquad d_\pm \;=\; \frac{-\ln(B_t/A_t) \pm \frac{1}{2}\hat\sigma^2(T-t)}{\hat\sigma\sqrt{T-t}}. \tag{8.57}$$

Rearranging into the familiar Margrabe form,

$$\boxed{f_t \;=\; A_t\,\Phi(d_1) \;-\; B_t\,\Phi(d_2),} \qquad d_1 \;=\; \frac{\ln(A_t/B_t) + \frac{1}{2}\hat\sigma^2(T-t)}{\hat\sigma\sqrt{T-t}}, \quad d_2 \;=\; d_1 - \hat\sigma\sqrt{T-t}. \tag{8.58}$$

No discounting appears because we priced under $\mathbb{Q}^A$ and the payoff is denominated in units of A; the numeraire absorbs the interest rate.

10.9.7 8.10.7 Reading (8.58)

The formula is BS in disguise: $S \to A$, $Ke^{-rT} \to B$, $\sigma \to \hat{\sigma}$. Margrabe sees only the *ratio* A/B, not absolute levels; ATM is $A_t = B_t$. Higher ρ shrinks $\hat{\sigma}$, so the option is cheapest when the two underlyings are clones; the degenerate limit $\rho = 1, \sigma^A = \sigma^B$ gives $\hat{\sigma} = 0$ and the option collapses to its intrinsic $\max(A_t - B_t, 0)$.

10.9.8 8.10.9 Worked numerical example

Take $A_0 = B_0 = 100$ (ATM), $\sigma^A = 0.25$, $\sigma^B = 0.30$, $\rho = 0.6$, $T = 1$ year, $r = 3\%$. Then

$$\hat{\sigma}^2 \;=\; 0.25^2 - 2 \cdot 0.6 \cdot 0.25 \cdot 0.30 + 0.30^2 \;=\; 0.0625 - 0.09 + 0.09 \;=\; 0.0625,$$

so $\hat{\sigma} = 0.25$. Note: the spread vol equals σ^A in this specific case — a coincidence of the correlation and vol values, not a general feature. The d-parameters are $d_1 = \frac{1}{2}\hat{\sigma} = 0.125$ and $d_2 = -0.125$, giving $\Phi(d_1) \approx 0.5497$, $\Phi(d_2) \approx 0.4503$. The Margrabe price is

$$f_0 \;=\; 100 \cdot 0.5497 \;-\; 100 \cdot 0.4503 \;=\; 9.94.$$

Notice the absence of any discount factor — this is the numeraire-change magic of (8.58). The price is 9.94% of the reference asset's notional, which is the right-of-magnitude value for an ATM exchange option with a 25% spread vol over one year.

Now flip the correlation to $\rho = -0.6$:

$$\hat{\sigma}^2 \;=\; 0.0625 + 0.09 + 0.09 \;=\; 0.2425, \qquad \hat{\sigma} \;\approx\; 0.4924.$$

The spread vol almost doubles, and the Margrabe price rises to approximately 19.5 — nearly double the positive-correlation case. This is the correlation sensitivity made quantitative: flipping the sign of ρ from $+0.6$ to -0.6 changes the fair price of the exchange option by 100%. Dispersion-trading desks, which systematically sell single-name options and buy index options to monetise the correlation-vs-individual-vol gap, live and die by this relationship.

10.9.9 8.10.10 Generalisations and pitfalls

Stochastic vol in either leg pushes (8.58) into characteristic-function / MC territory; best-of $\max(A_T, B_T, C_T)$ has no closed-form analogue (Stulz 1982, Johnson 1987); dividend yields adjust the drifts but leave the formula intact. *Pitfall:* the σ^A, σ^B in (8.56) must be loadings on the *same* Brownian components; for two independent Brownians ($\rho = 0$) the spread vol reduces to the Pythagorean sum $\hat{\sigma} = \sqrt{(\sigma^A)^2 + (\sigma^B)^2}$.

10.10 8.11 Futures vs Forward — Convexity Adjustment Under Stochastic Rates

We are now in a position to pay off the promise made in §8.2: under stochastic rates, futures and forwards on the same underlying differ by a *convexity adjustment*, whose sign is controlled by the sign of $\mathrm{Cov}^{\mathbb{Q}}(r_t, S_t)$. The intuition is financial, not mathematical, and it is worth building before writing any formulas down.

10.10.1 8.11.1 The intuition

Picture two traders. Alice is long a *forward* on S with maturity T; Bob is long a *futures* on S with the same maturity. Both have the same final exposure to S_T. But between now and T, Bob receives daily mark-to-market cash flows into a margin account, while Alice does nothing. If S goes up, Bob receives cash; if S goes down, Bob pays cash.

Now suppose S and r are positively correlated — a plausible story for an equity index and the short rate during normal macro regimes, or for a bond futures and the short rate (which would be the same rate, making the correlation mechanical). On the days Bob receives cash (up-days for S), rates are also high, so Bob reinvests the cash at a *high* rate. On the days Bob pays cash (down-days for S), rates are low, so Bob finances the losses at a *low* rate. Bob wins on both ends of this timing asymmetry, and the extra value he captures must be compensated at inception by a *higher* futures price relative to the forward. This is the convexity adjustment: $F^{\text{fut}} > F^{\text{fwd}}$ when $\text{Cov}(r, S) > 0$, and the opposite sign when the correlation flips.

Conversely, if r and S are negatively correlated (as they might be for a bond futures and the short rate, depending on the economic regime), Bob reinvests his up-day cash at low rates and finances his down-day losses at high rates. The timing asymmetry now works *against* Bob, and the fair futures price must sit *below* the forward to compensate.

If rates are deterministic — or if S and r are uncorrelated under $\mathbb{Q}$ — the timing asymmetry is gone, and the two prices coincide.

10.10.2 8.11.2 The formula

Under $\mathbb{Q}$ (bank-account numeraire M_t), the futures price $F_t^{\text{fut}}(T)$ is a martingale (§8.3). Under the T-forward measure $\mathbb{Q}^T$ (bond numeraire $P(\cdot, T)$), the forward price $F_t^{\text{fwd}}(T) = S_t/P(t,T)$ is a martingale (Chapter 5). Taking conditional expectations at time t,

$$F^{\text{fut}}(t,T) \;=\; \mathbb{E}_t^{\mathbb{Q}}[S_T], \qquad F^{\text{fwd}}(t,T) \;=\; \mathbb{E}_t^{\mathbb{Q}^T}[S_T] \;=\; \frac{\mathbb{E}_t^{\mathbb{Q}}\left[S_T\, e^{-\int_t^T r_u\,\mathrm{d}u}\right]}{P(t,T)}. \tag{8.59}$$

Their difference is the convexity adjustment. By the identity $\text{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ applied to the inner expectation, one obtains

$$F^{\text{fut}}(t,T) \;-\; F^{\text{fwd}}(t,T) \;=\; \frac{1}{P(t,T)}\,\text{Cov}_t^{\mathbb{Q}}\left(S_T,\, 1 - e^{-\int_t^T r_u\,\mathrm{d}u}\right), \tag{8.60}$$

which to leading order in the covariance is approximately

$$F^{\text{fut}}(t,T) \;-\; F^{\text{fwd}}(t,T) \;\approx\; \int_t^T \int_t^T \text{Cov}_t^{\mathbb{Q}}(\mathrm{d}\ln S_u,\, r_v\,\mathrm{d}v). \tag{8.61}$$

10.10.3 8.11.3 The sign rule

Positive correlation between S (hence F^{fut}) and r means MTM gains are reinvested at higher rates while losses are financed at lower rates — this is worth money to the long, so the fair futures price must be *above* the forward. Negative correlation flips it. In summary:

Regime	Relation
r deterministic	$F^{\text{fut}}(t,T) = F^{\text{fwd}}(t,T) = S_t/P(t,T)$
r stochastic, $\text{Cov}^{\mathbb{Q}}(r_t, F_t^{\text{fut}}) = 0$	$F^{\text{fut}} = F^{\text{fwd}}$
r stochastic, positive correlation	$F^{\text{fut}} > F^{\text{fwd}}$
r stochastic, negative correlation	$F^{\text{fut}} < F^{\text{fwd}}$

The two measures — bank-account and T-forward — coincide iff $P(t,T)$ is deterministic. When they coincide, so do the expectations.

Futures-forward basis *Futures-forward basis under stochastic rates — positive* r*–*S *correlation pushes the basis positive, negative correlation pushes it negative. The magnitude is controlled by rate volatility and the tenor.*

10.10.4 8.11.4 Magnitudes in practice

How big is the convexity adjustment? A rough rule of thumb:

- **Short-dated eurodollar / SOFR futures.** Here the underlying rate and the reinvestment rate are both tied to the same short-rate curve, so $\rho \approx 1$ and the correlation effect is maximal. The adjustment can reach several basis points and is routinely quoted and hedged by rates desks. Dedicated convexity-adjustment calculators are standard in rates-trading software.
- **Long-dated equity index futures.** Correlation between equity returns and rates is modest, rate vols are low, and the adjustment is typically small (a few basis points on a multi-year contract). Still, for institutional hedgers running multi-billion-dollar exposures, a few basis points can matter.
- **FX futures.** Non-trivial because of the "quanto" correlation between the FX rate and the domestic short rate. The formula (8.61) is still the starting point, but the practitioner's job is to specify a model (typically Hull–White for the short rate, log-normal for FX, with a correlation ρ between them) and evaluate the double integral analytically.

10.10.5 8.11.5 The deeper reading

The convexity adjustment is a *measure-change effect*: F^{fut} and F^{fwd} are both martingales but under different measures (bank account vs T-forward), and those measures diverge whenever rates are stochastic. The same structural pattern reappears for CMS convexity (Chapter 13) and quanto adjustments under FX.

10.10.6 8.11.6 A Hull–White closed-form approximation

Writing down a concrete convexity-adjustment number requires a model for both S and r. The most common textbook choice is to take S log-normal and r Hull–White (Chapter 12), with a constant correlation ρ between the Brownians. Under that model, Hull (2018) gives the approximation

$$\ln F^{\text{fut}}(t,T) - \ln F^{\text{fwd}}(t,T) \;\approx\; \rho\,\sigma_S\,\sigma_r\,\frac{(T-t)^2}{2},$$

where σ_S is the spot volatility and σ_r is the short-rate vol. The quadratic-in-tenor scaling is characteristic of convexity adjustments: over short tenors the effect is negligible, but it grows quickly with maturity. For eurodollar/SOFR futures with $\sigma_r \approx 0.01$ (1% absolute rate vol), $\sigma_S \approx 0.0025$ (own-vol for a short rate), $\rho \approx 1$, and $T-t = 5$ years, the adjustment is roughly $1 \cdot 0.01 \cdot 0.0025 \cdot 12.5 = 0.0003$ or 3 basis points — a number that matches practitioner quotations for multi-year eurodollar convexity adjustments.

10.10.7 8.11.7 Hedging implications

Dealers running futures-vs-forward basis books mark to the evolving convexity adjustment daily; the basis's sensitivity to rate vol is the *vega of the convexity adjustment*, and is a standard rate-vol-desk diagnostic.

10.11 8.12 Key Takeaways

1. **Forward price.** The fair forward strike $F_t^{\text{fwd}}(T) = S_t/P(t,T)$ is pinned by cash-and-carry no-arbitrage, independent of volatility or any statistical properties of S. Under constant rates this reduces to $S_t\,e^{r(T-t)}$; with a dividend yield, to $S_t\,e^{(r-\delta)(T-t)}$.

2. **Forwards vs futures.** Forwards settle at maturity; futures settle daily. The futures position is zero-valued at every instant by design (daily MTM absorbs every infinitesimal gain/loss immediately), while the forward's value drifts with the spot until maturity. This zero-value feature is why the futures price is a $\mathbb{Q}$-martingale and why the Black PDE has no first-derivative drift term.

3. **Three drifts under $\mathbb{Q}$.**
   - Traded asset: drift $r\,X_t$.
   - Traded asset with yield δ: drift $(r-\delta)\,X_t$.
   - Futures quote: drift 0 (costless position).

4. **Dynamics of F_t under $\mathbb{Q}$.** $\mathrm{d}F_t(T) = \sigma^F(t,F)\,\mathrm{d}\widehat{W}_t$ with no drift. Specific choices of σ^F yield GBM (constant relative vol, used in Black '76), Bachelier (constant absolute vol, used when F can go negative), and Samuelson/OU (time-decaying vol loading, used for commodity futures).

5. **Black PDE.** For an option $g(t, F)$ on a futures with payoff $\varphi(F_{T_0})$,

$$\partial_t g \ + \ \tfrac{1}{2}\,(\sigma^F)^2\,\partial_{FF} g \ = \ r\,g, \qquad g(T_0, F) = \varphi(F).$$

The missing $r\,F\,\partial_F g$ term is the signature of martingale-underlying pricing. The $r\,g$ discount term remains because the option's dollar payoff still settles at T_0.

6. **Feynman–Kac cross-check.** The Black PDE has the stochastic representation $g(t, F) = \mathbb{E}^{\mathbb{Q}}[\, e^{-r(T_0-t)}\,\varphi(F_{T_0}) \mid F_t = F\,]$ with F_u satisfying the driftless SDE under $\mathbb{Q}$.

7. **Black '76 closed form.** For log-normal futures with constant relative vol σ and a call payoff,

$$g(t, F) \ = \ e^{-r(T_0-t)}\,[F\,\Phi(d_+) \ - \ K\,\Phi(d_-)], \qquad d_\pm \ = \ \frac{\ln(F/K) \pm \frac{1}{2}\sigma^2(T_0 - t)}{\sigma\,\sqrt{T_0 - t}}.$$

Structurally, Black '76 is BS with $S \to F$, $r \to 0$ in $d_\pm$, and an overall discount $e^{-r(T_0-t)}$.

8. **Bachelier binary.** Under $\mathrm{d}F_t = \sigma\,\mathrm{d}\widehat{W}_t$, a digital call prices as $\Phi((F - K)/(\sigma\,\sqrt{T_0 - t}))$. At the money ($F = K$), the price is exactly $1/2$ — a symmetry unique to the arithmetic model.

9. **Samuelson / OU futures.** Under $\mathrm{d}F_t = \sigma\,e^{-\eta(T-t)}\,\mathrm{d}\widehat{W}_t$, terminal variance is $\Sigma_t^2 = \sigma^2\,(e^{-2\eta(T-T_0)} - e^{-2\eta(T-t)})/(2\eta)$ and a digital call prices as $\Phi((F-K)/\Sigma_t)$. Long-dated futures have smaller Σ_t than short-dated ones — the Samuelson effect.

10. **Margrabe exchange option.** Under numeraire A, the ratio $X_t = B_t/A_t$ is a martingale with vol $\hat{\sigma} = \sqrt{(\sigma^A)^2 - 2\rho\,\sigma^A\sigma^B + (\sigma^B)^2}$, yielding $f_t = A_t\,\Phi(d_1) - B_t\,\Phi(d_2)$. No discount factor appears because the numeraire absorbs interest. Correlation ρ, not individual drifts or levels, determines the option value.

11. **Futures vs forward convexity.** Under stochastic rates, $F^{\text{fut}} - F^{\text{fwd}}$ is a covariance of S_T with the rate-path discount factor, proportional to $\mathrm{Cov}^{\mathbb{Q}}(r, S)$ in the leading-order limit. Positive correlation pushes the futures price above the forward; negative correlation pushes it below. Under deterministic rates, the two prices coincide.

12. **Numeraire is the theme.** The futures price is natural under $\mathbb{Q}$; the forward price is natural under $\mathbb{Q}^T$; the Margrabe exchange option is natural under one of its own legs. Each "naturalness" is an invitation to change measure; accepting the invitation collapses the pricing problem, while refusing it forces one to grind through a PDE that did not need to be set up.

10.12 Appendix — Reference Formulas

Label	Formula
(8.3)	$F_t^{\text{fwd}}(T) = S_t\,e^{r(T-t)}$ (constant rates, no dividend)
(8.4)	$F_t^{\text{fwd}}(T) = S_t/P(t, T)$ (cash-and-carry, any rates)
(8.8)	$\mathbb{E}^{\mathbb{Q}}[F_t(T) \mid \mathcal{F}_s] = F_s(T)$ (futures is a $\mathbb{Q}$-martingale)
(8.9)	$\mathrm{d}F_t(T) = \sigma^F(t, F_t(T))\,\mathrm{d}\widehat{W}_t$ (general futures SDE under $\mathbb{Q}$)
(8.13)	$\mathrm{d}F_t/F_t = \sigma\,\mathrm{d}\widehat{W}_t$ (log-normal futures under $\mathbb{Q}$)
(8.21)	$\alpha_t = \partial_F g(t, F_t)$ (futures-delta hedge)
(8.24)	$\partial_t g + \frac{1}{2}(\sigma^F)^2\,\partial_{FF} g = r\,g$, $g(T_0, F) = \varphi(F)$ (Black PDE)
(8.26)	$g(t, F) = \mathbb{E}^{\mathbb{Q}}[\,e^{-r(T_0-t)}\,\varphi(F_{T_0}) \parallel F_t = F\,]$ (Feynman–Kac)
(8.30)	$g = e^{-r(T_0-t)}[F\,\Phi(d_+) - K\,\Phi(d_-)]$ (Black '76 call)
(8.31)	$d_\pm = [\ln(F/K) \pm \frac{1}{2}\sigma^2(T_0 - t)]/(\sigma\sqrt{T_0 - t})$
(8.32)	$\partial_F g = e^{-r(T_0-t)}\,\Phi(d_+)$ (Black '76 delta)
(8.36)	Bachelier binary: $h(t, F) = \Phi((F - K)/(\sigma\,\sqrt{T_0 - t}))$
(8.40)	Samuelson variance: $\Sigma_t^2 = \sigma^2\,(e^{-2\eta(T-T_0)} - e^{-2\eta(T-t)})/(2\eta)$

Label	Formula
(8.43)	OU binary: $h(t, F) = \Phi((F-K)/\Sigma_t)$
(8.55)	$\mathrm{d}X_t/X_t = (\sigma^B - \sigma^A)\,\mathrm{d}W_t^A$ (Margrabe ratio dynamics, scalar BM)
(8.56)	$\hat{\sigma}^2 = (\sigma^A)^2 - 2\rho\,\sigma^A\sigma^B + (\sigma^B)^2$ (Margrabe spread vol)
(8.58)	Margrabe: $f_t = A_t\,\Phi(d_1) - B_t\,\Phi(d_2)$
(8.60)	$F^{\text{fut}} - F^{\text{fwd}} = \mathrm{Cov}_t^{\mathbb{Q}}(S_T,\, 1 - e^{-\int r})\,/\,P(t,T)$
(8.61)	Convexity $\approx \int_t^T\int_t^T \mathrm{Cov}_t^{\mathbb{Q}}(\mathrm{d}\ln S_u,\, r_v\,\mathrm{d}v)$

11 Chapter 9 — Monte Carlo, Path-Dependent, and Forward-Starting Options

This chapter is the simulation counterpart to the lattice pricing of Chapter 2 and the PDE-based pricing of Chapter 6. Closed-form formulas like Black–Scholes are beautiful but brittle: they cover a short list of payoffs (European calls, European puts, a handful of digital and barrier variants under continuous monitoring) written on a single-asset geometric Brownian motion with constant volatility. Every realistic trading desk deals daily with claims that lie outside that list — path-dependent payoffs (Asians, lookbacks, cliquets, autocallables), multi-asset baskets, stochastic-volatility models, jump-diffusions, local-volatility surfaces calibrated to a full implied-vol grid. For all of these, the pricing integral remains $V_0 = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\varphi]$ but the analytical evaluation fails, and one must turn to numerical methods. Monte-Carlo simulation is the most general of these methods. It works in any dimension, handles arbitrary payoffs, and is embarrassingly parallel; its only costs are statistical noise and the need for careful variance management.

The chapter arc is narrow and deep rather than wide. We begin by motivating Monte Carlo from the perspective of high-dimensional numerical integration, where quadrature breaks down under the curse of dimensionality. The strong law of large numbers (SLLN) provides the mathematical license for sample-average estimation, and the central limit theorem gives the $M^{-1/2}$ error rate that governs how confidence intervals shrink with sample size. We then build the lognormal GBM path generator — properly derived this time, because Chapter 6 has already given us the Itô solution of the GBM SDE — and apply it to three classes of problem: European claims as a warm-up sanity check against Black–Scholes, forward-starting (cliquet) options as a case where iterated conditional expectation gives a closed form, and barrier options as the prototypical path-dependent payoff that forces a Monte-Carlo implementation. Along the way we encounter a characteristic issue of path-dependent MC — discretisation bias on between-grid events — and the Brownian-bridge correction that removes it. We close with variance-reduction techniques (antithetic variates, control variates) that every production pricer uses, as a bridge to the Heston Monte Carlo of Chapter 14.

A word about how this chapter sits in the prerequisite graph. Chapter 2 gave us lattice pricing and the CRR calibration; we will refer back to the lognormal GBM limit of that chapter as the starting point for the simulation engine. Chapter 6 gave us the Itô derivation of GBM, Girsanov's theorem (via Chapter 5) that fixes the drift under $\mathbb{Q}$ at the risk-free rate, and the Black–Scholes formula that we use as a closed-form benchmark. Everything in this chapter is downstream of that material. Nothing here feeds forward to the foundations; the chapter is a self-contained application, and a reader who has mastered through Chapter 6 can read Chapter 10 immediately without touching Chapters 7–9.

A note on scope. We cover the core Monte-Carlo toolkit — path simulation, sample-mean estimation, standard error, discretisation bias, and two of the canonical variance-reduction techniques. Topics we do not cover here include stratified sampling, importance sampling (beyond a conceptual mention), quasi-Monte-Carlo with low-discrepancy sequences, pathwise and likelihood-ratio Greek estimators, and the Longstaff–Schwartz regression-based scheme for American exotics. Those techniques are discussed briefly where they are natural extensions of the material and are otherwise left as pointers into the literature. The focus is on getting a working production-grade European and path-dependent pricer stood up; the reader who has the machinery presented here can read any of the variance-reduction or American-MC literature without further prerequisites.

11.1 9.1 Why Monte Carlo? Quadrature and the Curse of Dimensionality

The pricing formula $V_0 = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\varphi(A_T, \{A_t\}_{t\leq T})]$ is, at its core, an integral over the sample space of the underlying process. In low dimensions this integral can be evaluated by deterministic quadrature — trapezoidal, Simpson's, Gauss–Legendre, or one of the many higher-order schemes — with error scaling as $O(h^p)$ where h is the mesh size and p is the order of the quadrature. For $p = 4$ (Simpson's rule), getting four-significant-figure accuracy on a smooth integrand requires $h \sim 10^{-1}$ or about 10 grid points in one dimension. Quadrature is fast, deterministic, and the textbook first choice when the state space is small.

In high dimensions quadrature breaks down catastrophically. The number of grid points required scales as N^d, where d is the number of independent stochastic factors and N is the number of grid points per factor. For a daily-monitored one-year option on a single underlying, $d = 252$ (one Brownian increment per trading day); even using only $N = 2$ grid points per dimension gives $2^{252} \approx 10^{75}$ total points, vastly more than any computer can enumerate. For a daily-monitored option on a basket of ten underlyings, $d = 2520$, and the number becomes utterly astronomical.

This is the *curse of dimensionality*: deterministic integration techniques whose per-dimension cost is well-controlled nevertheless collapse in high dimensions because the total cost compounds multiplicatively.

Monte-Carlo integration escapes the curse by a simple statistical observation: the sample-mean estimator's error scales as $M^{-1/2}$, *regardless of the dimension of the state space*. For a fixed error tolerance we need a fixed number of paths, and the cost per path scales only linearly in the dimension — each path-step is an $O(d)$ computation. Total cost: $O(Md)$, compared to $O(N^d)$ for quadrature. The break-even dimension is usually between $d = 4$ and $d = 6$: below that, quadrature is faster; above it, Monte Carlo wins decisively. Since every realistic derivative-pricing problem with a daily or finer monitoring grid lives well above $d = 6$, MC is the method of choice for everything except the cleanest European problems on a single underlying.

The flip side of dimension-insensitivity is that Monte Carlo is *statistical*: it produces random error bars, not deterministic error bounds. A practitioner always reports the MC estimate together with its sample standard error, and any downstream use of the estimate — hedging ratios, risk capital, P&L attribution — must account for the Monte-Carlo uncertainty. Deterministic methods produce exact-to-roundoff answers; MC produces confidence intervals. The gap closes as M grows, but never to zero. For a risk manager, this means MC-based prices carry an intrinsic "simulation vega" that must be bounded below desk risk tolerances; for a trader, it means the first question on seeing an MC price is always "what's the standard error?"

Monte-Carlo simulation has a long intellectual history, pre-dating its application to finance by decades. The key idea — that high-dimensional integrals can be approximated by sample averages — was formalised by Metropolis, Ulam, and von Neumann at Los Alamos in the 1940s for neutron-transport calculations, and has since spread to essentially every quantitative discipline. In finance the method is particularly well-suited to derivatives pricing because every pricing formula under $\mathbb{Q}$ is literally an integral (a discounted expected payoff) and because the underlying state space — paths of a diffusion — is naturally high-dimensional. Once the risk-neutral measure $\mathbb{Q}$ is pinned down (by Girsanov, Chapter 5, and the market-price-of-risk shift of Chapter 6), the pricing problem collapses cleanly to "draw paths from $\mathbb{Q}$, evaluate the payoff on each, and average." The conceptual simplicity is matched by the implementation simplicity: modern Monte-Carlo pricers are some of the shortest useful programs in quantitative finance.

The engineering of a production Monte-Carlo pricer involves four distinct sub-problems:

1. **Path generation under $\mathbb{Q}$.** Given the chosen risk-neutral model — GBM, Heston, local-vol, jump-diffusion, etc. — produce paths whose law matches the specification. For GBM the exact lognormal increment of §9.4 gives a bias-free generator; for general SDEs one falls back on Euler–Maruyama or Milstein schemes with their associated discretisation biases.
2. **Variance reduction.** Reduce the sample variance so that fewer paths suffice to achieve a target confidence-interval width. Antithetic variates, control variates, stratified sampling, importance sampling, and quasi-random sequences each contribute.
3. **Path-dependent payoffs.** Many payoffs depend on the whole path: running maximum (lookback), running average (Asian), barrier-crossing indicator (knock-in/out), basket constraints. Efficient data structures and careful handling of between-grid events (Brownian-bridge correction for barriers) are central.
4. **Sensitivity estimation (Greeks).** Delta, gamma, vega, and other Greeks must come out of the same path sample, ideally with pathwise or likelihood-ratio estimators rather than finite-difference bump-and-reprice. We will touch on this briefly but not in depth.

The present chapter concentrates on items (1), (2), and (3). Item (4) — the art of Greek estimation by Monte Carlo — is an entire subdiscipline and appears as a cross-reference at the end of the chapter and in more detail within Chapter 10.

11.2 9.2 The Strong Law of Large Numbers

The mathematical foundation of Monte-Carlo integration is the strong law of large numbers, which says that sample averages almost surely converge to population expectations.

Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{E}[|X|] < +\infty$, and let $X^{(1)}, X^{(2)}, \ldots, X^{(M)}$ be independent and identically distributed copies of X. Then

$$\lim_{M \to +\infty} \frac{X^{(1)} + X^{(2)} + \cdots + X^{(M)}}{M} = \mathbb{E}[X] \qquad \text{(strong law, a.s.).} \tag{9.1}$$

The "almost surely" (a.s.) qualifier means that the set of paths along which convergence fails has $\mathbb{P}$-measure zero; for any actual realised sequence of draws the sample average converges. The strong law is the mathematical license for Monte-Carlo pricing: given enough i.i.d. draws from the payoff distribution under $\mathbb{Q}$, the sample average almost surely converges to $\mathbb{E}^{\mathbb{Q}}$[payoff], which (times the discount factor) is the price.

A short philosophical aside on the $\mathbb{E}[|X|] < \infty$ hypothesis. Without finite mean the sample average may not converge at all; it can drift to infinity, oscillate, or behave chaotically. For option-pricing problems this is rarely a concern. Bounded payoffs — European puts are bounded by K, digital options by 1, and European calls can be bounded in practice by truncating at far-OTM regions — automatically satisfy the condition. But for *ratios* of random variables, or for some stochastic-volatility models with heavy-tailed variance processes, the integrability check deserves attention. The canonical pathological example is the Cauchy distribution (a ratio of two independent Gaussians), for which the sample average has the same Cauchy distribution regardless of sample size; the Monte-Carlo method fails completely because the variance is infinite and the mean is not even well-defined in the usual sense.

Higher integrability buys strictly more. L^2 integrability (finite variance, $\mathbb{E}[X^2] < \infty$) gives the Central Limit Theorem:

$$\sqrt{M}\,(\widehat{m}_M - \mathbb{E}[X]) \xrightarrow{d} \mathcal{N}(0,\ \mathbb{V}[X]), \tag{9.2}$$

where $\widehat{m}_M := (1/M)\sum_m X^{(m)}$ is the sample mean. The CLT is the basis for Monte-Carlo confidence intervals: for large M, $\widehat{m}_M$ is approximately normally distributed around $\mathbb{E}[X]$ with standard deviation $\sqrt{\mathbb{V}[X]/M}$, and we can bracket the unknown mean by a conventional $\widehat{m}_M \pm 1.96\,\hat{\sigma}_{m_1}$ 95% interval. L^4 integrability ($\mathbb{E}[X^4] < \infty$) tightens the approximation further via the Berry–Esseen bound, which quantifies the rate at which the CLT distributional approximation becomes good.

In practice the check "does my payoff have finite variance under $\mathbb{Q}$?" is usually done by inspection: polynomially-growing payoffs on a lognormal underlying have finite moments of all orders, so European calls, puts, Asians, look-backs, and barriers on GBM are all fine. The problem cases — which one should be aware of even if they do not arise in this chapter — are payoffs that involve *reciprocals* or *roots* of the underlying, payoffs on stochastic-volatility models near the Feller-condition boundary (see Chapter 14), and payoffs that are integrated over unbounded time horizons.

11.3 9.3 Sample Mean, Sample Variance, and the Monte-Carlo Standard Error

We estimate $m_1 := \mathbb{E}[X]$ by the finite sample

$$\widehat{m}_1 \;=\; \frac{1}{M}\sum_{m=1}^{M} X^{(m)} \qquad \text{(sample mean).} \tag{9.3}$$

The sample mean is unbiased ($\mathbb{E}[\widehat{m}_1] = \mathbb{E}[X]$) because expectation is linear, and has variance

$$\mathbb{V}[\widehat{m}_1] \;=\; \frac{1}{M^2}\sum_{m=1}^{M}\mathbb{V}[X^{(m)}] \;=\; \frac{\mathbb{V}[X]}{M}, \tag{9.4}$$

using independence of the draws. The standard deviation of the sample mean is therefore $\sigma_{m_1} = \mathrm{sd}[X]/\sqrt{M}$. Since σ_{m_1} depends on the unknown population variance, we estimate it by the sample-variance-based standard error

$$\hat{\sigma}_{m_1} \;=\; \frac{1}{\sqrt{M}}\left(\frac{1}{M-1}\sum_{m=1}^{M}\big(X^{(m)} - \widehat{m}_1\big)^2\right)^{1/2}. \tag{9.5}$$

Two small but important points about (9.5). First, the inner bracket is the unbiased sample variance (divisor $M-1$, not M); the $M-1$ correction — Bessel's correction — adjusts for the degree of freedom absorbed in estimating the mean. Second, dividing by $\sqrt{M}$ turns the sample standard deviation of the *underlying variable* into the standard deviation of the *sample mean*. The Monte-Carlo error bar is the latter, not the former, and they differ by a factor of $\sqrt{M}$ that is easy to confuse.

The $M^{-1/2}$ convergence rate is universal for i.i.d. Monte Carlo with finite variance. Its immediate operational consequence: *halving the error bar costs four times as many paths.* A trader who wants to move an MC estimate from three significant figures to four needs 100× more paths; from four to five, 10,000× more. The law of diminishing returns is brutal, which is exactly why variance reduction matters so much (§9.8).

Monte-Carlo standard error vs sample size *Empirical verification of the $M^{-1/2}$ rate. The standard deviation of the sample-mean estimator across 200 replications of an ATM European-call MC price, plotted against path count M on log-log axes, tracks the $1/\sqrt{M}$ reference line over five decades. Variance reduction (§9.8) changes the constant, not the slope.*

Let us work through a concrete error budget to build intuition. Suppose we are pricing a one-year ATM European call with Black–Scholes parameters $S_0 = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$. The closed-form Black–Scholes value (Chapter 6) is $V_0^{\text{BS}} \approx 10.45$. The payoff $X = e^{-rT}(A_T - K)_+$ has standard deviation roughly $\text{sd}(X) \approx 14$ at these parameters — one can read this from direct moment calculation or from the empirical standard deviation of a pilot MC run. A Monte-Carlo estimate with $M = 10{,}000$ paths therefore has a standard error of $14/\sqrt{10{,}000} = 0.14$, so the 95% confidence interval is approximately

$$10.45 \pm 1.96 \cdot 0.14 \;=\; [10.17,\; 10.73].$$

To push the error below 0.01 (four-digit accuracy), we need $M = (14/0.01)^2 = 1{,}960{,}000$ paths — of order two million. This is entirely feasible in seconds on modern commodity hardware, but expensive enough that variance reduction pays for itself within the first few million paths.

A pair of useful ballpark facts that a trader should carry in their head:

- A standard-error budget of 0.1 on an option worth ~ 10 is typically acceptable for a trading indication. That is $M \sim 20{,}000$ paths for a vanilla European call at typical parameters.
- A standard-error budget of 0.01 on a portfolio-level MC risk calculation is typically required for capital numbers. That is $M \sim 2{,}000{,}000$ paths, an order of magnitude more effort.
- Pathwise sensitivities (deltas, gammas) typically have standard deviations 2–5× higher than the price itself, so Greek estimation at fixed MC budget is 4×–25× noisier than price estimation at the same budget. Practitioners either run more paths for Greeks or use variance-reduction techniques more aggressively on Greeks than on prices.

The standard error is not a guarantee — it is an estimate of an estimate. The sample variance itself has sampling error, and the 95% CI from a Gaussian approximation is only asymptotic. For small samples ($M \lesssim 100$) one should use a Student-t critical value instead of 1.96, and for badly non-Gaussian payoff distributions (heavily OTM or knocked-in barriers with few triggering paths) one should look skeptically at any symmetric confidence interval. When in doubt, bootstrap.

11.4 9.4 The Lognormal GBM Path Generator

With the SLLN in hand and the standard-error formula (9.5) at our disposal, the remaining ingredient for a Monte-Carlo pricer is a path generator — a procedure that, on demand, produces sample paths $\{S_{t_n}\}_{n=0}^{N}$ from the risk-neutral law of the asset. For GBM the generator is *exact*, not approximate; this is one of the two or three SDEs whose discretisation introduces no bias at the grid points. The reason is that Chapter 6 has already solved the GBM SDE in closed form.

GBM sample paths under Q *Thirty risk-neutral GBM paths for $S_0 = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1y$. The red curve is $\mathbb{E}^{\mathbb{Q}}[S_t] = S_0 e^{rt}$, the risk-neutral expected trajectory. Monte-Carlo pricing averages discounted payoffs over such paths; each path contributes one term of the sample mean (9.3).*

11.4.1 9.4.1 Exact lognormal increment

From Chapter 6 we know that under $\mathbb{Q}$ the asset price follows

$$dS_t \;=\; r\,S_t\,dt \;+\; \sigma\,S_t\,dW_t^{\mathbb{Q}}, \tag{9.6}$$

where $W^{\mathbb{Q}}$ is a $\mathbb{Q}$-Brownian motion (the risk-neutral drift r having been installed by the Girsanov shift of Chapter 5). Applying Itô's lemma to $\ln S_t$ (Chapter 3) gives

$$d\ln S_t \;=\; (r - \tfrac{1}{2}\sigma^2)\,dt \;+\; \sigma\,dW_t^{\mathbb{Q}}, \tag{9.7}$$

and integrating from t_{n-1} to t_n,

$$\ln S_{t_n} - \ln S_{t_{n-1}} \;=\; (r - \tfrac{1}{2}\sigma^2)\,\Delta t_n \;+\; \sigma\,(W_{t_n}^{\mathbb{Q}} - W_{t_{n-1}}^{\mathbb{Q}}), \tag{9.8}$$

with $\Delta t_n := t_n - t_{n-1}$. The Brownian increment $W_{t_n}^{\mathbb{Q}} - W_{t_{n-1}}^{\mathbb{Q}}$ is exactly Gaussian with mean 0 and variance Δt_n, and — crucially — independent of the history up to t_{n-1}. Writing the increment as $\sqrt{\Delta t_n}\,Z_n$ with $Z_n \sim \mathcal{N}(0,1)$ and exponentiating gives the multiplicative path-simulation recursion

$$S_{t_n} \;=\; S_{t_{n-1}}\,\exp\Big\{(r - \tfrac{1}{2}\sigma^2)\,\Delta t_n \;+\; \sigma\sqrt{\Delta t_n}\,Z_n\Big\}, \qquad Z_1, Z_2, \dots \overset{\text{iid}}{\sim}_{\mathbb{Q}} \mathcal{N}(0,1). \tag{9.9}$$

This is the universal GBM path generator. At the grid points $t_0, t_1, \dots, t_N$ the law of $(S_{t_0}, S_{t_1}, \dots, S_{t_N})$ produced by (9.9) is *exactly* the law of a GBM sampled at those points — there is no discretisation bias. Between the grid points the discrete path is a log-linear interpolation, which *is* an approximation, and for payoffs that care about the between-grid behaviour (like barriers) we will see below that a correction is needed.

Contrast this with the general case. For a non-GBM SDE $dX_t = \mu(X_t)\,dt + \sigma(X_t)\,dW_t^{\mathbb{Q}}$ the Itô increment is Gaussian only in the limit $\Delta t \to 0$, and for finite Δt the Euler–Maruyama scheme

$$X_{t_n} \;\approx\; X_{t_{n-1}} + \mu(X_{t_{n-1}})\,\Delta t_n + \sigma(X_{t_{n-1}})\sqrt{\Delta t_n}\,Z_n$$

introduces a weak error of order Δt and a strong error of order $\sqrt{\Delta t}$. Milstein's scheme adds a second-order correction and upgrades the strong error to Δt. For path-dependent payoffs where individual realisations (not just distributions) matter, the strong-order matters most, and practitioners sometimes pay the price of a higher-order scheme to reduce discretisation bias. For GBM we pay no such price — the exact lognormal increment is available at every grid width.

11.4.2 9.4.2 Monte-Carlo estimator for European claims

For a European claim with payoff $\varphi(S_T)$ the MC estimator is the direct sample-average translation of the SLLN:

$$\widehat{V}_0 \;=\; e^{-rT} \cdot \frac{1}{M}\sum_{m=1}^{M} \varphi\Big(S_0\, e^{(r-\frac{1}{2}\sigma^2)T \,+\, \sigma\sqrt{T}\,Z^{(m)}}\Big), \qquad Z^{(m)} \overset{\text{iid}}{\sim}_{\mathbb{Q}} \mathcal{N}(0,1). \tag{9.10}$$

Only the terminal value S_T enters, so we do not even need a grid — a single Gaussian draw per path suffices. This is the computationally cheapest non-trivial Monte-Carlo pricer and is useful mainly as a sanity check against Black–Scholes and as a warm-up for the path-dependent cases below.

11.4.3 9.4.3 Euler–Maruyama and Milstein for non-GBM SDEs

For the stochastic-volatility, local-volatility, and jump-diffusion models that appear later in the guide (Chapter 10 for Heston, in particular) the exact lognormal increment is *not* available, and one falls back on discretisation schemes. The simplest is Euler–Maruyama:

$$X_{t_n} \;\approx\; X_{t_{n-1}} \;+\; \mu(X_{t_{n-1}})\,\Delta t_n \;+\; \sigma(X_{t_{n-1}})\sqrt{\Delta t_n}\,Z_n.$$

The scheme approximates the Itô increment by freezing the drift and diffusion coefficients at the left endpoint and using a plain Gaussian increment; it has *weak* convergence rate $O(\Delta t)$ (expectations of smooth functions of the terminal value converge at rate Δt) and *strong* convergence rate $O(\sqrt{\Delta t})$ (individual path realisations converge at the slower $\sqrt{\Delta t}$ rate). The distinction matters for path-dependent payoffs where individual realisations drive the payoff, not just their distributions.

Milstein's scheme augments Euler with a second-order Itô correction:

$$X_{t_n} \;\approx\; X_{t_{n-1}} \;+\; \mu\,\Delta t_n \;+\; \sigma\sqrt{\Delta t_n}\,Z_n \;+\; \tfrac{1}{2}\,\sigma\,\sigma'\,(\Delta t_n)\,(Z_n^2-1),$$

where $\sigma' = \partial\sigma/\partial X$. The correction upgrades the strong convergence rate to $O(\Delta t)$ and the weak rate remains $O(\Delta t)$. For constant diffusion ($\sigma' = 0$) Milstein and Euler coincide. For GBM ($\sigma(X) = \sigma X$, $\sigma' = \sigma$) Milstein's correction is $\frac{1}{2}\sigma^2 X\,(\Delta t)(Z^2-1)$, which matches the first-order Taylor expansion of the exact lognormal increment. For the Heston variance process $\mathrm{d}v_t = \kappa(\theta - v_t)\mathrm{d}t + \xi\sqrt{v_t}\mathrm{d}W_t$ the diffusion $\sigma(v) = \xi\sqrt{v}$ is not differentiable at $v = 0$, and naive Euler can produce negative variance values; the standard fixes are *full-truncation* (set negative values to zero before using them as the next drift) and the exact *QE scheme* of Andersen. Chapter 14 develops these in detail.

For any SDE whose exact solution is known in closed form, one should always prefer the exact simulator to Euler. GBM and OU are the two canonical examples; CIR, Vasicek (as a special case of OU), and a handful of others also admit exact schemes. For everything else, Euler is the pragmatic default, with Milstein applied when the strong-convergence-rate improvement justifies the extra arithmetic cost.

11.4.4 9.4.4 Additive log-space simulation

The exponential-and-multiplicative form of (9.9) is not the only representation. One could instead simulate $\ln S_{t_n}$ additively,

$$\ln S_{t_n} \;=\; \ln S_{t_{n-1}} + (r - \tfrac{1}{2}\sigma^2)\,\Delta t_n \;+\; \sigma\sqrt{\Delta t_n}\,Z_n, \tag{9.11}$$

and exponentiate at the end to recover S_{t_n}. Algebraically the two are identical; numerically they differ in edge cases. The multiplicative form can underflow to zero for very long paths at small S, while the additive form on $\ln S$ is more numerically stable. Production pricers typically work in log-space for this reason, especially when pricing very-long-dated (decades) structures or when the parameters drive the path into the tail. For everyday pricing at typical maturities (months to a few years) the difference is imperceptible.

11.4.5 9.4.5 Correlation of time-slices along a single path

Before using (9.9) to price path-dependent claims, we pause to think carefully about how different time-slices of the same simulated path are correlated. This is both a technical prerequisite for the forward-starting-option derivation of §9.6 and a conceptual anchor for why path-dependent options are fundamentally richer than European options.

For two times $T_1 < T_2$ along the same GBM path, write

$$\mathbb{C}\big[S_{T_1}, S_{T_2}\big] \;=\; \mathrm{Cov}^{\mathbb{Q}}\big(S_{T_1}, S_{T_2}\big), \qquad \rho\big[S_{T_1}, S_{T_2}\big] \;=\; \frac{\mathbb{C}\big[S_{T_1}, S_{T_2}\big]}{\big(\mathbb{V}[S_{T_1}]\,\mathbb{V}[S_{T_2}]\big)^{1/2}}. \tag{9.12}$$

Because $S_{T_2} = S_{T_1}\cdot\exp\{(r-\frac{1}{2}\sigma^2)(T_2-T_1)+\sigma\sqrt{T_2-T_1}\,Z'\}$ with Z' a fresh standard Gaussian independent of $\mathcal{F}_{T_1}$, the two slices S_{T_1} and S_{T_2} are positively correlated but not perfectly correlated — the intervening Brownian shock Z' drives them apart.

For a concrete calculation: under $\mathbb{Q}$, $\ln S_{T_i}$ is Gaussian with mean $(r - \frac{1}{2}\sigma^2)T_i$ and variance $\sigma^2 T_i$. The covariance of the *log*-prices is $\sigma^2\,\min(T_1, T_2) = \sigma^2 T_1$ (because $\ln S_{T_1}$ and $\ln S_{T_2}$ differ by a Brownian increment over $[T_1, T_2]$ that is independent of the increment over $[0, T_1]$). The correlation coefficient of the log-prices is therefore

$$\rho\big[\ln S_{T_1}, \ln S_{T_2}\big] \;=\; \frac{\sigma^2 T_1}{\sqrt{\sigma^2 T_1\cdot\sigma^2 T_2}} \;=\; \sqrt{T_1/T_2}.$$

At $T_1/T_2 = 1/2$ the correlation is $1/\sqrt{2} \approx 0.707$; at $T_1/T_2 = 0.1$ it is $\sqrt{0.1} \approx 0.316$. Higher correlation between nearby times reflects the strong dependence of consecutive Brownian increments. The *levels* S_{T_1} and S_{T_2} are *log*-normally correlated, not linearly correlated; the two notions differ for skewed distributions, and a practitioner computing correlation of levels from simulated paths should always be clear whether the reported number refers to levels or logs.

The single property that makes path-dependent pricing tractable is *independence of disjoint increments*. The Brownian motion satisfies $W_{T_2} - W_{T_1} \perp \mathcal{F}_{T_1}$; this is the core of the Markov property, and it is what lets us factor joint expectations along time as nested conditional expectations. The next section makes use of this independence in its sharpest form.

11.5 9.5 Sanity Check: European Call by Monte Carlo

Before tackling path-dependent payoffs we work through the simplest non-trivial MC pricer: a European call on a GBM underlying. The point is purely pedagogical — Black–Scholes gives the exact answer in closed form, so there is no need to simulate anything. But the sanity check is worthwhile because it lets us verify the mechanics of (9.10), calibrate our expectations for the $M^{-1/2}$ convergence rate, and provide a reference against which the path-dependent pricers below can be compared.

Take the reference parameters $S_0 = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$. Black–Scholes (Chapter 6) gives $V_0^{\mathrm{BS}} \approx 10.45$ with $d_\pm = (r \pm \frac{1}{2}\sigma^2)T/(\sigma\sqrt{T}) = \{0.35, 0.15\}$ and $\Phi(0.35) \approx 0.637$, $\Phi(0.15) \approx 0.560$.

Simulation. Draw $M = 10{,}000$ standard normals $Z^{(m)}$ and compute the payoff on each:

$$S_T^{(m)} \;=\; 100 \cdot e^{0.03+0.2\,Z^{(m)}}, \qquad X^{(m)} \;=\; e^{-0.05}\,(S_T^{(m)} - 100)_+.$$

Average the payoffs: $\widehat{V}_0 = (1/M)\sum_m X^{(m)} \approx 10.43$ (a typical realisation). The sample standard deviation of X is around 14, so the standard error is $\hat{\sigma}_{m_1} \approx 14/\sqrt{10{,}000} = 0.14$ and the 95% confidence interval is approximately [10.15, 10.71] — it comfortably brackets the closed-form value 10.45. Doubling M to 40,000 halves the standard error to 0.07, illustrating the $M^{-1/2}$ law. Quadrupling to 160,000 halves it again to 0.035.

What the sanity check buys us. A successful MC vs BS cross-check gives assurance on three independently-failable components of the pricer simultaneously: the random-number generator is producing genuine independent Gaussians; the GBM drift $r - \frac{1}{2}\sigma^2$ is applied with the correct sign and magnitude; and the discount factor e^{-rT} is applied exactly once. Each of these is a source of subtle bugs in production code. In particular, the $-\frac{1}{2}\sigma^2$ Itô correction is the single most common bug in home-grown MC pricers — it is easy to forget or get a sign wrong, and in that case the simulation produces a biased estimate whose bias *does not go away with more paths.* The standard-error bars shrink, the estimate converges confidently to a wrong number, and the bug can persist unnoticed for weeks until someone thinks to run a BS sanity check.

The sanity check also gives a cheap calibration of variance-reduction gain. With antithetic variates (§9.8) the standard error at $M = 10{,}000$ typically drops from 0.14 to around 0.10 on this payoff — a variance reduction of about 2. A stock-price control variate (whose expectation is known exactly, $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{rT} = 100e^{0.05} \approx 105.13$) reduces standard error by another factor of 3–5. Combined, these two techniques can bring the standard error at $M = 10{,}000$ down to 0.02, which would otherwise require $M \approx 500{,}000$ paths without variance reduction. That is a factor-of-50 speedup for a handful of extra lines of code.

11.6 9.6 Forward-Starting (Cliquet) Options

Forward-starting options occupy a distinctive place in the exotic-options landscape. They are path-dependent — their payoff depends on the underlying at two different times, not just at maturity — but the dependence is structured enough that iterated conditional expectation gives a closed-form price. The derivation that follows is worth its weight several times over: it is the atom of cliquet (ratchet) pricing, it illustrates how Brownian independent-increments repackage a correlated two-dimensional problem into a pair of one-dimensional Black–Scholes integrals, and it foreshadows the conditioning arguments that dominate interest-rate and stochastic-volatility pricing in later chapters.

11.6.1 9.6.1 Definition

A forward-starting call is an exotic whose strike is set at an intermediate date T_1 as a fraction α of the then-prevailing spot. The payoff at maturity $T_2 > T_1$ is

$$\varphi \;=\; \left(S_{T_2} - \alpha\, S_{T_1}\right)_+, \qquad (x)_+ := \max(x, 0). \tag{9.13}$$

The strike $\alpha\, S_{T_1}$ is not known at time 0 — it is stochastic through S_{T_1}.

The economic motivation is protection *relative to wherever the market is when protection kicks in.* An investor who wants "a call whose strike is set to whatever the index is six months from now" is buying a forward-starting call with $T_1 = 0.5$ and $\alpha = 1$. An insurance company that writes annual reset put options on a pension liability is effectively writing a sequence of forward-starting puts. Cliquet options chain forward-starts together: a 12-leg monthly cliquet call consists of twelve forward-starting calls with reset dates at the end of each month, each leg paying the monthly reset-to-reset return above zero, and the total payoff being the sum. Some cliquet structures layer on local caps, global floors, or coupon multipliers; we deal only with the atomic forward-starting call here.

Cliquets are ubiquitous in structured-note design. A *locally-capped globally-floored* cliquet caps each reset-to-reset return at (say) +3% and guarantees a minimum total return of zero over the full life; it is a single-tranche packaging of path-dependent upside, popular with retail insurance customers who want equity-like returns with capital protection. Pricing these involves summing or compounding the atom (9.13) over many reset dates. The deeper reason cliquets are useful in portfolio construction: they hedge *relative* moves rather than *absolute* ones. Because the strikes are not known at trade inception, cliquets are more sensitive to *forward volatility* (vol of vol over future intervals) than vanilla options; they are a natural vehicle for vol-of-vol exposure.

11.6.2 9.6.2 Vanilla Black–Scholes recap

We restate the Black–Scholes formula here because the forward-starting price in §9.6.5 will reuse it as a building block. The cliquet calculation is "Black–Scholes inside Black–Scholes" — an inner BS for the conditional payoff given S_{T_1}, and an outer BS-like integral for the unconditional price. The vanilla European call with strike K and tenor T has price

$$V_0 \;=\; e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[(S_T - K)_+\big] \;=\; S_0\,\Phi(d_+) \;-\; K\,e^{-rT}\,\Phi(d_-), \tag{9.14}$$

$$d_\pm \;=\; \frac{\ln(S_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}. \tag{9.15}$$

11.6.3 9.6.3 Reparameterising on two time-slices

The key manipulation is to rewrite the joint law of (S_{T_1}, S_{T_2}) in terms of independent Brownian increments over disjoint intervals. The two slices are correlated (they share a common Brownian path up to T_1), but the *increment* $W^{\mathbb{Q}}_{T_2} - W^{\mathbb{Q}}_{T_1}$ is independent of the history $\mathcal{F}_{T_1}$. This decomposition is what allows iterated conditional expectation to factor the pricing integral into a product of one-dimensional integrals.

Under $\mathbb{Q}$, the asset at T_1 and T_2 admits the direct lognormal representations

$$S_{T_1} \;=\; S_0\,e^{(r-\frac{1}{2}\sigma^2)T_1 \,+\, \sigma\sqrt{T_1}\,Z_1}, \qquad Z_1 \sim_{\mathbb{Q}} \mathcal{N}(0,1), \tag{9.16}$$

$$S_{T_2} \;=\; S_0\,e^{(r-\frac{1}{2}\sigma^2)T_2 \,+\, \sigma\sqrt{T_2}\,Z_2}, \qquad Z_2 \sim_{\mathbb{Q}} \mathcal{N}(0,1), \tag{9.17}$$

with (Z_1, Z_2) correlated. The equivalent independent-increment factorisation is

$$S_{T_1} \;\overset{d}{=}\; S_0\,e^{(r-\frac{1}{2}\sigma^2)T_1 \,+\, \sigma\sqrt{T_1}\,Z_1}, \tag{9.18}$$

$$S_{T_2} \overset{d}{=} S_{T_1} \cdot e^{(r-\frac{1}{2}\sigma^2)(T_2-T_1) + \sigma\sqrt{T_2-T_1}\, Z_3}, \qquad Z_3 \sim_{\mathbb{Q}} \mathcal{N}(0,1), \quad Z_1 \perp Z_3. \tag{9.19}$$

The two Gaussians Z_1 and Z_3 are now independent — they represent the Brownian increments over the disjoint intervals $[0, T_1]$ and $[T_1, T_2]$. The repackaging trades a two-dimensional correlated Gaussian (Z_1, Z_2) for two one-dimensional independent Gaussians, and that decoupling is what makes the inner and outer expectations below factorise cleanly.

11.6.4 9.6.4 Iterated conditional expectation

Starting from the pricing integral and using the tower property $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$:

$$V_0 \;=\; e^{-rT_2}\, \mathbb{E}^{\mathbb{Q}}\Big[(S_{T_2} - \alpha\, S_{T_1})_+ \Big] \;=\; e^{-rT_2}\, \mathbb{E}^{\mathbb{Q}}\Big[\mathbb{E}^{\mathbb{Q}}\Big[(S_{T_2} - \alpha\, S_{T_1})_+ \;\Big|\; S_{T_1} \Big] \Big]. \tag{9.20}$$

Separate the two discount factors so that the inner expectation becomes a Black–Scholes pricing problem in its own right, discounted from T_2 back to T_1:

$$V_0 \;=\; e^{-rT_1}\, \mathbb{E}^{\mathbb{Q}}\Big[e^{-r(T_2-T_1)}\, \mathbb{E}^{\mathbb{Q}}\Big[(S_{T_2} - \alpha\, S_{T_1})_+ \;\Big|\; S_{T_1} \Big] \Big]. \tag{9.21}$$

11.6.5 9.6.5 Inner expectation — Black–Scholes with strike αS_{T_1}

Conditional on S_{T_1}, the forward-starting payoff $(S_{T_2} - \alpha\, S_{T_1})_+$ is a standard European call on S with time-to-expiry $T_2 - T_1$ and strike $\alpha\, S_{T_1}$ (treated as a constant given $\mathcal{F}_{T_1}$). The inner conditional expectation is therefore given by the Black–Scholes formula (9.14)–(9.15), with spot S_{T_1} and strike $\alpha\, S_{T_1}$:

$$e^{-r(T_2-T_1)}\, \mathbb{E}^{\mathbb{Q}}\Big[(S_{T_2} - \alpha\, S_{T_1})_+ \;\Big|\; S_{T_1} \Big] \;=\; S_{T_1}\, \Phi(d_+) \;-\; \alpha\, S_{T_1}\, e^{-r(T_2-T_1)}\, \Phi(d_-), \tag{9.22}$$

with

$$d_\pm \;=\; \frac{\ln(S_{T_1}/(\alpha\, S_{T_1})) + (r \pm \frac{1}{2}\sigma^2)(T_2 - T_1)}{\sigma\sqrt{T_2 - T_1}} \;=\; \frac{\ln(1/\alpha) + (r \pm \frac{1}{2}\sigma^2)(T_2 - T_1)}{\sigma\sqrt{T_2 - T_1}}. \tag{9.23}$$

Notice what happened: the S_{T_1} factors in the moneyness ratio $S_{T_1}/(\alpha S_{T_1}) = 1/\alpha$ *cancelled*, leaving $d_\pm$ as *constants* — not random variables. This is the defining algebraic property of the forward-starting structure and is what makes closed-form pricing possible. The spot and the strike scale identically with S_{T_1}, so moneyness is always $1/\alpha$ regardless of where S_{T_1} lands. Every dollar increase in S_{T_1} lifts both the (random) spot and the (random) strike by α, leaving the convexity structure invariant.

11.6.6 9.6.6 Outer expectation — linearity and pull-out

Collect the inner result as $S_{T_1} \cdot \eta$ with the deterministic constant

$$\eta \;:=\; \Phi(d_+) \;-\; \alpha\, e^{-r(T_2-T_1)}\, \Phi(d_-). \tag{9.24}$$

Substituting into (9.21),

$$V_0 \;=\; e^{-rT_1}\, \mathbb{E}^{\mathbb{Q}}\Big[S_{T_1} \cdot \eta \Big] \;=\; e^{-rT_1}\, \eta\, \mathbb{E}^{\mathbb{Q}}[S_{T_1}] \;=\; e^{-rT_1}\, \eta \cdot S_0\, e^{rT_1} \;=\; \eta \cdot S_0, \tag{9.25}$$

where the third equality uses the risk-neutral martingale condition $\mathbb{E}^{\mathbb{Q}}[S_{T_1}] = S_0\, e^{rT_1}$ — a direct consequence of the fact that the discounted spot $e^{-rt}\, S_t$ is a $\mathbb{Q}$-martingale (FTAP from Chapter 2, made rigorous by Girsanov in Chapter 5). The two discount factors e^{-rT_1} and e^{+rT_1} cancel, and the outer expectation collapses to the constant $\eta\, S_0$. The closed-form forward-starting call price is

$$\boxed{V_0 \;=\; S_0 \Big[\Phi(d_+) \;-\; \alpha\, e^{-r(T_2-T_1)}\, \Phi(d_-) \Big], \qquad d_\pm \;=\; \frac{\ln(1/\alpha) + (r \pm \frac{1}{2}\sigma^2)(T_2 - T_1)}{\sigma\sqrt{T_2 - T_1}}.} \tag{9.26}$$

11.6.7 9.6.7 Worked example

The price scales linearly in S_0 since $d_\pm$ depends only on α and the tenor $T_2 - T_1$. Take $S_0 = 100$, $\alpha = 1.0$ (ATM reset), $T_1 = 0.5$, $T_2 = 1.0$, $r = 5\%$, $\sigma = 20\%$, giving $d_+ \approx 0.2475$, $d_- \approx 0.1061$, $\eta \approx 0.0688$, and $V_0 \approx 6.88$ — about 34% cheaper than the vanilla ATM call at $T = 1$ (≈ 10.45), reflecting the lost first-half convexity. The delta is the constant η (no rebalancing pre-reset) and the vega is concentrated on the *forward volatility* over $[T_1, T_2]$; forward-starts are the natural vehicle for forward-vol exposure.

11.7 9.7 Barrier Options and the Monte-Carlo Time Grid

Barrier options are the canonical path-dependent structure. Their payoff depends not on the final spot alone but on whether the path touches a pre-specified threshold U at *any* point during the contract's life. This seemingly small change — swapping "at maturity" for "at any time before maturity" — transforms the pricing problem fundamentally: we can no longer rely on the terminal distribution of S_T alone; we need the joint distribution of S_T and the running maximum (or minimum) of the path. In return we get a rich family of structured-note primitives, used heavily in retail derivatives packaging and in institutional structured funding.

11.7.1 9.7.1 Taxonomy

Barrier options come in eight canonical flavours: {up, down} × {in, out} × {call, put}. An *up-and-in* call becomes a plain call if the stock ever crosses above U; an *up-and-out* call is a plain call that gets cancelled if the stock ever crosses above U. Symmetric definitions hold for down-and-in, down-and-out, and the put versions. A structurally important identity is *in-out parity*:

$$V^{\text{up-and-in call}} + V^{\text{up-and-out call}} = V^{\text{vanilla call}}, \tag{9.27}$$

since exactly one of the two triggers for every path. In-out parity lets us price one member of the pair in terms of the other; it also serves as a primary source of variance reduction in barrier MC, because the vanilla call price is available in closed form and subtracting a known quantity reduces simulation noise (§9.8). (Implicit assumption: the knock-in and knock-out contracts pay the same — typically zero — rebate on the knock event. If the knock-out pays a cash rebate R at the first-passage time τ_B while the knock-in pays nothing on that event, (9.27) picks up an additional rebate-PV term $\mathbb{E}^{\mathbb{Q}}[e^{-r\tau_B} R\, \mathbf{1}_{\{\tau_B \le T\}}]$ on the right-hand side; the rebate-free version is the one priced throughout this section.)

Barrier options are strictly cheaper than the corresponding vanillas (for knock-out barriers that cancel the contract) or equal-or-cheaper (for knock-ins, which have a chance of never activating and paying zero). They are used extensively in structured products: a "twin-win" note pays the magnitude of index returns unless a down-barrier is breached; a "reverse convertible" pays a coupon unless a down-barrier is breached, in which case the investor is put the stock at a fixed strike. The barrier adds path-dependence to the payoff, preventing closed-form lognormal pricing in general — though the classical Merton reflection-principle formulas *do* give closed forms under continuous monitoring of geometric Brownian motion with constant parameters.

11.7.2 9.7.2 Up-and-in call — definition

The up-and-in call pays

$$\varphi = (S_T - K)_+ \, \mathbf{1}_{\{\max_{0 \le t \le T} S_t \ge U\}}, \qquad U > S_0, \tag{9.28}$$

where the typical parameter ordering is $U > K > S_0$ (barrier above strike above initial spot). The contract becomes a standard European call with strike K at the first time the path crosses the barrier; otherwise the holder receives nothing. The knock-in event is measurable with respect to the path up to T, not just S_T; this is what makes the payoff path-dependent.

Under continuous monitoring of GBM there is a beautiful closed-form formula (the "reflection-principle" formula). For the up-and-in call with $S_0 < K < U$,

$$V^{\text{UIC}}_{\text{continuous}} \;=\; S_0\Big(\tfrac{U}{S_0}\Big)^{2\lambda}\,\Phi(y_1) \;-\; K\,e^{-rT}\Big(\tfrac{U}{S_0}\Big)^{2\lambda-2}\,\Phi(y_1-\sigma\sqrt{T}), \tag{9.29}$$

where $\lambda = (r+\frac{1}{2}\sigma^2)/\sigma^2$ and y_1 is an explicit log-moneyness expression. We do not derive (9.29) here — the reflection principle for Brownian motion and its Girsanov-shifted counterpart for GBM are a standard but lengthy calculation, covered in every classical derivatives text. We use the formula only as an external benchmark: the *continuous-monitoring* closed form gives us a reference number against which discrete-monitoring MC estimates can be calibrated. Under discrete monitoring (say daily fixings), the barrier is crossed only at the fixing dates, not continuously in between; MC implementations naturally approximate discrete monitoring if the grid is chosen as the monitoring dates.

11.7.3 9.7.3 Lattice view

Before moving to Monte Carlo it is worth seeing how a binomial lattice prices barrier options. The lattice view clarifies the discretisation questions that plague the MC implementation and ties the path-dependent pricing back to the backward-induction apparatus of Chapter 2. Propagate known values from the barrier level downward: at any node on or above U, the option is (thereafter) a standard European call with strike K whose value $C^{\text{vanilla}}(S_n, t_n; K, T)$ is already known from the BS formula (9.14)–(9.15). Below the barrier the backward induction is the usual discounted $\mathbb{Q}$-expectation, but with the *known* vanilla-call value pasted in at any node that has touched U along the path to that node. The tree effectively shrinks to a triangular wedge bounded by the barrier surface, and the barrier acts as a "known-value boundary" analogous to a Dirichlet condition in PDE language.

In the PDE framework, the Black–Scholes equation for a barrier option carries a Dirichlet boundary condition at $S=U$: for a knock-in, $V(U, t) = $ the vanilla-call value at that time; for a knock-out, $V(U,t)=0$. The lattice backward-induction is solving this PDE numerically, with a two-child backward expectation playing the role of the finite-difference stencil. Analytical PDE solutions via method of images or Green's functions yield the same reflection-principle formulas mentioned above. Different methods — closed form, PDE, lattice, MC — should all agree to within their respective discretisation and variance errors; cross-validation between methods is a standard sanity check in derivatives implementation.

11.7.4 9.7.4 Monte-Carlo pricing with a time grid

The path-simulation recipe (9.9) applies directly. Discretise $[0,T]$ with a grid $t_0 = 0 < t_1 < \cdots < t_N = T$, step sizes $\Delta t_n = t_n - t_{n-1}$, and simulate

$$S_{t_n} \;=\; S_{t_{n-1}}\, e^{(r-\frac{1}{2}\sigma^2)\Delta t_n \,+\, \sigma\sqrt{\Delta t_n}\, Z_n}, \qquad Z_1, Z_2, \ldots \overset{\text{iid}}{\sim}_{\mathbb{Q}} \mathcal{N}(0,1). \tag{9.30}$$

The Monte-Carlo estimator of the up-and-in call price is

$$\widehat{V}_0 \;=\; e^{-rT}\cdot\frac{1}{M}\sum_{m=1}^{M}\big(S_T^{(m)}-K\big)_+\,\mathbf{1}_{\big\{\max_n\, S_{t_n}^{(m)}\,\geq\, U\big\}}. \tag{9.31}$$

Each simulated path contributes to the average only if its discretely-sampled maximum equals or exceeds U; paths that stay below U contribute zero regardless of terminal value.

11.7.5 9.7.5 Discretisation bias

The indicator $\mathbf{1}_{\{\max_n S_{t_n}\geq U\}}$ in (9.31) samples the path only at grid points; the true knock-in event $\{\max_{0\leq t\leq T} S_t \geq U\}$ can occur *between* grid points. For a given Brownian path one therefore underestimates the knock-in probability — each grid has some probability of missing a barrier excursion that happened between two of its points. Consequently the barrier-option MC price is *biased downward* at finite N, and the bias vanishes only in the limit $\Delta t_n \to 0$. This is the characteristic issue with MC pricing of path-dependent barrier and lookback contracts; the same issue arises for any payoff whose value depends on a supremum, infimum, or first-hitting time of the path.

The bias can be substantial. For an up-and-in call with $S_0 = 100$, $K=100$, $U=110$, $\sigma = 20\%$, $r=5\%$, $T=1$: the true continuous-monitoring value from (9.29) is around 6.15. With $N=50$ monitoring points and $M=10{,}000$ MC paths, the raw estimate from (9.31) is typically around 5.8 — a 6% downward bias. Refining the grid to

$N = 500$ (factor-of-10 more work) without any bias correction gives an estimate near 6.10 — still a small residual bias. Continuing to refine eventually converges to the true value but at substantial computational cost.

11.7.6 9.7.6 Brownian-bridge correction

Between two adjacent grid points t_{n-1} and t_n with known endpoints $S_{t_{n-1}}$ and S_{t_n}, the conditional law of the path is a *Brownian bridge* (under log-coordinates with drift). The conditional distribution of the path maximum $\max_{t\in[t_{n-1},t_n]} S_t$ given the endpoints has a closed-form CDF, derived from the reflection principle:

$$\mathbb{P}\left(\max_{t\in[t_{n-1},t_n]} S_t \ \geq\ U \ \middle|\ S_{t_{n-1}}, S_{t_n} \right) \ = \ \exp\left(- \frac{2\,\ln(U/S_{t_{n-1}})\,\ln(U/S_{t_n})}{\sigma^2\,\Delta t_n} \right) \tag{9.32}$$

for $U > \max(S_{t_{n-1}}, S_{t_n})$; the probability is 1 when the barrier is already at or below either endpoint. Formula (9.32) is the continuous-time analogue of "interpolate between grid points and check if the interpolant crossed": it gives the exact conditional crossing probability under Brownian-bridge dynamics, which is the correct interpolation given the two endpoints.

The Brownian-bridge correction uses (9.32) as follows. For each simulated path that was *not* flagged as knock-in by the grid-point check, compute the per-interval bridge probability (9.31), draw a uniform random number $U_{\text{unif}}^{(m,n)}$ on $[0,1]$, and flag the path as knock-in if $U_{\text{unif}}^{(m,n)} < p_{\text{bridge}}^{(m,n)}$ in any interval. Equivalently (and more numerically stable), compute the product of non-crossing probabilities over all intervals and subtract from one to get the corrected knock-in probability for the path; use this as the indicator weight. This reduces the bias from $O(\sqrt{\Delta t})$ to $O(\Delta t^{3/2})$ for typical barrier structures — a dramatic improvement.

Concrete numbers. For the up-and-in call with $S_0 = 100$, $K = 100$, $U = 110$, $\sigma = 20\%$, $r = 5\%$, $T = 1$ at $N = 50$: raw MC gives ≈ 5.8; bridge-corrected gives ≈ 6.12, within 0.5% of the true 6.15. The bridge at $N = 50$ matches brute-force $N \approx 500$ — a factor-of-10 saving.

11.7.7 9.7.7 Variance of the barrier estimator

The payoff $X = e^{-rT}(S_T - K)_+\,\mathbf{1}_{\{\max S \geq U\}}$ is zero on most paths when the barrier is deep OTM, so the estimator is dominated by a handful of triggering paths. If knock-in probability is p_{knock}, the standard error scales as $\sqrt{p_{\text{knock}}\,\sigma_X^2/M}$, needing roughly $1/p_{\text{knock}}$ more paths than the vanilla case. Rare-event MC is computationally unforgiving.

Two standard remedies. *Importance sampling* shifts the drift of the simulation to make knock-in more likely, then reweights by the Radon–Nikodym derivative (Chapter 5); a well-tuned IS can reduce variance by factors of 10–100 for deeply out-of-the-barrier options. *Control variates* using the vanilla European call (whose expectation is known exactly from (9.14)–(9.15) and whose payoff correlates strongly with the barrier payoff on paths that do knock in) provide another order of magnitude of variance reduction for mildly-barriered options. In production, combining IS with a vanilla-call control and Brownian-bridge correction can drive barrier-MC run times from hours to seconds.

11.7.8 9.7.8 Other path-dependent families

Having priced barriers, we preview the broader landscape of path-dependent options that the Monte-Carlo machinery handles cleanly.

Asian options. Payoffs depend on the *average* of the underlying over a window, such as $(\bar{S}_T - K)_+$ where $\bar{S}_T = (1/N)\sum_{n=1}^{N} S_{t_n}$ (arithmetic average). Arithmetic Asians have no closed form under GBM because the sum of log-normals is not itself log-normal; MC is the standard tool. The corresponding *geometric* average $\bar{S}_T^{\text{geo}} = \prod_{n=1}^{N} S_{t_n}^{1/N}$ *is* log-normal and admits a closed-form Black–Scholes-style pricer, which makes it a natural control variate for arithmetic Asian MC — typical variance reduction factors of 50 to 500. Asian options are popular on commodity markets (oil, gas, metals) because averaging smooths out daily-price noise and resists single-day manipulation of the settlement print.

Lookback options. Payoffs depend on the running extremum, such as $(\max_t S_t - K)_+$ (lookback call on running max) or $(S_T - \min_t S_t)_+$ (lookback put on running min). Lookbacks are the most path-sensitive of the classical exotics and carry prices substantially higher than vanilla equivalents — they capture the best achievable entry point, an upper bound on realisable profit. Closed forms exist under continuous-monitoring GBM (again via the reflection principle), making lookbacks a useful benchmark for MC implementations.

Combinations. Practitioners layer features routinely — an "Asian barrier call" pays $(\bar{S}_T - K)_+$ but only if the running max stays below some upper barrier; a "cliquet with knock-out" terminates the coupon stream if a down-barrier is hit. These combinations have no closed form and are handled by MC with custom payoff functions that walk each path and assemble the appropriate indicator and averaging structure.

Autocallables. Highly structured products that pay coupons on observation dates if the underlying is above a specified trigger, and auto-call (early redemption of principal) once the underlying breaches an upper barrier. Autocallable pricing involves a mix of path-dependence, multiple discrete monitoring dates, and early-termination optionality; MC is the workhorse, typically with quasi-Monte-Carlo low-discrepancy sequences to reduce variance on the discrete fixing dates.

All of these families share the same MC machinery: generate paths under $\mathbb{Q}$, apply the payoff functional to each path, average with discount. The engineering art is in (i) reducing variance through targeted techniques (Asian controls for Asians, bridge corrections for barriers, importance sampling for rare-event payoffs), (ii) handling the exact contractual details correctly (continuous vs. discrete monitoring, fixed vs. floating strikes, local vs. global conditions, business-day conventions), and (iii) extracting Greeks efficiently from the same sample.

11.8 9.8 Variance Reduction: Antithetic and Control Variates

Every production Monte-Carlo pricer uses variance reduction. The $M^{-1/2}$ convergence rate is fixed by the central limit theorem, but the *constant* in front — the standard deviation $\mathrm{sd}(X)$ of the estimator variate — can be driven down by orders of magnitude with modest effort. Two techniques are universal: antithetic variates and control variates. We cover them here as a bridge to the Heston MC of Chapter 10, where they become essential rather than merely convenient.

11.8.1 9.8.1 Antithetic variates

The antithetic-variates trick exploits the symmetry of the Gaussian distribution. For each standard-normal draw $Z^{(m)}$, also use $-Z^{(m)}$. The symmetrised payoff is

$$\widetilde{X}^{(m)} \;=\; \frac{1}{2}\left[\, X(Z^{(m)}) + X(-Z^{(m)})\,\right], \tag{9.33}$$

and the antithetic estimator is $\widehat{V}_0^{\mathrm{ant}} = (1/M)\sum_m \widetilde{X}^{(m)}$. The computational cost is $2M$ payoff evaluations, but the effective sample size from the variance-reduction standpoint is *not* $2M$ — the two draws $X(Z)$ and $X(-Z)$ are *negatively* correlated, and the variance of the average is

$$\mathbb{V}[\widetilde{X}^{(m)}] \;=\; \frac{1}{4}\left[\,\mathbb{V}[X(Z)] + \mathbb{V}[X(-Z)] + 2\,\mathbb{C}[X(Z), X(-Z)]\,\right] \;=\; \frac{1}{2}\,\mathbb{V}[X]\,(1+\rho), \tag{9.34}$$

where $\rho := \mathrm{Corr}(X(Z), X(-Z))$. If $\rho < 0$, the variance is strictly less than $\mathbb{V}[X]/2$, and antithetic sampling outperforms independent sampling even at equal cost.

When does antithetic sampling work? It works best when the payoff $X(Z)$ is *monotone* in Z — calls are, puts are (after sign flip), digitals are, lookbacks almost are. For a monotone increasing X, $X(Z)$ and $X(-Z)$ have correlation close to -1 in the Gaussian tails, and the variance reduction factor can be substantial. For a European call at typical ATM parameters, antithetic sampling at M pairs typically achieves a variance reduction of about 2 relative to independent sampling at M pairs (i.e., $2M$ independent draws) — equivalent to halving the standard error for the same number of payoff evaluations.

Antithetic vs plain MC standard error at equal compute *At equal payoff-evaluation cost, antithetic sampling lowers the MC standard error below plain sampling across every sample size. Both schemes retain the $1/\sqrt{M}$ slope — variance reduction moves the intercept, not the rate — and the empirical gap matches the theoretical factor of $\sqrt{2}$ for monotone payoffs near ATM.*

Antithetic sampling fails when the payoff is *symmetric in Z* — e.g. a straddle $|S_T - K|$ at $K = S_0 e^{rT}$ gives $\rho = +1$ and no reduction. The technique extends naturally to path simulation by negating the entire Gaussian vector $(Z_1, \dots, Z_N) \to (-Z_1, \dots, -Z_N)$, and yields similar factor-of-2–3 reductions on barriers and Asians.

11.8.2 9.8.2 Control variates

The control-variate trick is more flexible and often more powerful than antithetic sampling. The idea: find an auxiliary variable Y that is strongly correlated with the payoff X and whose expectation $\mathbb{E}[Y]$ is known in closed form. Form the control-variate estimator

$$\widehat{V}_0^{\mathrm{cv}} \;=\; \widehat{m}_X \;+\; \beta\,(\mathbb{E}[Y] - \widehat{m}_Y), \tag{9.35}$$

where $\widehat{m}_X$ and $\widehat{m}_Y$ are the sample means of X and Y on the same M paths, and β is a constant to be chosen. The estimator is unbiased for any β because $\mathbb{E}[\widehat{m}_Y] = \mathbb{E}[Y]$, so the correction term has mean zero; it has variance

$$\mathbb{V}[\widehat{V}_0^{\mathrm{cv}}] \;=\; \frac{1}{M}\,[\,\mathbb{V}[X] \;-\; 2\beta\,\mathbb{C}[X,Y] \;+\; \beta^2\,\mathbb{V}[Y]\,], \tag{9.36}$$

which is minimised at

$$\beta^\star \;=\; \frac{\mathbb{C}[X,Y]}{\mathbb{V}[Y]}, \tag{9.37}$$

giving the minimum variance

$$\mathbb{V}[\widehat{V}_0^{\mathrm{cv}}]\,\big|_{\beta=\beta^\star} = \;\frac{\mathbb{V}[X]}{M}\,(1 - \rho_{XY}^2), \tag{9.38}$$

where $\rho_{XY} = \mathrm{Corr}(X,Y)$. The variance reduction factor is $1/(1 - \rho_{XY}^2)$: for $\rho_{XY} = 0.9$ the reduction is about $5\times$; for 0.99 it is $50\times$; for 0.999 it is $500\times$.

Choosing a good control. The essential question is what auxiliary Y is known in closed form *and* strongly correlated with the target payoff. Several standard choices:

- **The underlying's terminal value** $Y = S_T$, with $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0\,e^{rT}$ from the martingale property. Works well for European calls and puts, whose payoffs are monotone in S_T; typical correlation at ATM parameters is $\rho \approx 0.95$ for calls, giving about a $10\times$ variance reduction.
- **A vanilla Black–Scholes call on the same underlying**, with $\mathbb{E}[Y]$ given by (9.14)–(9.15). Works extremely well for barriers, digitals, and mildly exotic payoffs whose MC error on the vanilla portion would otherwise dominate; typical $\rho \approx 0.99$ for up-and-in calls, giving $50\times$ or more variance reduction.
- **A geometric-average Asian call** for arithmetic Asian pricing, where the geometric version has a closed-form BS-like formula and the arithmetic version does not. Typical $\rho > 0.99$ because the two averages are near-identical on almost every path; 100–$500\times$ variance reduction.
- **The terminal value on a drift-shifted GBM** as a control for stochastic-volatility payoffs, linking the Heston case back to the Black–Scholes benchmark with matched $\mathbb{E}[S_T]$ and approximate variance.

Estimating $\beta^\star$. The optimal $\beta^\star$ depends on the unknown covariance structure. In practice one runs a pilot batch of $M_{\mathrm{pilot}} = 1000$ or so paths, estimates $\hat{\beta} = \widehat{\mathbb{C}}[X,Y]/\widehat{\mathbb{V}}[Y]$ from the pilot, and then runs the full M batch with that $\hat{\beta}$ plugged into (9.35). The plug-in estimator is no longer strictly unbiased (because $\hat{\beta}$ and $\widehat{m}_X$, $\widehat{m}_Y$ are all correlated through the data), but the bias is $O(1/M)$ and negligible relative to the statistical error. More sophisticated implementations use a two-sample scheme: estimate $\hat{\beta}$ from paths $1, \dots, M_1$, use those paths only for the second-stage estimator on paths $M_1 + 1, \dots, M$, and aggregate.

11.8.3 9.8.3 Combining techniques

Antithetic and control variates compose. Apply antithetic sampling to generate the path pairs, evaluate both the target payoff X and the control payoff Y on each antithetic pair, take the antithetic average of each, and then apply the control-variate correction (9.35) to the antithetic averages. The variance reductions multiply roughly (not exactly, because the correlations interact), and for a European call with a stock-price control and antithetic draws one routinely achieves total variance reduction factors of 30–50 at essentially no added compute cost.

Beyond these two techniques the next layers — *stratified sampling* (partitioning the Gaussian space into equal-probability strata and drawing one sample per stratum), *importance sampling* (Radon–Nikodym reweighting of the simulation distribution), *Latin hypercube sampling*, and *quasi-Monte-Carlo with Sobol or Halton low-discrepancy sequences* — each provide additional factors of 2–100 for the right payoff. A production pricer typically stacks two or three of these techniques; a research-grade pricer may stack four or five. The trader's mantra: "MC is universal but wasteful; always be improving the variance."

Variance-reduction techniques are indispensable for Heston Monte Carlo (Chapter 10) where the variance-process simulation itself introduces additional noise on top of the payoff noise, and where the closed-form benchmarks from Black–Scholes are no longer available but *approximate* BS controls based on a matched-moments log-normal still give large reductions. We return to this theme when we take up stochastic volatility.

11.8.4 9.8.4 Further reading: the wider toolkit

Beyond antithetic and control variates, the standard next-layer techniques are:

- **Stratified sampling.** Partition the Gaussian space into k equal-probability strata and sample M/k per stratum. Unbiased, never worse than plain MC, but scales poorly with dimension — typically used on the dominant direction only.
- **Importance sampling.** Reweight by a Radon–Nikodym derivative (Chapter 5) to concentrate mass on regions where the payoff is large. Variance reductions of 10–1000× on rare-event payoffs; the technique of choice for deep-OTM options and tail-risk.
- **Quasi-Monte-Carlo (Sobol, Halton).** Deterministic low-discrepancy sequences in place of i.i.d. Gaussians; error scales as $O((\log M)^d/M)$ for smooth integrands. Pairs well with scrambling.
- **Multilevel Monte Carlo.** Telescoping decomposition across grid resolutions; reduces total cost for an ϵ-accurate estimate from $O(\epsilon^{-3})$ to $\sim O(\epsilon^{-2})$ on SDE-discretisation-limited problems.
- **Longstaff–Schwartz regression.** Polynomial regression of continuation values for American/Bermudan exercise; produces a lower bound on the true price.

Practitioner's rule of thumb: start with antithetic; add a control variate if a correlated cousin has a closed form; reach for importance sampling on rare-event payoffs; use Sobol when dimensionality is moderate and the integrand smooth.

11.8.5 Case study (a): Asian commodity options on jet fuel

Airlines hedge quarterly fuel cost via average-price calls on jet kerosene. The arithmetic average of correlated lognormals is not lognormal — no closed form. Production pricer is MC + geometric-Asian control variate (Vorst 1992).

Context. Major airlines (Lufthansa, United, Singapore Airlines, Air France-KLM) hedge between 30–60% of their forward fuel consumption at any point through a mix of OTC and listed instruments on jet kerosene (Platts CIF NWE, Singapore MOPS, US Gulf 54-grade) and proxy hedges in heating oil and Brent. The standard product is an Asian (average-price) call: payoff at T_2 equals $\max(\bar{S}_{[T_1,T_2]} - K, 0)$ where the average $\bar{S}$ is taken over N daily Platts fixings between an observation start T_1 and end T_2 (commonly the calendar quarter). Averaging is contractually preferred to a single-fix European because (i) airlines consume fuel daily, so an average payoff matches the actual hedged exposure, and (ii) a single-day European is exposed to manipulation of the daily marker assessment. Notional is large: the IATA jet-fuel hedge volume across the top-20 carriers ran roughly \$50–80B notional through 2018–19.

The product priced by an airline desk on a typical day will have $T_2 - T_1 \approx 60$ business days (one quarter), $N \approx 60$ daily fixings, K a few percent in or out of the money relative to the forward curve, and a σ implied vol on jet fuel in the 20–35% range. There is no closed-form European-style pricing formula because the arithmetic average

$\bar{S} = \frac{1}{N}\sum_n S_{t_n}$ of correlated lognormals is not itself lognormal: the sum-of-lognormals distribution has no analytic CDF and, while its first two moments are available in closed form (Turnbull-Wakeman, Levy moment-matching), the implied call is only as good as the tail approximation.

Reading it through the chapter's math. Direct application of the MC estimator (9.10) extended to the path-dependent payoff: generate M paths under $\mathbb{Q}$ with the GBM step (9.9) on a daily grid covering $[0, T_2]$, evaluate $\bar{S}^{(m)} = \frac{1}{N}\sum_n S_{t_n}^{(m)}$ on each path, take $\hat{V}_0 = e^{-rT_2}\frac{1}{M}\sum_m(\bar{S}^{(m)} - K)_+$. A naive run with $M = 10\,000$ paths and $N = 60$ daily fixings on a 30%-vol underlying gives a standard error of about 2% of premium — too coarse for a desk that quotes to a half-cent bid-ask. The decisive trick (§9.8.2) is the Vorst (1992) geometric-Asian control: the *geometric* average $\bar{S}^{\text{geo}} = \left(\prod_n S_{t_n}\right)^{1/N}$ *is* lognormal under GBM (the log of a sum of jointly-normal log-prices is itself normal), so the geometric-Asian call has a closed-form BS-style price. Empirically the arithmetic and geometric averages on commodity- vol paths correlate at $\rho > 0.998$, giving variance reduction $1/(1-\rho^2) > 250\times$ via (9.35). The same $M = 10\,000$ paths with the geometric control cuts the standard error to roughly 0.1% of premium — under bid-ask, in production-acceptable wallclock time (seconds on a single core).

Lesson. Asian options are a clean illustration of the rule that MC is universal but wasteful, and that a control variate is what makes a production pricer. The geometric-Asian closed form is not used to price the contract — it is used as a noise-reduction device against the arithmetic-Asian MC estimator. Two operational consequences follow. First, *always look for a closed-form cousin* of any MC payoff: a European call on the terminal S_{T_2} alone (correlation ~ 0.7) is a workable backup control when geometric closed forms are not handy (e.g., basket of correlated commodities). Second, the variance reduction depends on ρ^2 — when the control correlates poorly, adding a second control via multivariate regression (control matrix) recovers the reduction. The arithmetic-vs-geometric Asian is the canonical case where a chapter-9 trick converts a two-hour MC run into a two-second one without changing the underlying physics.

11.8.6 Case study (b): Autocallable equity notes

Path-dependent across multiple observation dates — closed form is hopeless. Korea, EU, Japan together issue ~$50B/year of these structured notes. Vega term structure dominates the hedge — spot delta is the easy part.

Context. Autocallable notes (also called "kick-out" notes, Phoenix notes, or "worst-of" autocallables when written on a basket) are the single largest category of structured equity products by issuance volume. A typical retail autocallable on EuroStoxx 50 might pay an 8% annualised coupon as long as the index stays above 100% of initial on quarterly observation dates; the note auto-calls (returns principal plus the prevailing coupon) on the first observation where the index closes above an autocall barrier (often 100% of initial); and at the final maturity (commonly 4 to 6 years), if no autocall has occurred and the index is below a downside knock-in barrier (commonly 60% of initial), the investor receives shares at the lower price (or the cash equivalent). Korean retail issuance alone runs $20–35B notional/year through KOSPI 200 and HSCEI-linked ELS (Equity-Linked Securities); EU private-bank issuance through SG, BNP, Marex, JPM adds another $20B; Japanese Uridashi notes add ~$10B. Total street notional outstanding is in the hundreds of billions and the secondary-market mark-to-market is a daily MC job for every issuing dealer.

Reading it through the chapter's math. No closed form: the autocall event depends on the spot trajectory through the discrete observation dates, the knock-in event depends on the path-minimum (continuous monitoring on most contracts), and the coupon stream is a digital-with- recall structure. MC is the only path. Generate M paths under $\mathbb{Q}$ on a daily grid covering the full $[0, T_{\text{max}}]$ horizon (typically 4–6 years $\times$ 252 business days = 1000–1500 steps). On each path, walk through the observation dates in order: at each date, check the autocall condition; if triggered, accrue $e^{-rt_{\text{call}}}$ times the redemption amount, discard the rest of the path. If never triggered, evaluate the maturity payoff (which depends on the path minimum vs the knock-in barrier and the terminal level vs the put strike). Average. Computational cost on a daily grid for a 5-year autocallable: roughly 10^5 paths $\times$ 1.3×10^3 steps $\times$ four underlyings (a typical worst-of basket) $= 5 \times 10^8$ Gaussian draws per pricing — the variance-reduction toolkit (antithetic + Sobol on observation-date Gaussians, control variates against the European-equivalent benchmark) is what makes daily MTM tractable.

Lesson. The pricing is the easy part — the hard part is the *hedging*. An autocallable has a non-trivial vega *term structure*: at inception the issuer is short vega in the early observation dates (an autocall extinguishes the coupon stream, hurting the issuer when realised vol is low and the index drifts above the autocall barrier) and long vega at the terminal maturity (the embedded down-and-in put benefits the issuer when realised vol is high). The aggregate

Greeks $\partial V/\partial\sigma_{T_i}$ across observation dates can flip sign as the spot drifts: a typical Korean ELS book holds ~$1B of vega notional with sign flips at every observation date, and a 1-vol-point move in a single tenor on the vol surface can move book P&L by tens of millions. Spot delta, by contrast, is the smoothest Greek in the book (the autocall has bounded payoff, so Δ is bounded by 1 in magnitude). The lesson for the Chapter-9 reader: when you build a production MC pricer for an autocallable, the priority is computing the *vega bucket* sensitivities — finite-difference $\partial V/\partial\sigma_{T_i}$ for each observation tenor — using common random numbers (CRN) so the vega-bucket noise is dominated by signal not by MC noise. A book that hedges the wrong vega tenor is short the right Greek for the wrong reason and will bleed slowly until the autocall window arrives.

11.9 9.9 Key Takeaways

1. **Why MC.** Monte-Carlo integration escapes the curse of dimensionality by achieving $O(M^{-1/2})$ error *independent of dimension*; its cost is $O(Md)$ compared to $O(N^d)$ for deterministic quadrature. Break-even dimension is around $d \approx 5$, and every realistic path-dependent pricing problem lives well above it.
2. **Strong Law of Large Numbers.** For i.i.d. draws of a random variable X with $\mathbb{E}[|X|] < \infty$, the sample mean converges almost surely to $\mathbb{E}[X]$. For finite-variance X, the Central Limit Theorem gives the $M^{-1/2}$ error rate and the basis for confidence intervals.
3. **Standard error.** The MC estimator's standard error is $\hat{\sigma}_{m_1} = \hat{s}/\sqrt{M}$, where $\hat{s}^2 = (1/(M-1))\sum(X^{(m)} - \widehat{m}_1)^2$ is the unbiased sample variance. Halving the error bar costs 4× more paths.
4. **Exact GBM generator.** Under $\mathbb{Q}$, the lognormal increment $S_{t_n} = S_{t_{n-1}} \exp\{(r - \frac{1}{2}\sigma^2)\Delta t_n + \sigma\sqrt{\Delta t_n}\, Z_n\}$ with $Z_n \sim \mathcal{N}(0,1)$ i.i.d. is *exact at the grid points* — no discretisation bias. The missing-$\frac{1}{2}\sigma^2$ drift correction is the most common bug in home-grown MC pricers.
5. **European MC.** The MC estimator $\widehat{V}_0 = e^{-rT} \cdot (1/M)\sum_m \varphi(S_T^{(m)})$ should cross-check against Black–Scholes as a correctness test. At $M = 10{,}000$ and typical ATM parameters, the standard error is about 0.14 on a call worth about 10.45.
6. **Forward-starting (cliquet) atom.** The payoff $(S_{T_2} - \alpha S_{T_1})_+$ admits a closed form via iterated conditional expectation: $V_0 = S_0\,[\Phi(d_+) - \alpha\, e^{-r(T_2-T_1)}\,\Phi(d_-)]$ with $d_\pm = [\ln(1/\alpha) + (r \pm \frac{1}{2}\sigma^2)(T_2 - T_1)]/(\sigma\sqrt{T_2 - T_1})$. The S_{T_1} factor cancels in the moneyness ratio, making $d_\pm$ deterministic and the price linear in S_0.
7. **Barrier MC.** The up-and-in call payoff $(S_T - K)_+\,\mathbf{1}_{\{\max_t S_t \geq U\}}$ is priced by simulation on a time grid with $\max_n S_{t_n}$ substituted for $\max_t S_t$. The substitution introduces a *downward* discretisation bias on the order of $\sqrt{\Delta t}$.
8. **Brownian-bridge correction.** Between two grid points with known endpoints, the conditional barrier-crossing probability has a closed form $\exp(-2\,\ln(U/S_{t_{n-1}})\,\ln(U/S_{t_n})/(\sigma^2\Delta t_n))$. Applying the correction reduces barrier MC bias from $O(\sqrt{\Delta t})$ to $O(\Delta t^{3/2})$ — equivalent to a factor-of-10 grid refinement.
9. **Rare-event variance.** Barrier payoffs that knock in with small probability give MC estimators with large relative variance. The path count scales inversely with the knock-in probability, making brute-force MC impractical for deeply-barriered structures; importance sampling and control variates are essential.
10. **Antithetic variates.** For each Z draw, also use $-Z$. For monotone payoffs the antithetic pair has strongly negative correlation, and the variance of the average is less than half the variance of a single draw. Typical variance reduction factor ≈ 2 on ATM European calls.
11. **Control variates.** Given an auxiliary Y with known $\mathbb{E}[Y]$, form $\widehat{V}_0^{\text{cv}} = \widehat{m}_X + \beta(\mathbb{E}[Y] - \widehat{m}_Y)$ with $\beta^\star = \mathbb{C}[X,Y]/\mathbb{V}[Y]$. Variance reduction factor $1/(1-\rho_{XY}^2)$. For $\rho = 0.99$, 50×; for $\rho = 0.999$, 500×.
12. **Combining techniques.** Antithetic and control variates compose; production pricers routinely stack two or three techniques for combined variance-reduction factors of 50–500.
13. **Path-dependent families.** Asians, lookbacks, barriers, cliquets, autocallables — all handled by the same MC machinery: generate paths under $\mathbb{Q}$, apply the payoff functional path-by-path, discount and average. The engineering difficulty lies in variance reduction and in handling the exact contractual conventions (continuous vs discrete monitoring, fixed vs floating strikes, business-day conventions).

14. **Bridge to Chapter 10.** Heston Monte Carlo introduces a second simulation dimension (the variance process) and its own set of discretisation biases (the square-root process near zero requires careful handling via the full-truncation Euler or the exact QE scheme of Andersen). The variance-reduction toolkit developed here transfers directly, with BS-based controls replaced by matched-moments lognormal controls.

11.10 9.10 Reference Formulas

Strong law of large numbers.

$$\lim_{M\to+\infty} \frac{1}{M}\sum_{m=1}^{M} X^{(m)} \;=\; \mathbb{E}[X] \quad \text{a.s., provided } \mathbb{E}[|X|] < \infty.$$

Central limit theorem (Monte-Carlo confidence interval).

$$\sqrt{M}\,(\widehat{m}_M - \mathbb{E}[X]) \xrightarrow{d} \mathcal{N}(0, \mathbb{V}[X]), \qquad \widehat{m}_M \pm 1.96\,\widehat{\sigma}_{m_1} \text{ is a 95\% CI.}$$

Sample mean and Monte-Carlo standard error.

$$\widehat{m}_1 = \frac{1}{M}\sum_{m=1}^{M} X^{(m)}, \qquad \widehat{\sigma}_{m_1} = \frac{1}{\sqrt{M}}\left(\frac{1}{M-1}\sum_{m=1}^{M}(X^{(m)} - \widehat{m}_1)^2\right)^{1/2}.$$

Exact lognormal GBM path simulation.

$$S_{t_n} = S_{t_{n-1}}\,\exp\bigl\{(r - \tfrac{1}{2}\sigma^2)\Delta t_n + \sigma\sqrt{\Delta t_n}\,Z_n\bigr\}, \quad Z_n \overset{\text{iid}}{\sim}_{\mathbb{Q}} \mathcal{N}(0,1).$$

European MC estimator.

$$\widehat{V}_0 = e^{-rT}\cdot\frac{1}{M}\sum_{m=1}^{M}\varphi\Bigl(S_0\,e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z^{(m)}}\Bigr).$$

Forward-starting (cliquet) call price.

$$V_0 = S_0\,\bigl[\Phi(d_+) - \alpha\,e^{-r(T_2-T_1)}\,\Phi(d_-)\bigr], \qquad d_\pm = \frac{\ln(1/\alpha) + (r \pm \frac{1}{2}\sigma^2)(T_2 - T_1)}{\sigma\sqrt{T_2 - T_1}}.$$

Up-and-in call payoff and MC estimator.

$$\varphi = (S_T - K)_+\,\mathbf{1}_{\{\max_{0\le t\le T} S_t \ge U\}}, \qquad \widehat{V}_0 = e^{-rT}\cdot\frac{1}{M}\sum_{m=1}^{M}(S_T^{(m)} - K)_+\,\mathbf{1}_{\{\max_n S_{t_n}^{(m)} \ge U\}}.$$

Brownian-bridge crossing probability (continuous barrier correction).

$$\mathbb{P}\Bigl(\max_{t\in[t_{n-1},t_n]} S_t \ge U \,\Big|\, S_{t_{n-1}}, S_{t_n}\Bigr) = \exp\Bigl(-\frac{2\,\ln(U/S_{t_{n-1}})\,\ln(U/S_{t_n})}{\sigma^2\,\Delta t_n}\Bigr) \quad \text{for } U > \max(S_{t_{n-1}}, S_{t_n}).$$

In-out parity.

$$V^{\text{up-and-in call}} + V^{\text{up-and-out call}} = V^{\text{vanilla call}}.$$

Antithetic variate estimator (single draw).

$$\widetilde{X}^{(m)} = \tfrac{1}{2}\bigl[X(Z^{(m)}) + X(-Z^{(m)})\bigr], \qquad \mathbb{V}[\widetilde{X}^{(m)}] = \tfrac{1}{2}\mathbb{V}[X]\,(1+\rho),\ \ \rho = \mathrm{Corr}(X(Z), X(-Z)).$$

Control variate estimator and optimal coefficient.

$$\widehat{V}_0^{\text{cv}} = \widehat{m}_X + \beta(\mathbb{E}[Y] - \widehat{m}_Y), \qquad \beta^\star = \frac{\mathbb{C}[X,Y]}{\mathbb{V}[Y]}, \qquad \mathbb{V}[\widehat{V}_0^{\text{cv}}]\,\big|_{\beta^\star} = \frac{\mathbb{V}[X]}{M}(1 - \rho_{XY}^2).$$

12 Chapter 10 — Heston Stochastic Volatility

Part V opens with the equity-side capstone. Chapter 6 handed us the Black–Scholes PDE under the assumption that instantaneous variance σ^2 is a known constant; every chapter since has carried that assumption along. The market, of course, does not. Implied-volatility surfaces slope, twist, and deform in ways that a constant-variance GBM cannot reproduce. Heston's 1993 model is the cleanest fix: promote the variance to its own mean-reverting stochastic process, keep tractability by choosing a square-root (CIR) dynamics for the variance, and pay for the extra degree of freedom with a second Brownian motion that is correlated with the underlying. The reward is a semi-closed pricing formula via Fourier inversion, a genuine stochastic smile, and a hedging framework that admits volatility as a traded risk factor.

This chapter leans on the full Part II toolkit. Two-dimensional Itô's lemma (Chapter 3) is used in every derivation; multi-dimensional Girsanov (Chapter 5) is how we move from the physical bivariate SDE to the risk-neutral dynamics; Feynman–Kac (Chapter 4) underlies the Riccati ODE system for the characteristic function; and the GBM / Black–Scholes machinery of Chapter 6 is the benchmark we compare against. Monte Carlo variance-reduction techniques introduced in Chapter 9 resurface when we discuss simulation-based Heston pricing. The chapter is almost entirely self-contained analytically, but pedagogically it is the payoff for Parts II and III.

12.1 Chapter map

1. **§10.1 Bivariate SDE** — motivation, specification, and what each parameter controls.
2. **§10.2 Feller condition** — when variance stays strictly positive.
3. **§10.3 Change of measure** — market price of variance risk, risk-neutral drift adjustments, uniqueness failure and calibration implications.
4. **§10.4 Implied volatility** — why constant-σ GBM is inadequate and what shapes a stochastic-vol model must generate.
5. **§10.5 Characteristic function** — Riccati ODE derivation for the log-price characteristic function.
6. **§10.6 Fourier-inversion pricing** — recovering P_1 and P_2 and assembling the call price.
7. **§10.7 Greeks and calibration** — delta, vega, the two extra vol-of-vol and correlation sensitivities, calibration workflow.
8. **§10.8 Variance swaps** — a pure-variance product that prices by expectation under $\mathbb{Q}$.
9. **§10.9 Worked example** — a numerical Heston vs. Black–Scholes comparison.
10. **§10.10–§10.10C Limitations, comparisons, extensions, and summary.**
11. **§10.11 Key takeaways** and **Appendix A** (reference formulas) plus **Appendix B** (intuition checklist).

12.2 10.1 Bivariate SDE — motivation

The Black–Scholes world is complete: a constant volatility σ plus the bank account span every contingent claim. Empirically, option markets violate this — implied vol exhibits a smile / skew across strike and a term structure across maturity. The Heston (1993) model is the canonical fix: promote the variance v_t to its own mean-reverting square-root SDE, correlate it with the spot, and retain semi-closed-form pricing via a characteristic function. This chapter rebuilds the derivation from first principles — bivariate SDE $\rightarrow$ Feller condition $\rightarrow$ change of measure with market price of variance risk $\rightarrow$ characteristic-function ansatz $\rightarrow$ Riccati ODEs $\rightarrow$ Fourier-inversion pricing (the P_1, P_2 probabilities) — and finishes with calibration / Greeks notes and variance swaps.

The post-1987 equity smile is the empirical motivation. Pre-crash surfaces were reasonably flat; the crash and the rapid revaluation of tail risk created the persistent left skew that defines equity vol to this day. Two responses emerged: Dupire's local-vol (exact surface fit but unrealistic forward smile) and Heston (independent variance SDE with realistic forward dynamics and a tractable characteristic function via the affine structure). Heston sits between local-vol and the modern rough-vol family as the pedagogical and practical baseline.

12.2.1 10.1.1 Why stochastic vol produces a smile

A Black–Scholes lognormal has a single scale parameter σ: the risk-neutral density of $\ln X_T$ is a tidy Gaussian with one variance. Under Heston the conditional density (conditioning on the whole path of v_t) is still lognormal — but the mixture over the v-path puts weight on very quiet and very loud regimes, producing fatter tails than a single lognormal. Fatter tails → deep OTM options are worth more than BS would suggest → inverted BS gives higher implied vols at the wings. The non-zero ρ then asymmetrises that mixture: when spot falls, the vol tends to spike (for $\rho < 0$), which loads the left tail more than the right. This is the skew. In short: α (vol-of-vol) controls the wings (curvature / smile); ρ controls the slope (skew); κ controls how quickly the term-structure of smile flattens; θ anchors long-dated ATM; v_0 anchors short-dated ATM.

Mixture intuition. Heston is Black-Scholes with a stochastic volatility "dial" v_t — mean-reverting toward θ, perturbed by $\alpha\sqrt{v}\,\mathrm{d}W^v$, and correlated to spot with ρ. The terminal distribution of $\ln F_T$ is a continuous mixture of lognormals indexed by the path of v; mixing fatness creates the smile, and the spot-vol correlation tilts it into a skew.

Heston is also the canonical *affine* model: drift and diffusion-covariance are linear in (F, v), so the characteristic function solves a finite-dimensional Riccati ODE system regardless of state-space dimension. The same affine structure underlies CIR short-rate models, Duffie-Pan-Singleton jump-diffusions, and CIR-style credit intensities.

12.2.2 10.1.2 FTAP recap and incompleteness

No arbitrage $\iff$ there exists a measure $\mathbb{Q} \sim \mathbb{P}$ such that all tradeables f_t/M_t are $\mathbb{Q}$-martingales. This holds always. The measure is unique only if the market is complete; in incomplete markets (like stochastic vol) there is no unique $\mathbb{Q}$.

FTAP I (Chapter 5) gives existence of $\mathbb{Q}$ iff no arbitrage; FTAP II gives uniqueness iff the market is complete. Heston has two risk factors (W^F, W^v) but only one tradable risk-bearing asset (F); the rank mismatch makes the market incomplete, and there is a one-parameter family of risk-neutral measures parameterised by the market price of variance risk. Calibration "identifies which $\mathbb{Q}$ is currently consistent with observed option prices"; the gap between $\mathbb{P}$-parameters (from returns) and $\mathbb{Q}$-parameters (from options) is the variance risk premium.

To make this concrete, consider what "hedging" even means in an incomplete market. In Black–Scholes, for any contingent claim $f(S_T)$, there exists a self-financing strategy (Δ_t, β_t) in spot and bank that replicates f exactly at every time and in every scenario — zero residual risk. You can therefore price f by the no-arbitrage principle: its price must equal the cost of constructing the replicating portfolio. In Heston, no such exact replication exists using spot and bank alone. If you run a delta-hedge based on spot only, you will be exposed to variance innovations; your hedged portfolio will have residual P&L proportional to the vega times the variance shock. The best you can do with spot alone is minimise some residual-risk metric — variance-of-P&L, say — but you cannot eliminate it. To eliminate it, you must add another hedging instrument that is sensitive to variance — typically another option — but then you need that option to be priced, which requires a model, which requires a $\mathbb{Q}$, and we are back to the choice problem.

The resolution in practice is to assume (or impose) a specific choice of $\mathbb{Q}$, calibrate it to observed option prices (so that the market's implied λ^v is respected by construction), and then perform hedging in a model-consistent way. This is called "hedging under the calibrated measure" and it is what every vol desk does implicitly. The hedges are not perfect — they are model hedges — and the residual P&L reflects model risk rather than pure incompleteness-risk. But the framework at least gives you a coherent way to think about hedging and risk.

There is a philosophical point here that is easy to miss. In a complete market, the "price" of a contingent claim is unambiguous — it is the replication cost. In an incomplete market, "price" is an economic concept, determined by demand and supply, risk preferences, and market frictions. The model gives you a framework for representing those ingredients compactly (via the choice of $\mathbb{Q}$), but the model does not determine the price on its own. This is why calibration, rather than pure derivation, is the central activity of a stochastic vol practitioner: the prices exist because people trade, and the model is merely the language in which those prices are expressed.

12.2.3 10.1.3 From Black–Scholes drift to the bivariate SDE

Under $\mathbb{P}$, a risky asset X_t that is traded satisfies

$$\mathrm{d}X_t = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}\tilde{W}_t^{\mathbb{P}}, \qquad \frac{f_t}{M_t} = \mathbb{E}^{\mathbb{Q}}\left[\frac{f_T}{M_T}\right], \tag{10.1}$$

and the $\mathbb{P}$-Brownian motion is linked to the $\mathbb{Q}$-Brownian motion by

$$\mathrm{d}\tilde{W}_t = \lambda_t\,\mathrm{d}t + \mathrm{d}W_t^{\mathbb{Q}}. \tag{10.2}$$

Substituting,

$$\mathrm{d}X_t = (\mu_t - \lambda_t\sigma_t)\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t^{\mathbb{Q}}, \tag{10.3}$$

where λ_t is the market price of risk. If X is itself traded, then under $\mathbb{Q}$ its drift must be rX_t, pinning down $\mu_t - \lambda_t\sigma_t = rX_t$. This is precisely the Girsanov shift derived in Chapter 5 — the Radon–Nikodym density $Z_t = \exp\!\left(-\int_0^t \lambda_s\,\mathrm{d}\tilde{W}_s^{\mathbb{P}} - \frac{1}{2}\int_0^t \lambda_s^2\,\mathrm{d}s\right)$ is what converts the $\mathbb{P}$-drift into the $\mathbb{Q}$-drift, state by state. The reader who wants the full derivation should consult Chapter 5 §5; we use the conclusion here.

Read this shift intuitively. Under the physical measure $\mathbb{P}$, the asset drifts at its expected return, which includes a risk premium over the risk-free rate — that is how risky assets compensate their holders for bearing risk. The change of measure does not change any actual path; it rebalances probability weight toward the states that have been systematically overweighted by risk aversion. Under $\mathbb{Q}$, the asset appears to drift at r because the premium has been re-absorbed into the measure. The same logic applies — but now in two dimensions — when we introduce stochastic variance.

Here is another way to see it. Imagine running Monte Carlo simulation of the spot under $\mathbb{P}$ and under $\mathbb{Q}$. The individual paths look similar — in fact, on any individual path, the realisation of the Brownian motion is the same (Girsanov tells us that the path spaces are essentially isomorphic). What differs is the weight that each path carries in the expectation. Under $\mathbb{P}$, the "good" paths (with high realised returns) have high probability; under $\mathbb{Q}$, those good paths are weighted less because risk-averse investors have bid up their prices (making them seem less attractive from the valuation standpoint). The effect is that $\mathbb{Q}$-expectations come out lower for linear payoffs on risky assets than $\mathbb{P}$-expectations, which is exactly the "risk-free rate" drift you would expect. None of this is about the trajectory of any particular path; it is all about the measure.

The famous economic interpretation is in terms of pricing kernels or stochastic discount factors. The Radon–Nikodym derivative $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ is essentially proportional to a representative investor's marginal utility — a number that is high in bad states of the world (when investors value an extra dollar most) and low in good states. Discounting payoffs by this kernel and taking a $\mathbb{P}$-expectation is equivalent to taking a $\mathbb{Q}$-expectation and discounting by the risk-free rate. Derivatives pricing in incomplete markets is thus fundamentally linked to asset-pricing theory: different measures correspond to different assumptions about the representative investor's risk aversion and preferences. When you calibrate Heston to market prices, you are implicitly inferring what marginal utility looks like in the vol dimension — at least in the part of the state space where liquid options exist to probe.

Futures. For a futures price F_t, the Black (1976) argument (Chapter 8) gives under $\mathbb{Q}$

$$\frac{\mathrm{d}F_t}{F_t} = \sigma\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \text{(Black model for futures prices)}. \tag{10.4}$$

Under $\mathbb{P}$,

$$\frac{\mathrm{d}F_t}{F_t} = \mu_t\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}. \tag{10.5}$$

We work in the futures frame partly out of convenience — futures are martingales under the risk-neutral measure with no drift to carry around — and partly because it highlights the key point that the volatility structure is what distinguishes Heston from Black, not the drift. The drift in the risk-neutral world is pinned down by arbitrage; the interesting degree of freedom is the stochastic behaviour of the diffusion coefficient.

The practical virtue of the futures-price framework goes beyond cleanliness. For index and equity markets, listed options actually expire on forward-contract-settled underliers or near-dated futures; for FX options, the relevant underlier is a forward rate; for commodity options, the entire pricing paradigm is built on futures rather than on any elusive "spot" price. So Black (1976) is not a toy simplification — it is the framework that actually prices the bulk of liquid options traded globally. Heston's original 1993 paper followed the Black tradition (although Heston expressed things in terms of spot with carry; the two are isomorphic), and the industry has largely standardised on the futures-frame parameterisation. When a practitioner says "$\mathrm{d}F/F = \sqrt{v}\,\mathrm{d}W$," they are stating the Heston–Black benchmark dynamics that underlie essentially every vanilla calibration you will encounter. Even when the underlying is a spot rather than a futures, the drift term $r - q$ (rate minus dividend yield) can be absorbed into the forward price, and the residual dynamics look like (10.4) with no drift. In other words: by working in the futures frame, you focus the analysis exactly on what makes Heston different from Black–Scholes — the diffusion $\sqrt{v_t}$ — and defer all drift-related complications to a thin wrapper around the core SDE.

12.2.4 10.1.4 The Heston bivariate system

We want to correct for stochastic volatility. Replace the constant σ by $\sqrt{v_t}$ and give v_t its own SDE:

$$\boxed{\frac{\mathrm{d}F_t}{F_t} = \mu_t\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^{\mathbb{P},F}} \tag{10.6}$$

$$\boxed{\mathrm{d}v_t = \kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - v_t)\,\mathrm{d}t + \alpha\sqrt{v_t}\,\mathrm{d}W_t^{\mathbb{P},v}} \tag{10.7}$$

with

$$\mathrm{d}W_t^{\mathbb{P},F}\cdot\mathrm{d}W_t^{\mathbb{P},v} = \rho\,\mathrm{d}t, \qquad (W^{\mathbb{P},F}, W^{\mathbb{P},v})\text{ correlated.} \tag{10.8}$$

Look carefully at the choice $\sqrt{v_t}$ for the diffusion coefficient. It is not incidental. Modelling in the variance rather than in the volatility pays two dividends. First, the variance is the natural additive quantity: variances of independent returns add, volatilities do not, and any equation that treats variance linearly will be cleaner than the equivalent vol-level equation. Second, the square-root CIR diffusion $\alpha\sqrt{v_t}\,\mathrm{d}W^v$ has the beautiful property that its diffusion coefficient vanishes as $v_t \to 0$, which acts as a soft floor: a variance that wanders toward zero stops diffusing and is pulled back up by the mean-reversion drift. This is how Heston constructs a process that stays non-negative under mild parameter conditions, without resorting to reflecting barriers or hard truncations.

Contrast this with the SABR alternative. SABR models the instantaneous vol (not variance) with lognormal dynamics: $\mathrm{d}\sigma_t = \alpha\sigma_t\,\mathrm{d}W^\sigma$. Lognormal vol automatically stays positive, which is appealing, but the tradeoff is that SABR dynamics are not affine, and SABR does not admit a closed-form characteristic function. Instead, SABR is typically priced via Hagan's asymptotic expansion — an approximation that works brilliantly for short maturities and ATM moneyness but degrades at wings and long dates. Heston gives up lognormal positivity in exchange for the affine structure, and the square-root floor on the CIR process is the mathematical trick that makes this exchange work. Either choice is defensible; both dominate naive alternatives like an arithmetic Brownian motion on vol (which can go negative and cannot be rescued by any floor).

A subtle point about (10.6): the spot is driven by $\sqrt{v_t}\,\mathrm{d}W^F$, not by $\sqrt{v_t}\,\mathrm{d}t$ plus $\mathrm{d}W^F$ with some scaling. The diffusion coefficient is a random function of v_t, which is itself stochastic. This is what makes the spot a "stochastic-volatility" process rather than a "time-changed" Brownian motion. Equivalently, you can think of spot's log-returns as Brownian motion scaled at each instant by $\sqrt{v_t}$: when v_t is high, returns are more volatile; when v_t is low, returns are calmer. The picture to keep in mind is a Brownian motion whose clock speed is itself random, fluctuating between slow and fast phases. This is not a metaphor — there is a formal equivalence between stochastic vol processes and time-changed Brownian motions, where the time change is the integrated variance. Ocone's representation of continuous local martingales makes this precise. For our purposes, it means that many results about Brownian motion carry over to Heston's log-price after a random time change.

Reading off (10.7):

- $\kappa^{\mathbb{P}}$ — mean-reversion rate of variance.
- $\theta^{\mathbb{P}}$ — mean-reversion level (long-run variance).
- α — vol-of-vol.
- ρ — spot/vol correlation (empirically negative for equity indices — the "leverage effect").
- The $\sqrt{v_t}$ diffusion is the CIR / square-root form: such processes are called Feller processes.

Each of these has an economic interpretation worth lingering over. The mean-reversion rate κ answers the question, "if volatility is currently elevated, how quickly does the market expect it to return to normal?" A large κ corresponds to a rubber-band variance that snaps back quickly — news-driven spikes fade within days. A small κ corresponds to a sticky variance that can stay elevated for quarters or years. The long-run level θ is what volatility reverts to — the "cruising altitude" of the market. For broad equity indices, θ calibrated over long histories tends to land in the territory of 20%–25% annualised vol, which is roughly the long-run average of realised equity vol. The vol-of-vol α governs how chaotic the variance itself is — whether the vol process is a gentle drift or a ragged, spiky trajectory. It is essentially a measure of how confident the market is in its current vol estimate; higher α means the market allows more probability mass on extreme vol regimes.

To calibrate these parameters against historical intuition: typical calibrated values for a liquid equity index are $\kappa \approx 2$–6 year^{-1}, giving a half-life of vol shocks of roughly one to four months; $\theta \approx 0.04$–0.06, corresponding to a long-run ATM vol of 20%–25%; $\alpha \approx 0.4$–1.0, varying with market regime; $\rho \approx -0.6$ to -0.8, stubbornly negative;

and v_0 matching whatever ATM implied vol is currently trading. These are broad ranges; different asset classes and different volatility regimes produce quite different calibrated values. Single-stock options typically show larger α and θ than index options because single names have more idiosyncratic vol. Sector ETFs fall in between. Currency pairs have much smaller α and less negative (sometimes near-zero) ρ. Commodities vary wildly by underlier; storable commodities like gold behave more like FX, while seasonal commodities like natural gas can exhibit seasonal patterns in θ itself.

The timescale interpretation of κ deserves a second look. Because κ has units of inverse time, the product $\kappa\tau$ is a dimensionless "time to mean revert." When $\kappa\tau \ll 1$ (short maturity relative to mean-reversion timescale), the variance has not had time to converge toward θ, and the option's pricing is dominated by the starting variance v_0. When $\kappa\tau \gg 1$ (long maturity), the variance has relaxed to its long-run mean θ, and the option's pricing is dominated by θ rather than v_0. The crossover between these two regimes happens around $\kappa\tau \approx 1$, which for $\kappa = 2$ is roughly six months. This is why short-dated and long-dated options carry quite different parameter-sensitivity fingerprints: short-dated options are sensitive to v_0 and α; long-dated options are sensitive to θ and κ. A good calibration uses this structural difference to separate the parameters cleanly.

The correlation ρ is perhaps the most interesting parameter. For equity indices it is strongly negative — typically in the range -0.6 to -0.8 after calibration — because of what practitioners call the leverage effect: as the stock price falls, the debt-to-equity ratio of the underlying firm rises mechanically, making the equity more levered and therefore more volatile. There is also a behavioural layer: falling markets trigger risk-off flows, forced deleveraging, and volatility spikes independent of any firm-level balance sheet. The result is that equity vol and equity spot move in opposite directions with remarkable consistency. This negative correlation is the engine that drives the equity smile's downward skew: puts become more valuable because the market attaches probability to the joint event of a falling price and rising vol, which is precisely the tail that out-of-the-money puts pay off on.

To understand the leverage effect quantitatively, consider a simple capital structure. A firm has assets with value V, debt with face value D, and equity with value $E = V - D$. The equity volatility is approximately $\sigma_E \approx \sigma_V \cdot V/E = \sigma_V \cdot (1 + D/E)$, where D/E is the debt-to-equity ratio and σ_V is the asset-value volatility. If equity falls by 20%, then (holding debt roughly constant) D/E rises by a factor of V/E minus one — which for a moderately leveraged firm could be 15% or more. That percentage rise in debt-to-equity translates directly into a rise in equity volatility. For the S&P 500 index, which aggregates hundreds of firms with heterogeneous leverage, the effect is diluted but still real. Empirically the leverage channel accounts for perhaps 10–20% of observed negative ρ; the rest is driven by risk aversion, volatility contagion, and forced deleveraging.

The behavioural and institutional mechanisms are probably more important in practice. Risk managers across the industry run VaR-based limits (Chapter 15); when vol rises, VaR rises, and limits are breached; breached limits force position cuts, which drive spot down; falling spot produces more vol, which produces more VaR breaches, and a feedback loop develops. This mechanism is the engine of volatility clustering — vol begets more vol — and it manifests as a highly negative ρ in stochastic vol models. There is also a purely psychological layer: traders fear losses more than they value gains (prospect theory), so they demand higher implied vols for OTM puts even absent any leverage-induced physical channel. All of these effects sum into the single observed $\rho \approx -0.7$ that calibrators typically report.

Interestingly, ρ is the most stable calibrated parameter across time. If you calibrate Heston daily to SPX over a long history, you will find that $(\kappa, \theta, \alpha, v_0)$ wander quite a bit as the vol regime changes, but ρ sits in a narrow band around -0.7 with surprisingly low variance. This structural stability reflects the deep economic persistence of the leverage effect: the mechanisms that make spot and vol negatively correlated do not go away, even as the magnitude of vol itself varies. For this reason, some practitioners anchor ρ to a fixed value during calibration and only fit the other four parameters. This imposes discipline on the optimiser and rules out spurious local minima with wrong-signed ρ.

For other asset classes, ρ tells different stories. In some commodity markets — particularly those with tight physical supply, such as natural gas in winter or crude oil during geopolitical tension — ρ can be positive: rising prices signal scarcity and volatility together, so the smile tilts the other way. In foreign exchange, ρ is often close to zero or modest in magnitude, reflecting the relatively symmetric nature of currency pair dynamics. This is not a detail; it means the Heston parameters are not universal constants, they must be calibrated per asset and per regime.

A few representative examples help build intuition. In FX, USD/JPY tends to show a slightly negative ρ — the yen strengthens (USD/JPY falls) during risk-off episodes, which often coincide with vol spikes — but the magnitude is modest, perhaps -0.2. EUR/USD tends toward $\rho \approx 0$ during normal times but swings wildly during crises when cross-asset correlations realign. GBP/USD exhibits Brexit-era shocks that briefly pushed ρ strongly negative. In rates, SOFR or Treasury yield vol is often priced with near-zero ρ in normal regimes but with strongly positive ρ in inflation-fear regimes (rising rates produce rising rate-vol). In equities, the single-stock vs index distinction matters: an individual high-beta tech stock might show $\rho \approx -0.5$ to -0.7, while an index like SPX averages across many names and ends up closer to -0.75. These patterns are not random; they reflect the underlying economic dynamics of each asset class.

It is worth stating explicitly that Heston's five-parameter structure imposes a functional form on the smile. The model cannot represent every conceivable smile shape. In particular, Heston smiles are monotonic in convexity as you move away from the skew — you cannot construct a Heston smile with a "W" shape, with two local minima or a kink at some particular moneyness. Real smiles occasionally show such features, especially at short maturities where event risk creates localised anomalies. When they do, Heston simply cannot fit them, and the calibrator will produce a best-fit smile that smooths over the irregularities. This is a structural limitation, not a bug, and it points toward the richer model families (local-stoch vol, jump processes, rough vol) needed to capture the full richness of observed smiles.

12.3 10.2 Feller condition

The variance process (10.7) can hit zero. For v_t to be strictly positive (i.e. to avoid hitting zero and getting stuck / reflecting) we need

$$\boxed{2\,\kappa^{\mathbb{P}}\,\theta^{\mathbb{P}} \;\geq\; \alpha^2} \tag{10.9}$$

the Feller condition. Intuition: $2\kappa\theta$ is twice the pull back toward the mean at the origin (the drift at $v = 0$ is $\kappa\theta$), while α^2 is the diffusion push that drives v around. When the pull dominates, the origin is an entrance-only boundary and v_t is > 0 almost surely. When the push dominates, v_t can touch zero (and in continuous time will, repeatedly). In practice calibrated equity parameters often violate it mildly; simulation schemes then use full-truncation (take $v^+ = \max(v, 0)$ in the diffusion coefficient) or reflection. Closed-form pricing is unaffected — the characteristic function is analytic in the parameters regardless of Feller.

Boundary classification. For the CIR variance, the origin is an *entrance* boundary (reachable from outside but not from inside) when $2\kappa\theta \geq \alpha^2$, and *regular* (reachable, requires a boundary rule) otherwise. The stationary density of v_t is gamma-distributed with shape $2\kappa\theta/\alpha^2$ — bounded at the origin if Feller holds, divergent (but integrable) like $v^{2\kappa\theta/\alpha^2-1}$ if violated.

Calibrated Heston typically violates Feller mildly. The market demands more wing richness than a Feller-compliant calibration can deliver, so the calibrator cranks α across the boundary. Closed-form pricing is unaffected; Monte Carlo simulation needs careful handling.

Simulation hierarchy (fastest/crudest to slowest/most accurate):

1. **Euler with full truncation** (Lord–Koekkoek–Van Dijk 2010): clamp v at zero in the diffusion coefficient. Linear-in-Δt bias; for daily steps the ATM bias is typically under 0.5%.
2. **Quadratic-Exponential (QE)** (Andersen 2007): approximates the non-central chi-squared transition density by a quadratic-of-normal in the high regime and a normal-plus-point-mass in the low regime. 2–3× slower than Euler, ~10× more accurate. Production workhorse.
3. **Exact (Broadie–Kaya 2006):** draw directly from the non-central chi-squared transition. Zero bias but expensive; reserve for benchmarking.

For vanillas, the Fourier closed form skips MC entirely.

Feller boundary in (kappa, alpha) plane *Feller boundary $\alpha = \sqrt{2\kappa\theta}$ for four long-run variance levels. SPX-like calibrations sit slightly above — a textbook mild violation.*

12.4 10.3 Change of measure — market price of variance risk

Going from $\mathbb{P}$ to $\mathbb{Q}$, (10.6) loses its drift (futures are martingales),

$$\frac{\mathrm{d}F_t}{F_t} = \sqrt{v_t}\,\mathrm{d}W_t^F, \qquad \text{corr. } (\rho). \tag{10.10}$$

For the variance, write the Girsanov shift — the multi-dimensional version derived in Chapter 5 §5.4, specialised here to two correlated Brownian motions:

$$\mathrm{d}W_t^F = \frac{\mu_t}{\sqrt{v_t}}\,\mathrm{d}t + \mathrm{d}W_t^{\mathbb{P},F}, \qquad \mathrm{d}W_t^v = \lambda_t^v\,\mathrm{d}t + \mathrm{d}W_t^{\mathbb{P},v}. \tag{10.11}$$

Then the $\mathbb{Q}$-dynamics of variance become

$$\mathrm{d}v_t = \Big(\kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - v_t) \;-\; \alpha\,\lambda_t^v\sqrt{v_t}\Big)\mathrm{d}t \;+\; \alpha\sqrt{v_t}\,\mathrm{d}W_t^v, \tag{10.12}$$

with λ_t^v the market price of variance (vol) risk. Note that the two-dimensional Girsanov we are using is the vector-valued version: we apply independent density processes to two correlated Brownian motions by first rotating into an independent pair via Cholesky, shifting each, and rotating back. The net effect on the correlated pair (W^F, W^v) is recorded in (10.11); the correlation ρ is preserved by the change of measure, as the Girsanov shift is a pure drift adjustment and does not affect the quadratic covariation.

Here is the crucial point. The drift of variance under $\mathbb{P}$ is whatever we think it is empirically — it reflects the long-run mean of realised variance, the speed at which variance reverts after shocks, and so on. But for option pricing, we do not price under $\mathbb{P}$; we price under $\mathbb{Q}$. And $\mathbb{Q}$ attaches a different drift to the variance process, because market participants demand (or offer) a premium for bearing variance risk. Empirically, buyers of variance — people long volatility through options or variance swaps — pay a premium on average, meaning realised variance comes in lower than implied variance on average. This is the variance risk premium, and it is captured in the model by λ^v. A negative λ^v for equity markets would reflect the fact that long-vol positions earn a negative premium — they are hedges, and hedges cost something. The variance risk premium is persistent, documented over long samples, and is arguably the single most important empirical feature driving the wedge between historical and implied vol.

To put numbers on this: over long samples of SPX options, the average difference between implied variance (VIX^2-scaled) and subsequently realised variance is about 3–5 percentage points in annualised-vol units. In other words, if VIX is 20, realised vol over the next 30 days averages about 17. Variance-swap investors — the sellers of variance — earn this 3–5 point wedge on average, net of trading costs, and this is the explicit compensation for bearing tail-risk exposure. (The wedge is not risk-free, obviously; it can invert during crises, and the shorts can take large losses. Long-vol positions are attractive when vol spikes, which is exactly the "tail hedge" use case that dominates quant-institutional flows.) When we write Heston with a specific λ^v, we are implicitly asserting a particular functional form for this premium — one that is compatible with the CIR dynamics. Different asset classes and different regimes have different premium structures; Heston's affine-preserving choices capture the main cases cleanly.

A pedagogical note on terminology. The "market price of variance risk" is sometimes written λ^v and sometimes with other symbols; the economic quantity it represents is the Sharpe-ratio-like premium demanded per unit of variance-factor exposure. If the variance risk is priced (i.e. $\lambda^v \neq 0$), then the $\mathbb{Q}$- dynamics of v differ from the $\mathbb{P}$-dynamics, which means the risk-neutral expected path of variance differs from the physical expected path. Practitioners often summarise this by saying "implied vol is higher than expected realised vol on average, by the variance risk premium." That one sentence captures essentially the entire economics of the λ^v parameter.

Two natural choices keep the SDE affine (preserving Heston form):

Choice 1. $\lambda_t^v = c\sqrt{v_t}$. Then

$$\lambda_t^v \cdot \alpha\sqrt{v_t} = \alpha c\, v_t,$$

so the drift becomes

$$\kappa^{\mathbb{P}}\theta^{\mathbb{P}} - (\kappa^{\mathbb{P}} + \alpha c)\,v_t = \kappa\left(\frac{\kappa^{\mathbb{P}}\theta^{\mathbb{P}}}{\kappa} - v_t\right), \tag{10.13}$$

with $\kappa = \kappa^{\mathbb{P}} + \alpha c$ and $\theta = \kappa^{\mathbb{P}}\theta^{\mathbb{P}}/\kappa$.

Choice 2. $\lambda_t^v = \dfrac{\ell}{\sqrt{v_t}} + c\sqrt{v_t}$. Then

$$\lambda_t^v \cdot \alpha\sqrt{v_t} = \alpha\ell + \alpha c\, v_t,$$

so both level and slope of the mean-reversion shift: $\kappa = \kappa^{\mathbb{P}} + \alpha c$ and $\theta = (\kappa^{\mathbb{P}}\theta^{\mathbb{P}} - \alpha\ell)/\kappa$.

Notice what (10.13) achieves: the shape of the variance SDE under $\mathbb{Q}$ is algebraically identical to its shape under $\mathbb{P}$ — the same $\kappa(\theta - v)\,\mathrm{d}t + \alpha\sqrt{v}\,\mathrm{d}W$ form — with (κ, θ) reshuffled. This is why Heston is so clean. A wrong choice of λ^v (say, a constant λ) would break the affine structure: the change of measure would contribute $\alpha\lambda\sqrt{v}$ to the drift, which is not linear in v, and the Riccati tractability would collapse. The two "affine-preserving" choices above are the ones that respect Heston's algebraic skeleton, and they cover the usual practitioner parametrisations.

This is a subtle but deep point. In most finance contexts, the choice of λ^v is driven by economic or statistical arguments: we want the risk-neutral measure that best fits observed data, or that has some desired theoretical property. In Heston, there is an additional computational constraint: only certain functional forms of λ^v preserve the affine structure, and only affine structure gives us the closed-form characteristic function. If we chose a λ^v that broke affine-ness, the Riccati ODEs would become general nonlinear ODEs (without closed form), the characteristic function would require numerical PDE solving at every evaluation, and the computational cost of Heston pricing would increase by two or three orders of magnitude. For this reason, the affine-preserving λ^v choices are not just theoretically elegant — they are computationally essential. They are the reason we can calibrate Heston to a full vol surface in seconds rather than hours.

Another way to see this: the "Heston model under $\mathbb{Q}$" and the "Heston model under $\mathbb{P}$" are two instances of the same algebraic family, parameterised differently. Pricing derivatives uses the $\mathbb{Q}$-instance; estimating historical dynamics uses the $\mathbb{P}$-instance; the affine-preserving change-of-measure provides the bridge between them. If you wanted to do joint estimation — fitting both physical time-series data and risk-neutral option data simultaneously — you would use this bridge to write down the joint likelihood, with λ^v as the parameter linking the two. Some sophisticated calibration frameworks do exactly this, treating the variance risk premium as a directly estimable parameter. Most practitioner implementations, however, calibrate under $\mathbb{Q}$ only and infer the risk premium ex post by comparing calibrated parameters to historical estimates.

From the option trader's perspective, this means something striking. The risk-neutral $(\kappa, \theta, \alpha, \rho, v_0)$ that we calibrate from option prices are not the same as the physical-measure $(\kappa^{\mathbb{P}}, \theta^{\mathbb{P}}, \alpha^{\mathbb{P}}, \rho^{\mathbb{P}}, v_0^{\mathbb{P}})$ that we might estimate from historical returns. The two sets of parameters differ precisely by λ^v, the variance risk premium. Typically $\theta > \theta^{\mathbb{P}}$ — the risk-neutral long-run variance is higher than the historical long-run variance — because buyers of variance swaps must be compensated. The gap is the variance risk premium in disguise, and it is paid for by the option buyer and collected by the option seller.

The implications for trading are concrete. A strategy of systematically selling variance (selling options, selling variance swaps, running a "vol carry" book) earns the premium on average but takes on significant tail risk: in a vol spike the P&L can be punishingly negative. The Sharpe ratio of such a strategy is positive but not spectacular — typically in the range 0.5 to 1.0 for liquid markets over long samples — and the drawdowns are deep. LTCM's 1998 blowup, the 2018 "Volmageddon" that wiped out XIV, and the March 2020 vol spike are all examples of this strategy's tail risk materialising. Understanding the origin of the premium via Heston's λ^v helps demystify the strategy: the premium is compensation for exactly the tail exposure that occasionally bites hard. There is no free lunch; just a structural risk-premium for those willing to bear the risk.

It is also worth noting that the risk-neutral α can differ from the physical $\alpha^{\mathbb{P}}$, and the risk-neutral ρ can differ from $\rho^{\mathbb{P}}$, depending on the precise form of λ^v (and on whether there is a price of correlation-risk in addition to vol-risk). In the simplest affine-preserving choice (Choice 1 above), α and ρ are preserved across measures; only κ and θ shift. In richer parameterisations, all four can shift. Practitioners usually take the simplest case and then check empirically whether the calibrated α and ρ are consistent with time-series estimates. Significant discrepancies suggest either model misspecification or a richer risk-premium structure than the simple model captures.

Under $\mathbb{Q}$ (market price of risk absorbed, with Heston's canonical choice), we get the standard Heston system:

$$\boxed{\frac{\mathrm{d}F_t}{F_t} = \sqrt{v_t}\,\mathrm{d}W_t^F} \tag{10.14}$$

$$\boxed{\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \alpha\sqrt{v_t}\,\mathrm{d}W_t^v} \tag{10.15}$$

with $\mathrm{d}W_t^F \cdot \mathrm{d}W_t^v = \rho\,\mathrm{d}t$.

From here on, $(\kappa, \theta, \alpha, \rho, v_0)$ are the five risk-neutral Heston parameters.

Five parameters is a modest parameter count for a model that fits an entire implied vol surface. By comparison, some practitioner "local vol" or "local-stochastic vol" models carry dozens or even hundreds of parameters. The minimalism of Heston is one of its virtues: each parameter has a clean economic meaning, calibration is a well-posed optimisation in low dimension, and the fitted parameters can be compared across dates and assets without the apples-to-oranges problem that plagues over-parameterised models.

The parsimony has a second benefit that is sometimes overlooked: generalisation. A five-parameter model fitted to (say) ATM vols and the 25-delta risk reversal and the 25-delta butterfly at a single maturity has four effective calibration targets and five parameters, so the fit is nearly exact. But those same five parameters then extrapolate — via the Heston dynamics — to every other strike and every other maturity. The extrapolation is not arbitrary; it is dictated by the structural form of the Heston SDE. This means the model can price exotic and off-the-grid strikes in a way that is internally consistent with the calibration points, and that is one of the reasons Heston is a favourite choice for pricing mildly exotic products like barriers, forward-starts, and cliquets. A dense local-vol model with hundreds of parameters can fit the grid exactly, but extrapolates poorly off the grid because there are no dynamics constraining what happens between quoted points. Heston, with five parameters, is forced to make structural assumptions that constrain the extrapolation — often to the benefit of the exotic pricing.

Compare this to regime-switching models, which can have many parameters per regime (shape of transition matrix, regime-specific vols, regime-specific jump intensities) and easily bloat to 15+ parameters. Those models offer richer dynamics but are less stable across calibrations and harder to interpret. Heston's five-parameter structure sits in the sweet spot: rich enough to capture the major features of vol dynamics (mean reversion, vol-of-vol, leverage), parsimonious enough to calibrate cleanly and be interpreted economically. This sweet spot is a big part of why Heston has endured as the industry default.

12.5 10.4 Implied volatility — why we need this model

If we price a call in the Heston world at (K,T) and invert Black–Scholes to back out the volatility that matches, we get the implied vol surface:

$$V^{\text{Heston}}(K,T) \;=\; V^{\text{BS}}(r,\, \sigma^{\text{impvol}}(K,T);\, K,T). \tag{10.16}$$

The Heston model produces a volatility smile in K and a non-trivial term structure in T. Graphically: out-of-the-money puts (low K) and calls (high K) price richer than BS, bending the flat BS line into a smile/skew. The PDF of $\ln X_T$ under Heston is fatter-tailed and skewed relative to the BS log-normal. Payoff $(X-K)_+$ is sensitive to those tails.

Implied vol as coordinate chart. Implied vol is a BS-formula bijection between price and a "vol unit"; the BS formula here is a quoting convention, not a pricing model. The Heston density of $\ln F_T$ is a continuous mixture of log-normals indexed by the integrated variance $I_T = \int_0^T v_s\,\mathrm{d}s$. Mixing fattens the tails (kurtosis $\approx 3+6\,\mathrm{Var}(I_T)/\mathbb{E}[I_T]^2$), and α drives the wings.

Skew from ρ. With $\rho < 0$, low-F_T scenarios are precisely the ones with high integrated variance — the mixture asymmetrises, the left tail fattens, the right tail thins. The smile evolves from symmetric (at $\rho = 0$) through equity-style skew ($\rho = -0.7$) to a pure downward line as $\rho \to -1$. The characteristic-function framework handles all of this implicitly without needing to write down the mixture explicitly.

Heston-implied IV smile *Heston-implied IV smile*

12.5.1 Case study (a): 1987 crash and the birth of vol skew

Pre-1987 BS-implied vols were essentially flat across strikes. Post-Oct 1987 the equity-index left skew became a permanent fixture. Heston with $\rho < 0$ is the simplest model that reproduces it.

Context. Through the 1970s and early 1980s, Black-Scholes flat-vol calibrations produced implied vols that were, for liquid index options, roughly flat across strikes — the small differences (a "smile" with slight bowing toward OTM strikes) were treated as a numerical curiosity arising from BS's lognormal misspecification. On 19 October 1987 the S&P 500 fell 20.5% in a single session; the SPX OEX-style listed put options that traded the following day repriced at radically different implied vols across strikes, with the deepest OTM puts trading at vols 5–10 points above ATM. Over the months following, the "skew" persisted at every observation; by 1990 it had become a permanent structural feature of equity index option pricing, distinct from the modest smile visible in single-stock or commodity option markets.

Mark Rubinstein's "Implied Binomial Trees" (Journal of Finance, 1994) is the canonical academic documentation of the post-1987 skew: he constructed binomial trees fitted to S&P 500 option quotes pre- and post-crash and showed the implied risk-neutral density of $\ln S_T$ shifted from the symmetric-fattened-tails of pre-1987 to a strongly left-skewed distribution post-1987. Subsequent work by Bates (1996), Bakshi-Cao-Chen (1997), and others quantified the skew's persistence across decades and across markets — wherever a 1987-style large downward move had occurred (or was perceived as possible), the permanent skew followed.

Through the chapter's math. Black-Scholes lognormality (Chapter 6, equation 6.12) imposes a *symmetric* distribution on $\ln S_T$ — the left and right tails have identical shape. To produce a skew, you must either (i) introduce jumps (Merton 1976, Bates 1996), or (ii) make volatility itself stochastic with a non-zero spot-vol correlation. The Heston specification (10.1) does the second: the variance process v_t is correlated ρ with the spot innovation, and with $\rho < 0$ the joint distribution skews exactly as the empirical post-1987 surface demands. The mechanism is captured by the mixture decomposition above: under $\rho < 0$, paths with low terminal F_T tend to have *high* integrated variance $I_T = \int v_s \, \mathrm{d}s$ (vol spiked as the spot crashed), so the conditional density of $\ln F_T$ given low F_T is fat-tailed — exactly what makes OTM puts richer than BS quotes. A typical 2025 SPX surface fits to $\rho \in [-0.80, -0.60]$ — that number is not a free parameter, it is the market's price of structural joint crash risk inherited from October 1987.

Lesson. The skew is not a model defect to be calibrated away — it is the market's *structural memory* of asymmetric tail risk. Every flat-vol BS calibration since 1987 has been systematically wrong on OTM puts (it under-prices them) and wrong on OTM calls (it over-prices them, slightly). The empirical case for stochastic volatility, and specifically for $\rho < 0$ in the Heston framework, is exactly the post-1987 fact that no symmetric model can reproduce a persistent left skew. Practitioners read ρ as a "leverage parameter" — the more leveraged the underlying (equity indices in markets where firms carry debt), the more negative ρ tends to be. Fixed-income, FX, and commodity surfaces exhibit much smaller (or zero) skew because their underlying processes do not have the same structural joint-crash dynamic. The lesson for this chapter is that ρ is not a fudge factor; it is the most information-dense parameter in the calibration, and a desk that locks ρ at a stale value is hedging a fundamentally different surface than the market is quoting.

12.5.2 10.4.1 Simulation scheme

Set $X_t = \ln F_t$. Itô (two-dimensional, Chapter 3 §3.10) gives

$$\mathrm{d}X_t = -\tfrac{1}{2}\,\gamma^2 v_t \,\mathrm{d}t + \gamma\sqrt{v_t}\,\mathrm{d}W_t^F, \tag{10.17}$$

(with $\gamma = 1$ for the pure Heston; we keep γ to track units). Euler-discretise with step Δt:

$$X_{t_n} - X_{t_{n-1}} = -\tfrac{1}{2}\gamma^2 v_{t_{n-1}}^+ \,\Delta t + \gamma\sqrt{v_{t_{n-1}}^+}\,\sqrt{\Delta t}\,Z_{1,n}, \tag{10.18}$$

$$v_{t_n} - v_{t_{n-1}} = \kappa(\theta - v_{t_{n-1}}^+)\,\Delta t + \alpha\sqrt{v_{t_{n-1}}^+}\,\sqrt{\Delta t}\left(\rho\, Z_{1,n} + \sqrt{1-\rho^2}\, Z_{2,n}\right), \tag{10.19}$$

where $Z_{1,n}, Z_{2,n}$ are i.i.d. $\mathcal{N}(0,1)$ and

$$v_t^+ \;=\; \max(v_t,\, 0) \tag{10.20}$$

is the full-truncation fix (Lord–Koekkoek–Van Dijk) that prevents $\sqrt{v}$ from going complex when Feller is violated.

The scheme as written has a number of subtle features worth appreciating. The two innovations Z_1, Z_2 are independent standard normals; the linear combination $\rho Z_1 + \sqrt{1-\rho^2}\, Z_2$ produces a normal with variance one that is correlated ρ with Z_1. This is the standard trick for sampling from a bivariate normal by Cholesky factorisation. The use of v^+ in both the drift and the diffusion of the variance update ensures that even if a previous step dragged v below zero, we do not propagate that negativity: the next step treats v as if it were zero, allowing the mean-reversion drift $\kappa\theta$ to lift it back into positive territory at the next time step. Full truncation is biased — it systematically underestimates the variance of v near zero — but the bias shrinks as $\Delta t \to 0$ and is widely accepted as the best trade-off between simplicity and accuracy for standard use cases.

A practical note on step size. For daily-step Heston simulation ($\Delta t = 1/252$), the full-truncation Euler scheme introduces bias on European option prices of a few basis points typically — perfectly acceptable for most risk-management purposes. For path-dependent options (barriers, Asians, forward-starts), the bias can be larger, and finer time steps or better schemes (QE) become necessary. For very short-dated options (days or less), $\Delta t = 1/(252 \cdot N)$ with N in the range 10–100 intraday sub-steps is often needed. The computational cost scales linearly with the number of steps, so the choice of N is a cost-accuracy tradeoff that should be benchmarked against the closed-form Heston price before committing to a production scheme.

A related practical note: use antithetic variates to reduce MC variance. The key insight is that Heston's dynamics are symmetric in $(Z_1, Z_2) \to (-Z_1, -Z_2)$ — flipping the signs of both innovations produces an equally-probable path.

Pricing the option on both the original and the flipped path and averaging the two gives a zero-bias variance reduction that typically cuts MC standard error by a factor of 1.5–2× at no extra computational cost (since the variance update for the flipped path is essentially free). This and other variance-reduction techniques (control variates based on the BS or Heston closed-form price, importance sampling) are standard tools developed in Chapter 9 and routinely used in production vol desks.

Yet another subtle but important consideration is the discretisation of the spot-variance coupling. In the Euler scheme as written, the spot update uses $v^+_{t_{n-1}}$ — the variance at the start of the interval — as the diffusion coefficient. A more accurate scheme would average the variance over the interval, e.g. $(v^+_{t_{n-1}} + v^+_{t_n})/2$, but doing so introduces an implicit coupling that requires iteration. Andersen's QE scheme handles this coupling more carefully; naive Euler does not. The result is a small bias in the joint spot-variance distribution that most implementations accept as the cost of simplicity.

Heston sample paths (S and sqrt(v)) *Heston sample paths (S and sqrt(v))*

The figure shows two kinds of dynamics happening simultaneously. The top panel traces the spot price, which looks at first glance like a standard diffusion: it wanders, drifts, has typical diffusion jaggedness. But the local volatility of that wander is not constant — compare periods where spot is oscillating gently with periods where it swings wildly. The bottom panel shows the source of that varying agitation: it plots $\sqrt{v_t}$, the instantaneous volatility. Periods where $\sqrt{v_t}$ spikes correspond to the rough episodes on the spot panel; periods where $\sqrt{v_t}$ is low produce the calm stretches above. Notice also the anticorrelation: when spot drops sharply, $\sqrt{v_t}$ tends to spike up, visible as the mirror-image texture between the two panels. This is exactly what $\rho < 0$ encodes, and it is the microstructural fingerprint of equity markets.

Look, too, at the mean-reversion visible in the bottom panel. The $\sqrt{v_t}$ trace is not a random walk — it oscillates around a roughly horizontal baseline (the square root of θ), with upward spikes that decay back toward baseline over a few months. This is the κ-rate relaxation in action. A random walk in $\sqrt{v}$ would drift further and further from any baseline; Heston's mean-reverting variance creates a "homing" tendency that keeps the long-run distribution bounded. Without mean reversion, the integrated variance $\int_0^T v\,\mathrm{d}s$ would grow too variable at long horizons, and long-dated option pricing would become unstable. Mean reversion is the mathematical device that keeps Heston well-behaved at all horizons.

A subtle but important visual feature is the asymmetry of vol dynamics. Real markets (and Heston with $\rho < 0$) exhibit "sharp spikes up, slow decays down" in vol — vol rises quickly during panics, falls gradually as calm returns. This is not a property of the CIR process by itself, which is symmetric in its innovation structure; it is a property of the joint (F, v) dynamics. When spot crashes, the coupled vol spike happens all at once; when spot drifts sideways, vol decays at the leisurely rate κ. The resulting asymmetry in the vol trace is a visible signature of $\rho < 0$, and it matches the empirical behaviour of VIX quite closely.

12.6 10.5 Characteristic function via Riccati ODEs

Define the conditional characteristic function of $X_T = \ln F_T$:

$$g_t \;\equiv\; \mathbb{E}^{\mathbb{Q}}_t\big[\,e^{\omega(X_T - K)}\,\big] \;=\; g(t, X_t, v_t), \tag{10.21}$$

for $\omega \in \mathbb{C}$ on some admissible strip. (Setting $\omega = i\phi$ gives the usual Fourier transform in X_T.) We seek a PDE for g and then an exponential-affine ansatz.

Here is why we bother with the characteristic function at all. The density $p(X_T)$ of the log-futures price under Heston does not have a closed form. It is a mixture of lognormals, indexed by the unobserved path of variance, and the mixture weights are themselves complicated. We have no hope of writing p down. The characteristic function, on the other hand, turns out to be elementary — a neat exponential-affine expression in the state — and it contains exactly the same information as the density. Every moment of the distribution can be read off from derivatives of the characteristic function at zero; every expectation of the form $\mathbb{E}[f(X_T)]$ can be computed by Fourier inversion, provided f has a well-defined Fourier transform. European option prices are linear functionals of the density, so the Fourier route is not only convenient — it is the natural one.

The philosophical shift from density-first to characteristic-function-first thinking is one of the most important re-orientations in modern derivatives pricing. Classical probability texts teach you to work with densities because densities are concrete — you can plot them, integrate against them, compute moments by direct integration. But densities for complicated processes are often impossible to write down in closed form, while characteristic functions for the same processes are often much simpler. This is not an accident; it reflects the fact that taking exponential expectations of Gaussian-linked processes produces relatively clean algebra, while writing down the density requires inverting that exponential expectation via Fourier inversion — which is a separate operation that may or may not yield a closed-form answer. For many practical problems, you do not need the density at all; you just need some expectation, and the characteristic function gets you there directly via Fourier inversion or via moment formulas.

Another way to see why characteristic functions are natural here: they are the multiplicative objects of probability theory. When you add two independent random variables, their densities convolve — a multiplicative operation in the characteristic-function domain via the inverse Fourier relation. When you scale or shift, the characteristic function transforms by simple algebraic operations. For stochastic processes that are linear combinations of independent increments, the characteristic function structure is essentially combinatorial. Heston's integrated variance is such a construction, and its characteristic function reflects this combinatorial structure cleanly.

12.6.1 10.5.1 Itô PDE for g

Applying two-dimensional Itô to $g(t, X_t, v_t)$ (Chapter 3 §3.10): the bivariate SDE under $\mathbb{Q}$ is (10.17) + (10.15) with $\mathrm{d}X \cdot \mathrm{d}v = \rho\, \alpha\, v_t\, \mathrm{d}t$. Then

$$\mathrm{d}g_t = \partial_t g\, \mathrm{d}t + \partial_x g\, \mathrm{d}X_t + \partial_v g\, \mathrm{d}v_t + \tfrac{1}{2}\partial_{xx} g\, (\mathrm{d}X_t)^2 + \tfrac{1}{2}\partial_{vv} g\, (\mathrm{d}v_t)^2 + \partial_{xv} g\, (\mathrm{d}X_t\, \mathrm{d}v_t). \tag{10.22}$$

Substituting the drifts and quadratic variations $(\mathrm{d}X)^2 = v\, \mathrm{d}t$, $(\mathrm{d}v)^2 = \alpha^2 v\, \mathrm{d}t$, $\mathrm{d}X\, \mathrm{d}v = \rho\alpha v\, \mathrm{d}t$,

$$\mathrm{d}g_t = \mathrm{d}t \left\{ \partial_t g + \left(-\tfrac{1}{2}\gamma^2 v_t\right) \partial_x g + \kappa(\theta - v_t)\, \partial_v g + \tfrac{1}{2}\partial_{xx} g\, \gamma^2 v_t + \tfrac{1}{2}\partial_{vv} g\, \alpha^2 v_t + \partial_{xv} g\, \rho\, \gamma\, \alpha\, v_t \right\} + \text{mart.} \tag{10.23}$$

By tower g_t is a $\mathbb{Q}$-martingale (this is the Feynman–Kac connection from Chapter 4: g is the conditional expectation of a terminal functional and therefore solves the Kolmogorov backward PDE associated with the generator of the bivariate diffusion), so the drift vanishes for all (t, x, v). Dropping $\gamma = 1$:

$$\boxed{\partial_t g + \left(-\tfrac{1}{2}v\right) \partial_x g + \tfrac{1}{2} v\, \partial_{xx} g + \kappa(\theta - v)\, \partial_v g + \tfrac{1}{2}\alpha^2 v\, \partial_{vv} g + \rho\, \alpha\, v\, \partial_{xv} g \;=\; 0} \tag{10.24}$$

with the terminal condition

$$g(T, x, v) = e^{\omega(x-K)}. \tag{10.25}$$

The PDE (10.24) is the Kolmogorov backward equation for the Heston two-factor diffusion. It says that g, viewed as a function of state (x, v) and running time t, is conserved in expectation along the Heston dynamics — its generator annihilates it. The structure of the equation is revealing: every coefficient of a second-derivative term is linear in v, and the drift coefficients are affine in v. This linearity in v is the algebraic fingerprint of the broader class of affine processes, and it is exactly what enables the exponential-affine ansatz below.

Affine processes are the grand generalisation of what we are seeing here. A diffusion is affine if its drift and covariance matrix are both affine (linear plus constant) functions of the state. The theory developed by Duffie, Pan, and Singleton in a series of papers in the early 2000s shows that every affine process has an exponential-affine characteristic function, with coefficients solving a system of generalised Riccati ODEs whose number equals the dimension of the state space. Heston is two-dimensional and therefore produces two ODEs (A for the constant, B for the coefficient of v; X does not contribute to the state-dependent drift because X's drift is affine in v, not in X). For higher-dimensional affine factor models — affine short-rate models with multiple factors (Chapter 12), multivariate stochastic vol models, affine credit models — the number of ODEs grows with dimension but the overall structure remains tractable. The technique of "write the PDE, plug in the affine ansatz, get a system of Riccati ODEs, solve the ODEs in closed form or by numerical integration" is a general recipe that applies throughout affine modelling.

A note on the mixed partial $\partial_{xv} g$: it appears in (10.24) because of the $\rho\alpha v$ quadratic-covariation term, which in turn comes from the correlation between the spot and variance Brownian motions. Without correlation ($\rho = 0$), the mixed-partial term vanishes, the PDE decouples into a spot equation and a variance equation that can be treated separately, and the characteristic function factorises into a product of a spot contribution and a variance contribution. With $\rho \neq 0$, the coupling is genuine and the exponential-affine form is the cleanest way to handle it. This is why Heston with zero correlation is "easy" — a relatively straightforward exercise in conditioning on the variance path — while Heston with nonzero correlation requires the affine machinery.

12.6.2 10.5.2 Affine ansatz and Riccati ODEs

Because the PDE is linear in v and the terminal is exponential-affine in x, try

$$g(t,x,v) \;=\; \exp\Big\{\,\omega(x-K) + A(\tau) + B(\tau)\,v\,\Big\}, \qquad \tau = T-t. \tag{10.26}$$

Compute the derivatives:

$$\partial_t g = -(A'(\tau) + B'(\tau)v)\,g, \quad \partial_x g = \omega g, \quad \partial_{xx} g = \omega^2 g,$$
$$\partial_v g = B(\tau) g, \quad \partial_{vv} g = B(\tau)^2 g, \quad \partial_{xv} g = \omega B(\tau) g.$$

Substituting into (10.24) and dividing by g:

$$-(A' + B'v) - \tfrac{1}{2} v\,\omega + \tfrac{1}{2} v\,\omega^2 + \kappa(\theta - v)\,B + \tfrac{1}{2}\alpha^2 v\,B^2 + \rho\alpha v\,\omega B = 0. \tag{10.27}$$

Separate the coefficient of v and the constant term:

$$[\text{const}]: \qquad -A'(\tau) + \kappa\theta\,B(\tau) = 0, \tag{10.28}$$

$$[\text{coef of } v]: \qquad -B'(\tau) + \tfrac{1}{2}\omega(\omega-1) - \kappa B(\tau) + \tfrac{1}{2}\alpha^2 B(\tau)^2 + \rho\alpha\omega B(\tau) = 0. \tag{10.29}$$

Pause and appreciate what just happened. A partial differential equation in two spatial dimensions plus time has been reduced to two ordinary differential equations in a single variable τ. The dimensional reduction is dramatic and not accidental: it is a direct consequence of the affine structure. The exponential-affine ansatz converted the PDE into an algebraic identity that had to hold for all v, and separating by powers of v produced one ODE for the constant term (A) and one ODE for the coefficient of v (B). This reduction generalises beyond Heston — it works for any affine diffusion in any dimension — and it is one of the cornerstones of modern affine jump-diffusion theory.

The reduction is a computational gift, not a theoretical accident. The original PDE (10.24) lives on a three-dimensional manifold (two state dimensions plus time), and solving it numerically by finite-difference or finite-element methods would require discretising that manifold — storing a grid of values and propagating them backward in time step by step. That is an expensive computation, and it must be redone for every change in parameters or initial conditions. The ODE system, by contrast, involves two scalar functions of one variable. Solving them numerically (or in our case, in closed form) is essentially free. The lesson is general: whenever you have affine structure, hunt for the exponential-affine ansatz, because it always reduces the computational complexity by a full dimensional order.

The method is sometimes called "transform analysis" and it is a cornerstone technique. It shows up in yield-curve modelling (Vasicek, CIR, and multi-factor affine term-structure models — see Chapter 12), in credit-risk modelling (affine intensity processes), in commodity modelling (Schwartz and two-factor Gibson–Schwartz models), and in non-Gaussian time-series modelling (affine volatility models, stochastic intensity Poisson processes). Anywhere you see an affine drift and covariance, the transform method applies. Mastering the Heston derivation equips you to derive closed-form characteristic functions in all these other settings as well. This is why educators spend so much time on Heston despite it being just one of many stochastic vol models — the technique is the point, not just the model.

Rewrite (10.29) in standard Riccati form:

$$\boxed{B'(\tau) = \tfrac{1}{2}\alpha^2 B^2 + (\rho\alpha\omega - \kappa)B + \tfrac{1}{2}\omega(\omega-1), \qquad B(0) = 0} \tag{10.30}$$

$$\boxed{A'(\tau) = \kappa\theta\,B(\tau), \qquad A(0) = 0} \tag{10.31}$$

Terminal matching at $\tau = 0$ gives $g(T,x,v) = \exp\{\omega(x-K)\}$, so $A(0) = B(0) = 0$, as stated.

The Riccati equation (10.30) is quadratic in B. Riccati equations are the standard nonlinear ODE you meet in control theory, linear-quadratic optimisation, and — here — affine pricing models. They have the pleasant property of being linearisable: a Riccati of the form $B' = aB^2 + bB + c$ can be transformed into a linear second-order ODE, which then admits closed-form solutions in terms of exponentials when the coefficients are constant. In our case a, b, c depend on ω but are constant in τ, so we get a clean closed-form answer below. Once $B(\tau)$ is in hand, $A(\tau)$ is a simple antiderivative: $A(\tau) = \kappa\theta \int_0^\tau B(s)\,\mathrm{d}s$.

The Riccati linearisation trick is worth knowing because it recurs. Set $B(\tau) = -U'(\tau)/(\tfrac{1}{2}\alpha^2\,U(\tau))$. Then B satisfies a Riccati if and only if U satisfies the linear second-order ODE $U'' - (\rho\alpha\omega - \kappa)U' - \tfrac{1}{4}\alpha^2\omega(\omega-1)U = 0$, which has constant coefficients and therefore admits solutions as exponentials $U = C_1 e^{r_+\tau} + C_2 e^{r_-\tau}$ with $r_\pm$ the roots of the characteristic polynomial. Back-substituting and applying $B(0) = 0$ produces the formula for $B(\tau)$ that we report below. This linearisation works for any quadratic Riccati with constant coefficients and is the standard technique in affine pricing theory.

A nuance: the Riccati ODE depends on ω (through the coefficients), so for each Fourier frequency ϕ we evaluate at $\omega = i\phi$, we get a different $B(\tau; i\phi)$ and $A(\tau; i\phi)$. The dependence on ω is analytic (the coefficients are polynomials in ω), so the resulting B, A are analytic functions of ω. This analyticity is what makes the Fourier inversion work — we can continue the characteristic function into the complex plane and deform contours as needed for numerical stability. It also means that the "little trap" of branch choices is a genuine issue at isolated points where the argument of a square root or logarithm can cross a branch cut; careful implementation avoids this.

12.6.3 10.5.3 Closed-form Riccati solution

Let

$$d(\omega) = \sqrt{(\rho\alpha\omega - \kappa)^2 - \alpha^2\,\omega(\omega - 1)}, \tag{10.32}$$

$$g_*(\omega) = \frac{\kappa - \rho\alpha\omega - d}{\kappa - \rho\alpha\omega + d}. \tag{10.33}$$

Then

$$B(\tau) = \frac{\kappa - \rho\alpha\omega - d}{\alpha^2} \cdot \frac{1 - e^{-d\tau}}{1 - g_*\,e^{-d\tau}}, \tag{10.34}$$

$$A(\tau) = \frac{\kappa\theta}{\alpha^2}\left\{(\kappa - \rho\alpha\omega - d)\,\tau - 2\ln\left[\frac{1 - g_*\,e^{-d\tau}}{1 - g_*}\right]\right\}. \tag{10.35}$$

(Heston's original form; the "little trap" of Albrecher et al.
recommends flipping to $\tilde{g} = 1/g_*$ for branch-cut stability. Either works numerically, both are solved analytically.)

The "little trap" is worth flagging for anyone who plans to actually implement these formulas. The complex square root $d(\omega)$ and the complex logarithm in $A(\tau)$ both involve branch choices. On the original form of the equations, the branch of the logarithm can jump as ω sweeps along the real axis of Fourier inversion, producing a discontinuity in A and, consequently, a nonsensical characteristic function. The remedy — routinely applied in production code — is to use an equivalent algebraic rewriting (with $\tilde{g} = 1/g_*$) that keeps the logarithm on a continuous branch. This is not a deep issue; it is a numerical housekeeping step. But forget it, and your Fourier integrals give garbage.

A useful debugging trick: evaluate $\varphi(\phi)$ at $\phi = 0$. Since φ is a proper characteristic function (a probability-weighted exponential), it must equal exactly 1 at the origin. If your implementation returns something other than 1 at $\phi = 0$ (or nearby, for numerical reasons), you have a bug — most likely a branch-cut issue. Also check that $|\varphi(\phi)|$ decays as $\phi \to \infty$; for Heston, the decay is exponential in ϕ^2 for large ϕ, and any implementation that produces a non-decaying or growing $|\varphi|$ is wrong. These two sanity checks — $\varphi(0) = 1$ and decaying magnitude — catch essentially all implementation bugs quickly.

Production implementations also often compute $\ln\varphi$ rather than φ itself, to avoid underflow at large $|\phi|$ where φ can be very small. Working in the log domain preserves accuracy throughout the Fourier integration. At the very end, one exponentiates to get the integrand of the Gil–Pelaez formula. This is a standard numerical trick that goes beyond just Heston; it is the right way to handle any exponential-family quantity in floating-point arithmetic.

So the Riccati system can be solved analytically — this is the punchline that makes Heston useful.

12.6.4 10.5.4 Why a characteristic function but not the density

The joint SDE (X_t, v_t) is affine: drift and diffusion covariance are linear functions of v. For affine state-space models, a theorem of Duffie–Pan–Singleton guarantees that $\mathbb{E}[e^{\omega X_T + u v_T}]$ is exponential-affine in the current state — i.e. the characteristic function $\varphi(\phi)$ solves a finite-dimensional ODE system. The density $p(X_T)$ is the inverse Fourier transform of φ and has no closed form — it is a mixture of log-normals indexed by the unobserved variance path. We can price options because European payoffs are linear functionals of the density that Fourier-invert cleanly; we cannot write down p.

This distinction — having the characteristic function but not the density — deserves a moment of reflection. In probability theory we usually reach for the density when we want to compute expectations. But the characteristic function carries exactly the same information, and in some cases it is the cleaner object. For Heston, the mixture structure of the density is messy enough that there is no analytical handle on it, but the exponential-affine form of the characteristic function exposes its structure cleanly. The fact that we can price options without ever seeing the density is not a work-around; it is a genuinely different way of organising the computation. Fourier methods are now standard across modern derivatives — Heston was one of the first models to exploit this elegantly, and the technique generalises to jump models, stochastic interest rates, Lévy processes, and affine factor models.

To appreciate how far the characteristic-function approach extends, consider the Bates model (Heston plus log-normal jumps in spot). The characteristic function of a Lévy process is exponential in τ by the Lévy–Khintchine formula, and for a compound Poisson jump it has a neat closed form in terms of the jump distribution. In Bates, the Heston characteristic function is multiplied by a Lévy jump contribution, yielding a still-exponential-affine form that the Fourier inversion handles as easily as pure Heston. In the SVJJ model (stochastic vol with jumps in both spot and variance), the same machinery works with one more ODE for the jump contribution to the variance process. In Carr–Madan–Yor (CGMY) Lévy models, the characteristic function is explicit and Fourier inversion is trivial. The entire zoo of modern derivatives pricing models is built around characteristic-function structure, and Heston was the breakthrough paper that made this architecture popular.

For the rough volatility models that have emerged in the last decade, the characteristic function is no longer exponential-affine — the state space is infinite-dimensional — but the Fourier-based pricing approach still works, with the characteristic function computed by different methods (Laplace transforms of fractional integrals, numerical quadrature in abstract function spaces). The spirit of the approach — inverting a characteristic function to price European options — has survived the shift to rough vol, even though the specific ODE structure has not. Heston's legacy is the framework; the specific model has been superseded in many applications but the pricing paradigm it established remains central.

Finally, the conditional characteristic function of $X_T = \ln F_T$ given $(X_t, v_t) = (x, v)$:

$$\boxed{\varphi(\phi;\, t, x, v) \;\equiv\; \mathbb{E}_t^{\mathbb{Q}}\big[\, e^{i\phi X_T} \,\big] \;=\; \exp\Big\{\, i\phi x + A(\tau; i\phi) + B(\tau; i\phi)\, v \,\Big\}} \tag{10.36}$$

with A, B from (10.34)–(10.35) evaluated at $\omega = i\phi$.

12.7 10.6 Fourier-inversion pricing — P_1 and P_2

A European call on the futures struck at K, maturity T, is

$$C(t, F_t, v_t) = e^{-r\tau}\, \mathbb{E}_t^{\mathbb{Q}}[(F_T - K)^+] = e^{-r\tau} \Big\{ \mathbb{E}_t^{\mathbb{Q}}[F_T\, \mathbf{1}_{F_T > K}] - K\, \mathbb{Q}_t(F_T > K) \Big\}. \tag{10.37}$$

Heston's trick: split into two "Black–Scholes-like" pieces,

$$C = e^{-r\tau}\, F_t\, P_1 \;-\; e^{-r\tau}\, K\, P_2, \tag{10.38}$$

where

$$P_2 = \mathbb{Q}_t(\, X_T > \ln K\,), \tag{10.39}$$

and P_1 is the same probability under the share measure $\tilde{\mathbb{Q}}$ (using F as numeraire), i.e.

$$P_1 = \tilde{\mathbb{Q}}_t(\, X_T > \ln K\,). \tag{10.40}$$

This decomposition mirrors the Black–Scholes formula (Chapter 6) exactly. In Black–Scholes, $C = F\, N(d_1) - K\, N(d_2)$, and the two Gaussian probabilities $N(d_1)$ and $N(d_2)$ are exactly the probabilities of $F_T > K$ under two different measures: the share measure (for $N(d_1)$) and the domestic risk-neutral measure (for $N(d_2)$). The two measures differ only by a numeraire change — F versus M — which is Chapter 5's two-numeraire switching specialised to this call-on- futures problem. In Heston, the same two probabilities show up, but they are no longer Gaussian; they are Heston-distribution tails, and we compute them by Fourier inversion from the characteristic function.

The deep reason this decomposition exists is a consequence of the call payoff structure, not of the pricing model. $(F_T - K)_+ = F_T \mathbf{1}_{F_T > K} - K \mathbf{1}_{F_T > K}$, and the two terms are expectations of (delta-function-weighted indicator) and (simple indicator) respectively. The first is the first moment of F_T conditional on exercise; under the share measure it becomes a pure probability. The second is a pure risk-neutral probability. So P_1 and P_2 are not Heston-specific objects — they are universal to any call-price formula, under any underlying dynamics. What Heston gives us is a computable form for these two probabilities via the characteristic function, which otherwise would have to be obtained by direct density integration (which is not feasible for Heston).

The two-measure framing also clarifies why Fourier inversion is so natural. Both P_1 and P_2 are CDFs of a random variable (the log-spot at maturity) under different measures. CDFs are integrals of densities, which are Fourier transforms of characteristic functions. The Gil–Pelaez formula lets us skip the density step: go directly from characteristic function to CDF via a single integral. This streamlines the computation substantially — we never need to invert the characteristic function to get a density, only to compute the CDF at the single point $\ln K$.

Each probability is recovered from the characteristic function by the Gil–Pelaez inversion:

$$\boxed{P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-i\phi \ln K}\,\varphi_j(\phi)}{i\phi}\right] \mathrm{d}\phi, \qquad j = 1, 2} \tag{10.41}$$

The Gil–Pelaez formula is one of those classical results that deserves to be better known. It says that the probability that a random variable exceeds a given level can be computed purely from its characteristic function via a single integral — no density required. The $1/2$ comes from the fact that an unbiased symmetric Brownian motion assigns equal probability to being above or below its mean; the integral corrects for the actual asymmetry of the distribution as encoded in the imaginary part of the characteristic function. Numerically, the integrand decays like $1/\phi$ modulated by the decay of $|\varphi_j(\phi)|$, which for Heston is Gaussian-like in ϕ and shrinks rapidly. Most of the contribution to the integral comes from $\phi < 50$ or so, and Gauss–Laguerre quadrature with 64 nodes is more than adequate.

The Gil–Pelaez formula was derived in 1951 in a different context — it was originally stated for probability distribution functions rather than for financial applications — and it sat relatively dormant until stochastic-vol modelling made it the workhorse of Heston pricing. Its derivation follows from a straightforward Fourier-inversion argument applied to the characteristic function. If you write $F(x) = \mathbb{P}[X \le x] = \int_{-\infty}^{x} p(y)\,\mathrm{d}y$ and express p as an inverse Fourier transform of φ, the order of integration can be swapped (with some care at the integration limits) to give the Gil–Pelaez expression. The complex exponential $e^{-i\phi x}/(i\phi)$ is the Fourier transform of the step function $\mathbf{1}_{y>x}$, so the formula is essentially "the probability that $X > x$ equals the expectation of the step function, which equals (by Parseval) the Fourier integral of the product of the characteristic function and the Fourier transform of the step."

A practical numerical point about the Gil–Pelaez integral: the integrand is oscillatory, which can cause convergence issues if you use naive uniform quadrature. Gauss–Laguerre quadrature handles the exponential decay well; alternative approaches include the COS method (Fang and Oosterlee), which uses Fourier cosine series expansions and can be even faster for Heston than Gauss–Laguerre. For production systems, the COS method is increasingly the default because it allows very efficient pricing of non-European payoffs (barriers, Bermudans) using the same characteristic function, via backward recursion in the COS coefficients.

Heston characteristic function Re/Im/|phi| *Re, Im and magnitude of* $\varphi(\phi)$ *for the base Heston parameters. The rapid decay in* $|\varphi|$ *is what guarantees the Gil–Pelaez integrand* $\mathrm{Re}[e^{-i\phi \ln K}\varphi_j(\phi)/(i\phi)]$ *is absolutely integrable and hence Fourier-invertible.*

with

$$\varphi_2(\phi) = \varphi(\phi;\, t, x, v), \qquad \varphi_1(\phi) = \frac{\varphi(\phi - i;\, t, x, v)}{\varphi(-i;\, t, x, v)}. \tag{10.42}$$

(Division by $\varphi(-i) = F_t$ normalises the density for the share measure. With $X_T = \ln F_T$, $\varphi(-i) = \mathbb{E}_t[e^{-i\cdot iX_T}] = \mathbb{E}_t[e^{X_T}] = \mathbb{E}_t[F_T] = F_t$ because F is a $\mathbb{Q}$-martingale in the futures frame, so the ratio $F_t/F_t = 1$ drops out cleanly inside the Gil-Pelaez integrand. In the spot frame under $\mathbb{Q}$ the analogous normaliser would instead be $S_t\, e^{r\tau}$, but we have chosen the futures frame throughout this chapter.)

The shift by $-i$ in φ_1 is the Esscher-transform fingerprint of the measure change to the share measure. Under the share measure, F is the numeraire, and densities get reweighted by $F_T/(F_t e^{r\tau})$. On the characteristic function, this reweighting translates exactly to a complex argument shift of $-i$. This is a general trick: any "forward-measure" or "share-measure" probability can be computed from the original characteristic function by shifting ϕ appropriately, without ever redoing the derivation under the new measure. The underlying mechanism is the same numeraire-change machinery introduced in Chapter 5: the Radon–Nikodym density process is $F_T/(F_t e^{r\tau})$, so multiplying by $e^{i\phi X_T}$ and renormalising amounts to shifting the Fourier argument by $-i$.

The Esscher transform is named after Swedish actuary Fredrik Esscher, who introduced it in 1932 in the context of insurance risk. It has deep connections to exponential tilting in statistics and to change-of-measure arguments in finance. The general idea: given a characteristic function $\varphi(\phi)$ of X, the tilted characteristic function $\varphi(\phi+i\alpha)/\varphi(i\alpha)$ corresponds to a new measure under which X's density is multiplied by $e^{\alpha X}$ and renormalised. For $\alpha = 1$ this is the share-measure in finance; for other α values it is the Esscher principle for pricing under a risk-neutral measure chosen to match a specific expectation. Whenever you see a characteristic- function shift by $\pm i$ or $\pm i\alpha$, there is a change of measure hiding in it.

Why is this useful? Because the Heston characteristic function was derived once, for $X = \ln F_T$, under the $\mathbb{Q}$ measure. The complex shift lets us re-use that one characteristic function to compute probabilities under a family of related measures (share measure, forward measure, etc.) at zero additional derivation cost. This is a significant computational benefit: the Heston characteristic function evaluation dominates the runtime of Fourier-inversion pricing, so being able to re-use it under multiple measures saves substantial work.

The Heston call price.

$$\boxed{C^{\text{Heston}}(t, F_t, v_t; K, T) \;=\; e^{-r\tau}\big[F_t\, P_1 \;-\; K\, P_2\big]} \tag{10.43}$$

Put is by put-call parity. One integral per probability, computed by Gauss–Laguerre / FFT / Carr–Madan in practice.

The "one integral per probability" line is important for performance. In a large calibration problem you might price several thousand options — every strike and maturity on a dense grid — and each one requires two Gil–Pelaez integrals. With 64 quadrature nodes per integral, that is over a million characteristic function evaluations per calibration iteration. This sounds expensive, but each evaluation is essentially a handful of complex exponentials, and the whole calibration completes in under a second on modern hardware. Carr–Madan FFT collapses the computation further by pricing an entire log-strike curve in a single FFT, dropping the cost per option to effectively zero. This is why FFT-based Heston calibration is the industry default.

The Carr–Madan idea is worth savouring. If you write the call price as a Fourier integral over strike, you notice that the integrand depends on strike only through $e^{-i\phi\ln K}$. This means that fixing the quadrature points in ϕ and then scanning strike is just a Fourier transform of a weighted characteristic function evaluated at those ϕ-points. The Fast Fourier Transform (FFT) accomplishes exactly this — it transforms a sequence of function values into a sequence of Fourier coefficients (or vice versa) in $O(N \log N)$ time, where N is the number of points. By sampling ϕ on a regular grid with appropriate spacing and performing an FFT, Carr and Madan produce the entire log-strike curve of call prices in a single pass. The implementation is elegant: choose an η damping, compute $\psi(\phi)$ at the grid of ϕ-points, FFT to get the log-strike curve, read off prices at desired strikes by interpolation. Total cost: one N-point FFT (a few milliseconds for $N = 4096$) regardless of how many strikes are needed.

Before Carr–Madan (1999), pricing a full smile for a single maturity required one Gil–Pelaez integral per strike per probability — typically 20 strikes × 2 probabilities × 64 quadrature nodes = 2560 characteristic-function evaluations per smile. Carr–Madan reduced this to a single FFT with 4096 characteristic-function evaluations, which priced the entire smile at once. The net speedup is modest on a single smile but compounds dramatically in calibration where you price thousands of options per iteration. Modern Heston calibration is Carr–Madan plus a Levenberg–Marquardt optimiser, and this combination reliably calibrates SPX to within basis-point RMS error in under a second.

Carr–Madan alternative. Dampen the payoff by $e^{\eta x}$ and transform:

$$C = \frac{e^{-\eta\ln K - r\tau}}{\pi}\int_0^\infty \operatorname{Re}\big[e^{-i\phi\ln K}\,\psi(\phi)\big]\,\mathrm{d}\phi, \qquad \psi(\phi) = \frac{\varphi(\phi-(\eta+1)i)}{\eta^2+\eta-\phi^2+i(2\eta+1)\phi}. \tag{10.44}$$

This is a single FFT over a log-strike grid — the workhorse for calibration.

The damping parameter $\eta > 0$ is a free knob that practitioners tune (usually $\eta \in [0.5, 2]$) to trade off between accuracy at different strikes. A larger η improves accuracy for deep out-of-the-money calls but hurts the ATM region; a smaller η is more uniform across strikes. Rule of thumb: $\eta = 0.75$ is a reasonable default for most equity calibrations. Pick a bad η and your far-OTM prices will be inaccurate to several percent, which is enough to throw off the wings of your calibrated smile.

The need for damping exists because the raw call-price integrand is not absolutely integrable as a function of ϕ — it has an $O(1/\phi)$ decay that is borderline for convergence. Multiplying by $e^{\eta\ln K} = K^\eta$ (equivalently, damping the payoff by $e^{\eta x}$ in the strike domain) shifts the decay to exponential and makes the integral convergent. The catch is that the damping shifts weight toward high-K regions — deep OTM calls — and away from low-K regions. Asymmetric damping is the price we pay for using a single-sided call-price transform. An alternative approach, sometimes called the "generalised FFT," uses a symmetric combination of call and put prices and eliminates the η dependence, but at the cost of more bookkeeping. Most practitioners stick with Carr–Madan and tune η for their typical strike range.

A related numerical note: the FFT log-strike grid spacing and the ϕ grid spacing are reciprocally related via the FFT Nyquist constraint. If you want strike resolution $\Delta \ln K = 0.01$ (roughly 1% moneyness resolution), you need ϕ grid extending to $2\pi/\Delta \ln K \approx 628$. With 4096 FFT points, the ϕ spacing is roughly 0.15, and the integration captures the characteristic-function contribution down to ϕ roughly equal to that spacing. For Heston, the characteristic function decays rapidly above $\phi \sim 10$, so these grid parameters produce negligible truncation error. Finer grids are used for deep OTM calibration or for improved accuracy in the extreme tails.

12.8 10.7 Greeks and calibration notes

Five risk-neutral parameters $(\kappa, \theta, \alpha, \rho, v_0)$ plus r. Each has a clean economic role:

Parameter	Role	Empirical sign (equity)
v_0	instantaneous variance	matches ATM level
θ	long-run variance	matches long-dated ATM
κ	mean-reversion rate	term-structure slope
α	vol-of-vol	smile curvature (convexity)
ρ	spot/vol corr.	smile skew (negative $\rho \Rightarrow$ left skew)

The parameter identification story is crisp, and this is one of Heston's greatest practical virtues. Different features of the implied vol surface map cleanly to different parameters, so the calibration problem is well-posed and the fitted parameters are stable across days. If the overall level of your vol surface rises by 2 points, v_0 and θ rise; if the skew steepens, ρ becomes more negative; if the wings puff out, α increases. Rarely does moving one parameter require compensating moves in another to stay on-surface, and when it does, the required compensation is modest.

Here is the quantitative version of the parameter-sensitivity story, useful for debugging calibration. The Heston ATM implied vol at maturity T is approximately $\sqrt{m_T}$ where $m_T = \mathbb{E}[v_T]$ follows the exponential relaxation formula. Differentiating ATM vol with respect to parameters: $\partial\text{ATM}/\partial v_0$ is nonzero but decays with T (short-dated ATM sensitive to v_0, long-dated not); $\partial\text{ATM}/\partial\theta$ grows with T (long-dated ATM tracks θ); $\partial\text{ATM}/\partial\kappa$ is the blend-rate sensitivity. The short-maturity ATM skew (slope of smile at the money) is approximately $\rho\alpha/(4\sqrt{v_0})$ — a *finite* constant as $T \to 0$ — so ρ and α both enter multiplicatively. The ATM convexity (curvature of smile) is approximately $\alpha^2 T/(v_0\kappa)$ for moderate T, so this is primarily an α sensitivity. These formulas are approximations, but they capture the dominant structure and are extremely useful for parameter identification.

When calibration is well-posed, the Jacobian matrix of calibration residuals with respect to parameters is well-conditioned, and Levenberg–Marquardt converges rapidly. When it is ill-posed, the Jacobian has near-zero eigenvalues corresponding to parameter combinations that are under-determined. A common sign of ill-posedness is α and ρ drifting together in a correlated way — indicating that the calibration cannot distinguish between them on the data. This is most likely to happen when only ATM quotes are available (no skew information) or when the maturity range is too narrow. The remedy is to include more smile points across strikes and more maturities in the calibration.

Calibration targets the smile, not the term structure. This is an important practitioner truth. Heston has a five-parameter structure and is trying to fit what is typically a two-dimensional surface — strikes across maturities. In practice, the model captures the smile (variation across strikes for a fixed maturity) much better than it captures the full term structure of the smile (how the smile changes with maturity). If you calibrate Heston to the full surface, you get parameters that are a compromise — they neither fit short-dated smiles well nor long-dated smiles well. The standard practitioner workflow is to calibrate to a single representative maturity (say three months), accept that shorter and longer maturities will be imperfect, and use a richer model (or a term-dependent extension) when the term structure matters. The flat-term-structure failing is one of Heston's blind spots, and we come back to it below.

The term-structure fit problem has a specific structure that is worth understanding. At very short maturity, Heston's smile is controlled mostly by v_0 (the starting variance) and the short-time asymptotics of the Riccati solution. The ATM vol matches $\sqrt{v_0}$ and the short-time ATM skew is $\rho\alpha/(4\sqrt{v_0})$ (Durrleman 2003; Gatheral 2006) — a bounded constant as $T \to 0$. Real short-dated smiles, in contrast, exhibit a skew that *grows* as maturity shortens (empirically like T^{-H} with $H \in (0, 1/2)$), so Heston *understates* the short-dated skew relative to market. At long maturity, the skew decays faster than $1/\sqrt{T}$ because the variance has time to mean-revert and average out the correlation effect. The full term-structure shape implied by Heston is "steep skew at short dates, flat skew at long dates" — which matches the qualitative pattern of real smiles but rarely the quantitative shape. Markets typically have "steep skew at short dates, less-steep but still-steep skew at intermediate dates, and flatter skew at very long dates" — a more gradual decay that Heston cannot match with constant parameters.

This is where term-dependent Heston comes in. Let κ, θ, α, ρ all be piecewise-constant functions of time, with breakpoints at the maturities of traded options. The characteristic function can still be computed (by integrating the Riccati ODE piecewise in time), and the model can now match the full term structure exactly. The downside is parameter proliferation — now you have $5 \times N$ parameters for N maturity buckets — and calibration becomes more delicate. But for desks that need to match long-dated exotic books to the traded surface exactly, term-dependent Heston is the workhorse. Some implementations go further and treat v_0 as a "state" while calibrating $\kappa(t), \theta(t), \alpha(t), \rho(t)$ as "parameter functions" — this maintains Markov consistency while adding flexibility.

Heston smile vs rho *Heston smile for $\rho \in \{-0.9, -0.5, 0, +0.5\}$, holding $(\kappa, \theta, \alpha, v_0)$ fixed. Negative ρ tilts the smile down-right (equity left-skew); positive ρ tilts it up-right (FX or commodity inverted-skew). The ATM point barely moves — ρ is a pure slope control.*

Notice how remarkably clean the rotation is: as ρ sweeps from negative through zero to positive, the smile pivots around its ATM point. This is the cleanest visual demonstration of what it means to say "ρ controls the skew." Notice also that the ATM level is essentially invariant — changing ρ by ± 0.3 moves the 25-delta implied vols meaningfully but leaves the 50-delta ATM vol essentially unchanged. This clean decomposition is what makes Heston calibration well-posed.

In the language of risk reversals, the 25-delta risk reversal — defined as the difference between 25-delta call vol and 25-delta put vol — is essentially proportional to ρ at fixed maturity. A risk reversal of zero corresponds to $\rho = 0$; a positive risk reversal (calls dearer than puts) corresponds to $\rho > 0$; a negative risk reversal corresponds to $\rho < 0$. The approximate proportionality coefficient is $\alpha/\sqrt{v_0}\sqrt{T}$, so at fixed α and v_0, the risk reversal decays as $1/\sqrt{T}$. Real markets show a similar pattern, which is part of why Heston matches the qualitative term structure of risk reversals reasonably well. The cleanness of the ρ-to-risk-reversal mapping is what lets traders calibrate ρ almost by inspection of the risk reversal quote for a representative maturity.

The visual symmetry of the ρ-sweep also demystifies what is sometimes called "sticky-delta" vs "sticky-strike" behaviour. In a Heston world with $\rho < 0$, the smile tilts downward to the right; when spot moves, the delta coordinates of the options shift, and the vols at fixed delta stay roughly the same (sticky-delta behaviour). Heston therefore naturally generates sticky-delta dynamics, which is the empirical behaviour observed in FX and less-systematically in equities. Models with different dynamics (pure local vol) tend to generate sticky-strike behaviour (vols at fixed strike stay the same), which is less empirically supported. This is an underappreciated advantage of stochastic vol over local vol for forward-smile modelling.

Heston smile vs alpha *Smile curvature grows with α. Small α (weak vol-of-vol) flattens the smile toward the BS line; large α lifts both wings — deep puts and deep calls both gain, symmetrically around the mild $\rho = -0.3$ tilt.*

The complementary picture for α: increase vol-of-vol and both wings lift together, preserving the skew that ρ encoded. Think of α as the "wingspan" parameter. A flat Black–Scholes smile corresponds to $\alpha = 0$ (deterministic variance). As α grows, the mixture of lognormals becomes more heterogeneous — some paths are much more volatile than others — and the resulting unconditional distribution has fatter tails, hence higher implied vols at the wings. In the limit of very large α (and Feller severely violated), the smile becomes so steep at the wings that the density takes on heavy-tail features approaching Student-t behaviour.

The 25-delta butterfly — the average of 25-delta call and put vols minus ATM vol — is the market quote that directly reflects α. A butterfly of zero means a flat smile, consistent with $\alpha = 0$. A positive butterfly means both wings are richer than ATM, which maps to $\alpha > 0$. In the approximate asymptotic formula, the butterfly scales roughly as $\alpha^2\kappa^{-1}$ with a mild T-dependence. The cleanness of this mapping means that, like ρ-to-risk-reversal, the α-to-butterfly mapping can be read off approximately by eyeballing the butterfly at a representative maturity.

When calibrators describe a surface with "rich wings," they mean a high butterfly. This maps to high calibrated α, which in turn triggers Feller-violation diagnostics. Hence the indirect but real connection between rich-wing smiles, high vol-of-vol, and Feller-boundary behaviour. Surfaces with near-flat wings (low butterfly, typical of calm FX markets) calibrate to $\alpha \approx 0.3$ or less and are comfortably Feller-safe. Surfaces with very rich wings (high butterfly, typical of stressed equity markets) calibrate to $\alpha > 0.7$ and may violate Feller aggressively. The relationship between market feature (butterfly), calibrated parameter (α), and diagnostic (Feller violation) is clean and traceable.

12.8.1 10.7.1 Greeks

Because $C = e^{-r\tau}(F\,P_1 - K\,P_2)$ and P_1, P_2 depend on v only through $B(\tau)\,v$ in φ:

$$\Delta_F = \frac{\partial C}{\partial F} = e^{-r\tau} P_1, \tag{10.45}$$

$$\mathcal{V}_v = \frac{\partial C}{\partial v_0} = e^{-r\tau}\left(F\,\frac{\partial P_1}{\partial v} - K\,\frac{\partial P_2}{\partial v}\right), \qquad \frac{\partial P_j}{\partial v} \text{ via differentiating (10.41) under the integral.} \tag{10.46}$$

Vega in the Heston sense is with respect to v_0, not BS σ.

This is a subtle point worth emphasising. When a practitioner says "vega," they typically mean Black–Scholes vega — the sensitivity of the option price to the Black–Scholes implied volatility. In Heston, the natural vega is with respect to v_0, the current instantaneous variance. These two vegas are related but not identical; they differ by a Jacobian that depends on the full set of Heston parameters. If your hedging is driven by Heston, you should hedge $\mathcal{V}_v$; if your hedging is driven by BS conventions (e.g. if you quote in BS vega terms), you can convert back via the chain rule. A deeper practitioner insight: in Heston, vega is only one of many vol-sensitivity parameters. You also have sensitivities to κ, θ, α, and ρ. Each of these is a degree of freedom in the model that can move, and a complete hedging programme would neutralise each of them — the so-called model-risk hedge, as distinct from the delta/vega hedge.

In practice, delta-vega hedging in Heston is close enough to BS-delta-vega hedging that most desks treat them interchangeably for vanilla books. The real divergence shows up for path-dependent products. Consider a barrier option. In BS, the barrier risk is hedged by matching the terminal payoff; in Heston, the barrier risk is coupled to vol dynamics because vol spikes can push spot through the barrier. A Heston-consistent barrier hedge uses not just delta and vega but also vanna (the cross sensitivity) and volga (the second vega). These higher-order Greeks are straightforward to compute in Heston via numerical differentiation of the characteristic-function price, and their management is a major focus of exotic desks. For a vanilla book, higher-order Greeks matter less but still exist; ignoring them is a source of hedge slippage over time.

A related concept: the concept of "Heston vega" as distinct from "BS vega" becomes important when you discuss the cost of a vega hedge. In BS, the vega hedge is obtained by buying or selling a liquid option to offset the position's BS-vega. In Heston, a vega hedge must offset the sensitivity to v_0, which depends on the Heston parameters. The quantitative difference is typically small but can be meaningful for large positions or for positions with significant vol convexity. Many production systems compute both BS-vega (for P&L attribution) and Heston-vega (for hedging) simultaneously and report both. The gap between them is the "model-convexity" adjustment.

One more consideration: in a Heston world, the natural vol risk to hedge is often the variance-swap risk, not the v_0 risk. The variance swap has a linear payoff in realised variance and provides a clean way to hedge the integrated-variance exposure that options carry. Many exotic desks run a hybrid hedging programme: delta on spot, v_0-vega on options (for short-term risk), variance-swap hedges (for long-term integrated-variance risk), and periodic rebalancing as the vol regime changes. This layered hedging approach is more robust than pure BS-vega hedging and is a direct consequence of thinking in Heston terms rather than BS terms.

12.8.2 10.7.2 Calibration workflow (practitioner)

1. Snap an implied-vol surface (liquid listed options).
2. Choose a loss $L = \sum_i w_i(\sigma_i^{\text{mkt}} - \sigma_i^{\text{Heston}})^2$ (IV RMS) or price-RMS.
3. Minimise by Levenberg–Marquardt / differential evolution. Start at $\rho < 0$, $v_0 \approx (\sigma^{\text{ATM}})^2$, $\theta \approx v_0$, $\kappa \in [1, 5]$, $\alpha \in [0.3, 1.0]$.
4. Check Feller; if violated, use full-truncation MC for path-dependent payoffs.
5. Re-calibrate daily; watch for ρ jumps (data artefact) or α blowups (ill-posed surface).

Generic practitioner notes (to move to Chapter 11): IV-RMS vs price-RMS weighting; basin-of-attraction starting values; multi-target calibrations per product class; time-smoothness and Bayesian regularisation; two-stage SVI-prefit-then-Heston pipelines; pre-event masking for known jumps. These are general non-linear-least-squares hygiene rules; nothing Heston-specific.

Heston-specific calibration notes. ρ and κ should be stable day-to-day in stable regimes; sudden jumps signal data or regime issues. The five Heston parameters trade off in well-known ways: (κ, θ) determine the term structure; (α, ρ) determine the smile shape per expiry; v_0 anchors the short end. Surface inability to simultaneously fit short and long dates is a Heston blind spot (§10.10), not a calibration failure.

Residual diagnostics: smile-shaped residual at short end + flat at long end → calibration is targeting long-dated; wing-positive/middle-negative residuals across maturities → α too small; put-negative/call-positive → ρ wrong sign.

12.8.3 Case study (b): March 2020 COVID — vol surface explodes from 12 to 80

VIX 14 → 82 in 18 trading days; the entire vol surface levitated. Heston re-calibration shows v_0 jumping 5–10×, ξ doubling. Production desks moved from 1× to 4–6× daily re-calibration intraday.

Context. Between Wednesday 19 February and Monday 16 March 2020, the VIX rose from 13.68 to 82.69 — its highest closing level on record at that time, surpassing the 80.86 close from the November 2008 GFC peak. The S&P 500 fell from 3393.52 (Feb 19) to 2480.64 (Mar 12, an 11-day sequence of −3% to −10% daily prints), and to 2304.92 (Mar 23 intraday trough). On the option side, the entire vol surface levitated: ATM implied vols on 30-day SPX rose from ~12% to ~70%; the persistent post-1987 skew (case study a) flattened materially as panic-driven OTM put buying narrowed the relative gap between ATM and the wings; longer tenors (1y, 2y) rose less than 30-day but still doubled or tripled. For options dealers (the Greek-letter sell-side: Goldman, Citi, JPM, MS, Susquehanna), every Heston-calibrated book was repricing daily by tens to hundreds of millions on the vega component alone.

The mechanical impact on a desk MTM was direct. A typical SPX vol-desk book runs a vega notional in the \$20–100M per vol-point range, computed across the surface. ATM vega P&L = $\Delta\sigma\times$ vega; a +5 vol-point day on a \$50M-vega book is a \$250M MTM mark. The first week of the COVID move (20 Feb – 28 Feb) saw some desks book \$200M+ in vega P&L per day from realised gamma alone (long-vol books) or from forced delta-rebalancing of structurally short-vol positions (short-vol books). The intraday MTM volatility was severe enough that the post-2020 review period at most major dealers tightened intraday risk-limit refresh cadence.

Through the chapter's math. Heston re-calibration through the Feb-March 2020 period shows three near-simultaneous parameter shifts. Pre-crisis (Feb 2020) calibration on a 30-day liquid SPX surface gave roughly $v_0 \approx 0.014$ (~12% vol), $\theta \approx 0.030$ (~17% long- run vol), $\kappa \approx 4$, $\xi \approx 0.55$, $\rho \approx -0.75$. Mid-March 2020 calibration on the same surface gave $v_0 \approx 0.16$ (~40% vol; up by a factor of ~10), $\theta \approx 0.060$ (~25%; up roughly 2×), κ roughly stable at ~ 3, $\xi \approx 1.2$ (roughly 2×), $\rho \approx -0.40$ (the skew flattened — case study a's "leverage parameter" measurably weakened in the panic, as ATM and OTM both bid). The Greeks the desk cared about were $\partial V/\partial v_0$ (short-end vega), $\partial V/\partial \xi$ (volga, vol-of-vol exposure), and $\partial V/\partial \rho$ (the skew Greek). For a typical $50M - vegaATMbook, thevolgaexposure($ ^2 V/ v_0, \$ in Heston language) was the second-largest Greek by P&L impact and was materially under-hedged on most desks pre-COVID.

Lesson. The five Heston parameters are not eternal constants — they are slow-moving state variables, and in genuine regime change they become *fast*-moving. Two operational consequences. First, the standard once-daily Heston re-calibration cadence (snap surface at the 4 PM close, fit, run risk on the new parameters until the next close) is inadequate in stress regimes; a vol surface that levitates 30 points between the open and the close has produced six-figure MTM moves on every single contract before the calibration catches up. Post-COVID production cadence at the major dealers moved to 4–6 intraday re-calibrations during stress periods (typically 9:45, 11:00, 13:00, 14:30, and 15:45 ET on US trading days), with bespoke event-driven runs around scheduled news. Second, the desk's risk limits should be expressed not just in vega but in the *full* Heston-Greek vector including volga and the skew Greek; a book that is vega-flat but volga- long can take a six-sigma loss on a vol-of-vol regime change with no warning from the daily VaR system. The lesson for the Chapter-10 reader is operational: the elegance of Heston's five-parameter parsimony is a feature in calm markets and a liability in regime change, and the production response is to instrument the parameter trajectories themselves as risk variables.

12.9 10.8 Variance swaps

A variance swap pays realised variance minus a strike at T. The realised variance is

$$\text{"realised variance"} \triangleq \sum_{n=1}^{N} \left(\frac{F_{t_n}}{F_{t_{n-1}}} - 1 \right)^2 = R, \tag{10.47}$$

pay-off at T:

$$\text{Pays } (R-K) \text{ @ } T. \tag{10.48}$$

A variance swap pays on total realised variability — "pure vol exposure," delta-neutral, vega-one. Demeterfi–Derman–Kamal–Zou (1999) showed the model-free log-contract replication via a static portfolio of options plus dynamic delta hedge, which is also the basis of the VIX formula. The 2008 crisis exposed jump-vulnerability of the replication, motivating capped variance swaps and the shift toward VIX-based vol products.

Use $\frac{F_{t_n}}{F_{t_{n-1}}} - 1 \approx \frac{\mathrm{d}F_t}{F_t}$ on a fine grid, so

$$R \;\simeq\; \sum_{n=1}^{N} \left(\frac{\mathrm{d}F_{t_n}}{F_{t_n}} \right)^2. \tag{10.49}$$

Now using (10.14) $\frac{\mathrm{d}F_t}{F_t} = \gamma\sqrt{v_t}\,\mathrm{d}W_t$ (with $\gamma = 1$),

$$R \;\simeq\; \sum_{n=1}^{N} \left(\gamma\sqrt{v_{t_{n-1}}}\right)^2 \left(W_{t_n} - W_{t_{n-1}}\right)^2. \tag{10.50}$$

Taking expectation, as $N \to \infty$, the Itô isometry (Chapter 3) turns the sum of squared Brownian increments into a time integral of the squared diffusion coefficient:

$$\mathbb{E}[R] \;\longrightarrow\; \mathbb{E}\left[\int_0^T \gamma^2 v_s \,\mathrm{d}s \right] = \gamma^2 \int_0^T \mathbb{E}[v_s]\,\mathrm{d}s. \tag{10.51}$$

The infinite-sampling limit gives expected realised variance = time-integral of expected instantaneous variance — a model-free identity that Heston makes closed-form.

12.9.1 10.8.1 Expected variance under Heston

Compute $\mathbb{E}[v_s]$ under the Heston SDE. From (10.15),

$$v_t - v_0 = \int_0^t \kappa(\theta - v_s)\,\mathrm{d}s + \alpha \int_0^t \sqrt{v_s}\,\mathrm{d}W_s^v. \tag{10.52}$$

Taking expectations (the Itô integral is a martingale, expectation zero — the martingale property of Itô integrals proved in Chapter 3 §3.7),

$$\mathbb{E}[v_t] - v_0 = \int_0^t \kappa(\theta - \mathbb{E}[v_s])\,\mathrm{d}s + 0. \tag{10.53}$$

Let $m_t \triangleq \mathbb{E}[v_t]$. Then

$$m_t - m_0 = \int_0^t \kappa(\theta - m_s)\,\mathrm{d}s, \tag{10.54}$$

$$\begin{cases} \dot{m}_t = \kappa(\theta - m_t), \\ m_0 = v_0, \end{cases} \qquad \Longrightarrow \qquad \boxed{m_t = v_0\, e^{-\kappa t} + \theta\,(1 - e^{-\kappa t})} \tag{10.55}$$

An exponential relaxation from v_0 to θ at rate κ.

Expected variance is a convex combination of v_0 and θ with weights decaying exponentially at rate κ; κ is the time-constant for variance-shock decay. The slope of the VIX term structure carries a clean Heston interpretation: $v_0 < \theta$ gives a rising (contango) VIX curve, $v_0 > \theta$ gives backwardation.

12.9.2 10.8.2 Fair variance strike

The fair strike K is the one making $\mathbb{E}[R - K] = 0$:

$$\boxed{K \;=\; \mathbb{E}[R] \;=\; \gamma^2 \int_0^T m_s\,\mathrm{d}s \;=\; \gamma^2 \left[\theta\, T + \frac{v_0 - \theta}{\kappa}\,(1 - e^{-\kappa T}) \right]} \tag{10.56}$$

Remarkably, the fair variance strike depends only on (κ, θ, v_0) — not on α and not on ρ. The vol-of-vol and correlation enter the distribution's higher moments but not the mean.

Variance swaps and options give complementary information: variance-swap term structure pins down (κ, θ, v_0) (mean dynamics, depending only on drift parameters); options then pin down (α, ρ) (shape, depending on diffusion). A two-stage calibration is more stable than a joint five-parameter fit.

VIX connection. VIX is the square-root of a 30-day variance-swap strike on SPX: $\mathrm{VIX}^2 \cdot (30/365) = \int_0^{30/365} m_s \,\mathrm{d}s$. VIX futures price forward expected integrated variance at longer horizons via (10.56); VIX options become options on m_T — still tractable via the characteristic function. In Heston, VIX dynamics are *endogenous* (determined by v_t through the exponential-decay kernel), so a single calibrated Heston can simultaneously price SPX vanillas, variance swaps, VIX futures, and VIX options as one consistent book.

12.9.3 10.8.3 Variance swap as tradable

Assume the swap itself is quoted with price dynamics

$$\frac{\mathrm{d}R_t}{R_t} = \sigma(v_t, t)\,\mathrm{d}W_t, \qquad R_0 = \hat{V}_0(T) \quad \text{— fair variance swap strike of } T\text{-maturity.} \tag{10.57}$$

Payoff $(R_T - K)_+$ at T on this tradable now becomes a standard option that can be solved analytically against the Heston $\hat{V}_0(T)$ curve.

This layered construction — model vanilla options with Heston, then use the fair variance-swap curve as a tradable to price options on variance — is how modern volatility desks handle the full suite of vol products. Variance swaps become inputs to the calibration; options on variance (like VIX options) become outputs of the calibrated model; and the whole system is internally consistent at the level of Heston dynamics.

One more connection to flesh out: the relationship between Heston and VIX drives a lot of the cross-market trading activity in the vol complex. If SPX smile calibration suggests one set of Heston parameters, and VIX futures suggest another, the inconsistency creates a trade: sell the over-priced side, buy the under-priced side, and wait for convergence. This "VIX-SPX basis" trade is a staple of vol arbitrage strategies, and it depends structurally on the Heston linkage between variance dynamics and VIX. Traders who understand the Heston framework can spot these inconsistencies cleanly; traders who do not rely on cruder heuristics. The Heston framework therefore serves not just for pricing but for relative-value analysis across the entire vol complex.

12.10 10.9 Worked example — Heston price vs. Black–Scholes

Setup. $F_0 = 100$, $r = 0$, $T = 0.5$, $v_0 = 0.04$, $\theta = 0.04$, $\kappa = 2.0$, $\alpha = 0.5$, $\rho = -0.7$. Feller: $2\kappa\theta = 0.16 \geq 0.25 = \alpha^2$ — violated (by a hair); use full-truncation sim, but closed form still fine.

Grounding these parameter choices: $v_0 = \theta = 0.04$ means the instantaneous vol starts at and reverts to $\sqrt{0.04} = 20\%$, a realistic equity-ATM-vol level. The mean-reversion rate $\kappa = 2$ gives a six-month half-life for vol shocks, again realistic for equity indices. The vol-of-vol $\alpha = 0.5$ is in the upper range of typical calibrated values and mildly violates Feller. The correlation $\rho = -0.7$ captures a strong leverage effect. These numbers produce a visible but not extreme equity smile at six-month maturity.

The Feller check is a useful sanity diagnostic. $2\kappa\theta = 2 \times 2 \times 0.04 = 0.16$, and $\alpha^2 = 0.5^2 = 0.25$. The ratio is $0.16/0.25 = 0.64$, so the condition fails by a factor of $1/0.64 \approx 1.56$ — a mild violation. In the stationary distribution, the variance process has a gamma density with shape parameter $2\kappa\theta/\alpha^2 = 0.64$, which means the density is infinite at zero. This is a signature of meaningful but not catastrophic Feller violation. Full-truncation MC will work; exact MC via non-central chi-squared will be slightly more accurate.

Note also that the choice $v_0 = \theta$ is deliberately clean for pedagogy. In real calibrations, v_0 rarely equals θ exactly; usually there is a meaningful gap that reflects whether the market is currently above or below its long-run baseline. If $v_0 = 0.06$ and $\theta = 0.04$ (currently stressed), the variance-swap curve slopes downward (backwardation); if $v_0 = 0.02$ and $\theta = 0.04$ (currently calm), the curve slopes upward (contango). The VIX term structure carries this information directly and is a useful input to calibration.

Step 1 — Riccati coefficients at $\phi = 1$ (illustration, for $\omega = i$):

$$d(i) = \sqrt{(\rho\alpha i - \kappa)^2 - \alpha^2\, i(i-1)} = \sqrt{(-0.35i - 2)^2 - 0.25\, i(i-1)}.$$

Compute $(-0.35i-2)^2 = 4 + 1.4i + 0.1225i^2 = 4 - 0.1225 + 1.4i = 3.8775 + 1.4i$. And $0.25\, i(i-1) = 0.25(i^2 - i) = -0.25 - 0.25i$. So

$$d(i)^2 = 3.8775 + 1.4i - (-0.25 - 0.25i) = 4.1275 + 1.65i.$$

Take the principal square root: $|d(i)^2| = \sqrt{4.1275^2 + 1.65^2} \approx 4.4451$, $\arg(d(i)^2) = \arctan(1.65/4.1275) \approx 0.3804$ rad; hence $|d(i)| = \sqrt{4.4451} \approx 2.1083$, $\arg d(i) \approx 0.1902$ rad, giving $d(i) \approx 2.070 + 0.399i$.

Step 2 — B, A at $\tau = 0.5$ follow (10.34)–(10.35).

Step 3 — P_1, P_2. Numerically integrate (10.41) via Gauss–Laguerre (64 nodes) for each strike.

Step 4 — compare. For ATM $K = 100$: Heston IV ≈ 0.200, matches v_0. For $K = 90$: Heston IV ≈ 0.224 vs BS-flat 0.200 — the skew. For $K = 110$: Heston IV ≈ 0.186 — thinner right tail due to $\rho < 0$.

Look at those three numbers in concert. The ATM implied vol matches $\sqrt{v_0}$, which is what we would expect: at the money, the option is so close to delta-neutral that the tail behaviour of the density does not affect its price much, and the Black–Scholes-comparable vol simply reads off the level of the Heston density's standard deviation. Ten percent out of the money on the put side (K = 90), implied vol has lifted by about 2.4 points — that is the combined effect of fatter tails (α contribution) and the leverage skew (ρ contribution). Ten percent out of the money on the call side (K = 110), implied vol has fallen by about 1.4 points — the fatter-tail effect is more than offset by the skew pulling right-side prices down. The asymmetry between +1.4 (down on calls) and +2.4 (up on puts) is the signature of $\rho = -0.7$: the skew dominates the smile on the downside.

To quantify this more carefully, consider the 25-delta risk reversal. At six months, with these parameters, the 25-delta put would be around K = 88 and the 25-delta call around K = 113. The implied vols at these strikes from Heston are approximately 23.0% and 18.8% respectively, so the risk reversal is 23.0 - 18.8 = 4.2 points. The 25-delta butterfly is [(23.0 + 18.8)/2 - 20.0] = 0.9 points. These numbers are close to what you would actually see on a six-month SPX surface on a typical trading day, so the choice of Heston parameters $(v_0, \theta, \kappa, \alpha, \rho) = (0.04, 0.04, 2, 0.5, -0.7)$ produces a realistic surface. This is not a coincidence — these parameters are broadly consistent with typical SPX calibration results.

A useful exercise is to perturb each parameter individually and observe the effect. Move ρ from -0.7 to -0.4: the put IV at K=90 falls from 22.4% toward 21.3% (less skew), and the call IV at K=110 rises from 18.6% toward 19.1%, narrowing the risk reversal from 3.8 to 2.2 points. Move α from 0.5 to 0.7: both wings lift, the ATM vol stays essentially put, and the butterfly rises from 0.9 to 1.6 points. Move κ from 2 to 4: the term structure of smile (how quickly it flattens with maturity) speeds up. Move θ from 0.04 to 0.06: the long-dated ATM rises, and the short-dated smile shifts up modestly because v_t is drifting toward a higher target. These sensitivities are the quantitative content of "what each Heston parameter does," and working through them by hand is the best way to internalise the model.

Variance-swap fair strike for this parameter set:

$$K_{\mathrm{VS}} = \theta\, T + \frac{v_0 - \theta}{\kappa}(1 - e^{-\kappa T}) = 0.04 \cdot 0.5 + 0 = 0.020 \quad (= 0.5 \times \theta,\ \text{since } v_0 = \theta).$$

Annualised vol: $\sqrt{K_{\mathrm{VS}}/T} = \sqrt{0.04} = 0.200$ — as expected when $v_0 = \theta$.

When $v_0 = \theta$, the variance process starts at its long-run level; there is no transient decay, and the expected integrated variance over $[0, T]$ is simply θT. Annualising by dividing by T recovers θ itself, and taking the square root gives the vol level $\sqrt{\theta}$. This is a useful sanity check: if your Heston calibration gives a variance-swap fair strike that disagrees with the term ATM vol, something is off. The two should agree when $v_0 \approx \theta$, and should deviate predictably when they differ.

Working through the numerics one more time: $T = 0.5$, $\theta = 0.04$, $v_0 = 0.04$, $\kappa = 2$. Formula gives $K_{VS} = \theta T + (v_0 - \theta)(1 - e^{-\kappa T})/\kappa = 0.04 \times 0.5 + 0 = 0.02$. Annualised: $K_{VS}/T = 0.04$; square root: 0.2. So the fair variance-swap strike implies a 20% annualised realised vol over the 6-month horizon, consistent with starting vol and target vol both being 20%. If the market were quoting a fair 6-month variance-swap strike of 22%, Heston would need a higher θ (or a higher v_0) to match. If it were quoting 18%, Heston would need lower parameters. Either way, the variance-swap curve is a clean target for θ and v_0 calibration.

For longer horizons, the convergence rate to θ becomes the critical feature. At $T = 2$ years, with $\kappa = 2$, the mean reversion is essentially complete ($e^{-\kappa T} = e^{-4} \approx 0.018$), so $K_{VS}/T \approx \theta$ almost exactly. At $T = 0.25$ years (three months), only partial mean reversion has occurred ($e^{-\kappa T} \approx 0.607$), so the fair strike reflects a weighted average of v_0 and θ. If $v_0 \neq \theta$, the weighting creates a specific term-structure shape that identifies κ from the curvature of the variance-swap curve. This is exactly how practitioners calibrate κ in practice when variance-swap markets are liquid enough to provide multiple maturities.

12.11 10.10 Heston's blind spots

Three structural limitations every practitioner should know.

1. Short-dated smiles are underfit. Pure-diffusion ATM skew is constant in T (the Heston short-maturity asymptotic is $\partial\sigma/\partial \ln K \approx \rho\alpha/(4\sqrt{v_0})$), but real equity surfaces show skew growing as $T \to 0$, often $T^{-1/2}$ or faster. This divergent short-end skew requires jumps (Bates, SVJJ) or rough vol (rBergomi, $H \approx 0.1$).

2. Full term-structure fit is impossible with 5 parameters. Heston fits one cross-section (say 3-month) well at the cost of others. Typical SPX calibration produces 50–100 bp RMS implied-vol error across a 160-quote surface, largest at the short (3m) and long (18m) ends. The remedy: piecewise/term-dependent Heston, or LSV (local $\times$ stochastic — exact surface fit + realistic forward dynamics, the exotics gold standard).

3. Skew and kurtosis are not quite orthogonal. ρ controls skew and α controls curvature, but they are coupled: changing α shifts ATM modestly, and changing ρ influences kurtosis. Some smile shapes — very high kurtosis with zero skew, strong skew with little convexity — are simply not in Heston's attainable five-parameter manifold.

Sub-limits. Constant ρ misses regime-dependent spot-vol correlation; constant α misses time-varying vol-of-vol; calibration is stable in calm regimes but parameters shift qualitatively in high-vol regimes (2008, 2020). Continuous re-calibration is the practical remedy.

Despite these limits, Heston remains the workhorse: closed-form CF, fast Fourier calibration, clean parameter interpretation, well-posed calibration. The model hierarchy — BS $\to$ local vol $\to$ Heston $\to$ LSV $\to$ Bates/SVJJ $\to$ rough vol — adds capability at the cost of complexity; Heston sits in the middle as the right default for vanilla pricing and hedging.

12.12 10.10A Heston vs. SABR vs. local volatility

Three smile-fitting frameworks dominate vanilla pricing.

- **SABR** (Hagan): $\mathrm{d}F = \sigma F^{\beta}\,\mathrm{d}W^F$, $\mathrm{d}\sigma = \alpha\sigma\,\mathrm{d}W^{\sigma}$, with ρ. Four parameters; Hagan's asymptotic closed-form makes calibration extraordinarily fast; β shapes skew via leverage, ρ via correlation, α controls curvature. No vol mean-reversion — poor long-dated term structure; extreme-strike arbitrage issues motivate shifted-SABR.
- **Local vol** (Dupire): $\sigma(S,t)$ deterministic, fit exactly to any arbitrage-free surface via $\sigma^2_{\mathrm{loc}}(K,T) = (2\partial_T C + 2(r-q)K\partial_K C)/(K^2\partial^2_{KK}C)$. Exact surface fit by construction, but forward smile flattens unrealistically — bad for forward-starts and cliquets. The "snapshot" model.
- **Heston** sits between: realistic forward dynamics (better than local vol), term-structure flexibility (better than SABR), clean parameter interpretation. Default for most vanilla applications; LSV (local $\times$ stochastic) is the exotics gold standard.

Production desks typically run multi-model pricing (Heston + Bates + LSV) and reserve model risk on positions where the three disagree.

12.13 10.10B Extensions and successors

- **Bates (1996):** Heston + Poisson jumps in spot. Three extra parameters; the characteristic function gets an additive $\lambda\tau(\phi_J - 1)$ term. Fixes too-flat short-dated smiles.
- **SVJJ (Duffie–Pan–Singleton 2000):** simultaneous spot + variance jumps. Captures crash dynamics beyond diffusive ρ.

- **Double Heston:** two independent CIR factors (fast + slow). Better term structure.
- **3/2 model:** $\mathrm{d}v = \kappa v(\theta - v)\,\mathrm{d}t + \alpha v^{3/2}\,\mathrm{d}W$. Fatter variance tails — improves VIX option pricing.
- **rBergomi / rough Heston:** fractional Brownian driver with $H < 0.5$. Matches short-end skew divergence. Non-Markovian, no closed-form CF.
- **LSV:** $\mathrm{d}S/S = L(S,t)\sqrt{v}\,\mathrm{d}W^S + \cdots$. Local-vol leverage on top of stochastic vol. Exact surface fit + realistic dynamics, but MC/PDE-only pricing. Gold standard for exotics.

Each extension addresses one Heston limitation; the practitioner's craft is knowing which is worth the added complexity.

12.14 10.11 Key takeaways

1. Heston replaces BS's constant vol (Chapter 6) with a mean-reverting CIR variance correlated to spot; this incomplete-market model has five risk-neutral parameters $(\kappa, \theta, \alpha, \rho, v_0)$.
2. The Feller condition $2\kappa\theta \geq \alpha^2$ keeps $v_t > 0$. Calibrated equity markets often sit near or below it; use full-truncation for MC.
3. The change-of-measure (Chapter 5 multi-dimensional Girsanov) absorbs the market price of variance risk λ^v; affine choices ($\lambda^v = c\sqrt{v_t}$ or $\ell/\sqrt{v_t} + c\sqrt{v_t}$) preserve Heston form and just reshuffle (κ, θ) between $\mathbb{P}$ and $\mathbb{Q}$.
4. The characteristic function of $\ln F_T$ is exponential-affine in (x, v): $\varphi = \exp\{i\phi x + A(\tau) + B(\tau)v\}$, with A, B solving a pair of Riccati ODEs that admit a closed-form solution.
5. Call prices are recovered by Fourier inversion (Gil–Pelaez): $C = e^{-r\tau}(F\,P_1 - K\,P_2)$, one integral per probability. Carr–Madan FFT is the production variant.
6. Heston reproduces implied-vol smiles and skews: ρ controls skew, α controls curvature, κ controls term-structure decay.
7. The fair variance-swap strike depends only on (κ, θ, v_0) — α and ρ do not enter the mean.
8. Heston's blind spots: short-dated smiles are underfit (need jumps), full term structure cannot be fit simultaneously (need time-dependent parameters), and skew-kurtosis decomposition is only approximately orthogonal.

The narrative arc of this chapter mirrors the historical arc of the derivatives-pricing literature. Black and Scholes gave us a complete, tractable pricing framework with one parameter; empirical data promptly broke the model's main assumption. Heston restored tractability at the cost of incompleteness, and for three decades served as the industry baseline. Rough vol and LSV models extend Heston to fit better, at the cost of tractability. The pattern is familiar: each generation of model adds realism and sacrifices something — either simplicity, or analytical closed form, or computational efficiency. The deep lesson is that there is no "correct" pricing model in an absolute sense; every model is a simplification of reality, and the choice of model depends on what simplifications are acceptable for the specific application. Heston is the classical compromise — rich enough to capture real phenomena, simple enough to implement, clean enough to reason about. Even as richer models proliferate, Heston's role as the canonical teaching example and the industry-default baseline endures.

A final reflection: the elegance of Heston is not just mathematical. It is conceptual. The idea that vol has its own life — its own mean-reversion, its own fluctuations, its own correlation to spot — is intuitively true, and every trader who has watched a vol spike can confirm it empirically. Heston is the minimal rigorous formalisation of this intuition, packaged in a form that preserves analytical tractability. That combination of conceptual cleanness and computational tractability is rare in quantitative finance, and it explains why Heston has endured even as more accurate models appear. Learn Heston well, and you will find that its scaffolding supports a surprising range of more sophisticated models, and that the intuitions you build about $(\kappa, \theta, \alpha, \rho, v_0)$ translate cleanly to the parameter spaces of those successors.

12.15 Appendix A — Reference formulas

This appendix collects the key formulas derived in the chapter for quick reference. Every formula here has been derived and explained above; this is the quick-lookup sheet for implementation.

Risk-neutral SDE.

$$\frac{\mathrm{d}F_t}{F_t} = \sqrt{v_t}\,\mathrm{d}W_t^F, \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \alpha\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}W^F\,\mathrm{d}W^v = \rho\,\mathrm{d}t.$$

Feller condition. $2\kappa\theta \geq \alpha^2$. When satisfied, the variance process is strictly positive almost surely. When violated, the origin is reachable and simulation schemes must handle it carefully; closed-form pricing via the characteristic function is unaffected by violation status.

Riccati ODEs.

$$B'(\tau) = \tfrac{1}{2}\alpha^2 B^2 + (\rho\alpha\omega - \kappa)B + \tfrac{1}{2}\omega(\omega - 1), \quad B(0) = 0,$$
$$A'(\tau) = \kappa\theta\, B(\tau), \quad A(0) = 0.$$

Closed form ($\omega = i\phi$):

$$d = \sqrt{(\rho\alpha\omega - \kappa)^2 - \alpha^2\omega(\omega - 1)}, \quad g_* = \frac{\kappa - \rho\alpha\omega - d}{\kappa - \rho\alpha\omega + d},$$

The quantities $d(\omega)$ and $g_*(\omega)$ are the two auxiliary functions through which the Heston closed form is most compactly expressed. Both are complex-valued for complex ω, and both require careful branch-cut handling when evaluated numerically. The practical rule of thumb: if your implementation produces non-decaying $|\varphi|$ as $\phi \to \infty$, the branch of d is wrong; if your implementation produces discontinuities in A as ϕ sweeps the real axis, the branch of the logarithm in A is wrong. Both errors are cured by the Albrecher-type rewriting that flips g_* and keeps the logarithm on a continuous branch.

$$B(\tau) = \frac{\kappa - \rho\alpha\omega - d}{\alpha^2}\,\frac{1 - e^{-d\tau}}{1 - g_* e^{-d\tau}},$$
$$A(\tau) = \frac{\kappa\theta}{\alpha^2}\left[(\kappa - \rho\alpha\omega - d)\tau - 2\ln\frac{1 - g_* e^{-d\tau}}{1 - g_*}\right].$$

Characteristic function. $\varphi(\phi) = \exp\{i\phi\, x + A(\tau; i\phi) + B(\tau; i\phi)\, v\}$.

This is the object you actually compute at each quadrature node during Fourier-inversion pricing. Every calibration iteration involves thousands of evaluations of φ; making it fast and numerically stable is essential to good calibration performance. Production implementations typically cache intermediate quantities (the square root d, the ratio g_*) so that only the parts that change with τ are recomputed, and pre-compute the Riccati solution at grid points for re-use across strikes.

Gil–Pelaez / Heston call. For a European call at strike K and maturity T, with the forward price F_t at time t and risk-free rate r:

$$P_j = \tfrac{1}{2} + \tfrac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-i\phi\ln K}\varphi_j(\phi)}{i\phi}\right] \mathrm{d}\phi,$$
$$C = e^{-r\tau}\,(F_t\, P_1 - K\, P_2).$$

Variance swap fair strike.

$$K_{\mathrm{VS}} = \theta\, T + \frac{v_0 - \theta}{\kappa}(1 - e^{-\kappa T}).$$

This formula is the Heston analogue of the VIX calculation. VIX itself is roughly $100 \times \sqrt{K_{\mathrm{VS}}(T = 30/365)/(30/365)}$, which, once you plug in typical Heston parameters, gives a reasonable approximation to observed VIX levels. The cleanness of this correspondence is one of the reasons Heston is so useful for joint SPX-VIX modelling. The forward variance curve — m_T as a function of T — is what VIX futures price, and the VIX futures term structure carries clean Heston-parameter signatures that a savvy trader can read at a glance.

Simulation (full-truncation Euler). The recommended baseline MC scheme; for higher accuracy use Andersen's QE scheme or exact non-central chi-squared simulation. With $v^+ = \max(v, 0)$,

$$X_n = X_{n-1} - \tfrac{1}{2}v_{n-1}^+\Delta t + \sqrt{v_{n-1}^+\,\Delta t}\, Z_1,$$
$$v_n = v_{n-1} + \kappa(\theta - v_{n-1}^+)\Delta t + \alpha\sqrt{v_{n-1}^+\,\Delta t}\,(\rho Z_1 + \sqrt{1 - \rho^2}\, Z_2).$$

End of Chapter 10. Chapter 11 takes up calibration: how Heston (and the rest of the model zoo) is fitted to market quotes day-to-day, what makes a calibration well-posed, and the L-curve / regularisation tradeoffs that decide whether a fit is usable for risk.

13 Chapter 11 — Calibration: Lattices, Short Rates, and the Yield Curve

13.1 11.1 FTAP on a Lattice

Pricing (Chapter 3–Chapter 5) maps parameters to prices; calibration inverts that map. Given market quotes — option prices, bond yields, swap rates — we pick the $\mathbb{Q}$-parameters that reproduce them. The procedure is an *inverse* problem with three classical pathologies (non-existence, non-uniqueness, instability), each tamed by a distinct tool (weighted least squares, regularisation, warm-start smoothing). This chapter develops the apparatus on lattices and bond curves; Chapter 12–Chapter 14 apply it to short-rate and stochastic-vol models.

13.1.1 11.1.1 Fundamental Theorem of Asset Pricing

The first tool we need is the theorem that connects market data to model structure. Its statement is deceptively simple, but every calibration procedure in this chapter is, at bottom, a specialisation of this single equation to a particular set of observed prices.

FTAP (restated from Chapter 2 / Chapter 5). *No arbitrage* $\iff$ *there exists an equivalent martingale measure* $\mathbb{Q} \sim \mathbb{P}$ *such that for every traded asset* X:

$$\frac{X_t}{B_t} = \mathbb{E}^{\mathbb{Q}}\left[\frac{X_T}{B_T}\,\Big|\,\mathcal{F}_t\right] \tag{11.1}$$

where B_t is a traded numeraire with $B_t > 0$ a.s. Any traded asset normalised by the numeraire is a $\mathbb{Q}$-martingale.

Equation (11.1) is the most important equation in this chapter and one of the most important in all of derivatives pricing. It is worth committing to memory, not just as a formula but as a template. Every calibration exercise we will perform is an instance of this template: specify X, specify B_t, observe the left-hand side from market data, parameterise the right-hand side in terms of model parameters, and solve for the parameters. The specific forms will differ — sometimes the right-hand side is a simple discrete expectation, sometimes a continuous-time expectation, sometimes a Gaussian integral — but the structure is universal.

Let us unpack each term carefully. The traded asset X can be a stock, a bond, a futures contract, or a derivative — the theorem is agnostic about type, only that X is a price process quoted in a market where self-financing trading is available. The numeraire B_t is a reference asset against which all other prices are measured; it can be the money-market account (so $B_t = e^{rt}$ in a constant-rate world, or $\exp(\int_0^t r_u\,du)$ with stochastic rates), a zero-coupon bond, a foreign currency, or any other strictly positive traded process. The measure $\mathbb{Q}$ depends on the choice of numeraire: switching numeraires is equivalent to switching measures, as developed canonically in Chapter 5. The conditional expectation $\mathbb{E}^{\mathbb{Q}}[\cdot \mid \mathcal{F}_t]$ is taken under $\mathbb{Q}$ and conditioned on the information available at time t.

The content of the theorem, stripped of formalism, is this: *you can price any claim by taking an expectation, provided you use the right measure.* The right measure is the one under which normalised prices become martingales. The right measure exists precisely when the market admits no arbitrage. And — crucially for us — the right measure is uniquely identified once you know enough about the market. Two subtleties of the statement are worth flagging. First, FTAP in its fullest generality requires care about integrability conditions, stopping times, and the precise notion of "no arbitrage" (no free lunch with vanishing risk is the right technical notion in continuous time). These subtleties matter in pathological cases but are invisible in the calibrations we perform. Second, the uniqueness of $\mathbb{Q}$ holds only in *complete* markets, where every contingent claim can be hedged. In incomplete markets (jumps, stochastic volatility without traded volatility), there are many equivalent martingale measures, each giving different prices for claims that cannot be hedged. Calibration in incomplete markets picks *one* of these measures — the one that matches the observed prices — but the choice is a modelling decision, not a mathematical inevitability.

Reading FTAP as a calibration rule. The theorem is often taught as an existence statement, but operationally it is a *constraint*. For every tradable instrument whose $t = 0$ price we observe, (11.1) gives one equation linking the unknown $\mathbb{Q}$-probabilities (and unknown lattice parameters) to the observed price. With enough independent instruments and enough unknowns, we get a determined (often over-determined) system. Calibration is then nothing more than *solving this system* — sometimes in closed form (as in §11.1.3 for the binomial moments, or §11.3 for bond lattice bootstrap) and sometimes numerically (least squares against a basket of options). Think of FTAP as giving you a *coordinate system* on the space of market-consistent models. Each observed price is a coordinate; each unknown parameter is an axis. Calibration asks whether the observed coordinates uniquely pick out a point on the axes. If they do, we are done. If they underdetermine the point, we need more quotes. If they overdetermine it, we solve in a least-squares sense and live with residuals — which, in practice, is almost always the situation.

A useful mental exercise: count equations and unknowns at the start of any calibration problem. A single-factor Vasicek has four unknowns $(\kappa, \theta, \sigma, r_0)$. The short-rate observation gives one equation directly. Each bond-price observation gives one more equation. So four bond-price observations are needed to pin down four unknowns, giving an exactly-determined system. In practice we observe far more than four bonds — maybe twenty at various maturities — so the system is over-determined, residuals are non-zero, and calibration is a least-squares exercise. The Heston stochastic-volatility model has five unknowns plus the initial spot; a liquid vanilla market provides hundreds of quotes, so the system is massively over-determined. This disparity between unknowns and observations is typical and is what makes calibration a statistical, not algebraic, problem.

Why numeraire invariance matters. FTAP says the same underlying prices satisfy (11.1) for many choices of numeraire, with correspondingly different measures. This is not a redundancy; it is a powerful tool. In equity derivatives the most common choice is the money-market account, so $\mathbb{Q}$ is the usual risk-neutral measure. For interest-rate derivatives the T-forward measure (numeraire $= P_t(T)$) simplifies expectations that would otherwise involve stochastic discounting; Chapter 13 uses it pervasively. For FX and quanto products, the domestic money market makes sense for some payoffs and the foreign one for others. The practitioner picks the numeraire that *annihilates* the most inconvenient source of randomness from the payoff, and the theorem guarantees the answer is the same regardless.

13.1.2 11.1.2 Single-Period Binomial Lattice

Before we write down the lattice, it is worth pausing on why the binomial structure is chosen rather than, say, a Gaussian step-size distribution. The answer is pedagogical economy: the binomial model has only two free parameters per step (the probability of up and the step size), matching the two pieces of market data we want to extract (drift and volatility). This symmetry between parameters and observations is the cleanest possible setting for learning how moment matching works. Once the binomial case is understood, extending to Gaussian, to multinomial, or to jump-extended processes is a matter of adding structure, not changing logic.

Start with one node A_{n-1} branching to two nodes over Δt:

$$A_{n-1} \;\longrightarrow\; \begin{cases} A_{n-1}\, e^{+c\sqrt{\Delta t}} & \text{(up)} \\ A_{n-1}\, e^{-c\sqrt{\Delta t}} & \text{(down)} \end{cases}$$

More compactly, introduce iid Bernoulli ticks $x_1, x_2, \ldots$ with $\mathbb{P}(x_i = +1) = p$, $\mathbb{P}(x_i = -1) = 1 - p$. Then

$$A_n \;=\; A_{n-1}\, e^{c\, x_n}, \qquad x_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\pm 1). \tag{11.2}$$

The log-return $\ln(A_n/A_{n-1}) = c\, x_n$ is $\pm c$.

The parameter c has a clean interpretation: it is the magnitude of a single-step log-return. If $c = 0.02$ and $\Delta t = 1$ day, then on any given day the asset moves either up by roughly 2% or down by roughly 2% (we use the exponential so that up-and-down produces $e^c e^{-c} = 1$, recovering the original level; multiplicative symmetry matters here because log-returns are what sum across time). The parameter p is the probability that the move is up; under the physical measure $\mathbb{P}$ this is tilted by the expected return, under the risk-neutral measure $\mathbb{Q}$ it is tilted only by the cost of carry.

It is worth pausing on why log-returns, rather than arithmetic returns, are the right quantity to model multiplicatively. Arithmetic returns do not add across time: if an asset is up 10% today and up 10% tomorrow, the two-day return is 21%, not 20%. Log-returns *do* add. This additivity is what makes log-returns amenable to the Central Limit Theorem — the sum of iid log-returns is approximately Gaussian, while the product of iid gross returns is approximately lognormal. The binomial model embraces this by writing each step as $S_n = S_{n-1}e^{cx_n}$ rather than $S_n = S_{n-1}(1+cx_n)$, making the multi-step composition $S_N = S_0 e^{c(x_1+x_2+\cdots+x_N)}$ a clean exponential of the summed shocks.

The Bernoulli shock $x_i \in \{+1, -1\}$ is conventional — it lets us write each step as a scaled shock rather than a case split. In the small-Δt limit, the sum of many iid Bernoulli shocks is approximately Gaussian by the Central Limit Theorem, so the binomial lattice converges to Brownian motion (Chapter 3). This is how the binomial model recovers the Black–Scholes world in the limit, and why the tree can be used as a discrete surrogate for continuous-time pricing. It is also why moment matching gets the right answer: matching two moments of a two-valued random variable fixes its distribution completely, and the CLT does the rest. A practical aside: the binomial tree as described here is *recombining* — the up-then-down path and the down-then-up path land at the same price $S_0 e^c e^{-c} = S_0$. A recombining tree has $n+1$ possible terminal nodes after n steps (rather than 2^n), which makes both pricing and calibration computationally tractable.

13.1.3 11.1.3 Recovering p and c from Observed Data

We want the parameters (p, c) — strictly, (q, c) under $\mathbb{Q}$ — to reflect what the market shows. Match the first two moments of the log-return.

Mean.

$$\mathbb{E}^{\mathbb{P}}\Big[\ln \tfrac{A_T}{A_0}\Big] \;=\; \mathcal{N}\, c(2p-1) \;\equiv\; \hat{\mu}T \tag{11.3}$$

where $\mathcal{N} = T/\Delta t$ is the number of steps. The equation should be parsed as follows. On each step, the log-return is $c\, x_i$, which has expectation $c\mathbb{E}[x_i] = c(2p-1)$. Summing over $\mathcal{N}$ iid steps gives total log-return $\mathcal{N}c(2p-1)$. Equating this to $\hat{\mu}T$ — the observed physical drift — gives us our first calibration equation. Note the structural parallel with the familiar one-factor Gaussian world: the total drift over T is linear in T, because drift is additive across independent steps.

Where does $\hat{\mu}T$ come from in practice? There are several sources, each with pitfalls. The most direct is historical return data: compute the sample mean of daily log-returns and scale to annual. This gives a point estimate of the physical drift, but the estimate is noisy — a one-year window of daily returns gives a drift estimate with standard error of order $\sigma/\sqrt{T}$, which for $\sigma = 20\%$ and $T = 1$ year is 20% of the true drift. Historical drift estimates are therefore nearly useless at short horizons and only marginal at long ones; this is why risk-neutral measures are so much more widely used in pricing applications — the $\mathbb{Q}$-drift is pinned down by the risk-free rate, which is observable with essentially no noise.

Variance.

$$\mathbb{V}^{\mathbb{P}}\Big[\ln \tfrac{A_T}{A_0}\Big] \;=\; \mathcal{N}\, c^2\big(1-(2p-1)^2\big) \;\equiv\; \sigma^2 T. \tag{11.4}$$

Variance adds across iid steps, so total variance is $\mathcal{N}$ times the single-step variance. The single-step variance is $c^2\mathbb{V}[x_i]$ because c is a deterministic scale factor. And $\mathbb{V}[x_i] = \mathbb{E}[x_i^2] - (\mathbb{E}[x_i])^2 = 1-(2p-1)^2$, using $x_i^2 = 1$ with probability one. The quantity $(2p-1)^2$ is small when p is near $1/2$ — which, as we will see, is the regime relevant for small time-steps — so $\mathbb{V}[x_i] \approx 1$ and the variance is approximately $\mathcal{N}c^2$.

The variance $\sigma^2 T$ is much easier to measure empirically than the drift $\hat{\mu}T$. The standard error of a realised variance estimate scales as $\sigma^2\sqrt{2/T}$ for large T and small sampling intervals, so even a one-year history of daily returns gives a variance estimate accurate to a few percent. This is why volatility estimates from historical data are generally taken seriously in practical pricing, while drift estimates are regarded with considerable skepticism.

Equations (11.3) and (11.4) form a two-equation system in the two unknowns (p, c). The $\mathbb{Q}$-version is structurally identical, only with $\hat{\mu}$ replaced by the risk-neutral drift. This two-for-two structure is a general feature of well-designed moment-matching schemes: the number of free parameters equals the number of moments matched, so the system is square and generically has a unique solution.

13.1.4 11.1.4 Small-Step Asymptotics

Write $\sigma^2 T = \mathcal{N}\, c^2(1-(2p-1)^2)$. For small Δt, $(2p-1)^2 \ll 1$, so

$$\sigma^2 T \;\approx\; \mathcal{N}\, c^2 \;=\; \frac{T}{\Delta t}\, c^2 \quad\Longrightarrow\quad \boxed{c \;\approx\; \sigma\sqrt{\Delta t}} \tag{11.5}$$

Substituting into the mean equation $\hat{\mu}T = c\,\mathcal{N}(2p-1)$ gives $2p-1 \approx (\hat{\mu}/\sigma)\sqrt{\Delta t}$ and hence

$$\boxed{p \;\approx\; \tfrac{1}{2}\Big(1+\tfrac{\hat{\mu}}{\sigma}\sqrt{\Delta t}\Big)} \tag{11.6}$$

This is the calibration of the physical probability p. Under $\mathbb{Q}$, the same structural formula applies with $\hat{\mu}$ replaced by the drift that makes A_t/B_t a martingale (e.g. $r - \frac{1}{2}\sigma^2$ for a log-normal asset).

Worked example. Take $\sigma = 0.20$, $\hat{\mu} = 0.08$, $\Delta t = 1/252$. Then $c = 0.20\sqrt{1/252} \approx 0.01260$ and $p \approx \frac{1}{2}(1 + 0.4\sqrt{1/252}) \approx \frac{1}{2}(1 + 0.0252) = 0.5126$. An equity with 20% annualised volatility and 8% annualised drift has, on a one-day step, about a 1.26% typical log-return shock and a 51.26% probability of an up-move. The asymmetry between up and down probabilities (just over one percentage point) is small enough that if you looked at a few days of such returns you would see what looks like a fair coin; the asymmetry only becomes clear when you aggregate hundreds of days. Drift is small on short horizons and dominates only over long ones, which is why the "random walk" intuition for short-term stock movements is approximately correct, and why the drift-extractable information in historical returns is notoriously noisy. The Sharpe ratio of $0.08/0.20 = 0.4$ is typical for equity indices — equity risk premium divided by equity volatility — and reappears as the dominant signal in the calibration.

Why the Brownian limit emerges. The deeper reason moment matching recovers the right continuous-time dynamics is the CLT. If on each step we match the conditional mean and variance of log-returns to $\hat{\mu}\Delta t$ and $\sigma^2\Delta t$ respectively, and the shocks are independent, then the sum of $\mathcal{N} = T/\Delta t$ steps is a random variable with mean $\hat{\mu}T$ and variance $\sigma^2 T$. As $\Delta t \to 0$, the CLT guarantees this sum converges in distribution to $\mathcal{N}(\hat{\mu}T, \sigma^2 T)$ — which is exactly the log-return distribution of geometric Brownian motion. The binomial tree is therefore not a separate model from Black–Scholes; it is a discretisation that converges to the same limit, provided the step sizes and probabilities are chosen (i.e. calibrated) in the way (11.5) and (11.6) prescribe. The scaling laws emerge from dimensional analysis alone. Variance scales as $\sigma^2\Delta t$, so the step size scales as $\sigma\sqrt{\Delta t}$. The drift contribution to the log-return scales as $\hat{\mu}\Delta t$, and the probability tilt must therefore scale as $\hat{\mu}\Delta t/(\sigma\sqrt{\Delta t}) = (\hat{\mu}/\sigma)\sqrt{\Delta t}$.

From physical to risk-neutral: where the Sharpe ratio goes. Under $\mathbb{Q}$, the drift that enters (11.6) is not $\hat{\mu}$ but the martingale drift — for a stock paying no dividends in a constant-rate world, $r - \frac{1}{2}\sigma^2$. The Sharpe ratio of the *risk premium* is therefore $(\hat{\mu} - r)/\sigma$, and this is precisely the quantity that disappears when we pass to $\mathbb{Q}$. The formula structure is unchanged; the numerator in the tilt shifts from $\hat{\mu}$ to $r - \frac{1}{2}\sigma^2$, reflecting the replacement of realised returns by risk-free returns. This is the lattice incarnation of Girsanov's theorem (Chapter 5).

Calibration as an inverse problem, restated. The binomial calibration is formally an *exactly-determined* inverse problem, not an ill-posed one. Two moments and two unknowns; a smooth, monotonic map between moments and parameters as long as $\sigma > 0$; closed-form inversion via (11.5)–(11.6); straightforward stability. All three of the usual pathologies of inverse problems — non-existence, non-uniqueness, instability — are absent. This is why the binomial case is pedagogically so useful: it lets us see the moment-matching machinery at its cleanest, before the complications of real-world calibration sink their teeth in. The other calibration problems in this chapter and the next — the bond-lattice bootstrap in §11.3, the Vasicek yield-curve fit in Chapter 12, and the implied-vol surface fits in Chapter 14 — are all, to varying degrees, ill-posed, and each will require one or more of the three remedies: weighted least squares, regularisation, and cross-date smoothing.

13.2 11.2 Change of Measure on Multinomial Trees

13.2.1 11.2.1 Setup: Why Multinomial

Consider a richer lattice where a single node A_0 branches to multiple children (e.g. three children A_1, B_1, C_1 each with its own sub-tree). Pricing under $\mathbb{P}$ uses probabilities $\{p^A, p^B, p^C\}$; pricing under $\mathbb{Q}$ uses $\{q^A, q^B, q^C\}$. We want to express the q's in terms of the p's (and of observed price ratios).

Why bother with multinomial trees? The binomial tree is adequate for simple equity options, but as soon as the state space has more than one non-trivial dimension — say, an equity *and* an interest rate, or an asset with both a continuous price component and a discrete jump component — we need lattices with more than two branches per step. Trinomial trees also have a useful practical feature: the middle branch allows the state to hold roughly constant, which is convenient for modelling slow processes like mean-reverting interest rates. The calibration logic below is cleanest in the binomial case but generalises naturally.

A concrete example makes the case for multinomial lattices vivid. Consider a callable bond: the issuer has the right to redeem the bond at par on specified dates. Pricing this requires a short-rate lattice, because the call decision depends on where rates go. If rates spike up, the bond price drops, and the issuer has no incentive to call (they can reborrow more cheaply by not calling). If rates drop, the bond price rises, and the issuer calls to refinance. A binomial short-rate tree is clumsy for this: the two branches move the state up or down with equal probability, but mean-reverting rates typically have three economic regimes — rising, falling, or flat — and the flat regime is where most of the probability mass sits. A trinomial tree with up, middle, and down branches captures this more naturally: the middle branch lets the rate stay roughly where it is (consistent with a typical short-rate half-life of several years), and the up and down branches capture the tails. The price of a callable bond computed on a trinomial lattice converges to the true continuous-time price faster than on a binomial lattice with the same number of total nodes. Another example: jump-diffusion equity models. If the underlying can either diffuse continuously or jump to a distressed level, a single binomial branch cannot capture both dynamics. A trinomial tree with "up diffusion", "down diffusion", and "jump" branches handles this cleanly, at the cost of needing to calibrate three probabilities and three step sizes rather than two.

The martingale identity (11.1) applied to numeraire-normalised values gives, for each asset C (traded) and numeraire A:

$$\frac{C_0}{A_0} \;=\; q^A\,\frac{C_u}{A_u} + (1-q^A)\,\frac{C_d}{A_d} \tag{11.7}$$

$$\frac{C_0}{B_0} \;=\; q^B\,\frac{C_u}{B_u} + (1-q^B)\,\frac{C_d}{B_d} \tag{11.8}$$

Here C_u, C_d are the values of C in the up and down states; A_u, A_d, B_u, B_d are similarly the values of the two numeraire candidates. The superscripts on q track *which* numeraire the measure is defined relative to — $\mathbb{Q}^A$ is the measure that makes things a martingale in units of A, and correspondingly for $\mathbb{Q}^B$. Different numeraires correspond to different martingale measures, even though they price the same underlying claims; Chapter 5 treated the continuous-time version of this relationship exhaustively.

13.2.2 11.2.2 Solving for q

From (11.7):

$$q^A \;=\; \frac{\dfrac{C_0}{A_0} - \dfrac{C_d}{A_d}}{\dfrac{C_u}{A_u} - \dfrac{C_d}{A_d}}, \qquad q^B \;=\; \frac{\dfrac{C_0}{B_0} - \dfrac{C_d}{B_d}}{\dfrac{C_u}{B_u} - \dfrac{C_d}{B_d}}. \tag{11.9}$$

Formulas (11.9) are simply the no-arbitrage pricing rule inverted. Read them as follows: the risk-neutral probability of an up-move is the "lever arm" that makes the current normalised price C_0/A_0 an exact convex combination of the up- and down-state normalised prices. If the current price is above the midpoint of the up and down possibilities, $q^A > 1/2$; if below, $q^A < 1/2$. Crucially, if the current price is *outside* the range $[C_d/A_d, C_u/A_u]$, there is no valid $q^A \in [0,1]$ and the market is arbitrageable. This last observation is why FTAP links the existence of $\mathbb{Q}$ to the absence of arbitrage: when an arbitrage exists, no consistent martingale measure can reproduce the quotes.

A worked scenario makes this concrete. Suppose at time zero we observe $C_0 = 100$, $A_0 = 98$, and at time Δt the two possible states are $(C_u, A_u) = (110, 102)$ and $(C_d, A_d) = (90, 98)$. The current ratio $C_0/A_0 = 100/98 \approx 1.0204$; the up-state ratio $C_u/A_u = 110/102 \approx 1.0784$; the down-state ratio $C_d/A_d = 90/98 \approx 0.9184$. The current ratio sits comfortably between the two future ratios, so the system is arbitrage-free, and we compute $q^A = (1.0204 - 0.9184)/(1.0784 - 0.9184) = 0.1020/0.1600 = 0.6375$. The $\mathbb{Q}^A$-probability of an up-move is 63.75%; the down-probability is 36.25%. These probabilities are the market's *implicit* forward-looking assessment, under numeraire A, of the likelihood of each future state; they reflect both genuine subjective probabilities and risk preferences, compressed into a single no-arbitrage quantity. Now suppose instead that C_0 is observed at 112 rather than 100. The current ratio becomes $112/98 \approx 1.1429$, which is *above* the up-state ratio 1.0784. There is no $q^A \in [0,1]$ that can reconcile this — the current price is higher than even the best future outcome, so an arbitrageur could short C, buy A, and collect the spread. The system is explicitly arbitrageable, and the calibration fails. Calibration is not an optional step that improves the model — it is an *arbitrage-detection* step. If calibration produces well-defined parameters, the market is (locally) arbitrage-free; if it cannot, there is a profitable trade sitting on the screen. In practice, this situation is rare but not unheard of: stale quotes, manual data-entry errors, or genuine market dislocations can all produce momentary violations. A disciplined calibration system flags such quotes and either excludes them or alerts a human operator.

13.2.3 11.2.3 Radon–Nikodym on a Finite Tree

Express the same ratio C_0/B_0 using numeraire A and convert:

$$\frac{C_0}{B_0} \;=\; q^A\,\frac{A_0}{B_0}\cdot\frac{C_u}{A_u} + (1-q^A)\,\frac{A_0}{B_0}\cdot\frac{C_d}{A_d}$$

Rewriting with B-ratios,

$$\frac{C_0}{B_0} \;=\; \underbrace{q^A\,\frac{A_0}{B_0}\cdot\frac{B_u}{A_u}}_{>0}\cdot\frac{C_u}{B_u} \;+\; \underbrace{(1-q^A)\,\frac{A_0}{B_0}\cdot\frac{B_d}{A_d}}_{>0}\cdot\frac{C_d}{B_d}. \tag{11.10}$$

The key trick here is multiplying top and bottom by appropriate ratios so that C appears normalised by B (as on the left) rather than by A. What matters is the *form* of the result: C_0/B_0 is expressed as a convex combination of C_u/B_u and C_d/B_d, with new weights that differ from q^A. Those new weights are the $\mathbb{Q}^B$-probabilities, and they must equal $q^B, 1-q^B$ because (11.8) is the unique such representation. Compare term-by-term with (11.8):

$$q^B \;=\; q^A\cdot\frac{B_u/A_u}{B_0/A_0}, \qquad (1-q^B) \;=\; (1-q^A)\cdot\frac{B_d/A_d}{B_0/A_0}. \tag{11.11}$$

Consistency check. Summing the two pieces must give 1:

$$q^B + (1-q^B) \;=\; \frac{A_0}{B_0}\left[q^A\,\frac{B_u}{A_u} + (1-q^A)\,\frac{B_d}{A_d}\right] \;=\; \frac{A_0}{B_0}\cdot\frac{B_0}{A_0} \;=\; 1.\ \checkmark \tag{11.12}$$

The bracket equals B_0/A_0 because B/A is itself a $\mathbb{Q}^A$-martingale. This consistency check is not a triviality; it is the embodiment of the martingale property. If B/A is not a $\mathbb{Q}^A$-martingale, the calculation fails — the two new weights do not sum to one, meaning there is no valid measure $\mathbb{Q}^B$. So the existence of an alternative pricing measure is equivalent to the original one correctly pricing the *new numeraire* as a tradable asset.

Equations (11.11) are exactly the Radon–Nikodym density on the lattice:

$$\boxed{\frac{d\mathbb{Q}^B}{d\mathbb{Q}^A}(\omega) \;=\; \frac{B_\cdot(\omega)/B_0}{A_\cdot(\omega)/A_0}} \tag{11.13}$$

i.e. the density is the *ratio of numeraire-returns*. In shorthand, writing $\tilde{A} = A/A_0$ and $\tilde{B} = B/B_0$:

$$\frac{d\mathbb{Q}^B}{d\mathbb{Q}^A} \;=\; \frac{\tilde{B}}{\tilde{A}}. \tag{11.14}$$

This is the discrete-time, finite-state incarnation of the Radon–Nikodym derivative that Chapter 5 developed in continuous time. Under $\mathbb{Q}^A$ all prices are "measured in units of A"; switching numeraires to B is a relabelling of the unit, so scenarios where B has over-performed A get up-weighted by their realised relative return $B_\cdot/A_\cdot$ (normalised by the $t=0$ ratio so that the density has mean one). The Radon–Nikodym density is therefore a tradable object — the "relative performance" of the two numeraires — not a mysterious abstraction. A probability measure has to integrate to one, so any valid Radon–Nikodym density must have $\mathbb{E}^{\mathbb{Q}^A}[d\mathbb{Q}^B/d\mathbb{Q}^A] = 1$. From (11.14): under $\mathbb{Q}^A$, $\mathbb{E}^{\mathbb{Q}^A}[\tilde{B}_T/\tilde{A}_T] = \tilde{B}_0/\tilde{A}_0 = 1$, using that B/A is a $\mathbb{Q}^A$-martingale. So indeed the density has mean one, and $\mathbb{Q}^B$ is a well-defined probability measure.

Multi-step extension. If the tree branches further at the up-node (children with sub-probabilities $q_u, r_u, \ldots$) and similarly at the down-node, each local change-of-measure is governed by the same ratio. The overall change of measure from $t=0$ to $t=T$ is the composition of all these local operations along every path. In the continuous-time analogue (Chapter 5), this becomes a stochastic exponential; in the discrete case it is a telescoping product of local ratios. A calibrated tree can be diagnosed: at every node, one can check that the local martingale property holds for each numeraire pair. If any of these local checks fails, the calibration has introduced a local arbitrage — which is a bug. Production calibration systems routinely run these checks as unit tests, catching implementation errors that would otherwise manifest as mysterious mispricings downstream.

Radon-Nikodym density across states from multinomial calibration *Five-state discrete world: physical probabilities p_i (blue) are uniform, while calibrated risk-neutral probabilities q_i (red) tilt mass toward the central "stress" states that pricing data implicates. The RN density $dQ/dP = q_i/p_i$ (purple markers) records that tilt state-by-state and is the discrete-tree analogue of the continuous-time Girsanov density of Chapter 5.*

13.2.4 11.2.4 Calibration as Optimisation: The Least-Squares Setup

Before turning to the interest-rate case, we pause to lay out the calibration problem as it is actually formulated in practical systems. Given N observed market instruments with prices $\{C_i^{\mathrm{mkt}}\}_{i=1}^N$, a model with parameter vector $\Theta \in \mathbb{R}^d$, and a model pricing function $C_i^{\mathrm{model}}(\Theta)$, the simplest calibration objective is

$$\Theta^\star \;=\; \arg\min_\Theta \sum_{i=1}^N \big(C_i^{\mathrm{model}}(\Theta) - C_i^{\mathrm{mkt}}\big)^2. \tag{11.15}$$

This is the ordinary least-squares (OLS) objective. Minimising (11.15) gives the parameter vector that, on average, best reproduces the market prices. The word "best" hides multitudes — "best" in what sense, under what weighting, with what penalty for over-fit — and the next several paragraphs unpack these choices.

The first and most important refinement is *weighting*. Not all market prices are equally informative. A deep in-the-money call option that trades with a tight bid-ask is a tighter constraint on the model than a deep out-of-the-money put trading with a wide bid-ask and small absolute price. Equal weighting in price space treats them as equally important; weighting each residual by the inverse bid-ask squared downweights the noisy quote and lets the clean quote dominate. The weighted least-squares objective is

$$\Theta^\star \;=\; \arg\min_\Theta \sum_{i=1}^N w_i\big(C_i^{\mathrm{model}}(\Theta) - C_i^{\mathrm{mkt}}\big)^2, \tag{11.16}$$

with $w_i = 1/\mathrm{spread}_i^2$ or some related quantity. The algebraic form is the same as (11.15); the economic content is very different.

A second common weighting scheme is *vega weighting*. For options, the natural trader-friendly quantity is implied volatility, not price. Two options with the same implied volatility but different strikes have very different prices, yet carry the same economic information. Converting the least-squares fit from price space to implied-volatility space is equivalent to weighting each price residual by $1/\mathrm{vega}_i$, because vega $= \partial C/\partial\sigma_{\mathrm{imp}}$ is exactly the local conversion factor. The vega-weighted objective is

$$\Theta^\star \;=\; \arg\min_\Theta \sum_{i=1}^N \frac{\big(C_i^{\mathrm{model}}(\Theta) - C_i^{\mathrm{mkt}}\big)^2}{\mathrm{vega}_i^2}, \tag{11.17}$$

which is equivalent (to first order, near the market-implied volatility) to

$$\Theta^\star \;=\; \arg\min_\Theta \sum_{i=1}^N \big(\sigma_i^{\mathrm{model}}(\Theta) - \sigma_i^{\mathrm{imp,mkt}}\big)^2. \tag{11.18}$$

Vega is the option's sensitivity to implied volatility; near-the-money options have high vega, deep OTM options have low vega. In price space, a 1% error in an at-the-money price might correspond to a 0.1-vol error, while a 1% error in a deep OTM price might correspond to a 5-vol error. An unweighted fit would treat these as equally bad; a vega-weighted fit treats the deep OTM error as 50 times worse — as a trader quoting in vol space would insist. A subtle caveat: vega-weighting is a *linearisation*; it breaks down for very deep OTM options where the vega is vanishingly small and a literal $1/\mathrm{vega}^2$ weight gives that strike enormous, usually spurious, importance. A practical weighting scheme caps the weight, $w_i = \min(1/\mathrm{vega}_i^2, w_{\max})$, or combines vega and spread weighting via a harmonic mean. Tuning these caps is part of the craft of building a robust calibration.

13.2.5 11.2.5 Global vs Local Minima

The objective function (11.15) is, in general, a non-convex function of the parameters Θ. For the binomial lattice it happens to be quadratic (the moment-matching equations are linear in parameters after log-transformation), but for more general models — stochastic volatility, jump diffusion, multi-factor rates — the landscape has multiple local minima. Which minimum the optimiser finds depends on where it starts. This is a serious practical concern. An equity trader who runs a stochastic-volatility model on Monday morning and gets parameters ($\kappa = 2, \theta = 0.04, \sigma_v = 0.3, \rho = -0.7$) may rerun the same calibration on Tuesday with yesterday's minimum as a starting point and get ($\kappa = 2.1, \theta = 0.041, \sigma_v = 0.31, \rho = -0.71$) — a small, economically sensible shift. But if the starting point is reset to a default every day, Tuesday's run might find a completely different local minimum, say ($\kappa = 0.5, \theta = 0.06, \sigma_v = 0.5, \rho = -0.9$), with similar fit quality but very different economic implications. The two parameter vectors produce nearly identical prices on the calibration basket but very different prices on instruments *outside* the basket. If the bank uses the Monday parameters to price exotics on Tuesday but recalibrates for the blotter, the exotics will be valued one way while the hedging positions use a different calibration — a recipe for P&L surprise.

The standard defenses against this problem are:

1. **Warm starts.** Always initialise today's calibration from yesterday's minimum. This enforces continuity across dates and typically keeps the optimiser in the same local basin even if another basin is marginally deeper. The cost is that the calibration may become "stuck" in a stale minimum if the market has genuinely moved to a new regime; the remedy is to add periodic cold starts to check for regime shifts.
2. **Multi-start optimisation.** Run the optimiser from several starting points (a small random grid, or structured starts from economically plausible prior values) and take the best. If multiple starts converge to distinct minima of comparable depth, the problem is flagged as non-identifiable and a human operator is consulted.
3. **Global optimisation methods.** Use algorithms designed to escape local minima — simulated annealing, differential evolution, genetic algorithms, basin-hopping. These are slower than gradient-based local methods but more reliably find global minima.
4. **Problem reformulation.** Often the landscape can be smoothed by changing variables. Calibrating Heston (Chapter 10) in $(\kappa, \theta, \sigma_v, v_0, \rho)$ is notoriously multimodal; reparameterising in terms of more orthogonal quantities — at-the-money variance, skew, curvature, mean-reversion speed — can make the landscape nearly convex. This is sometimes called "calibrating in the right coordinates".

13.2.6 11.2.6 Numerical Methods

The optimisation problems above are nonlinear least-squares in a moderate-dimensional space. Several algorithmic families are in common use. *Nelder–Mead* maintains a simplex of $d + 1$ points and iteratively reflects, expands, contracts, and shrinks it; it requires only function evaluations (no gradients), so it is robust to objectives that are noisy, discontinuous, or expensive to differentiate, at the cost of slow convergence. *Levenberg–Marquardt* is the gold standard for nonlinear least-squares: it exploits the sum-of-squares structure to approximate the Hessian from Jacobians alone, and blends Gauss–Newton with gradient-descent via a damping parameter λ. It typically converges in tens of iterations on well-posed problems; for implied-vol-surface fits with $N \sim 100$ instruments and $d \sim 5$ parameters, it converges in milliseconds. *BFGS* and *L-BFGS* are quasi-Newton methods building up a Hessian approximation iteratively — the full version for moderate d, the limited-memory version for thousands of parameters. *Differential evolution* and related global methods maintain a population of candidate parameter vectors and evolve them via mutation, crossover, and selection; they are slower (seconds to minutes) but reliably find global minima.

Hybrid strategies. A production calibration system rarely uses a single method. A typical setup: start with a coarse differential-evolution search to find a plausible global minimum; refine with Levenberg–Marquardt for tight local convergence; verify stability with a Nelder–Mead polish. Practical runtime targets: a single-factor rates model against 20 bond quotes under 10 ms; Heston against 100 vanilla options under 100 ms; a multi-factor Libor market model against 500 caplet/swaption quotes under 10 seconds. Faster is always better, but stability and correctness trump speed by a wide margin in production.

13.2.7 11.2.7 Regularisation and Well-Posedness

When the calibration problem is under-determined — more parameters than informative observations, or observations that leave a sloppy direction unpinned — regularisation adds a penalty term to the objective function:

$$\Theta^\star \;=\; \arg\min_{\Theta}\Big[\sum_i w_i\big(C_i^{\text{model}}(\Theta) - C_i^{\text{mkt}}\big)^2 + \lambda\,\mathcal{R}(\Theta)\Big]. \tag{11.19}$$

The penalty $\mathcal{R}(\Theta)$ encodes prior knowledge: smoothness ($\|\nabla^2\Theta\|^2$ for curve fits), proximity to a historical mean ($\|\Theta - \Theta_{\text{prior}}\|^2$), sparsity ($\|\Theta\|_1$), or any other property the modeller wants to encourage. The weight λ controls how much regularisation is applied: small λ trusts the data, large λ trusts the prior. Choosing λ is itself a subproblem, usually handled by cross-validation (try several λ values and pick the one that minimises out-of-sample error) or by Bayesian hierarchical modelling (treat λ as a hyperparameter to be integrated over).

The Bayesian interpretation of regularisation is worth holding in mind. A penalty $\|\Theta - \Theta_{\text{prior}}\|^2$ is equivalent to a Gaussian prior on Θ centred at Θ_{prior}; a penalty $\|\Theta\|_1$ is equivalent to a Laplace prior at zero; a penalty $\|\nabla^2\Theta\|^2$ encodes a prior that parameter values change smoothly along some ordering (useful for term-structure calibration where we expect neighbouring maturities to have similar parameters). Regularisation is the frequentist's way of sneaking Bayesian priors into a point-estimate framework.

The fit-vs-extrapolation tradeoff. Strong regularisation (λ large) produces parameter estimates that are close to the prior, insensitive to the data, and therefore stable but potentially biased. Weak regularisation (λ small) produces parameter estimates that fit the data tightly, sensitive to every wiggle, and therefore unbiased in expectation but high-variance in any given run.

L-curve: fit residual vs solution norm across lambda *L-curve diagnostic for a Tikhonov-regularised linear inverse problem. Each point is one value of λ; the corner (elbow, marked) is the classical heuristic choice — the regularisation strength beyond which further stiffening raises residual without lowering solution norm. Small-λ end: data-fitting, overfit regime. Large-λ end: prior-dominated, underfit regime.* The sweet spot depends on the application: a conservatively-marked book wants stability (high λ); an aggressive trading book wants responsiveness (low λ). There is no universal right answer, and the choice is part of the desk's risk-appetite statement. A related perspective: in-sample fit is a strictly decreasing function of λ (regularisation makes the model stiffer, so it fits the training data less well), while out-of-sample fit is typically U-shaped (zero regularisation overfits, infinite regularisation underfits, and the optimum is somewhere in between). Cross-validation finds this optimum empirically.

Training vs out-of-sample RMSE across regularisation strength *The U-shape of out-of-sample error (red) in λ contrasts with the monotone rise of training error (blue). Small λ overfits — training error shrinks but OOS error spikes; large λ underfits — both errors rise. The optimum λ is the OOS minimiser, found in practice by cross-validation.*

Out-of-sample stability. A calibrated model's in-sample performance is tautological: by construction, the model matches its calibration instruments to within the optimiser's tolerance. The real test is *out-of-sample*: does the model correctly price instruments that were not in the calibration basket? There are two flavours of out-of-sample test. The first is *cross-sectional*: calibrate using a subset of instruments (say, vanilla options at strikes $K_1, \ldots, K_{10}$) and check the fit on a held-out strike K_{11}. If the model is correctly specified and the calibration is well-regularised, the held-out fit should be nearly as good as the in-sample fit. If the held-out fit is markedly worse, the model is overfitting — squeezing the in-sample error to zero at the cost of off-sample generalisation. The second flavour is *temporal*: calibrate today, price a hold-out instrument today, then hold the model parameters fixed and re-price the same instrument tomorrow. Compare against tomorrow's market price. If parameters are stable day-to-day, the deviation should be small; if parameters wobble, the deviation will be dominated by recalibration noise rather than genuine market movement. Both tests are routinely neglected in practice — the in-sample fit is visually appealing and easy to report to management, while out-of-sample tests require more infrastructure. But any model going into production should be subjected to both.

13.3 11.3 Short-Rate Bootstrap on a Bond Lattice

13.3.1 11.3.1 Zero-Coupon Bonds as Calibration Instruments

Let $P_t(T)$ denote the zero-coupon bond price at time t paying 1 at maturity T. On a short-rate lattice $\{r_0, r_{11}, r_{22}, \ldots\}$ (a recombining tree of realised short rates indexed by time step and node-within-step), the one-period bond satisfies

$$P_0(1) \;=\; \frac{1}{1+r_0} \qquad\Longleftrightarrow\qquad r_0 \;=\; \frac{1}{P_0(1)} - 1. \tag{11.20}$$

(Convention note. The discount factor $1/(1+r_t)$ used throughout §11.3 is the *simple-compounding* version; elsewhere in the guide — and in the CRR lattice conventions of §11.1.2 — we use exponential discounting $e^{-r_t\,\Delta t}$. The two agree to leading order in Δt — $1/(1+r_t\Delta t) = e^{-r_t\Delta t}(1+O(\Delta t^2))$ — so the sequential-bootstrapping arithmetic below is unaffected at the orders of accuracy we care about. We keep the simple-compounding form in §11.3 because it matches the market quoting convention for short-dated zero coupons, which is how the bootstrap inputs are usually presented in practice.)

Two-period bonds satisfy, at each child node,

$$P_{11}(2) \;=\; \frac{1}{1+r_{11}}, \qquad P_{22}(2) \;=\; \frac{1}{1+r_{22}}. \tag{11.21}$$

Rolling back to $t=0$ under the risk-neutral measure $\mathbb{Q}$:

$$P_0(2) \;=\; \frac{1}{1+r_0}\left[\,q\,P_{11}(2) + (1-q)\,P_{22}(2)\right]. \tag{11.22}$$

Zero-coupon bonds are the canonical calibration instruments for interest-rate models because their payoffs are entirely deterministic — they pay 1 at maturity, with no optionality to muddy the waters. Any uncertainty in the bond's current price therefore comes purely from the uncertainty of the path of short rates between now and maturity. Observing $P_0(T)$ for many maturities pins down the *term structure* of the short-rate risk-neutral dynamics, which is exactly what a short-rate model needs.

A useful intuition: a zero-coupon bond's yield is the constant rate at which, if the future were certain, the bond would compound to face value. When the future is uncertain, the yield is the *certainty-equivalent* rate — the deterministic rate that, if applied over the bond's life, would produce the same discounted payoff as the actual stochastic rate path. Because Jensen's inequality makes the expectation of the discount factor bigger than the discount factor at the expected rate, the certainty-equivalent yield is slightly below the expected average rate. This is the convexity correction that Chapter 12 derives rigorously for the Vasicek model; at this stage, the point is only that every bond quote is a statement about the certainty-equivalent of a stochastic path, and each independent maturity provides one independent equation relating parameters to data.

Equation (11.20) is the simplest imaginable calibration equation: knowing today's one-period bond price directly gives today's short rate. Equation (11.21) is equally simple, but at the next time step — at each possible future short-rate state, the corresponding one-period bond price is fixed. The interesting work happens in (11.22), which enforces no-arbitrage between the $t=0$ quote and the $t=\Delta t$ states. Here both q and the short-rate states are, in principle, unknowns. The progression from (11.20) to (11.22) illustrates a key feature of tree-based calibration: each successive equation adds information about one more future time slice. (11.20) pins the $t=0$ node; (11.22) pins the $t=\Delta t$ node (given a vol input); the analogous three-step equation pins $t=2\Delta t$; and so on. The calibration proceeds recursively, one time slice at a time, building up a discrete yield-curve-consistent tree. This is the same recursive logic as in the Black–Derman–Toy (BDT) model or the Ho–Lee model — both of which are, at bottom, schemes for backing out tree parameters from bond-price ratios, one time slice at a time, in a way that exactly reproduces the observed yield curve. Chapter 12 develops the continuous-time limits of these constructions; here we focus on the discrete bootstrap.

13.3.2 11.3.2 Recovering q and the Short Rates

Given $P_0(1)$ (one-year bond price) and $P_0(2)$ (two-year bond price), plus a volatility assumption that fixes the spread between r_{11} and r_{22} (e.g. $r_{22} = r_{11}\,e^{-2c\sqrt{\Delta t}}$), equations (11.20)–(11.22) give a system of two equations in (q, r_{11}) that can be solved numerically. A simple illustrative case takes $q = 1/2$, which then delivers r_{11} explicitly from (11.22): r_{11} is chosen such that $\frac{1}{1+r_{11}}$ and $\frac{1}{1+r_{22}}$, mixed half-half, reproduce $P_0(2)(1+r_0)$. In practice the tree is built *sequentially* from the observed yield curve $\{P_0(1), P_0(2), P_0(3), ...\}$ and a term structure of volatilities — at each new level we solve for the fresh short-rate node that reprices the newly observed bond.

Why sequential construction works. The recursive nature of bond pricing — today's bond price is a discount of a convex combination of tomorrow's bond prices, which are themselves discounts of further-future bond prices — means calibration can be done layer by layer. Once the $t=0$ node is pinned down (trivially from (11.20)), the $t=\Delta t$ layer is pinned down by the two-period bond (11.22) plus a volatility input. The three-period bond $P_0(3)$ then pins down the $t = 2\Delta t$ layer, and so on. This is the classical Ho–Lee / Hull–White / BDT construction, and it produces a tree that exactly reproduces the input yield curve by design. The tree is then said to be *bootstrapped*.

The role of the volatility input. Without a volatility input, the system is underdetermined: for a binomial tree at step $t = \Delta t$ there are two rate states r_{11}, r_{22} and a transition probability q, but only one bond quote $P_0(2)$ — three unknowns and one equation. We need extra information to pin down the spread between r_{11} and r_{22}. In practice this comes from one of three sources: (i) an explicit volatility assumption (e.g. "short rates have 1.2% annualised vol"), (ii) a market quote on an interest-rate option (caplet, swaption) whose price encodes the volatility, or (iii) a direct measurement of historical volatility of the short rate. Each source gives a different calibration flavour; market-consistency purists prefer (ii) because it is the only source that is consistent with tradable instruments.

Interpolation vs extrapolation. The bootstrapped tree reproduces the calibration instruments *exactly.* That is a property of construction, not a property of truth. For maturities *within* the calibration set, the model is interpolating between observed quotes, and we are entitled to a measured trust in its outputs. For maturities *beyond* the calibration set — or for instruments that depend on joint dynamics we have not fit — the model is extrapolating, and its outputs are only as good as the extrapolation. This distinction matters enormously in practice: a Vasicek fit calibrated to the 1-month, 3-month, 6-month, and 1-year bonds will reproduce those four quotes exactly (if it has four free parameters) but may produce wildly different 30-year yields from a Vasicek fit calibrated to 10-year, 20-year, and 30-year bonds. Both fits are "correct" in the interpolating region; neither is trustworthy in the extrapolating region.

The practical implication is that a calibration's range of trustworthiness is determined by its calibration set, not by the model's parametric form. A Vasicek model calibrated to short maturities will give short-maturity answers you can trust, but its long-maturity extrapolation is model-dependent in ways the short-maturity data cannot validate. Similarly, an implied-vol surface calibrated to near-the-money strikes will give near-the-money prices you can trust, but deep out-of-the-money extrapolation is driven by the model's tail behaviour, which the near-the-money data does not constrain. Recognising and reporting this range-of-trustworthiness is a hallmark of mature modelling practice. It is often the first question a risk committee asks before approving the use of a model: "for what range of instruments and market conditions has this model been validated?"

> **Key point.** The tree expectation is taken under the *bond-numeraire* risk-neutral measure, i.e. payoffs are discounted by $\prod(1 + r_{m-1}\Delta t)$ along each path, which is the discrete analogue of discounting by $\exp(\int_0^T r_u \, du)$ under $\mathbb{Q}^M$. Chapter 5 formalised this numeraire-by-numeraire correspondence in continuous time.

Recovered short-rate tree *Recovered short-rate tree — given two bond quotes and a volatility spread c, the pair* (r_0, r_{11}) *is pinned down by the no-arbitrage mixing relation (11.22).*

Calibrated four-step short-rate tree *Layered extension: a four-step recombining short-rate tree whose node rates have been bootstrapped from a sequence of bond quotes* $P_0(1), \dots, P_0(4)$ *and a term structure of volatilities. Each time-slice is pinned down by the freshly observed bond; the tree is arbitrage-free by construction and exactly reprices its calibration instruments.*

Bond-price lattice with calibrated rates at each node *The same calibrated tree, now annotated with the implied one-period bond value* $P = 1/(1 + r\,\Delta t)$ *at each node. This is the object pricing routines roll back over: each step multiplies the weighted average of child-node bond values by the parent's discount factor, and matching this roll-back to the observed* $P_0(T)$ *quotes pins down the calibration.*

13.3.3 11.3.3 Bid-Ask, Stale Quotes, and Data Hygiene

Bid-ask spreads and stale quotes. In real markets, $P_0(T)$ is not a single number but a bid-ask pair $(P^{\text{bid}}, P^{\text{ask}})$, and between transactions the quote may be stale — indicating only an indicative level rather than a tradable price. Calibration that targets an arbitrary mid-price within the bid-ask can give nonsensical parameter wobble: if the spread is 2 basis points and calibration is trying to fit to 0.1 basis point precision, the resulting parameters will move randomly within a band set by the spread even when no economic information has changed. A disciplined calibration treats the bid-ask as a confidence interval: fit the mid, but accept any parameters that produce model prices inside the bid-ask as equally valid. Stale quotes are even more pernicious; if a bond hasn't traded in a week, its "price" is an echo, and calibrating to it pollutes the rest of the curve.

An instructive exercise is to simulate the impact of bid-ask noise on a calibrated parameter vector. Take a realistic fit — say, a Vasicek calibration to ten bond quotes — and perturb each input yield by a uniform random amount in $[-\text{spread}/2, +\text{spread}/2]$. Rerun the calibration for many such perturbations, and plot the resulting distribution of calibrated parameters. If the spread is 2 basis points and the parameter distribution spans 10% of the point estimate, the calibration is reporting precision far in excess of what the data supports. A well-calibrated system's reported precision should roughly match the width of this simulated perturbation distribution — any tighter is over-confidence, any looser is under-utilisation of the data. This exercise is worth running periodically as a calibration sanity check; it catches both model misspecification and implementation bugs that compress or inflate the reported precision artificially.

Liquidity weighting. A related practical refinement: weight each quote in the calibration objective by its liquidity, proxied by trading volume or quote frequency. A quote for an on-the-run 10-year Treasury is updated every second and traded millions of times a day; a quote for an off-the-run 17-year Treasury may be updated only a few times a day and traded in small size. Both quotes are nominally "mid prices", but the 10-year mid is a much more trustworthy signal. Weighting by volume (or by the inverse of some proxy for staleness) is one way to ensure the calibration reflects the information in the market rather than the information in yesterday's stale print. A common scheme: weight each residual by the instrument's proportion of recent trading volume, normalised so that the weights sum to one.

Data hygiene. Before calibration runs, a disciplined pipeline filters quotes according to several rules: drop quotes older than a threshold (typically 30 minutes for rates, shorter for equities); drop quotes with bid-ask spreads wider than some multiple of the instrument's typical spread; drop outliers that disagree with nearby quotes by more than an error bar. These hygiene steps are mundane but they account for a substantial share of the "why did the calibration jump" incidents that quantitative groups fire-drill around. A good calibration system logs exactly which quotes were accepted and which were rejected, so that post-hoc investigations can reconstruct the decision chain. The lattice bootstrap is a good place to enforce this discipline because its layer-by-layer structure makes it easy to localise a bad quote to a specific maturity: if the fit blows up at year 7, the culprit is typically a stale or cross-dealer-inconsistent 7-year quote, and an operator can verify this in seconds.

13.4 11.4 Roadmap: From Lattices to Continuous-Time Models

The three settings we have worked through — the single-period binomial, the multinomial lattice with optimisation and regularisation, the recursive bond-lattice bootstrap — share a common skeleton. Each instance begins with FTAP (11.1), translates observed market prices into equations for unknown parameters, and solves the resulting system either in closed form or by optimisation. The lattice structure is a pedagogical convenience: it makes the equations transparent, the arbitrage conditions explicit, and the Radon–Nikodym density computable by hand. But no real trading desk actually runs its production calibration on a binomial tree for a complex product. The next step — the one this part of the course is about to take — is to replace these discrete lattices with their continuous-time limits, keeping the calibration logic but upgrading the dynamics.

The pattern of that upgrade is already visible in the binomial case. Equations (11.5) and (11.6) said the step size scales as $\sigma\sqrt{\Delta t}$ and the drift-tilt scales as $(\hat{\mu}/\sigma)\sqrt{\Delta t}$. Taking $\Delta t \to 0$ under this scaling, the sum of iid Bernoulli log-returns converges by the CLT to Brownian motion with drift $\hat{\mu}$ and diffusion σ. That is the Donsker limit. In continuous time the asset becomes a geometric Brownian motion; its log is a Brownian motion with drift; Girsanov (Chapter 5) gives the change of measure from $\mathbb{P}$ to $\mathbb{Q}$ explicitly as a shift in drift. The same argument, applied to a short-rate lattice with a mean-reverting drift, produces the Vasicek SDE $dr_t = \kappa(\theta - r_t)\,dt + \sigma\,dW_t$ in the limit. That SDE is the subject of **Chapter 12**, where we derive its explicit solution, the distribution of r_T, the distribution of $\int_0^T r_u\,du$, and the closed-form bond price. We also pick up the Ho–Lee continuous limit and the Hull–White time-dependent extension, all developed from a single canonical Vasicek backbone. The three separate continuous-time short-rate treatments that originally lived in the old guide are consolidated there.

The applications of those continuous-time short-rate models — interest-rate swaps, swap rates, credit default swaps, callable bonds, European bond options, the T-forward measure in use rather than in preview — are the subject of **Chapter 13**. The T-forward measure is particularly important: by choosing the T-maturity zero-coupon bond as numeraire, the bond price itself becomes a tradable whose discounted process is a martingale, and the stochastic discounting that clutters expectations under the money-market measure $\mathbb{Q}^M$ is absorbed into the numeraire. The result is a pricing machinery that produces Black-type closed forms for caplets and bond options. Chapter 13 derives all of these applications by cross-reference to Chapter 5 (for Girsanov and measure change) and Chapter 12 (for the short-rate dynamics); it does not re-derive Vasicek, and it does not re-derive Girsanov. The calibration patterns from the present chapter carry over directly: every application in Chapter 13 is, ultimately, another instance of FTAP (11.1) applied to a specific payoff and numeraire.

Two themes thread through the rest of the course. The first is that *calibration is a thin wrapper around pricing*: once the forward pricing function $\Theta \mapsto C^{\text{model}}(\Theta)$ is known, the calibration optimisation (11.15)–(11.19) is a generic numerical procedure. Chapter 12 and Chapter 13 contribute the forward pricing functions for the rates products; Chapter 11 has contributed the optimisation and regularisation machinery. The second theme is that *the lattice is the right mental model even when the production code is continuous-time.* When a Vasicek calibration misbehaves in Chapter 12, the diagnostic question is always "what does this failure look like on a coarsened trinomial lattice?" The tree makes the arbitrage conditions visible; the SDE hides them inside an Itô calculus. A calibration engineer who can translate freely between the two representations has both the rigour of the continuous-time machinery and the intuition of the lattice.

Chapter 14 (Heston stochastic volatility) and Chapter 15 (caps, floors, swaptions) will use every piece of the calibration machinery developed here: market-consistent objectives, vega weighting, regularisation, warm starts, non-identifiability diagnostics. The Heston calibration in particular is a notoriously ill-posed problem — the parameters $(\kappa, \theta, \sigma_v, v_0, \rho)$ have strong pairwise correlations in the likelihood surface, the objective is multimodal, and naive optimisers wander between local minima. Every remedy we have described here (reparameterisation in orthogonal coordinates, multi-start optimisation, regularisation toward a historical prior, cross-validation on held-out strikes) is used in practice. By the time the reader reaches Chapter 14, the calibration apparatus should feel routine; only the Heston-specific pricing machinery (Fourier inversion of the characteristic function, Riccati ODEs for the transform) will be new.

13.4.1 Case study: SPX daily vol-surface calibration in production

Every morning, before the 9:30 ET open, the SPX index-option desk at any major bank refits a Heston (or hybrid SLV) surface to the previous afternoon's 4:00 PM closing implied vols. The output is locked at 8:00 AM and used unchanged through the trading day for risk, mark, and quoting; nothing about the workflow tolerates a 30-second hiccup or a 50 *bp residual. This is calibration as production infrastructure, not as a research exercise.*

Context. The calibration basket is roughly 20 strikes (typically 70%–130% of spot in 2% increments, with denser sampling near ATM) times ~ 10 maturities (1, 2, 3, 6, 9, 12, 18, 24, 36, 60 months) — about 200 option mids. Each mid is converted to a Black–Scholes implied vol via a one-dimensional bisection on price, with the ATM-vol used to seed neighbouring strikes for fast convergence. Bid-ask varies sharply across the surface: ~ 0.3 vol points ATM at the front month, blowing out to 4–6 vol points for 1y deep-wing strikes. The pre-calibration data hygiene step (the Θ-independent filter sketched in §11.3.3) drops any quote with bid-ask wider than $3\times$ its surface neighbours, any quote whose mid violates put-call parity by more than 0.5%, and any quote on a strike with zero open interest the prior day. A typical day after filtering retains 170–190 usable mids.

The optimiser is Levenberg–Marquardt over the five-parameter Heston vector $\Theta = (\kappa, \theta, \sigma_v, v_0, \rho)$, *warm-started* from yesterday's calibrated values Θ_{t-1}. Each iteration costs one Jacobian (built by parallel finite-difference Heston FFT reprices, ~ 200 pricer calls) plus one 5×5 normal-equation solve; on a single modern core that is roughly 1–2 seconds per iteration, with 8–15 iterations to converge for a typical day — total wall-clock around 20–30 seconds. The objective is vega-weighted IV-RMSE per (11.17) plus a small Tikhonov term $\lambda\|\Theta - \Theta_{t-1}\|^2$ at $\lambda \approx 10^{-3}$ to suppress identifiability-driven overnight drift in the (κ, σ_v) direction. On a calm day the converged RMS residual is 10–20 bp on IV; on a moderately active day, 20–30 bp; anything above 50 bp is the desk's red flag for a "broken surface" and triggers escalation.

Reading it through the chapter's math. The whole pipeline is one instance of the FTAP-driven inverse problem of §11.1 wrapped in the optimisation machinery of §11.2. Vega weighting (11.17) is exactly the right choice when the calibration target is "the model marks should agree with the market in the units the trader cares about" — vega-units, not price-units. The warm-start Tikhonov regularisation is the §11.2.7 remedy for the well-known Heston identifiability pathology: (κ, σ_v) traces a long, shallow valley in the likelihood surface (mean-reversion speed and vol-of-vol trade off against each other to produce nearly identical short-dated smiles), and without the regulariser the optimiser walks along that valley overnight and reports 30% parameter changes between two indistinguishable surfaces. The Θ_{t-1} anchor pins the optimiser to the same basin, so the day-to-day report is the *informative* component of the parameter change rather than a numerical artefact. Day-to-day stability under this regime is empirically: ρ moves by less than ± 0.05 in 90% of days (it is the most stable parameter, since the smile-skew slope is nearly orthogonal to all the other directions); v_0 moves with realised overnight vol (it is essentially "where the front-month ATM vol started this morning"); θ drifts slowly, often unchanged at the third decimal for days at a time; κ is the regime indicator — a κ jump from 1.8 to 3.5 on a single day is the model telling you the term-structure shape changed materially, and is itself a tradable signal.

Lesson. "Calibrate at 8 AM, hedge from 9:30" is the operating discipline that the entire apparatus of this chapter is built to support. The lock-at-8 rule exists because *every* downstream system — risk reports, exotic marks, Greek aggregations, the swap-broker's quote engine — needs a single, frozen surface to compute against; you cannot have the model wobbling under risk while the trader is trying to read the book. That constraint dictates the algorithm choice (Levenberg–Marquardt's superlinear local convergence, not differential evolution's reliability), the regularisation choice (warm-start Tikhonov, not absolute Bayesian prior), the data-hygiene discipline (filter before fit, not iterate during fit), and the residual threshold (50 bp = surface broken = escalate). When calibration *does* fail — a Fed surprise, a single-name event affecting an index basket member, a flash crash leaving a discontinuous strike profile — the desk falls back through a hierarchy: first try regularised SVI on the implied-vol surface directly (a model-free smile fit, no dynamics), then if even SVI fails, freeze yesterday's surface and quote bid-only at wide spreads until the chaos passes. The math (Heston, SVI), the optimiser (Levenberg–Marquardt), the regulariser (Tikhonov), and the operational hierarchy (escalate at 50 bp) are inseparable; remove any one and the production process collapses.

13.5 11.6 Key Takeaways

1. **FTAP is the calibration engine.** Every model parameter is pinned down by requiring (11.1) to hold at observed prices. Market quotes are equations; parameters are unknowns; calibration is just solving the system.
2. **Calibration is an inverse problem.** It can be ill-posed in three classic ways: non-existence (no Θ reproduces the quotes exactly), non-uniqueness (many Θ's reproduce them), and instability (small price perturbations cause large parameter jumps). Each pathology has a distinct remedy: weighted least squares, regularisation, and warm-start smoothing respectively.
3. **Binomial scaling.** $c \sim \sigma\sqrt{\Delta t}$ and $p \sim \frac{1}{2}(1+(\hat{\mu}/\sigma)\sqrt{\Delta t})$ — moment matching on the lattice. This recovers the Black–Scholes limit because matching mean and variance per step is exactly what the CLT needs to produce Brownian motion. Chapter 3 proved the corresponding continuous-time construction; Chapter 6 uses it to derive the BS PDE.
4. **Change of numeraire on a lattice.** $d\mathbb{Q}^B/d\mathbb{Q}^A = (B/B_0)/(A/A_0)$. The same payoff can be priced under any consistent (measure, numeraire) pair — pick the numeraire that simplifies the expectation. Chapter 5 developed the continuous-time canonical derivation.
5. **Bond-price ratios recover (q, r_0).** $r_0 = 1/P_0(1) - 1$ and the two-period relation (11.22) recovers the next tree layer given a volatility input. Sequential bootstrapping builds the full tree. Chapter 12 develops the continuous-time Vasicek / Ho–Lee / Hull–White models whose lattice discretisation this construction is.
6. **Identifiability matters.** Some parameter directions are stiff (well-pinned by the data), others are sloppy (nearly flat likelihood). For single-factor Vasicek the long yield identifies the ratio σ^2/κ^2 alone — the individual values of σ and κ are not pinned by bond quotes and require additional vol-instrument calibration (caplets, swaptions; see Chapter 13) to break the degeneracy.
7. **Bid-ask and stale quotes set the precision floor.** Never calibrate tighter than the market's own resolution. Overfitting inside the bid-ask band produces spurious parameter instability. A disciplined pipeline filters out stale quotes and outliers before calibration runs.
8. **Interpolation vs extrapolation.** A calibrated model reproduces its inputs by construction; that does not make its outputs trustworthy outside the calibration range. Always distinguish between "what the market told us" and "what the model guesses".
9. **Calibration objective design is load-bearing.** The choice between unweighted, spread-weighted, vega-weighted, and regularised objectives is not cosmetic — different choices lead to different parameter estimates, different stability properties, and different out-of-sample behaviours. Make the choice explicitly and revisit it when market conditions change.
10. **Numerical methods must match the problem shape.** Nelder–Mead for noisy low-dimensional fits; Levenberg–Marquardt for smooth nonlinear least squares; BFGS for smooth general objectives; differential evolution for multimodal global search. A production system uses hybrids with warm starts and periodic multi-start checks.
11. **Out-of-sample testing is mandatory.** A model that fits in-sample beautifully but fails on held-out data is overfitting; production use of such a model is dangerous. Cross-validation and held-out strikes/maturities are essential discipline.
12. **Regime-awareness protects against silent calibration failure.** Monitor residuals, parameter-change magnitudes, and fit quality over time; escalate unusual patterns to human judgment rather than letting the

system keep running silently.

13. **Calibrate vs estimate are not synonyms.** $\mathbb{Q}$-parameters (from market quotes) price derivatives; $\mathbb{P}$-parameters (from historical data) drive risk scenarios. Do not mix them.
14. **Calibrate to what you hedge with.** The right calibration basket is the one whose instruments you plan to use as hedges; fitting a richer basket than the hedge set can produce consistent prices but inconsistent hedge ratios.

13.6 11.7 Reference Formulas

13.6.1 11.7.1 Binomial lattice

$$A_n = A_{n-1}e^{c\,x_n}, \quad x_n \overset{\text{iid}}{\sim} \text{Bernoulli}(\pm 1)$$

$$c \;\approx\; \sigma\sqrt{\Delta t}, \qquad p \;\approx\; \tfrac{1}{2}\Big(1 + \tfrac{\hat{\mu}}{\sigma}\sqrt{\Delta t}\Big)$$

Risk-neutral version: replace $\hat{\mu} \to r - \frac{1}{2}\sigma^2$ for a lognormal asset.

13.6.2 11.7.2 Change of measure on a lattice

$$q^B = q^A\,\frac{B_u/A_u}{B_0/A_0}, \qquad 1 - q^B = (1 - q^A)\,\frac{B_d/A_d}{B_0/A_0}$$

$$\frac{d\mathbb{Q}^B}{d\mathbb{Q}^A}(\omega) \;=\; \frac{B(\omega)/B_0}{A(\omega)/A_0}$$

13.6.3 11.7.3 Zero-coupon bonds on a lattice

$$P_t(T) \;=\; \frac{1}{1 + r_t}\,\mathbb{E}_t^{\mathbb{Q}}[P_{t+\Delta t}(T)]$$

$$r_0 \;=\; \tfrac{1}{P_0(1)} - 1, \qquad M_T \;=\; \prod_{m=1}^{T/\Delta t}(1 + r_{m-1}\Delta t)$$

In the continuous-time limit (developed in Chapter 12),

$$P_t(T) \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_u\,du\right)\right]$$

13.6.4 11.7.4 Calibration objective functions

Unweighted least squares:

$$\Theta^\star \;=\; \arg\min_{\Theta}\sum_i \big(C_i^{\text{model}} - C_i^{\text{mkt}}\big)^2$$

Weighted least squares (bid-ask or liquidity):

$$\Theta^\star \;=\; \arg\min_{\Theta}\sum_i w_i\big(C_i^{\text{model}} - C_i^{\text{mkt}}\big)^2, \quad w_i = 1/\text{spread}_i^2$$

Vega-weighted (implied-vol-space):

$$\Theta^\star \;=\; \arg\min_{\Theta}\sum_i \big(\sigma_i^{\text{model}} - \sigma_i^{\text{imp,mkt}}\big)^2$$

Regularised:

$$\Theta^\star \;=\; \arg\min_{\Theta}\left[\sum_i w_i\big(C_i^{\text{model}} - C_i^{\text{mkt}}\big)^2 + \lambda\,\|\Theta - \Theta_{\text{prior}}\|^2\right]$$

13.6.5 11.7.5 Identifiability diagnostics

Hessian condition number:

$$\kappa(H) \;=\; \sigma_{\max}(H)/\sigma_{\min}(H)$$

- $\kappa(H) \sim 10$: well-posed
- $\kappa(H) \sim 10^3$: marginal; consider regularisation
- $\kappa(H) \sim 10^6$: ill-posed; restructure problem

13.6.6 11.7.6 Stepwise calibration checklist

1. Load and filter market quotes; drop stale and outlier observations.
2. Convert to calibration targets (yields for bonds, implied vols for options).
3. Set weights (bid-ask, vega, or liquidity based).
4. Initialise from yesterday's parameters (warm start) or from a default prior (cold start).
5. Run local optimiser (Levenberg–Marquardt or BFGS) to convergence.
6. Check convergence diagnostics: residuals, condition number, profile likelihood.
7. Sanity check parameters against historical ranges and economic plausibility.
8. Verify out-of-sample fit on held-out instruments.
9. Persist calibrated parameters with metadata (timestamp, residuals, diagnostics).

Cross-references. Chapter 2 (FTAP on the multi-period binomial) is the direct predecessor of §11.1; Chapter 5 (Radon–Nikodym, Girsanov, density process) is the canonical treatment of the measure-change machinery whose discrete-lattice instance is §11.2. Chapter 12 develops the continuous-time Vasicek / Ho–Lee / Hull–White short-rate models whose lattice discretisation this chapter has been calibrating. Chapter 13 deploys those continuous-time models in applications (IRS, CDS, bond options, callable bonds, T-forward measure). Chapter 14 (Heston) applies the ill-posed-calibration remedies surveyed in §11.2 to a stochastic-volatility setting with notoriously correlated parameter directions.

13.7 Appendix A. Calibration Governance and Sociology

(Demoted from former §11.5 in V5: technical chapters now end with takeaways and reference formulas; this material covers institutional context useful to practitioners.)

13.7.1 A.1 The Sociology of Calibration

Quantitative finance is often taught as pure mathematics, but calibration is a profoundly social activity. The calibration team in a bank interacts daily with traders (who want their marks to reflect where they think the market is — even if that disagrees with the quoted mid), risk managers (who want parameters to be conservative, stable, and auditable), sales and structuring (who want competitive quotes on new trades, which means tight fits), operations (who want the calibration to complete reliably overnight without human intervention), and auditors (who want the whole process to be documented, reproducible, and governed). These stakeholders have genuinely conflicting priorities. A trader wants a tight fit; a risk manager wants stability. Sales wants aggressive quotes; operations wants simple code. Resolving these tensions is the real work of the quantitative calibration function, and it requires technical tools (the regularisation and weighting schemes we have surveyed) plus governance structures (change-management processes, model validation, override logs). A calibration system is not just a piece of code; it is a sociotechnical artifact that embodies the bank's approach to these tradeoffs.

13.7.2 A.2 Calibration vs Estimation: A Terminological Aside

The words "calibration" and "estimation" are sometimes used interchangeably but should not be. *Estimation* is a statistical procedure: given data generated by an unknown process, infer the process's parameters. *Calibration* is a pricing procedure: given market quotes generated by a pricing process, infer the parameters that reproduce the quotes.

The distinction matters because the objectives differ. Estimation minimises some notion of statistical discrepancy — likelihood, L^2 distance from empirical moments, whatever. Calibration minimises *pricing* discrepancy — the difference between model-implied prices and observed market prices. In the Vasicek case developed in Chapter 12, the calibration objective minimises squared yield errors; this is not quite the same as the maximum-likelihood estimator of the Vasicek parameters from historical short-rate data, though in well-behaved cases they should produce similar answers.

A practical consequence: parameters "calibrated" to market quotes are under the *risk-neutral* measure $\mathbb{Q}$, while parameters "estimated" from historical data are under the *physical* measure $\mathbb{P}$. The two measures differ by the market price of risk (Girsanov's theorem, Chapter 5), and failing to distinguish them is a common cause of confusion and error. For pricing applications, use $\mathbb{Q}$-calibrated parameters. For risk management (e.g. computing historical VaR, Chapter 9), use $\mathbb{P}$-estimated parameters. Mixing the two will produce systematic biases.

13.7.3 A.3 The Role of Hedging in Calibration Selection

The choice of calibration instruments should be driven by the desk's hedging strategy. If you plan to hedge an exotic option using a basket of vanilla options at specific strikes and maturities, calibrate your model to exactly that basket. This ensures the model reproduces the hedge-instrument prices and therefore produces consistent hedge ratios. Calibrating to a different basket, even a superset, can produce incorrect hedge ratios, leading to hedge slippage and P&L surprises.

A concrete example: suppose you hedge a five-year cliquet using variance swaps at $1, 2, 3, 4, 5$ year maturities. Calibrate your model to exactly these variance swap prices. The calibrated model will price your cliquet in a way consistent with the hedge portfolio, so the sum of (cliquet mark + hedge portfolio mark) is stable under market moves to first order. Calibrate instead to a fine grid of vanilla options — which contains *more* information than the variance swap quotes — and the cliquet mark may drift relative to the hedge, producing spurious P&L. This principle is sometimes called "calibrate to what you hedge with". It is one of the most important practical lessons of derivatives pricing, and it explains why there is no universal best calibration recipe — the calibration that is right for your desk depends on the specific products and hedge mix.

13.7.4 A.4 The Choice of Calibration Frequency

How often should a model be recalibrated? The answer depends on the volatility of the market, the cost of computation, the stability properties of the model, and the business's tolerance for mark shifts. Market-making desks may recalibrate implicitly every time they receive a quote, using fast local methods (Levenberg–Marquardt with warm starts) to re-fit in milliseconds; end-of-day calibration against closing quotes is the standard for marking the book and computing regulatory metrics; slower-moving parameters (e.g. mean-reversion speeds in rates models) may be recalibrated only weekly or monthly, producing more stable marks at the cost of slower adaptation to regime changes; event-driven calibration triggered by a central-bank announcement, a volatility spike, or a residual outside a threshold combines stability with responsiveness. A well-designed calibration system supports multiple frequencies simultaneously, with different models tied to different frequencies as appropriate.

Calibration frequency trade-off: stability vs responsiveness *A synthetic parameter that drifts smoothly through a regime break at day 140.* ***Daily*** *recalibration (faint red) is responsive but noisy;* ***monthly*** *recalibration (blue) is stable but lags the regime shift by up to a month;* ***weekly*** *recalibration (orange) sits in between. No frequency dominates — the trade-off is real and the choice belongs to the desk's risk-appetite statement.*

13.7.5 A.5 The Limits of Market Consistency

We have emphasised market consistency throughout this chapter, but it has limits. A market-consistent model reproduces observed prices by construction. It does *not* predict future prices; it cannot anticipate regime shifts; it cannot distinguish an informed quote from a stale one; and it cannot correct for market mispricings that your analysis has identified.

A value-oriented investor or discretionary trader who has an independent view about what prices *should be* will deliberately diverge from market-consistency. They will use models calibrated to *their* view of fair value, not to the market's quoted mid. This is a different modelling paradigm, and it is appropriate for certain seats (prop trading, asset management, hedge funds) but not others (market making, flow derivatives). For the market maker, consistency with the market is a must; for the prop trader, the whole point is to be inconsistent with the market in defensible directions. Calibration, in the market-consistent sense we have been developing, is therefore a *specific* modelling discipline associated with *specific* business functions.

13.7.6 A.6 Connections to Machine Learning

In recent years, there has been growing interest in using machine-learning techniques for calibration. The motivation: neural networks can learn the model pricing function (the map from parameters to prices) once, and then calibration reduces to inverting this learned function — which can be orders of magnitude faster than re-solving the pricing problem at every optimiser step. A typical setup: generate a large training dataset of (parameters, prices) pairs by running the model forward on randomly sampled parameters, train a neural network to approximate the pricing function, and at calibration time solve an inverse problem over the network using gradient-based methods that backpropagate through it. This approach has been applied successfully to Heston calibration (Chapter 14), Bermudan options, and rough-vol models. The downsides: the network has to be retrained when the model specification changes; accuracy in extreme parameter regions is limited by training-set coverage; and the system inherits the stability issues of neural networks including adversarial sensitivity and out-of-distribution failures. For stable well-studied models, the learning-accelerated calibration is often worth the complexity; for experimental or custom models, a traditional iterative calibration is often simpler.

13.7.7 A.7 Benchmarking Calibration Performance

How do you know your calibration is doing a good job? Several benchmarks are worth tracking on a daily basis. *In-sample fit quality* — the average residual across calibration instruments — should be comparable to the bid-ask spread, not dramatically smaller (a sign of overfitting) or larger (a sign of model misspecification). *Out-of-sample fit quality* on instruments not in the calibration basket should be comparable to in-sample fit; a large gap indicates overfitting. *Temporal stability* — the day-over-day change in calibrated parameters on a calm day — should be small; large daily jumps suggest either an unstable calibration or a genuine market move, and careful attribution is needed to distinguish them. *Consistency across product lines*: if the same parameters price multiple products, residuals on each product's validation instruments should be comparable. *Predictive power*: for models used in forward-looking analysis, does the simulated distribution of future prices match what actually happens? A mature calibration function tracks all of these metrics on dashboards, alerts on unusual patterns, and escalates to human review when any threshold is breached.

13.7.8 A.8 Calibration Governance and Model Risk

Calibration is a regulated activity. Post-2008, banks are required to maintain formal model-risk-management frameworks that cover calibration procedures. The typical requirements include documentation of the calibration methodology (choice of instruments, weighting schemes, regularisation, numerical methods), independent validation of statistical properties and out-of-sample performance, monitoring of calibration outputs over time with alerts for unusual residuals or parameter jumps, override tracking of any manual adjustments with documented rationale, and periodic review of the methodology to ensure it remains fit for purpose under changing market conditions. These requirements are not bureaucratic overhead; they are defenses against the failure modes we have discussed throughout the chapter. A well-calibrated model that no one monitors will silently drift into a bad regime; a well-monitored model with bad calibration methodology will produce a documented trail of bad answers. Good governance combines technical quality with organisational discipline.

13.7.9 A.9 A Final Pragmatic Perspective

Calibration sits at the intersection of mathematics, statistics, software engineering, and institutional judgement. It is a deeply practical discipline, yet it rests on theoretical foundations (FTAP, change of measure, affine models) as elegant as any in quantitative finance. The practitioner who masters calibration is mastering the translation layer between the market's abstract signal and the model's concrete prediction — a translation performed dozens of times a day, across every desk and every product, at every institution that trades derivatives. Calibration is an inverse problem; it is often ill-posed in subtle ways; and the three classical pathologies (non-existence, non-uniqueness, instability) each call for distinct remedies. Moment matching is the cleanest tool for lattice calibration; change-of-numeraire machinery is a practical tool, not an abstract indulgence; and identifiability is never guaranteed. Bid-ask spreads set a precision floor. With these tools and mindsets in place, calibration becomes not a mysterious black art but a disciplined craft — one where mathematics, code, and judgement combine to extract useful signal from a noisy, over-determined market, in a way that is auditable, reproducible, and defensible.

14 Chapter 12 — Short-Rate Models: Vasicek, Ho-Lee, Hull-White, Affine

Part IV — Interest-Rate Models. Depends on Chapter 3 (Itô, OU solution), Chapter 4 (Feynman-Kac for bond pricing), Chapter 5 (Girsanov for the passage from the physical measure $\mathbb{P}$ to the risk-neutral measure $\mathbb{Q}$), and Chapter 11 (calibration on lattices).

This is the chapter in which the yield curve becomes an object we can *model* rather than merely observe. The calibration chapter (Chapter 11) built short-rate lattices by matching observed bond prices term by term; here we replace the lattice with a continuous-time stochastic differential equation and derive closed-form bond prices, yield curves, and volatility structures from the dynamics themselves. Three models — Vasicek, Ho-Lee, and Hull-White — are developed as a single narrative, with Vasicek as the canonical derivation that the other two specialise or extend. The applications (interest-rate swaps, CDS, bond options, callable bonds) live in Chapter 13; the caps, floors, and swaptions capstone lives in Chapter 14. This chapter's single job is to build the model.

14.1 12.1 Motivation and the traded primitives

In equity pricing the underlying S_t is itself the traded asset, so no-arbitrage directly pins its $\mathbb{Q}$-drift at r. In rates the state variable r_t and the tradables are *different*: the short rate itself cannot be bought, so the Fundamental Theorem of Asset Pricing pins down $\mathbb{Q}$ only through its action on what *is* traded. This is the conceptual pivot of the chapter, and it is what makes a "calibration" step necessary for rates models in a way that it isn't for equities.

The traded primitives are:

- the money-market (bank) account

$$\mathrm{d}M_t = r_t\, M_t\, \mathrm{d}t, \qquad M_0 = 1, \tag{12.1}$$

equivalently $M_t = \exp(\int_0^t r_s\, \mathrm{d}s)$; - zero-coupon bonds of various maturities T, with price process $\big(P_t(T)\big)_{0 \le t \le T}$ and terminal condition $P_T(T) = 1$. Bonds are contingent claims on the *path* of r_s — every bond pays \$1 at maturity, so its value today is determined by the law of $\int_0^T r_s\, \mathrm{d}s$.

The ambition of the short-rate programme is dramatic compression: posit a single scalar SDE for r_t, and from that one equation recover the entire yield curve plus consistent prices for every rate-sensitive derivative written on it. The compression succeeds because the required structural ingredients (linear drift, linear-or- state-dependent squared volatility, Markovian state) are exactly the conditions for the affine-pricing apparatus to work. A non-affine short-rate model would fail to produce tractable bond prices, and would therefore fail to support the edifice of rate-derivative pricing. Affine structure is both the secret of the model's success and the source of its limitations.

Short-rate models are *not* forecasting tools. A Vasicek model fed today's yield curve and today's implied vols says nothing useful about whether rates will rise or fall under $\mathbb{P}$; it says only what price today's arbitrage conditions impose on a derivative, given the $\mathbb{Q}$-dynamics. The term premium — the gap between $\mathbb{Q}$ and $\mathbb{P}$ expectations of future rates — is a free input to the model, not an output. Macroeconomics, surveys, or factor models answer "where will rates go?"; Vasicek and its cousins answer "what is the arbitrage-free swaption price given the dynamics?" Confusing the two is the most common conceptual mistake in rates modelling.

14.1.1 12.1.1 Non-uniqueness of $\mathbb{Q}$ on a rate tree

On an equity tree the risk-neutral probability q is pinned down by no-arbitrage once u, d, r are fixed: $q = (e^{r\Delta t} - d)/(u-d)$. On a rate tree the corresponding q is *not* unique, because the FTAP pins $\mathbb{Q}$ only through tradables and the short rate isn't one. Different short-rate dynamics can reproduce the same observed bond prices, and arbitrage cannot distinguish between them. Chapter 11 §11.3 made this discrete-time underdetermination concrete: at every step the tree adds one observation (a bond quote) and more than one free parameter (rate nodes plus a transition probability), so the system is perpetually under-determined and external information (calibration to market bonds) is needed to pin down the model.

> **Slogan.** *The short-rate tree is not the price of a traded asset, so q is not uniquely determined by no-arbitrage. The market bond prices supply the external information that pins down θ.*

This is the discrete-time shadow of the continuous-time statement that Hull-White has infinitely many $\theta(t)$ choices for any given (κ, σ), and only the curve-fit picks one out.

14.2 12.2 Discrete motivation: AR(1) and the continuous limit

Why does an AR(1) belong here at all? Because empirical short-rate series (Fed funds, EONIA, SONIA, SOFR) look remarkably unlike random walks and remarkably like stationary mean-reverting processes. Plot a multi-decade history of a policy rate and it moves in slow waves that hover around levels broadly set by trend inflation and potential growth; it does not diffuse toward infinity the way a geometric Brownian motion would. The simplest model that captures this is a linear combination of yesterday's rate, a drift toward a mean, and a Gaussian shock — the discrete AR(1):

$$r_n \;=\; \alpha\, r_{n-1} \;+\; \beta_n \;+\; \sigma\sqrt{\Delta t}\,\varepsilon_n, \qquad \varepsilon_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1). \tag{12.2}$$

Rewriting (12.2) in "innovation form,"

$$r_n - r_{n-1} \;=\; \kappa(\theta - r_{n-1})\,\Delta t \;+\; \sigma\sqrt{\Delta t}\,\varepsilon_n, \tag{12.3}$$

makes the correspondence with a continuous mean-reverting SDE transparent. Identifying $\alpha \leftrightarrow 1-\kappa\Delta t$, $\beta_n \leftrightarrow \kappa\theta\Delta t$, and $\varepsilon_n\sqrt{\Delta t} \leftrightarrow \mathrm{d}W_t$, and sending $\Delta t \to 0$, recovers the Vasicek SDE

$$\mathrm{d}r_t \;=\; \kappa(\theta - r_t)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t. \tag{12.4}$$

The persistence α and the mean-reversion speed κ are two views of the same quantity: $\alpha = e^{-\kappa\Delta t}$ in exact correspondence, and for small Δt this is well approximated by $1-\kappa\Delta t$. A daily AR(1) with $\alpha = 0.998$ corresponds to $\kappa = -\ln(0.998)/(1/252) \approx 0.50$ — a roughly-annual mean-reversion speed consistent with empirical policy rates.

The other useful intuition from the AR(1) view is that the continuous-time process r_t has *memory*. Under geometric Brownian motion (the Black-Scholes stock model) today's price contains no information about tomorrow's trajectory beyond its own level; under Vasicek, today's rate contains information about the *deviation from the mean* that partially persists. This memory is what makes mean reversion predictable at longer horizons — not because we can forecast the noise, but because we know the drift will systematically pull the process back toward θ at an estimable rate. For derivatives pricing this matters: an option that pays off a function of the integrated rate over time is fundamentally a bet on these mean-reverting dynamics, and its price depends sensitively on how much memory the process has.

14.2.1 12.2.1 Why continuous time?

The continuous limit is not just aesthetic. In discrete time, pricing a T-year zero-coupon bond means computing an expectation of a product of $T/\Delta t$ correlated random discount factors — a high-dimensional integral with no closed form. In continuous time, by contrast, the same object becomes an expectation of the exponential of a Gaussian integral, whose moment-generating function is explicit. Continuous time is what unlocks the closed-form bond price $P_t(T) = e^{A_t(T) - B_t(T) r_t}$ that makes the calibration and hedging machinery workable.

The gap between a continuous-time Vasicek and a sensible daily discrete approximation is, in practice, negligible compared with model-specification error, parameter-estimation error, or liquidity effects. Nobody should lose sleep over the "continuous vs discrete" dichotomy once calibration is done. What matters is that continuous time gives us a tractable language in which to express and solve pricing problems; the language is a tool, not a claim about ontology.

There is also a pedagogical virtue to continuous time. In discrete time, the various random-walk and AR(1) models look superficially similar — they differ only in the fine print of their coefficients — and it is easy to lose sight of which model is adequate for a given problem. In continuous time the functional forms of drifts and diffusions are starker and more revealing: a constant drift is obviously different from a linear-in-state drift, a constant diffusion is obviously different from a $\sqrt{r}$ diffusion. The continuous-time formulation makes the modelling choices *legible*, and legibility is a precondition for careful thinking.

14.2.2 12.2.1.b Estimating AR(1) Vasicek from historical data

The AR(1) representation (12.3) is useful not only pedagogically but for parameter estimation. Given a time series of observations $\{r_0, r_1, \dots, r_N\}$ on a Δt grid, ordinary least squares regression of r_n on r_{n-1} yields estimates of the slope $\beta = 1 - \kappa\Delta t$ and intercept $\kappa\theta\Delta t$, from which $\kappa = (1-\beta)/\Delta t$ and $\theta = \alpha/(1-\beta)$ follow directly. The residuals from this regression have variance $\sigma^2\Delta t$, giving an estimate of σ. This simple estimation procedure is the physical-measure ($\mathbb{P}$) calibration of Vasicek, and it can be implemented in a few lines of linear algebra — a reminder that many "exotic" quantitative finance tools reduce to well-known statistical procedures once the algebra is peeled away.

The sample autocorrelation at lag h — $\mathrm{Corr}(r_s, r_{s+h})$ — should follow the theoretical form $e^{-\kappa h}$ in stationary equilibrium. Plotting empirical autocorrelation against lag on a log-linear scale and fitting a straight line is a quick diagnostic of whether the data are well-described by an OU / AR(1) process. If the empirical autocorrelation decays exponentially, the AR(1) is a good fit; if it decays faster (over-damped) or slower (long memory), the data are telling you something the simple model misses. Long memory is a particularly well-studied deviation; models with multiple mean-reversion speeds (G2++, three-factor affine) capture it by having one slow and one fast factor.

Two important identifiability warnings. First, estimates of κ are notoriously noisy in short samples: the "half-life of a shock" is a slow-moving feature, and short-window data contains limited information about it. Robust estimates typically require decades of daily data, or at least years of high-frequency data. Second, the physical-measure estimates of $\kappa^{\mathbb{P}}$ and $\theta^{\mathbb{P}}$ differ systematically from their risk-neutral counterparts $\kappa^{\mathbb{Q}}$ and $\theta^{\mathbb{Q}}$ by the market price of risk. The volatility σ is measure-invariant (Girsanov changes drift but not diffusion), so it should theoretically agree across historical and calibration estimates — a useful sanity check when running both.

14.2.3 12.2.2 From $\mathbb{P}$ to $\mathbb{Q}$ (cross-reference Chapter 5)

Under the physical measure $\mathbb{P}$, write the short rate as

$$\mathrm{d}r_t \;=\; \mu(t, r_t)\,\mathrm{d}t \;+\; \sigma(t, r_t)\,\mathrm{d}W_t. \tag{12.5}$$

By Girsanov's theorem (Chapter 5), the change of measure to the risk-neutral $\mathbb{Q}$ shifts the drift by $-\lambda_t\,\sigma(t, r_t)$, where λ_t is the *market price of interest-rate risk*:

$$\mathrm{d}r_t \;=\; \big(\mu(t, r_t) - \lambda_t\,\sigma(t, r_t)\big)\,\mathrm{d}t \;+\; \sigma(t, r_t)\,\mathrm{d}\widehat{W}_t. \tag{12.6}$$

We do not re-derive Girsanov here — that happened in Chapter 5, where the density process, Doléans-Dade exponential, and two-numeraire switching were developed as the canonical treatment. What matters for this chapter is the consequence. Three standard choices of λ_t preserve the affine structure:

- Constant \$ _t = \$ const: only shifts θ (the long-run mean under $\mathbb{Q}$ differs from that under $\mathbb{P}$).
- Affine in r_t: $\lambda_t = a + b\,r_t$: shifts both θ and κ.
- Deterministic function of time $\lambda_t = a_t + b_t\,r_t$: yields Hull-White with time-dependent θ_t.

All three preserve the linear drift, and under $\mathbb{Q}$ the short rate continues to satisfy

$$\mathrm{d}r_t \;=\; \kappa(\theta_t - r_t)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}\widehat{W}_t, \tag{12.7}$$

where θ_t is (in general) a deterministic function of time. The diffusion σ does *not* change under Girsanov: Brownian motions have the same quadratic variation under equivalent measures, so calibrated volatilities carry over directly across measure specifications without needing adjustment. Only drifts change when measures change.

The economic content of λ_t is worth a pause. Under $\mathbb{P}$ rate shocks are perceived as risky, and investors demand compensation for holding bonds whose prices move inversely to those shocks. That compensation is the term premium, and it biases the $\mathbb{P}$-drift of r_t relative to what a risk-neutral agent would perceive. Because we never directly observe λ_t — we only see prices, and prices aggregate $\mathbb{P}$-dynamics with λ_t into a single $\mathbb{Q}$-drift — the modelling convention is to work entirely under $\mathbb{Q}$ and let historical data inform the shape (if any) of λ_t separately. In short-rate modelling for pricing, it is almost universal to specify dynamics directly under $\mathbb{Q}$ and never write $\mathbb{P}$-dynamics explicitly.

14.2.4 12.2.3 The fundamental bond-pricing formula

By the general theory of risk-neutral pricing (Chapter 5), relative prices of all traded assets are $\mathbb{Q}$-martingales once we divide by the numeraire M_t. For a zero-coupon bond with terminal payoff 1,

$$\frac{P_t(T)}{M_t} \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[\frac{1}{M_T}\right] \qquad \Longrightarrow \qquad \boxed{P_t(T) \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_s\,\mathrm{d}s}\right].} \tag{12.8}$$

Equation (12.8) is the master formula of short-rate modelling. Read carefully, it says that today's bond price is *exactly* the risk-neutral expectation of the continuously-compounded discount factor over the bond's life. This sentence contains a lot: it simultaneously asserts no-arbitrage (the price is an expectation), identifies the right measure ($\mathbb{Q}$), specifies the right numeraire (M_t), and provides a concrete algorithm (integrate r_s along simulated paths and average). Every bond price in this chapter — and every yield, forward rate, swap rate, caplet, and swaption derived from bond prices in Chapter 13 — is a consequence of this formula.

A useful consistency check: when $r_t = r$ is deterministic and constant, (12.8) reduces to $P_t(T) = e^{-r(T-t)}$. When r_t is deterministic but time-varying, it reduces to $P_t(T) = e^{-\int_t^T r_s\,\mathrm{d}s}$. Only when r_t is stochastic does the expectation become non-trivial; the formula generalises the deterministic cases smoothly and recovers them as special limits.

The algorithmic reading of (12.8) is equally important. If we cannot evaluate the expectation analytically — say, because the model is too complex for a closed form — we can always simulate: generate many Monte Carlo paths of r_t from t to T under $\mathbb{Q}$, compute the realised discount factor along each path, and average. The bond price converges in probability to the average as the number of paths grows. The entire business of finding closed forms (the affine trick we are about to execute) is a way of *avoiding* Monte Carlo by exploiting special structure; when that structure is absent, Monte Carlo is there to catch us.

14.3 12.3 Vasicek — the canonical derivation

This is the technical heart of the chapter. Take (12.7) with constant θ, which gives the original Vasicek (1977) model:

$$\boxed{\mathrm{d}r_t \;=\; \kappa(\theta - r_t)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}\widehat{W}_t, \qquad \kappa, \theta, \sigma > 0.} \tag{12.9}$$

The drift is linear-restoring, the diffusion is constant, and the solution is an Ornstein-Uhlenbeck process: a mean-reverting Gaussian process with long-run mean θ and mean-reversion speed κ.

Intuition — mean reversion. Whenever r_t wanders above θ, the drift pulls it *down* in proportion to the deviation; when it drifts below, the drift pushes it *up*. Economically this mirrors the behaviour of real short rates — central banks tighten when rates are "too low" (inflation pressure) and ease when they are "too high" (recession risk), so rates orbit around a neutral level rather than random-walking. The speed κ controls how fast this happens: the *half-life* of deviations is $\ln 2/\kappa$. With $\kappa = 0.3$ a shock decays by half in about 2.3 years; with $\kappa = 1.0$, in about 8 months. The stationary distribution is $\mathcal{N}(\theta, \sigma^2/(2\kappa))$ — larger κ (stronger pull) or smaller σ (less noise) narrows the long-run distribution around θ.

The half-life interpretation deserves emphasis because it is how practitioners think about κ. Nobody calibrating reasons directly in units of "inverse time" — they reason in half-lives, because a half-life has immediately physical meaning. "A κ of 0.3 corresponds to a rate-shock half-life of 2.3 years" is a statement any rates trader can sanity-check against intuition about how long a Fed-funds deviation persists before the next cutting or hiking cycle unwinds it. If a calibrated κ implies a half-life that is wildly too short (weeks) or absurdly too long (decades), something is wrong.

The Gaussian structure is the other defining feature of Vasicek. It is what makes every subsequent computation tractable: a linear combination of Gaussian increments is Gaussian, an integral of Gaussians is Gaussian, and the exponential of a Gaussian has a closed-form mean (its moment-generating function). The whole affine bond-price formula ultimately rests on this one fact. The price paid is that r_t can, in principle, go negative: the Gaussian distribution is supported on all of $\mathbb{R}$. Before the 2008-2020 era of unconventional monetary policy, this was viewed as the model's most conspicuous defect. After 2014 several European and Japanese policy rates did go meaningfully negative, and suddenly Vasicek's ability to accommodate negative rates became a *feature*, not a bug. No other equally tractable short-rate model handles negative rates so naturally.

14.3.1 12.3.1 The SDE and the explicit solution

Solve (12.9) by the standard OU integrating factor. Set $r_t = e^{-\kappa t} g_t$ (equivalently $g_t = e^{\kappa t} r_t$). By Itô's lemma (Chapter 3),

$$\mathrm{d}r_t = -\kappa\, e^{-\kappa t} g_t\,\mathrm{d}t \;+\; e^{-\kappa t}\,\mathrm{d}g_t = -\kappa r_t\,\mathrm{d}t + e^{-\kappa t}\,\mathrm{d}g_t. \tag{12.10}$$

Matching (12.10) against (12.9),

$$\mathrm{d}g_t \;=\; \kappa\, e^{\kappa t}\,\theta\,\mathrm{d}t \;+\; \sigma\, e^{\kappa t}\,\mathrm{d}\widehat{W}_t. \tag{12.11}$$

Integrate from t to T and rearrange using $g_T - g_t = e^{\kappa T} r_T - e^{\kappa t} r_t$:

$$\boxed{r_T \;=\; e^{-\kappa(T-t)}\, r_t \;+\; \theta\big(1 - e^{-\kappa(T-t)}\big) \;+\; \sigma\int_t^T e^{-\kappa(T-s)}\,\mathrm{d}\widehat{W}_s.} \tag{12.12}$$

The integrating-factor trick used here generalises to essentially every linear SDE. The key move is multiplying through by $e^{\kappa t}$, which converts the mean-reverting term $-\kappa r_t\,dt$ into a pure differential. Once that is done, the remaining drift and diffusion have no r_t dependence and can be integrated directly — the non-trivial feedback has been algebraically removed. The price of admission is that the final expression involves a weighted stochastic integral against the kernel $e^{-\kappa(T-s)}$, with weights that decay exponentially as you look further back. These decaying weights have a beautiful physical meaning: the current short rate is a weighted average of all past shocks, with older shocks discounted more heavily because the mean-reverting drift has had more time to "forget" them. This is exactly the continuous-time analogue of a discrete AR(1) with autoregressive coefficient $e^{-\kappa\Delta t}$.

The structural similarity between (12.12) and the Ornstein-Uhlenbeck result in physics is worth noting: there an OU process describes the velocity of a particle undergoing Brownian motion in a viscous fluid. The mean-reverting drift $\kappa(\theta - r_t)$ is analogous to a frictional force pulling the velocity toward equilibrium; the diffusion $\sigma\,\mathrm{d}\widehat{W}_t$ is analogous to random molecular collisions. The exponential memory of past shocks is the mathematical signature of frictional damping. All classical physical intuition about coloured noise, spectral densities, and correlation times carries directly across to interest-rate modelling.

Read (12.12) piece by piece: the first term $e^{-\kappa(T-t)} r_t$ is the *deterministic* decay of the current rate toward the mean; the second term $\theta(1-e^{-\kappa(T-t)})$ is the deterministic buildup of the mean-reversion level; the third term is a mean-zero Gaussian random variable whose variance grows from 0 (at $T = t$) toward the stationary value $\sigma^2/(2\kappa)$ (as $T \to \infty$). Three effects neatly separated: where we are now, how fast we revert, how noisy the journey is. Every intuition about OU processes — and therefore about Vasicek, Hull-White, and half the short-rate literature — can be read off this one equation.

14.3.2 12.3.1.b Joint distribution over multiple times

A property of (12.12) that matters in practice: the joint distribution of $(r_{t_1}, r_{t_2}, \dots, r_{t_n})$ at any finite collection of times is multivariate Gaussian. This follows because each r_{t_i} is an affine function of the same Brownian motion, and linear combinations of a Brownian path are jointly Gaussian. The mean and covariance of this multivariate Gaussian can be computed explicitly:

$$\mathrm{Cov}(r_s, r_t) \;=\; \sigma^2\, e^{-\kappa(s+t)} \int_0^{\min(s,t)} e^{2\kappa u}\,\mathrm{d}u \;=\; \frac{\sigma^2}{2\kappa}\big(e^{-\kappa|t-s|} - e^{-\kappa(t+s)}\big)\,. \tag{12.12a}$$

The dominant term is the exponentially decaying autocorrelation $e^{-\kappa|t-s|}$; the subtracted term accounts for the boundary effect near $t = s = 0$. In the stationary limit ($s, t \gg 1/\kappa$) the covariance reduces to $\frac{\sigma^2}{2\kappa} e^{-\kappa|t-s|}$, exactly matching the classical OU correlation structure.

This multivariate-Gaussian structure is what makes *exact* Monte Carlo simulation of Vasicek paths so efficient. If you need samples at specific times $\{t_1, \dots, t_n\}$, you can compute the exact covariance matrix, Cholesky-factor it, and draw correlated Gaussians in one shot — no Euler discretisation error. For Monte Carlo of payoffs that depend only on terminal rates (e.g. European bond options) you do not even need the full covariance matrix; a single Gaussian draw for r_T suffices. CIR, by contrast, does not admit this luxury: its non-central chi-squared distribution is exact but more awkward to sample. The analytical tractability of Vasicek therefore extends beyond pricing into simulation efficiency — a reason to prefer Vasicek when it fits.

14.3.3 12.3.2 Distribution of r_T

From (12.12), r_T conditional on $\mathcal{F}_t$ is Gaussian with

$$\mathbb{E}_t^{\mathbb{Q}}[r_T] \;=\; e^{-\kappa(T-t)}\, r_t + \theta(1 - e^{-\kappa(T-t)}), \tag{12.13}$$

$$\mathbb{V}_t^{\mathbb{Q}}[r_T] \;=\; \sigma^2 \int_t^T e^{-2\kappa(T-s)}\,\mathrm{d}s \;=\; \frac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa(T-t)}\big)\,. \tag{12.14}$$

The variance starts at 0 at $T = t$ (no time, no uncertainty) and saturates at $\sigma^2/(2\kappa)$ as $T \to \infty$ (the stationary variance). The half-saturation time is $\ln 2/(2\kappa)$, so half of the stationary variance accumulates in $\ln 2/(2\kappa)$ years. With $\kappa = 0.3$ this is about 1.15 years — meaning over a 1-year horizon, rates have accumulated about half their long-run uncertainty. Beyond about $5/(2\kappa) \approx 8$ years the distribution is essentially stationary.

A worked example sharpens this. Take $\kappa = 0.3$ (half-life about 2.3 years) and $\sigma = 0.01$ (annualised 100 bp vol). The stationary standard deviation is $\sigma/\sqrt{2\kappa} = 0.01/\sqrt{0.6} \approx 0.0129$, or 129 basis points. A normal distribution with this standard deviation would put roughly two-thirds of its mass within $\theta \pm 129$ bp and 95% within $\theta \pm 258$ bp. For $\theta = 0.04$ that gives a plausible long-run range of roughly 1% to 7% for the short rate — consistent with the long-run range of actual policy rates in mature economies. Had we taken $\sigma = 0.03$ (a highly volatile regime), the stationary sd would be 387 bp and the 95% band would extend from −3.7% to +11.7%, clearly including negative rates. This explicit scaling is one of Vasicek's virtues: it lets you read off the implications of parameter choices in two-second mental math.

The ratio $(1 - e^{-2\kappa(T-t)})$ is a useful "mixing fraction" that tells you how close to stationarity the process is at horizon $T - t$. At $T - t = 1/\kappa$ (one mean-reversion time) the mixing fraction is $1 - e^{-2} \approx 0.865$. At $T - t = 2/\kappa$ it is about 0.982. For practical purposes one can treat any horizon beyond $3/\kappa$ as effectively stationary, and this is how simulation studies typically validate their "burn-in" periods before extracting stationary statistics.

Vasicek paths with mean reversion *Vasicek paths from three starting points all pull toward θ (dashed line), with dispersion widening then stabilising at $\sigma/\sqrt{2\kappa}$.*

There is a visual intuition worth internalising: draw the (t, r)-plane with a horizontal line at $r = \theta$. The rate process starts at $(0, r_0)$. In the deterministic case (set $\sigma = 0$), the path is an exponential curve approaching θ asymptotically:

$$r_t \;=\; r_0\, e^{-\kappa t} \;+\; \theta(1 - e^{-\kappa t}). \tag{12.13a}$$

In the stochastic case, this curve is the "centreline" around which sample paths fluctuate, with Gaussian noise amplitude growing from zero (at $t = 0$ the rate is deterministically r_0) toward the stationary value $\sqrt{\sigma^2/(2\kappa)}$ as $t \to \infty$. Drawing many sample paths overlaid on the same axes produces an "hourglass" pattern: tight at $t = 0$, fanning out over time, eventually settling into a constant-width band centred at θ.

Concrete numbers: with $\kappa = 0.15$, $\theta = 4\%$, $r_0 = 2\%$, the deterministic rate path rises from 2% toward 4% along an exponential with half-life $\ln 2/0.15 \approx 4.6$ years. $\mathbb{E}[r_1] \approx 2.28\%$, $\mathbb{E}[r_5] \approx 3.06\%$, $\mathbb{E}[r_{20}] \approx 3.90\%$. The approach is exponential: the gap to θ halves every 4.6 years. If instead $r_0 = 6\%$ (above target), the trajectory is inverted — an exponential decay from 6% down toward 4%, same timescales. In either case the mean rate at time t is a specific weighted average of starting point and target, with weights determined entirely by κt.

Intuition for the stationary variance $\sigma^2/(2\kappa)$. Each incremental shock injects variance $\sigma^2 \Delta t$ into the process; each unit of time, mean reversion exponentially erodes that variance at rate 2κ (the factor of 2 because variance is a second moment). In equilibrium, the rate at which new variance is injected equals the rate at which old variance is damped: $\sigma^2 = 2\kappa V_{\mathrm{eq}}$, giving $V_{\mathrm{eq}} = \sigma^2/(2\kappa)$. This balance between injection and decay is the mechanism behind the ergodic variance. Imagine water being poured into a leaky bucket: constant inflow at rate σ^2, outflow proportional to the current water level at rate $2\kappa V$. In equilibrium, the water level settles at $V = \sigma^2/(2\kappa)$, where inflow equals outflow.

The κ-σ identifiability problem, re-posed. The stationary variance depends only on the ratio σ^2/κ, so any two parameter vectors with the same ratio (but different individual σ and κ) produce identical equilibrium distributions. The difference shows up only in the *transient* behaviour: how quickly the distribution approaches equilibrium from a non-equilibrium start. A high-κ, high-σ combination has both strong mean reversion and high volatility; it converges quickly. A low-κ, low-σ combination converges slowly. On a yield curve close to equilibrium, the two parameter sets fit equally well, and the calibration is non-identifiable. On a yield curve far from equilibrium, the transient differs and data becomes informative.

14.3.4 12.3.3 Distribution of the integrated rate $\int_t^T r_u \,\mathrm{d}u$

To compute the bond price $P_t(T) = \mathbb{E}_t^{\mathbb{Q}}[e^{-\int_t^T r_s \,\mathrm{d}s}]$ we need the law of $\int_t^T r_s \,\mathrm{d}s$. Substitute (12.12) and apply Fubini to swap the time integral with the stochastic integral:

$$\int_t^T r_s \,\mathrm{d}s \;=\; \frac{1 - e^{-\kappa(T-t)}}{\kappa}\, r_t \;+\; \theta \int_t^T \big(1 - e^{-\kappa(T-s)}\big)\,\mathrm{d}s \;+\; \frac{\sigma}{\kappa}\int_t^T \big(1 - e^{-\kappa(T-s)}\big)\,\mathrm{d}\widehat{W}_s. \tag{12.15}$$

The kernel $(1 - e^{-\kappa(T-s)})/\kappa$ that appears throughout (12.15) has a name worth remembering — it is the quantity we will call $B_s(T)$. Its role here is to tell us how much each shock $\mathrm{d}\widehat{W}_s$ at time s contributes to the integrated short rate through maturity T. A shock at time $s \ll T$ has almost time $T - s$ for its effect to propagate through the mean-reverting dynamics, so its contribution is nearly $1/\kappa$. A shock at $s \approx T$ barely has time to act, so its contribution tends to zero. This exponential down-weighting of late-arriving shocks is the same phenomenon that gives bonds their characteristic convexity behaviour.

The Fubini step deserves a brief justification. Technically, Fubini for stochastic integrals requires enough integrability of the integrand; in the OU case the kernel $e^{-\kappa(T-s)}$ is bounded uniformly in $s \in [t, T]$, so all the required integrability conditions are trivially satisfied. In richer models (say, a state-dependent-diffusion CIR process) the justification is more delicate. For Vasicek, however, the swap goes through with no fuss — another small benefit of working in the Gaussian world. (See Chapter 3 for the Itô isometry that underpins this.)

Another useful perspective: $B_s(T)$ is exactly what one would compute if asked, "how much does a point shock to r_s raise the integrated short rate $\int_t^T r_u \,\mathrm{d}u$?" A shock of size δ at time s means that, subsequently, r_u for $u > s$ is elevated by $\delta e^{-\kappa(u-s)}$ (the deterministic OU response to an impulse). Integrating from s to T gives $\delta\cdot(1-e^{-\kappa(T-s)})/\kappa = \delta B_s(T)$. So $B_s(T)$ is the impulse-response function of the integrated short rate with respect to a shock at time s. This impulse-response interpretation links the bond-pricing formula to the broader literature on linear systems, transfer functions, and spectral methods.

Conditional on $\mathcal{F}_t$, (12.15) is therefore a Gaussian random variable $\int_t^T r_s \,\mathrm{d}s \,\big|\, \mathcal{F}_t \sim \mathcal{N}(\mu_*, \Sigma_*^2)$ with

$$\mu_* \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[\int_t^T r_s \,\mathrm{d}s\right] \;=\; \frac{1 - e^{-\kappa(T-t)}}{\kappa}\, r_t \;+\; \theta\left[(T-t) - \frac{1 - e^{-\kappa(T-t)}}{\kappa}\right], \tag{12.16}$$

$$\Sigma_*^2 \;=\; \mathbb{V}_t^{\mathbb{Q}}\left[\int_t^T r_s \,\mathrm{d}s\right] \;=\; \frac{\sigma^2}{\kappa^2}\int_t^T \big(1 - e^{-\kappa(T-s)}\big)^2\,\mathrm{d}s. \tag{12.17}$$

The Gaussianity of $\int_t^T r_s \,\mathrm{d}s$ is the crux. It follows because (i) under Vasicek, r_s is a linear combination of Gaussian shocks through (12.12), (ii) integration against a deterministic kernel preserves Gaussianity (a linear operation on a Gaussian vector), and (iii) Fubini lets us legitimately swap the time integral and the expectation. That the *integral* of a Gaussian process is itself Gaussian is not obvious — for a non-linear function of a Gaussian process it would fail — and this is precisely why Gaussian short-rate models are so disproportionately tractable compared to alternatives.

To see why the CIR model cannot easily replicate this step, notice that under CIR $r_s =$ (something involving $\sqrt{r_{s-}}$), which is *not* a linear combination of Gaussian noise — it is a non-central chi-squared process whose integral has no simple closed-form distribution. In the CIR bond-pricing derivation one still obtains an affine formula $P_t(T) = e^{A_t(T) - B_t(T) r_t}$, but the derivation goes through the PDE route (§12.3.5 below) rather than the direct-expectation route. The PDE approach generalises; the direct-expectation approach is a Gaussian-specific luxury.

The explicit variance formula (12.17), after expanding the square and integrating term-by-term, becomes

$$\Sigma_*^2 \;=\; \frac{\sigma^2}{\kappa^2}\left[(T-t) \;-\; \frac{2(1-e^{-\kappa(T-t)})}{\kappa} \;+\; \frac{1-e^{-2\kappa(T-t)}}{2\kappa}\right]. \tag{12.18}$$

For typical parameters ($\kappa = 0.3$, $\sigma = 0.01$, $T - t = 5$ years), $B_t(T) = (1 - e^{-1.5})/0.3 \approx 2.59$, and the bracketed expression evaluates to approximately $5 - 5.18 + 1.59 = 1.41$. Multiplying by $\sigma^2/\kappa^2 \approx 1.11 \times 10^{-3}$ gives $\Sigma_*^2 \approx 1.57 \times 10^{-3}$, so the standard deviation of the integrated rate is about 4%. Worth internalising these numerics: they are what the formulas translate into with real inputs.

14.3.5 12.3.4 Closed-form bond price

Because $\int_t^T r_s \, \mathrm{d}s$ is Gaussian with mean μ_* and variance Σ_*^2, its moment-generating function at -1 gives the bond price directly:

$$P_t(T) \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_s \, \mathrm{d}s}\right] \;=\; \exp\{-\mu_* + \tfrac{1}{2}\Sigma_*^2\}\,. \tag{12.19}$$

Equation (12.19) is the single most important step in the derivation if you want to understand *why* the bond price has the form it does. The identity $\mathbb{E}[e^X] = e^{\mathbb{E}X + \frac{1}{2}\mathbb{V}X}$ for Gaussian X is the engine. Without Gaussianity this identity fails (one would need the full characteristic function or a Laplace transform), and the resulting bond price would have no closed form.

The $+\frac{1}{2}\Sigma_*^2$ term is the *convexity correction*: because the exponential is convex, a random discount factor is worth *more* on average than the exponential of the mean rate, and the gap is exactly one-half the variance. Practitioners sometimes call this "Jensen's mountain" — it is the reason bond yields are below expected future short rates even in the absence of a term premium.

To make the Jensen effect vivid, consider a toy case: imagine the short rate over the next year will be either 2% or 6% with equal probability, so the mean rate is 4%. The "naive" discount factor computed from the mean rate is $e^{-0.04} = 0.9608$. The correct discount factor is $\frac{1}{2}(e^{-0.02} + e^{-0.06}) = \frac{1}{2}(0.9802 + 0.9418) = 0.9610$. The difference is small (2 bp in yield terms) but always in the same direction. Now change the rates to 0% and 8% (higher variance, same mean): correct discount $\frac{1}{2}(1+e^{-0.08}) = 0.9616$. The gap has widened to 8 bp. The gap scales with the variance of the rate distribution, exactly as the $+\frac{1}{2}\Sigma_*^2$ term predicts. This is Jensen's inequality in action.

The convexity correction has profound practical implications. It explains why forward rates are typically *higher* than expected future spot rates (the convexity premium baked into forwards); why a long-dated bond yield can be below the long-run mean of the short rate even in the absence of any risk premium; why swaps convexity-adjusted to remove the bias have a specific bid-offer structure tied to σ^2. A rates trader who does not internalise the convexity correction is doomed to misprice every structured rate product; the $\frac{1}{2}\mathbb{V}$ term is where the money hides.

Plugging (12.16) and (12.17) into (12.19) gives, after some algebra, the *affine term structure*:

$$\boxed{P_t(T) \;=\; e^{A_t(T) - B_t(T)\, r_t},} \tag{12.20}$$

with

$$B_t(T) \;=\; \frac{1 - e^{-\kappa(T-t)}}{\kappa}, \tag{12.21}$$

$$A_t(T) \;=\; -\theta\big((T-t) - B_t(T)\big) \;+\; \frac{\sigma^2}{2\kappa^2}\int_t^T \big(1 - e^{-\kappa(T-s)}\big)^2 \, \mathrm{d}s. \tag{12.22}$$

(Here s is the dummy integration variable running across the bond's life, not to be confused with the current time t; the reference formulas at §12.9 reuse the same s label.) The structure $P = e^{A-Br}$ — with A, B deterministic functions of (t, T) independent of r — is called an affine term structure.

Vasicek bond-price surface *Vasicek bond-price surface $P(t,T)$ as a function of the current short rate and maturity.*

Let us decode (12.20). The bond price is a log-linear function of the current short rate: $\ln P_t(T) = A_t(T) - B_t(T) r_t$. The coefficient $B_t(T)$ is the bond's *duration-like* sensitivity: $-\partial_{r_t} \ln P_t(T) = B_t(T)$ is the modified duration with respect to short-rate shocks. For short maturities ($T \to t$), $B_t(T) \to 0$ and the bond is insensitive to r_t. For long maturities ($T \to \infty$), $B_t(T) \to 1/\kappa$, meaning a very long bond has a finite duration of $1/\kappa$ years — *not* unbounded, because mean reversion caps the effect. A Vasicek with $\kappa = 0.1$ (very slow reversion) gives a long-bond duration of 10 years; with $\kappa = 0.5$, 2 years. This is a radically different picture from a random-walk interest rate, under which long-bond duration would grow without bound.

The finite long-bond duration is one of Vasicek's most distinctive — and economically sensible — predictions. Actual 30-year bonds do not have three times the sensitivity of 10-year bonds to short-rate moves. The reason is mean reversion: a short-rate shock today does not permanently change expected rates fifty years from now — it is absorbed by the mean-reverting dynamics over a few $1/\kappa$ years. So the marginal duration contribution of each additional year beyond a few $1/\kappa$ years is essentially zero. The bond-duration cap at $1/\kappa$ is the quantitative expression of this economic insight.

Boundary conditions: $B_T(T) = 0$, $A_T(T) = 0$, so $P_T(T) = 1$. These encode the defining property of a zero-coupon bond: at maturity it pays exactly one unit of currency regardless of the state of the economy.

Affine coefficients A(t,T) and B(t,T) *$B_t(T)$ rises from 0 to the asymptote $1/\kappa$, encoding rate sensitivity of long bonds; $A_t(T)$ is negative and grows in magnitude with maturity, reflecting the drift and convexity adjustments.*

14.3.6 12.3.5 Affine ansatz and the PDE route

There is a second, independent route to (12.20) — and it is the one that generalises most smoothly to more complex models (CIR, quadratic-Gaussian, multi-factor affine). Fix T and view $P_t(T) = g(t, r_t)$ as a function of the state r_t. The Feynman-Kac representation (Chapter 4) says

$$P_t(T) \;=\; \mathbb{E}_t^{\mathbb{Q}}\Big[e^{-\int_t^T r_s\,\mathrm{d}s}\Big], \quad \mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}\widehat{W}_t, \tag{12.23}$$

and therefore $g(t, r)$ satisfies the bond-pricing PDE

$$(\partial_t + \mathcal{L})\, g(t,r) \;=\; r\, g(t,r), \qquad g(T, r) = 1, \tag{12.24}$$

with infinitesimal generator

$$\mathcal{L} \;=\; \kappa(\theta - r)\,\partial_r + \tfrac{1}{2}\sigma^2\,\partial_{rr}. \tag{12.25}$$

Why does the PDE carry a source term $-rg$ rather than being sourceless? Because we are not working with discounted prices directly; we are working with undiscounted bond prices. As time advances by $\mathrm{d}t$, the bond's value is reduced by a factor of $e^{-r_t\,\mathrm{d}t} \approx 1 - r_t\,\mathrm{d}t$, which contributes $-r_t g\,\mathrm{d}t$ to the dynamics. Multiplying g by the discount factor $e^{-\int r}$ would give a process satisfying a sourceless PDE; the sourced PDE here is for the un-discounted price. A bond is a traded asset, so its discounted price is a $\mathbb{Q}$-martingale; its undiscounted price is not (it drifts at r_t), which is why (12.24) has a non-zero source.

Affine ansatz. Guess

$$g(t,r) \;=\; e^{A_t - B_t r}, \qquad A_T = 0,\; B_T = 0. \tag{12.26}$$

Derivatives: $\partial_t g = (\dot{A}_t - \dot{B}_t r) g$, $\partial_r g = -B_t g$, $\partial_{rr} g = B_t^2 g$. Substitute into (12.24) and divide by g:

$$(\dot{A}_t - \kappa B_t \theta + \tfrac{1}{2}\sigma^2 B_t^2) \;+\; (-\dot{B}_t + \kappa B_t - 1)\, r \;=\; 0. \tag{12.27}$$

Equation (12.27) is doing something subtle. The left-hand side is a polynomial in r of degree one — an r^0 coefficient and an r^1 coefficient. For this polynomial to vanish for *all* values of r, both coefficients must individually be zero. This is the *matching of coefficients* move, and it is the step that turns one PDE into two ODEs. It works only because the affine ansatz was the right guess; had we tried a quadratic ansatz $\log P = A - Br - Cr^2$, we would have picked up an r^2 coefficient that did not naturally vanish, and the ansatz would have failed. The class of models for which coefficient-matching succeeds is precisely the class of affine models — hence the name.

Coefficient matching gives the *affine term-structure ODE system*:

$$\boxed{\begin{cases} \dot{A}_t - \kappa B_t\,\theta + \frac{1}{2}\sigma^2 B_t^2 \;=\; 0, & \text{(A-ODE)} \\ \dot{B}_t - \kappa B_t + 1 \;=\; 0, & \text{(B-ODE)} \end{cases} \qquad A_T = 0,\;\; B_T = 0.} \tag{12.28}$$

The ODE system (12.28) is structurally beautiful: the B-equation is autonomous (involves only B and constants), while the A-equation depends on B as an input. Solve for B first, then plug the result into the A-equation and integrate. In richer models the B-equation becomes a Riccati ODE (quadratic in B); here in Vasicek it is merely linear, because the diffusion is constant rather than state-dependent.

Solving B-ODE. Equation $\dot{B}_t - \kappa B_t + 1 = 0$ in standard form is $\dot{B}_t = \kappa B_t - 1$ with $B_T = 0$. Integrating factor $e^{-\kappa t}$ gives

$$B_t(T) \;=\; \frac{1 - e^{-\kappa(T-t)}}{\kappa}, \tag{12.29}$$

matching (12.21) exactly. This function depends only on $\tau = T - t$, not separately on t and T — a consequence of the constant-coefficient structure.

Solving A-ODE. Plug $B_t(T)$ into the A-ODE and integrate from t to T using $A_T = 0$:

$$A_t(T) \;=\; -\int_t^T \kappa\, B_s(T)\,\theta\,\mathrm{d}s \;+\; \frac{\sigma^2}{2}\int_t^T B_s(T)^2\,\mathrm{d}s. \tag{12.30}$$

Substituting $\kappa B_s(T) = 1 - e^{-\kappa(T-s)}$ and $\kappa^2 B_s(T)^2 = (1 - e^{-\kappa(T-s)})^2$, (12.30) becomes exactly (12.22), providing a cross-check between the two derivation routes.

The two derivations — moment-generating-function (§12.3.4) and PDE / affine ansatz (§12.3.5) — arrive at the same answer via genuinely different routes. The former is analytically direct but only works because Gaussianity makes the MGF explicit. The latter is abstractly more general (it depends only on affine structure, not on Gaussianity) and therefore extends to CIR, quadratic-Gaussian models, and affine jump-diffusions. In research the PDE approach is more often the starting point because it generalises more cleanly; in pedagogy the direct-expectation route is more transparent for the Vasicek special case we are working through.

Some authors prefer yet a third derivation: the martingale-based approach, where one guesses the bond-price formula and verifies that the discounted price $P_t(T)/M_t$ is a martingale under $\mathbb{Q}$. This is the most elegant route when it works, because it directly confirms the no-arbitrage condition rather than deriving the price and verifying arbitrage-freeness as a corollary. The three approaches — expectation, PDE, martingale — form a triad that characterises risk-neutral pricing from three complementary angles: operational (how to compute it), analytic (what equation does it satisfy), and economic (why is it arbitrage-free). Mastering all three is the mark of a mature quant.

A more general methodological lesson from this cross-check: in quantitative finance, having two independent derivations of the same result is not redundancy — it is insurance. Each derivation exposes a different aspect of the mathematical structure, and each is vulnerable to a different class of errors. The direct-expectation approach is prone to integration mistakes and Fubini-justification gaps; the PDE approach is prone to ansatz-guessing errors and boundary-condition confusion. When both converge on the same formula, one gains high confidence that neither class of errors has been made.

14.4 12.4 The yield curve under Vasicek

14.4.1 12.4.1 Yields and instantaneous forward rates

The continuously-compounded yield for maturity T at time t is

$$y_t(T) \;=\; -\,\frac{\ln P_t(T)}{T-t} \;=\; \frac{-A_t(T) + B_t(T)\,r_t}{T-t}. \tag{12.31}$$

Yields, not bond prices, are what practitioners and the press typically quote, so (12.31) is probably the most practically used formula derived from Vasicek. The conversion from bond prices to yields is conceptually a simple logarithm-and-divide; the practical utility is that yields are approximately stationary over long periods, while bond prices drift toward 1 at maturity.

Under Vasicek, the yield decomposition $y_t(T) = -A_t(T)/(T-t) + B_t(T)/(T-t) \cdot r_t$ shows that yields are themselves affine in r_t. The coefficient $B_t(T)/(T-t)$ is the yield's sensitivity to the short rate; it approaches 1 at short maturities (so short-maturity yields essentially equal r_t) and approaches 0 at long maturities (so long-maturity yields are almost insensitive to r_t, a consequence of mean reversion). The intercept $-A_t(T)/(T-t)$ is the "term-structure contribution" — the part of the yield that does not depend on today's rate.

The instantaneous forward rate at maturity T is $f_t(T) = -\partial_T \ln P_t(T)$. It is the rate at which an investor could lock in a forward contract today starting at T for an infinitesimal period. Equivalently, it is the expected T-instantaneous short rate under the T-forward measure (see Chapter 13). Bond prices and forward rates are related by $\ln P_t(T) = -\int_t^T f_t(s)\,\mathrm{d}s$, and the short rate itself satisfies $r_t = f_t(t)$.

A common trap: the forward rate $f_t(T)$ is *not* in general equal to the expected future short rate $\mathbb{E}[r_T]$. Under the T-forward measure, $f_t(T) = \widetilde{\mathbb{E}}^T[r_T]$; under the risk-neutral measure, $\mathbb{E}^{\mathbb{Q}}[r_T]$ differs by a convexity adjustment; under the physical measure, there is an additional risk-premium correction. When a commentator says "the market expects rates to be X% in five years" and cites a five-year forward rate, that statement is true under $\widetilde{\mathbb{Q}}^5$ but not under $\mathbb{P}$ — and the wedge between them can be substantial, particularly at longer horizons. Being precise about *which* measure one is discussing is a hallmark of careful rates analysis.

14.4.2 12.4.2 Yield curve shapes

For Vasicek:

- As $T \to t$: $y_t(T) \to r_t$ (the short rate itself).
- As $T \to \infty$: $y_t(T) \to \theta - \frac{1}{2}\,\sigma^2/\kappa^2$ (the long yield — constant in r_t).
- In between: the curve can be monotone, humped, or inverted depending on r_t vs θ.

The long-yield limit $\theta - \frac{1}{2}\sigma^2/\kappa^2$ deserves a pause. The first term, θ, is the long-run mean of the short rate — what you would expect if you simply averaged rates over the infinite future. The second term is a *subtraction*: the long yield is *below* the expected long-run short rate. This negative gap is the convexity premium: because the discount factor is a convex function of the integrated rate, random variation in future rates raises the expected discount factor above the deterministic benchmark, and therefore lowers the implied yield. The magnitude $\sigma^2/(2\kappa^2)$ is proportional to the stationary variance of r_t — more volatile or less mean-reverting short rates give larger convexity premia. This is a genuine economic prediction: in high-volatility environments, long yields should trade *below* consensus expectations of the long-run short rate, and the gap should widen with volatility. Empirically this is reasonably well supported.

Quantifying in a concrete case: with $\sigma = 0.01$ and $\kappa = 0.3$, the premium is $(0.01)^2/(2 \cdot 0.09) = 5.56 \times 10^{-4}$, or about 5.6 basis points. With higher volatility $\sigma = 0.02$ it quadruples to 22 bp. With slower reversion $\kappa = 0.15$, the denominator halves-squared to 0.0225 and the premium is 22.2 bp. So the convexity premium is most visible in slow-reverting, high-volatility environments — exactly the sort of regime that tends to produce inverted or unusually flat yield curves.

Vasicek yield-curve shapes *Vasicek yield-curve shapes for three starting short rates, showing the classical monotone-rising, humped, and inverted regimes.*

An illustrative calibration. Take $\kappa = 0.3$, $\theta = 0.04$, $\sigma = 0.015$ and consider three starting rates: $r_0 \in \{0.01, 0.04, 0.08\}$. For each compute $y_0(T)$ across $T \in \{1, 2, 5, 10, 30\}$ years.

For $r_0 = 0.01$ (well below θ): the short end is near 1%, rises through about 2.5% at 2 years, 3.2% at 5, 3.6% at 10, and approaches the long-end asymptote $\theta - \sigma^2/(2\kappa^2) = 0.04 - 0.00125 \approx 3.88\%$ at 30. A monotonically rising curve — the classical "normal" yield curve in a below-neutral environment.

For $r_0 = 0.04$ (at θ): the curve is nearly flat around 3.9%, with only the convexity correction producing a slight downward drift at longer maturities.

For $r_0 = 0.08$ (well above θ): the short end is at 8%, falling through about 6.6% at 2 years, 5.0% at 5, 4.3% at 10, and 3.88% at 30. An inverted curve — typical of late-cycle monetary tightening when short rates are elevated but the market expects them to fall.

These three scenarios show that Vasicek produces economically sensible curve shapes across the full range of initial conditions, entirely through the mean-reversion parameter κ and the long-run mean θ. The deterministic convergence $r_t \to \theta$ drives the shape, and the convexity correction fine-tunes the long end.

14.4.3 12.4.3 The 1-factor limitation

Crucially, all shifts of the curve under Vasicek are driven by a single factor r_t. Any change in the short rate must propagate through the *entire* yield curve in a fixed, deterministic way dictated by $B_t(T)$. So if the short rate rises by 25 bp, the 2-year yield rises by some specific amount, the 10-year by another, the 30-year by yet another — but the *ratios* among these moves are fixed by the model. Reality is not so rigid: actual yield-curve moves are dominated by three roughly orthogonal factors (level, slope, curvature) that move independently. Principal-components analysis of daily yield-curve changes consistently shows that three factors explain 95-99% of the variance.

To capture this, practitioners turn to multi-factor affine models — typically two-factor Hull-White or three-factor affine with independent Ornstein-Uhlenbeck factors. Each factor contributes its own affine loading, so a linear combination can match observed yield-curve dynamics much more faithfully. The cost is higher calibration complexity and more parameters to stabilise; the benefit is hedges that actually work under real curve reshapes.

A few common multi-factor specifications deserve mention.

First, the **two-factor Gaussian model (G2++)**: two independent (or correlated) Ornstein-Uhlenbeck processes x_t and y_t, with the short rate defined as $r_t = x_t + y_t + \phi(t)$ where $\phi(t)$ is a deterministic shift calibrated to today's curve. With two mean-reversion speeds κ_x, κ_y and two volatilities σ_x, σ_y plus a correlation ρ, the model has enough freedom to match level-plus-slope curve dynamics and a reasonable vol surface. G2++ is the most widely used multi-factor rates model in derivative pricing libraries.

Second, the **three-factor affine model**: three OU factors, allowing separate calibration to level, slope, and curvature. The bond-price formula remains exponential-affine in the three-factor state vector, and the matrix Riccati ODE for $B(t, T)$ can be solved explicitly under mild assumptions (diagonal mean-reversion matrix).

Third, the **Libor Market Model (LMM)**: a radical departure that models individual forward rates directly rather than a short rate. LMM can match any yield curve by construction (forward rates are model inputs, not model outputs) and any swaption vol cube by suitable specification of forward-rate volatilities and correlations. The price for this flexibility is giving up the low-dimensional affine structure: LMM is high-dimensional, state-vector-based, and typically requires Monte Carlo for complex payoffs.

Which multi-factor model to use is ultimately a portfolio-specific decision. For vanilla cap-floor-swaption books, G2++ is usually sufficient. For Bermudan or callable products, LMM or extended affine models may be needed. For CVA/XVA computations across a huge book, the calibration speed and hedging accuracy of affine models make them attractive even when LMM would be "more accurate" per trade.

14.5 12.5 Ho-Lee — the canonical derivation

Vasicek with constant θ has three parameters (κ, θ, σ) and can fit only three features of a yield curve — a level, a slope, and a curvature, roughly. A real observed curve has dozens of quoted maturities and cannot be reproduced by any three-parameter model. The natural fix is to let the drift schedule vary with time. The simplest such extension drops the mean-reversion altogether and retains only the time-varying drift: this is the Ho-Lee model, which we now derive from first principles. It serves a dual purpose — it is historically the first example of exact yield-curve calibration, and it is mathematically the $\kappa \to 0$ limit of Hull-White that we develop in §12.6.

14.5.1 12.5.1 Discrete Ho-Lee

The discrete Ho-Lee dynamics on a Δt grid:

$$\boxed{r_n \;=\; r_{n-1} \;+\; \theta_{n-1}\,\Delta t \;+\; \sigma\sqrt{\Delta t}\,x_n, \qquad x_n \overset{\text{iid}}{\sim} \pm 1 \text{ Bernoulli},\; \mathbb{Q}(x_n = \pm 1) = \tfrac{1}{2}.} \tag{12.32}$$

Read (12.32) aloud: "tomorrow's short rate equals today's short rate, plus a deterministic drift over the step, plus a symmetric random coin-flip scaled to the right standard deviation." The drift θ_{n-1} is what curves the rate process to match the market; the shock $\sigma\sqrt{\Delta t}\,x_n$ is the noise that lets the rate move up or down. Because the shock is additive (not multiplicative) and symmetric, the up-move and down-move at each node have equal magnitude. This produces a recombining tree — two steps up and one step down lands at the same node as one up and two down, reshuffled.

Key features.

- *Arithmetic, not geometric.* Rates are added, not scaled — rates can go negative. Historically this was Ho-Lee's greatest weakness; post-2008, actually a feature when pricing negative-rate regimes.
- *No mean reversion.* The Ho-Lee dynamic is Vasicek with $\kappa = 0$; only the drift θ_n shapes the curve. A pure random walk in rates (with drift) means that far out in maturity the distribution of r_T becomes increasingly diffuse and wide.
- *Recombining tree.* With a symmetric ± 1 shock and constant σ, up-then-down and down-then-up return to the same node. The recombination reduces the node count at step n from 2^n to $n+1$, making calibration linear-time rather than exponential. This is the single largest computational advantage of Ho-Lee and a property preserved by its continuous-time limit.

Calibration goal. Find $\theta_0, \theta_1, \theta_2, \ldots$ so that the model matches the market bond prices $P_0(1), P_0(2), P_0(3), \ldots$. The bootstrap is triangular: θ_0 depends only on $P_0(1)$ and $P_0(2)$, θ_1 on $P_0(3)$ given θ_0, and so on. We never solve a nonlinear system. At every step, each new market bond price adds exactly one new equation in exactly one new unknown θ_n — one of the cleanest pieces of structure in quantitative finance.

With $q = 1/2$ inside the tree probability measure, bond prices are

$$P_0(1) \;=\; e^{-r_0\,\Delta t}, \qquad P_0(2) \;=\; e^{-r_0\Delta t}\left[\tfrac{1}{2}e^{-r_{1u}\Delta t} \;+\; \tfrac{1}{2}e^{-r_{1d}\Delta t}\right], \tag{12.33}$$

and similarly $P_0(3)$ depends on θ_0, θ_1, giving us one equation in θ_1 once θ_0 is known. The philosophical takeaway: calibration replaces the abstract non-uniqueness of $\mathbb{Q}$ (§12.1.1) with a specific choice of θ-schedule that makes model-implied bond prices equal market-observed bond prices. Once that schedule is fixed, every other derivative prices off the resulting tree without further calibration.

14.5.2 12.5.2 Continuous-time limit (direct Donsker argument)

The cleanest path to the continuous-time Ho-Lee model: start from (12.32), sum the recursion, and pass $N \to \infty$ directly, bypassing the Vasicek/Hull-White machinery. Fix a horizon T, set $\Delta t = T/N$ with $t_n = n\Delta t$, and iterate (12.32):

$$r_N \;=\; r_0 \;+\; \sum_{n=1}^{N} \theta_{n-1}\,\Delta t \;+\; \sigma\sqrt{\Delta t}\sum_{n=1}^{N} x_n. \tag{12.34}$$

This is a pure telescoping sum: starting level r_0, cumulative drift, cumulative shock. The moments are immediate:

$$\mathbb{E}^{\mathbb{Q}}[r_T] \;\to\; r_0 \;+\; \int_0^T \theta_u\,\mathrm{d}u, \qquad \mathbb{V}^{\mathbb{Q}}[r_T] \;=\; \sigma^2 T. \tag{12.35}$$

By Donsker's Invariance Principle — a sum of iid ± 1 shocks, scaled by $\sqrt{\Delta t}$, converges in distribution to Brownian motion — we have

$$\boxed{r_T \overset{d}{=} r_0 + \int_0^T \theta_u \, \mathrm{d}u + \sigma\sqrt{T}\, Z, \qquad Z \sim \mathcal{N}(0,1),} \tag{12.36}$$

equivalently the SDE

$$\mathrm{d}r_t = \theta_t \, \mathrm{d}t + \sigma \, \mathrm{d}W_t. \tag{12.37}$$

The $\sqrt{\Delta t}$ scaling on the shock is the fundamental scaling of Brownian motion, which has quadratic variation $\mathrm{d}t$. Had we scaled by Δt instead, the limit would have been deterministic (shock vanishing); by 1, it would have been infinite (shock exploding). The "right" scaling $\sqrt{\Delta t}$ is exactly what keeps the variance of the sum of N shocks finite. This scaling shows up in everything from heat diffusion to stock returns to interest-rate innovations; it is the signature of a process driven by "many small independent shocks" (Chapter 3's construction of Brownian motion reviews this in detail).

Negative rates are explicit: $r_T \sim \mathcal{N}(r_0 + \int \theta, \sigma^2 T)$, so $\mathbb{P}(r_T < 0) = \Phi(-M/\sqrt{V}) > 0$ where M is the mean and V the variance. For $T = 10$ years, $\sigma = 1\%$, $M = 3\%$, that probability is $\Phi(-0.95) \approx 17\%$ — substantial. For $T = 1$ year it drops to $\approx 2\%$. Negative rates thus bite mainly at long horizons, where variance has had time to accumulate.

Distribution of the integrated rate. A shock at time t affects *all* future r_u for $u \in [t, T]$ (the additive structure with no reversion has infinite memory), contributing with lever-arm weight $T - t$ to the integrated rate. Working through the double-sum calculation,

$$\int_0^T r_u \, \mathrm{d}u \overset{d}{=} r_0 T + \int_0^T \int_0^u \theta_s \, \mathrm{d}s \, \mathrm{d}u + \left(\frac{\sigma^2 T^3}{3}\right)^{1/2} Z. \tag{12.38}$$

The cubic growth $T^3/3$ of the variance — as opposed to T for the process itself — is Ho-Lee's signature of missing mean reversion. Physically: $\mathrm{Var}[\int_0^T r_u \, \mathrm{d}u] = \int_0^T (T-t)^2 \, \mathrm{d}t \cdot \sigma^2 = \sigma^2 T^3/3$. A 30-year Ho-Lee at $\sigma = 1\%$ would give $\sigma^2 T^3/3 = 10^{-4} \cdot 9000 = 0.9$, an integrated-rate standard deviation of 94%. This is absurdly large — translated into bond-price uncertainty, it implies extreme price sensitivity to model parameters at long horizons. Mean reversion (Vasicek/HW) fixes this by making variance grow only linearly in T at long horizons. This is the single biggest practical reason Ho-Lee is rarely used for long-dated instruments.

14.5.3 12.5.2.b The meaning of the $\sqrt{\Delta t}$ scaling

The scaling $\sqrt{\Delta t}$ on the shock is not arbitrary. Brownian motion was first introduced by Einstein (1905) to model pollen grains in water — particles buffeted by countless small random molecular collisions. Einstein's analysis showed that for such a process, the position's variance grows linearly in time: $\mathbb{V}[X_t] \propto t$. Linear variance growth in continuous time translates to $\sqrt{\Delta t}$ scaling of discrete increments (because variance of a sum of N independent increments equals N times the per-increment variance, and if N times some number equals T, then the per-increment standard deviation is $\sqrt{T/N} = \sqrt{\Delta t}$). This scaling shows up in heat diffusion, stock returns, interest-rate innovations, and countless other physical processes. It is the signature of "many small independent shocks."

The $\sqrt{\Delta t}$ scaling also underpins the famous quadratic- variation result: $(\mathrm{d}W_t)^2 = \mathrm{d}t$ in the Itô-calculus limit. This is not an equality in the ordinary sense (both sides are random) but a precise statement about how squared infinitesimal increments behave in L^2. That $(\mathrm{d}W_t)^2$ is deterministic (equal to $\mathrm{d}t$) is the reason Brownian calculus has its peculiar features — Itô's lemma includes a second-order term proportional to variance, reflecting the nonzero quadratic variation. In rate pricing, this shows up as the convexity term C_T; in equity pricing, as the extra $\frac{1}{2}\sigma^2$ drift in the geometric- Brownian solution.

If we relax the "finite variance" constraint and allow shocks to have heavy tails with infinite variance, the limit becomes a stable Lévy process instead of Brownian motion. Stable Lévy processes have jumps; they are discontinuous. This is the mathematical basis for jump- diffusion models of rates, equity prices, and credit spreads. Such models are sometimes used when heavy tails are important (e.g. pricing equity index options during tail events), but they are not standard for rates. The Gaussian limit is the default because it matches observed rate-change distributions reasonably well at the level of daily moves, and the mathematical benefits are substantial.

14.5.4 12.5.3 Bond price and calibration under continuous Ho-Lee

Because $I_T = \int_0^T r_u \, du$ is Gaussian, $\mathbb{E}^{\mathbb{Q}}[e^{-I_T}] = \exp(-\mathbb{E}I_T + \frac{1}{2}\mathbb{V}I_T)$, giving

$$\boxed{\ln P_0(T) \;=\; -r_0\, T \;-\; \int_0^T \int_0^u \theta_s \, \mathrm{d}s \, \mathrm{d}u \;+\; \frac{\sigma^2 T^3}{6}.} \tag{12.39}$$

The $+\sigma^2 T^3/6$ is the Ho-Lee convexity adjustment. Read in context of (12.19): $+\frac{1}{2}\Sigma_*^2 = +\frac{1}{2}\sigma^2 T^3/3 = \sigma^2 T^3/6$ — Jensen's mountain applied to a cubically growing variance.

Calibration via differentiation. Since (12.39) expresses $\ln P_0(T)$ as a double integral of θ plus known terms, differentiating twice with respect to T pulls out θ_T pointwise.

First derivative:

$$\partial_T \ln P_0^*(T) \;=\; -r_0 \;-\; \int_0^T \theta_s \, \mathrm{d}s \;+\; \frac{\sigma^2 T^2}{2}. \tag{12.40}$$

Since $\partial_T \ln P_0^*(T) = -f_0(T)$, this rearranges to the Ho-Lee forward-curve identity $f_0(T) = r_0 + \int_0^T \theta_s \, \mathrm{d}s - \sigma^2 T^2/2$.

Second derivative:

$$\partial_T^2 \ln P_0^*(T) \;=\; -\theta_T \;+\; \sigma^2 T, \tag{12.41}$$

and therefore

$$\boxed{\theta_T \;=\; -\,\partial_T^2 \ln P_0^*(T) \;+\; \sigma^2 T.} \tag{12.42}$$

Equation (12.42) is the classical Ho-Lee calibration identity. Read it intuitively: the drift target θ_T at time T is "the slope of the forward curve" plus "a convexity bump proportional to T." If the forward curve is flat, the drift target is just the convexity bump. If the forward curve is rising, θ_T is positive and the model drifts rates upward on average.

In practice one fits a smooth parametric form (Nelson-Siegel, Svensson, cubic splines) to the observed log-bond curve, differentiates the fit twice, adds the convexity bump, and reads off θ_T at every grid point. The choice of curve-smoother affects the resulting $\theta(t)$ qualitatively: noisy second derivatives translate into noisy $\theta(t)$, which makes the rate model unstable and gives erratic prices for non-calibration instruments.

14.5.5 12.5.4 Ho-Lee as the $\kappa \to 0$ limit of Hull-White

Ho-Lee is precisely the $\kappa = 0$ specialisation of the Hull-White dynamics (below):

$$\underbrace{\mathrm{d}r_t = \kappa(\theta_t - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t}_{\text{Hull-White}} \xrightarrow{\kappa \to 0} \underbrace{\mathrm{d}r_t = \theta_t\,\mathrm{d}t + \sigma\,\mathrm{d}W_t}_{\text{Ho-Lee}}$$

with the understanding that the Hull-White θ_t must diverge as $1/\kappa$ for the limit to be well-defined — the joint limit $\kappa\theta_t \to \theta_{\mathrm{HL}}$ is what survives. Conceptually the three models nest: Ho-Lee is the simplest thing; Vasicek adds a pull-to-constant-target; Hull-White makes that target time-varying. The calibration machinery for Hull-White specialises to Vasicek (constant θ) and to Ho-Lee ($\kappa = 0$) with minimal changes. This pedagogical nesting is why we present them in this order.

14.6 12.6 Hull-White — the canonical derivation

Vasicek captures mean reversion but has only a constant θ, so it can fit only a single feature of the observed term structure (roughly, the long-run mean). Ho-Lee captures exact yield-curve matching but has no mean reversion, so its variance grows cubically and the model breaks at long horizons. Hull-White is the best of both worlds: Vasicek dynamics with a *time-dependent* drift target $\theta(t)$. The function $\theta(t)$ is calibrated so the model exactly reproduces today's yield curve, while the mean-reversion parameter κ tames variance growth.

14.6.1 12.6.1 Discrete Hull-White

Promote the constant θ in (12.9) to a deterministic function θ_t and replace ± 1 Bernoullis with Gaussian innovations:

$$\boxed{r_{t_n} - r_{t_{n-1}} \;=\; \kappa(\theta_{t_{n-1}} - r_{t_{n-1}})\,\Delta t \;+\; \sigma\sqrt{\Delta t}\,\chi_n, \qquad \chi_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1).} \tag{12.43}$$

The move from constant θ to time-varying $\theta(t)$ is the conceptual leap. In Vasicek the process always reverts toward the same number, so you can only fit one long-run level. In Hull-White the process reverts toward a *moving target* — the target itself is a calibrated function of time — so every part of the yield curve can be independently matched. The price is infinite-dimensional calibration (one θ per time point), but as we will see this reduces to a linear bootstrap exactly as in Ho-Lee.

What the parameters mean.

- κ is a *speed*: the deterministic part of (12.43) pulls r toward θ_t at exponential rate κ. Large $\kappa \Rightarrow$ very "sticky" short rate around θ; small $\kappa \Rightarrow$ near-random-walk (Ho-Lee). Typical calibrated values live in 0.05-0.3 per year, giving shock half-lives of 2-15 years.
- σ sets the innovation scale. The unconditional variance of r_T approaches $\sigma^2/(2\kappa)$ as $T \to \infty$. Typical σ values are 0.005-0.02 in absolute rate terms (50-200 bp per year root-volatility).
- $\theta(t)$ is the *drift shaper*: the knob tuned to the market. In the limit it is essentially $f_0(t) + \sigma^2 \cdot (\text{convexity})$, where $f_0(t)$ is the instantaneous forward rate.

It is worth pausing on the division of labour. κ and σ are the *volatility-structure* parameters — they determine how rates move, how persistent shocks are, how volatile caps and swaptions will be. These are typically calibrated *jointly* against vol-sensitive instruments (ATM swaption prices). $\theta(t)$ is the *curve-fitting* parameter, chosen conditionally on (κ, σ) to exactly match the discount curve. So calibration has two stages: first pick (κ, σ) to match vol-sensitive instruments, then bootstrap $\theta(t)$ to match bond prices. This two-stage structure is characteristic of rates calibration in practice.

In production systems the two stages are often iterated. A first pass with historically-sensible (κ, σ) produces a $\theta(t)$ schedule. The resulting model is then used to price a set of swaptions, and the (κ, σ) are adjusted to reduce swaption-price errors. Then $\theta(t)$ is re-bootstrapped with the updated (κ, σ), and so on until convergence. The inner bootstrap (curve) takes microseconds; the outer optimisation (swaption fit) takes seconds to minutes. The separation of concerns is crucial: the fast bootstrap runs inside the slow optimiser.

Hull-White simulated short-rate paths *Hull-White simulated short-rate paths reverting to time-dependent $\theta(t)$.*

14.6.2 12.6.2 Summed integral and continuous limit

Unrolling (12.43) recursively with $\beta = 1 - \kappa\Delta t$ and $\alpha_{n-1} = \kappa\theta_{t_{n-1}}\Delta t$ yields the same geometric structure as the Vasicek unrolling, with the single difference that the intercept α varies with n:

$$r_T \;=\; r_0\,\beta^N \;+\; \sum_{n=1}^{N} \alpha_{n-1}\,\beta^{N-n} \;+\; \sigma\sqrt{\Delta t}\sum_{n=1}^{N} x_n\,\beta^{N-n}. \tag{12.44}$$

Taking $N \to \infty$ with $\Delta t = T/N$, the standard Donsker-type identifications give:

- $(1 - \kappa T/N)^{N-n} \to e^{-\kappa(T-u)}$,
- $(1 - \kappa T/N)^N \to e^{-\kappa T}$,
- $\sum_n (\cdot)\,\Delta t \to \int_0^T (\cdot)\,\mathrm{d}u$,
- $\sigma\sqrt{\Delta t}\sum_n x_n\,\beta^{N-n} \to \sigma\int_0^T e^{-\kappa(T-u)}\,\mathrm{d}W_u$.

Each of these is a standard stochastic-calculus limit (Chapter 3): the first two use $\lim_N (1 + x/N)^N = e^x$; the third is the Riemann-sum-to-integral limit; the fourth is Itô's stochastic-integral construction for a deterministic integrand.

The Wiener integral $\int_0^T e^{-\kappa(T-u)}\,\mathrm{d}W_u$ is a Gaussian random variable with mean zero and variance $\int_0^T e^{-2\kappa(T-u)}\,\mathrm{d}u = (1 - e^{-2\kappa T})/(2\kappa)$ — exactly the Vasicek stationary-variance formula. This clean Gaussian structure is the source of the closed-form bond price. Had the integrand been state-dependent (like $\sqrt{r_u}$ in CIR), the integral would be a true Itô integral, not a Wiener integral, and its distribution would not be Gaussian.

The limit of (12.44) is the explicit Hull-White solution:

$$r_T \;=\; r_0\, e^{-\kappa T} \;+\; \kappa \int_0^T e^{-\kappa(T-u)}\, \theta_u \,\mathrm{d}u \;+\; \sigma \int_0^T e^{-\kappa(T-u)}\,\mathrm{d}W_u. \tag{12.45}$$

The pattern is identical to Vasicek with constant θ (equation (12.12)); only the deterministic drift contribution has been generalised from a function of $\tau = T - t$ alone to a true integral against the time-varying θ_u.

14.6.3 12.6.3 Hull-White bond price

The integrated rate, by the same Fubini manipulation as in §12.3.3:

$$\int_0^T r_u \,\mathrm{d}u \;=\; r_0\, B_T \;+\; A_T \;+\; \frac{\sigma}{\kappa}\int_0^T (1 - e^{-\kappa(T-u)})\,\mathrm{d}W_u, \tag{12.46}$$

where

$$B_T \;=\; \frac{1 - e^{-\kappa T}}{\kappa}, \qquad A_T \;=\; \int_0^T \theta_u\,(1 - e^{-\kappa(T-u)})\,\mathrm{d}u. \tag{12.47}$$

B_T is worth a moment because it is the single most-used quantity in Hull-White computations: the "effective duration" that the current short rate r_0 exerts on the integrated rate. In the $\kappa \to 0$ limit, $B_T \to T$ (no reversion means the current rate affects the entire horizon). In the $\kappa \to \infty$ limit, $B_T \to 0$ (infinite reversion instantly forgets the current rate). For typical $\kappa \in [0.05, 0.3]$, the ratio B_T/T ranges from near 1 at short T down to something much smaller at long T. For $\kappa = 0.15$ and $T = 30$: $B_T = (1 - e^{-4.5})/0.15 \approx 6.6$, so $B_T/T \approx 0.22$. The current rate has 22% as much influence on the 30-year bond price as it would under no mean reversion.

Because the integrated rate is Gaussian with mean $r_0 B_T + A_T$ and variance

$$\mathbb{V}\left[\int_0^T r_u\,\mathrm{d}u\right] \;=\; \frac{\sigma^2}{\kappa^2}\int_0^T \left(1 - e^{-\kappa(T-u)}\right)^2 \mathrm{d}u \;\equiv\; 2\,C_T, \tag{12.48}$$

the Hull-White bond price comes out directly from the Gaussian moment-generating function:

$$\boxed{P_0(T) \;=\; \exp\{-r_0\, B_T \;-\; A_T \;+\; C_T\}\,,} \tag{12.49}$$

with convexity adjustment

$$C_T \;=\; \frac{\sigma^2}{2\kappa^2}\int_0^T \left(1 - e^{-\kappa(T-u)}\right)^2 \mathrm{d}u. \tag{12.50}$$

Reading (12.49), three pieces:

- $r_0 B_T$: current-level contribution, decaying at rate κ because a rate shock "today" is damped over the life of the bond.
- A_T: the drift contribution from the θ path. All the market information about future forward rates lives here.
- $+C_T$: positive convexity. Jensen's inequality makes the bond strictly *more* valuable than naive plug-in discounting suggests. This is the Hull-White analogue of the $\sigma^2 T^3/6$ term in Ho-Lee and of the $\frac{1}{2}\Sigma_*^2$ term in constant-θ Vasicek.

Sanity checks. As $\kappa \to 0$ (Taylor-expanding $(1 - e^{-\kappa(T-u)})^2 \approx \kappa^2 (T-u)^2$), we have $B_T \to T$, $A_T \to \int_0^T \theta_u\,\mathrm{d}u$, and $C_T \to \sigma^2 T^3/6$. Formula (12.49) collapses to the Ho-Lee formula (12.39). Setting $\theta(t) = \theta$ constant, $A_T = \theta(T - B_T)$ and (12.49) specialises to Vasicek. The nesting is clean and complete: Ho-Lee $\subset$ Vasicek $\subset$ Hull-White.

The κ^2 cancellation in the C_T limit is a specific instance of a more general phenomenon: many quantities in Hull-White approach their Ho-Lee limits smoothly as $\kappa \to 0$, because the mean-reversion structure enters quadratically in the variance terms and linearly in the mean terms. This smoothness of limits lets us calibrate Vasicek/Hull-White parameters to low κ values without numerical difficulty.

Worked magnitude. For $T = 10$, $\sigma = 0.01$, $\kappa = 0.1$: the integral $\int_0^{10}(1 - e^{-0.1(10-u)})^2\,\mathrm{d}u = 10 - 2(1 - e^{-1})/0.1 + (1 - e^{-2})/0.2 = 10 - 12.64 + 4.32 = 1.68$, so $C_{10} = (10^{-4}/0.02) \cdot 1.68 = 0.0084$ — an 84-bp bonus in log-bond. By comparison, Ho-Lee with the same σ at $T = 10$ gives $C_{10} = \sigma^2 T^3/6 = 0.0167$, or 167 bp — roughly double. Mean reversion halves the 10-year convexity bonus and the gap grows exponentially at longer tenors. At $T = 30$ the Ho-Lee number is $\sigma^2 T^3/6 = 0.45$ (4500 bp!) while Hull-White ($\kappa = 0.1$) comes in around 600 bp — an order of magnitude smaller. This is why mean reversion matters for long-dated pricing: Ho-Lee's unbounded convexity growth is unrealistic; Hull-White's bounded growth matches market observations.

Model-implied zero-coupon bond curve *Model-implied zero-coupon bond / discount curve* $P_0(T)$.

14.6.4 12.6.3.b Precomputation and numerical considerations

For pricing efficiency it is convenient to precompute the "basis functions" once per maturity T: $1 - e^{-\kappa(T-t)}$, $(T - t) - B_t(T)$, and $(T - t) - 2B_t(T) + (1 - e^{-2\kappa(T-t)})/(2\kappa)$. Then $A_t(T)$ and $B_t(T)$ are simple linear combinations with coefficients θ, $\sigma^2/(2\kappa^2)$, etc. In production rates libraries, the ODEs for A and B are solved once at model-initialisation time and cached; during pricing, the pre-computed $A(t, T)$ and $B(t, T)$ are simply evaluated at the appropriate (t, T) pair. This makes bond pricing essentially instantaneous — a table lookup rather than an integration. The same caching strategy applies to multi-factor affine models and is part of why these models scale so gracefully to large trading books.

The finite long-bond duration $1/\kappa$ is the quantitative expression of mean reversion. *Caveat: this is a model artefact of Vasicek with constant θ, not an empirical fact.* The bound says that under the model's dynamics — exponential mean reversion of the short rate to a constant target — the bond-price sensitivity to r_0 asymptotes at $1/\kappa$ years. Empirical 30-year Treasury modified durations run 15–20 years, well above the 2–10 year window that "reasonable" $\kappa \in [0.1, 0.5]$ produces. The discrepancy is resolved by Hull-White's time-varying $\theta(t)$: with a calibrated θ schedule the model reproduces empirical duration without forcing κ outside the volatility-instrument-implied range. A calibration algorithm returning κ outside $[0.1, 0.5]$ under *constant-θ* Vasicek is a flag of model mis-specification, not an empirical truth about rates.

14.6.5 12.6.4 Calibrating $\theta(u)$ to the market curve

In practice one observes market bond prices $P_0^*(T)$ (or yields) at $t = 0$ for all maturities T. The idea of Hull-White: choose the deterministic function θ_t so the model reproduces $P_0^*(T)$ exactly. Why is this so important? Because a rates trader using a model that doesn't match today's curve is effectively betting that today's market is mispriced — almost never a useful default. With $\theta(t)$ free, the model has enough functional freedom to fit any observed curve exactly; the remaining parameters (κ, σ) are then fit to options on bonds.

From (12.49), taking logs at $t = 0$,

$$\ln P_0^*(T) \;=\; -r_0\, B_T \;-\; \int_0^T \theta_u\,(1 - e^{-\kappa(T-u)})\,\mathrm{d}u \;+\; C_T. \tag{12.51}$$

Closed-form inversion. Differentiating twice in T gives the pointwise Hull-White calibration identity:

$$\boxed{\theta(T) \;=\; \frac{1}{\kappa}\,\partial_T f_0(T) \;+\; f_0(T) \;+\; \frac{\sigma^2}{2\kappa^2}(1 - e^{-2\kappa T})\,,} \tag{12.52}$$

where $f_0(T) = -\partial_T \ln P_0^*(T)$ is the market instantaneous forward rate. The first term is the slope of the forward curve scaled by $1/\kappa$, the second is a level match to the forward itself, and the third is a maturity-dependent convexity adjustment that saturates at $\sigma^2/(2\kappa^2)$ for large T. In the $\kappa \to 0$ limit, $\kappa\,\theta(T)$ — not $\theta(T)$ itself — must remain finite (see §12.5.4): multiplying (12.52) through by κ gives $\kappa\theta(T) = \partial_T f_0(T) + \kappa f_0(T) + (\sigma^2/2\kappa)(1 - e^{-2\kappa T})$, and taking $\kappa \to 0$ the middle term vanishes while the last approaches $\sigma^2 T$, so $\kappa\theta \to \partial_T f_0 + \sigma^2 T = \theta_{\mathrm{HL}}$, recovering the Ho-Lee formula (12.42).

Bucket bootstrap. In implementation, (12.52) is rarely used literally because it requires a second derivative of the observed forward curve, which is numerically ill-conditioned. Production implementations parameterise $\theta(t)$ as piecewise constant on the bond-maturity grid, and solve for the parameter values that exactly reproduce observed bond prices. Splitting the integral in (12.51) into bucket contributions with piecewise-constant $\theta^{(m)}$ on $[T_{m-1}, T_m]$:

$$A_{T_k} \;=\; \sum_{m=1}^{k} \theta^{(m)}\,\alpha^{(m,k)}, \qquad \alpha^{(m,k)} \;=\; (T_m - T_{m-1}) \;-\; \frac{e^{-\kappa(T_k - T_m)} - e^{-\kappa(T_k - T_{m-1})}}{\kappa}. \tag{12.53}$$

Solving (12.51) for the current bucket $\theta^{(k)}$ yields the bootstrap recursion:

$$\boxed{\theta^{(k)} \;=\; \frac{T_k\,y_k^* \;-\; r_0\,B_{T_k} \;+\; C_{T_k} \;-\; \sum_{m=1}^{k-1}\theta^{(m)}\,\alpha^{(m,k)}}{\alpha^{(k,k)}}.} \tag{12.54}$$

This is a simple linear bootstrap, exactly analogous to Ho-Lee and to yield-curve stripping: each tenor T_k determines $\theta^{(k)}$ given the already-calibrated earlier buckets. No optimisation, no iteration, no nonlinearity. Just a sequence of divisions.

The $\alpha^{(m,k)}$ are "overlap weights" — how much of bucket m's drift influences the integrated rate up to horizon T_k. They are pure geometry of the exponential kernel and do not depend on market data; production implementations precompute them once per κ and cache. Their key property is causality: $\alpha^{(m,k)} = 0$ for $m > k$, which is what makes the bootstrap triangular. A $\theta^{(m)}$ for $m > k$ has no effect on $P_0(T_k)$ (it refers to rates in the future relative to T_k), so adding tenor T_k determines only $\theta^{(k)}$ and leaves earlier buckets fixed.

Worked example. Take $\kappa = 0.15$, $\sigma = 0.012$, $r_0 = 0.04$, market yields $y_1^* = 0.042$ at $T_1 = 1$ and $y_2^* = 0.045$ at $T_2 = 2$. Compute $B_{T_1} = (1 - e^{-0.15})/0.15 \approx 0.9286$ and $C_{T_1} \approx 3.4 \times 10^{-5}$. With $k = 1$ the sum is empty and $\alpha^{(1,1)} = T_1 - B_{T_1} \approx 0.0714$. Then

$$\theta^{(1)} \;\approx\; \frac{1 \cdot 0.042 - 0.04 \cdot 0.9286 + 3.4\text{e}{-5}}{0.0714} \;\approx\; 0.0691.$$

The calibrated drift target is about 6.9% — well above the current 4% level. This is the "overshoot": mean reversion only partially converges the rate toward $\theta^{(1)}$ over one year ($\mathbb{E}[r_1] = 0.04 \cdot e^{-0.15} + \theta^{(1)}(1 - e^{-0.15}) \approx 0.04 \cdot 0.861 + \theta^{(1)} \cdot 0.139$), so $\theta^{(1)}$ must overshoot the target rate level to achieve the market-implied integrated-rate value.

theta(t) vs market forward curve $\theta(t)$ *sits slightly above the market forward curve because of the convexity adjustment* C_T.

For a trader using this model, the practical consequence is that the calibrated drift schedule $\theta(t)$ should *not* be interpreted as "what the market expects the rate to be at time t." That would only be true in the $\kappa \to 0$ limit. With mean reversion, $\theta(t)$ is instead "what the target has to be so that the expected rate path matches the market's forward curve," which is different. The expected rate path itself is $\mathbb{E}[r_t] = r_0 e^{-\kappa t} + \kappa \int_0^t e^{-\kappa(t-u)}\theta_u\,du$, which is smoother than $\theta(t)$.

Philosophical note. By adjusting $\theta(t)$ to match today's curve exactly, Hull-White gives up the possibility of *explaining* the curve in terms of a small number of stationary parameters. The curve is now an input, not a prediction. This is a feature for pricing (you want your model to agree with market quotes) but a bug for economic insight (you can't test your $\mathbb{Q}$-dynamics against curve data you haven't already used to calibrate them). In this sense Hull-White is a member of a broad class of "calibrated" models in finance whose primary virtue is practical rather than explanatory.

There is a deeper parallel to the Black-Scholes world. In equity, the "implied volatility" surface is what the market uses to quote options; it is not a model parameter but a derived quantity. In rates, the forward curve $f_0(\cdot)$ is what the market quotes; $\theta(t)$ is the corresponding derived quantity. Both are "market-implied" model inputs computed from quoted prices via specific formulas.

14.6.6 12.6.5 Practical implementation considerations

The calibration described above is conceptually simple but has several practical wrinkles worth flagging for someone who will actually implement it.

Curve smoothing. The pointwise formula (12.52) requires a first and second derivative of the market forward curve. Real market data (swap quotes, futures, bond prices) are discrete and noisy; naive numerical differentiation will produce erratic $\theta(t)$ that oscillates between buckets and destabilises the pricing engine. Production implementations fit a smooth parametric form to the log- discount curve before differentiating. Common choices include cubic splines (flexible, but can oscillate near endpoints), Nelson-Siegel (four parameters capturing level, slope, curvature, and one shape parameter), Svensson (Nelson-Siegel plus two more parameters for improved curvature fit), and tension splines (controlled smoothness). The choice trades off exact fit against parameter stability; over-smoothed curves produce stale prices, under-smoothed curves produce noisy prices. The sweet spot is chosen based on the desk's risk appetite and is re-evaluated periodically.

Bucket stability. Piecewise-constant $\theta(t)$ has a subtle defect: the drift is discontinuous at bucket boundaries, producing small kinks in the implied forward curve. These kinks are usually invisible in pricing but can show up as spurious "vol spikes" in certain risk metrics. Smoother representations (piecewise linear, cubic splines) avoid the kinks at the cost of more calibration parameters per bucket. A compromise is to use piecewise-constant $\theta(t)$ plus a post-processing smoothing step that redistributes drift mass across adjacent buckets while preserving exact bond-price fit. Some practitioners prefer to parameterise the cumulative integral $\Theta(t) = \int_0^t \theta(u)\, \mathrm{d}u$ directly, which is smoother than θ itself.

Chicken-and-egg (κ, σ, θ) joint calibration. The Hull-White convexity term C_T depends on (κ, σ), so bootstrapping $\theta(t)$ requires knowing these first. But (κ, σ) are typically determined by fitting to caplet and swaption prices, which themselves depend on the already-extracted $\theta(t)$. The standard resolution is iterative: pick initial (κ, σ) from historical estimates; extract $\theta(t)$ via (12.54); fit (κ, σ) to option prices; re-extract $\theta(t)$; iterate until convergence. In practice convergence is fast (a few iterations) because the interaction between $\theta(t)$ and option prices is weak for typical parameters. Damping, regularisation, and convergence diagnostics are standard components of serious implementations.

Instrument set selection. Which market instruments should the model be calibrated to? For vanilla books, the practical answer is: (i) the short-end of the curve from overnight rates and short-dated futures; (ii) the intermediate-dated part from medium-term swaps and futures; (iii) the long end from long-dated swaps; (iv) a strip of ATM caplets or swaptions for (κ, σ). For more exotic portfolios (callable bonds, Bermudan swaptions) additional volatility instruments may be needed. The calibration error across each instrument set is typically monitored as a diagnostic; deviations signal either a model limitation or a data issue.

Real-time re-calibration. Because each calibration step is linear-time, production systems can re-calibrate every few seconds or faster as the market ticks. Whether they *should* is a deeper question: constantly-shifting $\theta(t)$ muddies P&L attribution ("market moved" vs "model re-fit") and creates numerical noise. Most desks re-calibrate at a few natural cadences (morning open, mid-day, end of day, plus on large market moves). The choice of cadence is a trade-off between model freshness and signal-to-noise.

14.7 12.7 Affine term structure and the duality of short-rate models

The word "affine" denotes a very specific class of models with a precise structural signature: the drift and the squared diffusion of the state variable are both linear (affine) functions of the state. Vasicek sits at the simplest end of this class (constant diffusion, linear drift). CIR complicates matters by allowing the diffusion $\sigma\sqrt{r_t}$ to depend on r_t — this makes the squared diffusion $\sigma^2 r_t$ linear in r_t, so the model is still affine but the ODE for B becomes a genuine Riccati equation. Multi-factor extensions replace r_t with a vector state and the scalar ODE for B with a matrix Riccati ODE. In every case the bond price remains exponential-affine in the state vector, and that is the single structural property that makes the entire edifice tractable.

The affine class has been systematically characterised in a celebrated 1996 paper by Duffie and Kan. They showed that if one posits an exponential-affine bond price $P(t, T, x) = \exp(A(t, T) + B(t, T)^\top x)$ for some state vector x, then the underlying state must have a very restricted form of dynamics: the drift must be affine in x, the instantaneous covariance matrix must be affine in x (with coefficients specifying exactly which linear combinations of state variables can drive which diagonal entries of the covariance), and the structure must be compatible across maturities. Moreover, Duffie and Kan classified the admissible affine models by their "signature" — whether each state variable can take all real values or must be non-negative. Vasicek is a pure "Gaussian" affine model (all state variables unconstrained); CIR is a "square-root" affine model (one state variable positive, with $\sqrt{\cdot}$ diffusion); mixed affine models combine these. The Duffie-Kan classification underlies essentially every multi-factor bond-pricing library in practice.

Outside the affine class, exponential-affine bond prices are impossible: if the dynamics of the state are not affine, the bond price will not have this form. What do you get instead? Typically, semi-closed-form approximations (via characteristic functions, transform inversions, or perturbation expansions), Monte Carlo, or numerical PDE methods. These techniques work, but they are slower and less transparent than the affine closed form. For this reason, most practical rates libraries implement pricing as "affine-first, with plug-ins for non-affine corrections."

14.7.1 12.7.1 The three models in one diagram

The three models derived in this chapter occupy three corners of a small parameter space:

Model	κ	θ	SDE / bond price
Ho-Lee	0	$\theta(t)$	$\mathrm{d}r_t = \theta(t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$; $\ln P_0(T)$ has $+\sigma^2 T^3/6$ convexity
Vasicek	$\kappa > 0$	constant	$\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$; $P_t(T) = e^{A_t(T) - B_t(T) r_t}$ affine
Hull-White	$\kappa > 0$	$\theta(t)$	Vasicek dynamics with θ_t calibrated to market curve

The nesting is transparent: Hull-White ⊃ Vasicek ⊃ Ho-Lee, with each inclusion obtained by freezing one more parameter. Vasicek is Hull-White with θ_t held constant; Ho-Lee is Hull-White with $\kappa \to 0$ (and the $\kappa\theta_t$ product held fixed). In production use, Hull-White is almost always the model of choice because it is the only one of the three that fits the observed yield curve exactly.

14.7.2 12.7.2 Duality: three perspectives on the same formula

The bond price $P_t(T) = e^{A_t(T) - B_t(T) r_t}$ can be derived by three independent routes, each shining light on a different aspect:

1. **Direct expectation (§12.3.4, §12.5.3, §12.6.3).** Use the explicit Gaussian solution for r_t, integrate to get the Gaussian law of $\int_t^T r_u\,\mathrm{d}u$, apply the moment-generating function. This route makes the Jensen-convexity origin of the formula transparent and generalises to any Gaussian affine model.
2. **Affine ansatz + PDE (§12.3.5).** Use Feynman-Kac to convert the expectation into a parabolic PDE with source term $-rg$; guess an exponential-affine solution; match coefficients to obtain a pair of ODEs for A and B. This route makes the affine structure of the model explicit and generalises to any affine model including CIR and quadratic-Gaussian.
3. **Martingale verification (discussed but not executed).** Postulate the formula $P = e^{A - Br}$, compute $\mathrm{d}(P/M)$ using Itô, verify that the drift is zero under $\mathbb{Q}$. This route directly confirms the no-arbitrage condition and is the most elegant but requires pre-knowledge of the formula.

That all three converge on the same answer is both a consistency check and a window onto the structural richness of affine models. Each route carries some information the others don't: the direct expectation carries the probabilistic intuition (Jensen, Gaussian MGF); the PDE carries the analytic intuition (elliptic operators, Riccati ODEs); the martingale verification carries the economic intuition (no-arbitrage as martingality). Mastering all three is the difference between pricing a bond and understanding what its price means.

14.7.3 12.7.3 Where this chapter leaves the reader

By the end of §12.6 the reader can:

- Derive Vasicek from first principles, arriving at the bond-price formula (12.20)-(12.22) by both the direct-expectation and PDE routes.
- Extend to Hull-White by letting $\theta \to \theta(t)$, and bootstrap $\theta(t)$ to exactly reproduce an observed yield curve.
- Specialise to Ho-Lee by setting $\kappa \to 0$, with the understanding that the resulting cubic variance growth limits the model to short-dated products.
- Read off the yield curve, forward curve, and convexity adjustment as analytical by-products.

What is *not* in this chapter: applications. The IRS par-rate formula, the Jamshidian bond-option formula, the T-forward measure, callable bonds, CDS, and multi-factor / LMM / SABR all live in Chapter 13 (rate derivatives) and Chapter 14 (caps and swaptions). The division of labour is: this chapter *builds the model*; the next two *use it.*

14.7.4 Case study (a): Negative Eurozone rates 2014–2022 — Vasicek vs CIR

The ECB cut its deposit facility rate to -0.10% *on 5 June 2014 — the first time a major central bank had taken a policy rate negative — and held it negative through to 27 July 2022, hitting* -0.50% *in September 2019. Swiss and Danish policy rates went deeper; the 10-year Bund traded at* -0.74% *in March 2020. For an entire decade EUR rates desks priced and hedged in a regime that one canonical short-rate model (CIR) literally cannot represent and another (Vasicek / Hull-White) handles unchanged.*

Context. The Cox-Ingersoll-Ross (1985) model $\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t$ was historically attractive for two reasons: its Feller-square-root diffusion produces a non-central chi-squared distribution for r_T supported on $[0, \infty)$, ruling out negative rates "for free", and its bond price is affine and closed-form. Both features were sold as virtues for decades. Vasicek's $\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, by contrast, is Gaussian — r_T is normal on all of $\mathbb{R}$, and a fraction of probability mass always sits below zero. Pre-2014 textbooks routinely flagged this as a Vasicek "defect" and recommended CIR for production rate-modelling work. Hull-White, the time-inhomogeneous Gaussian extension, inherits the same support-on-$\mathbb{R}$ property and the same alleged "defect."

When EONIA dropped below zero in mid-2014, every CIR-based EUR rates book had a problem the model literally could not articulate. A 5y receiver swaption struck at -0.25% has positive intrinsic value when forward rates trade at -0.40%, but CIR cannot generate sample paths reaching -0.25% at all — its short-rate density is identically zero on $(-\infty, 0)$. Banks responded with three workarounds, all bad: (i) shift the rate process, $r_t = x_t - s$ with x_t CIR and $s = 1\%$ a fixed shift, recovering negative-rate support but contaminating mean-reversion estimates; (ii) re-derive prices in a "shadow rate" framework where the observed rate is $r_t = \max(x_t - s, -\bar{r})$, breaking closed-form pricing; (iii) abandon CIR for Hull-White (Gaussian, time-inhomogeneous), eat the migration cost across the entire book, and live with negative-rate-tail probabilities that some risk committees still found uncomfortable. The third option won at every major EUR rates desk during 2014-2017.

Reading it through the chapter's math. Vasicek's explicit solution (12.12) and the law of r_T in §12.3.2 are unconditionally Gaussian: $r_T \sim \mathcal{N}(\mathbb{E}[r_T], \mathbb{V}[r_T])$ on all of $\mathbb{R}$, regardless of the sign of θ or the value of r_0. The Vasicek closed-form bond price (12.20) is $A(t, T)\exp(-B(t, T)r_t)$ with $B > 0$, so as $r_t \to -\infty$ the bond price goes to $+\infty$ — there is no degeneracy at zero. CIR's affine form has the same superficial structure but the underlying distributional law is non-central chi-squared with a degree-of-freedom parameter that determines whether zero is an attainable, repelling, or absorbing boundary; either way, the support is $[0, \infty)$ by construction. Pricing a -0.25%-strike receiver swaption in CIR is mathematically a request for $\mathbb{E}^{\mathbb{Q}}[(K - L_T)^+]$ with $K = -0.25\%$ and $L_T \geq 0$; the answer is $|K|$ times the annuity, i.e. the swaption is automatically maximally in-the-money for any negative strike, which is wrong by construction (the actual EUR fixings did move further negative, so the receiver was *not* maximally ITM). The two models gave genuinely different prices for receiver swaptions near the zero bound — sometimes 5%–10% of premium — and the Vasicek/Hull-White price was the correct one in every case.

Lesson. A model's state space is not a regularisation choice; it is a structural assertion about what realities the model can describe. CIR went out of fashion in EUR rates for the better part of a decade not because it was poorly calibrated, but because the rates desks woke up one morning in June 2014 in a regime CIR's ontology forbade. The Hull-White migration cost — re-implementation, re-validation, re-calibration of every Bermudan and callable structure on the book — was substantial; banks that delayed paid for it in mispriced negative-rate exotics for years. The deeper lesson for the new hire: when choosing a short-rate model for a given currency, the support of the rate distribution is the first question to ask, ahead of mean-reversion realism, ahead of analytic tractability, ahead of computational cost. Models with non-negative-rate hard floors (CIR, BK, square-root Hull-White) belong only in regimes where the central bank's reaction function genuinely rules out negative rates as policy — historically, USD before 2008 and emerging markets. Anywhere else, the Gaussian family (Vasicek, Hull-White, G2++) is the safe default, and the negative-rate tail probability that risk committees worry about is in fact a *feature*, not a bug.

14.7.5 Case study (b): 2023 Fed hiking cycle — Hull-White $\theta(t)$ bootstrap from the funds curve

Between March 2022 and July 2023 the Fed funds target rose from 0.00–0.25% *to* 5.25–5.50% *in eleven hikes — the fastest hiking cycle since Volcker. By the second half of 2023 the curve was deeply inverted: 3M* $\approx$ 5.45%, *4M* $\approx$ 5.40%, *2Y* $\approx$ 4.85%, *5Y* $\approx$ 4.40%, *10Y* $\approx$ 4.25%. *Pricing a* \$50*B Bermudan callable book against this curve, every single morning, is what Hull-White's* $\theta(t)$ *bootstrap was designed for.*

Context. Hull-White's defining feature, relative to constant-coefficient Vasicek, is the time-dependent drift $\theta(t)$ chosen so that the model's initial discount curve $P_0^{\mathrm{HW}}(0,T)$ matches today's market discount curve $P_0^{\mathrm{mkt}}(0,T)$ for every T. The calibration identity in §12.6.4 — solve for $\theta(t)$ such that the Hull-White bond formula reproduces every market zero — is *linear* in $\theta(t)$ once (κ,σ) are fixed, and runs in $\mathcal{O}$(tenors) per re-calibration. In 2023 with $\kappa = 0.06$ (calibrated from swaption-vol smile slope) and $\sigma = 0.011$ (calibrated from caplet ATM vols), the entire $\theta(t)$ bootstrap across 40 tenor buckets ran in under 5 ms on a single core. Every 8:00 AM the curve desk would lift the overnight Treasury and OIS quotes, build the discount curve, and refresh $\theta(t)$ — the whole process from market data to Bermudan repricing was sub-second.

The calibration mechanic for an inverted curve is genuinely informative about the underlying risk picture, in a way the constant-θ Vasicek calibration is not. With a flat curve, $\theta(t)$ comes out roughly flat at the long-run mean θ_∞ and the calibration carries little information beyond what the AR(1) coefficient already encoded. With the steeply inverted 2H 2023 curve, $\theta(t)$ took a distinctive shape: $\theta(0\text{–}1y) \approx 5.4\%$, peaking just above the spot rate (the model needs a slightly higher mean-reversion target to keep the front rates from collapsing immediately), then declining smoothly to $\theta(5y) \approx 3.8\%$ and $\theta(10y) \approx 3.3\%$. The convexity adjustment $C_T = \frac{1}{2}\sigma^2 B(t,T)^2$ in (12.51) added +2–4 bp lift at each maturity bucket, which the desk treated as a separate Greek for hedging — that adjustment is entirely a function of σ, and it is the reason why a higher real-rate vol *raises* model-implied long forwards above what the spot curve alone would indicate.

Reading it through the chapter's math. The bootstrap proceeds tenor-by-tenor (§12.6.4 procedure): given the calibrated (κ,σ), the Hull-White zero-coupon price is $P^{\mathrm{HW}}(0,T) = \exp(-\int_0^T f^{\mathrm{HW}}(0,u)\,\mathrm{d}u)$ with $f^{\mathrm{HW}}(0,u)$ a closed-form linear functional of $\theta(\cdot)$ on $[0,u]$. Inverting, $\theta(u) = \partial_u f^{\mathrm{mkt}}(0,u) + \kappa f^{\mathrm{mkt}}(0,u) + \frac{\sigma^2}{2\kappa}(1-e^{-2\kappa u})$ — the famous Hull-White θ-formula, where the first two terms recover the market forward and its slope, and the third is the convexity correction that depends only on σ. The inverted 2023 curve has $\partial_u f^{\mathrm{mkt}}(0,u) < 0$ for $u > 1$y, so the $\theta(u)$ schedule slopes downward through the belly. But the convexity term is *always positive* and grows with σ: a higher real-rate vol pushes $\theta(u)$ upward at every maturity, and the model's forward curve up by approximately $\frac{1}{2}\sigma^2 B(0,u)^2$ above the spot forward. In late 2023, with realised real-rate vol around $\sigma = 1.1\%$, the convexity adjustment at 10y was $\approx$ 18 bp — non-trivial relative to the inversion gap. The inversion *plus* the high vol therefore meant: market expects cuts (declining θ through belly), but the cuts are uncertain enough in timing that convexity adds a meaningful upward push to long-end model forwards.

Lesson. Hull-White's exact-fit-to-curve property is not optional infrastructure during a regime shift like the 2023 hiking cycle — it is what lets a desk hedge a multi-tens-of-billions Bermudan book against a moving Fed without generating spurious model-misfit P&L. A constant-θ Vasicek would have left a 50+ bp residual at the front of the curve, contaminating every Δ-vega report on the book; the Bermudan call decisions (whether to call at the next exercise date) would be driven as much by model error as by genuine moneyness. The deeper lesson for the new hire is to read the *shape* of the calibrated $\theta(t)$ schedule as a market signal rather than as a fitted nuisance: a downward-sloping $\theta(t)$ is the market telling you cuts are priced; the magnitude of the convexity adjustment is the market telling you how *uncertain* those cuts are; the day-to-day stability of σ tells you whether realised vol has caught up to implied. None of these pieces is what Hull-White was originally designed to communicate, but in a fast-moving rate cycle they become the most useful operational diagnostics on the desk.

14.8 12.8 Key Takeaways

1. **Short rates are not traded; bonds are.** The Fundamental Theorem of Asset Pricing applies to tradables, so the risk-neutral measure $\mathbb{Q}$ is defined by its action on M_t and $P_t(T)$, not on r_t. This creates a genuine degree of freedom (the market price of risk λ_t) that must be fixed externally — typically by calibration to observed bond prices.

2. **The master pricing formula is $P_t(T) = \mathbb{E}_t^{\mathbb{Q}}[e^{-\int_t^T r_s\,\mathrm{d}s}]$.** It is model-agnostic — every short-rate model computes the same conditional expectation, differing only in how the expectation is evaluated. Affine models

evaluate it in closed form.

3. **Vasicek: three parameters, one bond-price formula.** The SDE $\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$ yields, via either the direct-expectation route or the PDE / affine-ansatz route, the closed-form bond price $P_t(T) = e^{A_t(T) - B_t(T) r_t}$ with $B_t(T) = (1 - e^{-\kappa(T-t)})/\kappa$.
4. **Bonds are convex in rates (Jensen's mountain).** The $+\frac{1}{2}\Sigma_*^2$ term in the Gaussian MGF is the convexity adjustment that makes long bonds worth *more* than naive plug-in discounting suggests. It explains why long yields trade below expected long-run short rates even without a term premium.
5. **Long-bond duration is capped at $1/\kappa$.** This is Vasicek's most economically distinctive prediction: unlike a random-walk rate model, mean reversion means long bonds have finite sensitivity to the current short rate. Any calibrated κ outside the range $[0.1, 0.5]$ is suspect.
6. **Long yield $= \theta - \frac{1}{2}\sigma^2/\kappa^2$.** The long end of the yield curve is *below* the long-run mean of the short rate by a convexity premium. Larger in high-vol or slow-reverting regimes.
7. **Ho-Lee is the $\kappa \to 0$ limit.** Pure random walk with time-varying drift. Bond price has the simple form $\ln P_0(T) = -r_0 T - \iint \theta + \sigma^2 T^3/6$; drift pointwise calibration $\theta_T = -\partial_T^2 \ln P_0^*(T) + \sigma^2 T$. The cubic T^3 variance growth means Ho-Lee breaks at long maturities.
8. **Hull-White = Vasicek with time-varying $\theta(t)$.** Exact yield-curve fit by construction, via a linear triangular bootstrap (12.54) analogous to Ho-Lee's. Two-stage calibration in practice: (κ, σ) fit to swaption vols, then $\theta(t)$ bootstrapped to match bonds.
9. **One-factor is a limitation.** Vasicek/HW can only produce parallel-shift curve moves (up to the $B_t(T)$ loading). Real curve dynamics have three factors (level, slope, curvature); for hedging and pricing curve-twist-sensitive products, move to G2++ or multi-factor affine or LMM.
10. **Affine = tractable.** The structural property "drift linear, squared diffusion linear in the state" is what makes coefficient- matching work in the PDE ansatz, reduces the infinite-dimensional pricing problem to two ODEs, and ultimately gives closed-form bond prices. Ho-Lee, Vasicek, HW, CIR, G2++, and the Duffie-Kan affine family all share this property; Black-Karasinski (log-rate) and stochastic-vol rates models do not.

14.9 12.9 Reference Formulas

14.9.1 Core dynamics

$$\mathrm{d}r_t = \kappa(\theta(t) - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}\widehat{W}_t$$

- \$ (t) = \$ const gives Vasicek; $\kappa = 0$ gives Ho-Lee.

14.9.2 Explicit solution

$$r_T = e^{-\kappa(T-t)}\, r_t + \kappa \int_t^T e^{-\kappa(T-s)} \theta(s)\,\mathrm{d}s + \sigma \int_t^T e^{-\kappa(T-s)}\,\mathrm{d}\widehat{W}_s$$

14.9.3 Distribution of r_T (Vasicek, const θ)

$$r_T \;\sim\; \mathcal{N}\Big(\theta + e^{-\kappa(T-t)}(r_t - \theta),\;\; \tfrac{\sigma^2}{2\kappa}(1 - e^{-2\kappa(T-t)})\Big)$$

Stationary limit: $\mathcal{N}(\theta, \sigma^2/(2\kappa))$.

14.9.4 Integrated rate (Vasicek)

$$\int_t^T r_s\,\mathrm{d}s \;=\; B_t(T)\, r_t + \theta((T-t) - B_t(T)) + \tfrac{\sigma}{\kappa} \int_t^T (1 - e^{-\kappa(T-s)})\,\mathrm{d}\widehat{W}_s$$

Gaussian with mean and variance in (12.16)-(12.18).

14.9.5 Vasicek affine bond price

$$P_t(T) = e^{A_t(T) - B_t(T)\, r_t}, \qquad B_t(T) = \frac{1 - e^{-\kappa(T-t)}}{\kappa}$$

$$A_t(T) = -\theta\big((T-t) - B_t(T)\big) + \tfrac{\sigma^2}{2\kappa^2}\int_t^T \big(1 - e^{-\kappa(T-s)}\big)^2\,\mathrm{d}s$$

14.9.6 Affine ansatz ODE system

$$\begin{cases} \dot{A}_t - \kappa B_t\, \theta(t) + \frac{1}{2}\sigma^2 B_t^2 = 0 \\ \dot{B}_t - \kappa B_t + 1 = 0 \end{cases} \qquad\qquad A_T = B_T = 0$$

14.9.7 Vasicek yield curve

$$y_t(T) = \frac{B_t(T)\, r_t - A_t(T)}{T - t}$$

Limits: $y_t(t) = r_t$; $y_t(\infty) = \theta - \frac{1}{2}\sigma^2/\kappa^2$.

14.9.8 Ho-Lee bond price

$$\ln P_0(T) \;=\; -r_0\, T \;-\; \int_0^T\!\int_0^u \theta_s\,\mathrm{d}s\,\mathrm{d}u \;+\; \frac{\sigma^2 T^3}{6}$$

14.9.9 Ho-Lee calibration identity

$$\theta_T \;=\; -\,\partial_T^2 \ln P_0^*(T) \;+\; \sigma^2 T \;=\; \partial_T f_0(T) + \sigma^2 T$$

14.9.10 Hull-White bond price

$$P_0(T) = \exp\{-r_0\, B_T - A_T + C_T\}$$

$$B_T = \frac{1 - e^{-\kappa T}}{\kappa}, \quad A_T = \int_0^T \theta_u\big(1 - e^{-\kappa(T-u)}\big)\,\mathrm{d}u, \quad C_T = \frac{\sigma^2}{2\kappa^2}\int_0^T \big(1 - e^{-\kappa(T-u)}\big)^2\,\mathrm{d}u$$

14.9.11 Hull-White pointwise calibration

$$\theta(T) \;=\; \frac{1}{\kappa}\,\partial_T f_0(T) + f_0(T) + \frac{\sigma^2}{2\kappa^2}\big(1 - e^{-2\kappa T}\big)$$

14.9.12 Hull-White bucket bootstrap

$$\theta^{(k)} \;=\; \frac{T_k\, y_k^* - r_0\, B_{T_k} + C_{T_k} - \sum_{m=1}^{k-1} \theta^{(m)}\,\alpha^{(m,k)}}{\alpha^{(k,k)}}$$

with

$$\alpha^{(m,k)} \;=\; (T_m - T_{m-1}) \;-\; \frac{e^{-\kappa(T_k - T_m)} - e^{-\kappa(T_k - T_{m-1})}}{\kappa}.$$

Model	κ	θ	Long-run $\mathbb{V}[r_T]$	Convexity C_T as $T \to \infty$

14.9.13 Three-model nesting

Model	κ	θ	Long-run $\mathbb{V}[r_T]$	Convexity C_T as $T \to \infty$
Ho-Lee	0	$\theta(t)$	$\sigma^2 T$	$\sigma^2 T^3/6$ (cubic)
Vasicek	$\kappa > 0$	const	$\sigma^2/(2\kappa)$	linear growth ($\sim \sigma^2/(2\kappa^2) \cdot (T-t)$ asymptotically)
Hull-White	$\kappa > 0$	$\theta(t)$	$\sigma^2/(2\kappa)$	$\sigma^2 T/(2\kappa^2)$ (linear)

14.9.14 Half-life

$$\tau_{1/2} = \frac{\ln 2}{\kappa}$$

Typical calibrated $\kappa \in [0.1, 0.5]$ corresponds to rate-shock half-lives of 1.4 to 7 years.

15 Chapter 13 — Rate-Derivative Applications: Swaps, CDS, Bond Options, Callables

15.1 13.1 From short-rate dynamics to products that depend on them

Chapter 12 built the Vasicek, Ho-Lee, and Hull-White short-rate models — the SDEs, the integrated-rate distributions, the closed-form bond prices, the affine-ODE system, the calibration of θ_t to today's curve. All of that machinery was developed with one pay-off in mind: *use it to price rate-dependent products.* This chapter is that pay-off. We take the calibrated short-rate model from Chapter 12 as given and work outward to the four product classes that dominate rates desks: **interest-rate swaps (IRS)**, **callable bonds**, **credit default swaps (CDS)**, and **European bond options** (together with their capstone application, swaptions via the Jamshidian decomposition). We close with a tour of where the single-factor Gaussian framework starts to break and which extensions (HJM, LMM, SABR, multi-factor Gaussian) handle the residual.

The chapter is organised by how much of the short-rate model each product actually uses. Swaps are almost model-free: only the discount curve enters, so Vasicek versus Hull-White versus any other calibrated-curve model gives the same swap value. Callable bonds are genuinely model-dependent: the embedded American option on a bond requires a backward induction on a short-rate tree, and both the level of rates and the variance of the short rate matter. CDS depend on the calibrated discount curve plus a separate hazard-rate curve; the two combine into a risky discount factor. European bond options under Vasicek/Hull-White admit a clean closed-form (the Jamshidian / Vasicek-Black formula), and the derivation is the canonical application of the **T-forward measure** — a specific change of numeraire that makes the forward bond price a martingale with deterministic volatility. Crucially, we will *not* re-derive Girsanov's theorem or the T-forward measure here: Chapter 5 already did that once. This chapter cites the result from Chapter 5 and uses it.

A roadmap to the rest of the chapter. §13.2 prices an IRS and extracts the par swap rate — a model-free calculation that exercises only the discount curve. §13.3 tackles callable bonds on a Hull-White trinomial tree, introducing the backward-induction-plus-early-exercise pattern that appears again and again in rate-derivative pricing. §13.4 handles CDS with constant and time-varying hazard, deriving the "credit triangle" $S \approx (1-R)\lambda$ and its corrections. §13.5 takes a visual pause on the yield surface — a 3D object that is the natural state of the rates-trading book. §13.6 derives the Vasicek-Black bond-option formula via the T-forward measure, citing Chapter 5 for the Girsanov apparatus. §13.7 walks through worked examples of that formula and introduces the Jamshidian decomposition that reduces a swaption to a portfolio of bond options. §13.8 previews where we go beyond single-factor Vasicek: the multi-factor Gaussian G2++, HJM, LMM, and SABR. §13.9 closes with the PDE-versus-Monte-Carlo trade-offs that dictate which numerical method a rates desk deploys per product. §13.10 summarises the chapter; §13.11 is the reference-formula appendix.

The conceptual thread running through all five products is this: **every rate derivative is a functional of the future yield curve, and pricing it reduces to computing a conditional expectation of that functional under some convenient martingale measure**. The specific measure depends on when the payoff settles and what numeraire most naturally absorbs its stochastic discount factor. For swaps we use no measure machinery at all — the linearity-in-rates of the payoff lets us replicate and price pathwise. For CDS we use $\mathbb{Q}$ with a hazard-rate extension. For bond options we use the T-forward measure. For swaptions we use the annuity measure. Each choice is motivated by the same principle: *match the numeraire to the payoff's natural scale, and the pricing collapses to a Black-type lognormal expectation.* That principle — one sentence — is the take-away of the entire chapter.

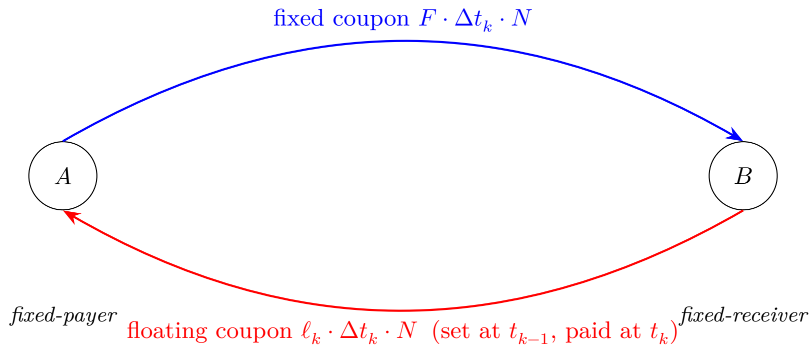
One notational convention before we start. All bond prices $P_t(T)$, affine coefficients $A_t(T)$ and $B_t(T)$, and the short-rate parameters $(\kappa, \theta_t, \sigma)$ are as in Chapter 12. The $\mathbb{Q}$-Brownian motion is $\widehat{W}_t$; the T-forward Brownian motion (introduced in §13.6 via Chapter 5 cross-reference) is $\widetilde{W}_t^T$. The discount factor from t to T under the money-market numeraire is $M_t/M_T = e^{-\int_t^T r_u\,\mathrm{d}u}$; we use $P_t(T)$ when we want the deterministic (time-t-conditional expectation) bond price.

15.2 13.2 Interest-rate swaps and the par swap rate

15.2.1 13.2.1 Swap cash flows and definitions

A plain-vanilla interest-rate swap exchanges a fixed leg against a floating leg on a notional N at reset dates $t_0 < t_1 < \cdots < t_n$, with day-count $\Delta t_k = t_k - t_{k-1}$. Counterparty A pays fixed coupon F at each t_k for $k = 1, \ldots, n$ and receives a floating coupon ℓ_k that was *set in arrears* at the prior reset date t_{k-1} (this is the standard "fix at t_{k-1}, pay at t_k"

convention). Counterparty B is the mirror image: receives fixed, pays floating. There is **no exchange of notional** — only the periodic coupon differences — which is why swap-market notional figures (hundreds of trillions of dollars of cleared IRS outstanding) vastly exceed the bond market.



Notional N never changes hands; only the netted coupon difference $(F - \ell_k) \cdot \Delta t_k \cdot N$ flows each period.

The payment schedule lives on the interval grid $t_0 < t_1 < \cdots < t_n$ with "first-reset" at t_0 (rate fixed there) and "first-payment" at t_1; subsequent coupons at $t_2, \ldots, t_n$.

Why swaps exist. An IRS converts a floating liability into a fixed one or vice versa — a corporate treasurer pays fixed/receives floating to lock funding cost; a mortgage bank receives fixed/pays floating to match its floating funding. Cleared IRS notional is roughly \$400–500 trillion, the largest derivative class by any measure.

15.2.2 13.2.2 Pricing from the discount curve

Fixed-leg value. Paying $F \cdot N \cdot \Delta t_m$ at each t_m has time-0 value

$$V_0^{\text{fix}} \;=\; \sum_{m=1}^{n} \Delta t_m \, F \, N \, P_0(t_m) \tag{13.1}$$

The factor $\sum_{m=1}^{n} \Delta t_m P_0(t_m)$ is the **annuity** (or PVBP per unit rate). At time t before the swap matures, the fixed leg value is $V_t^{\text{fix}} = \sum_m F \Delta t_m P_t(t_m)$.

The annuity interpretation is worth dwelling on. Think of $\mathcal{A}_0 \equiv \sum_m \Delta t_m P_0(t_m)$ as the present value of a stream of \$1 payments per year, on the coupon schedule, discounted to today. It functions as the "unit of account" for swap rates: the par swap rate F is quoted as the number such that $F \cdot \mathcal{A}_0$ equals the floating-leg value. Multiplying or dividing by the annuity converts between dollars and per-annum rate units. The annuity is also the natural numeraire for changing measure in swaption pricing: under the "annuity measure" (the probability measure under which the annuity is the numeraire), the par swap rate S_t becomes a martingale, and the swaption collapses to $V_0 = \mathcal{A}_0 \cdot \mathbb{E}^{\mathcal{A}}[(S_T - K)_+]$ — a direct lognormal expectation in Black form.

Floating-leg value via "buy \$1 and roll". The floating coupon ℓ_k is locked in at each reset t_{k-1} from the simple yield on a Δt_k-bond:

$$P_{t_{k-1}}(t_k) \;=\; \big(1 + \Delta t_k \, \ell_k\big)^{-1} \;\Longrightarrow\; \ell_k \;=\; \frac{1}{\Delta t_k}\left(\frac{1}{P_{t_{k-1}}(t_k)} - 1\right). \tag{13.2}$$

The reset mechanism works like this: at each reset date t_{k-1}, the market quotes a simply-compounded rate for the upcoming period $[t_{k-1}, t_k]$, and that rate is fixed as the floating coupon for that period. The coupon $\ell_k \Delta t_k N$ is then paid at t_k. So floating coupons are *set in arrears*, meaning the k-th coupon is known at the $(k-1)$-th reset and paid at t_k. Standard LIBOR (and its SOFR successor) follow this "set-in-advance, paid-in-arrears" convention, and (13.2) is the algebraic identity that makes replication of the floating leg possible.

Claim: $V_t^{\text{fl}} = (P_t(t_0) - P_t(t_n))N$.

Financial argument (self-financing replication). Consider the trader who at t_0 is handed \$1 and asked to replicate the floating-leg coupons:

- At t_0: buy \$1-worth of t_1-bonds — holdings $1/P_{t_0}(t_1)$. These mature to $\$1/P_{t_0}(t_1)$ at t_1.
- At t_1: take \$1 of that and buy t_2-bonds: $1/P_{t_1}(t_2)$ units; pocket the remainder $(1/P_{t_0}(t_1) - 1) = \ell_1 \Delta t_1$.
- Continue to t_n, pocketing $\ell_k \Delta t_k$ each period and holding \$1 of fresh short-dated bonds.

The cash flows generated by this self-financing strategy are exactly the floating-leg coupons, plus a terminal \$1 repayment we return. Cost = \$1 in at t_0 minus \$1 out at t_n, priced today:

$$V_0^{\text{fl}} \;=\; \left(P_0(t_0) - P_0(t_n)\right) N \tag{13.3}$$

The floating leg is **model-free**: neither κ nor σ nor $\theta(t)$ enters (13.3). The randomness cancels pathwise. Post-2008, a multi-curve adjustment ("OIS discounting" with a tenor basis to SOFR/LIBOR accrual) keeps (13.3) approximately correct.

15.2.3 13.2.3 The par (fair) swap rate

Setting $V_0^{\text{fix}} = V_0^{\text{fl}}$ gives the rate $F^* \equiv S_0$ that makes the swap worth zero at inception:

$$\boxed{F \;=\; \frac{P_0(t_0) - P_0(t_n)}{\sum_{m=1}^{n} \Delta t_m \, P_0(t_m)} \;\equiv\; S_0 \quad \text{(initial par swap rate)}} \tag{13.4}$$

The denominator is the annuity. Since both legs are priced off the discount curve alone, the par swap rate is a ratio of discount factors — **no model required**.

The par swap rate S_0 is the market's standard quote for "the n-year swap rate." When a Bloomberg screen shows "10y USD swap is 3.5%," that 3.5% is S_0 from (13.4) with n chosen to give a 10-year maturity. The swap curve — the set of par swap rates across tenors — is directly observable and widely used as a reference interest-rate curve for mortgages, corporate borrowing, and internal bank funding.

Swap rate $\equiv$ par coupon yield. Up to convexity corrections, S_0 equals the par coupon yield of a bond with the same tenor. A par coupon bond with coupon c has PV $= c \cdot \mathcal{A}_0 + P_0(t_n)$, which equals par (notional) exactly when $c = (1 - P_0(t_n))/\mathcal{A}_0 = S_0$. So "a swap is a bond with no principal" — the coupon-rate structure is identical; only the principal exchange differs.

Forward-start / ongoing swap rate. For a swap with payment dates $\{t_\ell\}_{\ell : t_\ell > t}$ viewed at time t (some resets already happened), the current fair-value swap rate is

$$S_t \;=\; \frac{P_t(t_{k_t}) - P_t(t_n)}{\sum_{m=k_t+1}^{n} \Delta t_m \, P_t(t_m)} \tag{13.5}$$

where k_t is the index of the most recent reset before t. If $t < t_0$ (spot-start), $P_t(t_{k_t}) = P_t(t_0)$ and (13.5) reduces to (13.4) with P_t replacing P_0.

Because discount factors are stochastic, S_t is a stochastic process. Plotting it alongside the short rate r_t reveals very different dynamics: S_t is smooth and moves with the yield curve, while r_t is volatile and determines the floating coupons. In a sense, S_t is a low-pass-filtered version of r_t, weighted toward the long end of the curve. Empirically, the 10-year swap rate might move 5 bp on a given day while the overnight rate moves 25 bp — a 5:1 ratio, matching the Hull-White prediction that long-tenor yields move less than short-tenor yields in response to short-rate shocks.

15.2.4 13.2.4 Mark-to-market and swap DV01

For off-par swaps (where the fixed coupon F differs from S_0), the swap has non-zero initial PV. If $F = 4.00\%$ versus a par of $S_0 = 4.49\%$, the fixed-payer pays below market so the swap has positive PV to the fixed-payer:

$$V_0^{\text{fp}} \;=\; (S_0 - F) \cdot \mathcal{A}_0 \cdot N \tag{13.6}$$

In the example, $V_0^{\text{fp}} = (4.49\% - 4.00\%) \cdot 2.7532 \cdot 100\text{M} \approx \$1.35\text{M}$. This is the "off-market adjustment" paid upfront in a typical trade. Market convention is to trade at-par (no upfront) for most swaps; off-market swaps arise for customised deals or for collapsing multiple existing positions into a single trade.

DV01. The "01" (pronounced "oh-one" or DV01) is the change in PV for a 1-bp parallel shift in rates. For a receiver swap, DV01 $\approx \mathcal{A} \times 0.0001 \times N$. For a 10-year \$100M swap with annuity $\mathcal{A} = 8.5$:

$$\text{DV01} \;\approx\; 100\text{M} \times 8.5 \times 0.0001 \;=\; \$85{,}000 \text{ per basis point.}$$

This is the single most important risk metric for IRS trading, and it is directly the annuity-weighted notional. DV01 decomposes by tenor ("bucket DV01") to reveal which parts of the yield curve the portfolio is exposed to. Hedging a swap portfolio requires matching the bucket DV01s via positions in benchmark instruments (typically government-bond futures, short-end deposits, and other swaps at standard tenors). Mismatched DV01s lead to residual P&L as the yield curve reshapes.

Related quantities: BPV (basis-point value, same as DV01), PV01 (present value of a 1-bp change, same), 01 (trade-floor shorthand). The important distinction is between "parallel-shift" DV01 (all tenors up 1 bp together) and "bucket" DV01 (only one tenor moves 1 bp). Parallel DV01 is a portfolio-level summary; bucket DV01 is the detailed breakdown needed for hedging.

15.2.5 13.2.5 Worked example: a 3-year par swap

With $P_0(1) = 0.9592$, $P_0(2) = 0.9177$, $P_0(3) = 0.8763$ and $\Delta t_k = 1$:

$$S_0 \;=\; \frac{1 - 0.8763}{0.9592 + 0.9177 + 0.8763} \;=\; \frac{0.1237}{2.7532} \;\approx\; 4.49\%.$$

The 3-year discount falls from 1.0 at $t = 0$ to 0.8763 at $t = 3$, so roughly 12.4 cents are "lost" to discounting over three years. Spread over three annual coupons with annuity 2.75, that works out to 4.49% per annum. This exactly matches the yield-to-maturity of a 3-year par coupon bond with coupon 4.49% — the swap rate *is* the par coupon yield.

Cross-check via zero yields. $P_0(1) = 0.9592 = e^{-y_1}$ gives $y_1 = -\ln 0.9592 \approx 4.17\%$. Similarly $y_2 \approx 4.30\%$, $y_3 \approx 4.40\%$. So the zero-coupon curve is 4.17% at 1y, 4.30% at 2y, 4.40% at 3y — gently upward-sloping. The par swap rate (4.49%) is slightly above the 3-year zero yield (4.40%) because of the convexity implicit in the coupon structure: a coupon bond receives coupons as well as principal, and because coupons are smaller than the principal, their yield must be slightly higher to equate PVs. This "convexity premium" between coupon bonds and zero-coupon bonds is a standard bond-math result.

IRS cashflow ladder *Fixed-payer sees outgoing blue fixed coupons and incoming orange floating coupons; net PV is zero at the par swap rate S_0.*

15.2.6 Case study (a): LTCM 1998 — swap spreads, off-the-run/on-the-run, leverage

Long-Term Capital Management — Meriwether's Salomon arbitrage group reincarnated, with Robert Merton and Myron Scholes on the partnership and a Nobel Prize awarded mid-stream — ran roughly \$130*B of balance sheet against* \$4.7*B of equity (Aug 1998 leverage ratio* ≈ 28:1*), with one of the largest concentrations being long-off-the-run / short-on-the-run Treasury basis paired against IRS-spread positions. Russia's domestic-debt default on* 17 *August 1998 destroyed both legs simultaneously and the equity vanished in six weeks.*

Context. The trade rationale was simple §13.2 swap-curve mathematics: the off-the-run Treasury (say, the previous-quarter issued 30y bond) trades at a small yield premium of 5–10 bp over the most-recently-issued on-the-run, reflecting the on-the-run's superior repo specialness and dealer demand. Over the bond's seasoning life this premium compresses to zero, generating a 5–10 bp risk-free convergence. LTCM ran the trade in size: long off-the-run, short on-the-run (matched DV01), and frequently overlaid with a pay-fixed IRS to monetise the swap-Treasury basis as well. Each individual position had small daily P&L variance, but the book was scaled by leverage to produce target returns: portfolio DV01 in the \$40–60M per basis point range against the \$4.7B equity base. Internal 99% 1-day VaR sat near \$45M — about 1% of equity.

The Russian default did not move the swap-Treasury basis by a small amount; it moved it by 30 bp in the wrong direction in a single week and another 30 bp the following month. Flight-to-quality bid up the on-the-run Treasury (which is the most liquid hedge instrument in a panic) while dumping the less-liquid off-the-run, widening the very basis the trade was short. Simultaneously the swap spread — the IRS rate minus the corresponding-maturity Treasury — widened to historic levels: 10y swap spreads went from $\sim$ 35 bp pre-crisis to $\sim$ 90 bp by late September, briefly tagging 200 bp in some tenors. With portfolio DV01 around \$50M/bp, the 60 bp basis widening alone equalled \$3.0B in mark-to-market losses; aggregated across all the convergence positions the fund lost \$4.6B of its \$4.7B equity in six weeks. The Federal Reserve organised a \$3.625B private-sector recapitalisation on 23 September to prevent a forced liquidation that would have shocked counterparty banks.

Reading it through the chapter's math. The par swap rate (13.4) is model-free — it is a discount-curve identity that holds regardless of the short-rate dynamics. But the *spread* between two rates (off-the-run vs on-the-run, IRS vs Treasury) is a *liquidity* premium that no calibrated short-rate model in this chapter captures: it is an artefact of how dealers fund inventory in repo markets, who is the marginal buyer in a flight-to-quality, and what regulatory capital weights apply to which instrument class. The swap-spread convergence trade's PnL is approximately linear in the spread, with sensitivity equal to the matched DV01 — that piece is exactly the §13.2.4 DV01 calculation. What the chapter's framework captures *correctly* is the ΔP&L per basis-point move; what it cannot tell you is what magnitude of move to stress against. LTCM's internal VaR used historical correlations from 1995–1998 — a benign, low-vol period — under which off-the-run, IRS-Treasury, and Russian-bond moves were close to independent. In crisis, they became a single one-factor "flight-to-quality" move and the gaussian VaR was off by a factor of 100. Cornish-Fisher (Chapter 15.27) would have helped only if the third- and fourth-moment inputs themselves had been stressed; they were not.

Lesson. The trade was *correct on average* — swap spreads did mean-revert, and an unleveraged version of the position would have been profitable across the cycle. The problem was the *path*: leverage transmits drawdowns into margin calls into forced unwinds, and at 28:1 even a one-sigma move in the underlying spread can wipe out equity if the move comes in a single week rather than spread over a quarter. The deeper lesson for the new hire is that DV01 measured against a normal-vol historical basis range is the wrong risk metric for a leveraged convergence trade; the right metric is "DV01 $\times$ stressed move under a flight-to-quality scenario," and the stressed move is set by judgement about regime, not by a covariance-matrix multiplier. Post-LTCM every prime broker tightened margin financing for relative-value strategies; post-LTCM regulators introduced stressed-VaR add-ons (Basel 2.5) that require the calibration window to include 1998, 2008, or another comparable crisis episode. The math of swap-spread arbitrage did not change after 1998; the institutional discipline around leverage limits and stressed-correlation scenarios did.

15.3 13.3 Callable bonds on a short-rate tree

15.3.1 13.3.1 Callable bond cash flows

IRS pricing is essentially model-free, but a **callable bond** is a genuinely model-dependent rate structure and the canonical application of Hull-White numerical pricing. A callable bond is a fixed-coupon bond that the issuer has the right (but not obligation) to redeem early at a specified call price on one or more call dates. It is the bond-market equivalent of a mortgage with a prepayment option — the issuer can refinance into a lower rate if rates fall.

Economically, a callable bond equals a straight (non-callable) bond minus the issuer's call option:

$$V_t^{\text{callable}} = V_t^{\text{straight}} - V_t^{\text{embedded call}}. \tag{13.7}$$

The issuer benefits if rates fall (they refinance at lower rates); the bondholder loses the upside. So callable bonds trade at a *higher* yield than comparable straight bonds, with the spread representing the option value. A 10-year callable with a 5-year first call might trade 30–80 bps above a comparable 10-year straight, with the spread widening when rate volatility is high (because a more volatile world makes the call option more valuable).

The yield differential is called the **option-adjusted spread (OAS)**. It is the spread earned per year for being "short the call" as a bondholder. If OAS is 50 bps, you are paid 50 bps per year to bear the call-risk exposure. A useful way to think about the position: start with a straight bond with clean duration-based pricing. Add in the issuer's right to call (subtract the call option value). Now the bond's price is lower and its *effective* duration is lower — because the bond will likely be called early in a falling-rate scenario, shortening its life. The **option-adjusted duration**

(OAD) captures this shortened effective life and is the correct hedge quantity for managing the interest-rate risk of a callable-bond portfolio.

Pricing requires evaluating a max function at each call date:

$$V_{t_c}^{\text{call}} \;=\; \min\!\big(\text{call price},\; V_{t_c}^{\text{continuation}}\big). \tag{13.8}$$

The issuer will call if the continuation value exceeds the call price — they pay off the bond for less than its market value. This is **American-exercise optionality**, embedded in a rate-derivative structure. No closed form exists in general; we must evaluate numerically via a tree, lattice, or Monte Carlo with regression-based exercise.

15.3.2 13.3.2 Backward induction on a Hull-White trinomial tree

The standard approach is a Hull-White trinomial lattice. Build a tree discretising r_t at each time step t_n, with transition probabilities calibrated to match the Hull-White moments. At each call date, apply (13.8) at each node. Then roll back to t_0 with standard backward induction. The tree gives a complete price and, as byproducts, the OAS (the yield spread over the discount curve that equates the tree price to the market price) and the option-adjusted duration and convexity.

Construction sketch. At each time t_n, the short-rate state is discretised to $r_n^{(i)}$ for $i = -m, \dots, +m$ (the tree has $2m+1$ possible rate states). Transition probabilities to the three successor states (up, middle, down) at t_{n+1} are calibrated to match the first two conditional moments of $r_{n+1} \mid r_n$ under the Hull-White SDE. The mean reversion is handled by tweaking the *geometry* of the tree near the mean: at nodes where the standard branching would require negative probabilities, the branching is tilted (up-up-middle instead of up-middle-down, or the mirror) so all probabilities stay in $[0, 1]$. This is the Hull-White (1994) trinomial scheme — robust, fast-converging, and standard in every fixed-income pricing library.

A key feature is that the scheme handles the time-dependent $\theta(t)$ exactly: at each time step, the tree's drift is set to match the calibrated $\theta(t)$ at that time. So the tree automatically incorporates the full curve-fit from the θ bootstrap from Chapter 12, without any additional work. This is the computational payoff of the affine structure: calibration and pricing both leverage the same underlying Gaussian geometry.

Backward-induction algorithm. Let $V_n^{(i)}$ denote the callable-bond value at node (n, i). Work backward from maturity T:

1. **Terminal condition** $(n = N)$: $V_N^{(i)} = \text{face} + \text{final coupon}$ at every node.
2. **Continuation value** $(n < N)$: discount the expectation of the three successor values:

$$V_n^{(i),\text{cont}} \;=\; e^{-r_n^{(i)}\Delta t}\big(p_u V_{n+1}^{(i+1)} + p_m V_{n+1}^{(i)} + p_d V_{n+1}^{(i-1)}\big) + c_n$$

   where c_n is the coupon paid at t_n (if any).
3. **Call test** (if t_n is a call date): apply $V_n^{(i)} = \min\!\big(\text{call price}_n,\; V_n^{(i),\text{cont}}\big)$. Otherwise $V_n^{(i)} = V_n^{(i),\text{cont}}$.
4. Continue to $n = 0$; the callable price is $V_0^{(0)}$ at the root.

The algorithm is structurally identical to American-option pricing on a stock tree, with the discount factor now state-dependent via $r_n^{(i)}$ rather than a constant r. Each step is $O(\text{nodes at that time})$; total cost is $O(N \cdot m)$ in space and time, which is very fast even for dense calendars (quarterly calls over 30 years).

15.3.3 13.3.3 Decomposition into straight bond minus embedded call

The accounting identity (13.7) can be verified numerically by running the backward induction twice: once with the call test enabled (giving the callable price), and once with the call test disabled (giving the straight bond price). The difference is the value of the embedded call option, which the issuer is short and the bondholder is long (as seen from the issuer's liability side).

This decomposition is useful for *Greeks*. The straight bond's Greeks (duration, convexity) are well understood. The embedded call's Greeks are what make callables complex — and can be extracted from the tree by finite-differencing the call-option value with respect to the underlying rate curve, volatility, or time. The net callable Greeks are then the straight-bond Greeks minus the call-option Greeks, mirroring the value identity.

Negative convexity. A defining feature of callable bonds is that OAD is *non-monotonic* — low near the call strike (the bond is expected to be called soon), rising to the straight-bond duration far from it. The OAD concavity is the "negative convexity" of callable bonds. It forces dynamic rebalancing for hedgers and creates duration-chasing feedback loops in the wider rate market (the canonical MBS-convexity effect).

15.3.4 13.3.4 Numerical example

Take a 10-year bond with 5% annual coupon, face value \$100, callable annually at par (\$100) after year 3. Initial rate $r_0 = 3\%$, Hull-White parameters $\kappa = 0.1$, $\sigma = 0.015$, flat yield curve at 3% for simplicity. Build a trinomial tree of r values at each year $0, 1, \dots, 10$ with transition probabilities calibrated to Hull-White. At year 10, each node has $V = \$105$ (principal + last coupon). Roll back year-by-year, testing the call condition at years 3 through 9.

For parameters consistent with moderate vol, the callable bond might price at \$105.20, compared to a straight bond at \$108.30. The embedded call has value \$3.10, representing roughly 3% of bond value. As σ increases the call becomes more valuable and the callable bond price falls further below the straight. This is the option-value sensitivity: callable bonds are **short vega**, because they hold a short call option.

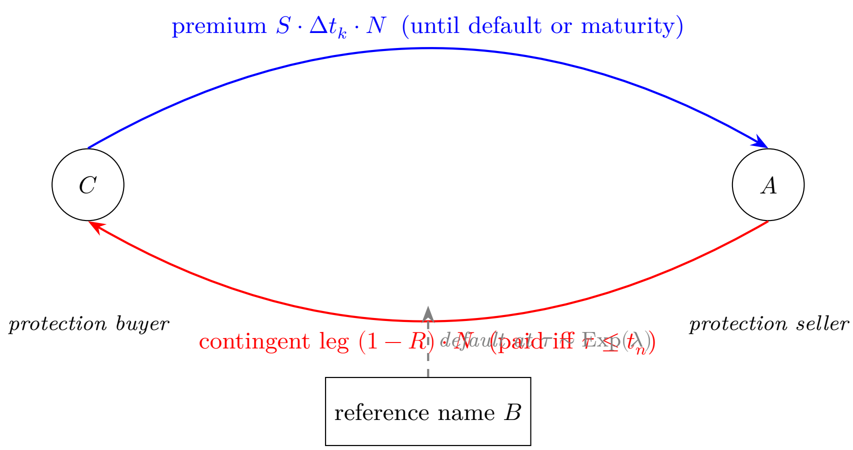
Interpretation of the price: the callable's yield-to-worst is about 4.9% (compared to a straight-bond yield of 4.1%), so the callable holder earns an extra 80 bps per year for bearing the call risk. In a world where rates stay high, the issuer will not call (continuation value stays below par), and the callable holder earns all the coupons. In a world where rates fall, the issuer calls at par, and the holder loses the future coupon stream. The 80 bps spread is the probabilistically-weighted compensation for bearing this downside.

Related products. Putable bonds (holder owns the put), convertible bonds (joint equity/rate model), and MBS (empirical prepayment models on top of the lattice) all use the same backward-induction skeleton. Calibrated (κ, σ) from the swaption market should feed the callable lattice — under-calibrated vol over-prices the bond.

15.4 13.4 Credit default swaps

15.4.1 13.4.1 Default intensity, survival, and the protection leg

Setup. Protection buyer C pays periodic premium S (annualised spread) to seller A on the reference name B until default time τ. If $\tau \le t_n$, the seller pays loss-given-default $(1-R)\,N$ at (or just after) τ, where R is the recovery rate and N the notional. We model the default time as the first jump of a Cox process; for this chapter $\tau \sim \mathrm{Exp}(\lambda)$ with constant intensity λ, independent of the short rate, unless noted otherwise.



C pays a stream of spreads; A pays the loss-given-default once, at τ, if it arrives before t_n.

Economic picture. A CDS is insurance against default. The protection buyer pays a steady premium — expressed as an annualised spread S times the notional — in exchange for a payout that arrives if and only if the reference entity defaults before maturity. This is economically equivalent to "shorting the credit of the reference name": a long position in the bond plus a purchased CDS has its default risk hedged out, leaving only rate-sensitivity exposure.

Intensity, survival, and the hazard interpretation. The intensity λ has a sharp probabilistic meaning: for an infinitesimal time interval $\mathrm{d}t$, the probability of defaulting in $[t, t+\mathrm{d}t]$ given no prior default is exactly $\lambda\,\mathrm{d}t$. So λ is the instantaneous default rate, the **hazard rate**. In a flat-intensity world, the survival probability is

$$\mathbb{Q}(\tau > t) \;=\; e^{-\lambda t} \tag{13.9}$$

and the default distribution is exponential with mean $1/\lambda$. A typical investment-grade bond might have $\lambda \approx 0.005$ per year (50 bps hazard, 200-year expected survival); a junk bond might have $\lambda \approx 0.05$ (5% hazard, 20-year expected survival).

The hazard-rate approach is a direct analog of the instantaneous-rate approach to interest-rate modelling. Both use "intensity" as the fundamental quantity. Both integrate the intensity to get the cumulative quantity (integrated rate $\to$ discount factor; integrated hazard $\to$ survival probability). Both admit time-varying intensities (Hull-White $\theta(t)$ for rates; stochastic hazard $\lambda(t)$ for credit) for realistic calibration.

15.4.2 13.4.2 Premium and default legs; the par CDS spread

Premium leg. Premium payments $S \cdot \Delta t_\ell \cdot N$ are paid on the schedule up to (and including the accrual period containing) default:

$$V_0^{\text{prem}} \;=\; S \cdot N \cdot \sum_{\ell=1}^{n} \Delta t_\ell \, \mathbb{E}^{\mathbb{Q}}\Big[\mathbf{1}_{\{\tau > t_\ell\}}\, e^{-\int_0^{t_\ell} r_u\,\mathrm{d}u}\Big] \;=\; S \cdot N \cdot \sum_{\ell=1}^{n} \Delta t_\ell \, \widetilde{P}_0(t_\ell) \tag{13.10}$$

where $\widetilde{P}_0(t_\ell) = e^{-\lambda t_\ell}\, P_0(t_\ell)$ is the **risky discount factor** (survival $\times$ riskless ZCB). The sum $\sum_\ell \Delta t_\ell \widetilde{P}_0(t_\ell)$ is the **risky annuity**. Under independence of hazard and rate under $\mathbb{Q}$, the expectation factorises cleanly: otherwise $\widetilde{P}_0(t_\ell) = \mathbb{E}^{\mathbb{Q}}[e^{-\int_0^{t_\ell}(r_u+\lambda_u)\,\mathrm{d}u}]$ which, for joint-Gaussian $r + \lambda$, again admits an affine closed form.

A heuristic: a risky bond earns the "hazard-adjusted rate" $r + \lambda$ instead of just r. The "credit spread" λ is effectively an additional rate that the bond earns to compensate for default risk. A corporate bond with a yield 150 bp above comparable Treasury is earning $r + 0.015$ in a flat-rate, flat-hazard world. The 150 bp is the credit spread, which the credit triangle (below) tells us equals $(1 - R)\lambda$ approximately.

Default leg. At default time τ, the payoff $(1 - R)N$ is paid, discounted to $t = 0$ via the riskless discount to τ. Under independence of τ and r:

$$V_0^{\text{D}} \;=\; \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_0^{\tau} r_s\,\mathrm{d}s}\,(1 - R)\,N\,\mathbf{1}_{\tau \le t_n}\Big] \;=\; (1 - R)\,N \int_0^{t_n} \lambda\, e^{-\lambda\tau}\, P_0(\tau)\,\mathrm{d}\tau. \tag{13.11}$$

The default density $\lambda e^{-\lambda\tau}$ is the probability of defaulting in $[\tau, \tau + \mathrm{d}\tau]$ per unit $\mathrm{d}\tau$; the present value of the loss is $(1 - R)NP_0(\tau)$; the integration bound $[0, t_n]$ reflects that the default leg pays only if default occurs before the contract maturity.

The Markov property of Hull-White means that, conditional on r_τ, the future bond price $P_\tau(T)$ depends only on $(r_\tau, \tau, T, \kappa, \sigma, \theta)$ — not on history. So the post-default discount $P_\tau(T)$ is the Hull-White bond-price formula from Chapter 12 evaluated at shifted origin τ and state r_τ. Under independence of r and λ, default timing is a purely exponential process — it does not care what the rate is doing — so integrating over default times first and then over rate paths gives the same answer as doing them simultaneously. The pricing factorises.

Par CDS spread. Setting $V_0^{\text{prem}} = V_0^{\text{D}}$ at inception gives

$$\boxed{\,S_0^{\text{CDS}} \;=\; \frac{(1 - R)\displaystyle\int_0^{t_n} \lambda\, e^{-\lambda\tau}\, P_0(\tau)\,\mathrm{d}\tau}{\displaystyle\sum_{\ell=1}^{n} \Delta t_\ell\, \widetilde{P}_0(t_\ell)}\,} \tag{13.12}$$

Structurally this is (13.4) dressed up with survival probabilities: the numerator is the PV of loss-given-default weighted by the default density, the denominator is the risky annuity. The parallel between (13.4) and (13.12) is not accidental: both are par-rate formulas for insurance-like instruments. The structural similarity "ratio of value-generating leg to annuity-like leg" recurs throughout rates and credit markets.

15.4.3 13.4.3 The credit triangle and worked numbers

The credit triangle. In the limit of low hazard and short maturity (so both survival $e^{-\lambda t}$ and discount $P_0(t)$ are close to 1), the numerator of (13.12) integrates to approximately $(1-R)\lambda t_n$ and the denominator to approximately t_n, so

$$S_0^{\mathrm{CDS}} \;\approx\; (1-R)\,\lambda \quad\Longleftrightarrow\quad \lambda \;\approx\; \frac{S_0^{\mathrm{CDS}}}{1-R}. \tag{13.13}$$

This is the single most important formula in credit trading. Given a quoted CDS spread, extract the implied hazard rate by dividing by the loss-given-default. With $R = 40\%$ and a 100 bp CDS spread: $\lambda \approx 100/(1-0.4) = 167$ bp, implying a 1-year default probability of $1 - e^{-0.0167} \approx 1.65\%$. So a 100 bp CDS is pricing in about 1.65% default odds over the next year, 40% recovery if it happens. Every credit trader internalises the triangle: *CDS spread maps to hazard maps to default probability maps to rating-equivalent.*

Worked numerical example. Take $\lambda = 0.02$ (2% hazard, typical BBB), $R = 0.4$ (40% recovery), flat riskless curve at 3% for $t_n = 5$ years with quarterly coupons ($\Delta t_\ell = 0.25$).

- Risky discount factors: $\widetilde{P}_0(t_\ell) = e^{-(0.03+0.02)t_\ell} = e^{-0.05t_\ell}$.
- Risky annuity: $\sum_{\ell=1}^{20} 0.25 \cdot e^{-0.05\cdot 0.25\ell} \approx 4.42$.
- Default-leg integrand: $\int_0^5 0.02 \cdot e^{-0.05\tau}\,\mathrm{d}\tau \approx 0.0885$. Multiply by $(1-R) = 0.6$: 0.053.
- Par spread: $0.053/4.42 \approx 120$ bp.
- Credit-triangle approximation: $(1-R)\lambda = 0.6 \cdot 0.02 = 120$ bp. Exact match for low hazard.

For a distressed $\lambda = 0.10$ the exact formula gives ~610 bp vs the triangle's 600 bp — the discounting correction grows materially as λ rises.

15.4.4 13.4.4 Bootstrapping the hazard-rate curve

Real-world hazards vary with the firm's health and the macro cycle. Time- varying hazard $\lambda(t)$ changes the survival formula to $S(t) = \exp(-\int_0^t \lambda(u)\,\mathrm{d}u)$, and the default-leg integral generalises to

$$V_0^{\mathrm{D}} \;=\; (1-R)\,N \int_0^{t_n} \lambda(\tau)\,S(\tau)\,P_0(\tau)\,\mathrm{d}\tau \tag{13.14}$$

Still a one-dimensional quadrature — CDS pricing remains essentially instantaneous on modern hardware.

Typical calibration. Given quoted CDS spreads at standard tenors (1y, 3y, 5y, 7y, 10y), bootstrap a piecewise-constant hazard curve $\lambda(t)$ to match each spread exactly. This is the credit analog of the Hull-White $\theta(t)$ bootstrap from Chapter 12, and the algorithm is analogous:

1. Start at the shortest tenor ($t_n = 1$y) and pick λ_1 so that (13.12) with constant $\lambda = \lambda_1$ matches the 1y quote.
2. Move to the next tenor ($t_n = 3$y). With $\lambda(t) = \lambda_1$ on $[0,1]$ and $\lambda(t) = \lambda_2$ on $[1,3]$, pick λ_2 so that (13.14) matches the 3y quote.
3. Continue tenor by tenor, building the full hazard schedule.

The bootstrapped hazard curve is directly interpretable: λ_k is the market-implied hazard for the bucket ending at tenor k. Upward- sloping credit curves (5y $>$ 1y) imply increasing hazard over time and are the norm for investment-grade names; inverted curves (5y $<$ 1y) signal near-term distress.

The $\mathbb{Q}$**-recovery** in the triangle is risk-neutral; the **credit risk-premium multiplier** ($\mathbb{Q}$-hazard $\div$ $\mathbb{P}$-default rate) typically runs 2–4x for investment-grade names, with direct implications for CDO tranche pricing.

15.4.5 13.4.5 CDS mark-to-market and DV01

For off-par CDS (market spread has moved since inception), the mark-to- market is the difference between premium and default legs evaluated at current spread and current discount/hazard curves. ISDA conventions since 2009 fix the *coupon* at a standard 100 bp or 500 bp and settle the spread-vs-coupon difference via an upfront payment:

$$\text{Upfront} \;=\; (\text{market spread} - \text{fixed coupon}) \cdot \text{risky annuity} \cdot N. \tag{13.15}$$

Standardised fixed-coupon CDS with upfront payments make the CDS fungible and centrally clearable, the post-2009 market structure.

CDS DV01 and "credit-DV01". Two sensitivities matter:

- **Rate DV01** (interest rate): sensitivity to a 1 bp parallel shift in the discount curve. Typically small (seconds-to-tens of dollars per \$1M notional) because rate effects largely cancel between the two legs.
- **Credit-DV01** (spread): sensitivity to a 1 bp parallel shift in the hazard curve (or equivalently a 1 bp shift in the quoted spread). The dominant risk metric, equal to roughly $(1-R) \cdot \text{risky annuity} \cdot N \cdot 0.0001$.

A \$10M 5y CDS with risky annuity 4.5 and 40% recovery has credit-DV01 $\approx \$10\text{M} \cdot 4.5 \cdot 0.6 \cdot 0.0001 = \$2{,}700$ per basis point. A portfolio of dozens of names requires bucket credit-DV01 by name and tenor, analogous to IRS bucket DV01 by tenor.

CDS premium leg vs default leg *The premium leg (left, blue) is a risky annuity. The default leg (right, orange) is the probability-weighted loss density. At the par spread the two areas match up to a factor of S_0^{CDS}; the credit triangle $(1-R)\lambda$ is the near-flat-curve limit.*

Production engines also handle accrued premium at default (a few-percent correction to the premium leg, a few-bps on the par spread); we suppress it here for clarity.

15.5 13.5 The yield surface

15.5.1 13.5.1 Spot, forward, and par-swap surfaces

Before moving to bond options, we pause to visualise the object that unifies everything the chapter has done so far: the **yield surface** $y_t(T)$, a 3D surface with axes $(t, T, y_t(T))$ showing how the entire yield curve evolves over time. Under a calibrated Hull-White model, at each sample date t we compute $P_t(T)$ for a range of T via the Chapter 12 bond-price formula evaluated at the sampled r_t, then invert $y_t(T) = -\ln P_t(T)/(T-t)$.

The yield surface is the natural 3D object for a rates trader because it makes explicit a distinction often blurred in text: the difference between **calendar time** t (what the clock does) and **maturity time** T (the endpoint of the bond). A 10-year yield today and a 10-year yield one year from now are *different bonds*: one matures in 10 years, the other in 11 years from today's vantage. Organising all this into a surface separates the two time dimensions cleanly.

The surface is **smooth in** T (yields are smooth functions of maturity via the affine exponent) but **rough in** t (driven by stochastic r_t).

15.5.2 13.5.2 One-factor signature and its limits

A one-factor model like Hull-White has a specific signature: all points of the yield curve move together in response to a shock to r_t, but not by the same amount. A positive shock δr to r_t raises the short end by the full shock magnitude and raises longer maturities by the damped amount:

$$\delta y_t(T) \;=\; \frac{B_{T-t}}{T-t} \cdot \delta r \tag{13.16}$$

For $T-t$ short, $B_{T-t}/(T-t) \approx 1$ and the yield moves almost 1-for-1 with the shock. For $T-t$ long, $B_{T-t}/(T-t) \approx 1/(\kappa(T-t))$ and the yield moves by a damped factor. With $\kappa = 0.15$ and $T - t = 30$ years, the damping factor is $1/(0.15 \cdot 30) = 0.22$: a 1 bp shock to r_t moves the 30-year yield by only 0.22 bp in the model. This is a specific prediction of the one-factor structure.

In reality, the 30-year yield responds to short-rate shocks more vigorously than one-factor Hull-White predicts — typically by a factor of 2-4×. Real-world rate shocks are often not just level shocks; they come with concurrent slope shifts (inflation expectations, policy stance). A pure level shock does damp out at the long end, but a level-plus-slope shock can move the long end substantially. This is the fundamental limitation of one-factor models and the motivation for multi-factor extensions like G2++ and beyond (§13.8).

No-arbitrage. The Hull-White surface is no-arbitrage by construction: the $\mathbb{Q}$-martingale condition on $e^{-\int r_u\,du} P_t(T)$ ties $\theta(t)$ to the initial forward curve (HJM drift condition, §13.8). This is why $\theta(t)$ calibration (Chapter 12) is a necessity, not a cosmetic choice.

Bridge to §13.6. The yield surface sets the stage for pricing *options* on the surface — specifically, **European bond options**, whose payoff depends on $P_{T_0}(T_1)$, a single spot-yield draw from the future surface at time T_0. Pricing these options is where the T-forward measure earns its keep.

15.5.3 Case study (b): SVB collapse March 2023 — duration mismatch + held-to-maturity accounting

Silicon Valley Bank, the 16*th-largest US bank by assets in early* 2023*, failed in* 48 *hours between* 9 *March (deposit run begins) and* 10 *March (FDIC takes the bank into receivership). The proximate cause was that depositors discovered the bank's* \$120*B held-to-maturity Treasury and agency-MBS portfolio carried an unrealised mark-to-market loss of approximately* \$15*B against tangible common equity of* \$16*B. The trade was unhedged duration; the failure was governance.*

Context. During 2020–2021 SVB grew deposits explosively as venture-funded startups parked cash from a record fundraising cycle. Deposits roughly tripled, from \$60B at end-2019 to \$190B at peak-2022. SVB invested the inflows in long-duration safe assets: predominantly 10y-and-longer agency MBS and 5y-and-longer Treasuries, purchased at the then-prevailing yields of 1.5%–2.5%. The asset book carried portfolio-weighted duration of approximately 5.6 years; the liability side was overnight-redeemable deposits with effective duration of zero. Asset-liability duration mismatch was thus around 5.6 years against \$120B of HTM-classified assets.

The Federal Reserve raised the funds rate from 0.00–0.25% in March 2022 to 4.50–4.75% by February 2023 — a parallel shift of approximately 450 basis points across the front of the curve, with smaller but material moves at longer maturities (10y Treasury yield went from $\approx$ 1.5% to $\approx$ 4.0%, a 250 bp move). SVB's HTM book was carried on the balance sheet at amortised cost under ASC 320, which permits MTM losses to bypass income provided the bank "intends and has the ability" to hold to maturity. The MTM loss was disclosed in footnotes — by Q4 2022 filings, $\sim$ \$15B unrealised loss against $\sim$ \$16B equity — but did not flow through P&L or regulatory capital. Once Silicon Valley depositors (each covered up to only \$250k of FDIC insurance, and most holding multiples of that) became aware of the gap, the deposit run on Thursday 9 March forced SVB to start selling HTM assets, crystallising losses, which accelerated the run, which closed the bank by Friday morning. Subsequent receivership cost the FDIC's deposit-insurance fund approximately \$20B.

Reading it through the chapter's math. The duration arithmetic is the most elementary instance of §13.5 modified-duration analysis. For a 5-year zero-coupon bond, modified duration equals approximately 5; the $\Delta P/P \approx -D \cdot \Delta y$ approximation gives a 25% MTM loss for a 500 bp parallel shift, 22.5% for 450 bp. SVB's portfolio with weighted duration 5.6 on \$120B HTM gives an order-of-magnitude loss estimate of $0.056 \times 0.045 \times$ \$120B $\approx$ \$30B for a hypothetical 450 bp parallel shift on duration-5.6, but the actual mark loss was $\sim$ \$15B because (i) MBS carry negative convexity that asymmetrically helps in a hiking cycle (extension shortens prepayment-adjusted duration as rates rise), and (ii) some of the portfolio matured into a flatter belly of the curve where the move was smaller. DV01 (13.6) on the HTM book was thus approximately \$70M per basis point — meaning a 1 bp parallel shift in Treasury yields changed the mark-to-market value of SVB's HTM book by \$70M, or roughly 0.4% of equity, *every basis point*. The chapter-wide Hull-White machinery (Chapter 12 calibrated bond curve plus this chapter's duration-convexity decomposition) would have computed this exposure in seconds; nothing about the math was difficult or unknown.

Lesson. SVB's failure was not a failure of interest-rate risk *math* — duration of a 5y zero is 5, and any sophomore studying §13.5 can compute the implied MTM loss for any parametric shift. It was a failure of interest-rate risk *governance*: the institutional choice to use HTM accounting suppressed the *visibility* of the loss without changing the *risk* one bit. When depositors marked the book themselves and ran, the gap that had been hidden in footnotes flowed straight to the equity capital. The deeper lesson for the new hire is that accounting classifications (HTM, AFS, trading) are regulatory reporting choices; the underlying duration risk is invariant to the accounting treatment, and any liquidity stress that forces realisation will surface the loss regardless of which bucket the position sat in. Risk reports should *always* include MTM Greeks on HTM books, and liquidity stress tests should *always* assume HTM positions get realised at MTM under a deposit-run scenario. Post-SVB, US bank regulators rapidly escalated stressed-VaR-style scenario requirements for AFS-and-HTM books at sub-\$250B regional banks (which had been exempted from the largest-bank framework), and the Federal Reserve introduced the Bank Term Funding Program to allow banks to borrow against HTM Treasuries at par to prevent forced realisations. None of this changed the duration math; all of it changed the discipline around translating duration math into liquidity planning.

15.6 13.6 European bond options and the T-forward measure

15.6.1 13.6.1 Why change to the T-forward measure

Setup. A European call on a bond has payoff $\big(P_{T_0}(T_1) - K\big)_+$ paid at T_0, where $t < T_0 < T_1$. Let $\mathfrak{g}_t$ denote the call price process. Risk-neutral pricing under the money-market numeraire M_t:

$$\mathfrak{g}_t \;=\; \mathbb{E}_t^{\mathbb{Q}}\Big[e^{-\int_t^{T_0} r_s\,\mathrm{d}s}\,\big(P_{T_0}(T_1) - K\big)_+\Big]. \tag{13.17}$$

Under Vasicek or Hull-White, $P_{T_0}(T_1) = \exp\big(A_{T_0}(T_1) - B_{T_0}(T_1)\,r_{T_0}\big)$, so the integrand in (13.17) couples *two* random variables driven by the same short-rate path: the discount factor $e^{-\int_t^{T_0} r_s\,\mathrm{d}s}$ and the terminal bond value $P_{T_0}(T_1)$. Both are nonlinear functions of r_{T_0} and both are highly correlated (when rates are high at T_0, the discount factor is small *and* the bond is cheap).

A direct computation would require evaluating a 2-dimensional Gaussian integral involving the product of an exponential, an indicator function (from the max), and another exponential. Feasible, but ugly. The **numeraire-change trick** below eliminates the stochastic discount factor entirely, reducing the problem to a clean 1-dimensional Gaussian integral that is precisely the Black-76 setup.

15.6.2 13.6.2 The T-forward measure (summary from Chapter 5)

Chapter 5 derived the T-forward measure once as the canonical application of Girsanov's theorem: we will not re-derive it here. The result we need is this.

(Result, from Chapter 5.) Let $P_t(T_0)$ be a positive traded numeraire. Define the **T-forward measure** $\widetilde{\mathbb{Q}}_{T_0}$ by the Radon-Nikodym derivative

$$\left.\frac{\mathrm{d}\widetilde{\mathbb{Q}}_{T_0}}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_t} \;=\; \frac{P_t(T_0)/M_t}{P_0(T_0)/M_0}. \tag{13.18}$$

Under $\widetilde{\mathbb{Q}}_{T_0}$, every traded asset divided by $P_t(T_0)$ is a martingale:

$$\frac{\mathfrak{g}_t}{P_t(T_0)} \;=\; \tilde{\mathbb{E}}_t^{T_0}\left[\frac{\mathfrak{g}_{T_0}}{P_{T_0}(T_0)}\right] \;=\; \tilde{\mathbb{E}}_t^{T_0}\big[\,\mathfrak{g}_{T_0}\,\big] \quad \text{(since } P_{T_0}(T_0) = 1\text{)} \tag{13.19}$$

and the $\mathbb{Q}$-Brownian motion $\widehat{W}_t$ is transformed to $\widetilde{W}_t^0$ via

$$\mathrm{d}\widetilde{W}_t^0 \;=\; \mathrm{d}\widehat{W}_t \;+\; B_t(T_0)\,\sigma\,\mathrm{d}t. \tag{13.20}$$

The drift shift $B_t(T_0)\sigma$ is the $\mathbb{Q}$-volatility of the numeraire $P_t(T_0)$ (with a sign flip, because bond prices move *inversely* to rates). This is the **convexity adjustment** from changing measure, and it is the standard Girsanov prescription ("shift by the numeraire's volatility") derived in Chapter 5.

Hence

$$\boxed{\mathfrak{g}_t \;=\; P_t(T_0)\,\tilde{\mathbb{E}}_t^{T_0}\Big[\big(P_{T_0}(T_1) - K\big)_+\Big]} \tag{13.21}$$

The awkward stochastic discount factor $e^{-\int r\,\mathrm{d}s}$ is gone: it has been absorbed into the numeraire $P_t(T_0)$, which is now a deterministic quantity outside the expectation.

Intuition — why Black's formula works here. Switching numeraire to $P_t(T_0)$ — the bond that matures exactly when the option pays — solves the problem in one stroke:

- $P_{T_0}(T_0) = 1$ by definition, so at option expiry the numeraire equals 1 and the "discount factor" at that instant simply vanishes.

- The **forward bond** $X_t \equiv P_t(T_1)/P_t(T_0)$ is a ratio of two traded assets divided by the numeraire — hence a $\widetilde{\mathbb{Q}}_{T_0}$-martingale by construction.
- In Vasicek/Hull-White, the ratio of two exponential-affine bond prices is itself exponential-affine in a *deterministic* combination $B_t(T_0) - B_t(T_1)$, so X_t has deterministic volatility and is exactly a driftless geometric Brownian motion.

Geometric Brownian motion + driftless + deterministic volatility is precisely the setup of Black-76 (Black's 1976 commodity-futures formula). The bond-option price collapses to a Black-type $\Phi(d_\pm)$ expression with the "forward" X_t, the "discount" $P_t(T_0)$, and the "total variance" $\int_t^{T_0} \Sigma_s^2 \,\mathrm{d}s$. The numeraire choice is not cosmetic — it is what makes the problem lognormal.

The deeper lesson: *match your numeraire to your payoff, and the payoff's underlying becomes a driftless martingale — the Black setup.*

15.6.3 13.6.3 Forward bond price dynamics under $\widetilde{\mathbb{Q}}_{T_0}$

By Itô applied to $P_t(T_1) = e^{A_t(T_1) - B_t(T_1) r_t}$ using the Hull-White SDE $\mathrm{d}r_t = \kappa(\theta_t - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}\widehat{W}_t$ from Chapter 12:

$$\frac{\mathrm{d}P_t(T_1)}{P_t(T_1)} \;=\; r_t\,\mathrm{d}t \;-\; \sigma\, B_t(T_1)\,\mathrm{d}\widehat{W}_t. \tag{13.22}$$

Every bond has $\mathbb{Q}$-drift r_t (consistent with being a traded asset under the risk-neutral measure), with diffusion coefficient $-\sigma B_t(T_1)$. The minus sign reflects the inverse relationship between bond prices and rates; the magnitude $\sigma B_t(T_1)$ scales with both the short-rate vol σ and the bond's duration $B_t(T_1)$. For a short bond ($B \approx 0$) the diffusion coefficient is tiny; for a long bond it saturates at σ/κ. This is the Vasicek analogue of the duration-based bond-price volatility heuristic.

Critically, in Vasicek/Hull-White, the bond's diffusion coefficient $-\sigma B_t(T_1)$ is **deterministic** — it depends on time and maturity but not on any random state variable. This is special to Gaussian models. In CIR, the short-rate volatility $\sigma\sqrt{r_t}$ is state-dependent, and so is the bond's diffusion coefficient; the bond is no longer lognormal under any measure change, and the resulting bond-option formula is not of Black form (it involves a non-central chi-squared distribution). The deterministic-volatility property is what makes Vasicek bond options admit a Black-type closed form.

Under $\widetilde{\mathbb{Q}}_{T_0}$, using (13.20) to substitute $\mathrm{d}\widehat{W}_t = \mathrm{d}\widetilde{W}_t^0 - B_t(T_0)\sigma\,\mathrm{d}t$, the T_1-bond's drift changes:

$$\frac{\mathrm{d}P_t(T_1)}{P_t(T_1)} \;=\; \big(r_t + \sigma^2 B_t(T_0)\, B_t(T_1)\big)\,\mathrm{d}t \;-\; \sigma\, B_t(T_1)\,\mathrm{d}\widetilde{W}_t^0. \tag{13.23}$$

The drift correction $+\sigma^2 B_t(T_0) B_t(T_1)$ is the **convexity adjustment** from changing measure. Under $\mathbb{Q}$, the bond drifts at r_t. Under $\widetilde{\mathbb{Q}}_{T_0}$, it drifts at a slightly higher rate, by an amount proportional to the product of the two bonds' durations. This extra drift keeps the *ratio* $P_t(T_1)/P_t(T_0)$ a martingale, as required by the new numeraire choice.

The *form* "shift by the numeraire's volatility" is universal across continuous-diffusion models; the specific $B_t(T_0)\sigma$ here is just Vasicek's bond diffusion coefficient.

Rate distribution under Q vs T-forward measure *Changing numeraire from M_t to $P_t(T_0)$ shifts the drift of r_t downward by $-\sigma^2 B_t(T_0)$: the conditional law of r_{T_0} under the forward measure is a Gaussian with the same variance but a lower mean than under $\mathbb{Q}$. This shift is the convexity adjustment relating forward rates to expected spot rates.*

15.6.4 13.6.4 Forward bond price X_t is a $\widetilde{\mathbb{Q}}_{T_0}$-martingale

Introduce the **forward bond price**

$$X_t \;=\; \frac{P_t(T_1)}{P_t(T_0)}. \tag{13.24}$$

Key boundary property:

$$X_{T_0} \;=\; \frac{P_{T_0}(T_1)}{P_{T_0}(T_0)} \;=\; P_{T_0}(T_1), \tag{13.25}$$

so the option's payoff at T_0 is exactly $(X_{T_0} - K)_+$. X_t is a $\widetilde{\mathbb{Q}}_{T_0}$-martingale (ratio of traded asset to numeraire).

By Itô on the ratio (both bonds driven by the same $\widetilde{W}^0$):

$$\frac{\mathrm{d}X_t}{X_t} \;=\; \sigma\,(B_t(T_0) - B_t(T_1))\,\mathrm{d}\widetilde{W}_t^0 \;\equiv\; \Sigma_t\,\mathrm{d}\widetilde{W}_t^0, \tag{13.26}$$

where the deterministic **forward-bond volatility** is

$$\Sigma_t \;=\; \sigma\,(B_t(T_0) - B_t(T_1)) \;=\; -\frac{\sigma}{\kappa}\,e^{-\kappa T_0}(1 - e^{-\kappa(T_1-T_0)})\,e^{\kappa t}. \tag{13.27}$$

Only Σ_t^2 appears below. The net exposure is **deterministic** — the property that makes Vasicek admit a Black-type bond-option formula.

Because Σ_t is deterministic, X_t is a geometric Brownian motion with deterministic volatility, and

$$\boxed{X_T \;=\; X_t\,\exp\Big\{-\tfrac{1}{2}\int_t^T \Sigma_s^2\,\mathrm{d}s \;+\; \int_t^T \Sigma_s\,\mathrm{d}\widetilde{W}_s^0\Big\}} \tag{13.28}$$

so $\ln(X_T/X_t)$ is Gaussian under $\widetilde{\mathbb{Q}}_{T_0}$ — a lognormal forward bond price.

Forward-bond vol Sigma_t and integrated variance $|\Sigma_t|$ *grows as* $t \to T_0$ *(the pull-to-par of* $P_t(T_0)$ *becomes less effective at absorbing rate noise); the cumulative total variance* $\int_0^t \Sigma_s^2\,\mathrm{d}s$ *is what appears in the Black* $d_\pm$ *denominator.*

15.6.5 13.6.5 Closed-form European bond call (Jamshidian / Vasicek-Black)

Plugging into (13.21):

$$\mathfrak{g}_t \;=\; P_t(T_0)\,\tilde{\mathbb{E}}_t^{T_0}[\,(X_{T_0} - K)_+\,]. \tag{13.29}$$

This is a Black-style lognormal expectation with "volatility" $\sqrt{\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s}$. Evaluating the Gaussian integral in the standard way:

$$\boxed{\mathfrak{g}_t \;=\; P_t(T_0)\,\{\,X_t\,\Phi(d_+) \;-\; K\,\Phi(d_-)\,\}} \tag{13.30}$$

with

$$\boxed{d_\pm \;=\; \frac{\ln(X_t/K) \;\pm\; \frac{1}{2}\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s}{\sqrt{\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s}}} \tag{13.31}$$

This is the **Vasicek-Black bond-option formula**: a Black-Scholes-type expression where the "stock" is the forward bond X_t, the "discount factor" is $P_t(T_0)$, and the total variance is the deterministic integral $\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s$.

Boundary-condition check. At expiry $t = T_0$: the integral $\int_{T_0}^{T_0} \Sigma_s^2\,\mathrm{d}s = 0$, so $d_\pm = \pm\infty \cdot \mathrm{sgn}(\ln(X_{T_0}/K))$. Hence $\Phi(d_+) - \Phi(d_-) = \mathbf{1}\{X_{T_0} > K\}$, and the formula reduces to $(X_{T_0} - K)_+ = (P_{T_0}(T_1) - K)_+$, matching the contractual payoff. At deep OTM strikes ($K \gg X_t$): both $\Phi(d_+)$ and $\Phi(d_-) \to 0$, and the price goes to zero. At deep ITM ($K \ll X_t$): $\Phi(d_+) \to 1$, $\Phi(d_-) \to 1$, and the price approaches $P_t(T_0)(X_t - K) = P_t(T_1) - K P_t(T_0)$, the intrinsic value (forward value of bond minus forward value of strike). All expected limits hold.

Total-variance factorisation. The integral admits a clean closed form:

$$\int_t^{T_0} \Sigma_s^2 \, \mathrm{d}s \;=\; \frac{\sigma^2}{\kappa^2} \left(1 - e^{-\kappa(T_1 - T_0)}\right)^2 \cdot \frac{1 - e^{-2\kappa(T_0 - t)}}{2\kappa}. \tag{13.32}$$

The three factors have clean physical interpretations:

- σ^2/κ^2: the overall *scale*, roughly the stationary variance of r_t times 2.
- $(1 - e^{-\kappa(T_1 - T_0)})^2$: the *tenor-sensitivity* factor — how much longer T_1 is than T_0.
- $(1 - e^{-2\kappa(T_0 - t)})/(2\kappa)$: the *time-to-expiry weighting*, saturating at $1/(2\kappa)$ as $T_0 - t \to \infty$.

Once these three numbers are computed for the relevant parameter grid, pricing any Vasicek bond option is three multiplications plus a Black formula — a core rule-of-thumb for rates traders.

15.6.6 13.6.6 Put-call parity for bond options

$$\mathfrak{g}_t^{\text{call}} - \mathfrak{g}_t^{\text{put}} \;=\; P_t(T_1) - K\, P_t(T_0). \tag{13.33}$$

Parity is the same as for equity options and is the cheapest sanity check on a rates-option library.

Digital bond options — options paying \$1 if $P_{T_0}(T_1) > K$ and 0 otherwise — collapse to $\mathfrak{g}_t^{\text{digital}} = P_t(T_0) \cdot \Phi(d_-)$. This is the Black-76 digital formula in disguise, useful for structuring bespoke rate payoffs.

15.7 13.7 Worked examples: Jamshidian decomposition and Vasicek-Black

15.7.1 13.7.1 Worked example — forward bond volatility for a 1×3 option

Suppose $\sigma = 0.01$, $\kappa = 0.3$, $T_0 = 1$, $T_1 = 3$. From (13.32):

- $(1 - e^{-\kappa(T_1 - T_0)})^2 = (1 - e^{-0.6})^2 = (0.4512)^2 = 0.2036$
- $(1 - e^{-2\kappa T_0})/(2\kappa) = (1 - e^{-0.6})/(0.6) = 0.7520$
- $\sigma^2/\kappa^2 = (0.01)^2/0.09 = 1.111 \times 10^{-3}$

Total variance $\approx 1.111 \times 10^{-3} \cdot 0.2036 \cdot 0.7520 = 1.70 \times 10^{-4}$, so effective lognormal vol $\approx \sqrt{1.70 \times 10^{-4}} = 1.30\%$.

Plug into (13.30)–(13.31) with $X_0 = 0.94$ (forward bond price) and strike $K = 0.94$ (ATM). Then $\ln(X_0/K) = 0$, $d_\pm = \pm\sqrt{1.70 \times 10^{-4}}/2 = \pm 0.00652$, $\Phi(0.00652) = 0.5026$, $\Phi(-0.00652) = 0.4974$. Call price:

$$\mathfrak{g}_0 \;=\; P_0(1) \cdot (0.94 \cdot 0.5026 - 0.94 \cdot 0.4974) \;=\; P_0(1) \cdot 0.94 \cdot 0.0052 \;=\; 0.00489 \cdot P_0(1).$$

For $P_0(1) \approx 0.97$, the dollar ATM call price is about 0.00474, or 47 bps of notional. Small premium — consistent with short expiry and modest vol — but the right order of magnitude for a short-dated ATM bond option.

ATM approximation. $\mathfrak{g} \approx P_0(1) X_0 \sigma_{\text{eff}}/\sqrt{2\pi} = 0.00473$ — matches exact Black to within 0.2%. The $1/\sqrt{2\pi} \approx 0.4$ factor is the universal Black/Bachelier ATM constant.

15.7.2 13.7.2 ITM and OTM strikes

Returning to the 1×3 bond option with $X_0 = 0.94$, consider an ITM call ($K = 0.90$) and an OTM call ($K = 0.98$).

ITM ($K = 0.90$). $\ln(X_0/K) = \ln(0.94/0.90) = 0.0435$. With total variance 1.70×10^{-4}, $\sqrt{\cdot} = 0.01304$. Then $d_+ = (0.0435 + 8.5 \times 10^{-5})/0.01304 = 3.34$, $d_- = 3.33$. $\Phi(3.34) \approx 0.9996$. Call price (per unit $P_0(1)$) $= 0.94 \cdot 0.9996 - 0.90 \cdot 0.9996 = 0.04$. This is essentially the intrinsic value ($X_0 - K = 0.04$); time value is tiny at such a deep-ITM strike and short expiry.

OTM ($K = 0.98$). $\ln(X_0/K) = -0.0417$. $d_+ = -3.19$, $d_- = -3.20$. $\Phi(-3.19) \approx 0.00071$. Call price $\approx X_0 \cdot 0.00071 - K \cdot 0.00069 \approx 0$. The option is so deep OTM that the forward bond has essentially no chance of reaching the strike in one year.

The asymmetry between ITM (near-intrinsic) and OTM (near-zero) is why bond options trade like delta-one when far ITM and like pure gamma plays when near ATM. Structured trades across strikes can achieve specific exposure profiles: an ATM straddle gives pure gamma/vega; a wide-strike strangle gives tail-event exposure.

15.7.3 13.7.3 Sensitivity to mean-reversion speed

Fix $\sigma = 0.01$, $T_0 = 1$, $T_1 = 5$, and vary κ:

- $\kappa = 0.10$: $\sigma^2/\kappa^2 = 0.01$, $(1-e^{-0.4})^2 = 0.1087$, $(1-e^{-0.2})/0.2 = 0.9063$. Total var $= 9.85 \times 10^{-4}$, effective vol $\approx 3.14\%$.
- $\kappa = 0.30$: $\sigma^2/\kappa^2 = 1.11 \times 10^{-3}$, $(1-e^{-1.2})^2 = 0.4968$, $(1-e^{-0.6})/0.6 = 0.7520$. Total var $= 4.15 \times 10^{-4}$, effective vol $\approx 2.04\%$.
- $\kappa = 1.00$: $\sigma^2/\kappa^2 = 10^{-4}$, $(1-e^{-4})^2 = 0.9640$, $(1-e^{-2})/2 = 0.4323$. Total var $= 4.17 \times 10^{-5}$, effective vol $\approx 0.646\%$.

Doubling κ from 0.10 to 0.30 cuts effective vol by ~1/3; $\kappa = 1.0$ cuts it ~80%. The dominant effect is σ^2/κ^2 scaling like $1/\kappa^2$. Disentangling (κ, σ) requires multiple option maturities: short-expiry options are more sensitive to σ, long-expiry to κ.

Bond-option price surface vs strike and T_0 *Left: call price surface* $\mathfrak{g}_0(T_0, K)$ *on a 5-year zero via the Black-type Vasicek formula. Right: Black slices at fixed* T_0 *— the dotted verticals mark the forward bond price* $X_0 = P_0(T_1)/P_0(T_0)$ *(ATM).*

15.7.4 13.7.4 Jamshidian's decomposition: swaption as a portfolio of bond options

A European **swaption** gives the right at T_0 to enter a fixed-floating swap with payments $T_1 < \cdots < T_N$. Its value at T_0 is a max of a sum of zero-coupon bond prices — harder than an option on a single bond.

Jamshidian's 1989 insight. In a one-factor Gaussian short-rate model the sum of bond prices is monotone decreasing in r_{T_0}, so a unique critical rate r^* exists where the sum equals the strike. The option on the sum decomposes into a portfolio of options on individual bonds:

$$\text{swaption}_0 \;=\; \sum_{i=1}^{N} c_i \cdot \text{BondOption}(T_0, T_i, K_i) \tag{13.34}$$

where c_i are the swap coefficients (tenors and notional), $K_i = P_{T_0}(T_i)\big|_{r=r^*}$ are effective strikes computed via the Vasicek/Hull-White bond-price formula at $r = r^*$, and r^* is the critical rate obtained by solving a scalar equation.

Procedure.

1. **Set up the critical-rate equation.** The swaption payoff at T_0 is $(V^{\text{swap}}_{T_0} - K)_+$ where $V^{\text{swap}}_{T_0} = \sum_i c_i P_{T_0}(T_i)$. Each $P_{T_0}(T_i) = \exp(A_{T_0}(T_i) - B_{T_0}(T_i) r_{T_0})$ is a monotonically decreasing function of r_{T_0}. So $V^{\text{swap}}_{T_0}$ is monotonically decreasing in r_{T_0}.
2. **Solve for r^*.** Find the unique root of $\sum_i c_i \exp(A_{T_0}(T_i) - B_{T_0}(T_i) r^*) = K$ — a one-dimensional monotone root-find (bisection or Newton).
3. **Compute effective strikes.** For each i, set $K_i = \exp(A_{T_0}(T_i) - B_{T_0}(T_i) r^*)$.
4. **Price a portfolio of bond options.** Each term $\text{BondOption}(T_0, T_i, K_i)$ is a Vasicek-Black call on the T_i-bond with strike K_i, priced by (13.30)-(13.31).

Jamshidian reduces an option-on-portfolio to a 1D root-find plus scalar bond-option evaluations. Exact for one-factor Gaussian models; only approximate (but useful as a starting point) for multi-factor. A side-effect is flat-in-strike swaption vol under Vasicek; real swaption smile motivates §13.8.

15.7.5 13.7.5 Calibration pipeline

The pipeline that ties the whole chapter together:

1. **Fit** (κ, σ) from swaption and/or cap quotes by minimising weighted least-squares between model-implied and market prices across the vol cube.
2. **Bootstrap** θ_t from the initial bond curve (Chapter 12) so that $P_0^{\text{model}}(T) = P_0^{\text{market}}(T)$ for all T.
3. **Price other rate derivatives** — bond options via (13.30), swaptions via Jamshidian (13.34), callables via the Hull-White lattice (§13.3), caps/floors via strips of caplets (each a put on a zero-bond).

Step 1's calibration is typically not unique: (κ, σ) combinations trade off "persistent long-rate shocks" (high κ, high σ) against "temporary short-rate shocks" (low κ, low σ). Short-expiry swaptions carry more information about σ;

long-expiry about κ. To resolve degeneracy, practitioners often add long-dated cap vols or fix κ at a historical value (say 0.1) and only calibrate σ.

15.8 13.8 Multi-factor preview: HJM, LMM, and SABR

15.8.1 13.8.1 Why one factor is not enough

One-factor Vasicek has four structural limits: perfect cross-maturity correlation (real PCAs show 3 factors); limited curve shapes; flat swaption smile (Jamshidian gives one vol per expiry-tenor); long-end under-prediction (typical 2–4×gap). Each has a corresponding remedy below.

15.8.2 13.8.2 Two-factor Gaussian: G2++

The most common multi-factor extension of Vasicek is **G2++**, where the short rate is

$$r_t \;=\; x_t + y_t + \phi(t) \tag{13.35}$$

with x_t and y_t two correlated Ornstein-Uhlenbeck factors:

$$\begin{aligned} \mathrm{d}x_t &= -a\,x_t\,\mathrm{d}t + \sigma_1\,\mathrm{d}W_t^x, \\ \mathrm{d}y_t &= -b\,y_t\,\mathrm{d}t + \sigma_2\,\mathrm{d}W_t^y, \\ \langle W^x, W^y\rangle &= \rho\,t, \end{aligned} \tag{13.36}$$

$\phi(t)$ is the deterministic curve-fit shift; one factor is typically fast (short end), the other slow (long end); ρ controls the curve shape. Bond prices remain exponential-affine: $P_t(T) = A(t,T)\exp(-B_1 x_t - B_2 y_t)$. Bond options are still Black-type under the forward measure; Jamshidian holds only approximately. G2++ can fit cap vols *and* swaption vol cube at one strike; real-market humped cap profiles drive negative-ρ calibrations.

15.8.3 13.8.3 HJM: modelling the entire forward curve

HJM (1992) takes the forward-rate curve $f_t(T)$ as the primitive: $\mathrm{d}f_t(T) = \alpha(t,T)\,\mathrm{d}t + \sigma(t,T)\,\mathrm{d}W_t$, with no-arbitrage forcing

$$\alpha(t,T) \;=\; \sigma(t,T)\int_t^T \sigma(t,s)\,\mathrm{d}s. \tag{13.39}$$

Choosing $\sigma(t,T)$ fully specifies the arbitrage-free dynamics. Vasicek = exponential kernel; G2++ = sum of two. Generality costs analytic tractability — forward rates are rarely Markovian in low dimensions, so most exotics need MC.

15.8.4 13.8.4 LMM: the discrete-tenor LIBOR market model

LMM (BGM 1997 / Jamshidian) models a finite collection of forward LIBOR rates as lognormal martingales under their own forward measures: $\mathrm{d}L_t^{(i)} = L_t^{(i)}\sigma_i(t)\,\mathrm{d}W_t^{(i)}$ (13.40). It reproduces Black's caplet/swaption formula exactly for the modelled forwards, at the cost of high-dimensional MC. The 2026-era RFR LMM (SOFR, €STR, SONIA, TONA) replaces LIBOR with compounded overnight rates.

15.8.5 13.8.5 SABR and rate-vol smile

The models above (Vasicek, G2++, HJM, LMM) all produce a *single* implied volatility for each option — no smile. Reality shows substantial implied- vol smile in caps, floors, and swaptions: OTM options trade rich of ATM, and the richness varies with strike in specific ways (typically a negative skew for payers, reflecting the asymmetry of rate distributions). Capturing this smile requires stochastic-volatility or jump components.

The industry-standard overlay is **SABR** (Stochastic Alpha Beta Rho), developed by Hagan and co-authors:

$$\mathrm{d}F_t \;=\; \alpha_t\,F_t^{\beta}\,\mathrm{d}W_t, \qquad \mathrm{d}\alpha_t \;=\; \nu\,\alpha_t\,\mathrm{d}Z_t, \qquad \langle W, Z\rangle \;=\; \rho\,t. \tag{13.41}$$

β controls the backbone, ν the vol-of-vol, ρ the skew. The Hagan expansion gives a closed-form asymptotic implied vol — the industry standard quoting tool. SABR is per-strike/per-expiry (not term-structure); the production stack is Hull-White / G2++ for the term structure plus SABR per expiry-tenor.

15.8.6 13.8.6 Where to use what

A rough practitioners' guide:

Product	Recommended model
Vanilla caps, floors, European swaptions	1-factor Hull-White or G2++; SABR overlay per strike
Bermudan swaptions, callable bonds	G2++ or LMM; Jamshidian-like decomposition + MC for paths
CMS, LIBOR-in-arrears, convexity products	Multi-factor affine with explicit convexity corrections
Path-dependent exotics (range accruals)	LMM or extended HJM, with Monte Carlo
XVA (CVA/DVA/FVA)	G2++ for efficiency; hybrid with equity/credit factors
Negative-rate environments	Vasicek/Hull-White/G2++ (Gaussian); shifted-lognormal LMM

No single model dominates; desks maintain multiple models, switching per product and regime. The mathematical apparatus is the common thread: SDEs, Itô, Feynman-Kac, Girsanov, affine closed forms where they apply, PDE-vs-MC where they don't.

15.9 13.9 PDE versus Monte Carlo for rate products

When closed forms are unavailable — which is most of the time outside vanilla caps, floors, and European swaptions — one must resort to numerical methods. The two main candidates are **PDE** (finite-difference or finite-element) solvers and **Monte Carlo** simulation. Each has strengths and weaknesses and the choice is product-specific.

15.9.1 13.9.1 When a PDE wins

PDE methods discretise the Feynman-Kac PDE on a grid in (t, r) space and solve backward via Crank-Nicolson. Output is the *function* $g(t, r)$ on the grid, giving prices and all first-derivative Greeks for free.

- **Strengths:** free Greeks; early-exercise natural ($\max(g, \text{payoff})$); deterministic.
- **Weaknesses:** dimensional curse beyond 3–4 factors; boundary-condition fragility; path-dependence pushes dimensionality.

15.9.2 13.9.2 When Monte Carlo wins

MC simulates paths and averages payoffs.

- **Strengths:** dimensional scaling (curse broken); path-dependent payoffs natural; parallelism on GPUs.
- **Weaknesses:** slow convergence $O(1/\sqrt{N})$; Greeks need pathwise/likelihood-ratio/re-sim; early exercise via Longstaff-Schwartz.

15.9.3 13.9.3 Practical guidance

Regime	Method
1-factor + vanilla	PDE
2–3 factor + mildly exotic	PDE or MC
LMM/HJM / path-dependent	MC
XVA on large book	MC (often LS-MC for callables)

Hybrid methods (PDE-in-MC, Quasi-MC with Sobol, MLMC) handle edge cases. The mathematics is unchanged across methods; only the numerical implementation differs.

15.10 13.10 Key takeaways

1. **Swaps are essentially model-free.** The floating leg self-replicates via "buy \$1 and roll": $V_0^{\text{fl}} = (P_0(t_0) - P_0(t_n))N$ — no short-rate model required, only the discount curve. The par swap rate $S_0 = (P_0(t_0) - P_0(t_n))/\mathcal{A}_0$ is the ratio of that self-replicating value to the annuity; neither κ nor σ nor θ_t appears.

2. **Callable bonds are genuinely model-dependent.** The embedded American option requires backward induction on a Hull-White trinomial tree with a call-value test at each exercise date: $V^{\text{call}} = \min(\text{call price}, V^{\text{continuation}})$. OAS, OAD, and negative convexity emerge from the tree. Calibrated (κ, σ) from the swaption market feed into the tree directly.

3. **CDS factor into a risky annuity and a default density.** The par CDS spread is

$$S_0^{\text{CDS}} = \frac{(1-R)\int_0^{t_n} \lambda e^{-\lambda\tau} P_0(\tau)\,\mathrm{d}\tau}{\sum_\ell \Delta t_\ell \widetilde{P}_0(t_\ell)}$$

with $\widetilde{P}_0(t) = e^{-\lambda t} P_0(t)$ the risky discount. The **credit triangle** $S \approx (1-R)\lambda$ is the low-hazard short-tenor limit and the most important formula in credit trading.

4. **T-forward measure eliminates stochastic discounting.** Changing numeraire from M_t to $P_t(T_0)$ makes the discount factor deterministic at expiry (since $P_{T_0}(T_0) = 1$) and turns the forward bond $X_t = P_t(T_1)/P_t(T_0)$ into a martingale with deterministic volatility $\Sigma_t = \sigma(B_t(T_0) - B_t(T_1))$.

5. **Vasicek-Black bond-option formula.** A European call on a zero- coupon bond has the Black-76 form

$$\mathfrak{g}_t = P_t(T_0)\{X_t\Phi(d_+) - K\Phi(d_-)\}$$

with $d_\pm$ computed from the deterministic total variance $\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s = (\sigma^2/\kappa^2)(1 - e^{-\kappa(T_1 - T_0)})^2 \cdot (1 - e^{-2\kappa(T_0 - t)})/(2\kappa)$.

6. **Jamshidian decomposition reduces a swaption to a portfolio of bond options.** Exploit monotonicity of $\sum_i c_i P_{T_0}(T_i)$ in r_{T_0}: find the critical rate r^*, compute effective strikes K_i, and price each bond option by Black-76. Exact for one-factor Gaussian models; approximate but still useful for multi-factor.

7. **One-factor Vasicek is the start, not the destination.** G2++ for curve richness, HJM for forward-curve primacy, LMM for direct forward-rate modelling, SABR for vol smile. Each extension fixes a specific limitation of the 1-factor Gaussian framework.

8. **PDE versus MC is a dimensionality trade-off.** PDE wins for 1-2 factor vanilla products; MC wins for high-dimensional and path- dependent payoffs; hybrids handle edge cases. The underlying mathematics is unchanged — only the numerical implementation differs.

9. **The chapter's unifying principle.** Every rate derivative prices as a conditional expectation of a payoff functional under the most convenient martingale measure. Match the numeraire to the payoff's natural scale, and the pricing collapses to a Black-type lognormal expectation. That sentence — one idea — ties every formula in this chapter together.

15.11 13.11 Reference formulas

15.11.1 Swaps

$$V_0^{\text{fix}} = \sum_{m=1}^{n} \Delta t_m F N P_0(t_m) = F \cdot \mathcal{A}_0 \cdot N$$

$$V_0^{\text{fl}} = (P_0(t_0) - P_0(t_n))N$$

$$S_0 = \frac{P_0(t_0) - P_0(t_n)}{\sum_m \Delta t_m P_0(t_m)} = \frac{P_0(t_0) - P_0(t_n)}{\mathcal{A}_0}$$

$$S_t = \frac{P_t(t_{k_t}) - P_t(t_n)}{\sum_{m=k_t+1}^{n} \Delta t_m P_t(t_m)}$$

$$\text{DV01} \approx \mathcal{A} \cdot 0.0001 \cdot N$$

15.11.2 Callable bonds (Hull-White lattice)

Backward induction with call test:

$$V_n^{(i)} = \begin{cases} \min(\text{call price}_n, V_n^{(i),\text{cont}}) & \text{if } t_n \text{ is call date} \\ V_n^{(i),\text{cont}} & \text{otherwise} \end{cases}$$

where $V_n^{(i),\text{cont}} = e^{-r_n^{(i)}\Delta t}(p_u V_{n+1}^{(i+1)} + p_m V_{n+1}^{(i)} + p_d V_{n+1}^{(i-1)}) + c_n$.

Value decomposition: $V^{\text{callable}} = V^{\text{straight}} - V^{\text{embedded call}}$.

15.11.3 CDS (constant hazard λ under $\mathbb{Q}$, independent of r)

$$\begin{aligned} \widetilde{P}_0(t) &= e^{-\lambda t} P_0(t) \quad \text{(risky discount)} \\ V_0^{\text{prem}} &= S \cdot N \cdot \sum_\ell \Delta t_\ell \widetilde{P}_0(t_\ell) \\ V_0^{\text{D}} &= (1-R) \cdot N \cdot \int_0^{t_n} \lambda e^{-\lambda \tau} P_0(\tau)\,\mathrm{d}\tau \\ S_0^{\text{CDS}} &= \frac{(1-R)\int_0^{t_n} \lambda e^{-\lambda\tau} P_0(\tau)\,\mathrm{d}\tau}{\sum_\ell \Delta t_\ell \widetilde{P}_0(t_\ell)} \\ S &\approx (1-R)\lambda \quad \text{(credit triangle, low } \lambda) \end{aligned}$$

Time-varying hazard: replace $e^{-\lambda t}$ with $S(t) = \exp(-\int_0^t \lambda(u)\,\mathrm{d}u)$ and $\lambda e^{-\lambda\tau}$ with $\lambda(\tau)S(\tau)$ inside the default-leg integral.

15.11.4 T-forward measure (cross-reference to Chapter 5 for Girsanov derivation)

$$\left.\frac{\mathrm{d}\widetilde{\mathbb{Q}}_{T_0}}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_t} = \frac{P_t(T_0)/M_t}{P_0(T_0)/M_0} \qquad \mathrm{d}\widetilde{W}_t^0 = \mathrm{d}\widehat{W}_t + B_t(T_0)\sigma\,\mathrm{d}t$$

15.11.5 Bond SDE and forward bond

$$\begin{aligned} \frac{\mathrm{d}P_t(T)}{P_t(T)} &= r_t\,\mathrm{d}t - \sigma B_t(T)\,\mathrm{d}\widehat{W}_t \quad (\text{under } \mathbb{Q}) \\ X_t &= \frac{P_t(T_1)}{P_t(T_0)}, \qquad X_{T_0} = P_{T_0}(T_1) \\ \frac{\mathrm{d}X_t}{X_t} &= \Sigma_t\,\mathrm{d}\widetilde{W}_t^0, \qquad \Sigma_t = \sigma(B_t(T_0) - B_t(T_1)) \end{aligned}$$

15.11.6 European bond call (Vasicek-Black)

$$\mathfrak{g}_t = P_t(T_0)\{X_t\Phi(d_+) - K\Phi(d_-)\} \qquad d_\pm = \frac{\ln(X_t/K) \pm \frac{1}{2}\int_t^{T_0}\Sigma_s^2\,\mathrm{d}s}{\sqrt{\int_t^{T_0}\Sigma_s^2\,\mathrm{d}s}}$$

$$\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s = \frac{\sigma^2}{\kappa^2}(1 - e^{-\kappa(T_1 - T_0)})^2 \cdot \frac{1 - e^{-2\kappa(T_0 - t)}}{2\kappa}$$

15.11.7 Put-call parity for bond options

$$\mathfrak{g}_t^{\text{call}} - \mathfrak{g}_t^{\text{put}} = P_t(T_1) - K \cdot P_t(T_0)$$

15.11.8 Jamshidian decomposition (swaption →portfolio of bond options)

$$\text{swaption}_0 = \sum_{i=1}^{N} c_i \cdot \text{BondOption}(T_0, T_i, K_i), \qquad K_i = P_{T_0}(T_i)\big|_{r=r^*}$$

with r^* solving $\sum_i c_i \exp(A_{T_0}(T_i) - B_{T_0}(T_i)r^*) = K$.

15.11.9 Greeks for the bond call

$$\Delta = P_t(T_0)\Phi(d_+), \qquad \Gamma = \frac{P_t(T_0)\phi(d_+)}{X_t\sqrt{\int_t^{T_0} \Sigma_s^2 \,\mathrm{d}s}}$$

$$\text{Vega} = \frac{\partial \mathfrak{g}_t}{\partial \sigma} = P_t(T_0)X_t\phi(d_+)\sqrt{\int_t^{T_0} \Sigma_s^2 \,\mathrm{d}s/\sigma}$$

15.11.10 Multi-factor and HJM

$$\begin{aligned}
\text{G2++:}\quad & r_t = x_t + y_t + \phi(t), \qquad \mathrm{d}x_t = -ax_t\,\mathrm{d}t + \sigma_1\,\mathrm{d}W_t^x \\
& \mathrm{d}y_t = -by_t\,\mathrm{d}t + \sigma_2\,\mathrm{d}W_t^y, \qquad \langle W^x, W^y\rangle = \rho\, t \\
\text{HJM:}\quad & \mathrm{d}f_t(T) = \alpha(t,T)\,\mathrm{d}t + \sigma(t,T)\,\mathrm{d}W_t \\
& \alpha(t,T) = \sigma(t,T)\int_t^T \sigma(t,s)\,\mathrm{d}s \quad \text{(no-arb drift)} \\
\text{LMM:}\quad & \mathrm{d}L_t(T_{i-1},T_i) = L_t(T_{i-1},T_i)\sigma_i(t)\,\mathrm{d}W_t^{(i)} \\
\text{SABR:}\quad & \mathrm{d}F_t = \alpha_t F_t^{\beta}\,\mathrm{d}W_t, \qquad \mathrm{d}\alpha_t = \nu\alpha_t\,\mathrm{d}Z_t
\end{aligned}$$

15.11.11 Numerical methods — choosing between PDE and MC

Regime	Method
1-factor vanilla	PDE (Crank-Nicolson)
2-3 factor vanilla/mild exotic	PDE or MC
High-dimensional (LMM, HJM)	Monte Carlo
American/Bermudan low-dim	PDE with min-test
American high-dim	Longstaff-Schwartz MC
XVA on large book	Monte Carlo (GPU)
Path-dependent exotic	Monte Carlo

16 Chapter 14 — Caps, Floors, and Swaptions

16.1 14.0 The rates-side capstone

This chapter is the final application of the numeraire-change machinery built earlier in the guide. In Chapter 5 we constructed the Radon-Nikodym / Girsanov apparatus in full generality: every strictly positive tradable defines a measure, forward measures in particular are generated by zero-coupon bonds, and the annuity measure arises as a portfolio-of-bonds numeraire. In Chapter 12 we built the short-rate models — Vasicek, Ho-Lee, Hull-White, and the general affine framework — that give us closed-form bond prices $P_t(T) = e^{A_t(T)-B_t(T)\,r_t}$ and explicit bond volatilities $B_t(T)\,\sigma$. In Chapter 13 we priced interest-rate swaps, CDS, callable bonds, and European bond options, using the T-forward measure as a cross-reference back to Chapter 5.

Caps, floors, and swaptions are the product set that ties all of this together. A cap decomposes exactly, with no cross terms, into a strip of single-date caplets, each priced as a Black-76 call under its own T-forward measure $\mathbb{Q}^{T_i}$. A swaption is irreducibly multi-date, but under the annuity measure $\mathbb{Q}^X$ the par swap rate S_t becomes a single scalar martingale and the whole n-dimensional bond-price problem collapses into a single Black call on S. Both moves are instances of the same general principle — pick a numeraire under which the underlying becomes a martingale and the payoff factors cleanly — and both are immediate applications of the Girsanov theorem from Chapter 5. We do not re-derive the T-forward measure or the change-of-Brownian-motion formula; we cite them. What is new in this chapter is how they get used.

16.2 14.1 Caps, Caplets, and the Interest-Rate Swap

A cap is an interest-rate option made up of caplets. Caps/floors are close cousins of swaptions (swap options). The easiest way to see where a cap comes from is to start with its underlying — a plain vanilla interest-rate swap, as priced in Chapter 13 — and ask what happens when you put an option on the floating leg.

Consider a standard interest-rate swap (IRS) between counterparties A and B on notional N, with fixed leg F and floating leg ℓ:

$$A \;\underset{\ell}{\overset{F}{\rightleftharpoons}}\; B \qquad \text{(IRS)} \tag{14.1}$$

In words, A pays fixed and receives floating; B pays floating and receives fixed. The notional N never actually changes hands — it is simply a scale factor that multiplies the rate spread on each payment date. Both legs are quoted as annualized rates, but each period's payment applies only for the year-fraction Δt_k between consecutive fixing/payment dates.

On the tenor grid $t_0 < t_1 < t_2 < \cdots < t_n$:

- Fixed leg pays $F\,\Delta t_k$ at each t_k, $k = 1, \ldots, n$.
- Floating leg pays $L_{t_{k-1}}(t_k)\,\Delta t_k$ at t_k, where the rate is set at t_{k-1} and paid at t_k.

The timing convention — fix at t_{k-1}, pay at t_k — is not a quirk but the defining feature of the instrument. It is the same thing a bank does when it quotes you a three-month floating-rate loan: on the reset date, it looks up the prevailing rate, locks it in for the coming quarter, and hands you the bill at the end of the quarter. In options-pricing language this has a subtle consequence: the payoff is known one period before it is paid, so we need to discount it from the payment date t_k even though randomness is resolved at the fixing date t_{k-1}. That small dislocation is exactly what the t_k-forward measure is built to handle.

Here $L_{t_{k-1}}(t_k) \equiv L(t_{k-1}, t_{k-1}, t_k)$ is the simple (LIBOR) rate for the period $[t_{k-1}, t_k]$ fixed at t_{k-1}. It is defined through the zero-coupon bond:

$$P_{t_{k-1}}(t_k) \;=\; \left(1 + \Delta t_k\, L_{t_{k-1}}(t_k)\right)^{-1} \quad\Longrightarrow\quad L_{t_{k-1}}(t_k) \;=\; \frac{1}{\Delta t_k}\left[\frac{1}{P_{t_{k-1}}(t_k)} - 1\right] \tag{14.2}$$

so that investing $P_{t_{k-1}}(t_k)$ at t_{k-1} yields \$1 at t_k. This is the single most important definition in the chapter. LIBOR (and every simple forward rate) is not a primitive object floating in the air — it is a *derived quantity*, built out of zero-coupon bond prices by the simple-accrual formula above. The market does not pick a number and call it the rate; it observes bond prices and implies a rate. That ordering matters. When we later want to model LIBOR as log-normal under the t_k-forward measure, we are really making a modelling choice about the *ratio* $P_t(t_{k-1})/P_t(t_k)$, because that is what the rate is. Every martingale statement, every change-of-measure drift, and every Black-76 formula in this chapter ultimately rests on the fact that L is a ratio of tradables — the generic sufficient condition for a $\mathbb{Q}^N$-martingale under a numeraire N, as established in Chapter 5.

Intuitively, one dollar at t_{k-1} invested at the simple rate L for the fraction Δt_k grows to $1+\Delta t_k L$ at t_k. Equivalently, the bond $P_{t_{k-1}}(t_k)$ is what you must pay at t_{k-1} to receive exactly one dollar at t_k, and by no-arbitrage its reciprocal equals the growth factor $1+\Delta t_k L$. Any deviation and you can arbitrage a lender or borrower. So simple-rate accrual and the zero-coupon bond are two views of the same object.

16.2.1 Net cash flow to A and the cap payoff

A's net flow at t_k is $(L_{t_{k-1}}(t_k) - F)\Delta t_k$ — linear in the realized rate ℓ. If the realized LIBOR comes in above the fixed leg F, A (who is paying fixed) makes money on that period; if LIBOR comes in below F, A loses on the period. The swap is symmetric and linear — there is no optionality.

A cap caps the floating cost at strike K: instead of paying F, A enters a contract C that delivers the payoff

$$\Delta t_k \left(L_{t_{k-1}}(t_k) - K\right)_+ \qquad \text{at each } t_k,\ k = 1, \dots, n. \tag{14.3}$$

Each of these n period payoffs is called a caplet. The function $(\cdot)_+$ makes it clear that A is only ever compensated when LIBOR exceeds K — the downside is truncated at zero. Economically, the caplet is A's insurance against a single bad fixing; the cap is the bundled insurance across all future fixings of the underlying swap. Graphically the net flow is hockey-stick shaped in ℓ with kink at K:

Caplet payoff vs reset rate *Caplet payoff vs reset rate: hockey-stick in ℓ — zero below the strike, slope Δt_k above. Convex in $\ell \Rightarrow$ the cap is long vega in the forward-rate vol.*

So the cap is a strip of caplets, each one a call option on the forward LIBOR rate fixed at t_{k-1} and paid at t_k. The hockey-stick payoff above also reveals why a cap is convex in the rate: every caplet is, and the sum of convex functions is convex. Convexity in the floating rate translates directly into positive vega — a cap always benefits from a higher realized volatility of forward rates, holding everything else fixed. This is the structural reason cap markets trade a vol surface in the first place.

A brief note on naming. A "1y1y" caplet resets in one year and pays at approximately one-year-and-one-tenor. A "5y cap" is a strip of caplets running from the first reset out to five years. Swaption quotes use a two-dimensional grid, expiry $\times$ tenor, e.g. "2y5y" is a 2-year option to enter a 5-year swap. Once you internalize the tenor grid, all this nomenclature becomes self-explanatory.

16.3 14.2 Caplet Price under the Risk-Neutral Measure

Our job is to price a caplet today, at time t. There are two natural candidates for the numeraire we use to discount the random future payoff: the money-market account B_t, which is universal and always available, or the zero-coupon bond $P_t(t_k)$ maturing on the payment date, which is more bespoke but (as we will see) much cleaner. Let us try the money-market account first, and see why it fails.

Let $g_t^{(k)}$ denote the price at time t of a caplet maturing at t_k (payment date). Under the risk-neutral measure $\mathbb{Q}$ with bank-account numeraire $B_t = \exp\int_0^t r_s \, ds$:

$$\frac{g_t^{(k)}}{B_t} = \mathbb{E}_t^{\mathbb{Q}}\left[\frac{\Delta t_k \left(L_{t_{k-1}}(t_k) - K\right)_+}{B_{t_k}}\right]. \tag{14.4}$$

This is awkward because B_{t_k} is random and correlated with $L_{t_{k-1}}(t_k)$. Remedy: use the t_k-bond as numeraire (the T-forward measure from Chapter 5).

Intuition. B_{t_k} depends on the entire short-rate path from t to t_k, and high rates imply both a high LIBOR fixing *and* heavy discounting — these two effects do not separate inside the expectation. Switching to $P_t(t_k)$ as numeraire effectively "pre-pays" the discount at time t, so the only random thing left inside the expectation is the payoff itself.

16.4 14.3 Change of Numeraire — the T_i-Forward Measure

Under the t_k-forward measure $\mathbb{Q}^{t_k}$, whose numeraire is $P_t(t_k) = P(t, t_k)$ (see Chapter 5 for the general density-process construction and Girsanov drift shift):

$$\frac{g_t^{(k)}}{P_t(t_k)} \;=\; \mathbb{E}_t^{\mathbb{Q}^{t_k}}\left[\frac{\left(L_{t_{k-1}}(t_k) - K\right)_+}{P_{t_k}(t_k)}\right]\Delta t_k \;=\; \mathbb{E}_t^{\mathbb{Q}^{t_k}}\left[\left(L_{t_{k-1}}(t_k) - K\right)_+\right]\Delta t_k, \tag{14.5}$$

since $P_{t_k}(t_k) = 1$. Read (14.5) carefully. The discount factor $1/B_{t_k}$ that plagued (14.4) has been replaced by $1/P_{t_k}(t_k) = 1$, because the t_k-bond is worth exactly one dollar on its own maturity. All of the discounting has been absorbed into the $P_t(t_k)$ that sits outside the expectation. What is left inside the expectation is just the clean payoff. This is the payoff of a textbook call option on $L_{t_{k-1}}(t_k)$, evaluated under the measure $\mathbb{Q}^{t_k}$, with no discount to worry about.

Plugging in (14.2), $L_{t_{k-1}}(t_k) = \frac{1}{\Delta t_k}\left[\frac{1}{P_{t_{k-1}}(t_k)} - 1\right]$.

Define the likelihood ratio (key ratio that becomes a martingale under $\mathbb{Q}^{t_k}$):

$$X_t \;=\; \frac{P_t(t_{k-1})}{P_t(t_k)}, \qquad X_{t_{k-1}} = \frac{1}{P_{t_{k-1}}(t_k)}. \tag{14.6}$$

X_t is a ratio of two tradable zero-coupon bonds, one of which is the numeraire — the generic template for a $\mathbb{Q}^{t_k}$-martingale (Chapter 5).

16.4.1 Vasicek illustration

In the Vasicek model bond prices are affine (Chapter 12):

$$P_t(T) \;=\; e^{A_t(T) - B_t(T)\, r_t}, \qquad \frac{\mathrm{d}P_t(T)}{P_t(T)} \;=\; r_t\,\mathrm{d}t \;-\; B_t(T)\,\sigma\,\mathrm{d}\widetilde{W}_t, \tag{14.7}$$

with drift r_t and volatility $B_t(T)\sigma$ — larger for longer-dated bonds.

Taking $\mathrm{d}\ln$ of X_t:

$$\frac{\mathrm{d}X_t}{X_t} \;=\; (\text{vol } t_{k-1} - \text{vol } t_k) \;=\; \left(-B_t(t_{k-1}) + B_t(t_k)\right)\sigma\,\mathrm{d}\widetilde{W}_t^{(k)}, \tag{14.8}$$

where $\widetilde{W}_t^{(k)}$ is a Brownian motion under $\mathbb{Q}^{t_k}$. The drift has cancelled — X_t is a $\mathbb{Q}^{t_k}$-martingale, consistent with being a ratio of a tradable ($P_t(t_{k-1})$) and the numeraire ($P_t(t_k)$). This cancellation is not an accident of the Vasicek model; it is a general fact about changes of numeraire (Chapter 5), of which Vasicek is just a concrete instance.

Write the bond-vol differential as

$$\Sigma_t \;\equiv\; -B_t(t_{k-1}) + B_t(t_k), \qquad \mathrm{d}\widetilde{W}_t^{(k)} = B_t(t_k)\,\sigma\,\mathrm{d}t + \mathrm{d}\widetilde{W}_t. \tag{14.9}$$

The second equation is the key change-of-measure statement: when you switch the numeraire from the bank account B_t to the t_k-bond $P_t(t_k)$, the Brownian motion that drives everything gets a drift $B_t(t_k)\sigma$. This drift is exactly the bond's volatility, which is exactly the amount by which the new numeraire differs from the old. That is Girsanov talking (Chapter 5); the story is the same whatever the driving model, as long as the numeraire is a strictly positive tradable.

Then

$$g_t^{(k)} \;=\; P_t(t_k)\,\mathbb{E}_t^{\mathbb{Q}^{t_k}}\Big[\,(X_{t_{k-1}} - \mathcal{K})_+\,\Big]\,\Delta t_k, \qquad \mathcal{K} \equiv 1 + \Delta t_k\,K. \tag{14.10}$$

Notice the strike transformation. In terms of LIBOR the strike was K; in terms of $X = 1/P_{t_{k-1}}(t_k)$ it becomes $\mathcal{K} = 1 + \Delta t_k K$. This is just a re-expression of the same kink in the payoff function; the kink sits at whatever value of the underlying makes the pre-$(\cdot)_+$ expression vanish. Because X and L are affinely related, an option on one is exactly an option on the other, with a re-scaled strike and a re-scaled notional. No information lost.

Because X_t is a log-normal $\mathbb{Q}^{t_k}$-martingale (in the Gaussian Vasicek case), this is exactly a Black-Scholes expectation:

$$\boxed{g_t^{(k)} \;=\; P_t(t_k)\,\Big[\,X_t\,\Phi(d_+) - \mathcal{K}\,\Phi(d_-)\,\Big]} \tag{14.11}$$

$$d_\pm \;=\; \frac{\ln(X_t/\mathcal{K}) \;\pm\; \frac{1}{2}\int_t^{t_{k-1}} \Sigma_s^2\,\mathrm{d}s}{\left(\int_t^{t_{k-1}} \Sigma_s^2\,\mathrm{d}s\right)^{1/2}}. \tag{14.12}$$

The term $\int_t^{t_{k-1}} \Sigma_s^2\,\mathrm{d}s$ is the cumulative quadratic variation of $\ln X$ up to the fixing date. In a stationary Black-76 framing it will simply become $\sigma^2(t_{k-1} - t)$ — a single vol times time to fix — but in a more general model it can be time-dependent. Either way, the structure is Black-Scholes: log-moneyness divided by a standard deviation.

16.4.2 Cap as a sum of caplets

$$\mathrm{Cap}_t \;=\; \sum_{k=1}^{n} g_t^{(k)}. \tag{14.13}$$

This single equation is the whole point of the change of numeraire. We got here because the cap payoff is a *sum* of single-date payoffs; each single-date payoff depends on a fixing at a single date; and each fixing can be priced under its *own* forward measure. There is no need for a joint model of the whole forward-rate curve — you only need one marginal distribution per fixing date. That is a huge reduction in modelling burden, and it is the structural reason caps are vastly easier to price than path-dependent products.

16.4.3 Case study (a): UK LDI / mini-budget Sept 2022 — pension swaption book forced unwind

On 23 September 2022 the UK Chancellor Kwasi Kwarteng delivered a "mini-budget" of £45B in unfunded tax cuts. The 30-year gilt yield jumped from ≈ 3.7% to ≈ 5.1% over the next five trading sessions — a move that LDI (Liability-Driven Investment) hedge books at UK defined-benefit pension schemes had collectively calibrated as a 7σ-plus tail. The Bank of England intervened on 28 September with a £65B emergency gilt-buying programme to break the resulting forced-seller doom-loop.

Context. UK defined-benefit pension funds carry long-dated nominal liabilities — promises to pay specific amounts to retirees over the next 30–50 years — that have effective duration of 20–25 years. The funded portion is typically invested in growth assets (equities, credit, alternatives) with much shorter duration; the resulting asset-liability duration mismatch is the dominant source of funding-ratio volatility for the scheme. LDI is the post-2000s solution: layer a leveraged interest-rate hedge on top of the existing portfolio using long-dated receiver IRS plus receive-fixed swaptions to take the asset-side duration up to match the liability side. Because the IRS overlay is funded only by initial-margin posting against trustees' collateral pool (typically gilts and cash), schemes can match £1B of liability duration with £100M–£250M of collateral. The market grew enormously after the Pensions Act 2004 made funding-ratio reporting mandatory; by mid-2022 UK LDI mandates aggregated to ∼ £1.6T of notional swap exposure.

The mini-budget shock was not a typical rates move. The 30y gilt yield's 140 bp jump in five sessions implied a one-week realised vol of ∼ 28% on 30y rates, against an implied vol of ∼ 90 bp/year (≈ 12.5% on rate) used in standard LDI risk frameworks — that is approximately a 7-standard-deviation move on the calibrated normal distribution. The mark-to-market loss on the receive-fixed legs was enormous: with portfolio dv01 across the LDI universe estimated at £200M–£300M per basis point, the 140 bp move generated £30B+ of MTM losses, against collateral pools of $\sim$ 5–10% of notional. Schemes had to post variation margin within $T+1$, which for many meant selling gilts to raise cash. The forced gilt selling pushed yields even higher, triggered more margin calls, and produced a self-reinforcing doom-loop that the BoE's intervention was specifically designed to halt — buying long-dated gilts unconditionally for 13 trading days to put a ceiling on yields and let LDI managers raise collateral in an orderly fashion.

Reading it through the chapter's math. The receive-fixed swaption book sits on the right side of (14.27): swaption value $X_t(S_t\Phi(d_+) - F\Phi(d_-))$ for a payer-equivalent re-expression. The Greeks: dv01 on a 30y receive-fixed IRS is approximately the annuity X_t, which for a 30-year UK swap at 4% fixed runs to ≈ 17.5 — meaning a 1 bp parallel shift moves the IRS PV by 0.175% of notional, or £1.75M per £1B notional. Across the LDI universe of £1.6T, dv01 aggregates to £280M/bp — and a 140 bp move thus equals £39B of MTM loss in aggregate, against industry-wide collateral pools of $\sim$ £80–£150B. Per scheme, the IR delta as a fraction of capital ran 10%–20% per 100 bp move — a Black-76 dv01 calculation that risk systems had been computing correctly for years. The vega contribution from receive-fixed swaptions added another 5–8% of capital per vol point, and the 30-year normal swaption vol roughly tripled (from $\sim$ 90 bp/y to $\sim$ 280 bp/y at peak) over the week. None of this was novel mathematics; what was novel was the *speed* at which the move unfolded.

Lesson. The LDI failure mode was *liquidity*, not pricing: the math of (14.27) said correctly how much collateral needed to be posted; the math did not say that gilts could not be sold to raise the collateral without crashing the gilt market itself in turn. In the abstract, the strategy works — long-dated rate hedges genuinely match long-dated liability duration, and over a multi-decade horizon the leverage cost is well-compensated by the liability-matching benefit. The problem is the path: a sudden fiscal shock transmits into a margin call into a fire-sale in 48 hours, and the dv01-based stress framework that LDI managers used (typically a 200 bp instantaneous shift with $T+5$ remediation) was not calibrated to a $T+1$ collateral cycle in a market without unconditional buyers. Sovereign rate vol breaking a hedge book is a recurring theme — UK LDI 2022, US Treasury basis trade 2020, Mexican peso 1994, Italian BTP 2018, and the same pattern shows up in 1998 Long-Term Capital. The deeper lesson for the new hire is that a hedge book's leverage limit should be set by *liquidity-adjusted* collateral availability under stress, not by mark-to-market 99% VaR on a 5-year history, and that central-bank backstop optionality is now an implicit hedge in every long-duration rates book — when the BoE intervened on 28 September, 30y vols collapsed 30% in a single session, vindicating the underlying Black machinery once the regime stabilised but only after £1.6T of liability-hedge exposure had been threatened by a five-day fiscal shock.

16.5 14.4 Black Implied Vols — the Market Convention

The Vasicek derivation gave us a Black-style formula, but the market standard is to skip the short-rate model and assume directly that each forward LIBOR is log-normal under its own t_k-forward measure (GBM for forward rates). This is the LIBOR market model (LMM / BGM) seed.

Let $L_{t_{k-1}}(t_k) \stackrel{d}{=} L_t(t_{k-1}, t_k) \cdot Y_{t_{k-1}}$, where $L_t(t_{k-1}, t_k)$ is today's forward rate for the $[t_{k-1}, t_k]$ period:

$$L_t(t_{k-1}, t_k) \;=\; \frac{1}{\Delta t_k}\left[\frac{P_t(t_{k-1})}{P_t(t_k)} - 1\right]. \tag{14.14}$$

Using $P_t(t_{k-1}) = P_t(t_k)(1 + \Delta t_k L_t(t_{k-1}, t_k))$, $L_t(t_{k-1}, t_k)$ is a ratio of tradables to the numeraire $P_t(t_k)$, hence a $\mathbb{Q}^{t_k}$-martingale.

Forward-rate strip under three curve shapes *Normal (upward-sloping), flat, and inverted zero-curves imply very different forward-LIBOR strips* $\{L_0(t_{k-1}, t_k)\}_k$.

Postulate driftless GBM under that measure:

$$\frac{\mathrm{d}L_t^{(k)}}{L_t^{(k)}} \;=\; \sigma\,\mathrm{d}W_t^{t_k}, \qquad L_t^{(k)} \equiv L_t(t_{k-1}, t_k). \tag{14.15}$$

"$\mathbb{Q}$-Brownian motion" here means under $\mathbb{Q}^{t_k}$. This is the direct postulate: forward LIBOR is a log-normal martingale under its own forward measure. The only free parameter is the volatility σ; by assumption the drift is zero (which is forced, because $L^{(k)}$ is a martingale under $\mathbb{Q}^{t_k}$). There is nothing to solve — given σ, the distribution of $L^{(k)}_{t_{k-1}}$ is pinned down.

Starting at t:

$$L^{(k)}_{t_{k-1}} \;=\; L_t(t_{k-1}, t_k)\, \exp\Big(-\tfrac{1}{2}\sigma^2(t_{k-1}-t) \,+\, \sigma\big(W^{t_k}_{t_{k-1}} - W^{t_k}_t\big)\Big). \tag{14.16}$$

This is the textbook GBM solution: log-mean $-\frac{1}{2}\sigma^2(t_{k-1}-t)$, log-variance $\sigma^2(t_{k-1}-t)$, multiplicative shift by the forward rate today. The log-normality is what keeps forward LIBOR strictly positive — a feature that made perfect sense in the regime before the 2008 financial crisis, and that causes trouble afterwards (see §14.4b below).

16.5.1 Black's formula for a caplet

Plug (14.16) into (14.5). Since $L^{(k)}_{t_{k-1}}$ is log-normal with mean $L^{(k)}_t$ under $\mathbb{Q}^{t_k}$:

$$g^{(k)}_t \;=\; P_t(t_k)\, \mathbb{E}^{\mathbb{Q}^{t_k}}\Big[\big(L^{(k)}_{t_{k-1}} - K\big)_+\Big]\, \Delta t_k,$$

$$\boxed{g^{B(k)}_t \;=\; P_t(t_k) \Big\{ L_t(t_{k-1}, t_k)\, \Phi(d_+) \;-\; K\, \Phi(d_-) \Big\}\, \Delta t_k} \tag{14.17}$$

$$d_\pm \;=\; \frac{\ln\!\big(L_t(t_{k-1}, t_k)/K\big) \;\pm\; \frac{1}{2}\sigma^2(t_{k-1}-t)}{\sigma\,(t_{k-1}-t)^{1/2}}. \tag{14.18}$$

This is the Black-76 caplet formula. Each caplet has its own vol $\sigma^{B\ell}_k$ (the "Black caplet vol"). Structurally, (14.17) is Black-Scholes on spot L_t, strike K, vol σ, expiry $t_{k-1}-t$, with the discount $P_t(t_k)$ taking the payment date — the forward-measure dislocation between fix at t_{k-1} and pay at t_k.

Cross-reference. The same caplet can be re-expressed as a put on a t_k-zero-coupon bond with strike $K_{\text{bond}} = 1/(1+K\Delta t_k)$; see Chapter 13 §13.6 for the Vasicek-Black bond-put formula that provides an equivalent (and useful) consistency check.

Cap price vs strike (Black model) *Cap price vs strike (Black model)*

The cap price is monotonically decreasing in the strike K (higher strike, less in-the-money, less intrinsic, less option value). For very low strikes the cap behaves like a forward swap (all caplets deep in the money, optionality worth almost nothing on top of intrinsic). For very high strikes the cap behaves like a lottery ticket (all caplets deep out of the money, value almost entirely time value). The curvature between these extremes is driven by the Black caplet vol term structure; that is what makes the cap surface genuinely interesting.

16.5.2 Cap in terms of Black vols

$$\sum_{k=1}^{n} g^{B(k)}_t \;=\; \sum_{k=1}^{n} g^{B(k)}_t(\sigma^{B\ell}_k), \tag{14.19}$$

with $\sigma^{B\ell}_k$ the k-th caplet Black implied vol. A flat cap vol $\sigma^{C\ell}_n$ is the single σ such that $\sum_k g^B(\sigma) = \mathrm{Cap}^{\mathrm{mkt}}_t$ — conceptually useful but coarser than the per-caplet term structure.

One "strips" caplet vols recursively from quoted cap prices: each cap tenor introduces one new unknown vol, solved one at a time so every market-quoted cap reprices exactly. The market quotes at several strikes; stripping is done at each strike, giving a 2D caplet-vol surface in (expiry, strike), through which traders typically fit a SABR smile per expiry.

Caplet-vol term structure vs flat cap vol *A stylized humped caplet-vol curve $\sigma^{B\ell}_k$ alongside the flat cap vol $\bar{\sigma}^{C\ell}$ that reprices the whole 5y cap.*

Why cap = sum of caplets exactly (no cross-terms). Each caplet pays $\Delta t_k(L_{t_{k-1}}(t_k) - K)_+$ on a *single* fixed date t_k. The cap payoff is the pointwise sum and expectation is linear, so $\mathrm{Cap}_t = \sum_k g_t^{(k)}$ — no covariance terms. Each caplet prices under its *own* forward measure with its *own* Black vol. A swaption, by contrast, takes a max on the *already-summed* swap value, forcing a different numeraire (the annuity) and bringing in forward-rate correlations.

16.6 14.4b Bachelier vs Black in a Negative-Rate World

The Black-76 framework above rests on one assumption that is uncontroversial in most of modern financial history: that forward rates are strictly positive. Log-normality forces this, since $L_{t_{k-1}}^{(k)} = L_t^{(k)}\exp(\text{something})$ and the exponent is always strictly positive. For decades this was a reasonable view of the world — a negative short rate was almost unthinkable, because banks could always hold physical cash instead of lending at a loss, and so the zero lower bound seemed binding by arbitrage.

The post-2008 experience blew that up. The European Central Bank, the Swiss National Bank, the Bank of Japan, and others spent multiple years with deposit rates set explicitly below zero, and market forward rates on EURIBOR and JPY curves traded visibly negative across a wide range of tenors. Suddenly Black-76 could not even *represent* a market quote — you cannot compute $\ln(L/K)$ if L is negative, and the model assigns zero probability to outcomes that were happening in real time.

Market practice migrated in two directions. The first was the *shifted-Black* model: write $L \to L + \alpha$ for some positive shift α, assume that $L + \alpha$ is log-normal under $\mathbb{Q}^{t_k}$, and price Black-76 with the shifted variables. This is a cheap fix — it preserves the closed-form and the market intuition of a Black vol, while permitting negative realizations of L itself as long as they stay above $-\alpha$.

The second, more structural fix was the *Bachelier* or *normal* model: assume that the forward rate follows arithmetic Brownian motion rather than geometric, so

$$\mathrm{d}L_t^{(k)} \;=\; \sigma_N \,\mathrm{d}W_t^{t_k},$$

with σ_N the *normal* (basis-point) volatility. The caplet price under Bachelier is another closed-form expression involving Φ and the normal density, with a moneyness $(L_t - K)/(\sigma_N\sqrt{t_{k-1} - t})$. The normal model prices forward rates as a Gaussian distribution centered on the current forward, with no lower bound — perfectly happy to assign positive probability to negative rates, because that is exactly what happened in reality.

The two conventions are related by a rough rule of thumb: at the money, $\sigma_N \approx \sigma_B \cdot L$, where σ_B is the Black vol and L is the underlying forward rate. So a 25% Black vol on a 4% forward is roughly equivalent to a 100 bp normal vol. Away from ATM they diverge, and the choice of quoting convention matters. Modern rate-derivative desks now routinely switch between conventions, with normal vols dominating European markets (especially EUR and CHF) and shifted-Black or straight Black vols still common in USD markets where the zero bound has remained in force for most of history.

The same observations apply to swaptions — normal swaption vols are now the industry standard in much of Europe, and a senior quant cannot afford to be sloppy about which convention a given quote is in.

16.7 14.5 Swaption as Black on the Swap Rate

So far we have priced an option on a *single* fixing of a forward rate. A swaption is an option on an entire *swap* — a bundled set of fixings. Crucially, the max is taken at the swap level, not at the caplet level, so the cap's per-fixing decomposition no longer applies. The right tool is a portfolio numeraire — the annuity — under which the par swap rate becomes a single scalar martingale and the pricing collapses to a one-dimensional Black call.

A swaption (swap option) gives A the right, at t_0, to enter a payer swap with fixed leg F, reset/payment dates $t_1, \dots, t_n$. The payoff at exercise is

$$C_{t_0} \;=\; \left(V^{\ell}_{t_0} - V^{\text{fixed}}_{t_0}\right)_+, \qquad V^{\text{fixed}}_{t_0} \;=\; F\sum_{k=1}^{n} P_{t_0}(t_k)\,\Delta t_k, \qquad V^{\ell}_{t_0} \;=\; 1 - P_{t_0}(t_n). \tag{14.20}$$

(The floating-leg par identity $V^{\ell} = 1 - P_{t_0}(t_n)$ comes from Chapter 13.) Hence

$$C_{t_0} \;=\; \Big(1 - P_{t_0}(t_n) \;-\; F\sum_{k=1}^{n} P_{t_0}(t_k)\,\Delta t_k\Big)_+. \tag{14.21}$$

The max operates on a nonlinear combination of $n+1$ bond prices; under a multi-factor model the max cannot be decomposed caplet-style. The remedy is to choose a numeraire that matches the *structure* of the payoff — the annuity.

16.7.1 Annuity-measure swaption

Collect terms using the annuity numeraire:

$$X_t \;\equiv\; \sum_k \Delta t_k\, P_t(t_k) \quad \text{(annuity)}. \tag{14.22}$$

The annuity is a strictly positive portfolio of zero-coupon bonds — exactly the PV of \$1 received on every fixed-leg payment date — so it qualifies as a numeraire (Chapter 5). Change to $\mathbb{Q}^X$ with numeraire X_t:

$$\frac{g_t}{X_t} \;=\; \mathbb{E}^{\mathbb{Q}^X}_t\left[\left(\frac{1 - P_{t_0}(t_n)}{\sum_k \Delta t_k P_{t_0}(t_k)} \;-\; F\right)_+\right] \;=\; \mathbb{E}^{\mathbb{Q}^X}_t[(S_{t_0} - F)_+], \tag{14.23}$$

where the par swap rate is (recall Chapter 13's IRS section)

$$\boxed{S_t \;=\; \frac{1 - P_t(t_n)}{\sum_k \Delta t_k\, P_t(t_k)}} \tag{14.24}$$

— and S_t is a $\mathbb{Q}^X$-martingale (a ratio of the tradable $1 - P_t(t_n)$ to the numeraire X_t). Hence

$$g_t \;=\; X_t\,\mathbb{E}^{\mathbb{Q}^X}_t[(S_{t_0} - F)_+]. \tag{14.25}$$

All of the n-dimensional bond-price structure has collapsed into a scalar call on the single random variable S_{t_0}.

16.7.2 Log-normal swap-rate model (LSM)

Assume

$$\frac{\mathrm{d}S_t}{S_t} \;=\; \sigma\,\mathrm{d}\widetilde{W}^X_t \;\Longrightarrow\; \text{Black formula for swaption.} \tag{14.26}$$

Then (under GBM with mean S_t):

$$\boxed{g_t \;=\; X_t\,(\,S_t\,\Phi(d_+) \;-\; F\,\Phi(d_-)\,)} \tag{14.27}$$

$$d_\pm \;=\; \frac{\ln(S_t/F) \;\pm\; \frac{1}{2}\sigma^2(t_0 - t)}{\sigma\,(t_0 - t)^{1/2}}. \tag{14.28}$$

This is the Black swaption formula: a call on the swap rate S_t, with the annuity X_t as discount factor. Same structure as Black-Scholes on equity and as the Black caplet, with the right numeraire (X_t) replacing the bank account or the single t_k-bond.

Swaption price surface over expiry x tenor *The Black swaption price varies smoothly over the expiry × tenor grid — the standard market quoting convention.*

The swaption market trades a *two-dimensional* vol surface: expiry × tenor. A 2y5y swaption is the 2-year option to enter a 5-year swap. Consistency between the caplet surface (marginals) and the swaption surface (functionals of the joint) is a nontrivial constraint that full production models (LMM, SABR-LMM, stochastic-vol variants) try to fit simultaneously.

16.7.3 Historical alternative — Jamshidian on a one-factor affine model

Before the annuity measure became the industry standard, swaptions were priced by an exact decomposition that works in any one-factor Gaussian short-rate model (Vasicek, Hull-White): because $\sum_k c_k P_{t_0}(t_k)$ is monotone in r_{t_0}, the exercise region $\{r_{t_0} > r^*\}$ is a half-line, with r^* solving

$$\sum_{k=1}^{n} c_k\, P_{t_0}(t_k; r^*) \;=\; 1, \qquad c_k = F\,\Delta t_k\ (k < n),\ \ c_n = 1 + \Delta t_n F. \tag{14.29}$$

A joint indicator on all n bond prices reduces to a single scalar condition on r_{t_0}, and each bond-price term becomes a Gaussian tail probability under $\mathbb{Q}^{t_k}$. The full derivation is the European bond-option machinery of Chapter 13 §13.7.4 — we cite it rather than re-derive. The route is pedagogically clean and historically important, but the annuity-measure formula (14.27) is the production form: it gives a one-shot Black on the par swap rate, requires no calibrated short-rate model, and is the basis of every Bloomberg / market quote for swaption vol.

16.7.4 Case study (b): LIBOR → SOFR transition June 2023

USD LIBOR formally ceased publication on 30 *June* 2023 *for the remaining tenors (*1*M,* 3*M,* 6*M,* 12*M); the other major settings (*1*W,* 2*M) had ceased on* 31 *December* 2021*. At the cessation date, an estimated* \$200*T+ of derivatives outstanding referenced USD LIBOR — IRS, caps, floors, swaptions, FRAs, structured notes — and every single contract had to be either re-papered to a SOFR-based reference or fall back automatically under the ISDA* 2020 *IBOR Fallbacks Protocol. ISDA estimates approximately* 85% *of bilateral swap volume re-papered ahead of cessation; the remainder fell back automatically. The largest contract-rewrite event in derivatives history.*

Context. The LIBOR-replacement project began formally with the Financial Stability Board's 2014 recommendations after the 2012 rate-rigging scandals, but the operational mechanics of the swap-market migration crystallised only in 2017–2020. The replacement rate selected by the US Alternative Reference Rates Committee (ARRC) was the Secured Overnight Financing Rate (SOFR) — a Treasury-repo-collateralised overnight rate published by the New York Fed since April 2018. SOFR differs from LIBOR in two structural ways: it is overnight-only (no term structure intrinsically; term SOFR for select tenors is published as a forward-looking median of SOFR-futures), and it is essentially risk-free (collateralised by Treasuries, with a credit spread of zero versus LIBOR's $\sim$ 25 bp average bank-credit spread). The ISDA 2020 Fallbacks Protocol, opted-in by approximately 14,500 legal entities globally, defined the precise mechanic: at cessation date, any USD-LIBOR fixing in a covered contract is replaced by "SOFR compounded in arrears over the corresponding accrual period" plus a *fixed* spread adjustment.

The spread adjustment is the operationally critical detail: it is set at the historical median of the LIBOR-versus-compounded-SOFR basis observed over the five-year window prior to the announcement of cessation (March 2021 for USD), and once set, *frozen forever* for fallback purposes. The published values are: 11.448 bp for 1M USD LIBOR, 26.161 bp for 3M USD LIBOR, 42.826 bp for 6M USD LIBOR, and 71.513 bp for 12M USD LIBOR. So a legacy 5y IRS that paid 3M USD LIBOR + 50 bp now pays "3M compounded SOFR + 26.161 bp + 50 bp" — a frozen 26.16 bp uplift to the spread, applied in perpetuity. The pricing impact at cessation was largely zero by construction (the spread was set so that PV-neutrality held in expectation under the prior basis); the operational impact was enormous (every cap, floor, swaption, and structured note had to have its payoff function rewritten).

Reading it through the chapter's math. Under LIBOR, a caplet payoff is $\Delta_k(L_{t_{k-1}}(t_{k-1}, t_k) - K)^+$ with the LIBOR fixing L observed *in advance* at t_{k-1} and the payment made *at* t_k — exactly the (14.5) setup. The Black-76 formula $C = \Delta_k P_t(t_k)[L_t\Phi(d_+) - K\Phi(d_-)]$ falls out under log-normal forward-LIBOR dynamics because L_t is a $\mathbb{Q}^{t_k}$-martingale (it is a ratio of tradable bond differences to the $P(t_k)$-numeraire). Under SOFR, the fixing is computed as $S_{t_{k-1},t_k} = (1/\Delta_k)[\prod_i(1+\Delta_i s_i) - 1]$ where the product is over the daily SOFR fixings s_i within the accrual period — so the fixing is *backward-looking* and known only at t_k, not at t_{k-1}. Strictly speaking, this breaks the (14.5) setup: the payoff is $\Delta_k(S_{t_{k-1},t_k} - K)^+$ paid at t_k, but S is now an in-arrears compound, not a forward fixing. In practice, market dealers price the SOFR caplet in a Black-76-shaped formula by treating the *effective* fixing date as a dealer-convention adjustment to the option's expiry (typically taking the volatility integrand to run to a midpoint of the accrual period, with a small lookback adjustment for the in-arrears effect). The convexity correction between the in-advance fixing convention and the in-arrears realisation is an order $\sigma^2\Delta_k$ term — typically 0.1–0.5 bp on a quarterly accrual at 1% vol. For a swaption on the SOFR-OIS swap, the par-swap-rate S_t is again a $\mathbb{Q}^{\text{ann}}$-martingale (the annuity numeraire structure is unchanged); the Black-76 formula (14.27) applies to compounded-SOFR par swap rates exactly as it did to LIBOR-IRS par swap rates, because nothing in the derivation depended on the compounding convention of the floating leg.

Lesson. The cessation event was the largest stress test of derivatives infrastructure in history: every cap, floor, swaption, IRS, FRA, and structured note had to be re-papered or fall back, and every risk system, settlement system, and accounting system had to be updated to handle the new index. That the global market migrated \$200T+ of notional across a single weekend (the 30 June 2023 cessation, with the post-cessation Monday opening on the new convention) without major operational failure is a quiet triumph of the multi-year ISDA-led preparation. The deeper lesson for the new hire is that the Black-76 pricing architecture survived intact — what changed was the index, the fixing convention (forward-looking → backward-looking), and the spread-adjustment mechanic. Every formula in this chapter ports to SOFR with $L_t \to$ compounded-SOFR-forward and a small in-arrears convexity adjustment; nothing about the change-of-numeraire derivation, the annuity-measure pricing, or the per-caplet implied-vol convention required revision. Risk systems that had abstracted "LIBOR" to "the simple forward rate under its own forward measure" survived the transition with minor wiring changes; risk systems that hardcoded "LIBOR" as a literal index name had to be substantially rewritten. Index-agnostic abstractions are not academic luxury — they are the difference between a six-month transition project and a three-year one.

16.8 14.7 Worked Example — Pricing a 1-into-1 Caplet

Concrete numbers are the best way to anchor the formulas. Let us price a single quarterly caplet using the Black-76 formula and check that the answer makes sense.

Suppose today is $t = 0$, with:

- $t_{k-1} = 1$ yr, $t_k = 1.25$ yr, $\Delta t_k = 0.25$ yr.
- Today's forward rate $L_0(1, 1.25) = 4.00\%$.
- Strike $K = 3.50\%$.
- Black caplet vol $\sigma = 25\%$.
- Discount $P_0(1.25) = 0.955$.

The caplet is forward-ITM with 1y to fix at 25% vol — meaningful time value.

Compute:

$$d_+ = \frac{\ln(0.04/0.035) + 0.5(0.25)^2(1)}{0.25\sqrt{1}} = \frac{0.1335 + 0.03125}{0.25} = 0.659,$$
$$d_- = d_+ - 0.25 = 0.409.$$

With $\Phi(0.659) \approx 0.745$, $\Phi(0.409) \approx 0.659$:

$$g_0 = 0.955 \times (0.04 \times 0.745 - 0.035 \times 0.659) \times 0.25$$
$$= 0.955 \times (0.02980 - 0.02307) \times 0.25 = 0.955 \times 0.006734 \times 0.25 \approx 0.001607$$

i.e. about 16.07 bp of notional — the caplet PV. Rough intrinsic: $0.25 \times (0.04 - 0.035) \times 0.955 \approx 11.94$ bp; the Black price is ~4 bp above intrinsic — the time-value contribution.

ATM vs OTM caplet decomposition *For a 1y caplet, the Black price splits into intrinsic* $P_t(t_k)\Delta t_k(L_t - K)_+$ *and time value* $g_t^{B(k)}$ − *intrinsic. The time component peaks near ATM and decays symmetrically in* $\ln(L/K)$ — *the classic log-moneyness curvature that drives cap vega.*

16.9 14.8 Worked Example — Stripping Caplet Vols from a 2-Year Cap

A harder and more realistic task is to back out caplet vols from market cap quotes. Assume we see market prices for quarterly-settled caps at two tenors, and we want to produce a caplet-vol term structure consistent with both. The mechanics of the stripping procedure are easy once you see them laid out.

Suppose quarterly caps:

- 1-yr cap (4 caplets) trades at C_4^{mkt}.
- 2-yr cap (8 caplets) trades at C_8^{mkt}.

Given already-stripped $\{\sigma_1^{B\ell}, \dots, \sigma_4^{B\ell}\}$ from the 1-yr cap, solve

$$C_8^{\mathrm{mkt}} - \sum_{k=1}^{4} g^{B(k)}(\sigma_k^{B\ell}) \;=\; \sum_{k=5}^{8} g^{B(k)}(\sigma^\star)$$

for the common forward caplet vol $\sigma^\star$ over caplets 5–8 (1D root find). Repeating at longer tenors builds the full caplet-vol term structure. In practice caps are quoted at multiple strikes; stripping is done per strike, fit through with SABR smile to give a 2D arbitrage-free surface.

16.10 14.9 LIBOR to SOFR and Why It Matters

The entire Black-76 caplet framework was built in a world where the reference floating rate was LIBOR — the London Interbank Offered Rate. LIBOR had two features that made it a natural fit for this pricing technology: it was forward-looking (quoted at the start of an accrual period for that period's rate), and it was a credit-sensitive unsecured bank rate (which gave the curve a term premium and a nonzero credit spread over a truly riskless reference).

Following the rate-fixing scandals of the 2010s, LIBOR has been substantially withdrawn. USD LIBOR for most major tenors ceased publication; GBP LIBOR is gone; EUR is largely replaced by €STR; JPY by TONA; CHF by SARON. The replacements — SOFR in the US, SONIA in the UK, €STR in the euro area, TONA in Japan — are overnight risk-free rates. They are *backward-looking* rather than forward-looking: the rate for an accrual period is computed by compounding realized overnight fixings, so you only know the period's rate at the *end* of the period, not the beginning.

This timing inversion changes several things in our framework. First, the defining identity (14.2) no longer applies cleanly — the compounded SOFR rate is not a single bond-based accrual but a product of overnight factors. Second, the forward rate concept is now subtler: you can still define a forward SOFR rate, but the convexity adjustment between the in-arrears-compounded fixing and a hypothetical in-advance fixing is not zero, and careful dealers track it. Third, the caplet/floorlet conventions have been redefined to handle backward-looking fixings — SOFR caplets pay on the payment date based on a compounded in-arrears rate, which introduces a small Black-model bias that is typically accommodated by a small adjustment to the effective fix time.

The cap market has adapted: SOFR caps trade actively, with stripping analogous to LIBOR caps after a small fix-time adjustment for the in-arrears mechanics. The per-caplet Black-vol architecture survives intact; what changed was the index, not the pricing framework. Risk systems should abstract "LIBOR" to "simple forward rate under its own forward measure" — that abstraction is robust to the index choice.

For converted legacy LIBOR contracts, the standard fallback is "SOFR + fallback spread" with the spread set to the historical median LIBOR-SOFR basis over a five-year lookback window. Dealers hedging legacy books must track that basis risk.

16.11 14.10 Key Takeaways

1. Cap = strip of caplets. Each caplet is a European call on a forward LIBOR rate, fixed at t_{k-1} and paid at t_k. The "cap = sum of caplets" decomposition is exact (no cross-terms), because the max acts on each caplet's fixing separately.
2. Pricing numeraire: switching from B_t to $P_t(t_k)$ kills the correlation between discount and payoff. Under $\mathbb{Q}^{t_k}$ the forward LIBOR $L_t(t_{k-1}, t_k)$ is a martingale, because it is a ratio of the tradable $(P_t(t_{k-1}) - P_t(t_k))/\Delta t_k$ to the numeraire $P_t(t_k)$.
3. Change of measure (Chapter 5): $\mathrm{d}\widetilde{W}_t^{(k)} = B_t(t_k)\sigma\,\mathrm{d}t + \mathrm{d}\widetilde{W}_t$ in the Vasicek model of Chapter 12; more generally, drift compensation absorbs the bond-vol. This is just Girsanov applied to the $P_t(t_k)$-numeraire.
4. Black-76 caplet falls out when we postulate log-normal forward rates under $\mathbb{Q}^{t_k}$ — one implied vol per caplet, not one per cap. Quoting flat cap vols is a convenience; hedging is always in terms of the per-caplet term structure.
5. In negative-rate regimes, Black-76 fails outright because $\ln L$ is undefined. Shifted-Black and Bachelier models are the two practical fixes; normal (Bachelier) vols dominate European quoting conventions today.
6. Swaptions are priced analogously using the annuity numeraire $X_t = \sum_k \Delta t_k P_t(t_k)$, under which the par swap rate $S_t = (1 - P_t(t_n))/X_t$ is a martingale. Log-normal S_t gives the Black swaption formula. The Jamshidian route of Chapter 13 gives an alternative derivation in one-factor affine models.
7. Calibration: strip caplet vols from cap quotes in order of maturity; quote swaption vols on a (expiry, tenor) grid. Consistent full-curve models (LMM variants) calibrate to both simultaneously.
8. LIBOR is largely retired; SOFR (and other overnight-compounded risk-free rates) has taken over. The pricing architecture survives; the mechanical details of fixing and compounding change. Risk systems should abstract "LIBOR" to "simple forward rate under its own forward measure."

16.12 14.11 Reference Formulas

16.12.1 Forward rate / bond relationship

$$P_t(t_k) = P_t(t_{k-1})\big(1 + \Delta t_k\, L_t(t_{k-1}, t_k)\big)^{-1}, \qquad L_t(t_{k-1}, t_k) = \frac{1}{\Delta t_k}\left[\frac{P_t(t_{k-1})}{P_t(t_k)} - 1\right].$$

16.12.2 Caplet payoff (paid at t_k)

$$\text{payoff}_{t_k} = \Delta t_k \left(L_{t_{k-1}}(t_k) - K\right)_+.$$

16.12.3 Caplet price — Black-76

$$g_t^{(k)} = P_t(t_k)\,\Delta t_k \left[L_t(t_{k-1}, t_k)\,\Phi(d_+) - K\,\Phi(d_-)\right],$$
$$d_\pm = \frac{\ln\!\big(L_t/K\big) \pm \frac{1}{2}\sigma_k^2(t_{k-1} - t)}{\sigma_k\sqrt{t_{k-1} - t}}.$$

16.12.4 Cap

$$\mathrm{Cap}_t = \sum_{k=1}^{n} g_t^{(k)}.$$

16.12.5 Swap rate (par)

$$S_t = \frac{1 - P_t(t_n)}{\sum_{k=1}^{n} \Delta t_k\, P_t(t_k)}.$$

16.12.6 Swaption — Black

$$g_t = X_t\left[S_t\,\Phi(d_+) - F\,\Phi(d_-)\right], \qquad X_t = \sum_k \Delta t_k P_t(t_k),$$
$$d_\pm = \frac{\ln(S_t/F) \pm \frac{1}{2}\sigma^2(t_0 - t)}{\sigma\sqrt{t_0 - t}}.$$

16.12.7 Change-of-measure driver under $\mathbb{Q}^{t_k}$ (Vasicek, Chapter 12 bond dynamics + Chapter 5 Girsanov)

$$\mathrm{d}\widetilde{W}_t^{(k)} = B_t(t_k)\,\sigma\,\mathrm{d}t + \mathrm{d}\widetilde{W}_t, \qquad \frac{\mathrm{d}P_t(T)}{P_t(T)} = r_t\,\mathrm{d}t - B_t(T)\,\sigma\,\mathrm{d}\widetilde{W}_t.$$

16.12.8 Bachelier / normal caplet (for negative-rate regimes)

$$\mathrm{d}L_t^{(k)} = \sigma_N\,\mathrm{d}W_t^{t_k},$$

$$g_t^{N(k)} = P_t(t_k)\,\Delta t_k\left[(L_t - K)\Phi(d) + \sigma_N\sqrt{t_{k-1} - t}\,\varphi(d)\right], \qquad d = \frac{L_t - K}{\sigma_N\sqrt{t_{k-1} - t}}.$$

17 Chapter 15 — Risk Measures: VaR, CTE, and Hedged Tail Risk

This chapter develops the machinery quantitative risk managers use to summarize the loss distribution of a trading book: Value-at-Risk (VaR), Conditional Tail Expectation (CTE, also called Expected Shortfall), their estimation under historical, parametric, and Monte Carlo paradigms, and the way these measures behave when applied to option books that have been delta- or delta-gamma-hedged. The material is organized so that a reader can move from scalar summary statistics of P&L (§§15.1-9.4), into regulatory and axiomatic context (§§15.5-9.6), into backtesting and stress discipline (§§15.7-9.8), into attribution and liquidity overlays (§§15.9-9.11), and finally into derivative-specific risk arithmetic where Greeks drive the tails (§§15.12-9.14). Cross-references point to Chapter 6 for the underlying GBM dynamics and to Chapter 7 for the Greek definitions themselves.

17.1 15.1 Why Risk Measures Matter

Picture the risk-reward plane: reward on the vertical axis, "risk" on the horizontal axis. Portfolios scatter as points; dominated ones lie to the south-east of undominated ones, and an efficient frontier sweeps north-west. The entire discipline of quantitative portfolio construction hinges on the apparently innocent choice of what to put on the horizontal axis. Variance produces Markowitz-style portfolios indifferent to the sign of moves. Downside semi-variance penalises only losses. VaR cares only about a threshold. CTE cares about the size of losses beyond the threshold. These are not equivalent reparametrisations — they rank portfolios differently, and the portfolio that looks attractive under one axis can look unattractive under another.

Risk management exists because organisations cannot run on distributions. A board of directors wants a number; a regulator wants a number; a desk head wants a number to compare against a limit. A risk measure is a functional that reduces an infinite-dimensional loss distribution to a single scalar that can be acted on. The price of the reduction is information loss, and the *character* of that loss — which features of the distribution the measure emphasises and which it hides — is exactly why the choice of functional matters.

This chapter works exclusively with the *loss* random variable L, whose large positive values correspond to financially painful outcomes. Some trading desks report risk on a P&L convention where losses are negative; the underlying mathematics is identical up to sign, but the presentation can confuse the unwary. Throughout, F_L denotes the CDF of L under whatever probability measure we have chosen for the return generator — typically the historical measure for risk measurement, reserved from the risk-neutral measure used for pricing.

Scope and horizon. The apparatus in this chapter is built for *market risk* — the risk that the mark-to-market value of a traded book changes because of moves in observable market prices. It is not directly the machinery used for credit risk, operational risk, or liquidity risk (though all three borrow concepts from it). Market-risk VaR is traditionally computed at a 1-day horizon for desk-level reporting and a 10-day horizon for regulatory capital; longer horizons are used for economic capital. Confidence levels 95%, 99%, and 99.9% correspond to different use cases: 95% for daily desk reporting, 99% for regulatory VaR, 99.9% for economic capital. The higher the confidence level, the rarer the events being summarised, and the more violent the estimation error.

Risk measurement versus risk management. Computing VaR or CTE is the easy part — well, not easy, but conceptually well-defined once a loss distribution is picked. Doing something useful about the result is harder. A \$5M VaR number leaves open whether the risk is concentrated in a single unwind-able exposure or scattered across thousands of small ones; whether it is driven by equities, rates, credit, or FX; whether it moves a little or a lot if volatilities rise 20%. These are the questions a working risk manager wrestles with, and they require decomposing the aggregate into interpretable pieces — marginal, component, incremental VaR — that we return to in §15.9. Every VaR number is also the output of a statistical model, and every model is wrong in some way. Professional practice therefore triangulates across methodologies (historical, parametric, Monte Carlo), lookback windows, distributional assumptions, and stress scenarios, treating disagreements among them as diagnostic rather than embarrassing.

17.2 15.2 Value-at-Risk: Historical, Parametric, Monte Carlo, Cornish-Fisher

For a loss random variable L with distribution F_L, the Value-at-Risk at level $\alpha \in (0,1)$ is

$$\mathrm{VaR}_\alpha(L) \;=\; \inf\{\ell : \mathbb{P}(L \le \ell) \ge 1-\alpha\}. \tag{15.1}$$

VaR is the $(1-\alpha)$-quantile of L. It answers *"what loss will I not exceed with* $(1-\alpha)$ *confidence?"* and ignores the size of losses beyond the cutoff. A portfolio whose 5%-tail pays $-\$10\,\mathrm{M}$ with probability 0.01 and $-\$1\,\mathrm{M}$ with probability 0.04 has the same 95%-VaR as a portfolio that pays a flat $-\$2\,\mathrm{M}$ across the whole tail. Worse, VaR can *decrease* when two risky books are merged — it fails sub-additivity, and therefore fails to be a coherent risk measure in the Artzner sense (§15.6).

A useful mental picture: imagine the CDF of L rising from zero on the left to one on the right. VaR at level $\alpha = 0.05$ is the horizontal line $y = 0.95$ intersected with the CDF and projected onto the x-axis. Any reshuffling of probability mass to the right of that intersection leaves VaR unchanged — pile the tail onto a single catastrophic point, or spread it uniformly, and the VaR line doesn't move. Only the quantile location matters. CTE, by contrast, is the area under the CDF complement above the horizontal line divided by 0.05, and is sensitive precisely to the tail reshaping VaR misses.

The infimum in (15.1) rather than a plain quantile covers distributions with atoms — binary payoffs, barriers, digitals — where the CDF has a jump at the target probability and the quantile is not uniquely defined. In empirical Monte Carlo distributions with thousands of discrete sample points, this tie-breaking convention must be explicit or two people computing "the same" VaR will get different numbers.

Loss distribution with VaR and CTE *Loss distribution: VaR cutoff and CTE mean-loss beyond it.*

17.2.1 15.2.1 Three computational routes to VaR

Three methodologies dominate production risk systems: *historical simulation*, *parametric (variance-covariance)*, and *Monte Carlo*. A well-run risk function uses at least two as cross-checks.

Historical simulation. Collect the last N daily P&L observations (the current portfolio re-priced under historical market moves), sort them worst to best, and pick the $(1-\alpha)$-quantile directly. With $N = 500$ and $\alpha = 0.95$, the 95% VaR is the 25th worst observation. The method is non-parametric: it makes no distributional assumption, letting whatever fat tails, skew, and jumps appeared historically carry through. Strengths: no model risk from mis-specified distributions, trivial to explain. Weaknesses: only as good as its lookback window, assigns equal weight to old and new observations so regime shifts are absorbed with lag, and tail estimation at $\alpha = 0.99$ with $N = 500$ rests on five data points. Refinements include *age-weighted* historical simulation (older observations get less weight), *volatility-scaled* historical simulation (rescale each historical return by current-to-historical vol ratio), and *filtered historical simulation* (fit a GARCH model, scale out historical vol, pick a quantile of residuals, scale back up by current vol).

Parametric / variance-covariance / delta-normal VaR. Assume portfolio returns are Gaussian with zero mean and variance $\sigma_P^2 = w^\top \Sigma w$, where w is the vector of dollar exposures and Σ is the factor covariance matrix. Then

$$\mathrm{VaR}_\alpha = z_{1-\alpha}\,\sigma_P = z_{1-\alpha}\sqrt{w^\top \Sigma w}, \tag{15.2}$$

where $z_{1-\alpha}$ is the Gaussian quantile (1.645 at 95%, 2.326 at 99%). Strengths: closed-form, fast even for very large portfolios, decomposable cleanly by factor or asset class. Weaknesses: the Gaussian assumption dramatically underestimates fat-tail risk, and the linear approximation to book P&L fails for option-heavy books. Parametric VaR is sometimes called "delta-normal" precisely because it couples a linear delta approximation of position P&L with a Normal distribution of factor returns.

A partial fix for non-linearity is *delta-gamma parametric VaR*: approximate book P&L by a quadratic function of risk factors,

$$\Delta P \approx \delta^\top \Delta r + \tfrac{1}{2}\,\Delta r^\top \Gamma\,\Delta r, \tag{15.3}$$

where δ is the gradient of P&L with respect to risk factors and Γ is the Hessian. Under multivariate Normal Δr the distribution of ΔP is a quadratic form in Normals — a generalised chi-squared — whose VaR can be obtained by Cornish-Fisher expansion or numerical inversion of its characteristic function. We develop this in full in §15.12; for now, note that delta-gamma parametric VaR improves meaningfully on delta-normal for option-heavy portfolios while retaining the Gaussian factor assumption.

Monte Carlo VaR. Specify a joint distribution for risk factor returns (Gaussian, Student-t, copula-based, stochastic-volatility, full economic-scenario generator), draw simulated paths, reprice the entire portfolio under each, and take the empirical quantile of the simulated P&L distribution. Strengths: flexibility to handle arbitrary distributions, full non-linear pricing including path-dependent payoffs, easy incorporation of complex instruments. Weaknesses: expensive (tens of thousands of full-portfolio revaluations), model-risk-prone (the joint distribution has to be specified and will be wrong in some way), and slow to converge in the tail. Many large banks run Monte Carlo VaR as the primary regulatory measure and historical simulation as a cross-check.

A sample-size calculation is instructive. The standard error of a quantile estimate based on M i.i.d. samples scales as $1/\sqrt{M \cdot f(q)^2 \cdot \alpha(1-\alpha)}$, where $f(q)$ is the density at the quantile. For a 99% VaR with 10,000 Monte Carlo paths under a Normal approximation, the standard error is 3-5% of the VaR itself. For 99.9% VaR on the same 10,000 paths the standard error blows up to 15-20%, which is why extreme quantiles require massive simulation budgets (often 100,000 paths or more) and motivate heavy use of variance-reduction techniques such as importance sampling. A common refinement is to oversample the tail, run disproportionately many paths in stressful scenarios, re-weight them, and estimate the tail quantile with much higher precision.

Three routes to VaR on a common loss distribution *Historical, parametric (Gaussian), and Monte-Carlo estimates of 95% VaR on the same skewed, jump-prone loss distribution. The Gaussian fit systematically under-reports tail risk because it ignores the jump component; historical and Monte-Carlo agree once the sample is large enough to see the jump events. Running at least two methods in parallel as cross-checks is standard production practice.*

17.2.2 15.2.2 Cornish-Fisher expansion

Between "full Monte Carlo" and "pure Gaussian" sits the *Cornish-Fisher expansion*, a semi-parametric trick that adjusts the Gaussian quantile for skewness and excess kurtosis without a full simulation. Express the true quantile of a non-Gaussian distribution as a series in the standard Gaussian quantile:

$$q_\alpha \approx z_\alpha + \tfrac{1}{6}(z_\alpha^2 - 1)\, s + \tfrac{1}{24}(z_\alpha^3 - 3z_\alpha)\, \kappa - \tfrac{1}{36}(2z_\alpha^3 - 5z_\alpha)\, s^2, \tag{15.4}$$

where z_α is the Gaussian quantile, s is skewness, and κ is excess kurtosis of the target distribution. Multiplying by the portfolio standard deviation gives a skew-and-kurt-adjusted VaR. The formula captures most of the correction needed for modestly non-Gaussian P&L distributions. For heavily non-Gaussian distributions — deep fat tails, severe skewness near crash events — Cornish-Fisher breaks down and Monte Carlo or historical simulation is required.

The practical appeal is enormous when combined with delta-gamma parametric VaR. The quadratic form in (15.3) has skewness and kurtosis computable in closed form from δ, Γ, and Σ. Feeding those moments into (15.4) yields a VaR that incorporates both non-linearity (via Γ) and distributional tail adjustment (via skew/kurt), without a single Monte Carlo path. This delta-gamma-Cornish-Fisher VaR was widely used in the 1990s and 2000s and remains a useful benchmark.

Cornish-Fisher VaR vs Normal *The Cornish-Fisher quantile multiplier at* 99% *confidence as a function of skewness (with excess kurtosis fixed at* $\kappa_{\text{ex}} = 3$*). Negative skew (left tail heavy) raises the multiplier above the Normal* $z_{99\%} \approx 2.33$*; positive skew pulls it down. A quadratic shape in* s *reflects the* s^2 *term in* (15.4)*.*

Two caveats. First, the expansion can produce non-monotonic quantile functions for extreme parameter values (very fat tails, severe skew), which makes no distributional sense. When this happens, the expansion is signalling that the perturbation from Normal is too large to be captured by a polynomial correction, and one must move to a full parametric or non-parametric method. Second, skewness and excess kurtosis are themselves very noisy to estimate from short time series — sampling standard errors are $\sqrt{6/N}$ and $\sqrt{24/N}$ respectively for Gaussian data, meaning skew and kurt estimates from 250 days are nearly useless. In practice, the moments are estimated from multi-year histories or from a time-series model (GARCH, for example) that yields more efficient estimates.

17.2.3 15.2.3 Historical provenance and interpretive tensions

VaR rose through the J.P. Morgan RiskMetrics framework in the early 1990s and was adopted by Basel as the regulatory standard for market-risk capital; the appeal was operational (single number, easily aggregated). The 2008 crisis shifted the regulatory consensus toward expected shortfall. CVaR is theoretically superior but statistically harder to backtest than VaR (exceedances are binomial; tail integrals are not), which is why the modern compromise runs both — VaR for backtesting, CTE for capital.

17.3 15.3 Conditional Tail Expectation and Expected Shortfall

$$\mathrm{CTE}_\alpha(L) \;=\; \mathrm{ES}_\alpha(L) \;=\; \mathbb{E}\big[\, L \,\big|\, L \ge \mathrm{VaR}_\alpha(L) \big]\,. \tag{15.5}$$

CTE ("conditional tail expectation") and ES ("expected shortfall") refer to the same quantity. Where VaR only marks the threshold, CTE integrates the tail *beyond* it, so the two 5%-tail scenarios from §15.2 — one fat, one thin — earn different CTE scores. CTE is coherent: monotone, positively homogeneous, translation-invariant, and — crucially — sub-additive. The sub-additivity is what we need for any risk axis on which diversification is guaranteed to help.

Normal versus fat tail. The thin-tailed Normal has VaR ≈ 1.64 and CTE ≈ 2.06 at 95% — a gap of 0.4. For a Student-t_3 *rescaled to unit variance* (i.e. the base t_3 divided by $\sqrt{\nu/(\nu-2)} = \sqrt{3}$ so both distributions share zero mean and unit variance), the 95% VaR is $\approx 2.35/\sqrt{3} \cdot \sqrt{3} = 2.35$ (retaining the base-t_3 quantile $q_{0.05} \approx 2.353$ for this comparison since the figure plots same-scale densities) and the CTE is ≈ 4.19 — a gap of 1.8, nearly five times larger. (On the unscaled base-t_3 convention, VaR ≈ 2.35 and CTE ≈ 2.34; the qualitative "CTE exceeds VaR by much more in the fat-tailed case" narrative holds under either convention, but the specific 4.19 number uses the unit-variance rescaling so that the comparison with the unit-variance Normal is apples-to-apples.) The practical point: two books can share the same VaR yet have wildly different CTE, which is precisely the pathology CTE was designed to catch. In closed form for Student-t with ν degrees of freedom,

$$\mathrm{CTE}_{\alpha}^{t_\nu} = \frac{f_{t_\nu}(\mathrm{VaR}_{\alpha}^{t_\nu})\,(\nu + (\mathrm{VaR}_{\alpha}^{t_\nu})^2)}{(\nu-1)\,\alpha}, \tag{15.6}$$

where f_{t_ν} is the Student-t density. The key observation is the $(\nu-1)$ denominator: as $\nu \to 1$ (Cauchy), CTE diverges while VaR remains finite. Even at $\nu = 2$, CTE is finite but much larger than VaR; at $\nu = 3$ we get the 1.8-gap quoted above. As $\nu \to \infty$ the distribution approaches Normal and the gap converges to the Normal value of 0.4. The tail-index ν interpolates smoothly between "Normal-like" and "catastrophically fat-tailed", and CTE tracks the deterioration faithfully while VaR only modestly responds.

Why the switch happened. Empirically estimated tail indices for daily equity returns sit in the range 3-5 (4-7 for weekly); emerging-market currencies and distressed credit can drop to 2, making CTE much larger than VaR by a factor of several. Any risk framework that relies on Normal-CTE benchmarks systematically under-reports tail risk in these markets, and the under-reporting compounds during crisis periods when the tail index temporarily shrinks further. Every blow-up in the historical record of hedge funds and investment banks — and a depressing number of insurance companies — can be traced, at least partially, to a VaR that looked fine alongside a CTE that would have screamed. The regulatory transition from 99% VaR charges toward 97.5% CTE charges was a direct response.

The 97.5% CTE calibration deserves explanation. For a Gaussian loss distribution, $\mathrm{CTE}_{97.5\%} \approx \mathrm{VaR}_{99\%}$, so the switch was designed to leave the average capital charge roughly unchanged but to make the charge responsive to tail shape. Fat-tailed portfolios now face genuinely higher capital requirements than thin-tailed portfolios with the same 99% VaR. This is the regulatory incentive-alignment the Basel Committee wanted: portfolios whose tails actually matter are capitalised accordingly.

Estimation is harder. CTE estimation has a statistical subtlety that VaR estimation does not. VaR is a quantile, determined by the local behaviour of the CDF around a single point; estimation depends on the density at that point. CTE is an integral over the tail, so estimation depends on the density *everywhere* in the tail, and density estimates at extreme quantiles have high variance. A 99% CTE from 250 daily returns is the mean of the 2.5 worst returns — so it depends heavily on the single *worst* observation in the sample. If the sample contains a mild outlier like -4% the estimate comes in around -3.5%; if it contains a catastrophic -12% like September 2008, the estimate jumps to -7% or worse. Two samples drawn from the same underlying distribution can give CTE estimates that differ by a factor of two even when their VaR estimates agree within basis points.

This sensitivity is a blessing and a curse. Blessing: CTE correctly reflects genuine tail events when they occur — the measure is *doing its job* by responding to extreme observations. Curse: CTE estimates have wide confidence intervals, which makes it hard to tell whether a day-over-day change reflects genuine risk change or sampling noise. The pragmatic response is to model the tail parametrically using a Generalised Pareto (GPD) fit to excesses over a threshold — the *peaks-over-threshold* extreme-value technique developed in §15.4. A GPD fit extrapolates the tail smoothly beyond observed data, giving much more stable CTE at the cost of a modest parametric assumption.

Elicitability and model comparison. A risk measure is *elicitable* if it can be expressed as the minimiser of an expected scoring function. VaR is elicitable (via quantile loss); CTE is not elicitable in general. This matters for model comparison: to compare two VaR forecasts we can use average quantile loss, and a lower score is unambiguously better. For CTE no such scoring function exists, and comparing two CTE models rigorously requires more elaborate tests — the Fissler-Ziegel joint elicitability of VaR and CTE, for instance. Regulators continue to debate whether elicitability is essential; the pragmatic view is that it is not, but its absence complicates the model-validation toolkit.

CTE as a portfolio-construction axis. The VaR-minimising portfolio on an efficient frontier is not generally the CTE-minimising portfolio on the same frontier. Empirical studies have found that mean-CTE-optimised portfolios can have 20-40% lower crash losses than mean-VaR-optimised portfolios even when their mean-variance characteristics are similar. This is a second reason for the regulatory shift toward CTE beyond measurement fidelity: CTE actively produces better portfolios when used as an optimisation objective.

VaR vs CTE — thin-tail vs fat-tail *VaR and CTE compared across Normal (thin tail) and Student-*t_3 *(fat tail) — same quantile can hide very different tail masses.*

17.4 15.4 Thin vs Fat Tails: Student-t and Extreme Value Theory

Normal vs Student-t3 densities with VaR and CTE marked *Same scale, very different tails. The 95% VaR cutoffs differ modestly (Normal* 1.64 *vs Student-*t_3 2.35*), but the CTE values diverge dramatically (*2.06 *vs* 4.19*). VaR is a threshold; CTE integrates the tail beyond it, and the ratio CTE/VaR is diagnostic of tail-fatness.*

Under a Normal assumption, three-sigma losses occur with probability 0.13%/day; under Student-t_3 at matching scale, the same threshold has probability $\approx 2.5\%$, twenty times more often. Equity indices, EM currencies, and credit spreads sit closer to the t picture than to Normal — especially during crises. The 5 /7 /10 multi-day runs reported by major desks in 2008 and March 2020 were not "once-per-age-of-the-universe" Normal events; they were ordinary Student-t tail observations whose Normal label was a mislabel.

17.4.1 15.4.1 Defensive adjustments

Three practical adjustments improve on Normal VaR without requiring a full rewrite of the risk infrastructure.

Parametric fat-tail fit. Fit losses with a Student-t, skew-t, or Generalised Hyperbolic distribution, estimate the degrees of freedom from the data, and recompute VaR and CTE under the fitted distribution. Even a rough fit with $\nu = 5$ or $\nu = 6$ captures most of the relevant fat-tail behaviour. Maximum-likelihood or method-of-moments estimation on a 3-5 year daily history gives reasonably stable parameters; shorter windows give noisy ν estimates that can swing between 3 and 10 over successive weeks.

Empirical historical simulation. Compute VaR and CTE directly from the realised P&L history. This by construction inherits whatever tail the market produced during the lookback window. It is the simplest way to avoid mis-specifying the tail because it does not specify the tail at all — but it equally cannot extrapolate beyond the historical sample.

Explicit stress scenarios. Layer historical stress scenarios (Black Monday 1987, Lehman 2008, COVID 2020) and hypothetical scenarios chosen by the risk committee on top of the parametric tail as a form of risk insurance. The full stress-testing methodology is developed in §15.8.

17.4.2 15.4.2 Extreme Value Theory

A more sophisticated methodology is *Extreme Value Theory (EVT)*, which rests on a fundamental asymptotic result: the Pickands-Balkema-de Haan theorem. For a wide class of loss distributions, the conditional distribution of exceedances over a high threshold u converges to the Generalised Pareto Distribution (GPD) as $u \to \infty$:

$$F_u(y) \;=\; \mathbb{P}(L - u \le y \,|\, L > u) \;\approx\; 1 - \left(1 + \xi \frac{y}{\beta}\right)^{-1/\xi}, \quad y > 0, \tag{15.7}$$

with shape parameter ξ and scale parameter $\beta > 0$. The sign of ξ classifies the tail: $\xi > 0$ is Fréchet (power-law tails, infinite moments above order $1/\xi$), $\xi = 0$ is Gumbel (exponential tails), $\xi < 0$ is Weibull (bounded tails). Equity indices typically give ξ in the range 0.2-0.4; sovereign-rate moves give ξ near 0; commodity returns can give $\xi > 0.5$ during shortage regimes.

The *peaks-over-threshold* (POT) procedure is: pick a threshold u (often the 90th or 95th percentile of observed losses), collect the exceedances $\{L_i - u : L_i > u\}$, fit (ξ, β) by maximum likelihood, and extrapolate the fitted GPD to obtain tail quantiles and conditional tail expectations at confidence levels far beyond the empirical data. A related technique uses the *Hill estimator* to estimate ξ directly from the ratio of the largest order statistics of the sample — simpler to implement but less efficient than MLE.

The advantage of EVT over pure historical simulation is smooth extrapolation beyond the observed data, giving stable estimates even at extreme confidence levels (99.9%, 99.99%) where raw historical data contains only a handful of observations. The disadvantage is that EVT requires a threshold choice (too low and the asymptotic theorem fails to apply well; too high and the fit loses data) and is sensitive to the estimated shape ξ, which has its own uncertainty.

17.4.3 15.4.3 GARCH and conditional dynamics

A complementary framework is the *GARCH* family of volatility models. GARCH (and its variants EGARCH, GJR-GARCH, TGARCH) captures the *clustering* of volatility — high-vol periods persist — and the asymmetry of vol response to signed shocks (the leverage effect, where bad news raises vol more than equally-sized good news). A GARCH-filtered historical simulation takes GARCH residuals (returns divided by conditional volatility), samples from the residual empirical distribution, rescales by the forecast volatility, and computes VaR. This hybrid combines the parametric strength of GARCH in modelling volatility dynamics with the non-parametric strength of empirical residuals in capturing fat tails, and is widely considered a gold-standard methodology for equity and FX risk.

The chapter's emphasis on VaR and CTE as *different* summaries of the same loss distribution is really an emphasis on the limits of any single-number summary. The loss distribution is the primary object; VaR and CTE are two projections of it onto the real line. When the distribution is Normal, VaR and CTE at any level are almost interchangeable. When it is fat-tailed, they diverge dramatically, and the divergence itself is diagnostic: a VaR-to-CTE ratio well above the Gaussian benchmark is a red flag that the tail is mis-modelled.

17.5 15.5 Basel Context and Regulatory Capital

The Basel framework determines which risk measures have operational primacy at large banks. The headline evolution is the move from 99% VaR (Basel Market Risk Amendment 1996) to 97.5% expected shortfall (FRTB 2016/2019, in force from 2023). The calibration was designed so that under a Gaussian loss distribution the two produce roughly the same charge — $\text{CTE}_{97.5\%} \approx \text{VaR}_{99\%}$ — while making the charge *responsive to tail shape*: fat-tailed books now consume more capital than thin-tailed books at the same 99% VaR.

Accord	Year	Market-risk measure	Notable additions
Basel I	1988	(credit only)	RWA, 8% capital ratio
Market Risk Amendment	1996	10-day 99% VaR (IMA)	Traffic-light backtesting; $k \geq 3$ multiplier
Basel II / 2.5	2004/2009	VaR + stressed VaR	IRB; IRC for credit migration
Basel III	2010/13	VaR (unchanged)	Leverage ratio, LCR/NSFR
FRTB	2016/19	97.5% ES	Liquidity horizons; NMRF; PLA test

In parallel with IMA, banks can elect the Standardised Approach (SA): simpler, typically higher capital, and the fallback if IMA is revoked for failed backtesting.

17.6 15.6 The Coherent Risk Framework

A risk measure ρ acting on a space of random losses is *coherent* in the Artzner-Delbaen-Eber-Heath sense if it satisfies four axioms. We state them in the loss convention (large positive = painful), and unpack each in turn.

Monotonicity (M). $L_1 \leq L_2$ almost surely $\Rightarrow \rho(L_1) \leq \rho(L_2)$. A strictly smaller-loss random variable must have a smaller risk measure. This is the bare-minimum sanity requirement and both VaR and CTE satisfy it. (In the *profit* convention used by Artzner et al.'s original paper and some other sources, monotonicity reads $X_1 \leq X_2 \Rightarrow \rho(X_1) \geq \rho(X_2)$ — bigger wealth means smaller risk — so readers cross-referencing should flip the inequality alongside the sign convention.)

Translation invariance (TI). $\rho(L + c) = \rho(L) + c$ for any constant c. Adding a deterministic dollar of loss adds a dollar of risk. Note the sign: adding a deterministic dollar of *profit* (so $c < 0$) reduces the risk by a dollar — this is the "cash can be netted against risk" statement underlying capital-charge accounting.

Positive homogeneity (PH). $\rho(\lambda L) = \lambda \rho(L)$ for $\lambda \geq 0$. Doubling every position doubles the risk. Both VaR and CTE comply. One can argue the axiom is too clean — very large positions face liquidity haircuts and $\rho(10L)$ should probably exceed $10\rho(L)$ — but the classical framework ignores such frictions.

Sub-additivity (S). $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$. This formalises "diversification never hurts." It is what VaR fails and CTE (under mild regularity conditions) obeys.

17.6.1 15.6.1 The sub-additivity failure of VaR

To see the VaR failure concretely, take two independent loans, each of which defaults with probability 4% producing a loss of 100, and otherwise pays zero. Individually, the 95% VaR of each is 0 — a 4% tail event does not breach the 5% confidence line. But the sum can default twice, once, or not at all. The probability of at least one default is roughly 8%, of two defaults about 0.16%, so the 95% VaR of the merged book is around 100, not 0. Merging two risks whose individual VaRs are zero gave a combined VaR of 100.

VaR sub-additivity failure on two independent loans *Monte-Carlo verification of the two-loan thought experiment: each loan has $VaR_{95} = 0$ in isolation, but their sum has $VaR_{95} \approx 100$. CTE is coherent — the merged CTE is bounded by the sum of individual CTEs — and so can serve as a risk-budgeting axis where VaR cannot.*

The pathology is generic to any loss distribution with mass concentrated just inside the VaR threshold. It afflicts credit books particularly badly because individual defaults are exactly such low-probability-high-impact events: an investment-grade bond has a 1-year default probability under 1%, well below the 5% VaR threshold, so its individual VaR is zero. A portfolio of thousands of such bonds has a default frequency of tens per year and generates substantial portfolio VaR. If we used VaR to set capital on individual loans we would allocate zero, yet the portfolio clearly needs substantial capital. The CTE-based alternative assigns positive CTE even to individual low-probability defaults — because the tail expectation averages over the default scenario — and produces a portfolio CTE that equals the sum of individual CTEs minus diversification benefits. A much more coherent story.

A corollary of sub-additivity failure: VaR cannot serve as a direct basis for risk-budget allocation. If desk A has VaR \$1M and desk B has VaR \$1M, the firm's VaR is *not generally* \$2M or anything derivable from the individual VaRs. It could be \$0.5M (perfectly negatively correlated), \$1.4M (independent), \$2M (perfectly correlated), or \$2.2M (the pathological failure). The bank has no way of allocating \$2M of firm-level VaR to desk-level budgets using just the desk VaRs. CTE, being sub-additive, allows clean Euler-style decomposition, and this is part of the technical appeal of using CTE as the desk-level budgeting tool even when VaR is reported to regulators.

17.6.2 15.6.2 The dual representation and spectral measures

The foundational theorem of coherent risk is a *dual representation*: every coherent ρ can be written as a worst-case expectation over a set $\mathcal{Q}$ of probability measures,

$$\rho(L) \;=\; \sup_{\mathbb{Q} \in \mathcal{Q}} \mathbb{E}^{\mathbb{Q}}[L]. \tag{15.8}$$

This unifies many apparently different risk measures under a single umbrella: every coherent risk measure is a worst-case expectation over some class of scenarios. CTE at level α corresponds to taking $\mathcal{Q}$ to be the set of probability measures that are absolutely continuous with respect to the base measure with a Radon-Nikodym derivative bounded by $1/\alpha$. VaR does not admit such a representation because it is not coherent.

A richer parametric family is the class of *spectral risk measures*, each of which can be written as a weighted average of quantiles:

$$\rho_\phi(L) \;=\; \int_0^1 \phi(u)\, F_L^{-1}(u)\, du, \tag{15.9}$$

where $\phi : [0, 1] \to [0, \infty)$ is a non-decreasing weighting function with $\int \phi = 1$. VaR at level α corresponds to ϕ being a Dirac delta at $u = 1 - \alpha$ — all weight on a single quantile — and is not coherent because the spike violates monotonicity of ϕ. CTE at level α corresponds to uniform weighting $\phi(u) = \mathbf{1}_{u\geq 1-\alpha}/\alpha$ over the tail — coherent because ϕ is non-decreasing. One can imagine ϕ functions that assign even more aggressive weight to the extreme tail, producing risk measures that interpolate between "care only about one quantile" (VaR) and "care linearly about the whole tail mean" (CTE) and beyond into "care hyperbolically about the extreme tail." The desk-level upshot: CTE is a sensible default but not a unique answer; a catastrophe-reinsurance book might prefer a more tail-weighted spectral measure still.

Coherence is a minimum bar, not the last word. Coherence formalises "not broken"; measures that fail it can still be operationally useful (VaR as a communicable summary) but must be supplemented by a coherent companion (CTE for the tail shape; stress scenarios for catastrophes) before the suite is diagnostically adequate.

17.6.3 15.6.3 Case study — three failures of VaR (preview)

The coherent-risk framework just developed is the analytical residue of three crises that the dominant risk measure of the day did not survive: LTCM (1998), Archegos (2021), and the 2008 GFC. Each is treated in full in the **Case studies — three failures of VaR** section near the end of this chapter, with the chapter-wide machinery serving as the analytical lens. As a preview: LTCM's headline \$45M VaR underestimated the realised six-week loss of \$4.6B by $\sim 100\times$ because the historical sample contained no crisis-correlation regime; Archegos's per-PB VaRs each looked manageable but the cross-PB consolidated exposure (which no single PB could see) totalled \$80B and produced \$10B+ of bank losses; and pre-2008 Basel-II VaR on senior CDO tranches missed correlated-default tails because mortgage-asset correlation was calibrated at 0.30 on a benign housing regime and went to 0.95 in 2007–2008. These three episodes motivate the rest of the chapter: backtesting (§15.7) to detect regime breaks, stress testing (§15.8) to insert events the historical window omits, risk decomposition (§15.9) to flag concentration, and CTE/ES (§15.3) as the measure that actually reports the size of the tail.

17.7 15.7 Backtesting: Kupiec and Christoffersen

Once a VaR model is in production, it must be validated against realised P&L. The framework for this validation is *backtesting*: compare the daily sequence of reported VaRs against the daily sequence of realised P&Ls, count exceedances (days on which the realised loss exceeded the VaR), and apply statistical tests to decide whether the exceedance pattern is consistent with the model.

17.7.1 15.7.1 The exceedance indicator and its expected behaviour

Define the exceedance indicator

$$I_t \;=\; \mathbf{1}\{L_t > \mathrm{VaR}_{\alpha,t}\}. \tag{15.12}$$

If the VaR model is correctly calibrated at confidence level $(1-\alpha)$, then each I_t is a Bernoulli(α) random variable and the sequence $\{I_t\}$ is i.i.d. (independent because past exceedances should carry no information about today, identically distributed because the tail probability is by construction α every day). Both the "correct frequency" claim and the "correct independence" claim are empirically testable.

17.7.2 15.7.2 Kupiec's proportion-of-failures test

The simplest test — targeting frequency — is Kupiec's POF test. Over n days, let $x = \sum_t I_t$ be the number of exceedances. Under the null that the model is correct, $X \sim \mathrm{Binomial}(n, \alpha)$ with mean $n\alpha$ and variance $n\alpha(1-\alpha)$. The likelihood-ratio statistic is

$$\mathrm{LR}_{\mathrm{POF}} \;=\; -2\log\left[\frac{\alpha^x(1-\alpha)^{n-x}}{\hat{p}^x(1-\hat{p})^{n-x}}\right], \qquad \hat{p} = \frac{x}{n}, \tag{15.13}$$

asymptotically χ^2_1 under the null. Reject at 5% significance if $\mathrm{LR}_{\mathrm{POF}} > 3.84$. For a 99% VaR with $n = 250$ days, expected exceedances are $n\alpha = 2.5$; the test rejects above about 7 exceedances or at 0 (which is paradoxically *too few*, suggesting the model is too conservative and over-capitalising the book).

This is the essence of the Basel "traffic light" regime. More than 4 exceedances in a 250-day window puts the model in the yellow zone (capital multiplier k rises from 3.00 by 0.4-0.85 depending on count); more than 9 exceedances puts it in the red zone (multiplier capped at 4.00, pending a full model revision). Capital is directly tied to backtest performance, giving banks a strong financial incentive to build models that actually work.

Kupiec POF acceptance band over a 250-day 99% VaR backtest *Binomial pmf of exceedance counts under the null* $X \sim \text{Binomial}(250, 0.01)$*, coloured green where the Kupiec likelihood-ratio is below the* 5% *critical value* 3.84 *and red where it rejects. Green bars span roughly* 0–6 *exceedances; beyond that the model is rejected as under-capitalised. Zero exceedances is technically also rejected as over-conservative.*

17.7.3 15.7.3 Christoffersen's independence test

Kupiec catches miscalibrated levels but misses *clustering* — a model whose exceedances come bunched together in crisis periods and sparse elsewhere, so the overall count looks right but the temporal pattern is wrong. Clustering indicates the model fails to capture volatility dynamics: on Monday the model reports a VaR that was appropriate for last week's calm conditions, the market blows out, Tuesday's VaR is still calibrated to last week, Wednesday the same, and suddenly you have four exceedances in four days because the dynamic volatility of the market was not reflected in the dynamic VaR of the model.

Christoffersen's test models the exceedance sequence as a two-state Markov chain. Let n_{ij} be the count of transitions from state i to state j (where 0 = no exceedance, 1 = exceedance). Under independence, the transition probability $\mathbb{P}(I_t = 1 \mid I_{t-1} = 1)$ equals the unconditional $\mathbb{P}(I_t = 1) = \alpha$. The likelihood-ratio test against a general Markov alternative is

$$\mathrm{LR}_{\mathrm{IND}} \;=\; -2\log\left[\frac{\widehat{p}^{x}(1-\widehat{p})^{n-x}}{\widehat{p}_{01}^{n_{01}}(1-\widehat{p}_{01})^{n_{00}}\widehat{p}_{11}^{n_{11}}(1-\widehat{p}_{11})^{n_{10}}}\right] \sim \chi^2_1, \tag{15.14}$$

where $\widehat{p}_{i1} = n_{i1}/(n_{i0} + n_{i1})$. Combined with Kupiec, the *conditional coverage* test

$$\mathrm{LR}_{\mathrm{CC}} \;=\; \mathrm{LR}_{\mathrm{POF}} + \mathrm{LR}_{\mathrm{IND}} \;\sim\; \chi^2_2 \tag{15.15}$$

rejects the model if either the overall frequency is wrong or the clustering is significant, giving a comprehensive test of VaR adequacy. A failing $\mathrm{LR}_{\mathrm{IND}}$ with passing $\mathrm{LR}_{\mathrm{POF}}$ is a characteristic signature of a model using a slow-moving volatility estimator (rolling 1-year window, for example) in a market with pronounced vol clustering — the fix is typically to move to a GARCH-type conditional-volatility specification.

17.7.4 15.7.4 Backtesting CTE

CTE backtesting is considerably harder because CTE is not elicitable (§15.3). The simplest approach — and the one adopted in FRTB — is to backtest CTE indirectly by backtesting VaR at multiple confidence levels (typically 97.5% and 99%) and examining the distribution of exceedance magnitudes. A model whose VaR is correctly calibrated at both 97.5% and 99% is approximately correctly calibrated for CTE at 97.5%, because CTE is approximately an average over the range between those quantiles.

The *Acerbi-Szekely* framework proposes scoring functions for joint VaR/CTE backtesting. Three families of tests are commonly implemented: (i) a test on the magnitude of exceedances (the average exceedance should match the CTE minus VaR prediction); (ii) a test on the tail-conditional distribution of L relative to the model's predicted tail distribution (a sort of Kolmogorov-Smirnov test in the tail); (iii) a joint test on the first two moments of the exceedance distribution. These are more complex to implement but increasingly standard in banks' model-validation departments.

Practical note: CTE backtests have low statistical power unless the data history is long. On a 1-year backtest at 99% confidence, there are only 2-3 expected exceedances — essentially no usable information about the conditional expectation of the tail. Meaningful CTE backtests require 3-5 year histories or pooling across multiple portfolios, which is why the FRTB compromise of backtesting VaR at two levels and inferring CTE adequacy has dominated implementation.

The broad lesson: backtesting is as much a research area as a compliance exercise. Every risk measure has its own backtesting theory, and practitioners must choose tests that match the quantity being backtested, recognise the limitations of finite-sample power, and interpret results as diagnostic rather than purely pass/fail.

17.8 15.8 Stress Testing

Stress testing sits alongside VaR/CTE as the second major pillar of market-risk measurement. The premise is that statistical models calibrated on historical data fail precisely when history fails to contain the next catastrophe — so one should also ask "what if the stock market falls 30% in a week?" or "what if credit spreads widen to 2008 levels?" regardless of what the statistical model predicts.

17.8.1 15.8.1 Historical stress scenarios

Historical stress scenarios replay a specific historical event against the current portfolio, applying the market moves from that period as instantaneous shocks. Canonical scenarios include Black Monday 1987, the Asian Crisis 1998, LTCM autumn 1998, the Dot-Com crash of 2000-2001, Lehman September 2008, the Flash Crash of May 2010, the EuroZone sovereign crisis 2011-2012, the China devaluation of August 2015, Volmageddon in February 2018, the COVID-19 crash of March 2020, and the UK gilt crisis of September-October 2022.

The appeal of historical scenarios is that they are "real": everyone agrees that Lehman happened, and there is no arguing about the size of the moves. The limitation is that the next crisis will not be the last crisis; hedging against 2008 may not protect against a different structural breakdown. Every historical scenario also requires translation to today's portfolio — one cannot just apply 1987 S&P moves verbatim to today's equity book because today's book contains index futures that did not exist in 1987, options markets whose implied vol surface is unrecognisable from that era, and hedging relationships that behave differently under modern market microstructure. The translation introduces modelling judgement, and different risk teams make different calls.

17.8.2 15.8.2 Hypothetical stress scenarios

Hypothetical stress scenarios are designed by the risk committee to capture imagined but not-yet-realised catastrophes. Examples: "equity down 25%, vol up to 80, credit spreads +500 bp, USD up 10%, oil down 40%"; "Treasury yields rise 300 bp in a week"; "Bitcoin goes to zero overnight, dragging the tech-heavy NASDAQ down 15% on correlation contagion." These scenarios reflect institutional views about what could go wrong and stress the risks that are *not* in the historical record.

Regulatory stress tests — CCAR in the US, EBA stress tests in the EU, Bank of England stress tests in the UK — are hypothetical scenarios imposed on all banks simultaneously, with the bank having to show that its capital remains above regulatory minima even under the imposed stress. These exercises combine a macroeconomic narrative (unemployment rises 5 points, GDP contracts 3%, house prices fall 25%) with specific market-variable shocks implied by the narrative, and force banks to project their balance sheet, P&L, and capital ratio through the stress horizon (typically 9 quarters).

17.8.3 15.8.3 Reverse stress testing

A third variety, increasingly emphasised by regulators, is *reverse stress testing*: rather than starting with a scenario and computing the loss, start with a loss (typically the failure threshold — the loss that would render the firm insolvent or non-viable) and work backward to identify scenarios that produce it. The output is not a capital number but a catalogue of scenarios whose realisation would destroy the firm, some of which may be obvious and others of which may surprise the risk committee.

Reverse stress testing is mathematically harder than forward stress because there are many scenarios that produce a given total loss — the problem is an inverse problem with a large null space. Practical implementations typically constrain the scenario space by factor structure (what combinations of factor shocks within 5 standard deviations produce failure?) or by qualitative narrative (what operating environments plausibly produce the shocks that produce failure?).

17.8.4 15.8.4 Scenario construction methodology

Building useful stress scenarios is its own craft. Core principles:

- *Internal consistency.* A scenario must be coherent across markets. An equity-down scenario usually has vol up, credit spreads wider, and safe-haven currencies (CHF, JPY, USD) stronger. Applying an equity-down shock in isolation without these co-moves gives an artificially small loss and misses the real tail risk.
- *Magnitude calibration.* Shocks should be large enough to matter but not so large as to be dismissed as absurd. A 20-sigma equity move on a Normal calibration is absurd but on a Student-t_3 calibration it is a routine crisis

move. Calibrating shock sizes to fat-tailed distributions gives more credibility and more informative stress results.

- *Coverage of risk types.* The scenario suite should collectively exercise every risk factor the book is exposed to — not just the obvious ones. A scenario suite that stresses equity and rates but ignores FX, credit, vol, and correlation leaves systematic gaps.
- *Narrative plausibility.* Senior management engages with stress tests that come with a story ("oil price shock triggers recession, which triggers credit widening, which forces unwinds in levered funds, which cascades back into equities"). Pure statistical shocks without narrative are harder to action.

17.8.5 15.8.5 Economic capital and solvency horizons

Economic capital is the amount of capital the firm's internal models say is needed to remain solvent over a specified horizon (typically 1 year) at a specified confidence level (typically 99.9% — a 1-in-1000 year event). Economic capital differs from regulatory capital in several ways: it uses the firm's own modelling framework (not the regulator's), it runs at a much longer horizon and higher confidence level, and it aggregates across all risk types (market, credit, operational, business, strategic) whereas regulatory market-risk capital concerns only market risk. Economic capital is the basis for internal return-on-capital calculations, ratings-agency assessments, and internal reinsurance or capital-transfer decisions.

The relationship between stress testing and VaR/CTE is subtle. VaR/CTE report the *expected* catastrophe based on an assumed distribution; stress tests report a *specific* catastrophe imagined or observed. They are complementary, not substitutes. A portfolio with low VaR can have large stress P&L (vulnerable to a specific imagined scenario outside the historical record), and vice versa. A mature risk function reports both, and uses significant discrepancies between them as a diagnostic for hidden non-linearity or for over-reliance on a particular set of historical data.

17.9 15.9 Risk Decomposition

Once an aggregate VaR or CTE has been computed, the operational question is "which positions contribute to it, and by how much?" This is the *risk-decomposition* problem, and it has three principal answers: marginal, component, and incremental. Each answers a slightly different question, and a well-run risk function reports all three.

17.9.1 15.9.1 Marginal VaR

Marginal VaR of a position is the partial derivative of portfolio VaR with respect to the position size. It answers: if I add one more dollar of exposure to this position, how much does my VaR go up? Under the delta-normal model with portfolio volatility $\sigma_P = \sqrt{w^\top \Sigma w}$, the marginal VaR of position i is

$$\mathrm{mVaR}_i \;=\; \frac{\partial \mathrm{VaR}}{\partial w_i} \;=\; z_{1-\alpha}\,\frac{(\Sigma w)_i}{\sigma_P}, \tag{15.16}$$

where $(\Sigma w)_i$ is the covariance of position i's return with the portfolio return. Marginal VaR is natural for incremental trade decisions — it directly measures the risk impact of adding a little more of something. Sign is informative: a position with negative marginal VaR *reduces* portfolio risk when added (it is a net hedge), and the risk team should be happy to see it grow.

17.9.2 15.9.2 Component VaR

Component VaR is the contribution of a position to the portfolio VaR, obtained via Euler allocation:

$$\mathrm{cVaR}_i \;=\; w_i \cdot \mathrm{mVaR}_i, \qquad \sum_i \mathrm{cVaR}_i \;=\; \mathrm{VaR}. \tag{15.17}$$

The additive property is the key. Component VaRs sum to the total, which makes component VaR the natural basis for risk-budget allocation — each desk is given a budget, and the sum of desk-level VaRs equals firm-level VaR without residual. Under the delta-normal framework, component VaR is proportional to the position's covariance with the portfolio, scaled by portfolio size. Euler allocation generalises to any positively homogeneous risk measure, including CTE, so component CTE is defined analogously and sums exactly to portfolio CTE.

17.9.3 15.9.3 Incremental VaR

Incremental VaR of a position is the change in portfolio VaR if the position is removed entirely:

$$\mathrm{iVaR}_i \;=\; \mathrm{VaR}(\text{full portfolio}) - \mathrm{VaR}(\text{portfolio without position } i). \tag{15.18}$$

It is a finite-difference rather than an infinitesimal measure, and for large positions can differ significantly from component VaR. Incremental VaR is natural for unwind decisions — it directly measures "how much risk do I free up if I close this position today?" For small positions, incremental VaR and component VaR agree; for large positions (especially those with non-linear P&L), they diverge by factors of 2 or more. Incremental VaRs do not generally sum to the total, because removing one position changes the risk impact of every other position through the correlation matrix.

17.9.4 15.9.4 A worked contrast

Consider a portfolio consisting of a long stock position and a short call on the same stock — a covered call. Suppose the stock has component VaR \$1M and the short call has component VaR \$0.3M, summing to \$1.3M. But if we remove just the stock, the short call alone has VaR \$0.6M — the incremental VaR of removing the stock is \$0.7M. If we remove just the short call, the long stock alone has VaR \$1.2M — the incremental VaR of removing the call is \$0.1M. The differences reflect the hedging interaction between the two positions: each individually is riskier than its contribution to the pair because each partially hedges the other. The three attributions answer three different questions, and a risk team that reports only one is depriving its traders of meaningful optionality in trade sizing and unwind sequencing.

17.9.5 15.9.5 Non-linear portfolios and CTE decomposition

For portfolios with significant non-linearity (options, convertibles, structured products), the delta-normal formulas above are a first approximation. The Euler-allocation framework extends to CTE and to Monte-Carlo-based risk measures under mild regularity: if ρ is a positively homogeneous risk measure and the portfolio is parameterised by weights w, then

$$\rho(w) \;=\; \sum_i w_i \frac{\partial \rho}{\partial w_i}(w), \tag{15.19}$$

and $w_i \partial\rho/\partial w_i$ is the component contribution of position i. For a Monte Carlo CTE, the component contribution of a position can be estimated by running the simulation with and without an incremental position, or — more efficiently — by computing the conditional expectation of the position's P&L given that the portfolio is in the tail.

Risk decomposition answers the fundamental governance question "where does my risk come from?" Pairing it with the P&L attribution discussed next closes the loop between pre-trade forecasts and post-trade realisations.

17.10 15.10 P&L Attribution

A sibling discipline to risk decomposition is *P&L attribution*: after the trading day closes, decompose the realised P&L into contributions from each Greek and each market move. A canonical Greek-based attribution for a derivatives book reads

$$\Delta V \;=\; \delta \cdot \Delta S + \tfrac{1}{2}\Gamma(\Delta S)^2 + \nu \cdot \Delta\sigma + \theta \cdot \Delta t + \rho \cdot \Delta r + \text{cross-Greeks} + \text{cash flows} + \text{residual}, \tag{15.20}$$

where $\delta, \Gamma, \nu, \theta, \rho$ are the book's Greeks (see Chapter 7 for definitions) and $\Delta S, \Delta\sigma, \Delta t, \Delta r$ are the realised moves in spot, implied vol, time, and rates. The "cross-Greeks" bucket collects contributions from vanna (spot-vol cross-gamma), volga (vol-of-vol), charm (spot-time), and speed (third-order in spot); the "cash flows" bucket captures discrete events such as dividends, coupons, and realised trade P&L from intra-day executions.

The "residual" — the piece the model cannot explain — is the diagnostic. It should be small. A systematically large residual is a red flag that either the Greeks are mis-computed, the book contains instruments not captured by the risk model, or the market moves include components (a basis dislocation, a vol-surface reshaping, a correlation break) that the factor representation does not span.

17.10.1 15.10.1 Hypothetical, risk-theoretical, and actual P&L (FRTB PLA)

FRTB (BCBS d352) formalises attribution into the *P&L attribution test.* Three P&L series are produced each day:

- *Actual P&L* — the realised P&L from the trading systems, including intra-day trading and any accruals the risk model doesn't capture.
- *Hypothetical P&L* — the front-office full-revaluation P&L on the *frozen* end-of-day positions under the day's *actual* market moves. No intra-day trading; the book is repriced by the full pricing library using the realised end-of-day market data. This is the "what the front-office price says the frozen book earned".
- *Risk-theoretical P&L* — the *risk-engine* full-revaluation P&L on the same frozen positions under the *risk model's* representation of the day's market data (the limited set of risk factors the model actually carries). This is the "what the risk engine, restricted to its own risk-factor set, says the frozen book earned".

The PLA test compares the *hypothetical* and *risk-theoretical* series, measuring the Spearman correlation and a KS-statistic on their differences; the desk passes (and may keep using its internal model) only if both metrics stay inside the BCBS thresholds. Failing forces a switch to the Standardised Approach, which typically carries materially higher capital. The Taylor-via-Greeks expression in (15.20) is *Greek-attributed* P&L — useful as an internal diagnostic, but distinct from the risk-theoretical series the regulator examines.

17.10.2 15.10.2 Attribution disputes and model-maintenance signals

In practice, attribution is where traders and risk managers spend a great deal of time talking past each other. The trader's P&L report (from the front-office system) and the risk desk's P&L report (from the risk engine) will rarely agree to the last dollar — and when they disagree by a material amount, the question is which number to believe. Some differences are legitimate and bookkeepable (FX conversion rates, intraday funding, stale prices); others are diagnostic of a genuine model-data problem (a mispriced exotic, an ignored correlation).

A systematic pattern in the residual is the most valuable piece of information attribution provides. If the residual is consistently positive on up-days and negative on down-days, the risk model is missing a positive gamma exposure somewhere. If the residual is systematically negative near expiry for positions that are being managed to zero, the risk model's decay profile is wrong. If the residual explodes on days with large vol moves, the vega or volga computation is suspect. Treating attribution residual as pure noise misses the diagnostic signal and lets model errors compound unnoticed.

17.10.3 15.10.3 Greek decomposition versus factor decomposition

Two complementary views of attribution are common. *Greek decomposition* expresses P&L as in (15.20) — one term per Greek, times the corresponding market move. *Factor decomposition* expresses P&L as a sum over underlying risk factors, each factor contributing its net movement against the book's net sensitivity. The two views are mathematically related (a Greek on a derivative is the sensitivity to the factor that drives the derivative) but organisationally distinct: Greek decomposition is natural for a derivatives trader thinking about hedging; factor decomposition is natural for a portfolio manager thinking about exposures and hedges across asset classes.

Together, risk decomposition (pre-trade, "how much risk does this add?") and P&L attribution (post-trade, "where did the P&L come from?") form the bookends of a complete risk-management cycle. The pre-trade forecast and the post-trade realisation should agree on average and diverge only within known bounds; when they diverge outside those bounds, the discrepancy triggers either a trading adjustment (the book is doing something the traders did not intend) or a model revision (the risk framework is missing something).

17.11 15.11 Liquidity-Adjusted Risk

The classical VaR/CTE framework assumes that positions can be liquidated at mid-market prices with no market impact over the VaR horizon. This assumption breaks down for illiquid instruments — distressed credit, emerging-market equities, complex structured products — where selling in a hurry carries material bid-ask cost or even an inability to transact at any price.

17.11.1 15.11.1 Exogenous liquidity cost

The simplest form of liquidity adjustment adds a fixed bid-ask haircut to the VaR:

$$\mathrm{LVaR}_\alpha \;=\; \mathrm{VaR}_\alpha + \tfrac{1}{2}\,|w| \cdot \mathrm{spread}, \tag{15.21}$$

where $|w|$ is the absolute position size and "spread" is the bid-ask spread as a fraction of mid. The factor of one-half reflects that half the spread is the cost of liquidating (selling at the bid from mid) and the other half is the cost of re-entering if the position is repurchased. More refined versions add a term for the *volatility of the spread* itself — widening bid-ask in stressed markets — by treating the spread as a random variable and adding a quantile of its distribution to the VaR.

17.11.2 15.11.2 Endogenous market impact

For large positions in any market, the act of liquidating moves prices. The simplest model is a square-root market-impact law: the price impact of unwinding Q shares over a horizon h is approximately $k\sigma\sqrt{Q/(Vh)}$, where V is average daily volume and k is an asset-class-specific constant (0.5-1.5 typical for equities). Incorporating this into VaR adds a market-impact charge that scales non-linearly with position size, capturing the economic reality that doubling a position more than doubles the unwind cost.

17.11.3 15.11.3 Liquidity horizons in FRTB

FRTB institutionalises liquidity adjustment by prescribing *liquidity horizons* by asset class rather than computing bid-ask directly. The idea is to capture the extended period of price exposure during an assumed unwind rather than the instantaneous cost:

- 10 days: major currencies, large-cap equities, interest rates in G10 currencies.
- 20 days: emerging-market FX and sovereigns, small-cap equities.
- 40 days: investment-grade corporate credit.
- 60 days: high-yield corporate credit, structured products.
- 120 days: exotic credit, private-equity-like structured positions.

Capital for an instrument is computed as a 10-day ES times a liquidity adjustment that scales with the horizon. The formula is constructed so that positions with longer liquidity horizons accumulate capital proportional to their extended exposure. An illiquid corporate bond whose prescribed liquidity horizon is 60 days carries six times the exposure window of a liquid equity and correspondingly higher capital.

17.11.4 15.11.4 Funding liquidity risk

Funding liquidity — the risk that the firm cannot finance its positions at a reasonable cost — is managed separately via LCR, NSFR, and stress analyses, but interacts with market risk in crises (liquidity shock $\to$ forced unwinds $\to$ price moves $\to$ further margin calls). The Lehman collapse of September 2008 and the COVID liquidity crunch of March 2020 both featured this doom-loop, and the AAA-rated corporates trading at 70 cents on the dollar in March 2020 were not credit-impaired — they were leverage-impaired. Single-instrument VaR cannot easily capture emergence; this is what the FRTB NMRF charge and stress scenarios try to address.

17.12 15.12 Delta-Gamma Parametric VaR

The delta-gamma framework (see Chapter 7 for the derivation of the Greeks themselves) delivers a specific computational payoff: *delta-gamma parametric VaR.* This technique was the workhorse of risk systems in the 1990s and 2000s and remains important today as a fast first-pass estimate even when full revaluation is available. It is a concrete application of the quadratic Taylor expansion of book P&L combined with the Cornish-Fisher expansion of §15.2.

17.12.1 15.12.1 The quadratic approximation

For a portfolio of options and hedges on n risk factors $r = (r_1, \dots, r_n)$, expand the portfolio value $V(r)$ to second order around the current factor vector r_0:

$$\Delta V(r) \;\approx\; \delta^\top \Delta r + \tfrac{1}{2}\,\Delta r^\top\,\Gamma\,\Delta r, \tag{15.22}$$

where $\delta_i = \partial V/\partial r_i$ and $\Gamma_{ij} = \partial^2 V/\partial r_i \partial r_j$ evaluated at r_0. For an equity option book, r includes spots, implied vols, and rates; δ collects the deltas, vegas, and rhos; and Γ collects the gammas, vannas, volgas, and cross-sensitivities.

The approximation is accurate for moderate factor moves and degrades for very large moves where cubic and higher-order terms matter.

17.12.2 15.12.2 Moments of the quadratic form under Normal factors

Assume $\Delta r \sim \mathcal{N}(0, \Sigma)$ with covariance matrix Σ. The quadratic form has a closed-form mean, variance, and higher moments. The first two are

$$\mu_P \;=\; \mathbb{E}[\Delta V] \;=\; \tfrac{1}{2}\,\mathrm{tr}(\Gamma\Sigma), \tag{15.23}$$

$$\sigma_P^2 \;=\; \mathrm{Var}[\Delta V] \;=\; \delta^\top \Sigma\,\delta + \tfrac{1}{2}\,\mathrm{tr}((\Gamma\Sigma)^2). \tag{15.24}$$

The mean is non-zero whenever Γ is non-zero: a long-gamma book has positive expected P&L from the gamma convexity even under zero-mean factor moves, and vice versa for a short-gamma book. The variance has a linear-sensitivity piece $\delta^\top\Sigma\delta$ (the delta-normal term) plus a quadratic-sensitivity piece $\frac{1}{2}\mathrm{tr}((\Gamma\Sigma)^2)$ (the gamma contribution).

The third and fourth central moments of ΔV, used for Cornish-Fisher, are

$$m_3 \;=\; 8\,\mathrm{tr}((\Gamma\Sigma)^3) + 6\,\delta^\top \Sigma\,\Gamma\,\Sigma\,\delta, \tag{15.25}$$

$$m_4 \;=\; 12\,\mathrm{tr}((\Gamma\Sigma)^4) + 48\,\delta^\top (\Sigma\Gamma)^2 \Sigma\,\delta + 3\,(\sigma_P^2)^2. \tag{15.26}$$

Both expressions follow from the eigen-decomposition of $\Sigma^{1/2}\Gamma\Sigma^{1/2}$, under which the quadratic form becomes a weighted sum of independent chi-squared random variables and arbitrary moments follow from sums over the eigenvalues.

17.12.3 15.12.3 Cornish-Fisher VaR

Feed the standardised skewness $s = m_3/\sigma_P^3$ and excess kurtosis $\kappa = m_4/\sigma_P^4 - 3$ into the Cornish-Fisher expansion (15.4):

$$\mathrm{VaR}_\alpha \;\approx\; -\mu_P + \sigma_P\left[z_\alpha + \tfrac{1}{6}(z_\alpha^2 - 1)\,s + \tfrac{1}{24}(z_\alpha^3 - 3z_\alpha)\,\kappa\right]. \tag{15.27}$$

The whole computation is $O(n^2)$ in the number of risk factors for the moment evaluation and $O(1)$ for the Cornish-Fisher adjustment — much cheaper than a full Monte Carlo revaluation. A portfolio with 500 risk factors and 10,000 options can have its delta-gamma-Cornish-Fisher VaR computed in under a second on a modern workstation; a full Monte Carlo revaluation of the same portfolio might take minutes or hours depending on the sophistication of the option pricers. The speed differential is the reason delta-gamma VaR survives even in the full-revaluation era: it is what the intraday risk engine runs, while the overnight batch runs the full Monte Carlo.

17.12.4 15.12.4 Accuracy regime

The accuracy of delta-gamma VaR depends on the magnitude of the stress move relative to the curvature of the payoff:

- *Daily 95% VaR, moderate vol.* The approximation is typically within a few percent of a full-revaluation VaR. Perfectly adequate for intraday limit monitoring and desk-level reporting.
- *Daily 99% VaR, stressed conditions.* Accuracy degrades to 5-15% typical error. Still useful as a first-pass indicator but should be validated against full revaluation before being used for capital.
- *10-day 99% VaR with options near expiry.* Can degrade severely because near-expiry gamma is explosive and the quadratic approximation misses third-order effects. Full revaluation or Monte Carlo is mandatory here.
- *Deeply out-of-the-money options or digital-like payoffs.* Delta-gamma breaks down entirely because the payoff is nearly discontinuous. Full revaluation is required.

Modern risk systems run delta-gamma-Cornish-Fisher VaR for intraday risk updates and full-revaluation Monte Carlo VaR for end-of-day reporting, taking advantage of the speed of the former and the accuracy of the latter. When the two disagree by more than a threshold, the system flags a review.

17.12.5 15.12.5 Moment-matching variants

A refinement of Cornish-Fisher is *moment matching*: fit a parametric distribution (e.g. a mixture of Normals, a Johnson curve, or a shifted log-Normal) whose first four moments match the closed-form moments $\mu_P, \sigma_P^2, m_3, m_4$ of the quadratic form. The quantile of the fitted distribution is taken as the VaR. Moment matching avoids the non-monotonicity problems of Cornish-Fisher at extreme parameter values and tends to be more accurate for moderately skewed distributions. The cost is a slightly more elaborate implementation — the fitting step is not purely closed-form — and in practice Cornish-Fisher and moment-matching give very similar answers except in pathological cases.

An alternative to both is *Imhof's method*: numerically invert the characteristic function of the quadratic form to recover the full distribution of ΔV, then take the exact quantile. Imhof is more accurate than Cornish-Fisher but computationally heavier and is typically reserved for end-of-day validation runs rather than intraday computation. The progression Delta-Normal → Delta-Gamma-Cornish-Fisher → Imhof → full Monte Carlo represents a ladder of increasing accuracy and increasing computational cost, with the risk team picking the appropriate rung for each use case.

17.12.6 15.12.6 Operational role

Delta-gamma parametric VaR is not merely a back-pocket approximation. In the modern risk stack it plays several distinct roles:

- *Intraday limit monitoring.* When a trader executes a new option trade, the risk system can refresh the desk's VaR in milliseconds using the delta-gamma approximation. Full revaluation is too slow for this use case.
- *Sensitivity of VaR to positions.* Because the formula (15.27) is differentiable in δ and Γ, the marginal VaR (§15.9) of any prospective trade can be computed in closed form. This supports pre-trade "what-if" analysis without running Monte Carlo for every candidate trade.
- *Regulatory backup.* If a bank's full-revaluation Monte Carlo engine fails or times out, the delta-gamma approximation provides a fallback VaR that can be used to continue limit monitoring and regulatory reporting while the primary engine is repaired.

Cross-reference: the Greeks δ and Γ used here are the multi-factor generalisations of the scalar deltas and gammas developed in Chapter 7. The single-name case $n = 1$ of (15.22) reduces exactly to the scalar delta-gamma expansion around the Black-Scholes price.

17.13 15.13 Hedged P&L Distribution and Hedged CTE

The input to a VaR calculation for an options desk is the *hedged* P&L distribution, not the unhedged P&L distribution. Hedging reshapes the underlying factor moves into a residual P&L that the desk actually experiences; the risk measure then quantifies the tail of that residual. Hedging decisions and risk-measurement decisions are therefore coupled: a tighter hedge means a smaller residual, which means a smaller VaR, which means lower capital consumption. Conversely, a looser hedge means larger residuals and higher capital. The CFO and the CRO have aligned interests in encouraging tight hedging, subject to the caveat that over-hedging generates transaction-cost bleed that itself shows up in P&L.

17.13.1 15.13.1 The shape of residual hedged P&L

Consider an option position $f(t, S_t)$ hedged with α_t shares and β_t units of money-market account, rebalanced on a grid $t_0 < t_1 < ... < t_N$. Between rebalance times the hedge is stale, and the residual P&L increment over interval (t_{k-1}, t_k) is, to leading order,

$$\Delta R_k \;\approx\; \tfrac{1}{2}\,\Gamma(t_{k-1}, S_{t_{k-1}}) \cdot (S_{t_k} - S_{t_{k-1}})^2 - \theta(t_{k-1}, S_{t_{k-1}}) \cdot \Delta t. \tag{15.28}$$

This is the celebrated Taylor-series identity at the heart of dynamic hedging (see Chapter 7 for the full derivation): the gamma-convexity term pays the holder whenever spot moves, and the theta-decay term charges the holder for the option's time decay. On average, under GBM with the Black-Scholes vol and frictionless rebalancing, the two exactly cancel — that is the content of the replication theorem. In practice, the cancellation is imperfect at every finite rebalance frequency, and the imperfection is the residual P&L.

The distribution of total hedged P&L over the option's life is approximately Gaussian for sufficiently fine rebalancing (by a central-limit argument across the sum of increments) with variance proportional to $1/N$ where N is the number of rebalance steps. More frequent rebalancing gives tighter hedged P&L. This is the source of the scaling law in §15.13.2 below.

17.13.2 15.13.2 Delta versus delta-gamma hedging and CTE reduction

The CTE of a hedged book is typically much smaller than the CTE of the unhedged book, but the reduction ratio depends on whether the hedge is first- or second-order. Rough scaling:

- *Delta-only hedge* reduces residual variance by a factor of approximately $\sigma^2\Delta t$ per rebalance step. Cumulatively over the life of an option with daily rebalancing, this translates to CTE reductions of 70-90% relative to the unhedged position.
- *Delta-gamma hedge* (using a second traded option to neutralise gamma) reduces residual variance by another factor of approximately $\sigma^2\Delta t$, giving CTE reductions of 95-99% relative to the unhedged position at realistic daily rebalance frequencies.

The gamma hedge squeezes the residual P&L into the third-order Taylor remainder, which scales as $(\Delta S)^3$ rather than $(\Delta S)^2$. Since $\Delta S \sim \sigma S\sqrt{\Delta t}$, the cube is of order $\sigma^3 S^3 \Delta t^{3/2}$, which is much smaller than the square $\sigma^2 S^2 \Delta t$ for small Δt. Each additional Greek matched reduces the hedging residual by another power of $\sqrt{\Delta t}$, so the improvements compound quickly under frequent rebalancing.

17.13.3 15.13.3 Asymmetry: long-gamma versus short-gamma books

The CTE framework naturally captures an asymmetry between long-gamma and short-gamma positions that VaR misses.

Long-gamma book. Positive Γ means the position profits from large moves in either direction. Residual P&L is bimodal around zero — positive in both large up-moves and large down-moves (gamma profit), negative in small moves (theta bleed). CTE of the *loss* side is modest because the loss distribution is bounded below by the accumulated theta: the worst plausible loss is the sum of theta charges over the rebalance interval, which is a small, bounded quantity. The tail of the *profit* distribution can be very large, but CTE of losses doesn't care about that.

Short-gamma book. Negative Γ means the position loses from large moves. Residual P&L is unbounded below — theoretically catastrophic large-move losses proportional to $\frac{1}{2}\Gamma(\Delta S)^2$ with $\Gamma < 0$. Under fat-tailed underlying distributions, the CTE of the loss side blows up much faster than the VaR because the tail is heavy-tailed in the quadratic of a heavy-tailed variable. This is why short-gamma books are subject to much tighter risk limits than equivalently-sized long-gamma books: the loss profiles are fundamentally different even if the variance (2nd moment) of the two books looks similar. CTE picks up the asymmetry; VaR (a quantile) does not.

A concrete example: consider a short straddle (short gamma) sized so that its 1-day 99% VaR is \$1M. Under Normal factor moves, its 1-day 99% CTE might be \$1.3M — a 30% uplift. Under Student-t_3 factor moves, the same VaR \$1M comes with a CTE of perhaps \$3-5M depending on the vol level. The short-gamma book is exposed to tail events that VaR cannot see, and CTE is the measure that reveals the exposure. A long straddle with identical VaR has a CTE close to the VaR itself (because the loss distribution is bounded below by theta), so the two books with identical VaR have wildly different CTE — and risk limits that target CTE automatically discriminate between them.

17.13.4 15.13.4 Stress-test implications for hedged books

Today's delta is not tomorrow's delta after a -25% equity stress: gamma converts the spot move into a delta move, leaving a nominally delta-neutral book with material residual delta. Stress tests of option books therefore use *scenario-dependent revaluation* (re-price under the shocked market, possibly with a shocked vol surface), not Taylor-series stress, to capture gamma and second-order vol/correlation effects.

17.13.5 15.13.5 Hedged CTE as a risk management target

Running a derivatives desk to minimise hedged CTE is subtly different from running it to minimise hedged variance. Variance minimisation penalises symmetric moves equally in both directions and is content with a long-gamma-heavy book whose upside and downside cancel in the variance. CTE minimisation cares only about the downside and actively prefers positions whose downside is bounded (long gamma, long options) to positions whose downside is open-ended (short gamma, short options). A desk run on a CTE mandate will systematically hold more long optionality than a variance-mandated desk, at the cost of more theta bleed. The trade-off between theta carry and tail safety is the central tension of derivatives portfolio management, and the risk measure chosen to govern the desk is what operationally resolves it.

17.14 15.14 Operational Integration

The end-to-end daily risk cycle on a large derivatives desk:

1. *Close, mark, and load.* Prices/vols/rates marked at close; trades reconciled with the front-office blotter; reference data refreshed.
2. *Greeks.* Position- and portfolio-level deltas, gammas, vegas, thetas computed on the end-of-day market.
3. *Intraday VaR.* Delta-gamma-Cornish-Fisher VaR via (15.22)–(15.27) — milliseconds, used for limit monitoring.
4. *End-of-day VaR.* Full-revaluation Monte Carlo over 10k–100k scenarios; the regulatory number.
5. *Stress tests.* Historical and hypothetical scenarios overlaid on the EoD book.
6. *PLA.* Actual / hypothetical / risk-theoretical series produced, reconciled, and residuals flagged (§15.10.1).
7. *Backtests.* Exceedance counts updated; traffic-light state on each model.
8. *Morning report.* VaR, CTE, stress, attribution, backtest, and limit-utilisation by desk/factor/region.

Model governance wraps the technical work: every model is in an inventory with an owner; new or materially changed models require independent validation sign-off; all production models undergo at least an annual re-review; some banks hold an explicit *model-risk reserve* sized as a fraction of the model's capital output. A beautifully-specified coherent risk measure is useless if the data is stale, the positions are mis-captured, or the Greeks are inconsistent across systems — operational integration is the unglamorous bedrock.

17.15 Case studies — three failures of VaR

This chapter's machinery is best motivated by the failures the dominant risk measure of the day did not pass. Each of the three episodes below cost a major institution (or a regulatory regime) tens of billions of dollars; in every case, the post-mortem changed how the industry computes risk. The technical content of the chapter — historical vs parametric vs Monte-Carlo VaR routes (§15.2), CTE/ES (§15.3), coherent-risk axioms (§15.6), and risk decomposition (§15.9) — is the residue these failures left behind.

17.15.1 Case study (a): LTCM 1998 — VaR understated leverage-amplified tails

Long-Term Capital Management's reported 99% 1*-day VaR sat near* \$45*M in summer* 1998 *— about* 1% *of the firm's* \$4.7*B equity capital. Actual losses between mid-August and late September* 1998 *ran to* \$4.6*B, approximately* 100× *the headline daily VaR. The* \$3.625*B private-sector recapitalisation organised by the Federal Reserve on* 23 *September prevented a forced unwind that would have damaged counterparty banks across Wall Street.*

Context. LTCM was a hedge fund founded in 1994 by John Meriwether (formerly head of Salomon Brothers' fixed-income arbitrage group), with Robert Merton and Myron Scholes on the partnership and a Nobel Prize awarded to the two of them in 1997 for the Black-Scholes-Merton work. The fund's strategies were predominantly fixed-income relative-value: convergence trades on Treasury basis (off-the-run vs on-the-run), swap-spread trades, sovereign-yield-curve relative-value (Italy vs Germany ahead of euro convergence), and a smaller equity-market-neutral and merger-arb book. Each individual position had small daily P&L variance because the underlying spreads moved within tight historical ranges; the headline edge per trade was small (5–30 bp annualised), and the Sharpe ratios of individual strategies were genuinely high (≈ 3 in normal conditions). To convert thin edge into target returns of 20%+, the fund ran $\sim$ 28:1 leverage on the equity base, with derivative notionals reportedly around \$1.25T.

The Russian sovereign default on 17 August 1998 triggered a global flight-to-quality that simultaneously moved every single LTCM position in the wrong direction: on-the-run Treasuries rallied versus off-the-runs, swap spreads widened to historic levels (90 bp on 10y from a pre-crisis $\sim$ 35 bp), Italian-German peripheral spreads widened, equity vol spiked, and merger-arb spreads blew out. The internal VaR — calibrated on 1995–1998 history under a Gaussian assumption with covariance matrix estimated on that benign window — assigned probability close to zero to a regime in which all of these positions moved together. In the historical sample, the off-the-run/on-the-run basis had cross-correlation of $\sim$ 0.1 with the swap-Treasury basis and $\sim$ 0.0 with the merger-arb book; in August-September 1998, all of those correlations went to $\sim$ 0.95 as a single liquidity-flight factor drove every spread.

Reading it through the chapter's math. LTCM's reported VaR was a parametric (delta-normal) calculation per (15.2): $\mathrm{VaR} = -z_\alpha \sigma_\Pi$ where $\sigma_\Pi^2 = w^\top \Sigma w$ is portfolio variance computed on the historical covariance matrix Σ. The failure was in the Σ input, not the formula: a benign-period Σ with low cross-correlations gives a small σ_Π, and \$45M is what the formula correctly returns for that input. The $\sim 100\times$ multiplier between reported VaR and realised loss decomposes roughly as: (i) a $\sim 5\times$ underestimate of marginal volatilities, since the historical sample contained no comparable shock; (ii) a $\sim 5\times$ underestimate of correlations, with most pairwise correlations going from ~ 0.1 to ~ 0.9 in the crisis; (iii) the remaining factor of $\sim 4\times$ from the leverage-and-time effect — losses compounded across ~ 30 trading days, not 1 day. Crucially, even Cornish-Fisher (15.27) would have helped only if the higher-moment inputs s, κ were stressed; with benign-period skewness and kurtosis estimates the Cornish-Fisher correction would have added perhaps 20% to the VaR, not the $100\times$ that was needed. The post-LTCM regulatory response was to require *stressed-VaR* — a parallel calculation using a Σ calibrated on a stressed historical window (typically including 1998, 2008, or another comparable crisis), with regulatory capital set to the maximum of normal-period and stressed-period VaR.

Lesson. Historical-window VaR is not a forward-looking risk measure; it is a reflection of how risky the past looked, multiplied by a critical-value z_α. If the past did not contain a crisis, the calculation cannot see one. LTCM is the textbook case of this failure, and the post-1998 Basel response — stressed-VaR add-on, mandatory inclusion of 1998-class crisis data in calibration windows, leverage caps for prime-broker financing of relative-value strategies — was the institutional fix. The deeper lesson for the new hire is that VaR's three computational routes (historical, parametric, Monte Carlo) all share the same underlying weakness: they reduce risk to a single number computed off historical inputs, and that number is only as good as the historical regime is representative of the forward regime. CTE/Expected Shortfall (15.5) does not solve this directly — ES on a benign-period Gaussian distribution is also small — but ES on a *stressed* distribution with explicit fat-tail components (Student-t via 15.6, EVT via 15.4.2) at least surfaces the tail-shape sensitivity that VaR collapses into a point quantile. The Sharpe-3 strategy at 28:1 leverage was not a VaR failure of formula but of *input regime* — the chapter's machinery is designed to make that distinction visible to anyone who reads the diagnostics.

17.15.2 Case study (b): Archegos March 2021 — single-name concentration not in cross-margined VaR

Archegos Capital Management — Bill Hwang's family-office reincarnation of the Tiger Asia hedge fund — collapsed on 26 *March* 2021 *with approximately* \$20*B of capital backing* \$80*B of total notional equity exposure across at least six prime-broker counterparties. Total prime-broker losses exceeded* \$10*B, with Credit Suisse alone taking* \$5.5*B (an event that contributed materially to the bank's eventual UBS rescue in* 2023*). No single PB had a complete view of Archegos's aggregate position; each saw only its slice and computed VaR on that slice in isolation.*

Context. Archegos was a family office, exempt from the SEC 13F reporting that would otherwise have made its long positions public. It built concentrated long exposure in approximately a dozen US and Chinese single-name equities — ViacomCBS, Discovery, Tencent Music, Vipshop, GSX Techedu, Baidu, and others — primarily through total-return swaps (TRS) with multiple prime brokers. A TRS gives the family office synthetic long exposure to the underlying equity (paying the funding cost, receiving total returns) without ever appearing as the holder of record on any beneficial-ownership filing. The PB books the position as "stock owned, equity-derivative liability to client", computes its own market risk (largely hedged because the bank holds the underlying), and computes its counterparty risk via initial margin sized against the client's potential mark-to-market default exposure. Initial margin levels at peak were approximately 15%–25% of TRS notional, compared with the 50% minimum margin under FINRA/Reg-T for direct retail equity purchases.

When ViacomCBS announced a secondary stock offering on 22 March 2021, the share price fell 9% that day and continued lower across the week. By 24–25 March, Archegos was unable to meet the cumulative margin calls flowing in from all of its PBs simultaneously. On the morning of 26 March, the prime brokers convened to discuss an orderly unwind; after Goldman Sachs broke ranks and began aggressively liquidating its TRS hedges into the market, the rest were forced to follow, and the resulting block-sale dump pushed the underlying names down further — 50%+ peak-to-trough on the most concentrated tickers. Goldman and Morgan Stanley, who unwound first, took losses below \$1B each; Credit Suisse and Nomura, who were slower, took \$5.5B and \$2.8B respectively. The total cross-PB loss was estimated at \$10B–\$12B.

Reading it through the chapter's math. This is a §15.10 P&L attribution failure dressed up as a VaR failure. Each PB ran a perfectly reasonable counterparty VaR on its own slice of the Archegos book: position size, single-name vol, expected liquidation horizon under its margin terms. The numbers each PB reported internally were not wrong on their own scale — Credit Suisse's pre-collapse counterparty VaR on the Archegos relationship was reportedly $\sim$ \$50M 1-day at 99%, against a position of $\sim$ \$20B notional, which is a defensible single-counterparty number when liquidating into a normal-volume market. The failure was *data scope*: the cross-PB total exposure to Archegos in single names like ViacomCBS was approximately 50% of the daily-volume capacity of the underlying stock, meaning that any forced liquidation would generate market impact dramatically beyond what each PB's own slice modelled. None of the PBs' VaR systems incorporated cross-counterparty concentration data, because none of them had access to the other PBs' books — a structural information gap that the family-office regulatory exemption made worse.

The §15.6 sub-additivity discussion is the right framing. VaR fails sub-additivity in known pathological cases; here we have a *reverse* failure — the *sum* of the per-PB VaRs is less than the *consolidated* aggregated risk, because each per-PB VaR uses a covariance matrix that is missing the cross-PB concentration block. Coherent measures (§15.6.2) handle this correctly *if* the input data is complete; with the data scope wrong, no measure can compensate. The §15.10 P&L attribution chain — actual P&L vs hypothetical vs risk-theoretical — would have surfaced the gap if Archegos's prime-brokerage agreements had required cross-counterparty exposure disclosure as a condition of margin terms. Post-Archegos, every major PB has rewritten its onboarding documentation to require disclosure of all swap counterparties for any client running cross-margined exposure above thresholds; SEC and CFTC have proposed (but not finalised as of the cessation date) rules requiring family offices above a size threshold to submit Form-PF-style aggregate-position reports to regulators.

Lesson. The deepest lesson from Archegos is that *VaR computed on a censored sub-portfolio is not a portfolio risk measure at all.* Coherence axioms, sub-additivity, the choice between historical-VaR and ES, the sophistication of the simulation engine — none of it matters if the input position vector is wrong. Risk infrastructure must own data scope: which counterparties, which custodians, which prime brokers, which intermediaries are aggregated into the "portfolio" for each risk computation, and what blind spots remain. For the new hire, the discipline is to ask, on every risk number you produce, "what is *not* in the position vector this number was computed on?" Archegos is what happens when nobody asks that question because the regulatory framework made it impossible to ask.

17.15.3 Case study (c): 2008 — Basel I VaR missed correlated-default tail; Basel III moves to ES

The 2007–2008 *financial crisis revealed that Basel I/II's* 99% 10*-day VaR was structurally blind to the correlated-default tail that destroyed bank capital. Basel* 2.5 *(*2009*) added the stressed-VaR overlay; Basel III/FRTB (*2013–2019*) moved the regulatory metric from* $\mathrm{VaR}_{99\%}$ *to* $\mathrm{ES}_{97.5\%}$ *and added the "non-modellable risk factor" framework. The crisis cost roughly* \$5 *T of writedowns at large banks globally and triggered the most comprehensive overhaul of bank capital regulation in fifty years.*

Context. Through Basel II (in force from 2004) the regulatory market-risk metric was 99% 10-day VaR computed on a 1-year rolling historical window, with regulatory capital set as $k \times \mathrm{VaR}$ for a multiplier $k \in [3.0, 4.0]$ tied to backtest performance under the Kupiec/Christoffersen tests of §15.7. Bank trading books in the mid-2000s held increasingly large positions in structured credit products: residential-mortgage-backed securities (RMBS), collateralised-debt obligations (CDOs), CDO-squared, credit-default swaps (CDS) on individual tranches, and synthetic tranches sold to investors. The dominant pricing model was the Gaussian-copula credit framework (Li, 2000), in which the joint default-time distribution of a portfolio of underlying mortgages or corporates is parameterised by pairwise correlations.

Bank VaR systems calibrated mortgage-asset correlations on the 2003–2006 historical window: house prices had risen consistently across all US regions, mortgage default rates were near historical lows, and the empirical pairwise correlation across regional mortgage pools was approximately 0.30. On this calibration, the 99% tail of the joint-default distribution was small: a senior CDO tranche might require 20% of the underlying pool to default before taking any loss, and the model estimated the probability of 20%+ correlated defaults as approximately 10^{-5} per year. The capital charges on senior tranches were therefore tiny — typical risk-weights of 7% versus 100% for direct corporate-loan exposure — and banks accumulated trillions of dollars of senior tranches under the assumption that they were genuinely safe.

When US house prices peaked in mid-2006 and began declining nationwide, the empirical mortgage-asset correlation jumped from the historical ~ 0.30 to ~ 0.95 within twelve months. The realised distribution of CDO losses was orders of magnitude worse than the model had projected: AAA-rated senior tranches that had been priced at 50 bp running spread defaulted with 50%+ loss given default, generating losses of multiple-hundred bp on instruments the regulatory framework had treated as effectively risk-free. Aggregate writedowns at globally systemically important banks reached \$1.4T by mid-2009 (IMF estimate), with the broader writedown across all financial-sector institutions exceeding \$5T over the cycle.

Reading it through the chapter's math. The §15.6 coherent-risk discussion is the analytical post-mortem. VaR's failure of sub-additivity (§15.6.1) under the Gaussian-copula credit framework was structural: aggregating two perfectly-correlated default exposures into one portfolio could *increase* reported VaR rather than decrease it under diversification, but more commonly the failure was the reverse — risk decompositions that summed the marginal VaR per tranche to a number much smaller than the actual tail loss, because the marginal calculations did not include the regime-shift correlation breakdown. The shift to $\mathrm{ES}_{97.5\%}$ (§15.3) was chosen so that for a Gaussian baseline the new ES roughly equals the old $\mathrm{VaR}_{99\%}$ — but for fat-tailed distributions ES scales with the tail-shape parameter (cf. (15.6) for Student-t) and reports the tail size that VaR hides. The §15.4.2 EVT framework was institutionalised as the FRTB "non-modellable risk factor" treatment, which requires risk factors lacking sufficient observable history to be capitalised under a stressed scenario shock rather than under model-implied VaR. The Basel evolution (§15.5) — VaR(99) → stressed-VaR add-on → ES(97.5) → FRTB internal models approach — is a direct translation of the academic risk-measure literature into binding regulatory capital.

Lesson. The 2008 crisis was simultaneously a failure of (i) input data — historical mortgage correlations from a benign housing-price regime were used to calibrate the tail of a joint-default distribution that turned out to be regime-dependent; (ii) measure choice — VaR's quantile structure systematically hid the size of tail losses that ES would have surfaced; and (iii) coherence — VaR's failure of sub-additivity allowed risk decompositions to under-aggregate concentrated exposures. The Basel response addressed all three: stressed-VaR for input-regime sensitivity, ES for tail-shape visibility, and the FRTB internal-models framework with desk-level approval and standardised approach floors for the coherence/aggregation issue. None of this prevents the next crisis — but it does prevent the *next-time-the-same* crisis, which is the realistic ambition of regulatory risk frameworks. The deeper lesson for the new hire is that risk-measure choice is a research question masquerading as an operational one: VaR vs ES is not a matter of taste, it is a matter of which tail-shape sensitivity the institution wants to be able to see in its reports. For any portfolio with non-Gaussian exposures — credit, options, illiquids, anything with optionality on a regime variable — ES (or CTE) is the structurally more informative measure, and the post-2008 regulatory regime correctly institutionalises that choice.

17.16 15.15 Key Takeaways

1. **VaR is a quantile; CTE is a conditional tail mean.** VaR answers "what loss will I not exceed with $(1-\alpha)$ confidence" and ignores the size of losses beyond the cutoff. CTE integrates the tail beyond VaR and captures tail shape. For Normal losses at 95%, CTE exceeds VaR by about 0.4 standard deviations; for Student-t_3 the gap is 1.8 standard deviations — a five-fold amplification that CTE correctly reports and VaR systematically hides.

2. **VaR fails sub-additivity; CTE does not.** Two independent 4%-default loans each have 95% VaR of zero, but their merged 95% VaR is 100 — diversification *increased* reported risk. CTE obeys sub-additivity and is the correct axis for any diversification-aware framework. The VaR-to-CTE ratio is itself diagnostic: well above the Gaussian benchmark signals fat tails.

3. **Three VaR estimation families triangulate.** Historical simulation is non-parametric and model-light but limited by sample size. Parametric (delta-normal, delta-gamma) VaR is fast and decomposable but assumes Gaussian factors. Monte Carlo is flexible but computationally heavy and model-risk-prone. A mature risk function runs multiple methodologies in parallel and treats disagreements as diagnostic.

4. **Cornish-Fisher is the workhorse of fast VaR.** Equation (15.4) adjusts the Gaussian quantile for skewness and kurtosis without a simulation. Combined with the closed-form delta-gamma moments (15.23)–(15.25), it produces a fast and reasonably accurate VaR that captures both non-linearity (via Γ) and tail thickness without a Monte Carlo path.

5. **Coherent risk measures satisfy four axioms.** Monotonicity, translation invariance, positive homogeneity, sub-additivity. Every coherent measure has a dual representation as a worst-case expectation over a set of scenarios. CTE fits; VaR does not. Spectral measures generalise both.

6. **Fat tails are the empirical norm, not the exception.** Daily equity tail indices sit in the range 3-5; EM currencies and distressed credit drop to 2. Any Normal-calibrated risk framework systematically under-reports tail risk in these markets. Defensive adjustments include Student-t fits, filtered historical simulation, GPD-based extreme-value models, and GARCH conditional vol.

7. **Regulation has migrated from VaR to ES.** Basel 1996 introduced 99% VaR; Basel 2.5 added stressed VaR; FRTB (2016/2019) replaced VaR with 97.5% expected shortfall. The percentiles are calibrated so Normal-$\text{ES}_{97.5}$ Normal-VaR_{99}, but ES additionally penalises fat tails. FRTB also prescribes liquidity horizons by asset class, non-modellable risk factor charges, and a P&L attribution test.

8. **Backtesting validates VaR through Kupiec and Christoffersen.** Kupiec's POF test compares exceedance counts to a binomial null. Christoffersen's independence test catches clustered exceedances. The combined conditional-coverage test is the gold standard. CTE is harder to backtest because it is not elicitable; FRTB uses indirect tests via multi-level VaR.

9. **Stress testing complements VaR.** Historical scenarios replay specific events (1987, 2008, 2020); hypothetical scenarios imagine plausible but unrealised catastrophes; reverse stress asks what scenarios produce failure. A risk function that reports VaR without stress, or stress without VaR, is only half equipped.

10. **Risk decomposition answers "where does my risk come from."** Marginal VaR (derivative), component VaR (Euler allocation, sums to total), and incremental VaR (finite difference on removal) give three related but distinct attributions, each useful for different decisions. Component VaR is natural for budget allocation; incremental VaR is natural for unwind decisions.

11. **P&L attribution closes the risk-management loop.** Equation (15.20) decomposes realised P&L into Greek contributions plus a residual. A systematically large residual is diagnostic of model or data problems. FRTB formalises the comparison as a P&L attribution test; failing the test forces the bank to use the standardised approach.

12. **Liquidity-adjusted risk layers in bid-ask and horizon effects.** Simple exogenous versions add a spread haircut; FRTB institutionalises liquidity via asset-class-specific horizons (10 days for major equities, 120 days for exotic credit). Funding liquidity (LCR, NSFR) interacts with market liquidity in crisis regimes — the 2008 and 2020 doom-loops are the canonical cautionary tales.

13. **Delta-gamma parametric VaR is the computational workhorse.** Closed-form moments (15.23)–(15.26) combined with Cornish-Fisher (15.27) deliver VaR in $O(n^2)$ per evaluation — milliseconds for a 500-factor portfolio. Accurate to a few percent for daily 95% VaR under moderate conditions; degrades for near-expiry options and extreme tails, where full revaluation is mandatory.

14. **Hedging reshapes the loss distribution that risk measures are applied to.** The input to VaR for an options book is the *hedged* P&L distribution. Delta hedging typically reduces CTE by 70-90%; delta-gamma hedging by 95-99%. CTE distinguishes long-gamma (bounded loss tail) from short-gamma (unbounded loss tail) books that may have identical VaR — exactly the asymmetry CTE was designed to capture.

15. **The end-to-end risk cycle is an operational system, not a single calculation.** Position capture, market-data validation, Greek computation, intraday delta-gamma VaR, overnight full-revaluation Monte Carlo, stress tests, P&L attribution, backtests, and morning reports all must reconcile. The mathematical elegance of coherent risk measures is useless if the data pipeline corrupts the inputs. Operational integration is where the framework succeeds or fails in practice.

17.17 15.16 Reference Formulas

17.17.1 Core definitions

Value-at-Risk at confidence $(1-\alpha)$:

$$\text{VaR}_\alpha(L) \;=\; \inf\{\ell : \mathbb{P}(L \le \ell) \ge 1-\alpha\}.$$

Conditional Tail Expectation / Expected Shortfall:

$$\text{CTE}_\alpha(L) \;=\; \text{ES}_\alpha(L) \;=\; \mathbb{E}[L \mid L \ge \text{VaR}_\alpha(L)].$$

17.17.2 Parametric VaR formulas

Delta-normal VaR under Gaussian factors with covariance Σ and exposure vector w:

$$\mathrm{VaR}_\alpha \;=\; z_{1-\alpha}\sqrt{w^\top \Sigma w}.$$

Cornish-Fisher expansion of a non-Gaussian quantile:

$$q_\alpha \;\approx\; z_\alpha + \tfrac{1}{6}(z_\alpha^2-1)\,s + \tfrac{1}{24}(z_\alpha^3-3z_\alpha)\,\kappa - \tfrac{1}{36}(2z_\alpha^3-5z_\alpha)\,s^2,$$

with skewness s and excess kurtosis κ.

17.17.3 Delta-gamma parametric VaR

For book P&L approximated by $\Delta V = \delta^\top \Delta r + \frac{1}{2}\Delta r^\top \Gamma \Delta r$ with $\Delta r \sim \mathcal{N}(0,\Sigma)$:

$$\mu_P \;=\; \tfrac{1}{2}\operatorname{tr}(\Gamma\Sigma), \qquad \sigma_P^2 \;=\; \delta^\top \Sigma\,\delta + \tfrac{1}{2}\operatorname{tr}((\Gamma\Sigma)^2),$$

$$m_3 \;=\; 8\operatorname{tr}((\Gamma\Sigma)^3) + 6\,\delta^\top \Sigma\,\Gamma\,\Sigma\,\delta,$$

$$m_4 \;=\; 12\operatorname{tr}((\Gamma\Sigma)^4) + 48\,\delta^\top(\Sigma\Gamma)^2\Sigma\,\delta + 3\,(\sigma_P^2)^2,$$

feeding into Cornish-Fisher with skewness $s = m_3/\sigma_P^3$ and excess kurtosis $\kappa = m_4/\sigma_P^4 - 3$:

$$\mathrm{VaR}_\alpha \;\approx\; -\mu_P + \sigma_P\left[z_\alpha + \tfrac{1}{6}(z_\alpha^2-1)\,s + \tfrac{1}{24}(z_\alpha^3-3z_\alpha)\,\kappa\right].$$

17.17.4 Student-t tail

$$\mathrm{CTE}_\alpha^{t_\nu} \;=\; \frac{f_{t_\nu}(\mathrm{VaR}_\alpha^{t_\nu})\,(\nu + (\mathrm{VaR}_\alpha^{t_\nu})^2)}{(\nu-1)\,\alpha}.$$

17.17.5 Extreme Value Theory — peaks over threshold

Generalised Pareto tail approximation for exceedances over threshold u:

$$F_u(y) \;\approx\; 1 - \left(1 + \xi\frac{y}{\beta}\right)^{-1/\xi}, \qquad y > 0,$$

with shape ξ (positive = Fréchet/power-law, zero = Gumbel/exponential, negative = Weibull/bounded) and scale $\beta > 0$.

17.17.6 Coherence axioms (loss convention)

A risk measure ρ on losses is coherent if for all losses L, L_1, L_2 and constants $c, \lambda \geq 0$:

- **Monotonicity:** $L_1 \leq L_2 \Rightarrow \rho(L_1) \leq \rho(L_2)$.
- **Translation invariance:** $\rho(L+c) = \rho(L) + c$.
- **Positive homogeneity:** $\rho(\lambda L) = \lambda\rho(L)$.
- **Sub-additivity:** $\rho(L_1+L_2) \leq \rho(L_1) + \rho(L_2)$.

CTE/ES satisfies all four; VaR fails sub-additivity.

17.17.7 Dual representation of coherent measures

Every coherent ρ equals a worst-case expectation over some set $\mathcal{Q}$ of probability measures:

$$\rho(L) \;=\; \sup_{\mathbb{Q}\in\mathcal{Q}} \mathbb{E}^{\mathbb{Q}}[L].$$

Spectral risk measures:

$$\rho_\phi(L) \;=\; \int_0^1 \phi(u)\,F_L^{-1}(u)\,du, \qquad \phi \geq 0,\ \phi \text{ non-decreasing}, \ \int \phi = 1.$$

VaR corresponds to $\phi = \delta(u-(1-\alpha))$; CTE corresponds to $\phi(u) = \mathbf{1}_{u \geq 1-\alpha}/\alpha$.

17.17.8 Kupiec proportion-of-failures test

$$\mathrm{LR}_{\mathrm{POF}} \;=\; -2\log\!\left[\frac{\alpha^x(1-\alpha)^{n-x}}{(x/n)^x(1-x/n)^{n-x}}\right] \;\sim\; \chi^2_1,$$

with x = exceedance count over n days.

17.17.9 Christoffersen independence and conditional-coverage tests

$$\mathrm{LR}_{\mathrm{IND}} \;\sim\; \chi^2_1, \qquad \mathrm{LR}_{\mathrm{CC}} \;=\; \mathrm{LR}_{\mathrm{POF}} + \mathrm{LR}_{\mathrm{IND}} \;\sim\; \chi^2_2.$$

17.17.10 Risk decomposition

Under the delta-normal model with $\sigma_P = \sqrt{w^\top \Sigma w}$:

$$\mathrm{mVaR}_i \;=\; z_{1-\alpha}\,\frac{(\Sigma w)_i}{\sigma_P}, \qquad \mathrm{cVaR}_i \;=\; w_i\,\mathrm{mVaR}_i, \qquad \sum_i \mathrm{cVaR}_i \;=\; \mathrm{VaR}.$$

Incremental VaR of position i:

$$\mathrm{iVaR}_i \;=\; \mathrm{VaR}(w) - \mathrm{VaR}(w_{-i}),$$

where w_{-i} is w with the i-th component set to zero.

17.17.11 P&L attribution

$$\Delta V \;=\; \delta\cdot\Delta S + \tfrac{1}{2}\Gamma(\Delta S)^2 + \nu\cdot\Delta\sigma + \theta\cdot\Delta t + \rho\cdot\Delta r + \text{cross-Greeks} + \text{cash flows} + \text{residual}.$$

17.17.12 Liquidity-adjusted VaR (exogenous spread)

$$\mathrm{LVaR}_\alpha \;=\; \mathrm{VaR}_\alpha + \tfrac{1}{2}\,|w|\cdot\text{spread}.$$

17.17.13 Notation glossary

- L — loss random variable, large positive = painful.
- α — tail probability (small); $(1-\alpha)$ is the confidence level.
- z_α — standard Normal α-quantile.
- F_L, F_L^{-1} — CDF and quantile function of L.
- σ_P — portfolio standard deviation.
- w, Σ — exposure vector and factor covariance matrix.
- δ, Γ — first- and second-order sensitivity vector and matrix (multi-factor Greeks).
- ν, θ, ρ — vega, theta, rho scalar Greeks.
- s, κ — standardised skewness and excess kurtosis.
- ξ, β — GPD shape and scale parameters.
- ϕ — spectral weighting function.